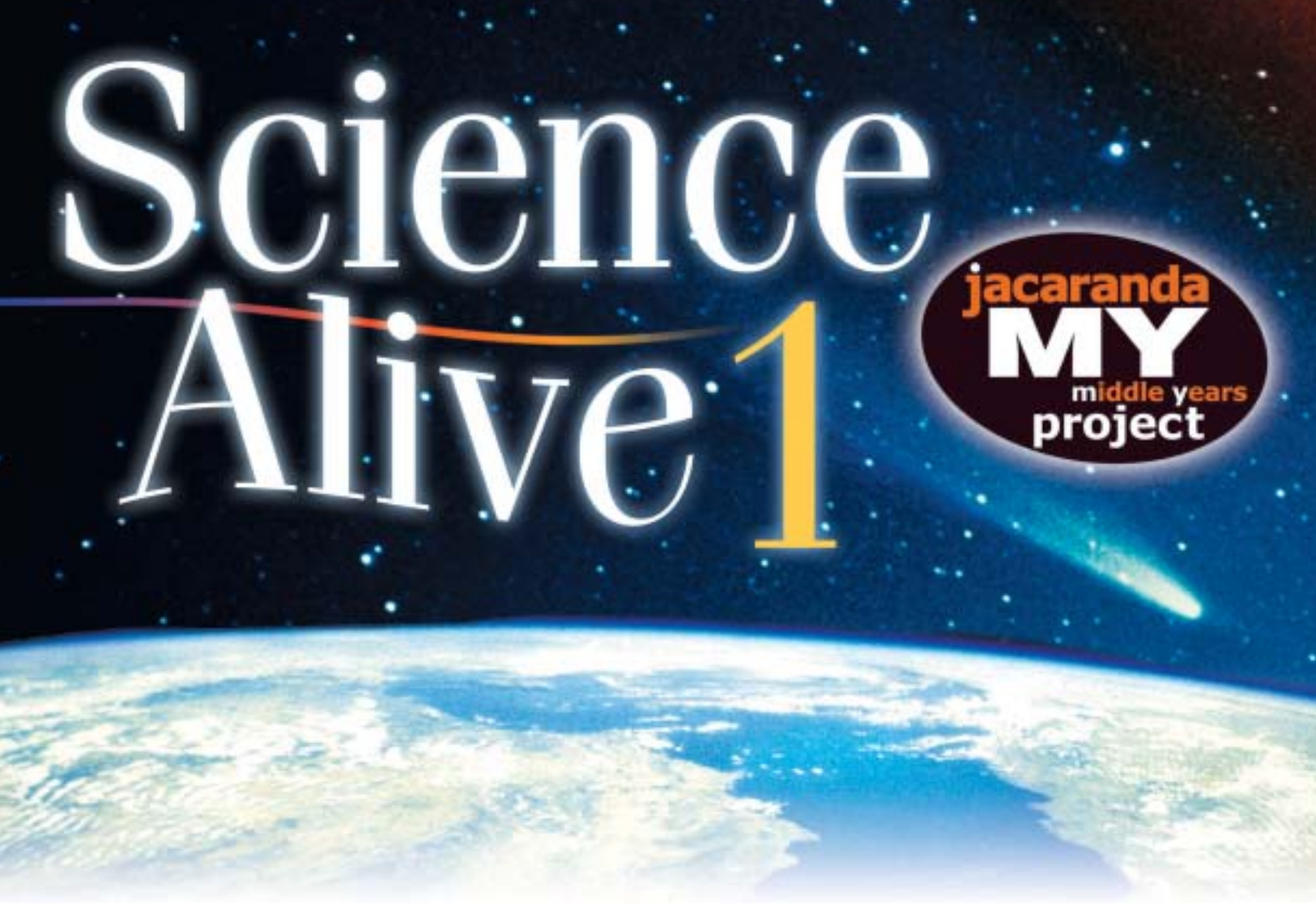
Science
Alive 1
jacaranda
MY
middle years
project

Science Alive 1

jacaranda MY middle years project

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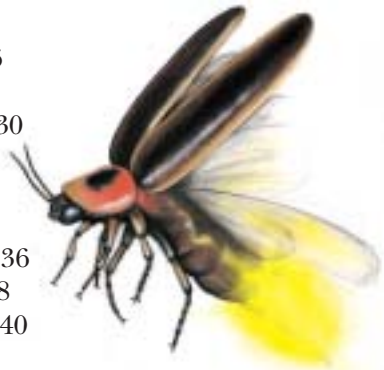

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Coverage of Multiple Intelligences

Activities in *Science Alive 1*, and in its companion text *Science Worksheets for Multiple Intelligences 1*, cater for a wide range of learning abilities and styles. Many address more than one learning style, and almost all require the exercise of some degree of verbal/linguistic intelligence.

The following grid lists the activities in these two texts that have been specifically written to engage a predominant intelligence.

Chapter	Intelligence type	Activity in *Science Alive 1**	Worksheet in *Science Worksheets for Multiple Intelligences 1***
1. Science alive!	Verbal/linguistic	5(10), 23(1f)	
	Visual/spatial	5(9), 7(5), 11(7), 15(8), 5(HOS***)	
	Bodily/kinaesthetic	7(HOS)	1
	Intrapersonal	13(8), 17(5)	4
	Interpersonal	3(8), 13(7), 17(4), 19(4), 21(7)	3
	Logical/mathematical	9(7), 11(6), 13(4, 5, 6), 15(5), 8(HOS), 22(8), 23(11)	2
	Musical/rhythmic		
	Naturalist	19(5, 6), 21(5, 6)	5
2. Separating mixtures	Verbal/linguistic	45(7)	7, 9
	Visual/spatial	39(6), 43(6)	6
	Bodily/kinaesthetic	27(HOS), 32(HOS), 33(8), 35(HOS), 37(HOS, 7), 39(HOS)	8
	Intrapersonal		
	Interpersonal	29(8), 31(9), 39(8)	10
	Logical/mathematical	27(6, 7, 8), 35(6, 7), 39(7), 44(HOS), 46(4b), 47(13, 1c)	
	Musical/rhythmic	45(7)	
	Naturalist	29(9), 41(5, 6)	
3. Machines everywhere	Verbal/linguistic	67(1), 71(4)	14
	Visual/spatial	59(6), 61(5), 63(8, 9), 65(7), 69(1, 6, 7), 70(3), 71(14, 3)	13
	Bodily/kinaesthetic	57(5), 61(HOS, 6)	12, 15
	Intrapersonal		
	Interpersonal		
	Logical/mathematical	53(5), 55(2, 4, 5, HOS), 59(5)	11
	Musical/rhythmic		
	Naturalist		
4. The changing Earth	Verbal/linguistic	77(8), 79(7), 93(9)	
	Visual/spatial	95(1, 4)	19
	Bodily/kinaesthetic	75(7), 79(HOS), 83(HOS), 85(HOS), 91(7)	16
	Intrapersonal		17
	Interpersonal		
	Logical/mathematical	75(8), 77(7), 87(7), 89(7), 81(7), 94(16), 95(4)	20
	Musical/rhythmic		
	Naturalist	85(7, 8)	18
5. Forensic science	Verbal/linguistic	99(10), 101(4, 9), 103(6), 115(7)	
	Visual/spatial	102(HOS), 103(3), 105(7), 118(10, 18)	22, 23
	Bodily/kinaesthetic	105(HOS), 107(HOS), 109(HOS), 111(HOS), 113(HOS)	24
	Intrapersonal	99(11), 103(8), 110(HOS)	
	Interpersonal	109(8), 111(6)	21
	Logical/mathematical	99(9, 10), 101(5), 107(7), 109(7), 113(5), 117(6, 8), 119(1–3)	25
	Musical/rhythmic		
	Naturalist		

Chapter	Intelligence type	Activity in *Science Alive 1**	Worksheet in *Science Worksheets for Multiple Intelligences 1***
6. Magnetic and electric effects	Verbal/linguistic	123(7), 133(6), 135(9)	29
	Visual/spatial	125(7), 129(6), 141(8), 142(8)	26
	Bodily/kinaesthetic	123(HOS), 124(HOS), 125(HOS), 128(HOS), 129(HOS), 136(HOS), 137(HOS)	
	Intrapersonal		
	Interpersonal		28
	Logical/mathematical	123(8), 129(8), 131(5, 6), 137(6), 139(HOS, 9), 141(7), 142(11), 143(5, 6)	27
	Musical/rhythmic		30
	Naturalist		
7. Cells	Verbal/linguistic	165(13)	
	Visual/spatial	155(8), 157(10, 12), 159(10), 161(9)	32, 34
	Bodily/kinaesthetic	152(HOS), 153(HOS, 9), 157(HOS), 159(HOS)	35
	Intrapersonal	167(5, 6, 7)	
	Interpersonal	147(11)	
	Logical/mathematical	153(5, 8), 159(9), 163(10)	31
	Musical/rhythmic		
	Naturalist	165(12)	33
8. Classification	Verbal/linguistic		36
	Visual/spatial	171(8), 173(3), 181(6)	
	Bodily/kinaesthetic	181(7), 191(16)	
	Intrapersonal		40
	Interpersonal	191(3)	
	Logical/mathematical	171(4, 5, 6), 173(2, 3), 175(8), 185(8, 9), 187(8), 190(10, 11)	39
	Musical/rhythmic		
	Naturalist	171(7), 177(9), 179(4, 5, 6), 181(5, 8), 184(HOS), 185(10, 11), 187(7, 9), 189(HOS, 5)	37, 38
9. Neighbours in space	Verbal/linguistic	201(7, 8), 207(8), 213(8), 214(5), 215(3b,c)	41
	Visual/spatial	199(HOS, 7, 8), 203(6, 7), 207(5, 6), 209(6, 7), 211(5), 214(1), 215(3a)	45
	Bodily/kinaesthetic	197(6), 205(HOS), 207(HOS), 209(HOS), 211(8), 214(6), 215(2)	42, 43
	Intrapersonal	195(3), 201(4, 6), 214(7)	
	Interpersonal	197(6), 213(9), 215(9)	
	Logical/mathematical	195(7), 197(4, 5), 199(6), 201(5), 211(5, 7)	44
	Musical/rhythmic		
	Naturalist		
10. Light and sound	Verbal/linguistic	219(9)	46
	Visual/spatial	225(4, 5), 227(7), 238(1, 3, 6)	47
	Bodily/kinaesthetic	219(HOS), 221(HOS), 222(HOS), 227(HOS), 233(9), 239(1)	
	Intrapersonal		
	Interpersonal		50
	Logical/mathematical	219(8), 229(6, 7, 8), 231(9), 239(2, 10, 13)	48
	Musical/rhythmic	229(9), 232(HOS)	49
	Naturalist	219(10)	

* Page number followed by activity number in brackets
** Worksheet number
*** Hands on Science

Coverage of outcomes

Science Alive 1 (with its companion text *Science Alive 2*) has been designed to allow students to achieve Science Level 5 CSF II outcomes. The learning outcomes listed below are extracts from the Curriculum and Standards Framework II document. (The Victorian Curriculum and Assessment Authority recommends that teachers and students address the entire text of the document.) The page references cited are those on which there are activities related to the outcome indicator.

Biological science

Living together: past, present and future

Learning outcome and chapter	Indicators. Evident when student can:	Page references
5.1 Explain the biological basis of classification of organisms into major groups. *Science Alive 1* Chapter 8: Classification	■ identify patterns of similarities and differences between a range of living things ■ define the major characteristics used in the 5-Kingdom system of classification ■ explain why particular sets of features, for example, colour, movement and structural features, are useful or not useful, to sort organisms using dichotomous keys.	171, 173, 175, 177, 179, 181, 184, 185, 190, 191 175, 187, 189, 190, 191 173, 177, 179, 181, 185, 190
5.2 Describe interactions between living things and between living things and their non-living surroundings. *Science Alive 2* Chapter 7: Living together Chapter 8: Changing times	■ describe different interactions in an ecosystem, including competition, predation, collaboration, parasitism, pollination, reproduction and parenting ■ construct a food web of organisms in an ecosystem ■ show graphically relationships between members of food chains, including a parasite-host relationship and producer-consumer relationships ■ describe the effect of changes in the environment on interactions in an ecosystem.	*Science Alive 2*

Structure and function

Learning outcome and chapter	Indicators. Evident when student can:	Page references
5.3 Relate the structure and organisation of different cells to their function. *Science Alive 1* Chapter 7: Cells	■ identify major structural components of cells as viewed at light microscope level ■ determine the functions of cells from their observable features ■ describe organisational relationships between organs, tissues, cells and systems.	152, 153, 155, 157, 159, 161, 166, 167 147, 149, 159, 161, 166, 167 163, 165, 167
5.4 Explain how plants and animals obtain, transport and use nutrients. *Science Alive 2* Chapter 3: Nutrients	■ describe the mechanical and chemical processes of digestion ■ describe the role of the circulatory system in transporting the products of digestion to cells ■ illustrate the pathway taken by water and minerals from the soil to the leaves of a flowering plant ■ identify the categories of inorganic and organic nutrients required by plants and animals and the uses to which they are put.	*Science Alive 2*

Chemical science

Substances: structure, properties and uses		
Learning outcome and chapter	**Indicators. Evident when student can:**	**Page references**
5.1 Use a simple particle model to explain the structure and properties of solids, liquids and gases. *Science Alive 2* Chapter 1: Substances: A look inside	■ describe the structure of solids, liquids and gases in terms of the arrangement and motion of particles in each physical state ■ relate readily observable properties of matter to particle bonds and energy.	*Science Alive 2*
5.2 Relate the safe use and disposal of common substances to their physical and chemical properties. *Science Alive 1* Chapter 2: Separating mixtures Laboratory toolbox (Also in *Science Alive 2* Chapter 3: Chemical reactions Chapter 8: Changing times)	■ describe and demonstrate safe laboratory techniques in the handling of substances used in the classroom when heating, cooling, transferring and cleaning up ■ distinguish between common substances that are safe for sink disposal, and those that are not, in terms of solubility in water and effect on the environment ■ describe alternative methods of disposal of common substances that are not safe for sink disposal ■ outline and analyse several environmental problems caused by the inappropriate disposal of toxic substances.	5, 7, 8, 22 5, 41 43 *Science Alive 2*

Chemical reactions		
Learning outcome and chapter	**Indicators. Evident when student can:**	**Page references**
5.3 Describe ways of producing a chemical change and influencing its rate. *Science Alive 2* Chapter 4: Chemical reactions	■ describe the particular conditions needed for some chemical reactions to occur ■ fully explain slow and fast chemical reactions ■ describe factors which affect the rate of a reaction.	*Science Alive 2*
5.4 Relate simple procedures for preparing and separating mixtures to medical and industrial procedures. *Science Alive 1* Chapter 2: Separating mixtures	■ describe a range of techniques for separating and concentrating mixtures ■ apply an appropriate method for separating the components of a mixture ■ describe the formation of colloids, including emulsions ■ describe medical and industrial applications of separating techniques.	27, 29, 31, 32, 33, 37, 39, 41, 43, 45, 46, 47, 111 27, 32, 33, 37, 39, 44, 45, 46, 47, 111 35, 46 37, 39, 41, 43, 45, 47

Earth and space sciences

The changing Earth

Learning outcome and chapter	Indicators. Evident when student can:	Page references
5.1 Describe the formation, composition and cycling of rocks. *Science Alive 1* Chapter 4: The changing Earth	■ identify the lithosphere as the region of the Earth where rocks are formed	75, 77, 79, 81, 83, 87, 94
	■ distinguish between sedimentary, igneous and metamorphic rocks on the basis of their formation and composition	77, 79, 81, 83, 87, 89, 91, 94, 95
	■ describe ways to estimate the age of rocks	87, 91, 94, 95
	■ explain the rock cycle.	83, 85, 87, 89, 95
5.2 Relate the properties of rocks to the ways in which they are used. *Science Alive 2* Chapter 5: What's in a rock?	■ describe properties of igneous, sedimentary and metamorphic rocks in terms of composition, grain size, colour and texture	*Science Alive 2*
	■ describe the relationship between rocks, ores and minerals	
	■ explain why igneous, sedimentary and metamorphic rocks are used for particular purposes.	

Our place in space

Learning outcome and chapter	Indicators. Evident when student can:	Page references
5.3 Describe how the positions of the planets, moon, sun and stars affect natural phenomena. *Science Alive 1* Chapter 9: Neighbours in space	■ describe how the tides are affected by the positions of the Earth, moon and sun	209, 211, 214
	■ account for the different phases of the moon in the lunar cycle	209, 211, 214
	■ explain the phenomenon of the seasons in terms of relative positions of the Earth and sun	201, 205, 207, 214
	■ account for the apparent annual movement of the stars and constellations across the sky.	199, 201, 203, 215
5.4 Describe major components of the universe. *Science Alive 2* Chapter 9: The universe	■ define, in simple terms, what is meant by 'the universe'	*Science Alive 2*
	■ summarise major characteristics of selected features of the universe beyond our solar system	
	■ describe different kinds of evidence which contribute to knowledge of the universe.	

Physical science

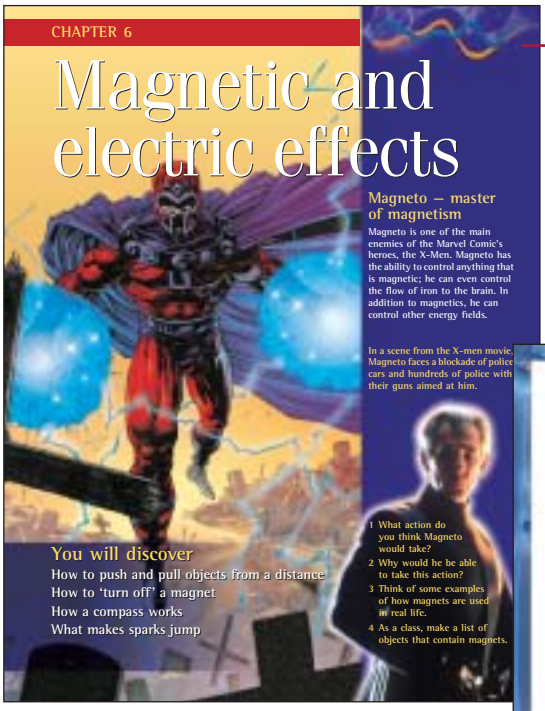
Energy and its uses		
Learning outcome and chapter	**Indicators. Evident when student can:**	**Page references**
5.1 Describe the characteristics and applications of the transmission and reflection of energy in the form of heat, light and sound. *Science Alive 1* Chapter 10: Light and sound (Also in *Science Alive 2* Chapter 2: Heat)	■ use the terms conduction, radiation and absorption to describe heating and cooling effects ■ describe how insulated containers maintain the temperature of the contents	*Science Alive 2*
	■ recognise patterns of reflection of light from plane and curved mirrors and relate these to applications of mirrors	225, 226, 227, 238, 239
	■ describe similarities and differences in the way light, heat and sound are transmitted.	219, 221, 222, 223, 225, 229, 231, 232, 233, 235, 237, 238, 239 and *Science Alive 2* Chapter 2
5.2 Describe the operation of direct current (DC) series and parallel circuits in terms of current and voltage. *Science Alive 2* Chapter 6: Electric circuits	■ correctly connect commonly used components in series and parallel circuits ■ describe the operation of series and parallel circuits, using terms such as current and voltage ■ identify from circuit diagrams those circuit elements that are connected in series and those that are connected in parallel ■ relate the brightness of a torch globe to the magnitude of the current in it and the voltage across it ■ link the brightness of two or more globes connected in series and in parallel circuits to the magnitude of the voltage and current.	*Science Alive 2*

Forces and their effects		
Learning outcome and chapter	**Indicators. Evident when student can:**	**Page references**
5.3 Describe simple magnetic and electrostatic effects in terms of a field model. *Science Alive 1* Chapter 6: Magnetic and electric effects	■ describe field patterns surrounding differently shaped magnets and simple combinations of magnets	128, 129, 142
	■ describe attraction and repulsion of magnets and objects near magnets as effects of the magnetic field	123, 124, 125, 127, 129, 142, 143
	■ explain the action of a compass as the movement of a magnet in the magnetic field of the Earth	129, 131, 142
	■ explain attraction and repulsion of charged objects in terms of the electric field around them	136, 137, 139, 141, 143
	■ describe the behaviour of objects in an electric field.	136, 137, 139, 141, 143
5.4 Explain how mechanical systems can direct and modify force and motion. *Science Alive 1* Chapter 3: Machines everywhere	■ describe the action of mechanical parts, such as gears, pulleys and levers in transmitting and modifying forces	51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 70, 71
	■ describe the action of simple machines that change the size or direction of movement	51, 53, 55, 57, 59, 61, 63, 65, 67, 69, 70, 71
	■ define the operation of one or more examples of simple mechanical systems	53, 55, 57, 59, 63, 65, 67, 69, 70, 71
	■ construct a model to demonstrate the operation of an everyday piece of equipment, tool or appliance	55, 57, 59, 61, 71
	■ describe how the model produces a mechanical advantage.	51, 55, 57, 59, 61, 69, 71

Skills, processes and procedures
The science skills, processes and procedures described in the Victorian Curriculum and Standards Framework II as appropriate to Level 5 are addressed in the 'Hands on Science' activities throughout the text, and especially in the Laboratory Toolbox and Chapter 5: Forensic Science.

How to use this book

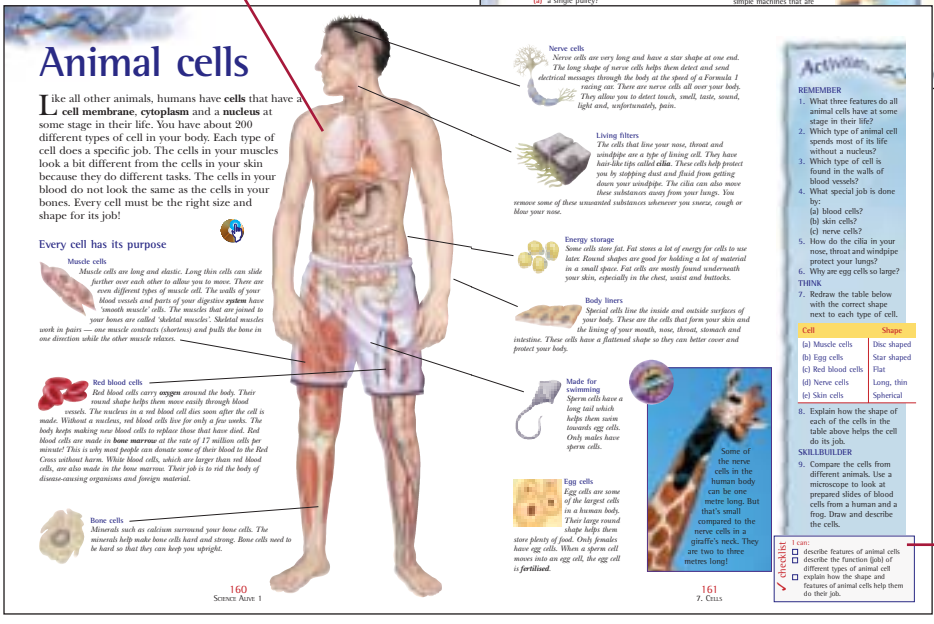
Science Alive 1 has been designed to engage and challenge young adolescent students of science. The features outlined here demonstrate how to use this stimulating resource most effectively.



Stunning opening pages build on what students know and think now.

Concluding page in each chapter is both a glossary and a summary of key terms.

Spectacular labelled illustrations will engage both the visual and linguistic learner.



'Check and challenge' spread — the last full spread in each chapter — allows students to revise, apply and extend their learning.

Checklists at the end of every spread allow students to check and acknowledge their progress.

'How about that!' features provide snippets of special interest to motivate and arouse curiosity.

The 'Then and now' features link the science of today with the science of the past.

Many spreads are linked to exciting worksheets that test the full range of multiple intelligences.

Practical multiple intelligence worksheets link to icons in the textbook.

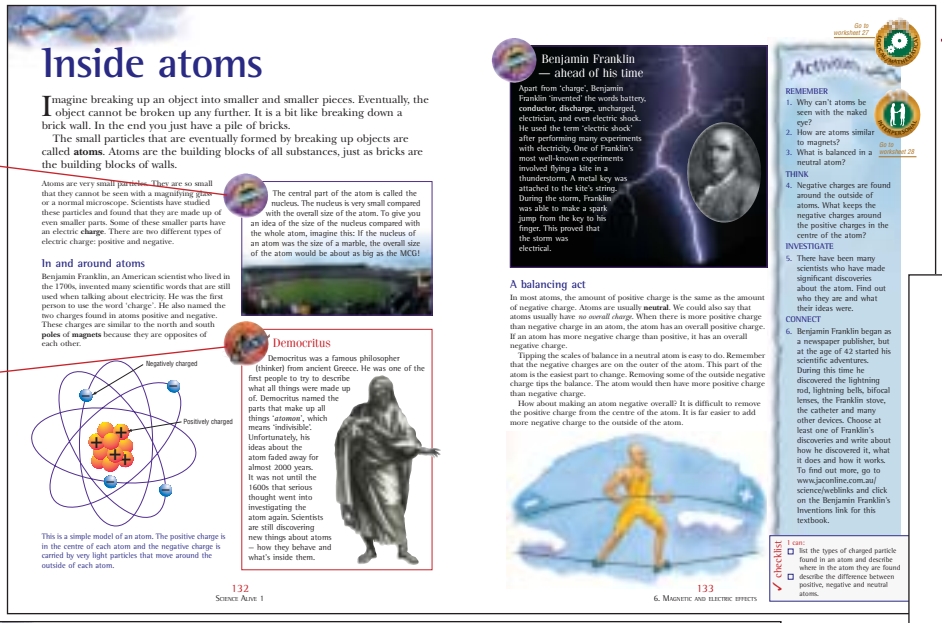
The 'How it works' features show how scientific knowledge is applied to technology or how important scientific processes occur.

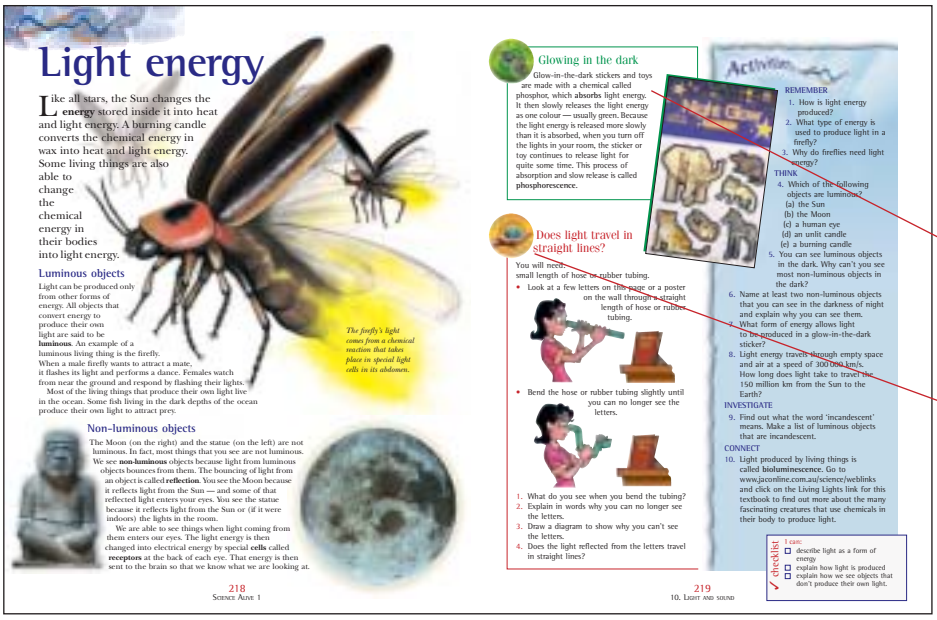
'Hands on science' activities and experiments support concept development and provide opportunities to practise science skills, processes and procedures.

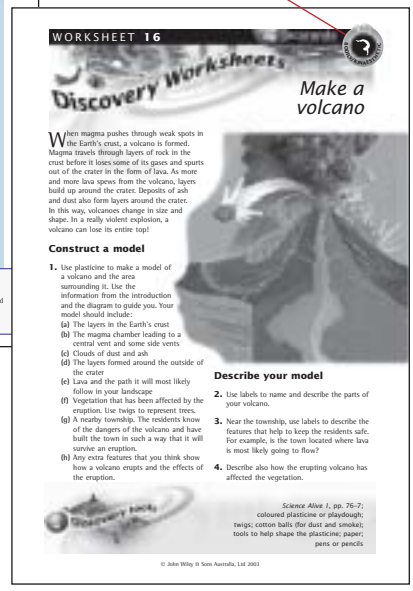
Newspaper spreads appearing in each chapter present information in a vibrant and interesting way.

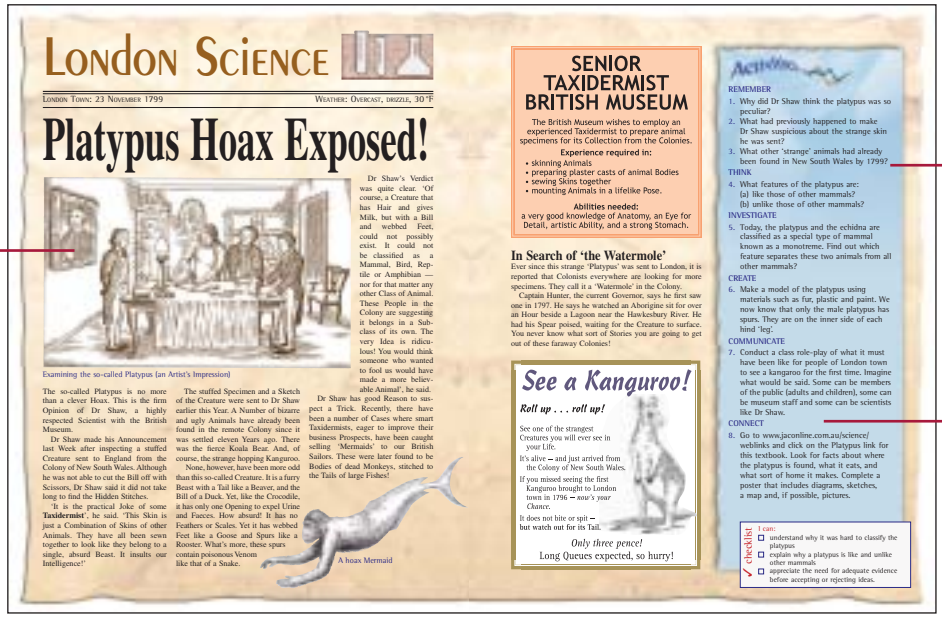
Graded activities test a range of abilities and intelligences.

A variety of 'Connect' questions allow students to explore fascinating web sites via Jacaranda's web site: www.jaconline.com.au









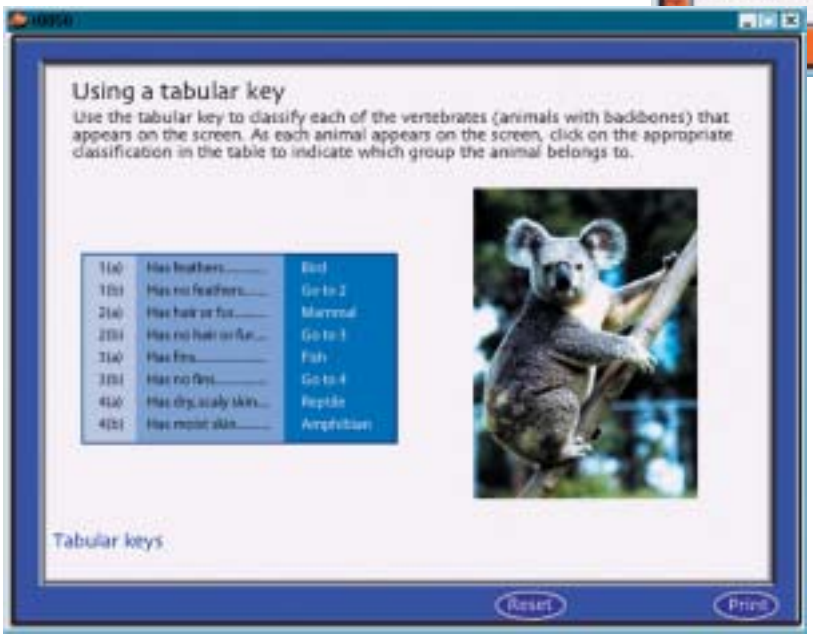
About the CD-ROM

The CD-ROM accompanying this textbook includes the text in pdf format. The text is linked to the following features that are designed to assist learning, review concept development at critical places within the text, and help students assess their own learning.

- Learning objects bring diagrams and other illustrations to life with the use of animation and provide a level of support for concept development that is not possible with a still picture. In turn, each learning object is linked to an e-tivity that tests understanding of the concept developed or re-inforces the learning object.
- Checkpoints are summaries of chapters, or parts of chapters. The checkpoints link all relevant learning objects and provide students with immediate feedback through five quick questions.
- Test yourself questions, placed at the end of each chapter, provide a link to tests that consist of 20 multiple-choice questions. Students can use these to review the chapter and/or prepare for class tests.

Running the CD-ROM

To run this CD-ROM — on both Windows 95/98 or NT, and Macintosh — simply place it in the CD-ROM drive.



Minimum system requirements

Windows 95/98 or NT
Processor: Pentium
CD-ROM drive speed: 4x
16Mb RAM
speakers

Macintosh
Macintosh OS 7.6
Processor: PowerPC
CD-ROM drive speed: 4x
6Mb RAM

Troubleshooting

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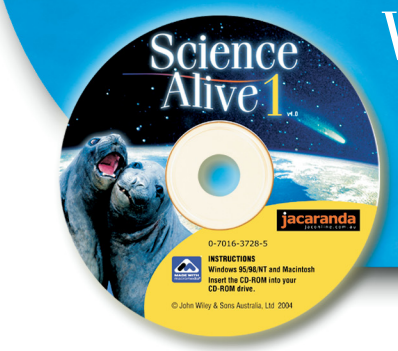

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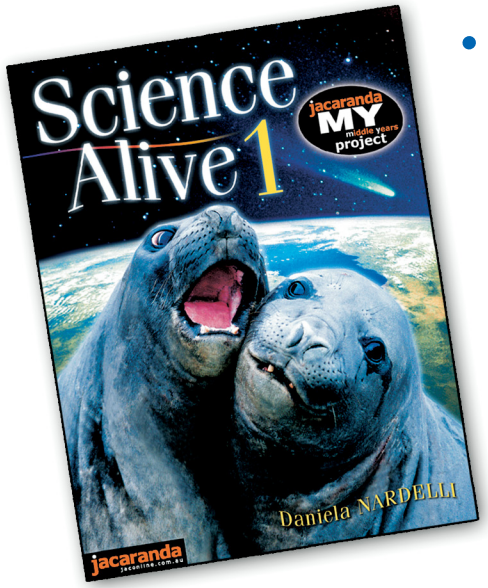

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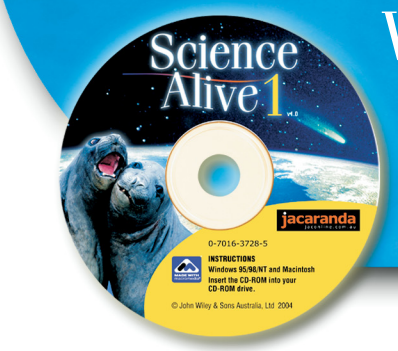

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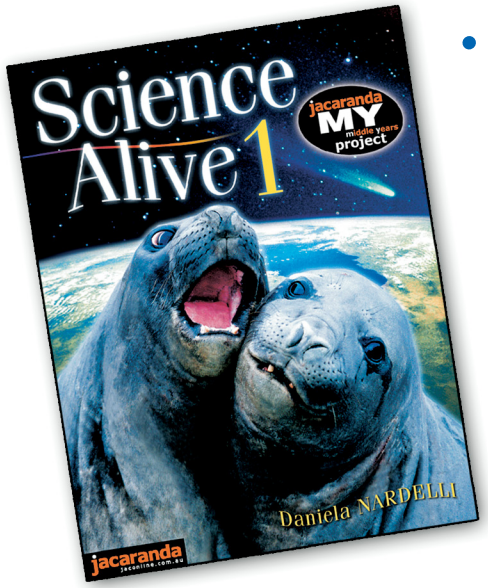

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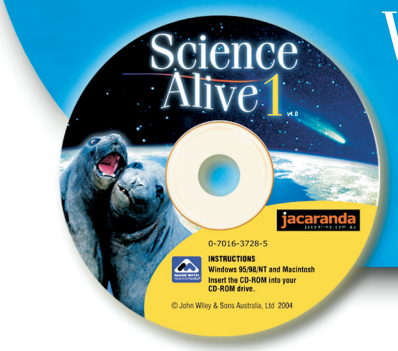

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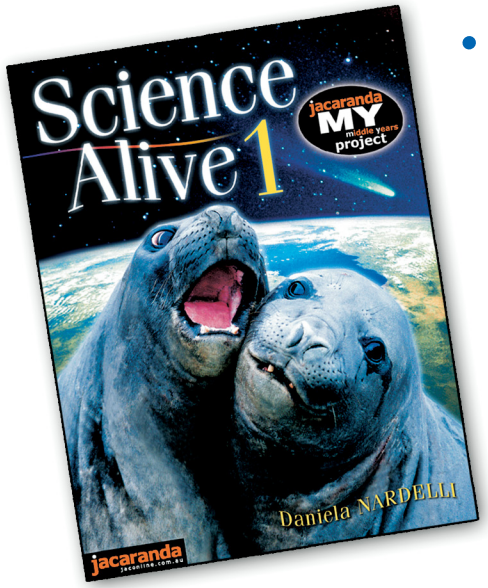

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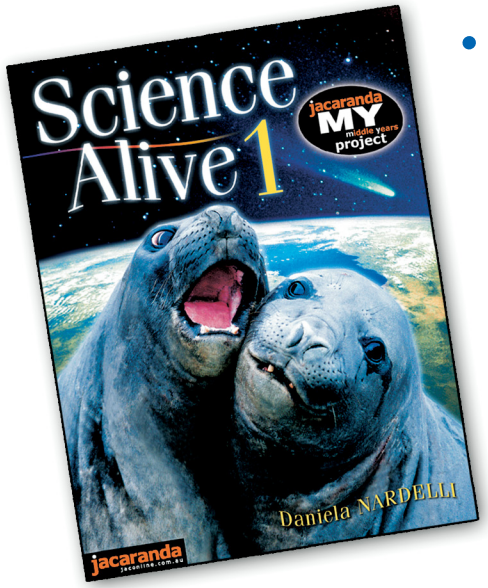

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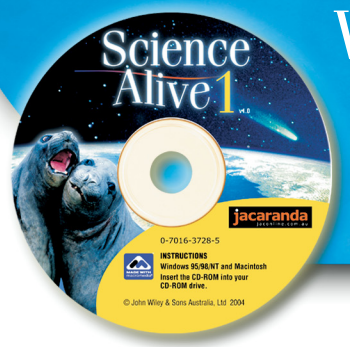

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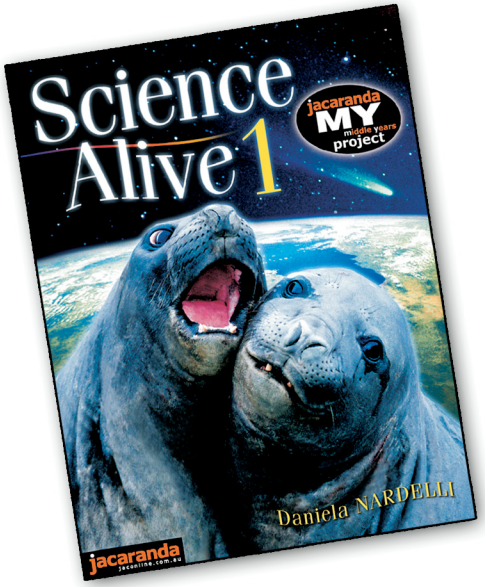

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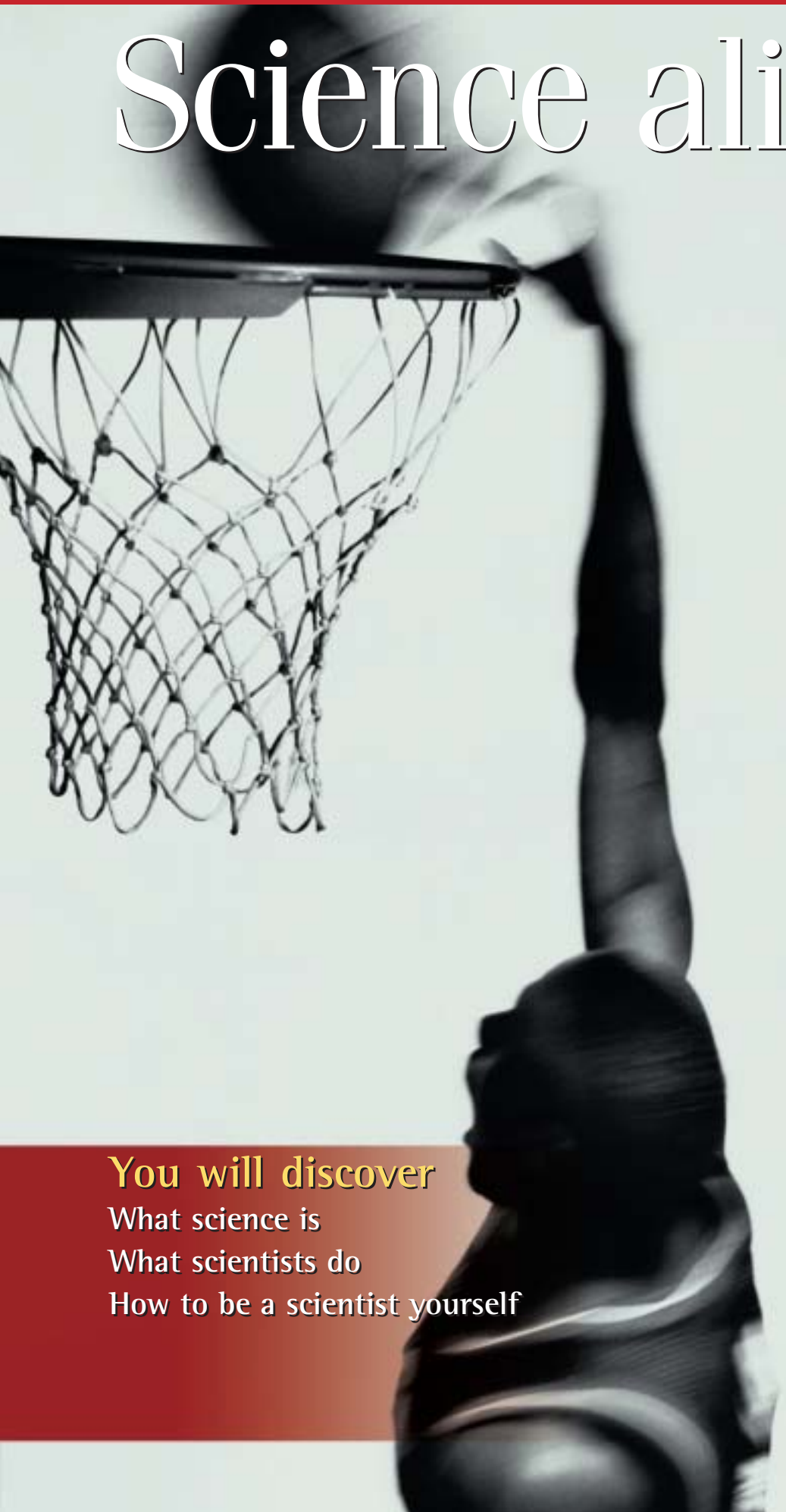
CHAPTER 1

Science alive!



You will discover

What science is
What scientists do
How to be a scientist yourself

Air time

When basketball players leap into a slam dunk, they hang in the air before sinking the ball through the hoop. Air time seems to vary for different slam-dunk artists. But scientists who study how bodies move tell us that most air times are between 0.8 and 0.9 seconds. Some players' air times are sightly longer because of the way their bodies are built or because they train to be better jumpers.

Scientists also help design the shoes worn by athletes and the surfaces they play on. They study the flight of balls through the air and the performance of different materials and clothing. Scientists and engineers create devices that measure the movement — how high, how long and how fast — of athletes and their sporting equipment.

It's not just sport that benefits from science. Scientists are at work in places you might not think of right away.

1 List three ways in which a scientist may have been involved in the design of a sports shoe.
2 What equipment might scientists use in a laboratory?
3 As a class, list 10 everyday objects. For each object, suggest how a scientist may have been involved in its design.

Where are the scientists?

You can find scientists just about anywhere. They could be in a desert finding out how plants survive without water. They could be digging deep into the ice in Antarctica. You might find a scientist searching for fossils on a rocky shore, counting rare animals in a rainforest or measuring electricity in a power station. Some scientists work in laboratories, searching for a cure for a disease. Others work with chemicals. You might even find a scientist in space.

Humble beginnings for famous scientists

Some of the most famous scientists in history started out as average students! Albert Einstein was slow to learn to speak and hated going to class. He passed his university exams by reading notes taken by his friends. Much of Einstein's knowledge and understanding of physics came from reading and experimenting.

Isaac Newton started out studying law at university. His love for maths and science began only after he started reading about them in his own time. Newton is most famous for his explanations of gravity and colour.

Marie Curie was the first person to be awarded two Nobel Prizes. She is famous for her work on radioactivity. Marie's love for science began when she attended an unofficial university. When she was accepted into a proper university, Marie's skills and knowledge were below that of the other students. Her determination and hard work led to her success.

Branching out

Earth science

*Earth scientists, or **geologists**, study the Earth. They investigate and explain how rocks and mountains form. Some specialised geologists, called seismologists, study earthquakes. Palaeontologists study fossils and ancient rocks. Vulcanologists study volcanoes.*

Biology

***Biologists** study living things. They investigate how living things function and how they live together. Some biologists, like botanists and horticulturists, study plants. **Zoologists** and veterinarians study animals. Microbiologists study very small living things. People like doctors and dentists use their knowledge of biology to help keep people and their teeth healthy.*

A mix of science

The boundaries between the different sciences are often crossed. Biophysicists and biochemists work in more than one field. Scientists often work together to solve problems. A physicist worked with medical staff to design the bionic ear. Physicists and geologists work together to locate underground mineral deposits using sound. And chemists work with biologists to find cures for diseases.

Physics

Physicists *study different types of energy. They investigate and explain things like movement, heat, nuclear energy, light and electricity. Some* ***engineers*** *use their knowledge of physics to make sure buildings are strong and cars are safe. Physics knowledge is also used in electronics, computer design and even special effects in movies.*

Astronomy

Astronomers *study the sky. They are concerned with planets, moons, stars, comets and other objects in space.*

Chemistry

Chemists *study how substances react with other substances. They investigate and explain why some substances behave differently from others and how they can best be used. Industrial chemists might look for ways to make better paints, or special plastics. Pharmacists are chemists too. They work with chemicals that are used to treat illness and disease.*

Activities

REMEMBER

1. Which science is concerned with comets?
2. State whether the following are true or false:
   (a) A biologist is more likely to study the human body than the speed of a motor car.
   (b) Isaac Newton was the first person to be awarded two Nobel Prizes.
   (c) Chemistry is the study of different types of energy.
   (d) Computer design requires knowledge of physics.

THINK

3. What type of scientist would examine rocks to see how old they are?
4. Give an example of what you think a biophysicist and a biochemist might do.
5. Look at the photograph of the basketballer on page 1. Suggest how each of the following scientists might be involved in improving his performance:
   (a) biologist
   (b) chemist
   (c) physicist.

INVESTIGATE

6. Find out what biomechanics is. Where would these scientists work? Who else would they work with?
7. Leonardo da Vinci was an early scientist who learnt a lot about science by experimenting. Find out about some of the machines that Leonardo da Vinci invented.

IMAGINE

8. Work in groups of three or four. Imagine that you are a group of scientists. Write and perform a short play or mime that tells the audience about what you do.

✓ checklist

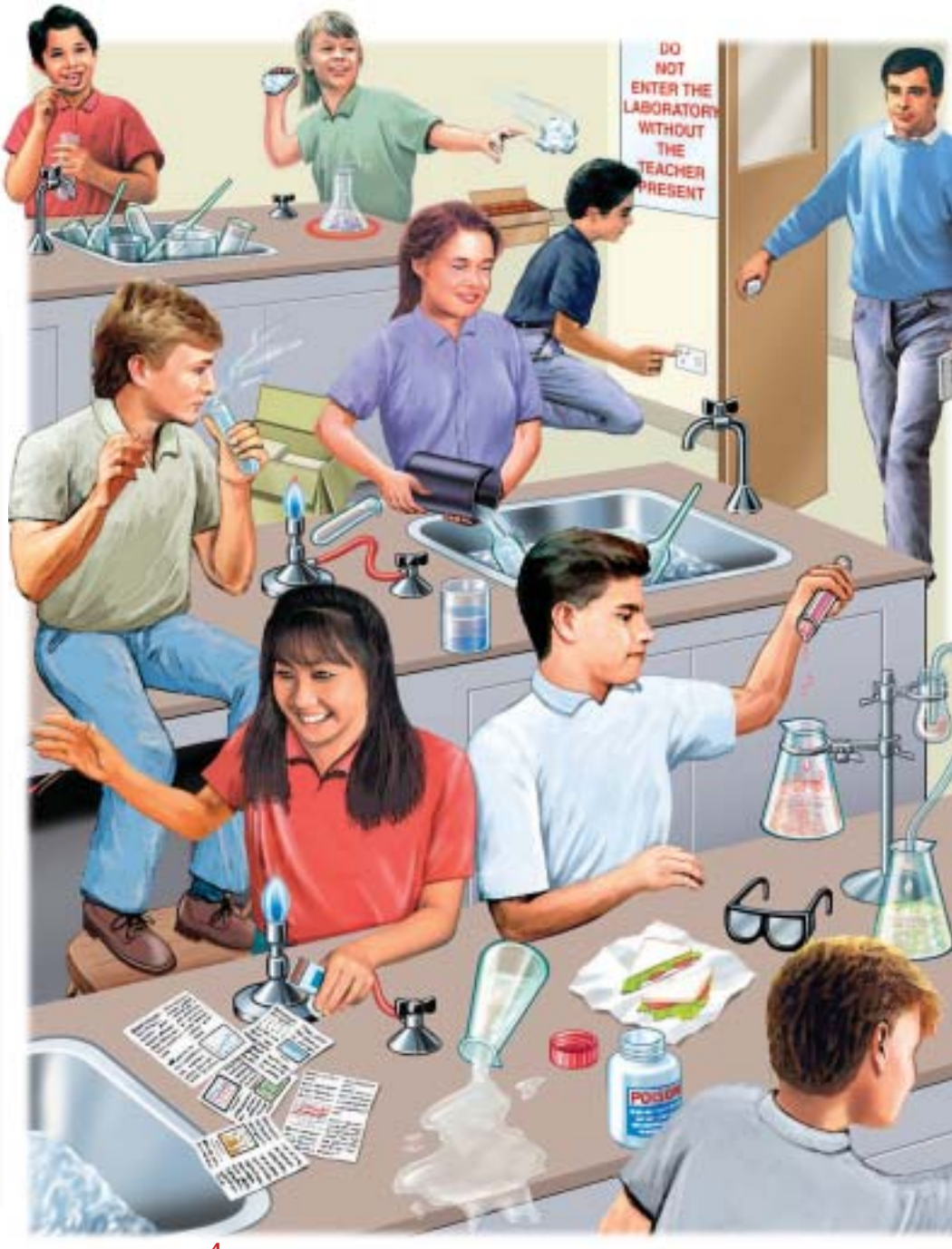
I can:

- ☐ appreciate that scientists work in a variety of situations
- ☐ list different areas of science
- ☐ describe what is involved in each science area.

Play it safe

The equipment and tools that scientists use can be unsafe if they are not used properly. It is also important that the correct equipment is used in each experiment, as each item is designed for a particular purpose.

Before you begin working in a laboratory, it is important to know the safety rules and how to use the equipment correctly. A full list of safety rules and equipment appears in the Laboratory Toolbox on pages 241–2. The safety rules are designed to keep you and your classmates safe. Look for any dangerous activities taking place in this picture.



Warning signs

Your teacher will tell you how to handle the chemicals in each experiment. At times you may come across warning labels on the substances you are using. These are some of the labels you are likely to come across in a school laboratory.

Always wear gloves when using chemicals with this symbol. Corrosive substances can cause severe damage by eating away at living things and objects like metal containers. Acid is an example of a corrosive substance.

Chemicals with this label can cause death or serious injury if swallowed or breathed in. They are also dangerous when touched without gloves because they can soak into the skin. Mercury is a toxic substance.

These substances are easily set on fire. Methylated spirits is flammable.

Fire alert!

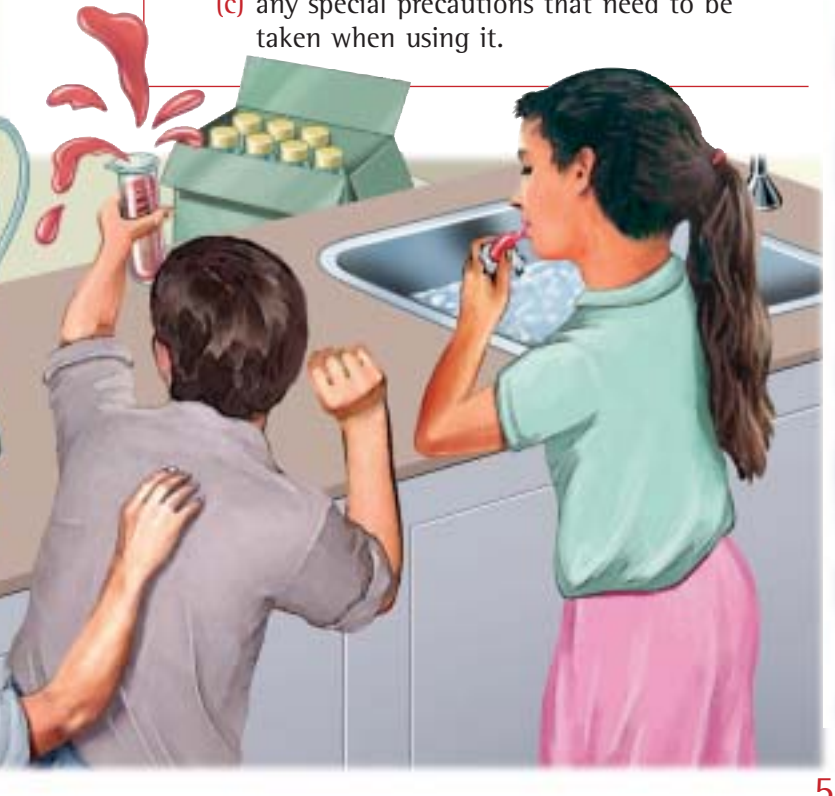
All schools have a fire-safety program. Your teacher will be able to tell you the quickest and safest way to leave the building in case of a fire. If a fire does break out in the science laboratory, don't panic:

- If equipment is on fire, let your teacher know immediately. Your teacher will use sand, a fire blanket or a fire extinguisher to put out a small fire.
- If the fire is too big to be put out, you will need to follow your school's fire-exit plan. Your teacher will direct you to the safest exit and may ask you to close doors and windows, and turn off the lights. Move quickly, without running, to the assembly area.
- If you or your clothes are on fire, STOP, DROP and ROLL. Call out for help. Someone will smother the flames with a fire blanket.

Using the right stuff, safely

You will need:
all of the equipment listed on page 242 of the Laboratory Toolbox
pencil
ruler.

- Draw a two-dimensional scientific diagram of each piece of equipment. You may need to refer to page 251 of the Laboratory Toolbox.
- For each piece of equipment, list:
  (a) the name of the equipment
  (b) what it is used for
  (c) any special precautions that need to be taken when using it.



Activities

REMEMBER

You may need to refer to pages 243–7 to answer some of these questions.

1. What colour should the Bunsen burner flame be when you are not using it for heating?
2. What protective equipment or clothing should you always wear when using a Bunsen burner?
3. State whether the following are true or false. Correct any false statements by re-writing them so that they are true:
   (a) Long hair must be tied back when using a Bunsen burner.
   (b) Matches can be safely washed down the sink.
   (c) Always point a test tube towards you when heating, so you can see what is happening inside it.
   (d) Safety glasses need to be worn only when heating over a blue Bunsen-burner flame.
   (e) Use a heatproof mat under all equipment and chemicals.
   (f) Water spills do not need to be cleaned up because they are not dangerous.

THINK

4. If the teacher says it's safe to smell a chemical, how should you do it?
5. Why is there a sand bucket and a fire blanket in the laboratory?
6. Why should safety glasses be worn during an experiment?
7. Why is a heatproof mat placed under a Bunsen burner when heating?
8. When a mixture is being filtered, the filter funnel can be placed in the mouth of a conical flask. What is the advantage of using a conical flask rather than a beaker?

CREATE

9. Design a poster that outlines one of the laboratory safety rules on page 241.
10. Use a multimedia tool to give a presentation to the class explaining the fire safety procedure at your school.

checklist

I can:
- ☐ list the school laboratory rules
- ☐ recognise warning labels on chemicals
- ☐ follow the school evacuation plan in case of a fire
- ☐ label equipment and know what it is used for.

Heating in the laboratory

In many scientific experiments, substances need to be heated. A **metallurgist** might want to find out how heat affects metals used to build a space rocket. **Chemists** sometimes need to use heat to speed up chemical reactions. A **biologist** might use heat to separate some of the chemicals from plants.

In school laboratories, heating is usually done with a Bunsen burner or a hotplate. Each of these 'tools' has advantages and disadvantages.

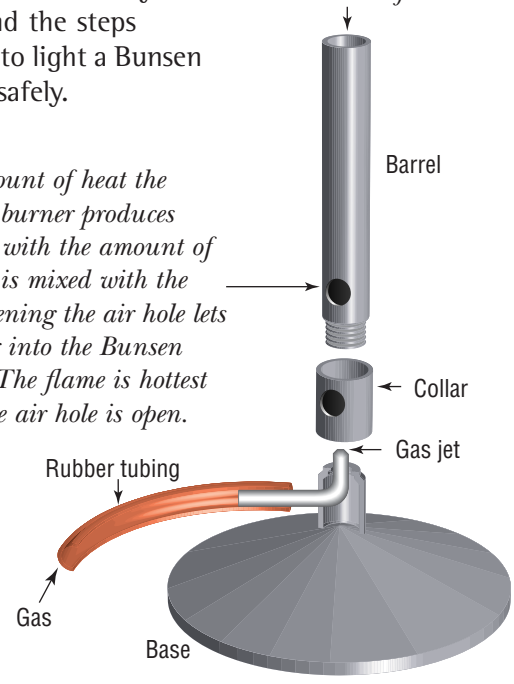
Electric hotplate
A hotplate is used when substances need to be heated slowly. Hotplates are also used when heating substances that are highly flammable. Electric hotplates take some time to warm up.

Bunsen burner

It is most likely that you will use a Bunsen burner when you need to heat a substance in the laboratory. A Bunsen burner heats objects with a naked flame, so care needs to be taken when using it. Make sure you know the safety rules and the steps needed to light a Bunsen burner safely.

Lighting a mixture of air and gas produces the Bunsen-burner flame.

Air hole
The amount of heat the Bunsen burner produces changes with the amount of air that is mixed with the gas. Opening the air hole lets more air into the Bunsen burner. The flame is hottest when the air hole is open.



Important equipment

Whenever heating equipment is used, there is a risk of fire. Fire extinguishers should be located in every school laboratory. The fire extinguishers in laboratories work by stopping oxygen from 'feeding' the fire. A fire cannot burn without oxygen.

Sand and fire blankets work in a similar way to an extinguisher. They starve the fire of oxygen. Sand is used on small equipment or bench fires. Use a fire blanket to smother flames on a person who is on fire.

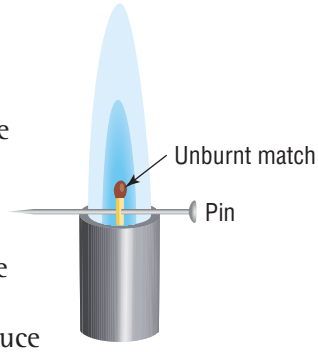
Go to worksheet 1

MI BODILY/KINAESTHETIC

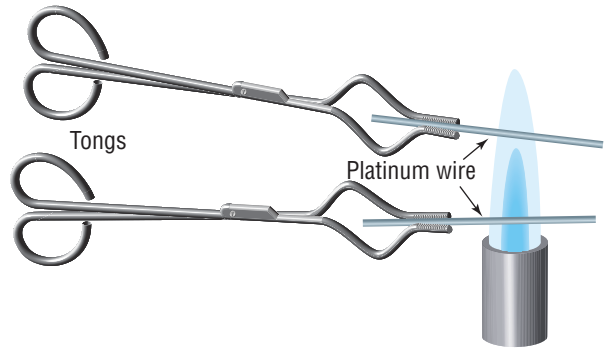
Where is the hottest part of the flame?

You will need:
Bunsen burner
heatproof mat
matches
platinum or nichrome wire
tongs
pin
safety glasses.

- Use a pin to hang an unburnt match over the barrel of a Bunsen burner.
- Light the Bunsen burner according to the rules on page 243.
- Turn the collar to produce a blue flame.



1. What happens to the match hanging over the barrel?
- Turn the Bunsen burner off and remove the match and pin with tongs.
- Re-light the Bunsen burner and turn the collar to produce a blue flame again.
- Use the tongs to hold the wire across the flame, close to the barrel of the Bunsen burner.



2. What colour does the wire become?
- Move the wire up a little.
3. Is the colour of the wire different now?
4. Draw a diagram of the Bunsen-burner flame, labelling the parts that are hottest.

Which flame is hotter?

You will need:
Bunsen burner
matches
pieces of porcelain
clock or watch.
heatproof mat
tongs
safety glasses

- Light the Bunsen burner according to the rules on page 243.
- Open the air hole.

1. Describe the flame. What colour is it? Does it make a noise?
- Hold a piece of porcelain over the flame with the air hole open.
- Take notice of roughly how long it takes for the porcelain to turn red-hot.
- Let the porcelain cool down on the heatproof mat.
- Close the air hole.

2. Describe the flame. Is it easy to see?
- Hold the porcelain in the flame.
3. Does it turn red-hot?
4. Do you notice anything else about the porcelain after heating in this flame?
5. Which flame is the hottest flame?

Activities

REMEMBER

1. Which is the safest flame when the Bunsen burner is not being used?
2. Which is the hottest part of a blue flame?

THINK

3. Make a list of the safety equipment needed when using a Bunsen burner.
4. Why would a hotplate be better than a Bunsen burner for heating methylated spirits?

CREATE

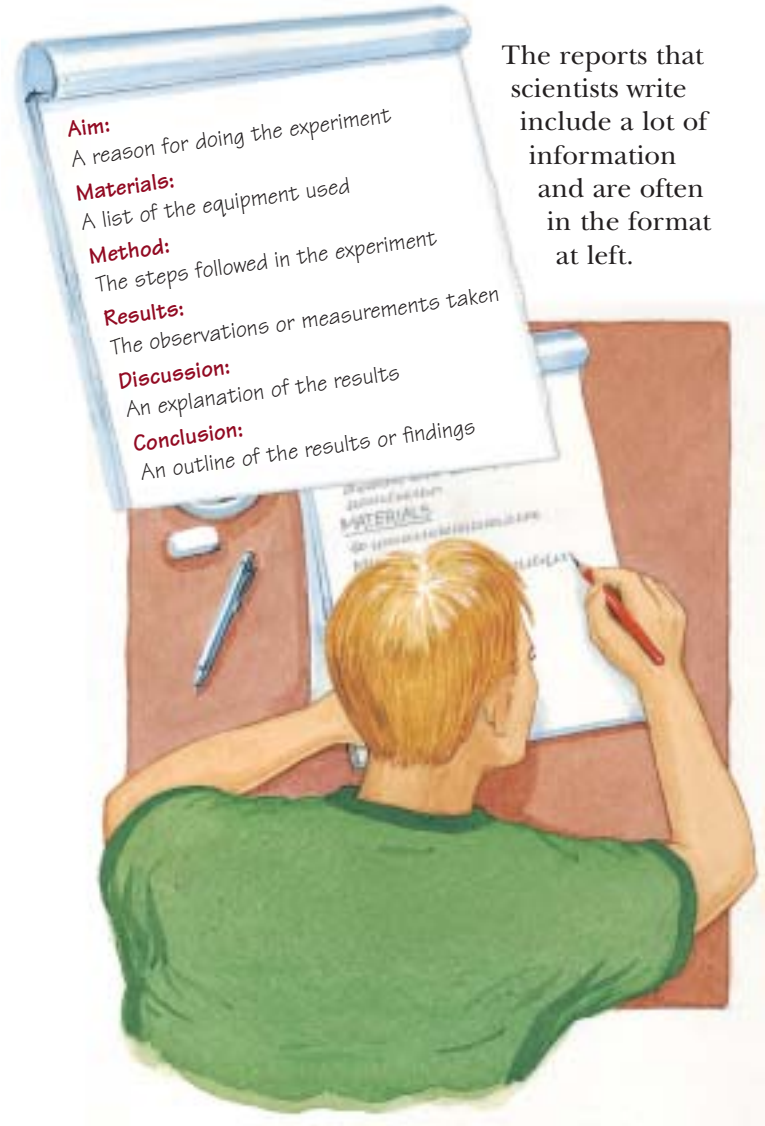
5. Draw a map of the laboratory. On your map, make sure to include benches, sinks, exit, eye-wash, fire blanket, fire extinguisher, gas taps, glass bin, glassware, power points, rubbish bins and sand bucket.

✓ checklist

I can:
- ☐ label the parts of a Bunsen burner
- ☐ use a Bunsen burner safely
- ☐ locate fire-safety equipment in the classroom.

Reporting back

Scientists perform experiments to find out answers to their questions. When their experiments are completed, a report is written to tell others about their work.



The reports that scientists write include a lot of information and are often in the format at left.

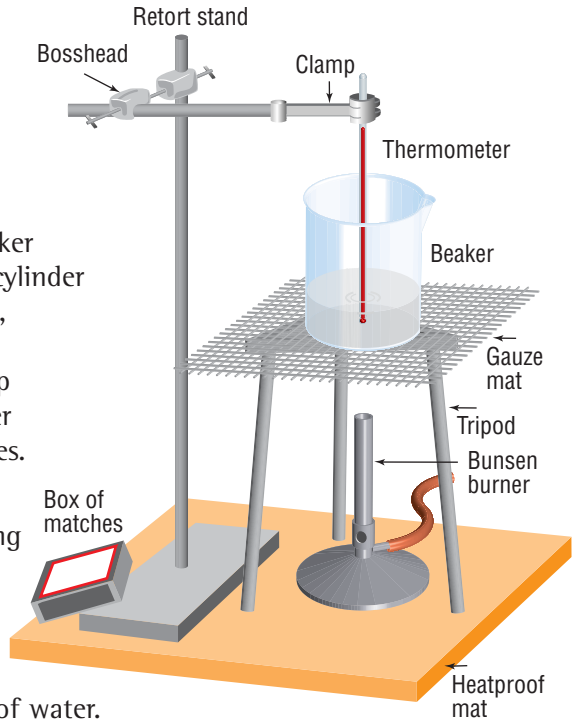
A more detailed explanation of what a laboratory report must include is on page 250 of the Laboratory Toolbox.

Temperature graphs

A graph is a useful way to present the results of an experiment. A graph gives an overall picture of the results. It can also be used to predict values that occur between, or after, those measured during an experiment.

The aim of this experiment is to show how the temperature of water changes while it is heated over a Bunsen burner.

You will need:
Bunsen burner
heatproof mat
matches
tripod
gauze mat
250 mL beaker
measuring cylinder
retort stand, bosshead and clamp
thermometer
safety glasses.

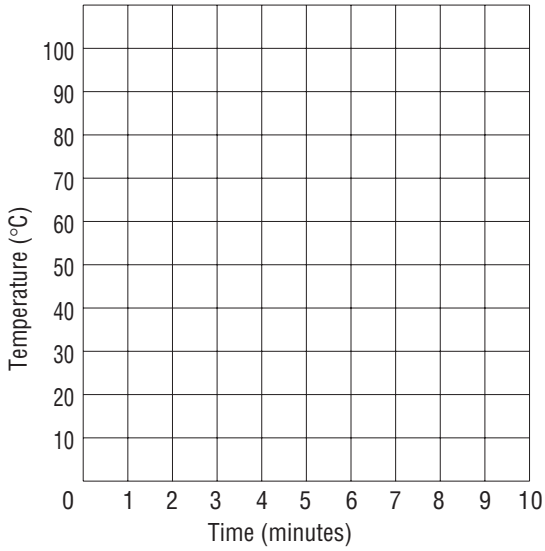


- Use a measuring cylinder to measure out 100 mL of water.
- Pour the water into the beaker.
- Set up the equipment as shown in the diagram. Make sure that the bulb of the thermometer is not on the bottom of the beaker or out of the water.
- Wait for a minute to allow the thermometer to adjust to the water temperature.
- Measure the temperature of the water. Record the starting temperature in a table. The starting temperature is recorded when time is 0 minutes.

Time (min)	Temp (°C)	Time (min)	Temp (°C)
0		6	
1		7	
2		8	
3		9	
4		10	
5			

Go to worksheet 2

- Put your safety glasses on.
- Light the Bunsen burner according to the rules on page 243.
- Open the air hole and heat the beaker over a blue flame.
- Measure and record the temperature of the water every minute for ten minutes.
- Turn off the Bunsen burner and allow the equipment to cool down.
- Plot a line graph of the data you have collected.



- Join the data points with a smooth line.

1. Describe how the temperature of the water changes over time.
2. How does your graph compare with the graphs of other students?
3. Use your graph to estimate the temperature of the water four and a half minutes after heating started.
4. Predict the temperature that the water would have reached 11 minutes after heating started.

Extension

- Set up data-logging equipment with a temperature probe. Your teacher will help you with this.
- As soon as the Bunsen burner is turned off, place the temperature probe in the water.
- The data logger will record the temperature of the water as it cools.

1. Describe the cooling graph.
2. List the advantages of using a data logger instead of a thermometer to measure temperature.

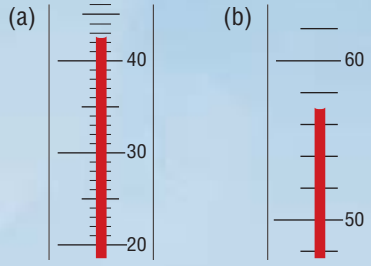
Hot and cold

The hottest air temperature ever measured on Earth is 58 °C. The measurement was taken in 1922 in Libya. The lowest temperature ever measured was in 1983 in Antarctica. That temperature was –86.6 °C.

Activities

REMEMBER

1. Why do scientists write reports about the experiments they conduct?
2. In which part of a laboratory report would a graph of distance against time be drawn?
3. What is the difference between the results of an experiment and the conclusions made?
4. What temperatures are shown on the following thermometers?



THINK

5. Why shouldn't a thermometer rest on the bottom of the beaker during heating?
6. Why would you not use a thermometer with a scale from 0 °C to 30 °C for measuring the temperature of your body?

INVESTIGATE

7. Does adding salt to water change the way the temperature rises when the water is heated? Design an experiment to test your answer.

checklist

I can:

- ☐ write a laboratory report
- ☐ measure temperature with a thermometer
- ☐ use tables and graphs to record results.

Science on the move

Who invented the car? Well, it depends on what you mean by 'car'. Since the ancient Greeks, people have had ideas about moving without the help of animals. A steam-driven vehicle was invented in 1792. But was it really a car? In the 1880s, cars with more modern motors were built.

Cars have been changed bit by bit by many people over hundreds of years. For this reason, it is hard to say who invented the type of car that people drive today.

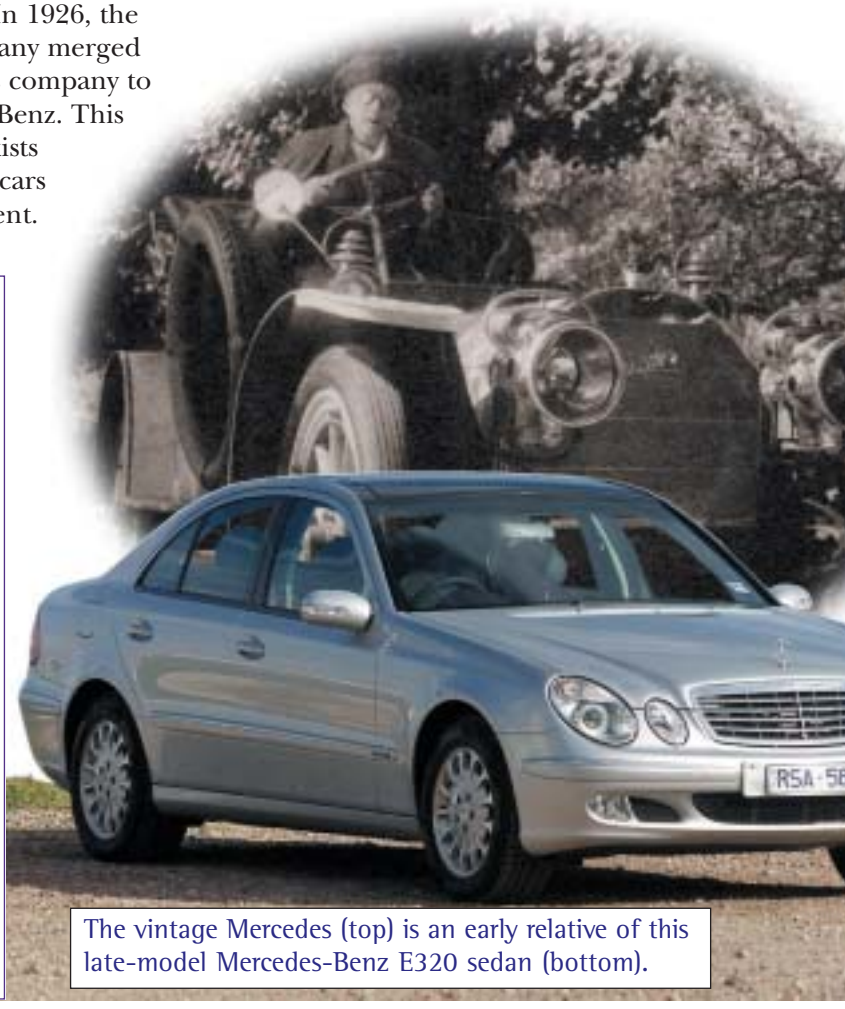
Steaming ahead

The first cars were powered by steam. They were big, heavy and very noisy. Many people complained about the safety of 'road steamers'. New laws had to be passed. In 1865, a law was written to limit the speed of road steamers in Britain to 3 km/h in the city. A man waving a red flag had to walk in front of the steamer to warn people of its approach.

New ideas

While some inventors worked on steam-driven cars, others worked on different types of motor. Through their experiments, they found that battery cars were quieter than the steam cars. But the battery cars needed constant re-charging. Then, in the 1880s, Karl Benz and Gottlieb Daimler designed the earliest relatives of the modern car. Benz and Daimler each had their own designs and **patented** them at about the same time. Karl Benz and Gottlieb Daimler were not scientists. They were **engineers**. Engineers use scientific ideas to build new **technology** and make it work.

When Daimler died in 1900, his company took his name off the car he had created. They replaced it with 'Mercedes'. The new name came from the daughter of a businessman who sold Daimler's cars. In 1926, the Mercedes company merged with Karl Benz's company to form Mercedes-Benz. This company still exists today, but their cars look very different.

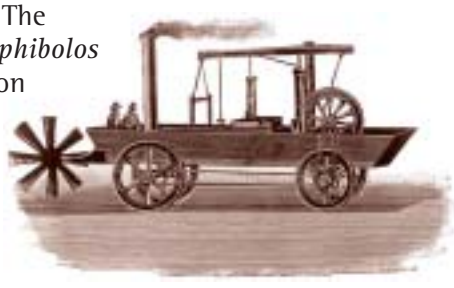


The vintage Mercedes (top) is an early relative of this late-model Mercedes-Benz E320 sedan (bottom).

What's in a name?

In 1792, Oliver Evans applied for a patent on a steam carrier, his early version of a car. (A patent gives an inventor permission to use and sell his or her ideas without competition from someone else.) Evans called his machine the '*oruktor amphibolos*'. Luckily, other people had similar inventions with better names. Otherwise, what we call the car or automobile today, could have been called the '*oruktor amphibolos*'!

Fact: The word *amphibolos* comes from amphibian. The *oruktor amphibolos* could drive on land as well as propel through water.



The people who make and test Formula One racing cars are also engineers. The engineers work together to make the cars safer, lighter and faster.

Go to worksheet 3

MI INTERPERSONAL

Activities

REMEMBER

1. What was an advantage of battery cars over steam cars?
2. (a) What was the speed limit in British cities during 1865?
   (b) Why was a speed limit needed?
3. How does the work of engineers like Jacquie Fox benefit the community?

THINK

4. Why do inventors patent their inventions?
5. What is the difference between a scientist and an engineer?
6. A four-wheel drive is built differently from a sedan.
   (a) What are four-wheel drives designed to do?
   (b) List some of the features of a four-wheel drive that allow it to do this.

PREDICT

7. Draw what you think cars will look like in 20 years from now. Think about the shape and size of the car and what energy source it might use.

CAREER PROFILES

Engineering cars

Engineers design all cars, not just Formula One cars. Car companies employ many different types of engineer. Each engineer has his or her own part to design and test.

Jacquie Fox is a product engineer with Toyota. She assists with the design and development of **resin** (plastic) parts like exterior lights. The parts she designs have to fit in with parts designed by other engineers. Jacquie must make sure that her designs work properly, are safe and improve the look of the car.

Jacquie says the most exciting part of her job is '... seeing the finished product on a vehicle'. Jacquie and her team spend long hours solving problems and finetuning designs; however, she says that '... the rewards far outweigh the hours worked'.

Jacquie loved physics at school, but didn't enjoy chemistry. After studying engineering at university, she became more interested in chemistry. Now Jacquie uses her understanding of both physics and chemistry in her designs.

'Our jobs benefit the community because we are continually improving the safety and performance of cars. Some obvious examples are airbags, ABS brakes and **traction** control. We are also continually finding ways to reduce exhaust **emissions** for cleaner air.'

✓ checklist

I can:

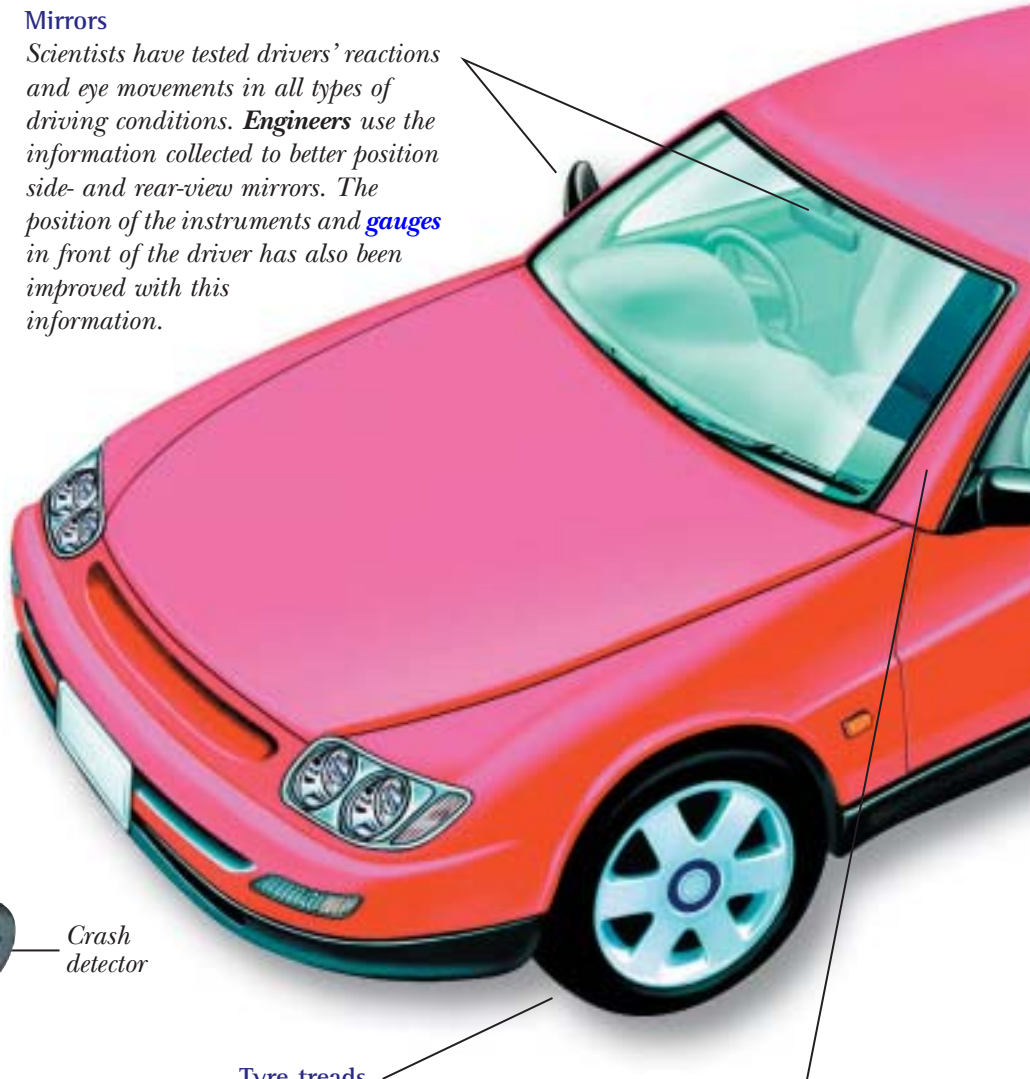
- ☐ describe some early cars
- ☐ understand the role of an engineer
- ☐ discuss the difference between a scientist and an engineer.

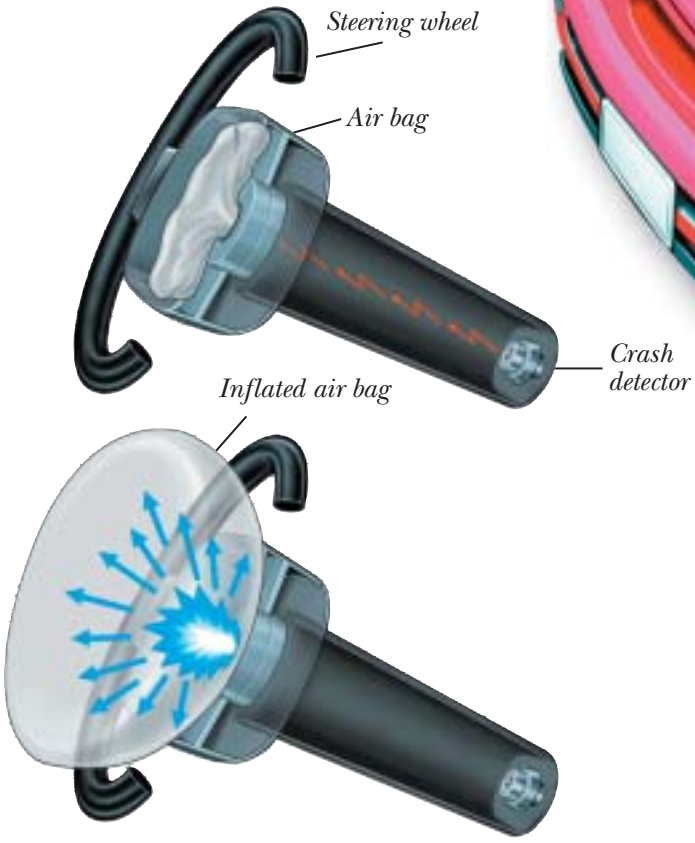
High-tech cars

The inventors of the car could never have imagined some of the latest developments in cars brought about by science. Scientific knowledge is used to improve the design of every new car. The cars we drive today are safer, stronger, lighter and faster than ever before.

Mirrors

*Scientists have tested drivers' reactions and eye movements in all types of driving conditions. **Engineers** use the information collected to better position side- and rear-view mirrors. The position of the instruments and **gauges** in front of the driver has also been improved with this information.*





Airbags

Airbags are designed to stop a person colliding with the inside of the car. If the car is involved in a crash, the airbags fill within one twenty-fifth of a second and burst from their storage area at about 300 km/h. Soon after, the air escapes from the airbag through tiny holes in the fabric.

Tyre treads

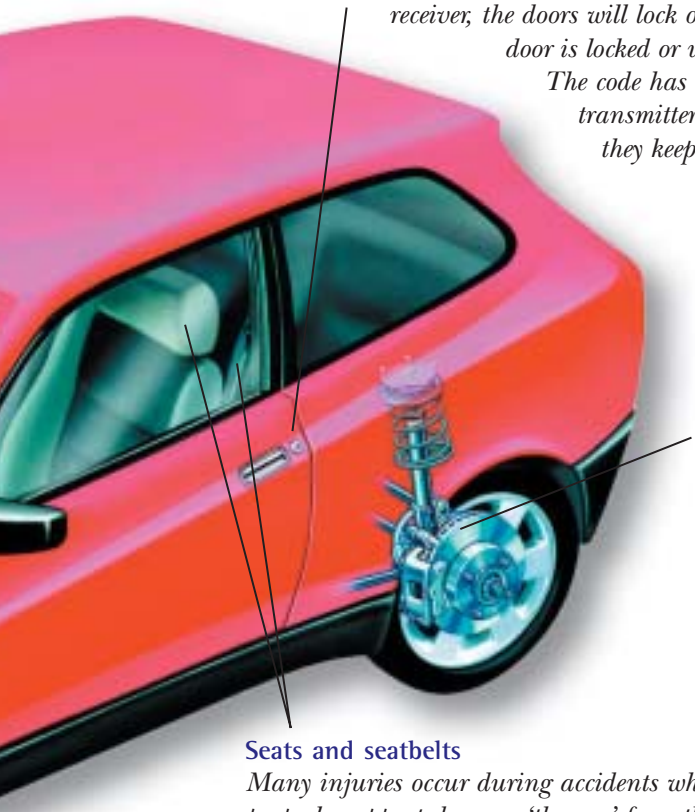
The lines and grooves in tyres are called treads. Treads help keep the car under control in wet weather. Engineers design treads so that, in wet weather, water can squirt out from under the tyre. If this did not happen, the tyres would have no grip on the road and would slide over the water instead.

Better vision

Engineers have designed stronger windscreens and pillars, so that the thickness of the pillar can be made smaller. The size and the position of the pillars give the driver a better view of the road, making the car safer to drive.

Remote keyless entry

Most new cars are locked and unlocked by pressing a button on a key ring. These key rings have ***transmitters*** *inside them that send out signals. A* ***receiver*** *picks up the signal. If the signal matches a code in the receiver, the doors will lock or unlock. Each time the door is locked or unlocked, the code changes. The code has to change in both the transmitter and the receiver, so that they keep recognising each other.*



ABS brakes

Anti-lock Braking Systems (ABS) help cars stop more safely. If the wheels of a car lock up (stop turning) on a slippery road, then it is almost impossible to stop. ABS brakes work by stopping the wheels from locking up. They make sure that the wheels keep turning and slow down with the car, rather than sliding.

Seats and seatbelts

Many injuries occur during accidents when passengers are not properly supported or are 'thrown' from their seats. Headrests are designed to stop the passenger's head from 'whipping' backwards during an accident. To stop passengers from being thrown from their seat, all cars are fitted with seatbelts. Seatbelts are designed to lock during sudden forward movements.

Cars with wings

The same **technology** that helps aeroplanes to fly keeps racing cars on the ground. The wings of an aeroplane are designed to lift it off the ground and keep it in the air. A racing car uses the same idea, but in reverse. The wing or **aerofoil** on a racing car is upside down compared to an aeroplane. The aerofoil on a racing car keeps the car on the ground. Without it, a racing car could take off. Racing cars often travel faster than light aircraft just before take-off.

Activities

REMEMBER

1. What is the purpose of the treads on tyres?
2. Why are cars fitted with airbags?
3. List two ways that scientists and engineers have improved the driver's vision from a car.

THINK

4. Why are airbags made with fabric that has tiny holes in it?
5. What might happen to a racing car if its aerofoil was turned upside down?

BRAINSTORM

6. What might be the aim of the experiment below?

7. In groups of three or four, make a list of other technology used in cars.

IMAGINE

8. Imagine you could design any type of technology. What technology would you put into a car of the future? Explain why you chose this.

checklist

I can:

- ☐ list new technology that makes cars safer
- ☐ describe how scientific knowledge has improved cars
- ☐ understand how technology is important in motor cars.

Science in the swim

Australian swimmers are among the best in the world. The facilities they use have been especially designed by scientists. Scientists also assist with training methods and keeping our swimmers healthy.

Gliding through the water

Scientists at the Australian Institute of Sport use a video connected to a computer to analyse every part of an athlete's swim. The information they collect is used to tell a swimmer exactly where and how they can improve. The area of science that studies how animals (including humans) move is called **biomechanics**. **Marine biologists** study the animals in our seas and oceans. Their knowledge of the skins of marine animals was used to help design the FastSkin suit.

Keeping the head down creates a lot less drag. It also helps keep your hips and legs higher in the water, creating a more streamlined shape.

*In the past, swimmers shaved their bodies to reduce **drag**. Now, swimmers cover up as much as possible. This suit produces less drag than human skin.*

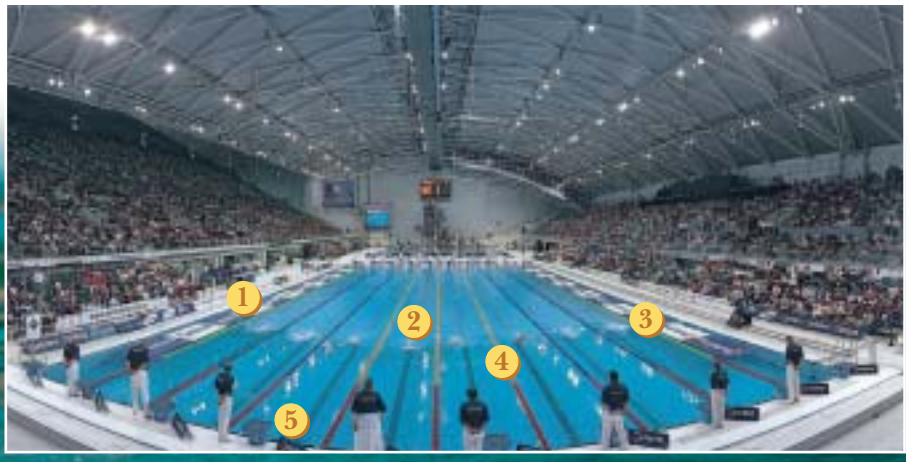
After years of studying different techniques, scientists agree that a bent arm, pulled through the water in an S shape, helps a swimmer swim faster. Swimmers also keep their fingers slightly spread to push against a greater amount of water.

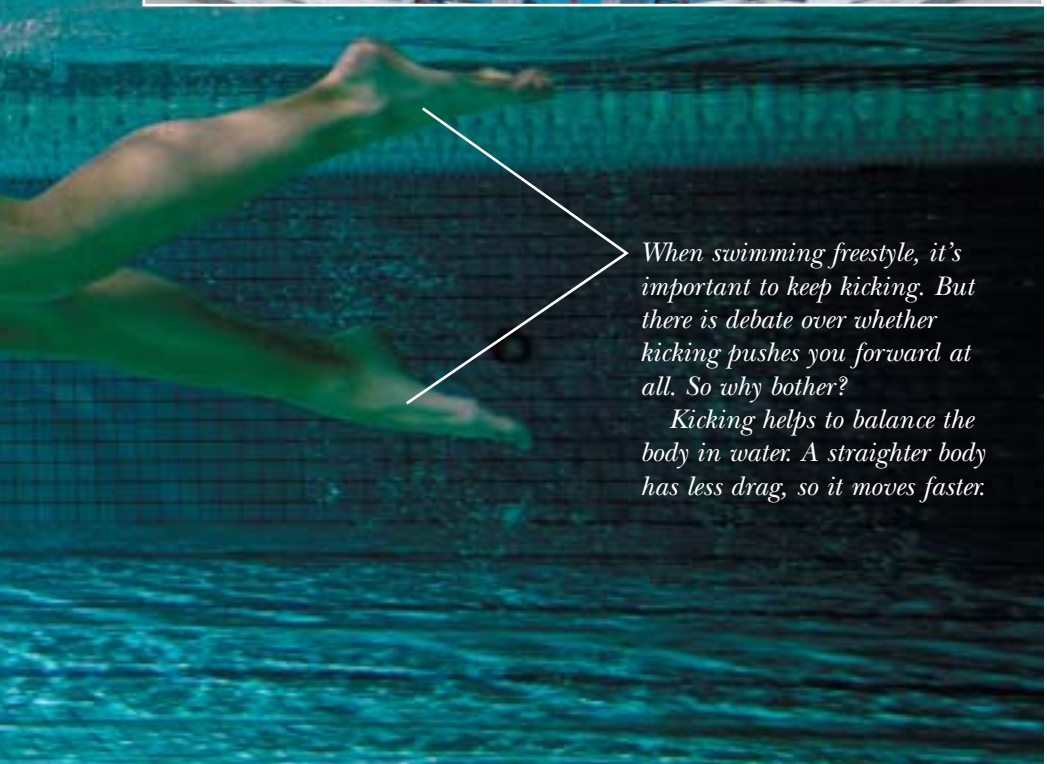
Go to worksheet 4

The fast pool

Australian scientists and engineers designed a 'fast pool' for the 2000 Sydney Olympics. Many records were broken. However, scientists had to make sure that the pool's design complied with Olympic rules.

1. *The concrete pool had to be built to very accurate lengths. The rules allow a pool to be up to 3 cm longer than the 50 m official length. If Ian Thorpe had raced the 200 m freestyle in a pool that was 3 cm too long, his time could have been 0.015 s longer. That's long enough to rob a swimmer of a world-record time!*
2. *Deep pools do not reflect waves as much as shallow pools. The Olympic pool has a minimum depth of 2 m.*
3. *Wavy water is more difficult to swim through than still water. The sides of a pool are the waviest parts, because the waves crash into the sides and bounce back again. The gutters on the side of the Olympic pool are designed to stop water waves bouncing back to the centre of the pool.*
4. *These anti-wave lane ropes were developed by scientists to help keep the water still. The slotted discs are attached to a steel cable that stops waves crossing into the next lane.*
5. *These starting blocks are on a slight slope. The design of the hand- and foot-grips give the swimmer the best chance for a good start.*





When swimming freestyle, it's important to keep kicking. But there is debate over whether kicking pushes you forward at all. So why bother?

Kicking helps to balance the body in water. A straighter body has less drag, so it moves faster.

Activities

MI INTRAPERSONAL

REMEMBER

1. How long is an Olympic-sized swimming pool?
2. What type of scientist helped design the FastSkin suit?
3. Why do swimmers keep their fingers slightly spread?

THINK

4. How does a knowledge of marine biology help with the design of suits like the FastSkin?

CALCULATE

5. If an Olympic-sized swimming pool were 3 cm too long, how much further would a competitor swim in an 800 m race?

INVESTIGATE

6. What role do scientists play in checking for drug use among athletes?

CONNECT

7. Go to www.jaconline.com.au/science/weblinks and click on the Australian Institute of Sport link for this textbook to answer the following questions.
   (a) What are the important physical characteristics of swimmers?
   (b) Apart from swimming, what other types of training do swimmers do?

DESIGN

8. Create your own swimming suit that is designed to be fast in the water.

✓ checklist

I can:

- ☐ understand how science helps improve swimming times
- ☐ list ways that scientists contribute to sporting achievement and fairness.

Healthy science

It is easy to see how studying biology could lead to a career in health science. However, other studies in science are also very useful. Much of the **technology** used in the health industry is designed and tested by scientists who have backgrounds in a combination of biology and other sciences.

CAREER PROFILES

Name: Arianne Lee

Job title: Clinical trials assistant

Company: Medtronic Australasia

Field of science: Medical clinical research

What first interested you in science?

Arianne's interest in science was first sparked in grade four, after learning about famous scientists from the past. She has always been fascinated by how relevant science is to everyday life.

Some time ago, Arianne was a patient in clinical trials for asthma medication and has since taken an interest in the health sciences. Her father's work in the **pharmaceutical industry** has also given her an interest in this field of science.

Study

Arianne has a Bachelor of Science from Monash University. Her major areas are **physiology** and **pharmacology**. These areas are related to human biology and chemistry.

The nature of her work

Arianne is a link between a research company that tests **pacemakers** and the hospitals that run the tests. She decides who will test the pacemakers and in which hospitals. Arianne seeks permission from the Ethics Committee before any tests are conducted. These committees decide whether the tests are appropriate and also make sure that proper procedures will be followed. After collecting data from the clinical tests, she reports on the results.

Arianne's job as a clinical trials assistant allows her to work for a company whose technology saves lives. She says, 'Every day I am able to experience the feeling that my scientific knowledge makes a real difference in the lives of everyday people. Not only does this technology improve the quality of people's lives, but it saves lives too'.

Name: **Paula Fisher**

Job Title: Laboratory Scientist

Company: Vision BioSystems, Ltd

Field of science: Histology

Paula Fisher is a **histologist**. Histology is the study of the **cells** and tissues that make up animals, including humans. Histologists look at small samples of cells and tissues under microscopes.

How does your job benefit the community?

Cancer is a condition where cells grow abnormally. Each year, many people are **diagnosed** with cancer. Experts agree that the earlier treatment starts, the better the chances of survival.

Paula and her team help engineers to design machines. Their machines make diagnosing cancer a much faster process.

Paula's job requires her to run the machines and make sure that the cells are clear and easy to see. This makes diagnosing cancer more accurate as well as faster.

What is the most exciting part of your job?

Aside from helping people, Paula sees many benefits to her job. She says, 'The company is new and still growing. There are lots of opportunities available and the work varies as the company changes. There is also the possibility of travel'.

Study

Paula enjoyed biology and chemistry at high school. She studied Applied Science at university and followed this up with a Graduate Diploma in Medical Laboratory Science. Many universities require students wanting to get into this field to have studied chemistry and one other science, usually physics or biology.

A controlled pace

An artificial pacing system is a device that is used to control the rate of a heartbeat. A pacing system is used for conditions where a heart has an irregular or slow beat. A pacing system has three parts: a pacemaker, leads and a programmer. The pacemaker and leads are inserted into the patient during an operation.

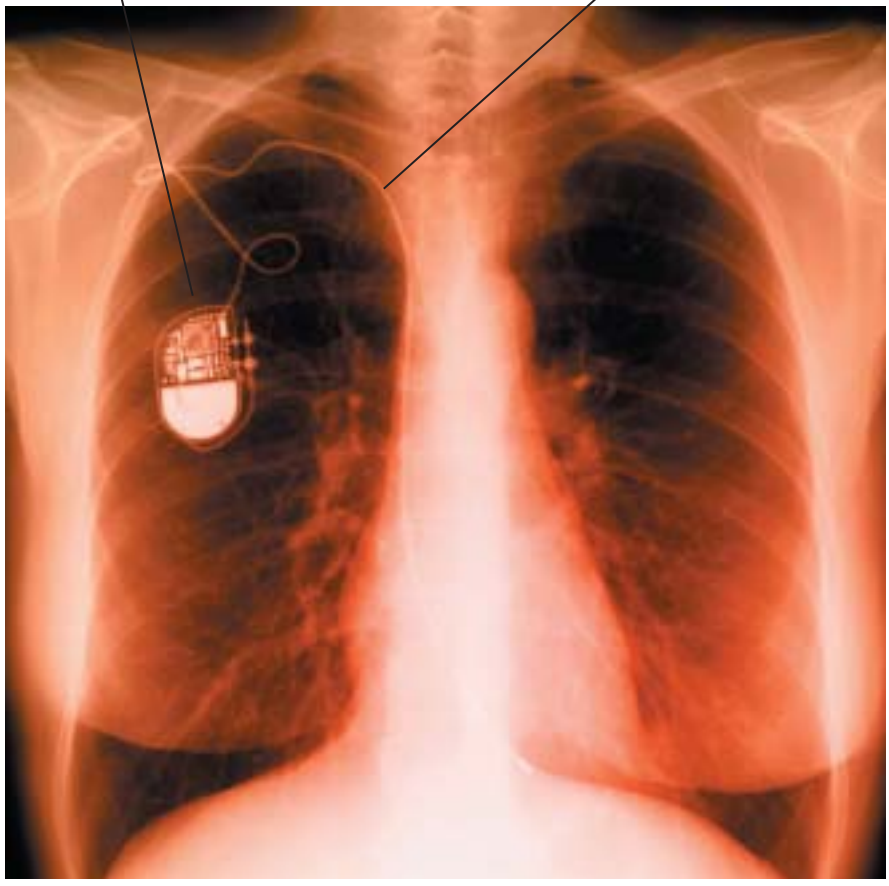
Pacemaker

Some pacemakers send a regular electrical pulse to the heart to tell it when to beat. Others remind the heart to beat only if it begins to beat too slowly. Some pacemakers can even detect the amount of activity the patient is doing and will change their heartbeat to match the activity level.

Pacemakers run on batteries that last from 7–12 years.

Leads

Electrical wires, called leads, connect the pacemaker to the heart. The leads carry electrical impulses to the heart and send messages from the heart back to the pacemaker. The leads are inserted through veins into the heart.



Programmer

The programmer is a computer system used by doctors and nurses to send and receive messages from the pacemaker. A doctor or nurse can change the settings of a pacemaker without surgery by using a magnetic-programming head. The programming head is placed over the pacemaker during a regular visit to the hospital or clinic. Some patients can even have their pacemakers checked over the phone through special transmitters.

Activities

REMEMBER

1. What medical trials did Arianne participate in herself?
2. Which sciences would you need to study at high school to become a histologist?

THINK

3. What do the two professions on this double page have in common?

BRAINSTORM

4. In groups of two or three, make a list of the skills and other qualities people in the health sciences need to have.

IMAGINE

5. Imagine you have just completed a design for the first pacemaker. List three issues you will need to research before you connect the pacemaker to a human body.

CONNECT

Use the Internet or other resources to answer the following questions.

6. Find out what the medical name for a slow-beating heart is.
7. Find out how the batteries in a pacemaker are changed.
8. On average, how many times per minute does a heart beat?

checklist

I can:

- ☐ describe some scientific jobs in the health industry
- ☐ describe how a pacemaker works
- ☐ appreciate some of the issues faced by the inventors of the pacemaker.

Out and about

When some people think of science and scientists, they think of people in laboratory coats, mixing substances in test tubes. Some scientists do exactly that. But many scientists don't spend much time in laboratories.

The scientists on this double page work outside. They are concerned about the environment, and the plants and animals that share it.

CAREER PROFILES

Name: **Peter Dann**

Job title: Research Manager

Company: Phillip Island Nature Park, Victoria

Field of science: Zoology

The purpose of the Phillip Island Nature Park is to protect the wildlife in the area. That means keeping the animals safe from threats to them and their environment. It also means managing the tourists that visit the park every day.

The job

Peter works with the penguins that nest on Phillip Island. He counts the birds and observes their behaviour. He uses **technology** to track the diving depths of the penguins and writes articles about his findings. At other times, Peter helps other people conduct research on penguins. He decides what will be researched and who will do it.

High-school science

Peter enjoyed science at school, but his interest in the environment mainly came from regularly camping out in the bush. His parents inspired and encouraged his interest in nature. He is now a **zoologist**. Zoology is a field of biology. He says, 'One of the good things about being in this job is that you work with interesting people. The people working around me and with me are a source of inspiration . . . You don't do this for the money. It suits my nature; I'm a curious sort of person. I like working outside and away from the big cities. There are also benefits for the community'.

Name: **Roger Kirkwood**

Job title: Research Biologist

Company: Phillip Island Nature Park, Victoria

Field of science: Marine biology

Roger also works at the Phillip Island Nature Park. He takes care of the fur seals in the area. Roger travels out to Seal Rocks every month to collect samples of animal wastes. From this he can tell what the seals eat. He collects data on the population of the seals and uses satellites to track their paths. Roger is often called on to rescue seals that become entangled in fishing nets.

Go to worksheet 5

The best and worst

Roger says that, 'The best part of the job is that there are no typical days. I get to work outdoors on something that is worthwhile and valuable. There is freedom to research and go to places others dream of going to. I scuba dive all the time and travel to the islands in Bass Strait. I've been to Antarctica nine times and to South America four times'.

Roger finds that the most difficult part of his job is sitting at his desk, looking at data. He says, 'It can also be frustrating to start new projects. A new project has to be accepted by so many people that it can take several months before the project can actually start'.

Study and skills

Roger has a Bachelor of Applied Science. He completed extra studies in zoology. People who work in this field often have a background in biology. They need to be able to get along with people, but also work in isolated situations.

Name: Tony Ladson

Job title: Project Leader (river restoration)

Company: University of Melbourne, Cooperative Research Centre for Catchment Hydrology

Field of science: Environmental engineering

Our waterways are important. They supply us with water, animals live in them and plants grow along them. But, sometimes, the waterways get 'sick'.

Tony has been described as a stream doctor. He works in the field to collect data about the flow of rivers. He studies materials along the beds and banks of rivers and is part of a large effort to improve their health. Tony says that the most exciting part of his job is '... coming up with new ideas, working with committed people and influencing what happens in the real world'. He also has the opportunity to write books, reports and manuals about the data he collects.

As a student

Tony enjoyed studying science — especially the 'hands on' parts. He has an engineering degree from the University of Melbourne, a Master of Science from the University of Minnesota, USA and a PhD from the University of Melbourne.

Looking to the future

Environmental engineering combines biology with geography and environmental studies. Tony says that, 'There are many jobs available in this field because it is so new. Improving the environment is essential for our future. Many of the jobs that will be in this field in the near future don't even exist now'.

Activities

REMEMBER

1. List two animals that the Phillip Island Nature Park looks after.
2. Why is Tony Ladson sometimes called a stream doctor?

THINK

3. Which area of science is common to all three of the jobs on these pages?

BRAINSTORM

4. In groups of two or three, discuss and write down answers to the following:
   (a) Outline the reasons why the three jobs on this page are important.
   (b) What would the world be like if these jobs did not exist?

INVESTIGATE

5. Investigate why fishing nets and pollution are dangerous to seals. Design a poster that aims to stop the problem.
6. What is the major natural predator of fur seals?

checklist

I can:

- ☐ appreciate that not all scientists work in laboratories
- ☐ list some of the jobs that environmental scientists do
- ☐ discuss how science can help to improve our environment.

The Eastern Shores Weekly

26 May 2007 Weather: 17°C

New road threatens koalas

Environmental experts and koala protection groups are concerned about a proposed new highway that is to be built through several koala colonies. The information was leaked a week ago to a member of the Eastern Shores Koala Protection Group. The highway is to extend into a previously protected area of the Eastern Shores Forest.

Garry Michaels, an **ecologist** and spokesperson for the koala protection group, says that members have been observing the area closely for the past twelve months. 'We have attached collars with radio-tracking devices to the koalas in the area to monitor their movements. Our results show that building a section of the highway through the forest will destroy the homes of 30 koalas', he said.

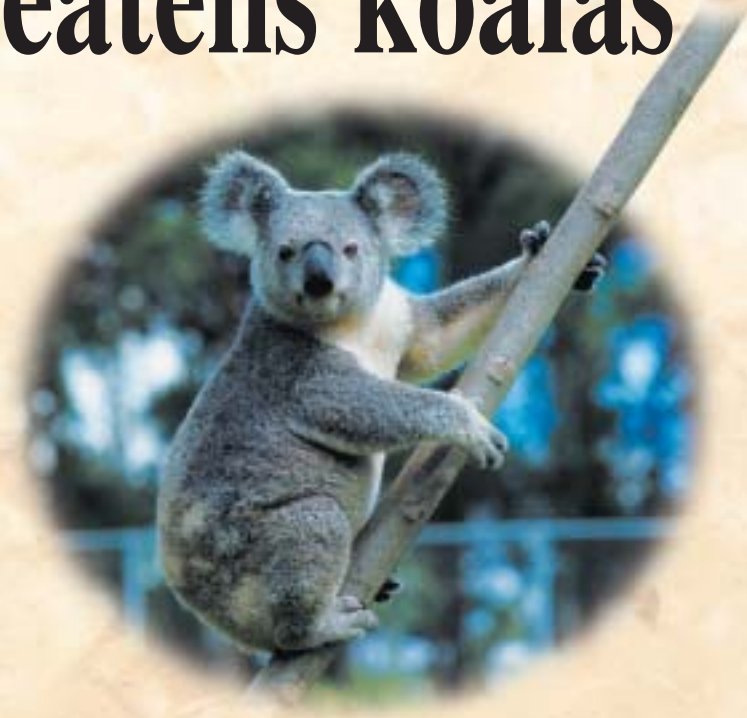
The future for the Eastern Shores koalas

The highway is the first part of a new development plan for the area. Land along the new highway will be turned into house blocks, parks, shopping areas and restaurants. Council members say the development will boost the number of tourists to the area and become an inviting, new place to live. New homes will have fantastic ocean and forest views.

A local council member was available for comment and said, 'We have done our homework. The koalas can be relocated to another part of the forest, where currently there are no koalas. They will have a larger area than the space they occupy now. We can fence off the area with new koala-proof fences so that they will not venture onto the new road'.

Garry Michaels and his team have thoroughly researched the diets of the local koalas by tracking them and by analysing their droppings. He claims that the koalas will not be able to move to another part of the forest because the trees that grow there are not the type the koalas prefer.

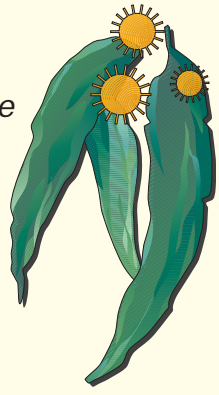
Experienced **zoologist** Carrie Taylor agrees that moving the koalas is not the answer. 'Koalas are fussy eaters. Each individual koala has its own preference for food and shelter. Moving a koala could be difficult, as you would need to find the same type of trees in another spot. Changing the social mix within a group of koalas could also be disastrous', she said.



Pathologists working with Carrie Taylor have gathered their own data from relocated koala colonies. Their research shows that many koalas have died of starvation when moved to other areas. Others have also become ill from the stress of being handled. Mr Michaels is afraid the same thing will happen to the Eastern Shores koalas.

K-Fence

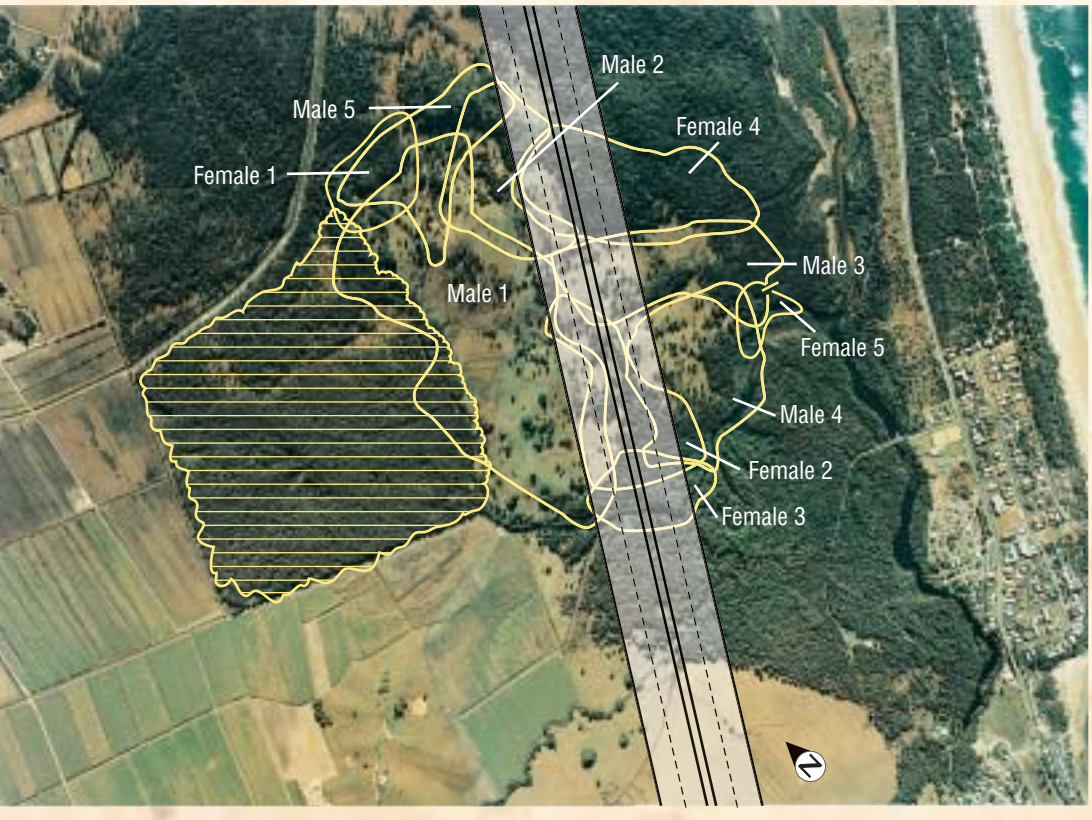
Research suggests that cars are one of the biggest killers of koalas. Do your bit to save the koalas on your property from reaching the roads.
Install a K-Fence NOW!!



Our fences are proven to be completely koala-proof. They are strong, environmentally friendly and easy to install.

Note: K-Fence is only for use along roadways.
It is not to be installed between joining properties.

The secret life of koalas



This map shows one of the koala colonies in the Eastern Shores Forest that will be affected by the proposed new highway. Each area shows the home range of a single koala. The shaded area is where the koalas could be relocated.

Each koala moves and feeds in its own home range until it's time to mate. The overlap between some regions is where the shared 'social' trees are located. These are the trees where the koalas mate.

The home ranges of males are larger than those of females. The size of the range gives a male more access to female koalas.

Activities

REMEMBER

1. Why are trees being cut down in the Eastern Shores Forest?
2. How does the koala protection group keep track of the koalas' movements?
3. How does Garry Michaels know what the koalas eat?

THINK

4. The council's solution to the koala problem is to relocate them. State the arguments against the council's solution.
5. Look carefully at the map of Eastern Shores Forest.
   (a) Which is the dominant male?
   (b) With which females does the dominant male share social trees?
   (c) Which social trees are in danger from the proposed highway?
   (d) What could happen if a single shared tree was removed from a colony of koalas?
   (e) How would this affect the overall population of koalas in the area?

BRAINSTORM

6. Make a list of the various scientists that could help people to decide whether the road should go through the forest.
7. In groups of three or four, brainstorm some solutions to the problem of the proposed highway. Take into account the research done by the local council, the koala protection group, Carrie Taylor and her team.

✓ checklist

I can:

- ☐ see how research is important in science
- ☐ explain how scientists can be involved in solving community issues
- ☐ discuss some of the threats to koalas.

Check and challenge

THE WORLD OF SCIENCE

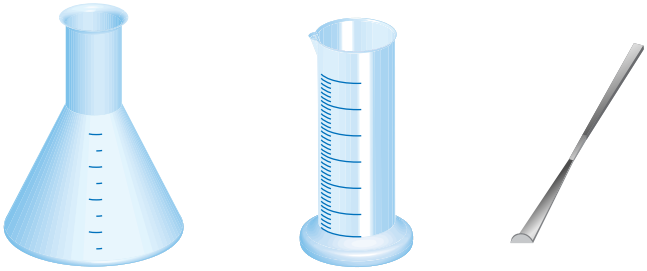
Scientists

1. Match the following scientists with their work.

Physicist	Investigates how rocks and mountains form
Chemist	Studies living things
Biologist	Explains things like movement, heat and light
Astronomer	Studies how substances react with others
Earth scientist	Studies the sky

Equipped for science

2. Draw and label the parts of a Bunsen burner. At the top of the Bunsen burner, draw in the safety flame. Show whether the air hole is open or closed.
3. Name these pieces of equipment and describe what they are used for.

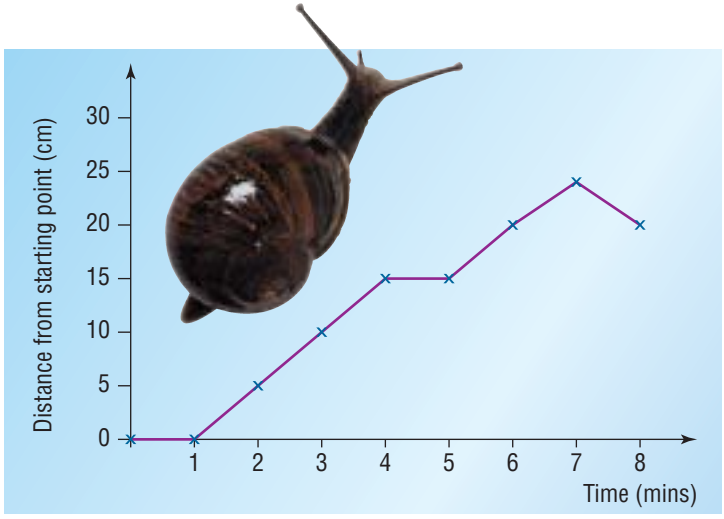


Hands on science

4. List two safety rules and explain why they are important.
5. (a) Give an example of a substance you might use in a laboratory that is:
   (i) corrosive
   (ii) flammable.
   (b) What precautions should be taken when using these substances?
6. Re-write the following sentences correctly by selecting the appropriate words in *italic*.
   (a) When lighting a Bunsen burner, light the match *before / immediately after* turning on the gas.
   (b) When using a thermometer to measure the temperature of a liquid as it is heated, place the bulb of the thermometer on the *bottom / near the centre* of the beaker.
7. Four students each measured the temperature in the classroom using a thermometer. Their results were:

Student	Temperature (°C)
1	23.5
2	24.0
3	25.0
4	22.0

   (a) Enter the results into a spreadsheet.
   (b) Draw a bar graph of the results.
   (c) What are some possible reasons for the difference between measurements?
8. The following graph shows how far from the starting point a snail moves in an experiment.



   (a) How far from the starting point was the snail after seven minutes of timing?
   (b) During what times did the snail not move at all?
   (c) What does the graph tell us about the snail's movement between the seven- and eight-minute marks?
9. Make a list of the information that a laboratory report should include.

Sports science

10. Up until 1964, a hand-held stopwatch was used to time events at the Olympics and other games. Since then, electronic and computerised timing have been introduced. Stopwatches can measure accurately to 0.1 s and electronic timers to 0.01 s. Computerised timers measure to 0.001 s.
    (a) Why is this type of technology important?
    (b) Why is it more important today than 50 years ago?
11. The following table shows the winning times for the men's 400 m freestyle swimming event. The data are from various Olympic games from 1896 until 2000.

Year	Name, country	Time (min/s)
1896	Paul Neumann, Austria	8:12.6
1908	Henry Taylor, Great Britain	5:36.8
1920	Norman Ross, USA	5:26.8
1932	Buster Crabbe, USA	4:48.4
1948	Bill Smith, USA	4:41.0
1960	Murray Rose, Australia	4:18.3
1972	Bradford Cooper, Australia	4:00.27
1984	George DiCarlo, USA	3:51.23
1996	Danyon Loader, New Zealand	3:47.97
2000	Ian Thorpe, Australia	3:40.59

    (a) Are all of the data over equally spaced time intervals?
    (b) Plot a graph of the times for the men's 400 m freestyle. Take into account your answer to question (a).
    (c) Use your graph to estimate what the winning time for this event was in the 1956 Melbourne Olympic games. Go to www.jaconline.com.au/science/weblinks to check your answer.
    (d) How have the winning times changed over the 104-year period?
    (e) Suggest some reasons for the change in winning times.
    (f) Discuss how you believe the times for the men's 400 m freestyle might change over the next 40 years.

Science: driving careers ahead

1. Read Laura Thomas's profile. Use the profile and any other resources to answer these questions and statements:
    (a) Make a list of some of the jobs that people with a science background can do in the automotive industry.
    (b) Which of these skills would be useful for engineers in the automotive industry: physics, biology or chemistry? Explain your answer.
    (c) What types of emission do cars produce?
    (d) What measures have car manufacturers taken to reduce emissions?
    (e) Make a list of the safety features that car manufacturers include in modern cars.
    (f) Imagine you are a reporter for a science magazine. What questions would you ask Laura Thomas and members of her team?

CAREER PROFILES

Name: Laura Thomas
Job title: Monaro program manager
Company: Holden Ltd
Field of science: Program management (engineering)

Laura has worked as a manufacturing engineer, design and development engineer, value analysis engineer, estimator, business analyst and program manager at various automotive companies.

Laura's current job is as a program manager with Holden Ltd. She says, 'I lead a team that turns ideas in people's minds into real cars driving on our roads. My team members come from styling, engineering, manufacturing, purchasing, quality, marketing and other departments. My job is to make sure that the car produced at the end of the project meets our customers' expectations, makes money and inspires our employees to come back for more ... With every new car introduced, we make some progress in the areas of **technology**, **emissions** and safety which benefits road users and the community at large'.

2. Look at the photograph of the scientist. What do you think the scientist is doing?

Summary of key terms

aerofoil: a wing designed to assist the lift in an aeroplane. It is used in reverse to stop lift in fast-moving cars.

astronomer: a scientist who studies the sky beyond the Earth's atmosphere

biologist: a scientist who studies living things

biomechanics: the study of how animals, including humans, move

cells: the small parts that make up living things

chemist: a scientist who studies how substances react with other substances

diagnose: to identify a disease or condition

drag: a push from air or water against an object. The push acts against the movement of the object.

ecologist: a scientist who studies the relationship between a living thing and its environment

emissions: chemicals released into the air. Cars release emissions.

engineer: a person who uses scientific ideas to build new technology and make it work.

gauge: an instrument for measuring pressure, volume, dimensions, petrol, speed etc.

geologist: a scientist who studies the Earth

histologist: a scientist who studies the **cells** and tissues that make up animals — including humans. Histologists look at small samples of cells and tissues under microscopes.

marine biologist: a scientist who studies **organisms** that live in the sea

metallurgist: a scientist who works to shape metals or improve its quality. A metallurgist may also be involved with mining for metals.

pacemaker: a device used to control the rate of a heartbeat

patent: permission to use or sell ideas without competition from someone else

pharmaceutical industry: an industry that manufactures medicines and other medical treatments

pharmacology: the study and manufacture of medicines and other drugs

physicist: a scientist who studies the laws of nature. For example, movement, heat, light, gravity and sound.

physiology: a study of how the parts of living things work together

receiver: a device which receives electrical or other signals

resin: a type of plastic

technology: the application of science to the design of a device or process

traction: the grip of an object (like a wheel) to a surface (like a road)

transmitter: a device which sends out electrical or other signals

zoologist: a scientist who studies animals

CHAPTER 2

Separating mixtures

You will discover

The properties of different mixtures

How to separate mixtures

How sewage is treated

How mixtures are separated in industry and for medical purposes

Pure chocolate?

This mixture of chocolates can be easily separated into colours. We can make up piles of red ones, blue ones and so on. But each pile is not pure chocolate. Each sweet is a mixture of a chocolate centre and a sugary coating. If we separate the centre from the coating we still won't have pure substances. Chocolate is a mixture of ingredients: cocoa, butter, sugar, milk and flavours blended together to give a great taste. The sugary coating is a mixture too. Even the colouring can be a mixture of colours.

Most substances on Earth are not pure. They are mixtures, like these chocolates. Some mixtures, like the water in the ocean, can be separated into pure substances quite easily. Others, like different gases in the air, are very difficult to separate.

1 Are the ingredients in chocolate pure or are they mixtures too?
2 How can we separate the different colours in the sugary coatings?
3 Apart from chocolate, air and water, what other substances are mixtures?
4 Why is it important to know how to separate mixtures?

Which are my shoes?

You're at a party — but your host won't let you wear shoes inside the house. So you take off your shoes and throw them on the pile beside the door.

When it's time to leave, you have to find your own shoes again. How do you do it? You probably recognise your shoes by their colour, shape, size and maybe even odour! These features are all **properties** of shoes.

The key to being able to find your shoes is knowing the properties of your own shoes. If you don't know their properties, you can't tell the difference between your shoes and all of the others in the mixture of shoes.

Properties are worth knowing

Properties of this gold ingot that we can see include:

- *Colour is yellow or gold.*
- *Base is rectangular with a slightly smaller rectangular top surface.*
- *The surface looks shiny.*

Other properties that we can't observe just by looking at the photo include:

- *It is heavy for its size.*
- *It is hard.*
- *It feels smooth.*

Anything that we can observe with our five senses is a property.

Some properties, such as the ability to conduct electricity, or to dissolve in water, or melt at a particular temperature, can be observed and measured in a laboratory.

Knowing the properties of different substances enables us to separate them from other substances in mixtures. This process is known as **separation**.

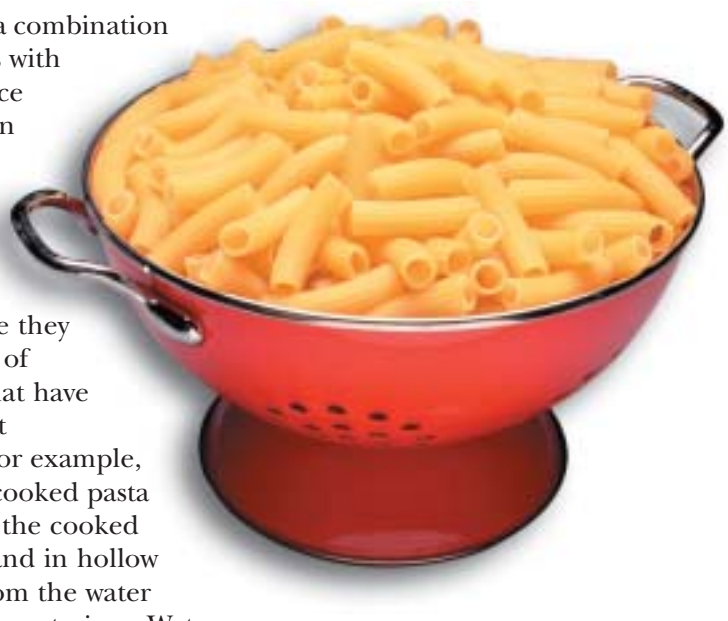
Big foot

The largest human foot belongs to an American, Matthew McGrory. His shoes are a size (US) 28.5. He wouldn't have too much trouble picking his shoes out of a pile!

Go to worksheet 6

Easy separating

A **mixture** is a combination of substances with each substance having its own properties. Some mixtures can be separated easily because they are made up of substances that have very different properties. For example, after you've cooked pasta you separate the cooked pasta (solid and in hollow cylinders) from the water (liquid) using a strainer. Water passes easily through the strainer, but the pasta gets caught.



If you are doing woodwork and you drop some nails in the sawdust there are several ways to separate them, because their properties are so different.

Separating a mixture of iron and sand

You will need:
mixture of sand and iron filings (prepared by your teacher)
magnet
plastic wrap or small plastic bag.

- Cover the magnet with the plastic.
- Pass the magnet slowly over the mixture of sand and iron filings.
- Observe what happens as the magnet gets closer to the mixture.

1. Can you separate the two substances?
2. Why was the magnet covered with plastic?

Not-so-easy separating

Not all mixtures are as easy to separate as iron and sand, or pasta and water. If you dropped some fine sand in sawdust, it would be much harder to separate the mixture. The differences between the properties of sand and sawdust are not as easy to see as the differences between nails and sawdust.

It's a bit like trying to find your shoes in that giant pile. Imagine how hard that would be if all the shoes were the same colour and shape.

This chapter describes techniques used for separating other mixtures. All of the techniques make use of the different properties of the substances in the mixture.

Activities

REMEMBER

1. What is a mixture?
2. Why are some mixtures easier to separate than others?

THINK

3. List all the properties you can think of, for:
   (a) salt
   (b) sand
   (c) water.
4. How would you separate the parts of a mixture of salt, sand and water? Use your lists in the previous question to find a method.
5. Imagine you dropped nails in the sawdust at woodwork. Suggest two reliable ways of separating the nails from the sawdust.

INVESTIGATE

6. Design and carry out an experiment to separate a mixture of sand from salty water. Ask your teacher before you carry out the experiment. *Hint:* Separate one substance at a time.
7. How would you separate the sand from a mixture of sand and sawdust? Draw a flow chart to show the steps you would use. Check your method with your teacher before trying out your experiment.

BRAINSTORM

8. Make a list of the properties of shoes; for example, water resistance.

✓ checklist

I can:

- ☐ explain what mixtures are
- ☐ identify the properties of substances that allow simple mixtures to be separated.

Filtering at home

On a hot summer's day, the last thing you want is a dirty swimming pool. Even though the windows in your house are all open, you certainly don't want flies and mosquitoes inside. And when it's too hot to be outside, you wouldn't want to come inside to find the airconditioner blowing out dust.

Keeping things apart

When leaves land in a swimming pool, a **mixture** is formed. The flies, mosquitoes and dust in the air form another uncomfortable mixture. But the parts of each mixture can be separated using **filtration**.

Filters allow some substances through while blocking others out. Particles that are smaller than the holes in the filter will pass through. Bigger objects will be blocked.

Airconditioners filter air as well as cool it. They have very fine filters that are designed to stop small dust particles getting through. Air is drawn into the airconditioner and filtered before it is cooled.

Screens are placed on windows and doors to keep insects out. They filter the air that enters your home and prevent insects from entering.

A basket is used to filter leaves and other large objects from the water. The water is then pumped through sand or other fine filters to remove smaller particles from the water.

A dust bag in a vacuum cleaner has holes in it that allow air to pass through, but trap dust and dirt.

Go to worksheet 7

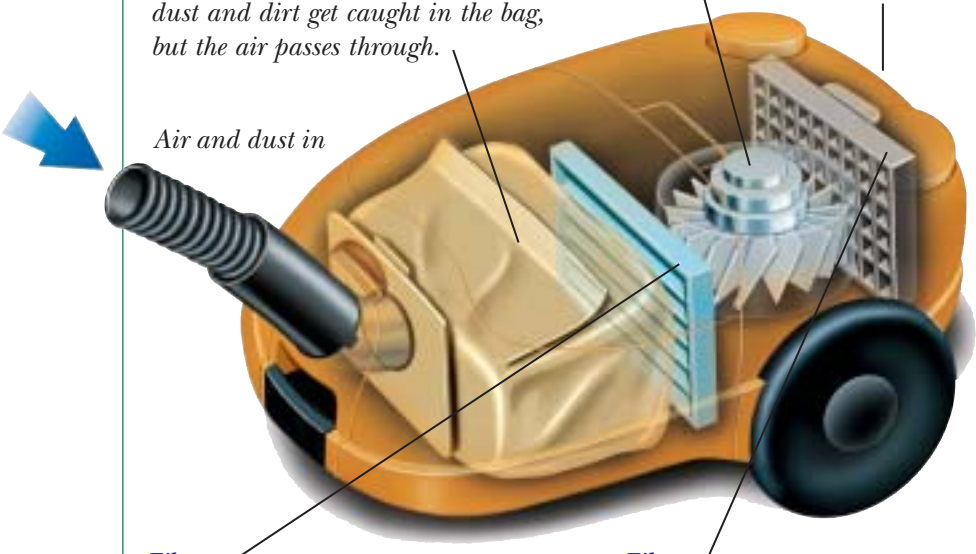
Cleaning up

Motor and fan
The motor turns a fan. The spinning fan creates the suction to draw in dirt, dust and air through the hose.

Dust bag
The dust bag in a vacuum cleaner is a filter. Dust particles, dirt and air are sucked into the dust bag. The bag has holes in it that are too small for dust and dirt to pass through. The dust and dirt get caught in the bag, but the air passes through.

Clean air out
Clean, dustless air leaves the vacuum cleaner.

Air and dust in



Filter
Some very small particles of dust may get through the dust bag. If dust passes through to the motor, the motor could be damaged. This filter catches small particles of dust that were not blocked by the dust bag.

Filter
Some vacuum cleaners have another filter to make sure that the air leaving the vacuum cleaner is clean.

Kitchen filters

The water that goes down our drains passes through many filters; the first is in the sink. The filter in the plughole stops food and scraps from the sink going down the drain.

The size of a filter depends on what needs to be blocked or stopped. The sink plughole filter has quite large holes. It is meant to stop only large food scraps getting through.

You wouldn't sift flour in a pasta strainer. The flour lumps would pass straight through. Tea strainers and coffee filters are much finer than pasta strainers. They need to stop small tea-leaves and ground coffee from flowing into the drink.

Activities

REMEMBER

1. What is the purpose of the dust bag in a vacuum cleaner?
2. Why does a sink have a plastic or metal insert in the plughole?
3. Why are holes in a pasta strainer bigger than the holes in a coffee filter?

THINK

4. Apart from insects, what else do screen doors keep out?
5. What property does filtration depend on?
6. The air that is drawn into a vacuum cleaner needs to pass through the dust bag before it leaves the vacuum cleaner at the other end. Why does a vacuum cleaner work better when the dust bag is empty?
7. Many kitchens have an exhaust fan. What mixtures would the filters above the stove separate?

BRAINSTORM

8. In groups of two or three, make a list of filters in each room of your home. What substances does each filter let through? What substances do they block?

INVESTIGATE

9. Find out what materials are used in fish-tank filters. Which ones are the best to use?

✓ checklist

I can:
- ☐ explain how filters work
- ☐ describe some uses for filters
- ☐ understand the need for differently sized filters.

Indonesia Review

28 August 2003 Weather: Sunny breaks, 18°C

Asia chokes

Every year a thick, hazy cloud moves over South East Asia. It's not a rain cloud. The dirty mix of smoke and fumes moves from Indonesia and Malaysia and spreads across to neighbouring countries.

The haze is the result of illegal forest fires lit by farmers, squatters and plantation companies. The fires pollute the air and threaten the survival of wildlife, including endangered species such as the Sumatran tiger and the orang-utan. Fires are a major cause of deforestation in South East Asia. In 1997 alone, over one million hectares of vegetation was destroyed by fire.

In 1997, *The Year of the Great Haze*, suffocating smoke affected 70 million people. The World Health Organisation has warned that haze-related deaths will increase. Although the fires have been less intense this year, officials have sent out health warnings to all residents on the Borneo and Sumatran islands. Hospitals are on alert, ready for an increase in breathing-related illnesses.

Residents are advised to:

- stay indoors as much as possible
- keep windows closed
- block all doorways and air vents with wet towels.

Those willing to risk going outside are buying face or gas masks to **filter** the air they breathe. Officials have warned that many of the masks are poor quality and do not protect the wearer's lungs.

Indonesian and neighbouring governments are trying to save the environment and protect people's health. Fire fighters have been sent to fight the fires instead of allowing the blazes to burn out. The Indonesian environment minister released the names of the plantation companies involved. But will tougher laws and stricter enforcement stop the illegal fires? Stan Ward, an Australian fire-fighting volunteer in Indonesia, doesn't think so. 'These guys will just light another fire across the road from where you are putting a fire out', he said.

Learning from Mexico City

High **pollution** figures released by Enviro Asia today revealed that of the 15 most polluted cities in the world, 12 are located in Asia. Much of the air pollution comes from cars, factories and power stations. As industries expand through Asia, governments are looking for help.

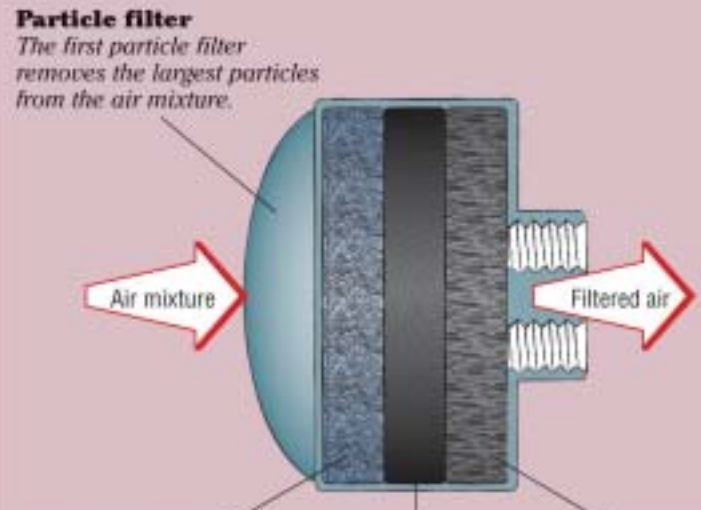
Mexico City has a long history of pollution and is still one of the most polluted cities in the world. Residents and governments of Mexico City are working together to create cleaner air and want to pass on what they have learned.

The government of Mexico City has set strict guidelines for industries and cars. All new cars in Mexico City are fitted with **catalytic converters**. These filters can be fitted to car exhaust systems. Harmful chemicals such as carbon monoxide and unburned hydrocarbons are released when fuel burns. As they pass through the catalytic converter, some of the harmful chemicals are trapped and others react to form less dangerous emissions.

Mexican officials will assess the situation in Asia. They have warned all industrial nations that without careful planning and strict guidelines, many more cities could become just like Mexico City, or even worse.

Gas mask sale!

Our gas masks are guaranteed to thoroughly purify air to meet strict standards.



Remember: Replace your filters on a regular basis.

Activities

REMEMBER

1. List three causes of air pollution.
2. List three filters mentioned in this newspaper article and advertisement.
3. Briefly describe a device used to **purify** the air we breathe.

THINK

4. Why are **concentrations** of gases like carbon monoxide greater in cities than in country areas?
5. Why do the filters from gas masks need to be replaced regularly?
6. Why were the residents of Borneo and Sumatra asked to block doorways and air vents with wet towels?
7. Suggest a reason why old cars have been banned from use in cities like Beijing and Mexico City.
8. The cloth masks that the people in the photograph are wearing do not filter many harmful chemicals from the air. Explain why the harmful chemicals can get through, but the dust particles cannot. Draw a diagram to help with your explanation.

BRAINSTORM

9. In groups of three or four, make a list of the ways in which we can reduce pollution from cars. In your list, include ways to reduce the number of cars on the road.

✓ checklist

I can:

- ☐ appreciate the problems caused by air pollution
- ☐ describe ways to purify air
- ☐ understand how a gas mask works.

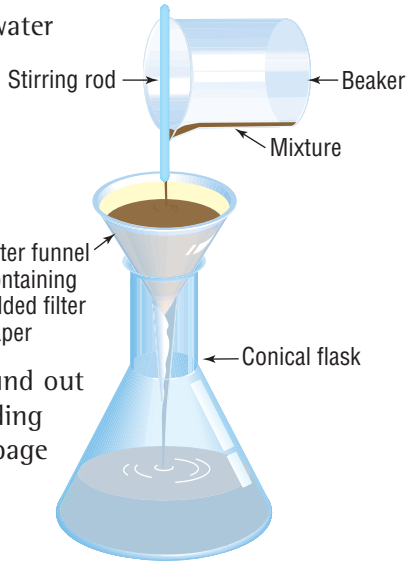
Going bush

After a long hike in the bush, a cup of billy tea would be delicious — except for the floating tea-leaves. A bottle of water collected from the creek would be thirst quenching — except for the muddy bits!

If you've ever been camping, you may have found yourself without the comforts of home. You probably didn't bring a tea strainer or a water purifier. How could you **filter** out the bits that you don't want?

Muddy water

You will need:
sample of muddy water
beaker
filter funnel
filter paper
stirring rod
conical flask.



- Fold the filter paper into quarters to form a cone. You can find out more about folding filter paper on page 244 of the Laboratory Toolbox.
- Place the filter paper in the filter funnel. Wet the paper a little to hold it in place.
- Set up the equipment as shown.
- Slowly pour the muddy **mixture** down the stirring rod and into the filter funnel.

!CAUTION: Do not allow the mixture to flow over the top of the filter paper.

1. Describe the sample of muddy water. For example, did the mud float on top of the water? Was it spread throughout the water?
2. The substance that passes through the filter, in this case the water, is called the **filtrate**. Describe the appearance of the water after it passes through the filter paper.
3. The parts left in the filter paper are called the **residue**. Describe the residue left on the filter paper.

Getting rid of mud

Muddy water is an example of a **suspension**. The dirt is suspended in the water, where it is still visible and doesn't **dissolve**. If you leave the muddy mixture to stand for a while, the dirt particles will sink to the bottom of the container. This important property helps us to separate dirt from water, even if we don't have a filter.

The muddy water can be **decanted** to separate the dirt from the water. Firstly, the mixture is left to settle. The dirt particles form a **sediment** on the bottom of the container. Secondly, the clean water from the top is poured off, or decanted carefully.

Decanting does not separate the muddy mixture as well as filtering. Some water is left in the residue. Some particles of dirt will be washed into the decanted water.

!CAUTION: Drinking creek water after filtering out the mud is only safe if you can be certain that no industrial wastes, harmful chemicals or germs have entered the water.

A camp cuppa

Billy tea is also a suspension. The tea-leaves are suspended in water. The tea-leaves are **insoluble**; that is, they won't dissolve in water. If left to stand, they will settle to form a sediment. To drink a cup of tea, the mixture could be decanted or filtered to remove the tea-leaves. Spinning the mixture very quickly also separates the leaves from the tea.

The method of spinning a mixture to separate its parts is called **centrifuging**. Solid suspensions are often separated from a liquid using a centrifuge. Two liquids can also be separated this way. The heavier parts of the mixture are forced to the outer edge of the centrifuge. The lighter parts can then be decanted off.

The tea is ready to drink after a few swings of the billy. The tea-leaves form a sediment on the bottom of the billy.

Making billy tea

No camping trip would be complete without billy tea. Billy tea was traditionally made in a metal can that was heated over a campfire.

Then a handful of dry tea-leaves was tossed into the boiling water and left to brew for a few minutes. For extra flavour, early settlers sometimes added gum leaves to the water as it boiled. To make the leaves settle to the bottom, the billy was swung in full circles at arm's length. Three anticlockwise spins and the tea was ready to drink. But they had to be careful. A timid swing or one ending in the wrong spot could result in a hot and painful soaking … and no billy tea!

Activities

REMEMBER

1. When **filtration** separates a mixture of muddy water:
   (a) which part is the filtrate?
   (b) which is the residue?
2. What happens to a suspension if it is left to stand for a long time?
3. Which method of separation uses spinning to separate the parts of the mixture?

THINK

4. What properties of water and dirt make them ideal to separate using filtration?
5. What properties of tea-leaves and water make them ideal to separate by centrifuging?
6. Early settlers would spin the billy three times in an anticlockwise direction before drinking their tea. Would it make any difference if the billy was spun in a clockwise direction? Explain your answer.
7. During filtration, why is it important that the mixture is poured carefully?

CREATE

8. Make your own billy tea. Instead of centrifuging the tea, use another method to separate the tea-leaves from the tea. Write down your method for separating out the tea-leaves. Was it an effective method? Explain your answer.

✓ checklist

I can:

- ☐ use filtration to separate the parts in a suspension
- ☐ explain how sedimentation, decanting and centrifuging can separate the parts in a suspension.

Mixtures to consume

Most of the food we eat and most of the liquids we drink are **mixtures**. Some, like fresh orange juice, form **sediments**. They are **suspensions**. But others are different types of mixture.

Drinking solutions

Most fizzy drinks are not suspensions. They are **solutions**. These solutions are made from carbon dioxide, sugar and flavours dissolved in water. Substances that **dissolve** are said to be **soluble**. Soluble substances do not settle to the bottom of a container. They spread evenly throughout the solution and can't even be seen.

Filtering cannot separate the substances in a solution. The dissolved substances pass straight through very fine **filters**. You can tell if a mixture is a solution by holding it up to the light. If it is a solution, you can see through it.

Cordial is a solution that you can easily make yourself. You can alter the flavour of cordial by changing the **concentration** of the solution. The more syrup you add, the more concentrated the solution becomes. If you add too much syrup, you can **dilute** it by adding more water.

Parts of a solution

Both cordial and fizzy drinks contain water. The water is a **solvent**. The sugars, flavours and carbon dioxide are all **solutes**. Solutes dissolve in solvents to form solutions.

Water is a good solvent, because many substances dissolve in it. Eucalyptus oil is a solvent for grease stains, because grease dissolves in it.

More edible mixtures

Another type of mixture is a **colloid**. Particles of one substance are spread evenly throughout another. Substances that form colloids can be solids, liquids or gases. These mixtures are not clear, like solutions, but do not settle to form sediment like suspensions.

An **emulsion** is a special type of colloid. Emulsions are formed when one liquid is spread evenly through another liquid and does not settle in a layer.

Fresh pumpkin soup is a colloid. Small bits of pumpkin are spread through water and do not settle in a layer.

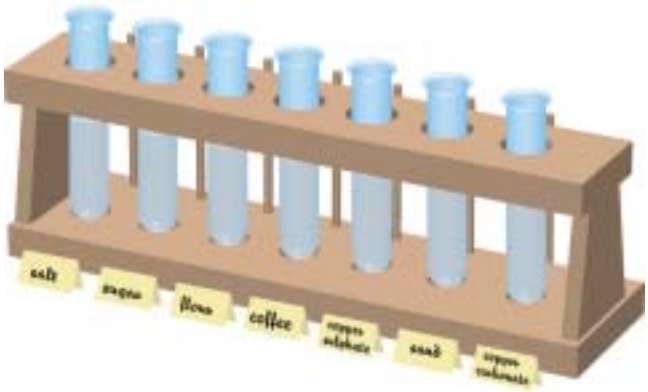
Marshmallows are colloids. They are mixtures of air spread through sugar and other ingredients.

***Homogenised* milk** *is an emulsion. It is made from butterfat (oil) spread through water. Both parts of this mixture are liquids.*

Go to worksheet 8

Soluble or insoluble?

You will need:
safety glasses and laboratory coat
heatproof mat 7 test tubes
test-tube rack spatula
samples of:
salt sugar flour coffee sand
copper sulfate copper carbonate.



- Half fill each of the test tubes with cold water.
- Label the test tubes salt, sugar, flour, coffee, and so on.
- Use a spatula to add a very small amount of each substance to its labelled test tube. Do not use more than a quarter of a spatula full.
- Shake each test tube. Refer to page 244 of the Laboratory Toolbox for instructions on how to do this.
- Draw up a table of your results like this incomplete one:

Substance mixed with water	Clear or cloudy?	Solution (soluble) or suspension (insoluble)?
Salt Sugar Flour ...		

- Hold each test tube up to the light. Decide whether the mixture is clear or cloudy. Record your results in the table.
- Decide whether each mixture is a solution or a suspension. Record this in the table.

1. Which of the substances dissolved in water?
2. How can you tell if a substance has dissolved?
3. Which of the mixtures could be separated by **filtration**?

Activities

REMEMBER

1. What solutes are present in fizzy drinks?
2. Which two types of substance mix to form emulsions?
3. Label the solvent, solute and solution in this photo:

THINK

4. Is fog a solution, suspension or colloid? Explain your answer.
5. Non-homogenised milk separates into two parts if it is left to stand. But homogenised milk does not settle. It is an emulsion.
   (a) What type of mixture is untreated milk?
   (b) Cream and butter are both made from milk. What types of mixture are cream and butter?

INVESTIGATE

6. Some substances dissolve better in hot water than in cold water. Design an experiment to measure the amount of sugar that can be dissolved in tap water that is cold and tap water that is hot.

CLASSIFY

7. In groups of three or four, decide whether the following substances are suspensions, solutions, colloids or combinations of these. Write down a reason for each decision. Share your results with the class.
   muddy water cup of coffee
   mayonnaise whipped cream
   hot chocolate
   cup of tea with tea-leaves in it

checklist

I can:
- ☐ describe how solutions are formed
- ☐ distinguish between suspensions, solutions and colloids
- ☐ explain how to dilute concentrated solutions.

Salty solutions

The oceans and seas hold most of the world's water. So, castaways stranded on an island in the ocean should have no trouble finding a drink. But sea water is far too salty to drink! So, how can pure water be **separated** from a salty-water **mixture**?

A solution to the problem

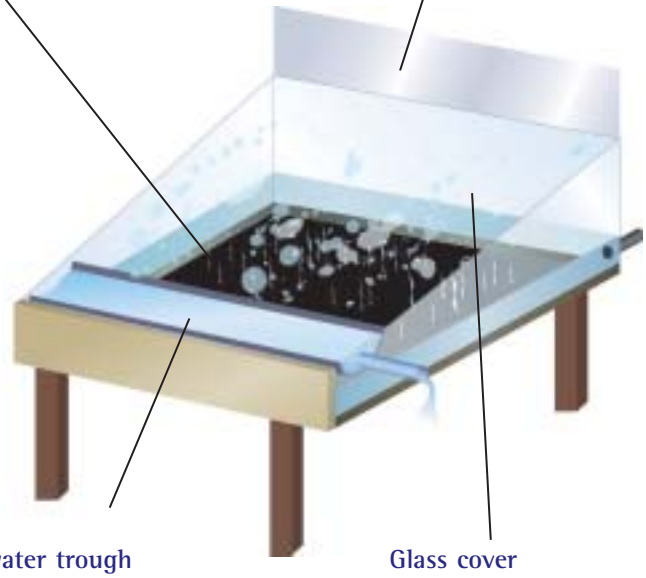
Sea water contains salt and many other substances. Larger objects, like seaweed and bits of shell, can be removed by filtering the mixture. But the filtered salty water is a solution. The salt will not separate from the water during the filtering process. Another method, called **distillation**, can be used to **purify** the water.

Solar distillation

Life-saving water stills have been built in many places where fresh water is scarce. Water stills heat water, evaporate vapour from the mixture and collect the pure water.

Black tray
The black tray warms up when the sun shines on it. The salty water in the tray heats up as well. The water begins to evaporate, leaving the salt behind.

Reflector
The reflector helps to direct sunlight onto the tray.



Clean water trough
The liquid water trickles down along the glass cover and falls into a trough. This water is free of salt and other impurities. The salt remains in the black tray, where it can be collected and used for other purposes.

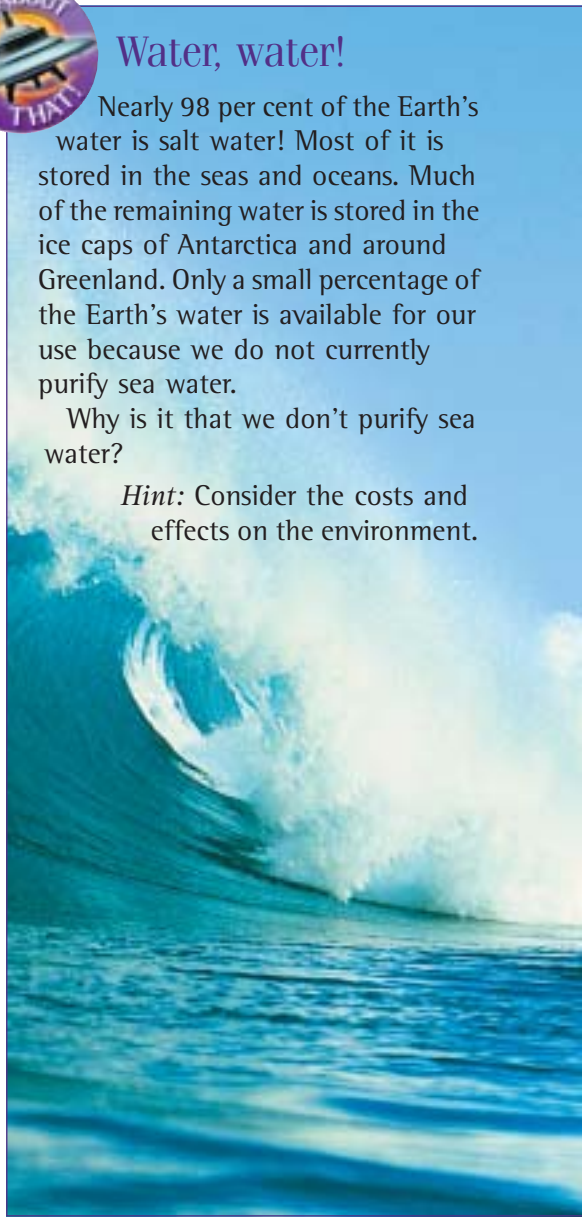
Glass cover
*The glass cover stops the evaporated water from escaping. When the **water vapour** reaches the glass, it begins to cool down. The vapour turns back into liquid water.*

Water, water!

Nearly 98 per cent of the Earth's water is salt water! Most of it is stored in the seas and oceans. Much of the remaining water is stored in the ice caps of Antarctica and around Greenland. Only a small percentage of the Earth's water is available for our use because we do not currently purify sea water.

Why is it that we don't purify sea water?

Hint: Consider the costs and effects on the environment.



Go to worksheet 9

Crystallisation and distillation

You will need:

Bunsen burner, matches and a heatproof mat
tripod and gauze mat
evaporating dish
salt-water mixture
safety glasses and laboratory coat
distillation kit (such as Quickfit apparatus).

What is distillation?

Distillation separates substances by heating them and collecting the vapour that is boiled off.

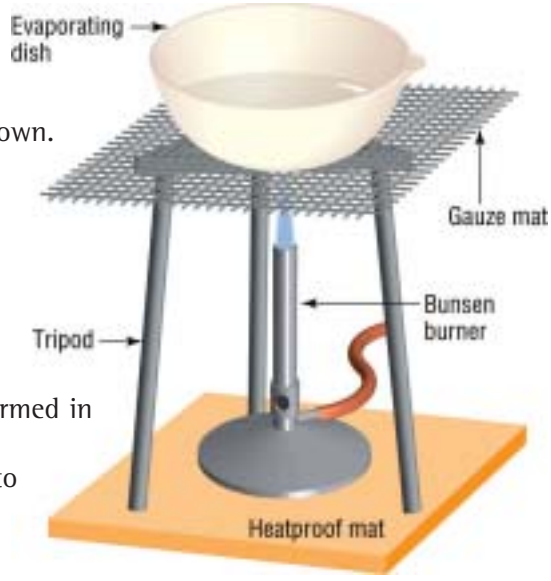
This method separates:

- solutions where a solid has been dissolved in a liquid
- two liquids that have different boiling points.

After distilling, the two substances in the mixture are separated into different containers. One remains in the original container. The other is collected as a vapour and cooled to become a liquid again.

Part A Crystallisation

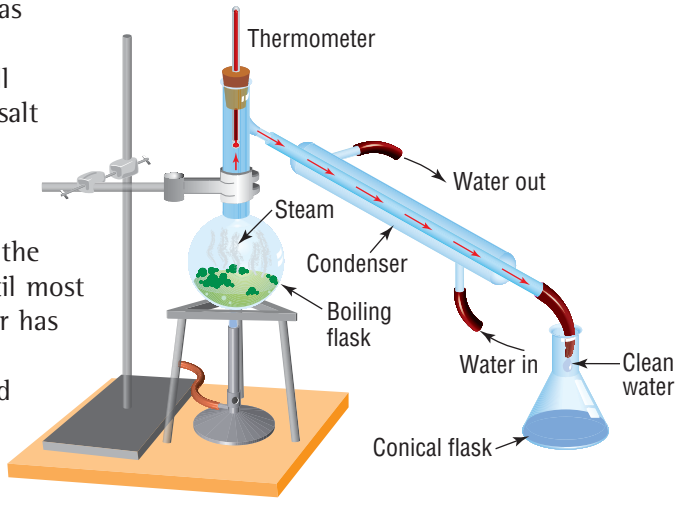
- Pour a small amount of salt water into the evaporating dish.
- Place the evaporating dish on the gauze mat as shown.
- Gently boil the salt water until most of the water has evaporated.
- Once the dish has cooled, observe its contents.



1. What **residue** has formed in the dish?
2. What has happened to the water?
3. This method of separation is called crystallisation. What could it be used for?

Part B (teacher demonstration) Distillation

- Set up the distillation equipment as shown.
- Pour a small amount of salt water into the boiling flask.
- Gently boil the mixture until most of the water has evaporated.



4. When would distillation be used instead of crystallisation?
5. What is the purpose of the water surrounding the condenser?

Activities

REMEMBER

1. Which methods of **separation** can be used to separate the parts of a solution?
2. What is the purpose of the glass cover of a solar water still?

THINK

3. What difference in properties does distillation rely on?
4. Why is crystallisation not suitable for purifying water?
5. Where does table salt come from?
6. Find out how distillation is used in industry.

CONNECT

7. Go to www.jaconline.com.au/science/weblinks and click on the Perfume link for this textbook to find instructions on how to distil perfume. Try using orange peel, flower petals or leaves.

checklist

I can:

- ☐ describe the processes involved in distillation and crystallisation of mixtures
- ☐ use crystallisation to separate the parts of a solution.

Separating colours

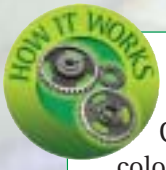
For thousands of years, people have added colours to food. The colours made food more appealing. Early civilisations used saffron and other spices to colour some foods yellow. Some of the colours added to the taste, but some were poisonous.

Today, colours are added to food for many reasons. Some colours protect vitamins that are affected by sunlight. Other colours replace colours lost during storage. Some fruits and vegetables are coloured to give them a more even appearance. Other foods, like lollies, are coloured just for fun!

Safe colours

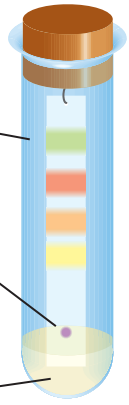
Because some colours added to food in the past were actually poisons, colours and other additives must pass strict tests before they can be used in foods. Foods with added colours can be tested at any time to identify the particular additives that have been used.

One method used for testing is called **chromatography**. Even foods that appear to be just one colour could be a mixture of colours. Chromatography separates out the different colours in the **mixture**.



Chromatography

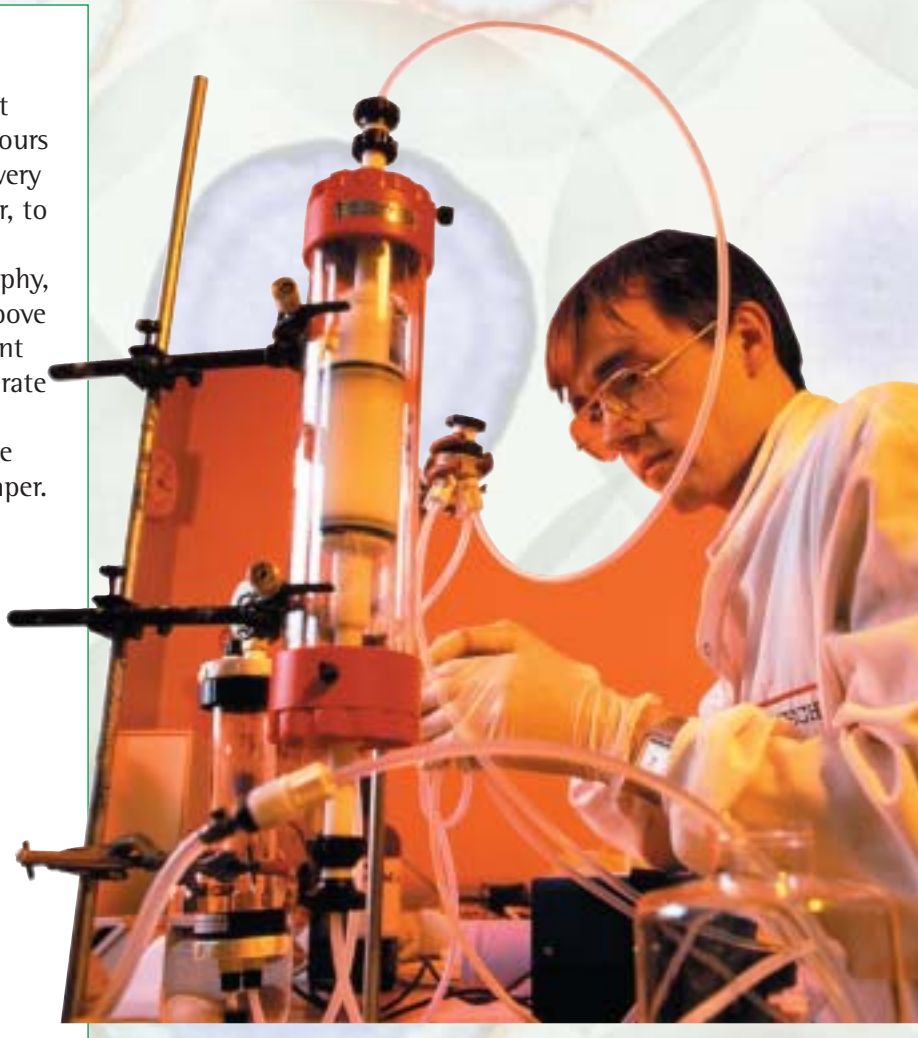
Chromatography works because different colours have different **solubilities**. Some colours **dissolve** more easily than others. Water is a very good **solvent** for many food colours. However, to **separate** the colours, they are not all placed straight into the water. For paper chromatography, the food colouring is placed on paper just above the solvent. The colours dissolve as the solvent soaks up the paper column. The colours separate because they are washed along the paper at different rates. The least soluble colours move more slowly and travel less distance up the paper.



Separated colours
The colours that dissolve more easily are carried furthest up the filter paper by the solvent. The colours become separated along the paper strip.

Sample of food-colour mixture
A small amount of food colour is placed on the paper, above the level of the solvent.

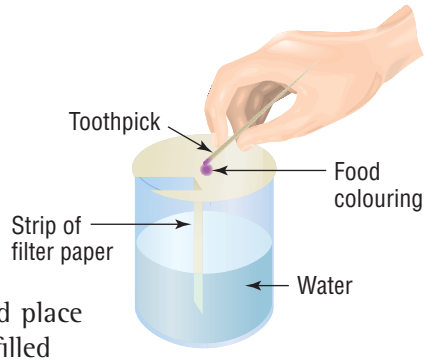
Solvent
The filter paper is hung so that it just dips into the solvent. The solvent soaks up the strip of filter paper, taking the food colours with it.



A chromatograph automatically separates mixtures by chromatography.

Separating colours with chromatography

You will need:
food colouring
filter paper
beaker
toothpick
scissors.



- Cut a strip into the filter paper and place it over a half-filled beaker of water. The strip should just dip into the water.
- Use the flat end of a toothpick to place a small amount of food colouring onto the centre of the filter paper.
- Wait for the water to soak up into the filter paper. Observe the mixture as it separates into different colours.
- Try this experiment with other food colourings.

1. Are the colours used in food colouring single colours or mixtures of colours? Explain your answer.
2. How many colours in total are needed to make each of the different food colourings that you tested?

Chromatography in industry

Chromatography is used in the food industry to detect more than just food colours. Food scientists can tell us what other ingredients have been added to food. Chromatography can also identify pesticides and harmful chemicals that have entered our food from the water in creeks and dams, or from soil **pollution**.

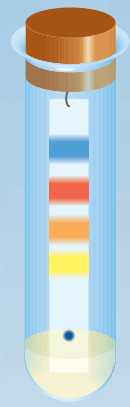
Forensic scientists use gas chromatography to detect a range of substances, including traces of illegal drugs. They can also use chromatography to compare mixtures found at crime scenes with those found on suspects. Many mixtures contain a unique combination of substances. For example, ink from different pens is slightly different, even if the colours *look* the same.

The separating technique of chromatography is used to detect substances in blood and urine. In medical laboratories, samples of blood or urine are tested for drugs and alcohol. Abnormal levels of vitamins and hormones in a person's blood can also be detected using chromatography.

Activities

REMEMBER

1. Complete the sentence: Chromatography is a method used to separate parts of a mixture with different _____.
2. Why is water used as a solvent to separate food colours?
3. List the colours found in this ink, from:
   (a) the fastest moving to the slowest moving
   (b) the most soluble to the least soluble.
4. List three uses of chromatography in industry.



THINK

5. Why is the mixture placed above the level of the solvent in chromatography?
   Hint: What would happen if the mixture was placed in the solvent?
6. Zoe performs a chromatography experiment on waterproof markers using water as a solvent. Will her experiment work? Explain your answer.

CREATE

7. Use chromatography to create colourful designs that can be displayed as scientific art. Fold the filter paper and use different colours to make your designs unique.

INVESTIGATE

8. Investigate different types of solvent that could be used to separate pen ink and waterproof markers. Before running the experiment, have your choice of solvents approved by your teacher. As a starting point, you may wish to use methylated spirits.
9. Does the colour of food affect whether or not people choose to buy and eat it? Design an experiment to test your answer.

checklist

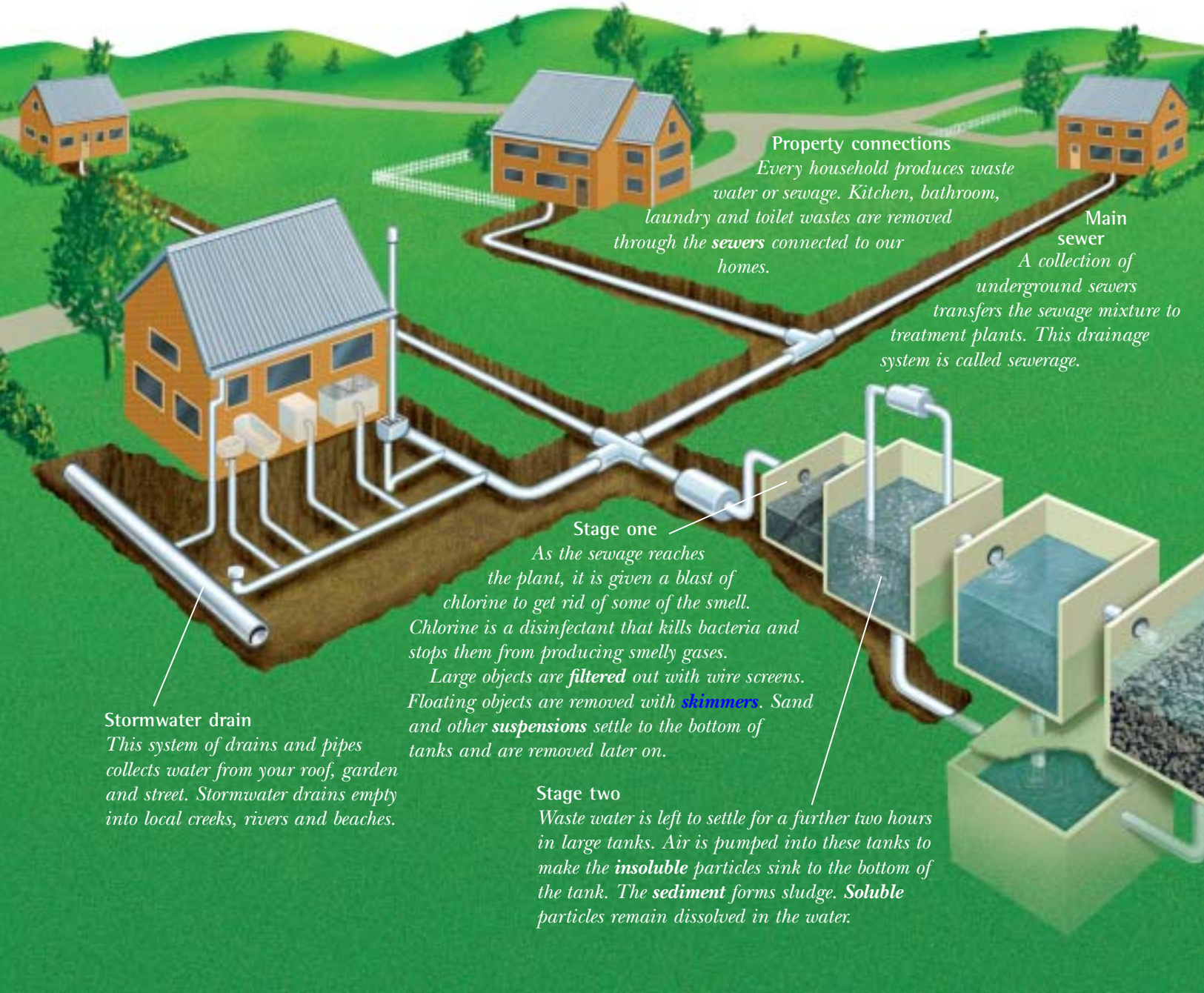
I can:
- ☐ explain, in basic terms, how chromatography works
- ☐ use chromatography to separate mixtures of colours
- ☐ describe examples of how chromatography is used in industry.

Flushing wastes

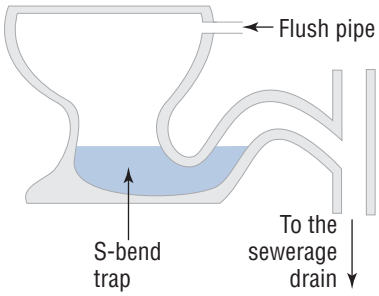
The waste water that flows from toilets, laundries, bathrooms and kitchens must be purified before the water is used again or released into the sea. **Bacteria**, which can cause diseases, also needs to be removed before the waste water is safe to reuse or release into the environment.

Separating and treating

The introduction of a **sewerage** system through Melbourne stopped the spread of diseases like typhoid. A system of drains now transports **sewage** to water-treatment plants. One water-treatment plant is in the suburb of Werribee, south-west of Melbourne. The plant separates the water from the other parts of the sewage mixture. Purified water from the sewage mixture can be released back into Port Phillip Bay and major rivers. It is also used for irrigation of crops. This important separating process keeps our waterways, beaches and drinking water clean and free from **contamination**.



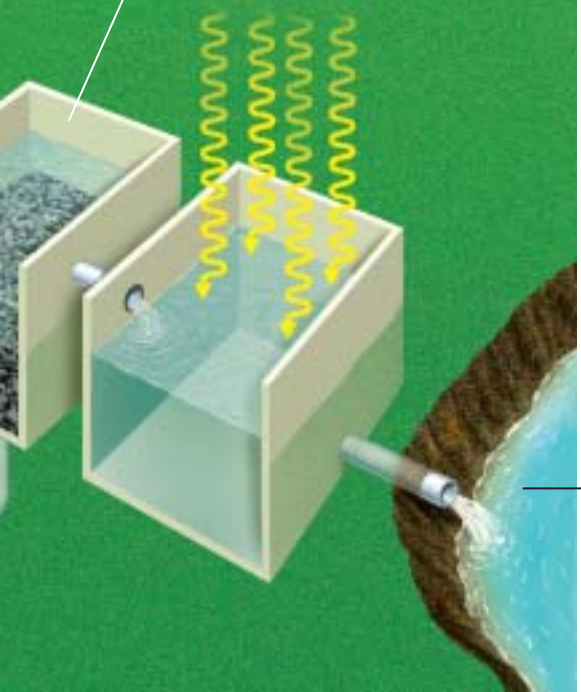
Filtering out bad smells



Bacteria and substances that are washed down with waste water can have a bad odour. The sewer pipes that remove waste water from your home fill up with smelly gases. A simple filter keeps the smells from entering your house.

The pipes that carry waste water from toilets and sinks have bends in them called S-bends. The bends fill with water, which acts as a filter. Gases cannot pass through the water back into the house, so our homes stay odour free!

Stage three
'Good' bacteria feed on materials that are still in the waste water. The water is trickled through layers of stone, where the bacteria remove the last traces of waste from the water.



Outlet to sea
The water is given one last dose of chlorine, to kill the bacteria. The water then passes through ponds, where sunlight ***disinfects*** *the water before it is released to Port Phillip Bay or used for irrigation.*

Smelly old Smellbourne

If you walked the streets of Melbourne in the 1880s you would understand how Melbourne earned the name 'Smellbourne'. Kitchen, bathroom and laundry waste water was poured into open drains along the streets. Even human toilet wastes floated along these drains. Industry and animals added to the murky mixture that flowed from the drains into the rivers and out into Port Phillip Bay. Clean drinking water wasn't easy to find, because the rivers and lakes were becoming contaminated. Bacteria which cause diseases, such as typhoid, were spread by these open drains.

Activities

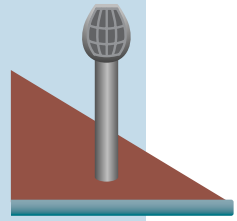
REMEMBER

1. How was clean drinking water being contaminated in Melbourne in the 1880s?
2. From which rooms of a house does the sewer remove waste water?
3. Which separating method removes large objects from sewage mixtures?

THINK

4. Explain the difference between sewage and sewerage.
5. How does the environment benefit from separating water from the other substances in sewage?
6. Why should harmful chemicals and rubbish never be poured down stormwater drains?

OBSERVE



7. Vent pipes and gully traps allow any smelly gases to escape from the sewer pipes.
   (a) Where are the vent pipes and gully traps in your home?
   (b) What would happen if the gully trap was covered?

CONNECT

8. Go to www.jaconline.com.au/science/weblinks and click on the Think Before You Flush link for this textbook. Make a list of items or substances that you should not pour down the kitchen sink or flush down the toilet. In a two-column table suggest why each item should not enter the sewerage system.

checklist

I can:

- ☐ explain how sewage travels from my home to the treatment plant
- ☐ understand the different separating methods used with sewage.

Recycling products

Most local councils have a **recycling program**. Items such as paper, some plastics, glass, aluminium and steel can be recycled and made into new products.

Recycling reduces the amount of rubbish created and saves precious resources such as trees and bushland. Many **manufacturing processes** pollute the environment. Recycling and reusing materials reduces the need for manufacturing processes.

Separating by sight

Household rubbish is usually a **mixture** of food scraps, recyclable materials and other waste. The first step in recycling is to **separate** the recyclable items from other household rubbish. We can see the differences between the types of rubbish, and we know which items can be recycled. Big recycling plants use this knowledge to separate the tonnes of recycled goods they receive.

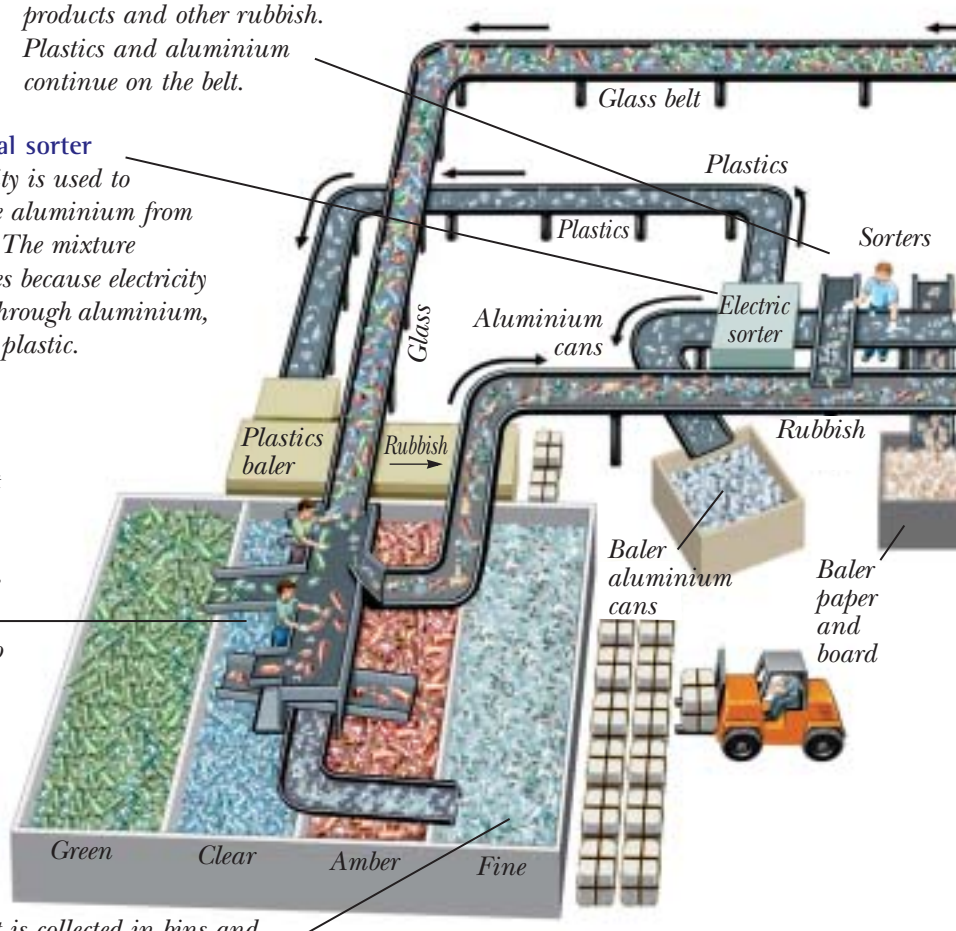
Magnet
Steel cans are separated using a magnet. The steel is collected in a separate container, ready to be sent to steel manufacturers. Material that is not attracted to a magnet continues along the conveyor belt.

Air classifier
Plastic, aluminium and paper cartons are lighter than glass. A blast of air blows these lighter materials to a separate conveyor belt.

Sorters
People sort out paper products and other rubbish. Plastics and aluminium continue on the belt.

Electrical sorter
Electricity is used to separate aluminium from plastic. The mixture separates because electricity passes through aluminium, but not plastic.

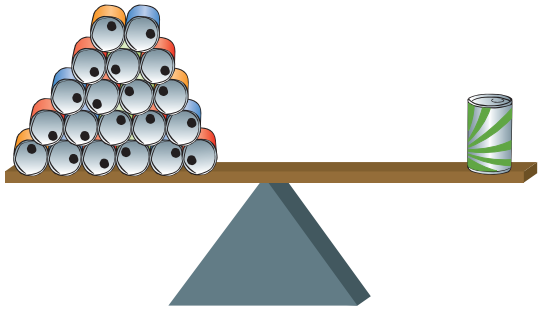
Sorters
People separate different coloured glass by hand. Any other rubbish that has made it through the conveyor belt is also removed. Glass is sent to processing plants where each separate colour is crushed, melted and mixed with raw materials to make new bottles and jars.



Fines bin
Finely broken glass cannot be separated by sight. It is collected in bins and sent to a glass processing plant. At the processing plant, lasers are used to separate very small pieces of differently coloured glass.

Saving energy

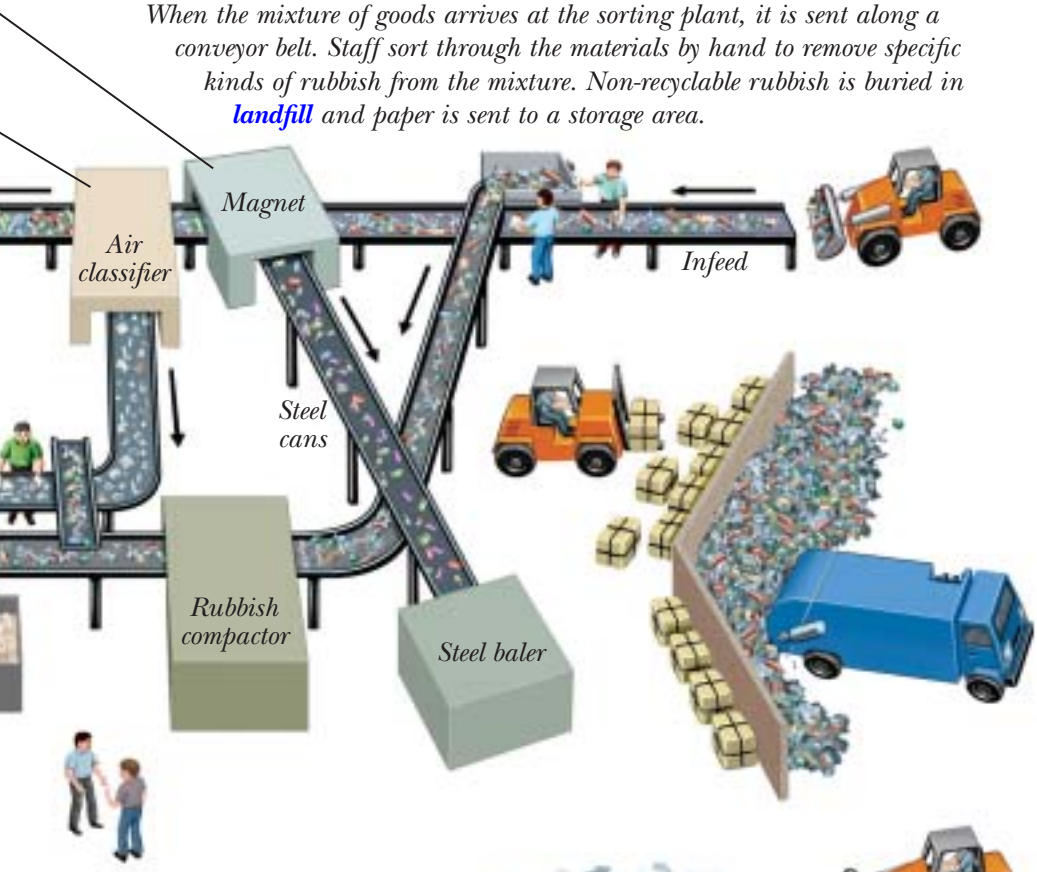
How does recycling bottles save oil, coal and gas? It takes energy from fossil fuels to run the equipment that mines and processes the materials that make new products. Recycling also reduces the amount of **raw materials** that need to be extracted from the ground and processed.



It takes the same amount of energy to make one new aluminium can from raw materials as it does to make 20 new cans from recycled ones.

Pre-sort

When the mixture of goods arrives at the sorting plant, it is sent along a conveyor belt. Staff sort through the materials by hand to remove specific kinds of rubbish from the mixture. Non-recyclable rubbish is buried in ***landfill*** *and paper is sent to a storage area.*



Paper storage

Paper makes up a large part of our household rubbish and we usually sort it out at home. It is often collected separately from other recyclable materials. From the recycling plant it is sent to ***paper mills*** *for re-processing. At the mill, paper is shredded and mixed with water to make new paper products.*

Activities

REMEMBER

1. Give two reasons why recycling is good for the environment.
2. Why do recyclable materials need to be separated?

THINK

3. Prepare a table like the one below and fill it out with information on separating rubbish.
   (a) Record the methods used to separate different types of material in a recycling plant.
   (b) For each method, record which material is removed from the flow of rubbish.
   (c) What properties of this material allowed it to be separated from the mixture?

Method	What is removed?	Properties

4. Why aren't magnets used to separate out aluminium cans?
5. Explain why people, rather than machines, need to manually separate some of the recycling mixture.

CREATE

6. Design a poster or brochure that explains which items can be recycled. Check with your local council about how they prefer recycling materials to be separated ready for collection. Include this in your brochure or poster.

checklist

I can:

- ☐ understand how a recycling plant operates
- ☐ see how different separation techniques suit different materials.

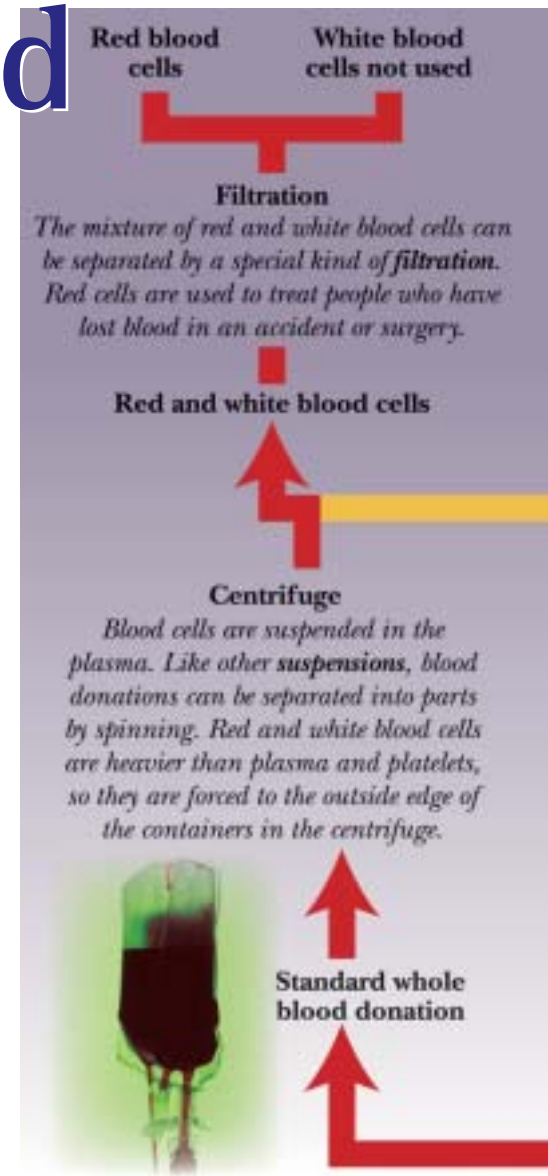
Separating blood

About one million donations of blood are made in Australia each year. Some of the donations are given to people who have lost blood during surgery, accidents or disasters. Blood is also given to people during the treatment of many diseases, including cancer. These people need to be given a regular supply of blood.

The blood mixture

Blood is a life-giving **mixture**. It can be separated into four parts: **plasma**, a clear, yellowish liquid; **red blood cells**, to carry oxygen; **white blood cells**, to fight disease; and **platelets**, to clot blood.

Because each part of the blood has a special job to do in our bodies, different problems can be treated with different parts of the blood. In Australia, blood is collected and separated by the Australian Red Cross Blood Service. **Separation** allows doctors to treat a larger number of patients and save many lives.



HOW ABOUT THAT!

The average human body contains between 5 and 7 litres of blood. Each single drop of blood contains about 250 million red blood cells, 400 000 white blood cells and 15 million platelets. During exercise, each of these drops of blood passes through the heart about 8 times every minute.

The blood travels through blood vessels. The total length of a person's blood vessels is about 800 000 km! Compare this to the circumference of the Earth which is about 40 000 km. How amazing is that!

HANDS ON SCIENCE

Using a centrifuge (teacher demonstration)

You will need:
centrifuge
mixture containing iron oxide, lead oxide (red lead) and water.

!CAUTION: Use red lead in a well-ventilated room. Avoid contact with skin and eyes. Do not dispose of down the sink.

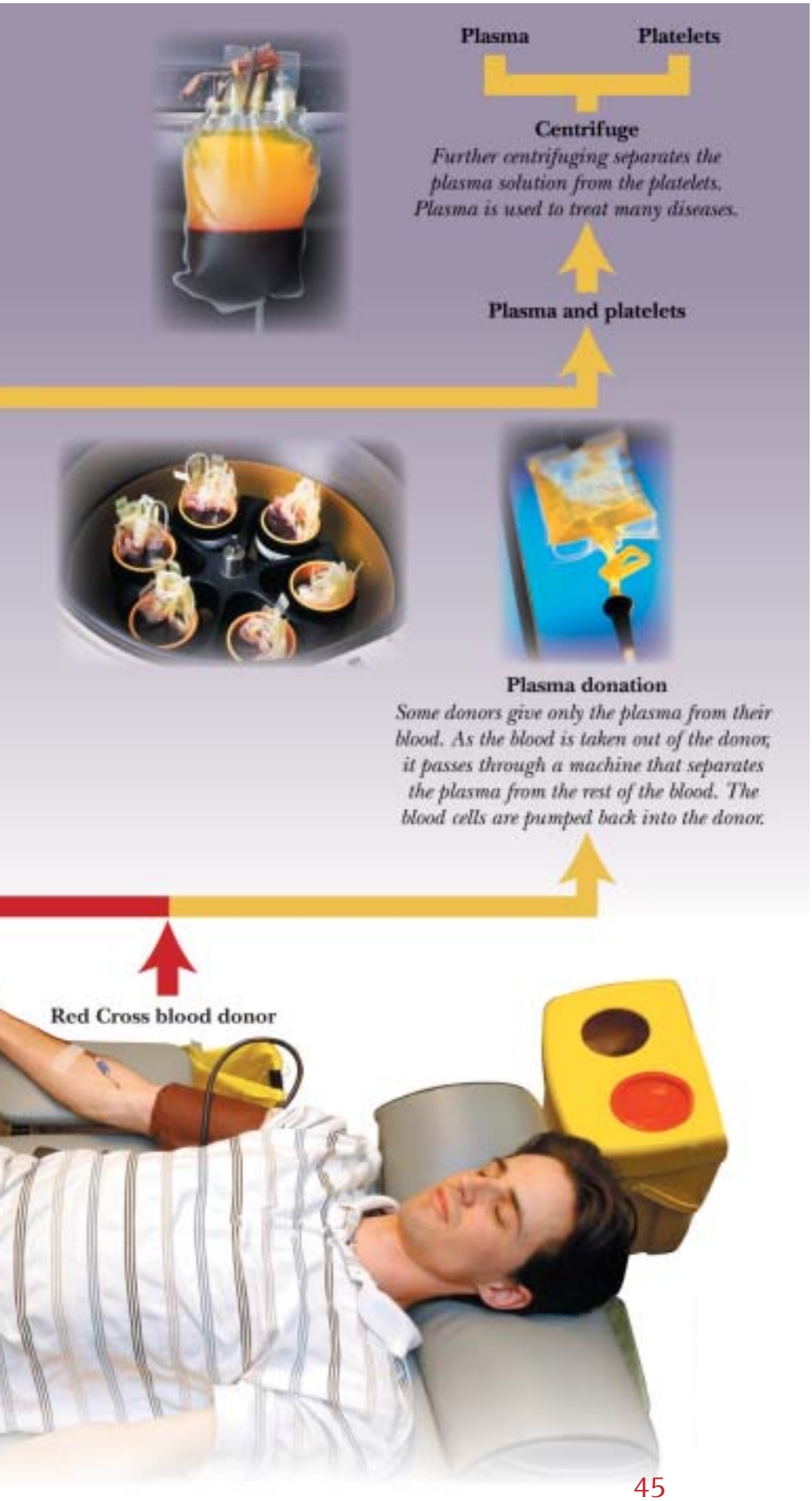
- Stir the mixture then pour equal amounts into two separate centrifuge test tubes.
- Place the test tubes on opposite sides of the centrifuge.
- Allow the centrifuge to spin for about a minute.
- Observe the mixture after centrifuging.

1. Describe the mixture after **centrifuging**.
2. Why must the test tubes be placed on opposite sides of the centrifuge?
3. Could the separated substances form a mixture again? Explain your answer.
4. What type of mixture was the iron oxide, lead oxide and water before centrifuging?

Once blood is separated, each part has to be stored differently.

- Red blood cells can be stored for 42 days at 2–6 °C.
- Plasma can be frozen for 12 months at –40 °C.
- Platelets are stored for 5 days at 20–24 °C. During this time they have to be moved at least every 12 hours, to stop them clumping together. (Platelets seal wounds in our bodies by sticking together.)

Go to worksheet 10



Activities

REMEMBER

1. Why is blood separated into different parts?
2. Why do blood clots not form in a blood donation?
3. Which technique is used to separate the different parts of blood?
4. Turn to page 39 to answer this question. Which separation technique is used to detect traces of drugs in a patient's blood or urine?

THINK

5. Why is blood separated in a centrifuge rather than left to settle by itself?
6. What property of plasma and platelets allows them to be separated with a centrifuge?

CREATE

7. Create an advertisement to encourage people to donate blood. The advertisement could be in the form of a poster, a song, a set of digital photos or part of a multimedia presentation.

CONNECT

8. Go to www.jaconline.com.au/science/weblinks and click on the Australian Red Cross Blood Service link for this textbook.
   Find out:
   (a) how old you need to be to donate blood
   (b) the minimum weight blood donors must be.

✓ checklist

I can:

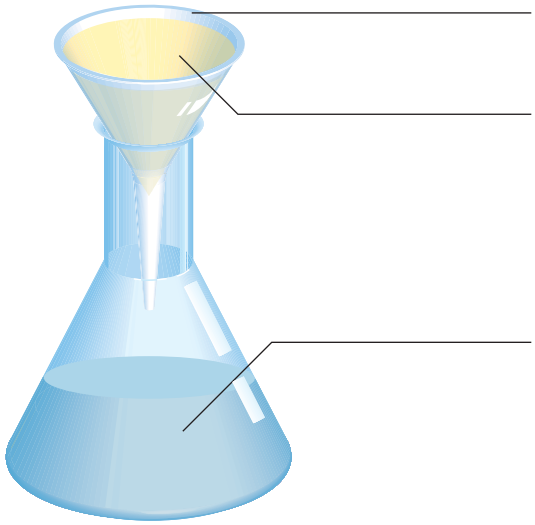
- ☐ describe medical applications of separating techniques
- ☐ understand that blood is made up of four different parts.

Check and challenge

Separating mixtures

Separating and concentrating techniques

1. Give an example of a mixture that could be separated using a filter.
2. Decide if the word in *italic* makes the statement true or false. If the statement is false, replace the *italic* word with the correct word.
   (a) To increase the concentration of a solution such as cordial, add more *solvent* to the solution.
   (b) *Chromatography* can be used to separate substances with different solubilities.
   (c) When a mixture of salt and water is heated, the water begins to evaporate. The salty solution becomes *diluted* in this process.
   (d) The *heavier* parts of a mixture are forced to the outer edge of a centrifuge when it spins.
3. (a) Label the following diagram:



   (b) Explain how this equipment separates mixtures.
4. (a) List the properties of iron filings, sugar and water.
   (b) Describe a method for separating a mixture containing iron, sugar and water. Each of the three substances needs to be collected into a separate container.
5. Some filters have smaller holes in them than others. Why are the holes in coffee filters smaller than the holes in pasta strainers?

Types of mixture

6. Complete this sentence:

   A mixture is a combination of _____, each having its own _____.
7. Decide if the word in *italic* makes the statement true or false. If the statement is false, replace the *italic* word with the correct word.
   (a) Suspensions contain *soluble* particles in a liquid.
   (b) A suspension can be separated in a *centrifuge*.
   (c) Milk is a *solution*.
   (d) Emulsions are a type of *colloid*.
8. Complete this sentence:

   Chromatography separates mixtures whose substances _____ at different rates.
9. Oil floats on water. When detergent is added, the oil forms droplets in the water that do not settle out. What type of mixture has been formed?

10. How are emulsions and colloids similar and how are they different?
11. Homogenised milk is a:
    A suspension?
    B emulsion?
    C colloid?
    D solution?
12. Make a list of some of the substances that would be present in a mixture that is usually separated with a catalytic converter.

Appropriate methods

13. Draw and label the equipment needed to separate a mixture of sugar and water if:
    (a) only the sugar needs to be kept
    (b) both the sugar and water need to be kept.
14. Pasta is placed in water to cook. The pasta sinks to the bottom when it is left to stand.
    (a) What type of mixture is the pasta and water?
    (b) Describe two techniques that can separate this type of mixture.
    (c) Which technique is best for pasta and water? Explain your answer.
15. Spin cycles of washing machines separate clothes and water. What techniques of separation do they use?

Separating in medicine and industry

16. Some medicines form suspensions. Why do they need to be shaken before taking?
17. Why does blood collected from the Red Cross Blood Service need to be separated before it is used?
18. What is the purpose of an S-bend in a kitchen sink pipe?
19. Which of the following separation techniques are used in a water-treatment plant? You may select more than one answer.
    A Filtration
    B Chromatography
    C Centrifuging
    D Sedimentation
    E Crystallisation
20. List some of the substances that a gas mask separates from air to purify it.
21. What is the purpose of magnets in a recycling plant?

Concentrate

1. Many cleaning solutions can be bought in concentrated form. These concentrated solutions need to be diluted in water before they are used.
    (a) In this case, what is the solvent?
    (b) If the cleaning concentrate is a bright blue colour, how could you tell the difference between the diluted and concentrated solutions?
    (c) The concentrated cleaning solution needs to be diluted so that there is one part of concentrate to five parts water. How many litres of water need to be added to half a litre of concentrate?
    (d) If you wanted to make a concentrated solution, would you add large or small amounts of solute? Explain your answer.
2. Suggest two ways of making a solution of salt and water more concentrated.

Medical and industrial separation

3. Food workers are unsure of whether a particular artificial colour has been added to a new batch of lollies. The colour is believed to affect certain medical conditions. The new batch of lollies cannot be distinguished from other lollies by looking at them. How could the workers check for the presence of the colour in the lollies?
4. (a) What is the purpose of an oil filter in a car?
    (b) Why does this oil filter need to be changed regularly?
5. Could water be used as a solvent to find out which pigments are in a sample of oil paint? Explain your answer.
6. Why do surgeons wear masks and gloves during an operation? Explain your answer in terms of separating substances.

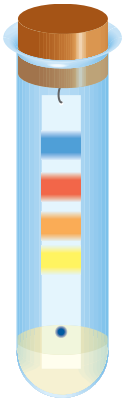
Summary of key terms

bacteria: the smallest life form found on Earth. Some types of bacteria are responsible for decay and disease.

catalytic converters: filters fitted to the exhaust systems of cars. Catalytic converters trap harmful chemicals released by cars to reduce the amount of harmful emissions entering the atmosphere.

centrifuging: a separation technique that spins heavier substances to the outside of the centrifuge

chromatography: a separation technique that separates small traces of substances using differences in solubility



circumference: the distance around the outside of a circle

colloid: a mixture in which the particles of one substance are spread evenly throughout another

concentration: the amount of solute compared to the amount of solvent in a solution

contaminated: a useful substance that contains another substance or substances that we don't want

crystallisation: a separation technique that uses evaporation to separate the parts of a solution. The solvent is evaporated and leaves behind the solute.

decanted: the liquid part of a suspension is carefully poured off, leaving the solid sediment behind

dilute: (*noun*) a solution that contains a small amount of solute compared to solvent; (*verb*) to make weaker by adding solvent

disinfect: to remove harmful bacteria

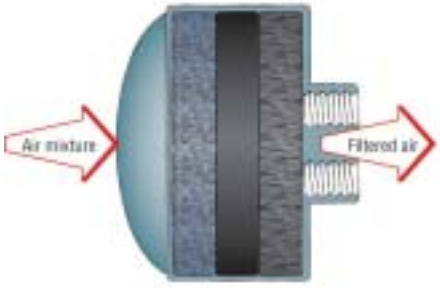
dissolve: to mix one substance (the solute) completely with another (the solvent) so that it is no longer visible

distillation: a separation technique that uses evaporation to separate substances. One substance is evaporated. The vapour is collected and then condensed back into a liquid.

emulsion: a colloid with droplets of one liquid spread evenly through another

evaporate: to change from a liquid to a gas; for example, from water to steam

filter: a device that allows some materials to pass through. It blocks those too large to fit through.



filtrate: the part of a mixture that passes through a filter

filtration: a separation technique that separates objects of different sizes

forensic scientists: scientists that collect, examine and compare evidence to solve mysteries and crimes

homogenised milk: processed milk. The butterfat (oil) is broken up into droplets and spreads evenly through the rest of the milk. This milk is an emulsion, so the butterfat will not settle out.

insoluble: unable to be dissolved

landfill: an area set aside for the dumping of rubbish

manufacturing process: process used to make products

mixture: a combination of substances, each with its own properties

paper mill: place where paper products are manufactured

plasma: the yellowish liquid part of blood that contains water, minerals, food and wastes from cells

platelets: living cell fragments found in blood. They are responsible for healing wounds. They do this by clumping together.

pollution: any substance that has an undesirable effect on the environment. For example, carbon monoxide is a pollutant produced by cars.

properties: characteristics or features of an object or substance

purify: to remove unwanted substances from a mixture

raw material: material in its natural form; such as minerals in the ground, forests, and materials found in water or in the air

recycling program: a program often run by local councils, to collect recyclable materials from people's homes

red blood cells: living cells that transport oxygen to all other living cells in the body

residue: the substance that does not pass through a filter. A residue is also left behind after evaporation.

sediment: the material that collects when suspensions are left to stand. Insoluble substances that collect at the bottom of a container are sediments.

separation: the process of dividing a mixture into its parts

sewage: a mixture of water and substances that flows from laundries, bathrooms, kitchens and toilets

sewer: pipes that remove sewage from a property

sewerage: the system of drains and pipes that takes the sewage away from a property

skimmer: a device that passes along the surface of a mixture and removes floating substances

solubility: a property of a substance that describes how it dissolves. Substances with different solubilities dissolve at different rates.

soluble: able to be dissolved

solute: substance that is dissolved in solvent to form a solution

solution: a mixture of a solute dissolved in a solvent. Solutions are clear, but can be coloured.

solvent: the substance in which a solute dissolves to form a solution

suspension: a mixture of a gas or liquid and an insoluble substance. The insoluble substance will settle out when the mixture is left to stand.

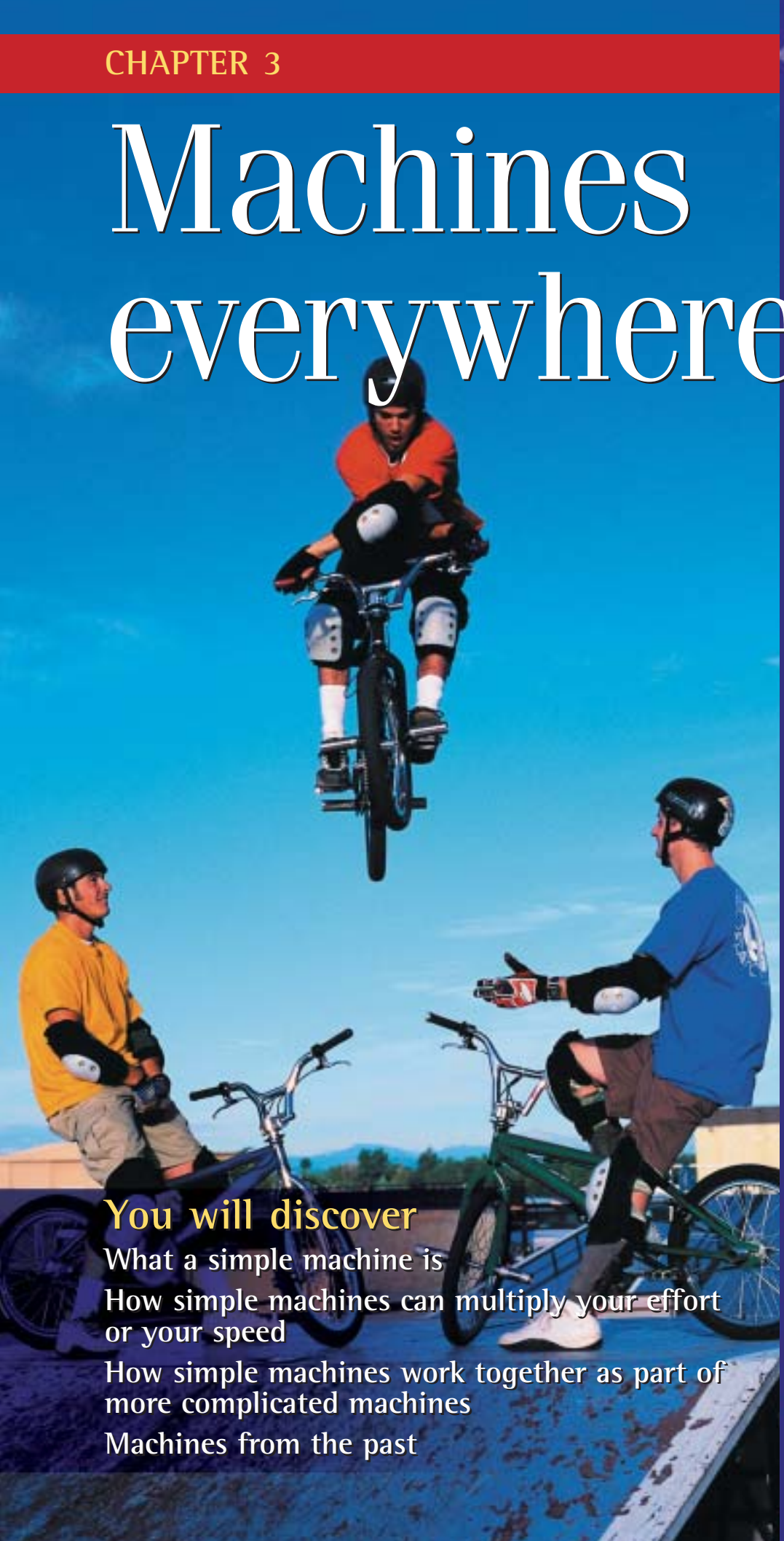
water still: a device used to distil water

water vapour: water in a gaseous state

white blood cells: living cells that fight bacteria and viruses. They are part of the human body's immune system.

CHAPTER 3

Machines everywhere



You will discover

What a simple machine is

How simple machines can multiply your effort or your speed

How simple machines work together as part of more complicated machines

Machines from the past

Flying high – trick machines

Imagine you're about to catch some serious air or you're jamming through the dirt in a high-speed BMX race. Maybe you're about to commit to a tail-end transfer or a backflip barspin. You could be doing it hard in the uphill leg of a road race or going through your paces in a velodrome, or just getting around town on your wheels.

Your bike is a machine that can get you going fast enough to do some major tricks. It can help you up a backbreaking hill or just make it easier to get to the local shops. A bike is a machine, but it is actually made up of many simpler machines.

1 How does a bike get me going faster?

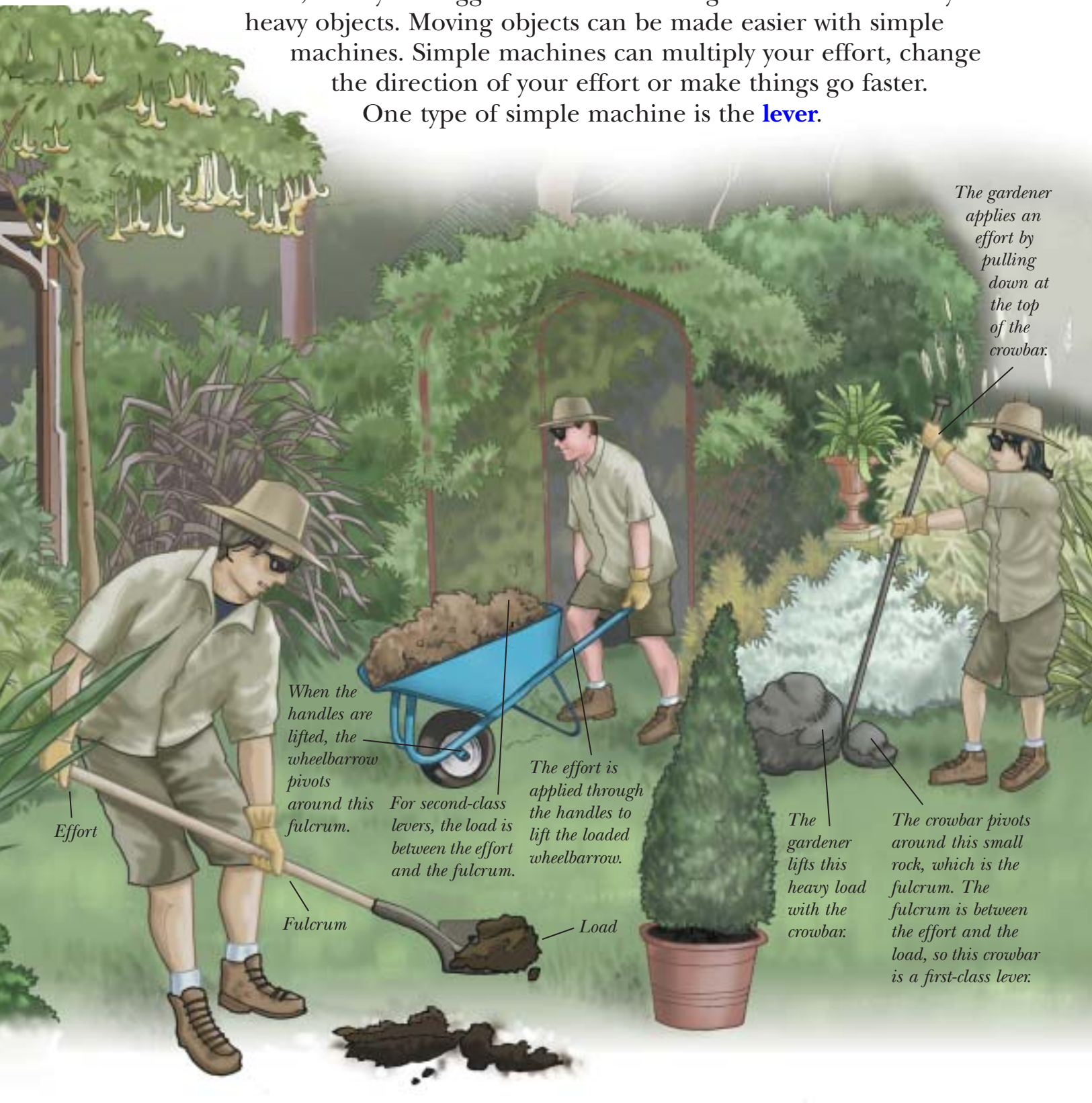
2 How does a bike tackle hills?

3 What simple machines are found in bikes?

In the garden

It would be difficult to move a pile of rocks from one end of the garden to the other by hand. It takes much less **effort** using a wheelbarrow. The effort you use is actually a **force**.

Sometimes, even your biggest effort isn't enough to move some very heavy objects. Moving objects can be made easier with simple machines. Simple machines can multiply your effort, change the direction of your effort or make things go faster. One type of simple machine is the **lever**.



Using levers

A lever is usually a long, rigid object that moves around a turning point called a **fulcrum**. You need to put in an effort to make the lever move a **load**. Levers are named according to where the fulcrum, load and effort are positioned along the lever.

First-class levers

First-class levers, such as the shovel, secateurs and shears below, turn around a fulcrum that is between the effort and the load. All first-class levers are **force multipliers**. They magnify your effort. This means that you may be able to move loads that you couldn't move without the lever.

Second-class levers

Second-class levers are also force multipliers. They turn around a fulcrum that is at the end of the lever. For second-class levers, the load is always between the effort and the fulcrum.

The wheelbarrow is a second-class lever. It is used to move objects that would otherwise be too heavy to carry. The *load* in a wheelbarrow is between the fulcrum and the effort.

Activities

REMEMBER

1. What do you call the force you apply to a lever to move an object?
2. Explain how a first-class lever is different from a second-class lever.
3. Why are first- and second-class levers called force multipliers?

THINK

4. Two wheelbarrows are identical, except for the length of their handles. Which would you prefer to use? Why?
5. Sketch and label the effort, fulcrum and load in each of the following diagrams. State whether each is a first- or second-class lever.

(a)

(b)

(c)

✓ checklist

I can:

- ☐ compare first- and second-class levers
- ☐ identify the effort, fulcrum and load in a lever system.

Living levers

The long bones in our arms and legs act as **levers**.

Our joints form fulcrums and our muscles apply a **force** to the bones to make them work. Many of the levers in our bodies are **third-class levers**.

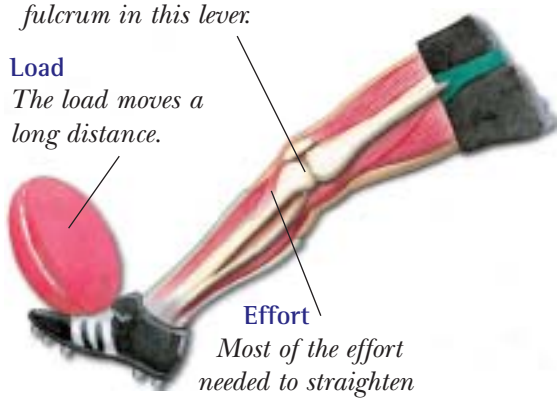
Third-class levers are those in which the effort is between the fulcrum and the load. All third-class levers are **speed multipliers**. A big effort needs to be applied, but the load moves over a greater distance, at a higher speed.

Some sports make use of more than one type of simple machine. Woodchopping uses the arm and axe together as a third-class lever. The axe blade strikes the wood block at a high speed. The blade itself is another simple machine called a **wedge**. A wedge pushes objects apart. In this case, the axe blade splits the piece of wood into two parts.

The leg below is acting as a third-class lever. The effort is between the fulcrum and the load.

Fulcrum
The lower part of the footballer's leg pivots around the knee. The knee is the fulcrum in this lever.

Load
The load moves a long distance.



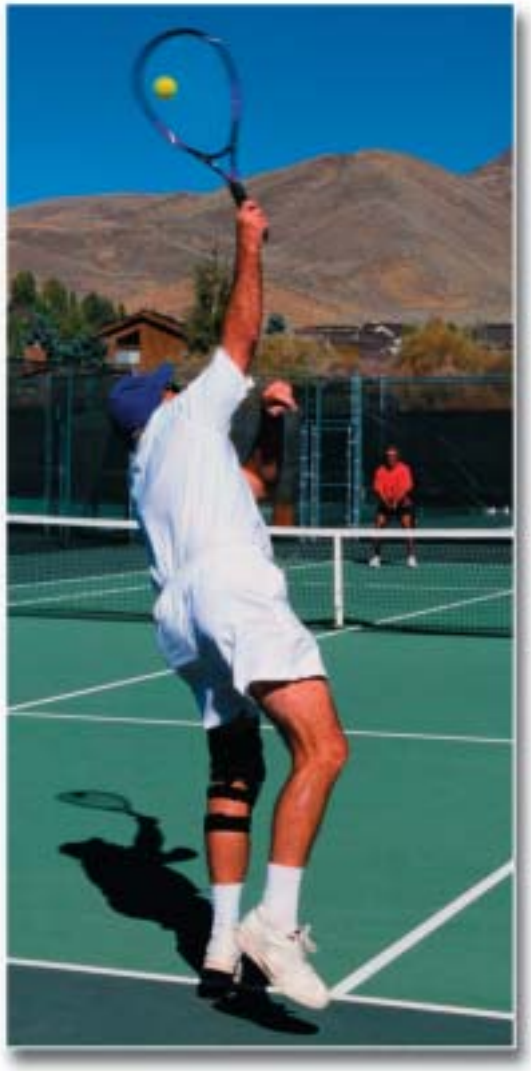
Effort
Most of the effort needed to straighten your leg when kicking a ball comes from the muscles in your legs. The effort to kick a ball is applied from muscles that attach to the top of your lower leg.

Levers in sport

In sport, levers are usually used to increase the speed of a load and the distance over which it travels. Tennis racquets, cricket bats and softball bats are all third-class levers. They are used to make objects move faster and further.

The longer the distance between the load and the effort, the faster the load will move. Tennis players reach up high when they serve. This way, the lever formed by the arm and the racquet is much longer.

A fast serve can measure over 200 km/h. Greg Rusedski has served at a speed of 239.8 km/h. Mark Philippoussis has served up to 226.9 km/h.

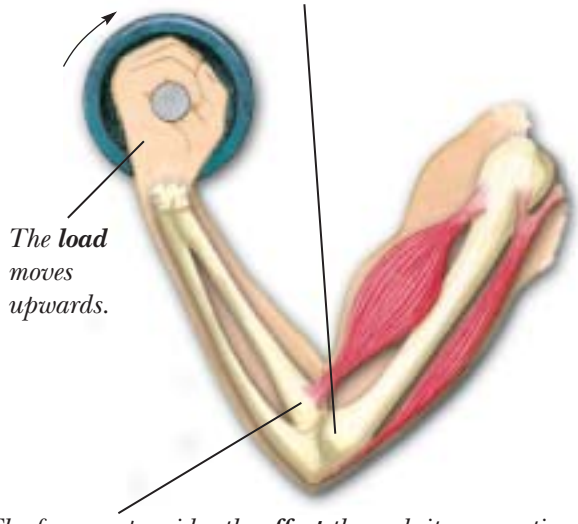


Different classes of lever in the arm

Bending and straightening the arm uses two different classes of levers.

Bending the arm

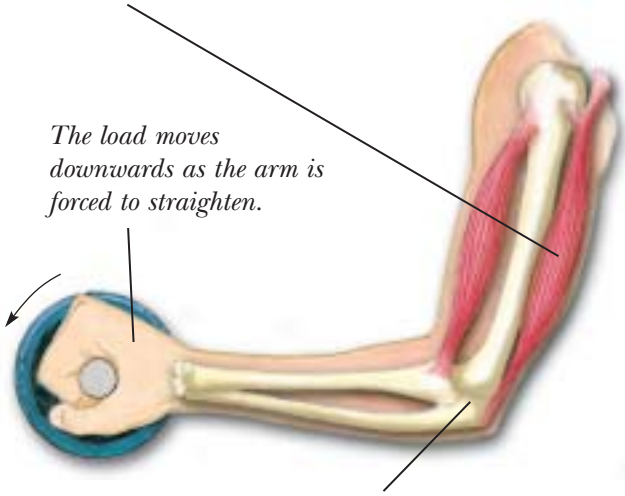
*The elbow is the **fulcrum** because it forms the pivot point for the movement of the arm.*



*The forearm provides the **effort** through its connection to the biceps muscle below the elbow. Therefore, the effort is between the fulcrum and the load, making the forearm into a third-class lever.*

Straightening the arm

This muscle is called the triceps muscle. It provides the effort used to straighten the arm. You use your triceps muscle when you push a heavy load down. The triceps are also used to shoot goals in basketball and netball.



*The triceps muscle attaches to the elbow, which is the fulcrum. The fulcrum is between the effort and the load, so your forearm acts as a **first-class lever**.*

Activities

REMEMBER

1. Explain how third-class levers are different from first- and second-class levers.
2. (a) Label the effort, fulcrum and load along the fishing rod.
   (b) What class of lever does the fishing rod act as?

THINK

Some sports, like rowing, do not rely on speed alone. It is very difficult to move through water, so a **force multiplier** is needed to help the rowers.

3. (a) What class of lever do these oars form?
   (b) What other sports use force-multiplying levers?
4. Do you think that humans are built for speed or for strength? Explain your answer.

INVESTIGATE

5. Choose a sport not already mentioned on this page.
   (a) Explain how levers are used in this sport.
   (b) What classes of levers are used?
   (c) Why are these types of levers used?
   (d) Draw a labelled diagram to show how the levers are used in your chosen sport.

checklist

I can:

- ☐ distinguish between speed-multiplying levers and force-multiplying levers
- ☐ explain how levers are used in a variety of sports.

Playing with machines

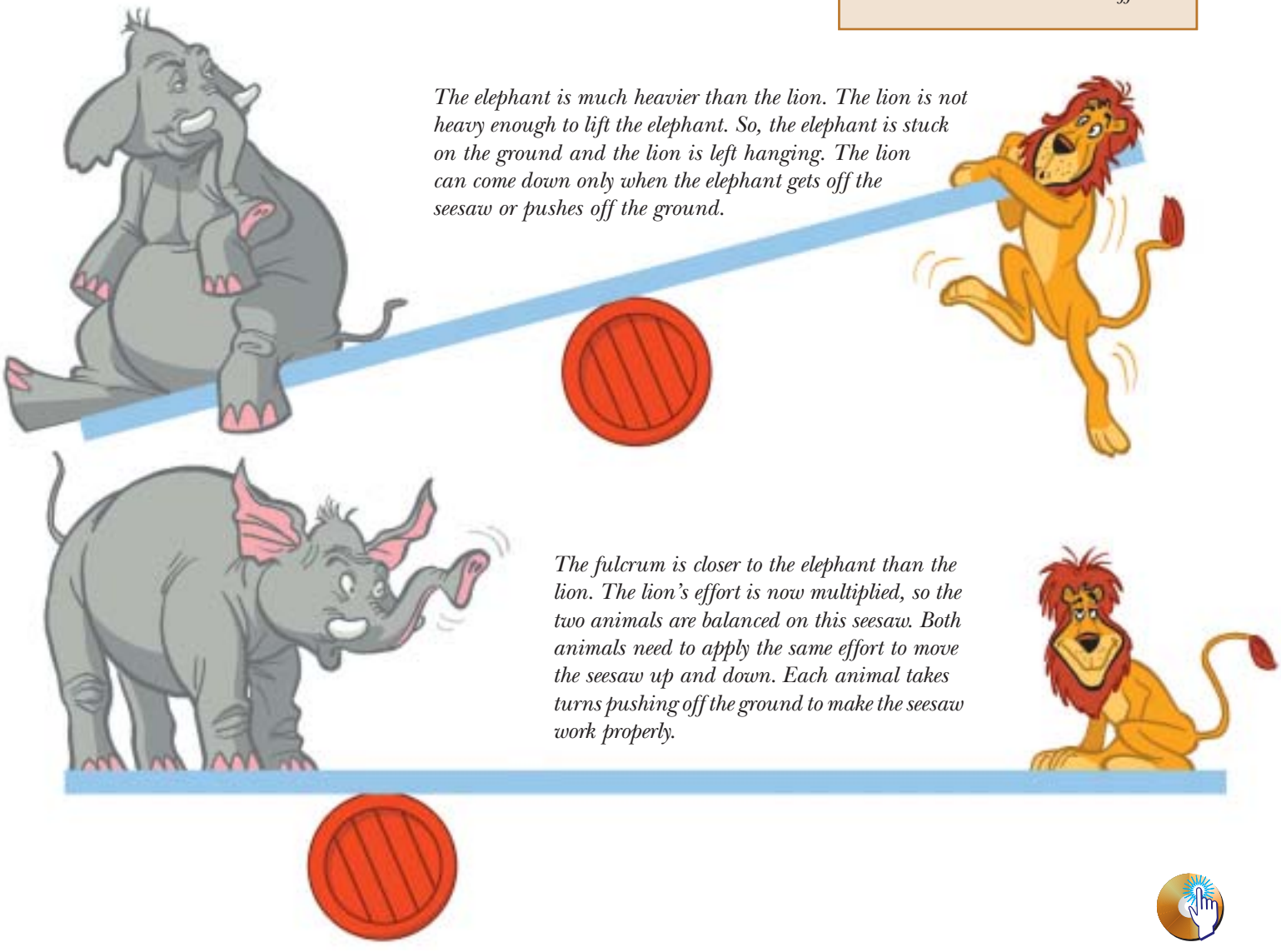
A seesaw can be a lot of fun if the two people on it are about the same size. A seesaw works best this way because both people need to apply the same **effort** to make the seesaw work. On an unbalanced seesaw, one person is left to do all the work.

Placing the fulcrum

A seesaw is a special type of **first-class lever**. The **fulcrum** of a seesaw is usually exactly in the middle of the effort and the **load**. The two people on a seesaw take turns in being the effort and the load, depending on their movement. First-class levers are **force multipliers**, so a small person *should* be able to lift a larger person on a seesaw without using much effort.

Mechanical advantage measures how much easier a lever makes a task. Mechanical advantage is calculated by dividing the load by the effort. In the example of the seesaw, the effort of the lion was able to lift the elephant. The mechanical advantage in this case is the weight of the elephant divided by the weight of the lion.

$$\textit{Mechanical advantage} = \frac{\textit{load}}{\textit{effort}}$$



The elephant is much heavier than the lion. The lion is not heavy enough to lift the elephant. So, the elephant is stuck on the ground and the lion is left hanging. The lion can come down only when the elephant gets off the seesaw or pushes off the ground.

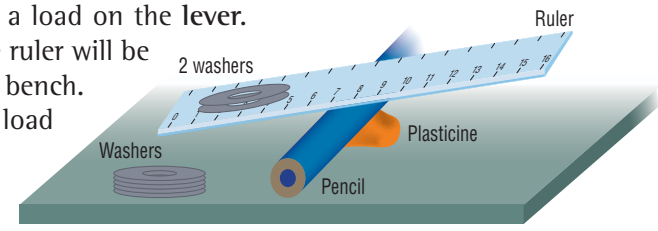
The fulcrum is closer to the elephant than the lion. The lion's effort is now multiplied, so the two animals are balanced on this seesaw. Both animals need to apply the same effort to move the seesaw up and down. Each animal takes turns pushing off the ground to make the seesaw work properly.

Go to worksheet 11

Balancing a lever

You will need:
30 cm ruler
pencil or triangular piece of wood in the shape of a prism
plasticine
6 identical washers.

- Set up the ruler with the pencil or wood as a fulcrum. The fulcrum needs to be exactly in the centre of the ruler.
- Use the plasticine to attach the pencil to the bench.
- Place two of the washers 5 cm from the fulcrum. These washers act as a load on the lever. The end of the ruler will be resting on the bench.



- Try lifting the load by using two washers placed close to the fulcrum. Slide the washers away from the fulcrum in the opposite direction of the load, until they *just* lift the load. These washers act as the effort in the lever system. What is the closest distance between the effort and the fulcrum that allows you to lift the load washers?

Record your results in a table like this one:

Number of washers used as the load	2	2	3	4
Distance between the load and the fulcrum (cm)	5	5	5	5
Number of washers used as the effort	2	1	1	2
Distance between the effort and the fulcrum (cm)				
Mechanical advantage (load/effort)				

- Remove the two effort washers. Now use just one of the washers to lift the original two load washers. Record the smallest distance between the effort and the fulcrum that allows you to *just* lift the load washers.
- Predict where a washer needs to be placed if the load is increased to three washers. Record your prediction.

1. Test your prediction and record your results.

- Increase the load to four washers. Use two washers as the effort to just lift the load. Record your results.

2. Look for patterns in your table of results. Write a simple rule that explains how to place the effort washers so that they *just* lift the load washers.
3. Predict the largest load that you could lift with an effort of just one washer. Borrow some identical washers from another group and test your prediction.
4. Was your prediction correct? If not, explain why.
5. What happens to mechanical advantage when the distance between the effort and the fulcrum is increased?

Activities

REMEMBER

1. True or false?
   (a) First-class levers have the load between the fulcrum and the effort.
   (b) Levers are better force multipliers when the fulcrum is closest to the load.
   (c) Only people of the same size can balance out a seesaw.

THINK

2. (a) Calculate the mechanical advantage of a seesaw that allows a 40 kg person to lift a 60 kg person.
   (b) Draw a scaled diagram to show how these two people can balance a 6 m long seesaw.
3. Explain how the distance between the fulcrum and the effort affects the mechanical advantage of a lever.

CREATE

4. Draw or design a piece of playground equipment that makes use of at least one lever. Be sure to label the fulcrum, load and effort.

CONNECT

5. Go to www.jaconline.com.au/science/weblinks and click on the Balancing Frogs link for this textbook to balance some frogs on a seesaw.

checklist

I can:
- ☐ understand what occurs if the fulcrum is moved
- ☐ relate the mechanical advantage of a lever to the distances between the effort, fulcrum and load.

On the road

Imagine what it would have been like to travel long distances without cars, motorbikes and bicycles. A few hundred years ago you would not have been able to use any of those wheeled vehicles. In fact, the **wheel** itself was not invented until about 3500 BC. How different would our lives be without wheels?

Levers in a spin

Wheels are levers that rotate around the fulcrum. They are attached to a smaller wheel called an **axle**. The wheel and the axle turn in circles around the **fulcrum**, but the wheel moves through a larger circle than the axle. Like other **levers**, a wheel and axle can be used as a **force multiplier** or a **speed multiplier**. A wheel and axle act as a force multiplier when the **effort** is applied to the big wheel. If the effort is used to move the small axle, the machine acts as a speed multiplier.

*This wheel and axle acts as a force multiplier. The driver can turn the steering wheel with very little effort. The outer edge of the steering wheel turns through a bigger distance than the steering column. The steering column moves a smaller distance than the steering wheel, but applies a much greater **force** on the heavy tyres of the car.*

This wheel and axle acts as a speed multiplier. A big force from the engine moves the axle a short distance. The wheel turns in a much larger circle than the axle. The outside edge of the wheel spins very quickly compared to the axle.

Compound machines

Cars and bicycles are **compound machines**. They are made up of many simple machines working together. As well as the wheels and axles that move them along, they have levers like handlebars, handbrakes and brake pedals. A special kind of wheel, called a **gear**, is another simple machine found in cars and bikes.

Handbrake

*This handbrake is a **first-class lever**. When the rider squeezes the handbrake, their effort is transferred along the cable to the brake pads that press against the wheel.*

Bike chain

It is possible to join wheels or axles together with chains or belts. Bikes have chains rather than belts because they are less likely to slip off. The pedals' axle and the rear axle have teeth that mesh together with the chain. Wheels with teeth on them are called gears.

Pedals

*The pedals turn in a larger circle than the axle they are attached to. The axle can move a bigger **load** than the pedals. This part of the bike is acting as a force multiplier.*

Front wheel

Front axle

One of the first bicycles invented was called the 'swiftwalker'. It didn't have pedals. The 'walker' had to push off the ground to make the bike go forwards. Both feet were then lifted off the ground and the bike coasted forwards. Another push sent the bike a little further forwards. Bicycle walkers had to wait over 100 years before pedals were invented. The invention of the pedal turned walkers into 'riders', but, without rubber tyres, it was a very bumpy ride!

Rear axle
The pedals' axle is joined to this axle by a chain. Turning the pedals causes this axle to turn as well. This rear axle is much smaller than the back wheel. It makes the back wheel turn very quickly. The back wheel and axle acts as a speed multiplier.

Back wheel

Activities

REMEMBER

1. Which part of a wheel is like the fulcrum of a lever?
2. Explain the difference between compound and simple machines.

THINK

3. Complete these sentences:
   (a) When a big wheel is used to turn a small wheel, the _____ is multiplied.
   (b) When a small wheel turns a bigger wheel, the _____ is multiplied.

COMPARE

4. (a) How does the size of a steering wheel in a bus compare with the size of a steering wheel in a racing car?
   (b) Suggest a reason for the difference in size.

CREATE

5. Make a **belt drive** using Lego® or another building kit. Use appropriate wheels to make:
   (a) a force-multiplying belt drive
   (b) a speed-multiplying belt drive
   (c) a belt drive that does not multiply speed or force.

CONNECT

6. Find out who invented the rubber tyres that are used on bicycles. Also find out when rubber tyres were first used.

✓ checklist

I can:
- ☐ describe a compound machine as a device made up of many simple machines
- ☐ distinguish between speed-multiplying and force-multiplying wheels and axles.

On the road again

The **gears** on a bike usually act as **speed multipliers**. Depending on the type of bike you own, they can make a hill a little easier too. The gears on a bike are called **sprockets**. A chain links the sprockets around the pedal with those on the back wheel. In most cases, the front sprockets of a bike are bigger than the back ones.

Getting into gears

The gear that moves first is called a **driving gear**. The front sprocket of a bike is a driving gear. The front sprocket makes the rear sprocket turn. The rear sprocket is a **driven gear**.

When a large driving gear turns a small driven gear, the small gear turns faster. These gears are speed multipliers. Most bicycle gears are speed multipliers because the front driving sprockets are usually larger than the rear driven sprockets.

What if you are going up a hill? You would need your bike to act more like a **force multiplier** than a speed multiplier. Some bikes have several sprockets. Choosing a different combination of gears makes hills easier to tackle.

Bike gears

When a small gear drives a big gear, the big gear turns with less speed, but greater **force**. This combination of gears is a force multiplier. But not many bikes allow you to choose a front sprocket that is smaller than the rear sprocket.

The best combination for going up hills is when the smallest front sprocket is used with the biggest rear sprocket. The pedals need to be turned more, but the reward is an easier uphill ride. This combination of gears is called low gear.

Along a flat road, high gears are used to move a bike quickly. The largest front sprockets are linked to the smallest rear sprockets. In high gear, you have to push harder, but the reward is that you don't have to pedal as fast.

Notice that the smallest front sprocket is used with the biggest rear sprocket. This combination of gears makes it easier to go up hills.

The combination of a large front sprocket and a small rear sprocket makes the bike go faster.

Machine on a lean

The gears in a car can be used as force multipliers to help a car make it up a steep incline. But the shape of a road can help too. Steep hills and mountains have roads that wind around them. Winding roads make the trip much easier, but the distance to the top is much greater than if the road climbed straight up the hill. A winding road is a simple machine called an **inclined plane**. Inclined planes reduce the effort needed to raise objects to a higher level.

Getting going again

A car jack is an inclined plane. The winding 'road' on the jack forms a **thread**. The closer the turns of the thread, the more the jack magnifies your **effort**. The threaded part of the jack is a special inclined plane, called a **screw**. The screw brings the sides of the jack closer together when the jack handle is turned. As the sides move in, the top of the jack lifts the load.

Go to worksheet 12

Go to worksheet 13

Activities

REMEMBER

1. What is a sprocket?
2. Complete these sentences:
   When a bigger gear turns a smaller gear, the driven gear turns with increased _____.
   When a smaller gear turns a larger one, the driven gear turns with increased _____.
3. When would you most likely ride a bike in low gear?
4. Apart from changing the speed and force of turning, what else can gears do?

THINK

5. Answer the following questions about the four different gear combinations listed.
   - A A bike whose chain links a front sprocket of 56 teeth with a rear sprocket of 24 teeth
   - B A bike whose chain links a front sprocket of 40 teeth with a rear sprocket of 24 teeth
   - C A bike whose chain links a front sprocket of 40 teeth with a rear sprocket of 32 teeth
   - D A bike whose chain links a front sprocket of 56 teeth with a rear sprocket of 32 teeth

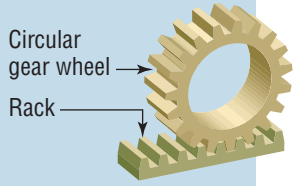
   (a) Explain which of the combinations above would be best for uphill riding.
   (b) Explain which of the combinations above would be best for riding along a flat road.

CREATE

6. Use Lego® or any other building kit to set up two gears that mesh together. Starting with a gear with 16 teeth and a gear with 8 teeth, investigate the number of turns the driven gear makes for one turn of the driving gear. Investigate gears with different numbers of teeth.

INVESTIGATE

7. Find out what **rack and pinion gears** are and how they work in a car.
8. Find out what an **idler gear** is used for.



checklist

I can:
- ☐ combine gears to form speed multipliers and force multipliers
- ☐ use an inclined plane as a force multiplier
- ☐ recognise different types of inclined planes.

On the building site

Inclined planes can be as big as a road winding around a mountain. They can also be as small as the tip of a nail. They can split objects apart or hold them together. Many inclined planes are found on building sites.

It takes months to build a house: cutting timber frames to size, nailing the frame together and carting bricks. Many simple machines are used by builders and construction workers to make their jobs easier. Without these simple machines a house would take much longer to build, and some tasks could never be done at all. Some of the simple machines used by builders are inclined planes.

*The inclined plane around the screw is called the **thread**. The distance between two threads on a screw is called the **pitch**.*

*It would be almost impossible to remove this nail without some help from the hammer. The hammer is being used as a **first-class lever**.*

*The chisel is a type of inclined plane called a **wedge**. Wedges are **force multipliers** that can be used for shaving objects.*

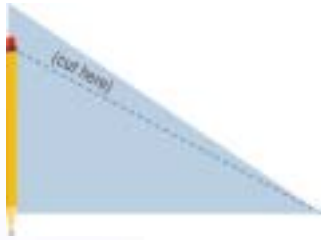
*This screwdriver is a **wheel** and **axle**. The handle turns in a larger circle than the tip.*

The tip of a nail is shaped like an inclined plane. It would be much more difficult to hammer a flat-tipped nail into this wood.

The length of a screw thread

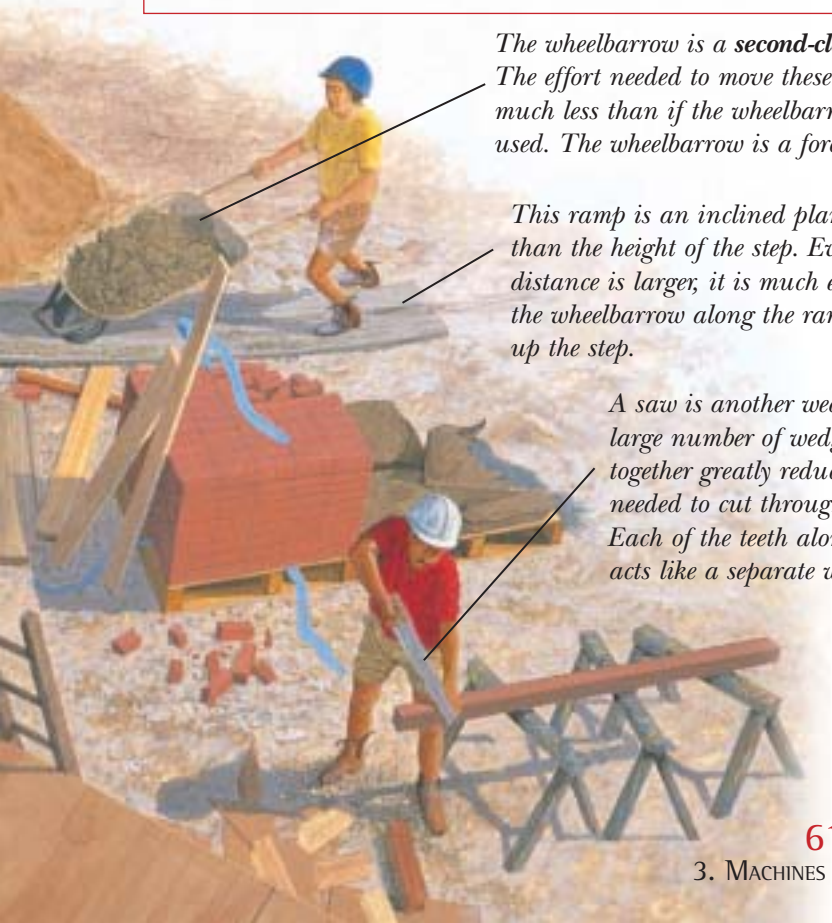
A screw is a curved inclined plane. When a screw is used to hold objects together, it needs to be turned through the length of the thread. The **effort** is applied over a much larger distance than if the screw was driven straight into the objects. The distance a screw actually travels through the timber is longer than the length of the screw.

You will need:
a pencil
A4 piece of paper
scissors.



- Fold the paper in half, diagonally, to make a right-angled triangle.
- Cut along the fold. You will need only one of the triangles.
- Cut the triangle so that its height is less than that of the pencil.
- Measure the height of the triangle.
- Measure the length of the longest side of the triangle.
- Wrap the paper triangle around the pencil.

1. If the pencil were a screw, how deeply could the screw reach into a piece of timber?
2. How far has the screw actually turned?



*The wheelbarrow is a **second-class lever**. The effort needed to move these bricks is much less than if the wheelbarrow was not used. The wheelbarrow is a force multiplier.*

This ramp is an inclined plane. It is longer than the height of the step. Even though the distance is larger, it is much easier to push the wheelbarrow along the ramp than pull it up the step.

A saw is another wedge. Having a large number of wedges working together greatly reduces the effort needed to cut through this timber. Each of the teeth along the blade acts like a separate wedge.

Activities

REMEMBER

1. Make a sketch of a screw, and add labels to show why it is an inclined plane.
2. Give two examples of wedges used on a building site.
3. List an advantage and a disadvantage of using a ramp as an inclined plane.

THINK

4. Imagine two screws that are identical, except for the height of their pitch. What difference would you notice using each of the screws?

FIND

5. List all of the following types of machine that you can find at the building site under the following headings:
   (a) speed multipliers
   (b) first-class levers
   (c) second-class levers
   (d) wheels and axles
   (e) inclined planes.

INVESTIGATE

6. A spring balance can be used to measure forces. Use a spring balance to measure the effort force needed to pull an object up an inclined plane. Compare this with the effort needed to lift the object straight up the height of the inclined plane.

✓ checklist

I can:

- ☐ describe some simple machines used on building sites
- ☐ explain what the pitch of a screw is
- ☐ understand why inclined planes are useful.

PAPYRUS PRESS

27 JUNE 2500 BC WEATHER: HOT

THE GREAT PYRAMID IS COMPLETED!

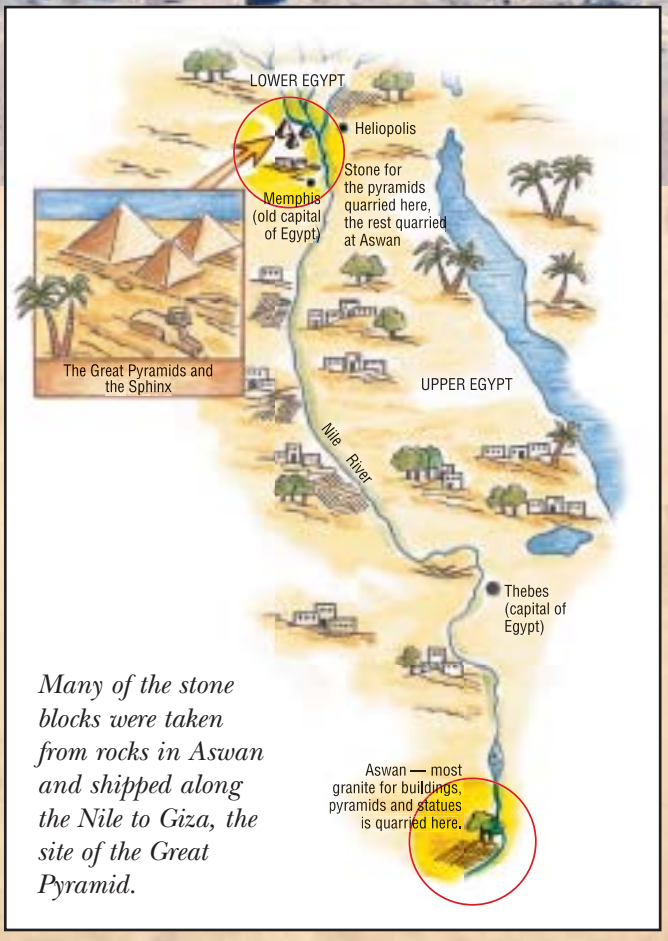
Pharaoh Khufu has rejoiced at the sight of his Great Pyramid! He has publicly praised the 100 000 men who have worked for him for the past 20 years. The pharaoh has promised the workers that the gods will look down on them and guide them.

The Great Pyramid will be an impressive burial place for our great pharaoh. It stands almost 150 metres high and covers 5 hectares of land. About two million stone blocks have been precisely cut and placed to make the pharaoh's tomb. Each stone block weighs about 2 tonnes.

The stonecutters have worked continually for years. They used their chisels and drills to cut the stones to fit together perfectly. In fact, the stones fit together so closely that a knife blade cannot be wedged between them.

The surveyors and engineers have made sure that the base rocks are as flat as possible. There is only about 2 centimetres difference between diagonal corners that are over 300 metres apart.

Our Great Pyramid will stand for many years and tell our children and their children of the brilliant skills of the Egyptians. And it will tell our future generations of the magnificent machines that we have mastered.



Many of the stone blocks were taken from rocks in Aswan and shipped along the Nile to Giza, the site of the Great Pyramid.

Engineering a miracle!

The building project is a credit to all the workers, their skills and their machines. We spoke to Chefre, the head **engineer** working on the Great Pyramid. He explained to us how the Great Pyramid was built.

Papyrus Press (PP): *What great machine was used to place the stone blocks on the water vessels at Aswan?*

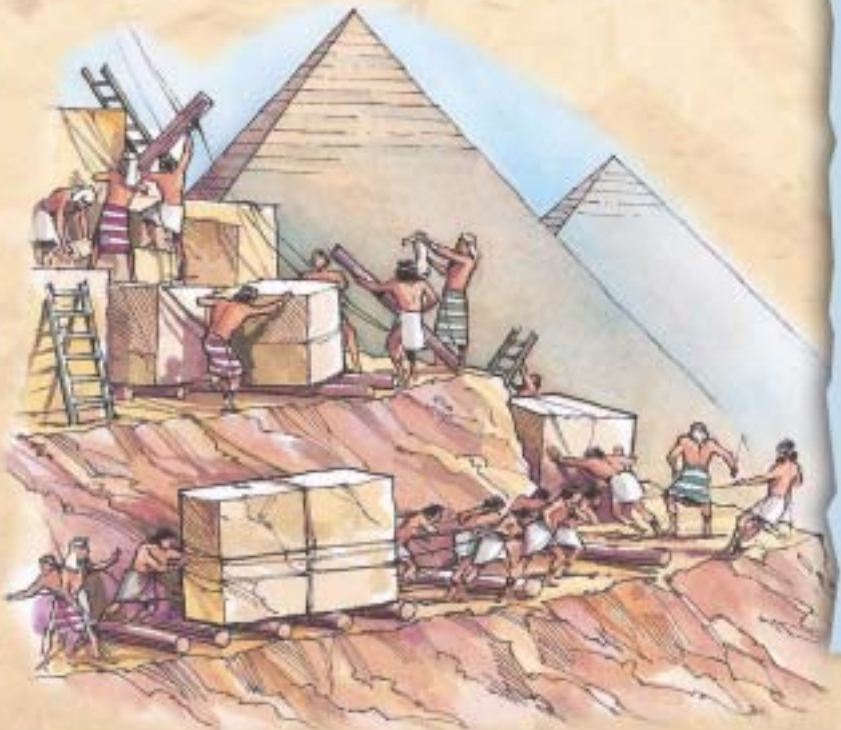
Chefre (C): *The engineering teams have shown the workers how to use **levers** to lift heavy stones. We gave them long levers made of wood and **fulcrums** made of stone. The levers make it possible for five men to do the work of ten! The workers moved the fulcrum and changed the length of the levers to multiply their **effort** as much as possible.*

PP: *What about moving the stones from the river to the building site? Surely they were heavy to drag along?*

C: *Again, a very simple machine made the job much easier. The workers used their levers to lift the heavy stones onto large wooden rollers. The rollers move across hard ground much more easily than dragging the stones without rollers.*

PP: *What sort of magical machine did you use to lift the blocks high up the sides of the pyramid?*

C: *We made long, giant ramps that led up the sides of the pyramid. We used the rollers to haul the stones along the ramps. As the pyramid grew taller, we made the ramps bigger and higher. My assistant suggested wetting the ramps. The water made the ramps slippery, so the rollers moved more easily up the slope.*



Activities

REMEMBER

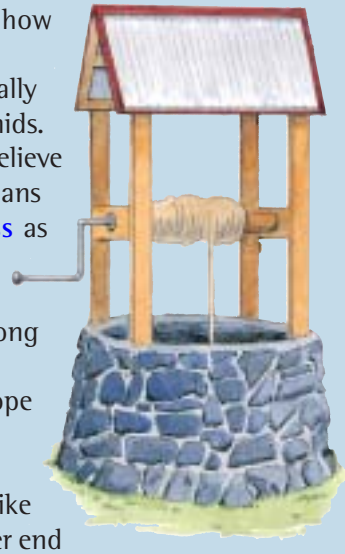
1. How long did it take for the Great Pyramid to be built?
2. How many men are believed to have worked on the construction of the Great Pyramid?
3. What did the ancient Egyptians use levers for?

THINK

4. What type of simple machine was used to move the large stone blocks?
5. What type of simple machine is a ramp?
6. Are the ramps the Egyptians used **speed multipliers** or **force multipliers**? Explain your answer.
7. What simple machines did the stonecutters use?

CREATE

8. Draw a picture to show how ancient Egyptian workers might have used a lever to raise a massive stone onto rollers.
9. There are many theories about how the ancient Egyptians actually built the pyramids. Some people believe that the Egyptians used a **windlass** as well as the rollers to haul large stones along the ramps. A windlass is a rope attached to a pole with a handle, much like a well. The other end of the rope was tied to the stone block. Draw what you believe an Egyptian windlass would look like. Write a short explanation of how it would work.



✓ checklist

I can:

- ☐ describe how simple machines may have been used in the construction of the pyramids in ancient Egypt.

Way back when …

During the 1850s, visitors from all over the world flocked to the Victorian goldfields in search of riches. Many of the visitors brought with them new skills and new machines to help them find their fortunes.

At first the gold was easy to find because it lay in streams of water. Soon almost all of the surface gold was gone. It became necessary to start digging for gold. The technology that we have today was not available, so simple machines were used to dig for the gold, haul it to ground level and separate it from rock.

Windlass

Mining shafts were dug to about 40 m deep. The problem of lifting the rocks and soil to the top of the shaft was solved with a wheel and axle called a **windlass**.

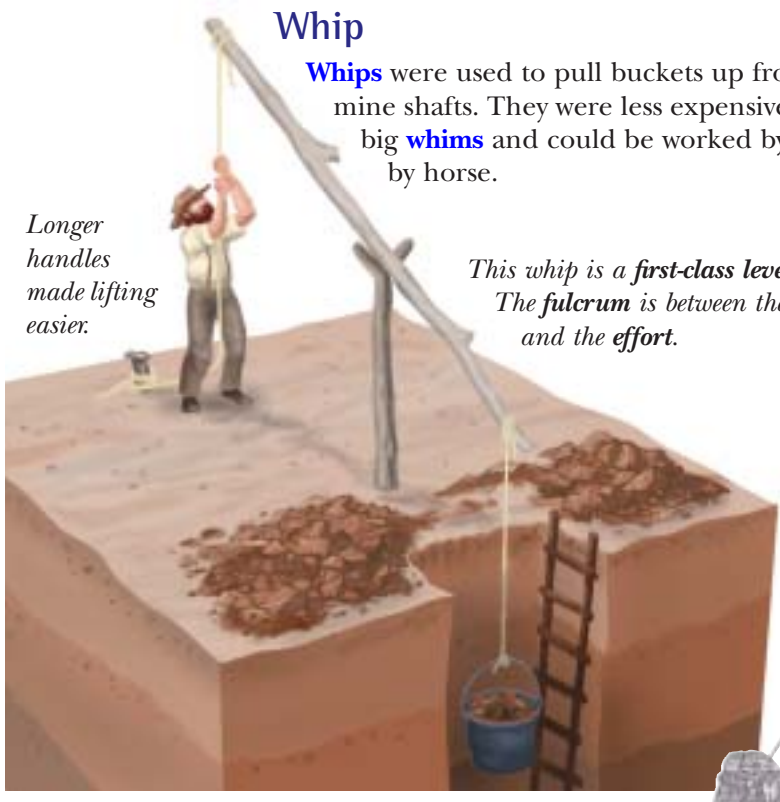
The rope is tied to a bucket. The bucket is filled with rocks and soil at the bottom of the mine. When it is full, the miner at the top turns the handle (the wheel). The rope winds around the axle, raising the bucket. After the bucket is emptied at the top of the mine, the windlass is also used to return the bucket to the miners at the bottom of the shaft.

Whip

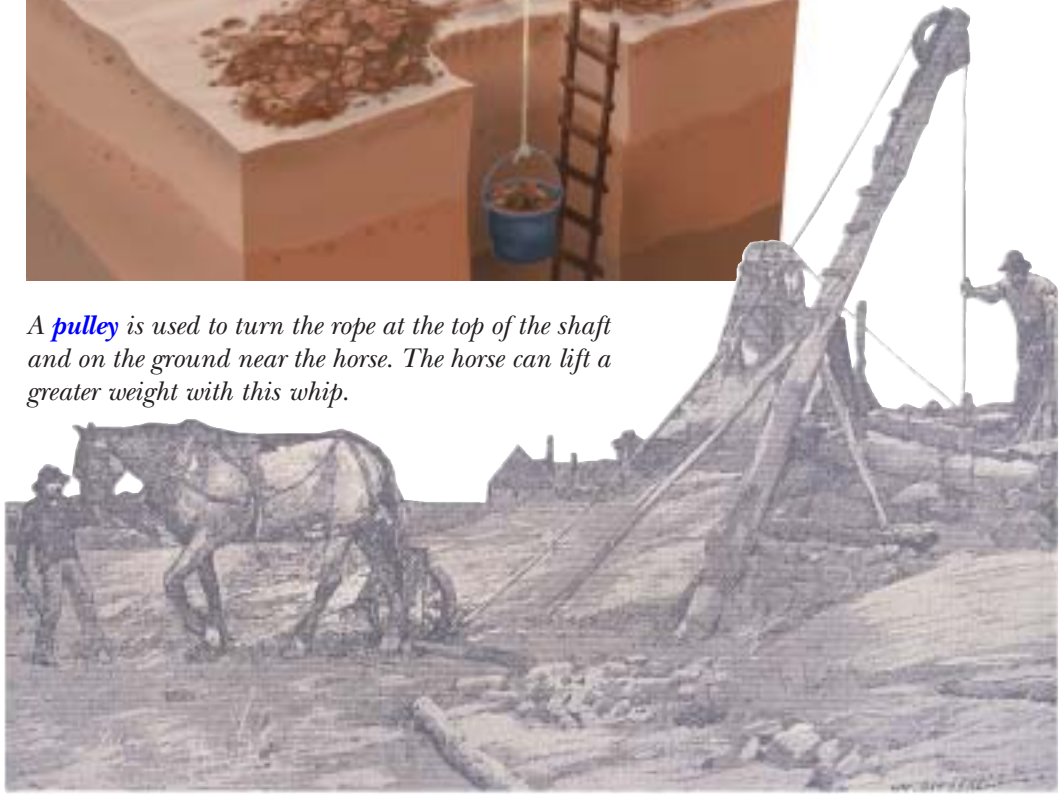
Whips were used to pull buckets up from deep mine shafts. They were less expensive than the big **whims** and could be worked by hand or by horse.

Longer handles made lifting easier.

*This whip is a **first-class lever**. The **fulcrum** is between the **load** and the **effort**.*

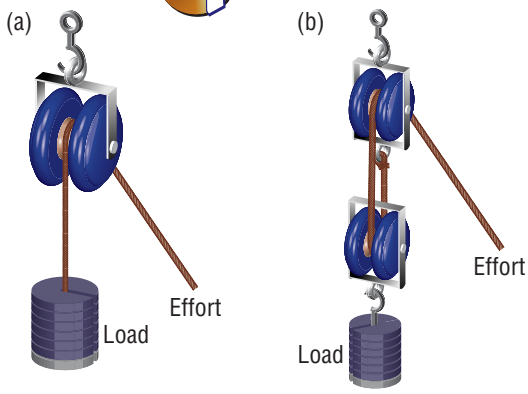


*A **pulley** is used to turn the rope at the top of the shaft and on the ground near the horse. The horse can lift a greater weight with this whip.*



Go to worksheet 14

A pulley is useful for lifting something upwards by pulling downwards. A block and tackle is made up of many pulleys joined together. Two or more pulleys used together are force multipliers. For each extra pulley, the machine magnifies your effort even more. The only disadvantage of a block and tackle is that the effort needs to be applied over a long distance.



Whim

When most of the gold near the surface had been removed from the shallow mines, the miners had to dig deeper to search for gold. The whim was used to haul two large buckets up very deep shafts.

*The horse walks in a very large circle to make the drum turn. As the drum turns, the rope lifts one bucket and lowers the other. The horse walks in a large circle to turn the drum. The horse's effort is multiplied, so it can move the buckets easily. This machine is a **wheel** and **axle**.*

These grooved wheels turn so that the rope moves smoothly over them. They are called pulleys.

Activities

REMEMBER

1. Why are two pulleys better than one?
2. What is a block and tackle?
3. What is a disadvantage of the block and tackle?

THINK

4. A pulley is a type of wheel and axle. Where is the fulcrum of a pulley wheel?
5. How could long-handled shovels be a disadvantage to miners?
6. Describe how the hand-operated whip works.

OBSERVE

7. A **Californian pump** was often used to pump water from a stream into a **puddling machine**. By looking at the photo, explain how it might work.

✓ checklist

I can:

- ☐ explain how pulleys can be used to increase effort
- ☐ describe some uses of simple machines on the goldfields.

Life on the goldfields

The Victorian gold rush started in the early 1850s and lasted about ten years. When 'diggers' first arrived, they had tents and clothes and a few simple tools. As time passed, more and more people, including women and children, moved to the goldfields. With the increasing population came the need for more industries and machines in the goldfield areas.

Travel across the goldfields

Horses were one of the main methods of transport on the goldfields. A horse and cart could be used to transport a single person or a group of people, much like a car does today.

The cart itself is a **compound machine**; it is made up of more than one simple machine.

*Horse and cart drivers sometimes carried whips. Whips are **third-class levers**. A quick, short flick of the whip makes the tip of the whip move a big distance, very quickly. Like all third-class levers, the whip is a **speed multiplier**.*

*The brake handle is a big **lever**. The long distance between the top of the handle (**effort**) and the **fulcrum**, makes the lever an excellent **force multiplier**. Pushing the brake lever forces large pads to press against the wheels, stopping the cart.*

*The **wheels** on the cart are simple machines too. They make it easier for the cart to move. Without turning wheels, the cart would drag or slide across the ground.*

Doctor or blacksmith?

Tooth decay was common on the goldfields because of poor diet and dental hygiene. The decay was so bad that teeth began to rot away. The doctor was usually responsible for pulling out rotten teeth. But if the doctor wasn't about, the blacksmith had the tools! The blacksmith's pliers, used for pulling out the nails in horseshoes, were often given a quick rinse and used as doctor's tools.

Pliers were used as levers. The long handles magnified the doctor's (or blacksmith's) effort for a better grip on the teeth.

The candleworks

There was no such thing as electricity on the goldfields. Kerosene lamps and candles provided light once the sun had set. Candles were made with the help of a simple machine at the candleworks known as the **nodding donkey**.

*This 'nodding donkey' is a **second-class lever**. The candle maker pulls down on the handle to dip the wicks into the hot wax. After seven seconds, the wicks are removed from the wax and left to cool. The process is repeated until the candles are of the correct thickness.*

Activities

IMAGINE

1. Life was very different in the 1850s. Imagine you lived in a goldfields township. How would your life differ from your life today? What would you have to go without?

THINK

2. Why does the brake on a horse-drawn cart need to be a force multiplier?
3. Draw a diagram of the whip that the horse and cart drivers used. Label the effort, fulcrum and **load**.
4. What class of lever are the doctor's pliers?
5. Is the nodding donkey a force or a speed multiplier? Explain your answer.
6. Spinning tops were popular toys on the goldfields and are still popular today. These toys are an example of a wheel and axle.
   (a) Draw a sketch of the spinning top and label the wheel and **axle**.
   (b) To which part of the spinning top do you need to apply the effort to make it spin quickly – the wheel or the axle? Explain your answer.

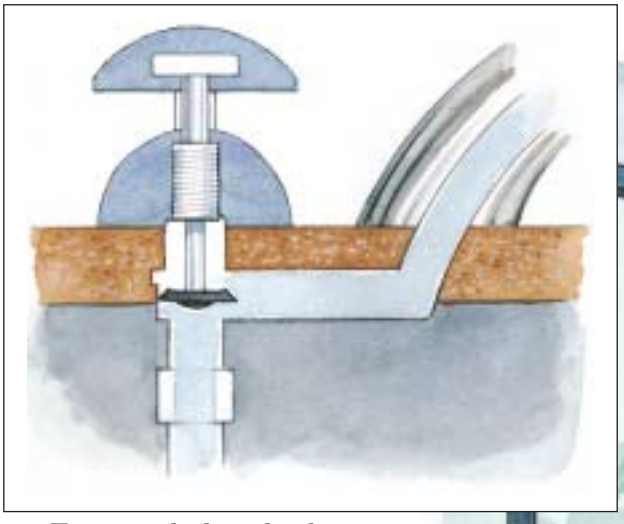
checklist

I can:

- ☐ explain how some simple and compound machines work
- ☐ describe some uses of simple and compound machines on the goldfields.

Machines in the kitchen

Even some everyday items in the kitchen are simple machines. Without them, we'd find certain tasks almost impossible — like slicing a piece of meat, prising the lid off a coffee tin or stopping the flow of water into the sink. Kitchen machines can be used to multiply our effort. Tap handles and can openers are examples of **force multipliers** in our homes. Other machines, like eggbeaters, speed up our **effort**. Eggbeaters also change the direction of our effort.



Taps are wheels and axles. Larger tap handles are easier to turn than smaller ones. When the handle is turned, the axle turns, moving a small washer inside the tap to let the water flow through it.

*The blade of a knife is a sharp **wedge**. It makes chopping much easier and so does the long handle. The handle acts as a **lever**. Longer handles are needed to cut through harder foods.*

*A ceiling fan is a speed-multiplying **wheel** and **axle**. Electricity is used to apply a huge effort to the small axle in the fan's shaft. The fast-spinning blades form the wheel of the simple machine.*

*Try closing a door by pushing it close to the hinge. It is much harder than closing the same door from the handle. The door is a **second-class lever**.*

Go to worksheet 15

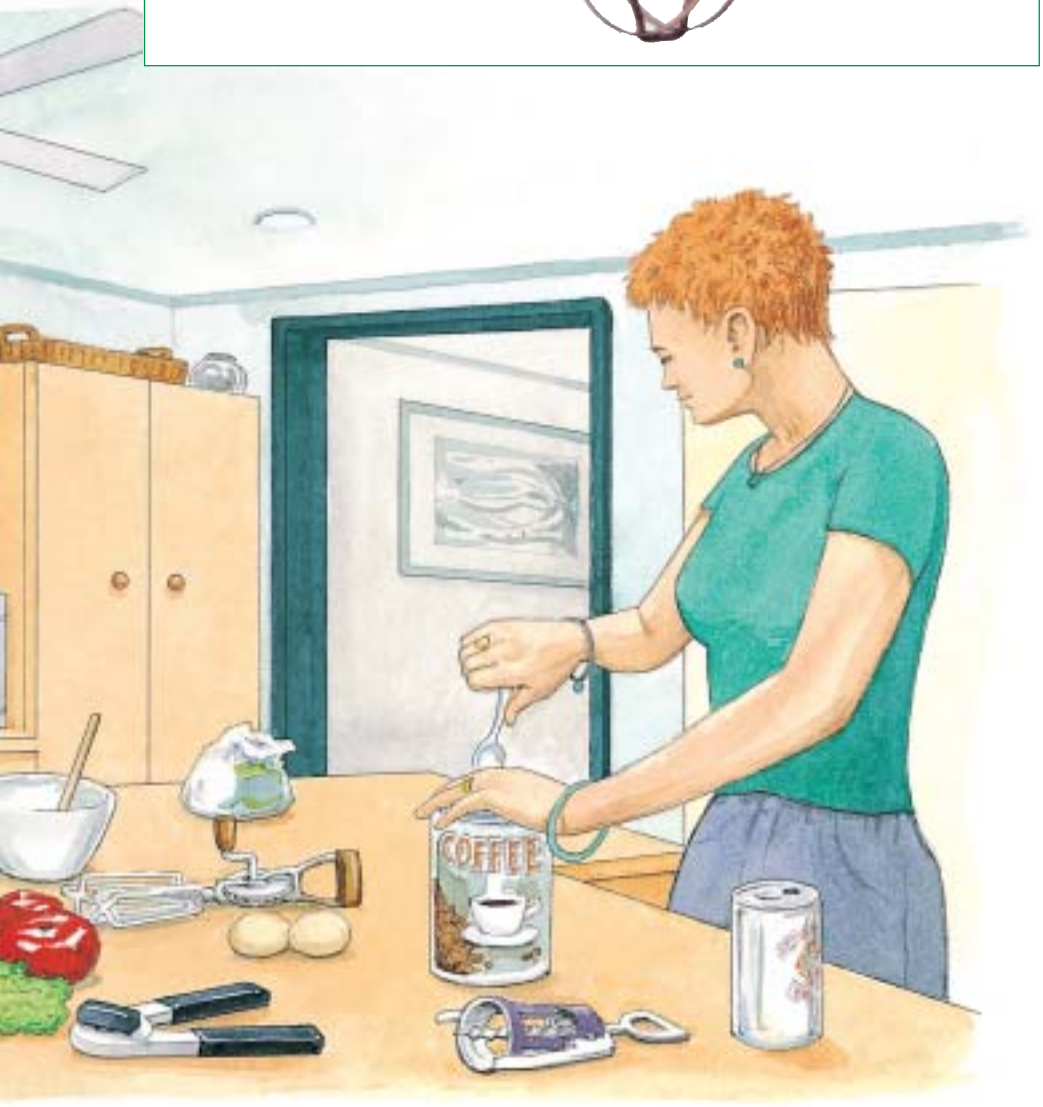
The eggbeater

Eggbeaters are **compound machines.** The gears used in eggbeaters are designed to change the direction of the effort. They also act as **speed multipliers,** making it easy to turn the beaters quickly.

*These gears are **bevelled gears**. They change the vertical turn of the handle into the horizontal spin of the beaters.*

This handle turns in a big circle. It makes the smaller bevelled gears turn with greater speed.

The beaters spin much faster than the handle turns.



Activities

SEARCH

1. Look carefully at the kitchen scene on this page. List the objects that work as simple machines. For each object, describe the simple machine and how the machine creates an advantage.

REMEMBER

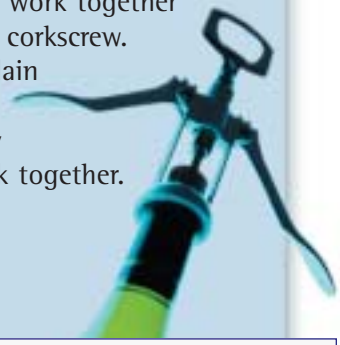
2. Why are bigger tap handles easier to turn than smaller ones?
3. Label the effort, **fulcrum** and **load** on the cupboard door.
4. What class of lever is the knife when you are chopping?
5. What are bevelled gears?

THINK

6. The tap uses more than one simple machine. By looking at the diagram, find one other simple machine inside the tap. Describe how it works as a simple machine.

OBSERVE

7. Take a close look at this corkscrew. List at least three simple machines that work together in a corkscrew. Explain how they work together.



checklist

I can:

- ☐ describe how bevelled gears work
- ☐ explain how some simple machines work together in compound machines.

Check and challenge

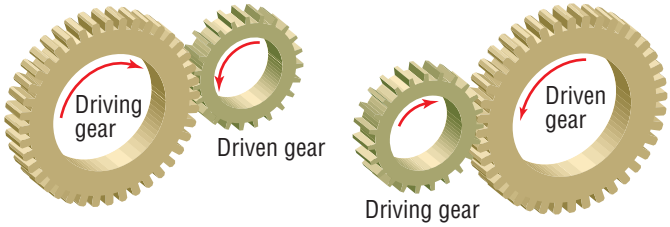
Machines everywhere

Check

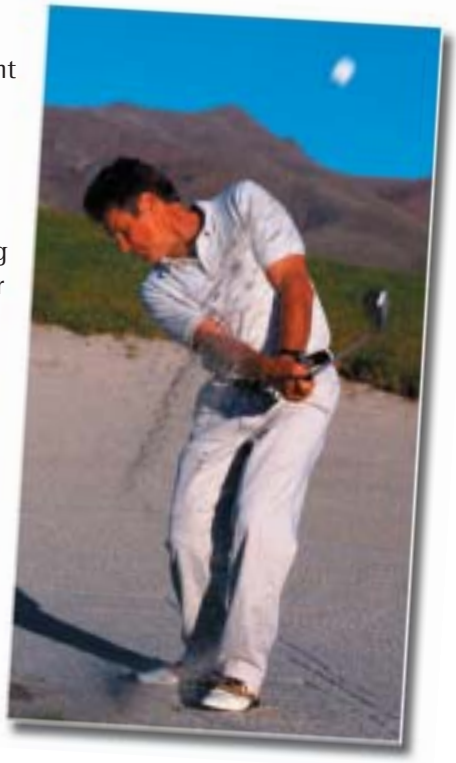
Changing force and speed

1. List two classes of levers that magnify effort.
2. Softballers use a softball bat to hit the ball a long distance at very high speeds.
   (a) Sketch the softballer and her bat. Label the effort, fulcrum and load on your diagram.
   (b) What class of lever is the softball bat?
   (c) Is the softball bat acting as a speed or force multiplier?

3. Which of the following is a speed multiplier and which is a force multiplier?



4. Draw a diagram to show how a pulley can be used to change the direction of a force.
5. Which of the following machines can act as speed multipliers:
   (a) a single pulley?
   (b) wedge?
   (c) first-class lever?
   (d) third-class lever?
   (e) inclined plane?
6. Why is it important for this golfer to use his club as a third-class lever?
7. In golf, a driver is used to make long shots and a putter for making short shots. Why are drivers usually longer than putters?



Mechanical systems

8. Describe what sprockets are and where they can be found.
9. What is a compound machine?
   A A machine that squeezes objects to make them smaller
   B A machine that multiplies your force
   C A machine that multiplies your speed
   D A machine made up of two or more simple machines
10. How is a block and tackle different from a single pulley?

Everyday machines

11. (a) What simple machines are stairs most similar to?
    (b) Do they act as force or speed multipliers?
12. Name and describe the simple machines that are needed to turn a door handle and push open a door.
13. Why aren't door handles in the *middle* of doors?

Mechanical advantage

14. (a) Suggest a way to increase the mechanical advantage of this lever.
    (b) Suggest a different way of using the plank to help load the box onto the truck.

15. Explain why the roads around hills and mountains are winding roads rather than straight roads.
16. A first-class lever is used to lift 12 coins.
    (a) What is the mechanical advantage of the lever when it uses four coins to lift the 12 coins?
    (b) If the 12 coins are placed six centimetres from the fulcrum, how far should the four coins be from the fulcrum to lift the heavier pile of coins?

Getting into gear

17. (a) How is a rack and pinion gear *similar* to a bevelled gear?
    (b) How is a rack and pinion gear *different* from a bevelled gear?

Machines of all types

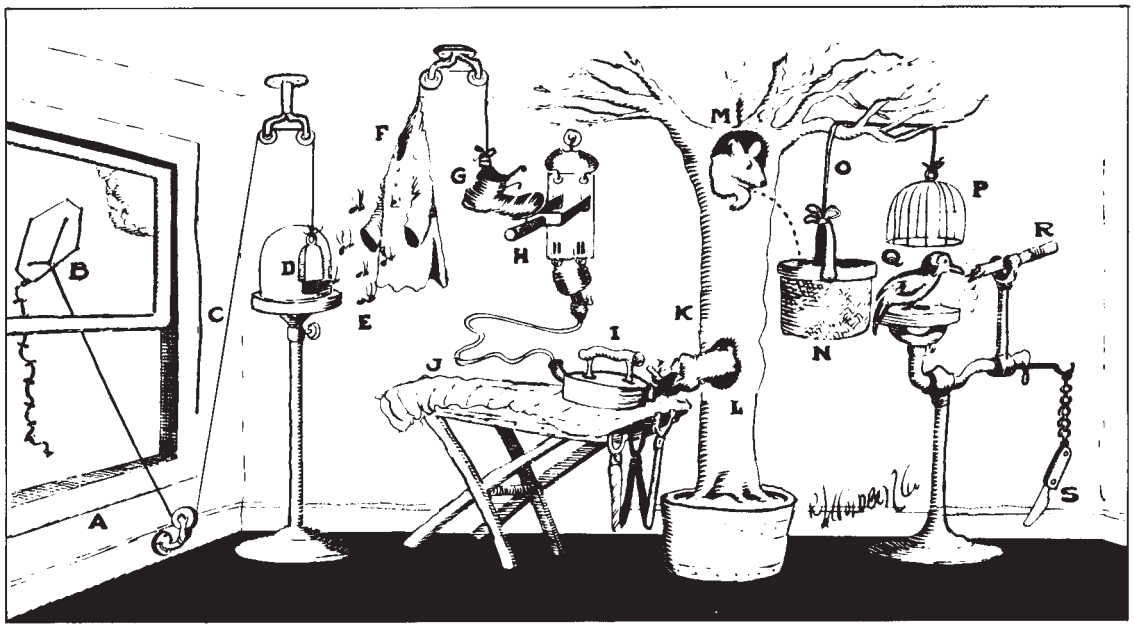
1. Would a lever work without a fulcrum? Explain your answer.
2. Imagine that you wanted to drive a screw into a piece of wood. You have the choice of using a screwdriver with a large handle or one with a smaller handle. Which would you use? Explain your choice.

Magnificent machines

3. Your task is to design a machine that will take five seconds to burst a balloon. Follow these steps to design your machine:
    (a) Look at the cartoon by Rube Goldberg (below).
    (b) In a group of three or four, brainstorm some ideas for simple machines that could be used to construct the balloon-bursting machine.
    (c) As a group, draw a rough copy of a design for the balloon-bursting machine.
    (d) As a group, present a poster of your machine and a brief description of how it works.

How machines work

4. Design a multimedia presentation that outlines how one of the following machines works. Make sure you discuss whether the simple machines used are speed or force multipliers.
    (a) corkscrew
    (b) eggbeater
    (c) car steering system
    (d) bicycle gears
    (e) water tap



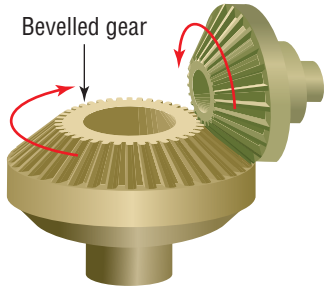
Open window (A) and fly kite (B). String (C) lifts small door (D) allowing moths (E) to escape and eat red flannel shirt (F). As weight of shirt becomes less, shoe (G) steps on switch (H) which heats electric iron (I) and burns hole in pants (J). Smoke (K) enters hole in tree (L), smoking out opossum (M) which jumps into basket (N), pulling rope (O) and lifting cage (P), allowing woodpecker (Q) to chew wood from pencil (R), exposing lead. Emergency knife (S) is always handy in case opossum or the woodpecker gets sick and can't work.

Summary of key terms

axle: a small wheel at the centre of a larger wheel. The axle turns with the larger wheel.

belt drive: a belt drive is much like a sprocket. The difference is that the wheels in a belt drive don't have teeth whereas the wheels in sprockets do.

bevelled gear: a gear that changes the direction of the effort by 90°



Californian pump: a hand-operated compound machine used to collect water from a creek or stream. The Californian pump was widely used on the goldfields to pump water into a puddling machine.

compound machine: a machine made up of more than one simple machine

driven gear: a gear that is moved by another gear

driving gear: a gear that causes another to move

effort: the push or pull used to try to make an object move

engineer: a person who uses scientific ideas to design and build new technology and make it work

first-class lever: a lever with the fulcrum between the effort and the load

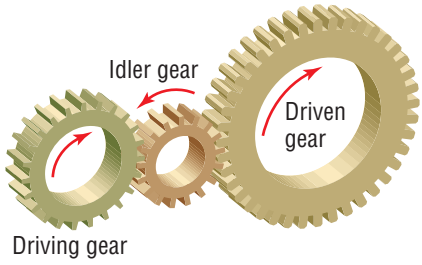
force: a push, pull or twist

force multiplier: a simple machine that multiplies the effort you use

fulcrum: the point around which a lever turns

gear: a toothed wheel that can turn other toothed wheels

idler gear: a third gear added between a driving and driven gear to keep them moving in the same direction



inclined plane: a simple machine that reduces the effort required to raise objects to a higher level

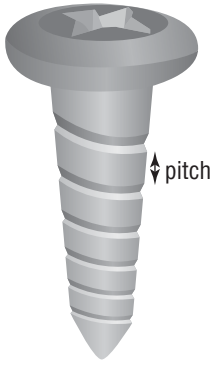
lever: a long, rigid object that moves around a turning point called a fulcrum

load: an object or force that resists motion

mechanical advantage: a measure of how much easier a simple machine makes a task

nodding donkey: a hand-operated compound machine used in the 1800s to dip candle wicks into hot wax during the candle-making process

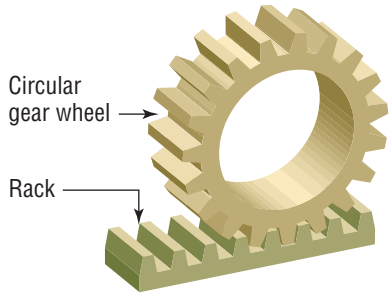
pitch: the vertical distance between the adjacent parts of the thread of a screw



puddling machine: a horse-driven compound machine from the goldfields used to break up rocks and clay. Diggings from mines were placed in this machine with water. Wooden rakes were dragged through the machine to break up the clay so that gold could sink to the bottom of the puddling machine. The muddy water was then drained away so that the gold could be removed.

pulley: a simple machine consisting of a grooved wheel over which a rope can move freely

rack and pinion gears: gears that change circular motion into straight line motion



screw: a curved, inclined plane

second-class lever: a lever with the load between the effort and the fulcrum

speed multiplier: a simple machine that increases the speed of an object

sprocket: gears joined by a chain, as in a bicycle

third-class lever: a lever with the effort between the load and the fulcrum

thread: the curved ridge of a screw

wedge: an inclined plane that pushes objects apart

wheel: a lever that moves in a complete circle. The fulcrum is at the centre of the wheel.

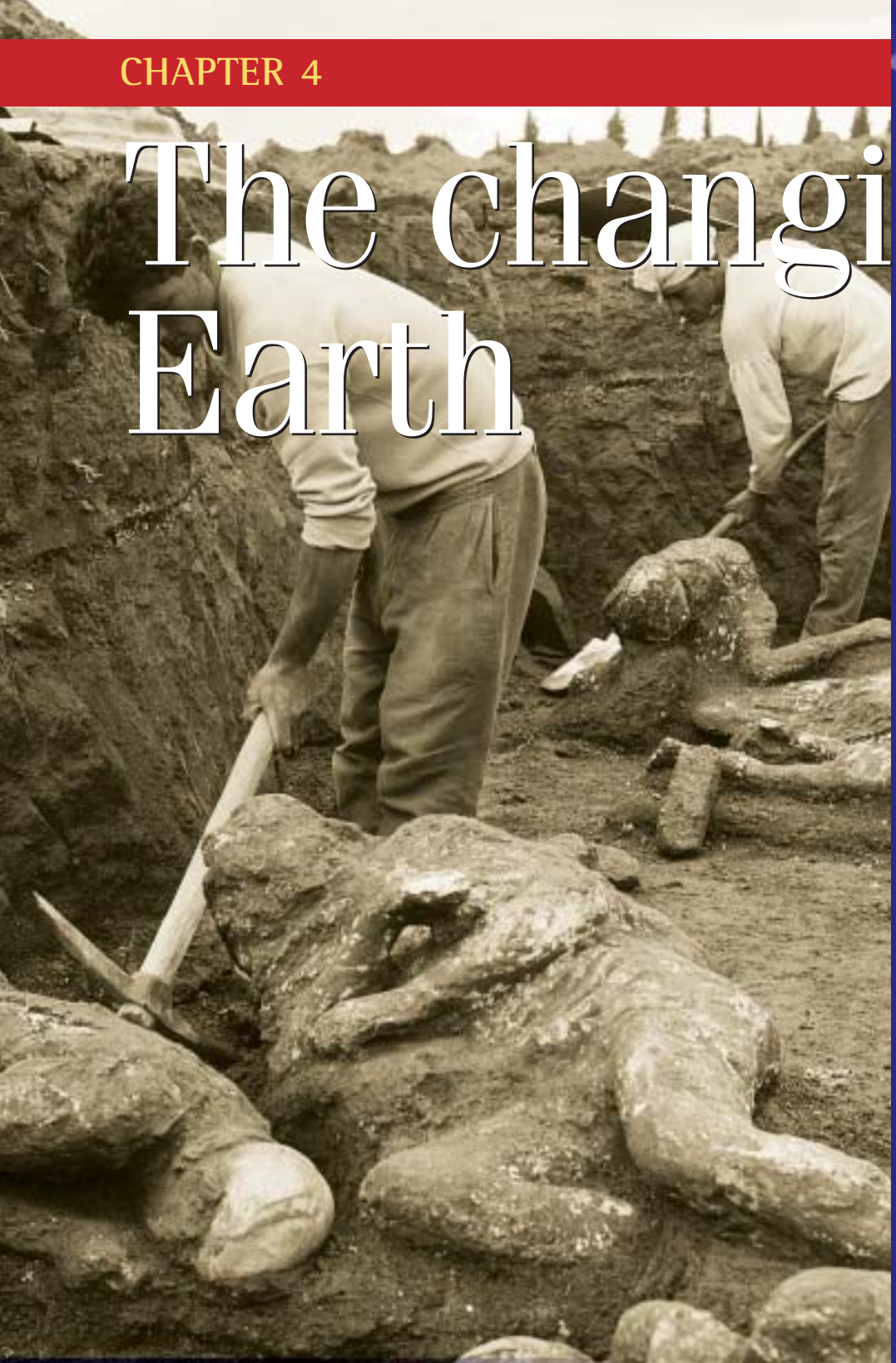
whim: a large wheel, axle and pulley system used to haul large buckets from very deep mine shafts. A whim is operated by horse power.

whip: a lever or pulley system to lift and lower buckets from deep mine shafts. A whip can be operated by hand or by a horse.

windlass: a wheel and axle used to lift and lower buckets from shallow mine shafts

CHAPTER 4

The changing Earth



You will discover

That the Earth is constantly changing
How different types of rock form
How rocks can change from one type to another
Ways to estimate the age of rocks

Death of a city

Surveying the ruins of Pompeii a modern-day traveller observes, '... and to the left, to the right, wherever I look, grim reminders of the AD 79 disaster stare back at me. I am surrounded by the ruins of an ancient resort town that was once home to 15 000 people. Now all that remains are the plaster casts of the bodies caught in the mountains of hot ash that buried an entire city.'

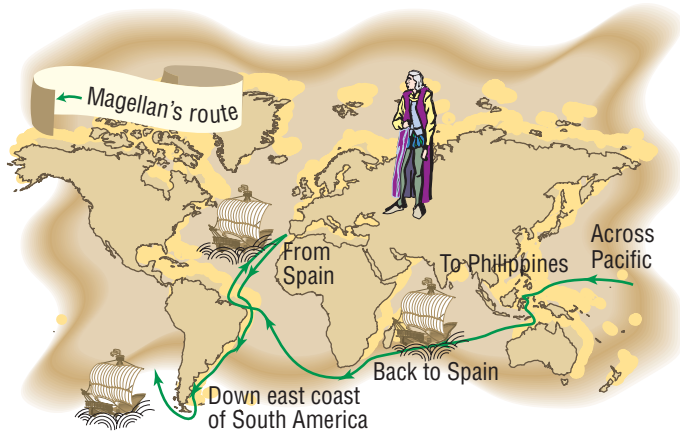
The citizens of Pompeii would have been taken by surprise by the exploding volcano. Mount Vesuvius had been asleep for centuries. In fact, only a few scholars would have known that the mountain in the distance was actually a volcano.

For over 100 years, excavations at Pompeii have been slowly uncovering more and more pieces of the past. Every day many tourists visit the site and walk the streets that the ancient residents of Pompeii once walked. And many of them glance at the distant mountain that last erupted in 1944, and which some scientists predict will erupt again soon.

1 Why are there now only plaster casts of the bodies that were caught in the disaster?
2 How long can volcanoes 'sleep' before they erupt again?
3 What can we learn from looking at fossils and the ruins of buried cities?

What's in the Earth?

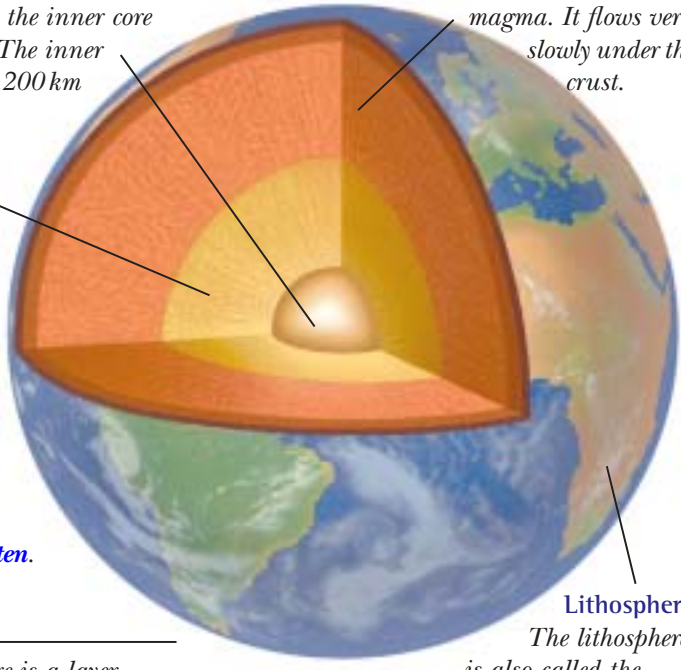
Until the sixteenth century, many people believed that the Earth was flat. In 1519, Ferdinand Magellan and his crew set off from Spain to the Spice Islands (Indonesia) in a westerly direction. Most people believed that eventually they would reach the end of the Earth and fall off. Instead, by continuing in the same direction, the crew arrived back in Spain and showed that the Earth could not be flat.



Inner core
The inner core is very hot. Its temperature is about 7000 °C. Even though it is hotter than the outer core, the inner core is solid. Scientists believe that the weight pushing in on the inner core keeps it solid. The inner core is about 1200 km thick.

Outer core
The outer core is about 2200 km thick and very hot. Temperatures from 4000 °C to 6000 °C make the rock in the outer core ***molten****.*

Mantle
The mantle is a 2900 km thick layer under the crust. It is made mostly of solid rock. But the upper part of the mantle, just under the crust, is not so solid. This 'slushy' rock is called magma. It flows very slowly under the crust.



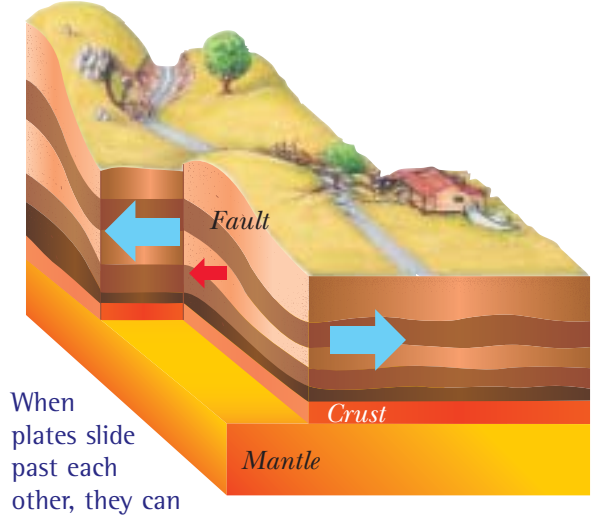
Atmosphere
The atmosphere is a layer around the surface of the Earth. It is over 80 km thick and made up of gases that are needed for life on Earth. The mixture of gases in the atmosphere is called ***air****. Most of the air mixture is made up of nitrogen. Oxygen makes up about one-fifth of the air.*

Lithosphere
The lithosphere is also called the crust. It is the layer that forms the hard surface of the Earth. The crust ranges in thickness from 8 km to about 40 km. The thinnest parts are under the oceans. The crust is mostly made up of soil, water and rocks.

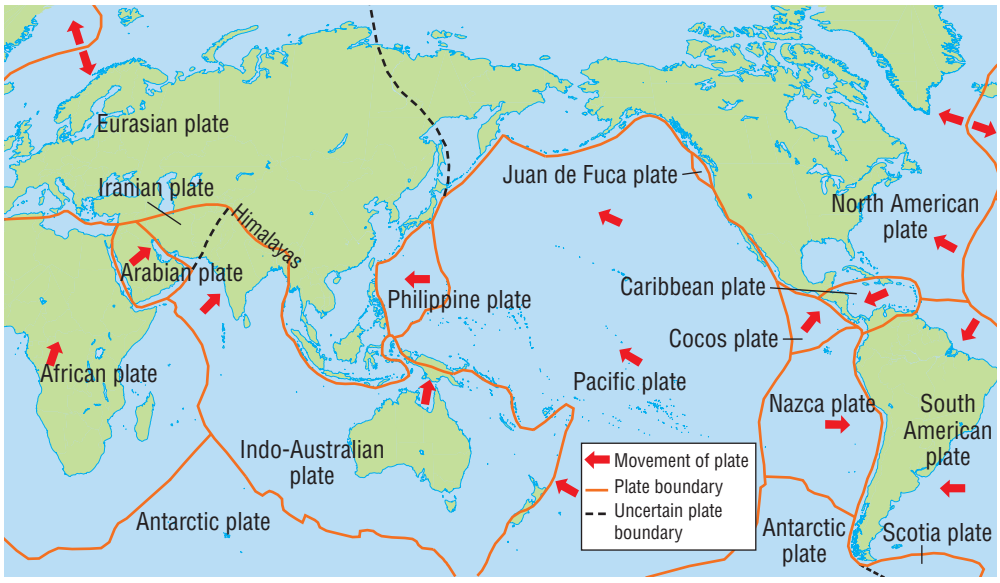
Floating on the mantle

The Earth's **crust** floats on semi-liquid **magma** at the top of the **mantle**. The crust is made up of large pieces called **plates**. The Earth's plates move very slowly over the top of the mantle.

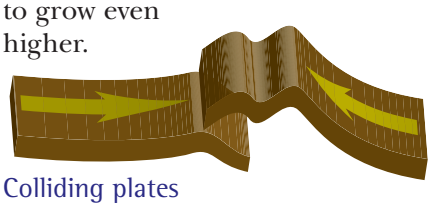
When plates slide past each other, earthquakes occur. Severe earthquakes happen when the sliding plates don't slide smoothly and get jammed together. This is happening in California. The **pressure** between the plates builds up. Suddenly the plates jolt past each other and the sliding starts again.



When plates slide past each other, they can get jammed and cause earthquakes.



Sometimes the Earth's plates crash into each other. When this happens, one plate can sink under the other. It's been happening between the Indo-Australian and Eurasian plates. The Indo-Australian plate is sinking under the Eurasian one. The world's highest mountain range, the Himalayas, is on the Eurasian plate. This plate buckles as the Indo-Australian plate slides under it and causes the Himalayas to grow even higher.



Colliding plates

Activities

REMEMBER

1. List the layers within the Earth from:
   (a) thinnest to thickest
   (b) the centre outward.
2. Where is magma found?
3. Why is the inner core solid even though it is hotter than the molten outer core?
4. How are the Himalayas being formed?
5. Describe one reason why severe earthquakes occur.

THINK

6. In 1961, geologists started a project that attempted to drill through the crust into the mantle. Where in the crust would the hole have been drilled? Explain your answer.

CREATE

7. Build a model of the Earth's layers using plasticine. When you have finished, cut a segment out of your model so that the different layers can be seen. Make sure that your model shows which layers are thickest and which are thinnest.

GRAPH

The deepest hole drilled into the Earth is only about 14 km deep. Even though this is a short distance compared with the distance to the centre of the Earth, the temperature increases dramatically. The distance to the centre of the Earth is about 6370 km. The table above shows the temperature at different depths in a drill hole.

Depth (km)	Temperature (°C)	Depth (km)	Temperature (°C)
0	15	5	158
1	44	6	187
2	73	7	215
3	102	8	242
4	130		

8. Use graph paper or a spreadsheet to plot a line graph of temperature against depth. Label the vertical axis 'Temperature' and the horizontal axis 'Depth'.
   (a) Use your graph to predict the temperature at 4.5 km.
   (b) At approximately what depth is the temperature of the Earth equal to 200 °C?
   (c) Estimate the temperature at the centre of the Earth.
   (d) How close is your estimate to the actual temperature (7000 °C)? Explain why your estimate might be different from the temperature that scientists have predicted.

✓ checklist

I can:

- ☐ label and describe the layers of the Earth
- ☐ explain earthquakes and mountain building in terms of the movement of the plates that make up the Earth's crust.

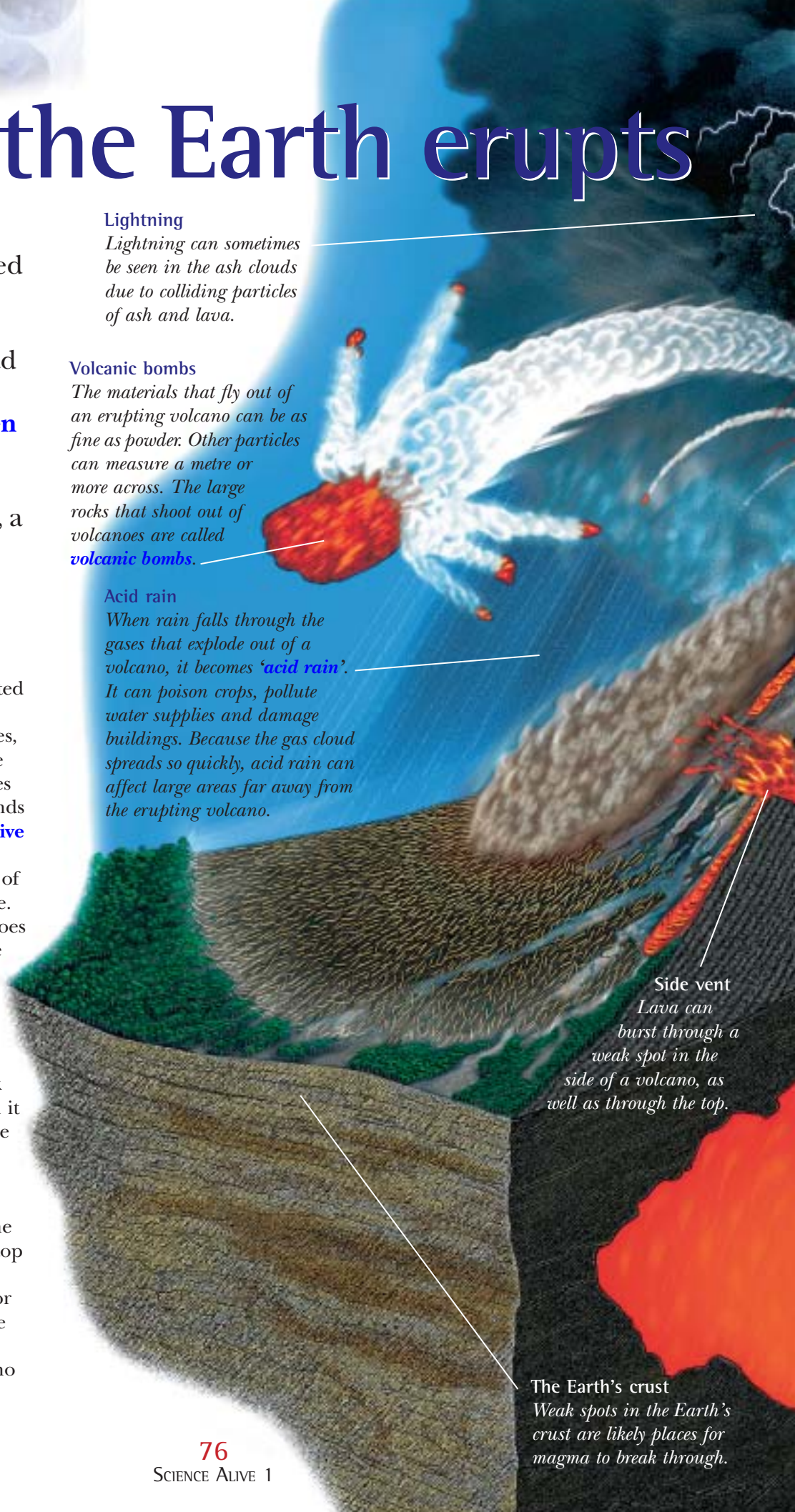
When the Earth erupts

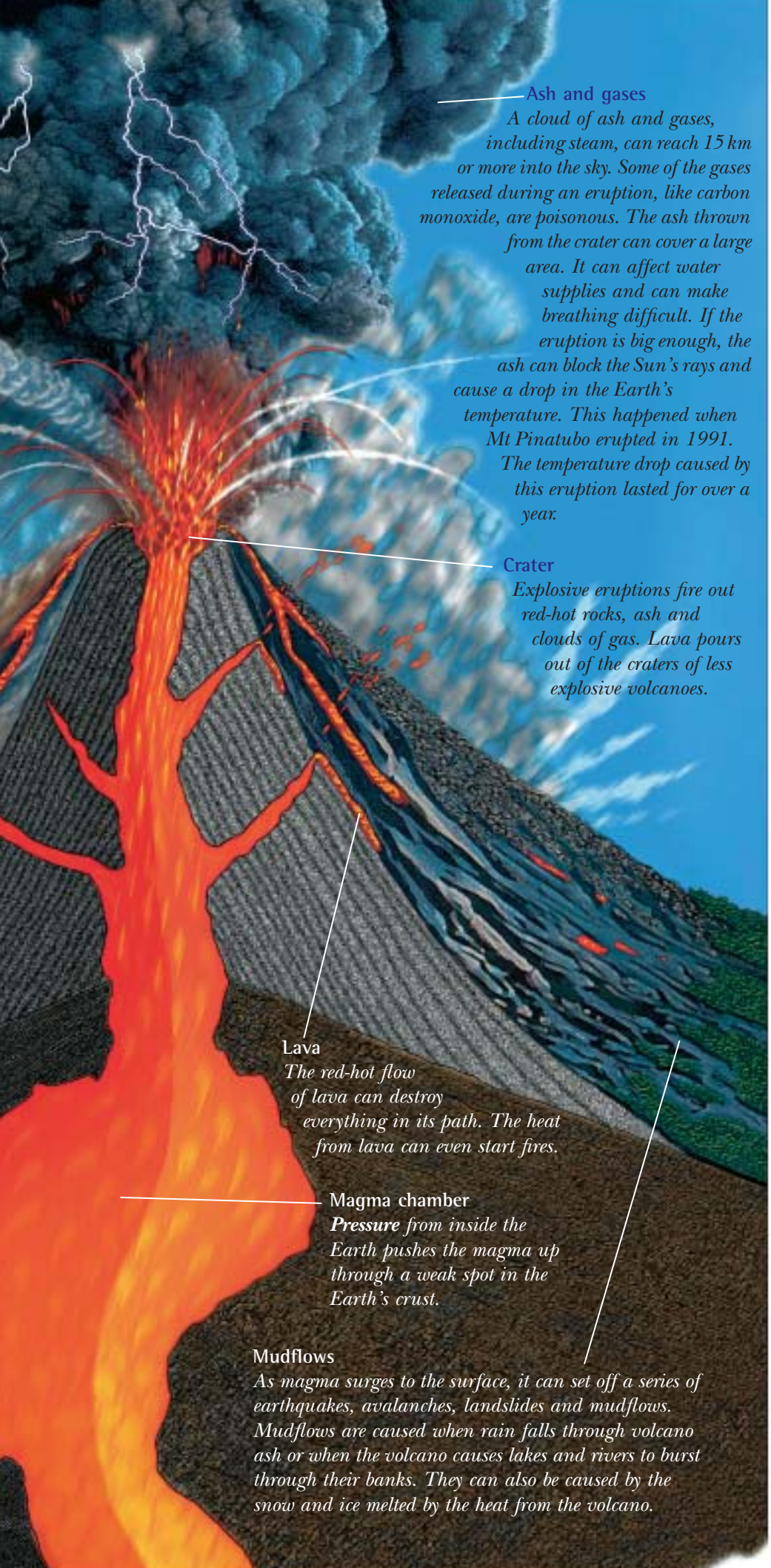
In 1943, a crack mysteriously appeared in the ground in a flat cornfield in Mexico. A week later the crack had become a 150 m high volcano. Red-hot **molten lava** spewed from deep inside the growing volcano. Ten years later, a 400 m high, rocky mass stood where a peaceful cornfield once lay.

This volcano is now described as **dormant** because it has not erupted for over 20 years. Dormant volcanoes are 'sleeping' volcanoes, but they can wake up at any time without much warning. Volcanoes that have not erupted for thousands of years are said to be **extinct**. **Active** volcanoes are those that have erupted recently and show signs of erupting again in the near future. There are about 500 active volcanoes around the world. None of these active volcanoes are in Australia.

What is a volcano anyway?

Magma can break through weak spots in the Earth's **crust**. When it does, a volcano is formed. As the magma reaches the Earth's surface, it releases some of its gases. The molten rock at the Earth's surface is called lava. The lava can ooze gently out of the top of a volcano or it can burst out explosively. Lava can be runny or really thick. The gases inside the volcano can explode out so violently that the top of a volcano can blow right off!





Ash and gases
A cloud of ash and gases, including steam, can reach 15 km or more into the sky. Some of the gases released during an eruption, like carbon monoxide, are poisonous. The ash thrown from the crater can cover a large area. It can affect water supplies and can make breathing difficult. If the eruption is big enough, the ash can block the Sun's rays and cause a drop in the Earth's temperature. This happened when Mt Pinatubo erupted in 1991. The temperature drop caused by this eruption lasted for over a year.

Crater
Explosive eruptions fire out red-hot rocks, ash and clouds of gas. Lava pours out of the craters of less explosive volcanoes.

Lava
The red-hot flow of lava can destroy everything in its path. The heat from lava can even start fires.

Magma chamber
***Pressure** from inside the Earth pushes the magma up through a weak spot in the Earth's crust.*

Mudflows
As magma surges to the surface, it can set off a series of earthquakes, avalanches, landslides and mudflows. Mudflows are caused when rain falls through volcano ash or when the volcano causes lakes and rivers to burst through their banks. They can also be caused by the snow and ice melted by the heat from the volcano.

Go to worksheet 16

MI BODILY/KINAESTHETIC

Activities

REMEMBER

1. Describe the difference between a dormant and an active volcano.
2. When is a volcano extinct?
3. Where did the lava erupting from this volcano come from?

THINK

4. Explain how the eruption of a single volcano can affect the whole world.
5. How is lava different from magma?
6. How might people living in river valleys near a volcano be affected if the volcano erupted?

IMAGINE

7. Imagine that you live in a small town near a volcano. A warning has been issued that the volcano is about to erupt. What measures could be taken by the residents to protect the town and its people?

CREATE

8. Write a newspaper report that describes the eruption from one of these famous volcanoes:
   Mt Vesuvius
   Mt Pinatubo
   Mt St Helens
   Mt Krakatoa.

✓ checklist

I can:

- ☐ describe how volcanoes form
- ☐ describe some of the hazards associated with volcanoes
- ☐ understand what can happen during a volcanic eruption.

When lava cools

Lava surges out onto the Earth's surface at temperatures of 1000 °C or more! At that temperature, **lava** could take weeks to cool down. But sometimes, when volcanoes erupt under water for instance, the lava cools much faster. When lava cools down it becomes solid rock. Looking closely at rocks that formed from red-hot lava tells a story about the way they cooled.

Frothy rocks

Some violent volcanic eruptions shoot out lava filled with gases. The lava cools quickly, while it is still in the **air**, and traps the gases inside. Rocks that form this way are full of holes. Two examples of these rocks are **pumice** and **scoria**. The substances in the **magma** affect the colour and hardness of the rock that is formed when lava cools. Scoria has more iron in it than pumice, so it is darker than pumice.

*Pumice is a pale-coloured rock. It is so full of holes that it is very light. It floats on water and sometimes washes up on beaches. Powdered pumice is used in some **abrasive** cleaning products.*

Scoria is heavier than pumice. It is usually found closer to the volcano's crater than pumice. Scoria is a red-brown or grey rock that can be used in garden paths or around drainage pipes.

Growing crystals

Rocks that form from lava cooling above the Earth's surface are called **extrusive rocks**.

Basalt is an extrusive rock that can take on many appearances. One big difference between samples of basalt is the size of the crystals that make up the rock. The size of the crystals gives us clues about how or where the rock formed. The longer it takes for the lava to cool down, the bigger the rock crystals grow.

Basalt with bubbles
The crystals in this basalt are large because they formed from lava on the ground. The crystals had time to grow before the rock became solid. Notice the holes. The lava was filled with gases when it began to cool. The gases have since escaped.

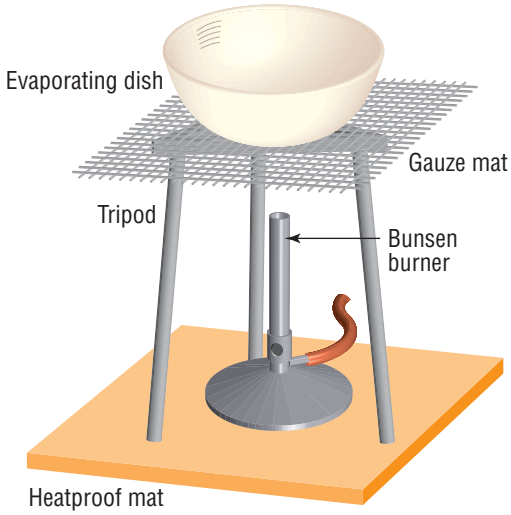
Pillow basalt
This rock formation came from a volcano that was once under water. The rocks formed from underwater volcanoes are smooth and round. The crystals in this basalt are so small that they are difficult to see.

Obsidian
Obsidian is a smooth, black rock that looks like glass. It is formed when lava cools almost instantly. This rock is different from basalt because it cooled so quickly that no crystals formed. Sometimes very fine air bubbles are trapped in the rock, which give it a coloured sheen.

What affects crystal size?

You will need:
Bunsen burner, matches and heatproof mat
tripod and gauze mat
alum solution
2 Petri dishes
evaporating dish
safety glasses and lab coat.

- Pour roughly equal amounts of alum solution into the evaporating dish and the Petri dishes.
- Put one of the Petri dishes in the refrigerator.
- Put the other Petri dish on a windowsill.
- Place the evaporating dish on the gauze mat.



- Gently heat the evaporating dish without allowing the solution to boil rapidly.

!CAUTION: Rapid boiling may result in spitting. Wear safety glasses at all times.

- Continue heating the solution until nearly all of the water has evaporated.
- Observe the size of the crystals formed in the evaporating dish.
- After two days, observe the size of the crystals formed in the two Petri dishes.
- Observe the crystals formed in the refrigerator again after four or five days.

1. Draw a labelled diagram of the crystals formed in the evaporating dish and in the two Petri dishes. Your diagram needs to show the difference in size between the crystals.
2. Each of these crystals grew over a different time span. How does the time allowed for the crystal to form affect the size of the crystals?

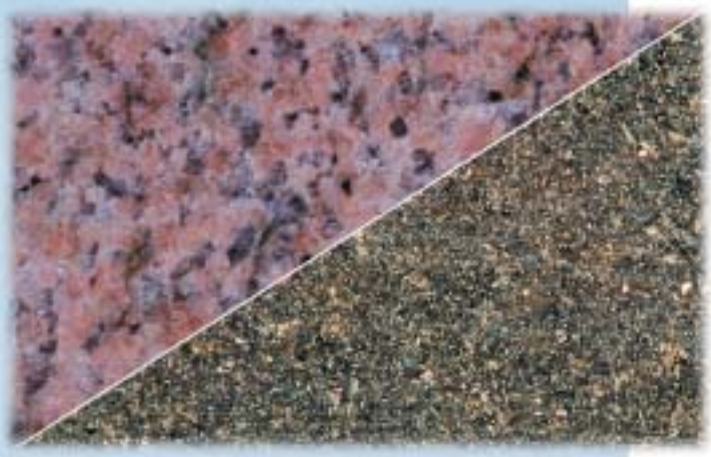
Activities

REMEMBER

1. Where do extrusive rocks form?
2. Why are the crystals in pillow basalt smaller than the crystals in basalt which form on the ground?
3. Scoria and pumice are formed in a similar way. Why are their colours different?
4. What type of extrusive rock could easily be mistaken for glass?

THINK

5. In which of these two rocks did the lava cool fastest? Explain your answer.



6. What does the presence of 'bubbles' in rocks tell us about the lava?

IMAGINE

7. Imagine you are a mass of magma. Describe your journey from the depths of the Earth's crust, through to its surface.

CONNECT

8. Go to www.jaconline.com.au/science/weblinks and click on the Guinness Book of Records link for this textbook. Mt Everest is said to be the highest mountain in the world, but there is a volcano that is higher.
   (a) What is the name of the volcano?
   (b) Where is the volcano?
   (c) Why do we call Mt Everest the highest peak, when this volcano is taller than Everest?

checklist

I can:
- ☐ compare crystal sizes in rocks
- ☐ explain why some rocks have air bubbles
- ☐ describe how extrusive rocks are formed.

Cooling under ground

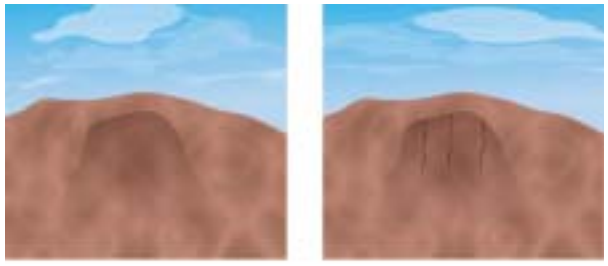
Trapped deep within the Earth lie bodies of **magma**. Far away from the cooling effects of **air** and water, magma can take millions of years to turn into solid rock.

Rocks that form when magma cools in the Earth's **crust** are called **intrusive rocks**. Many of these rocks are hidden beneath the surface and discovered only when the ground around them is blown or washed away. Intrusive rock bodies can be as small as a few centimetres across or larger than 100 kilometres across.

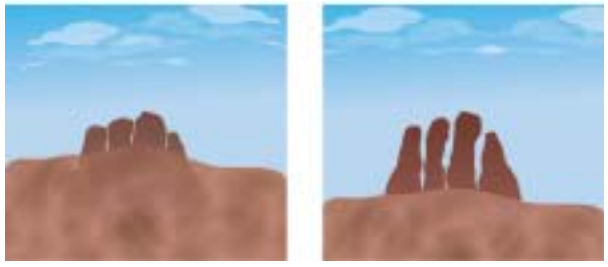
Intruding rocks

Batholiths are intrusive bodies of rock that measure more than 100 km across. If it weren't for the **weathering** process, many of these rocks would not ever be seen. **Granite** and other hard rocks often form batholiths.

The batholith forms under the Earth's surface when magma cools down.



Cracks form in the batholith while it cools down.

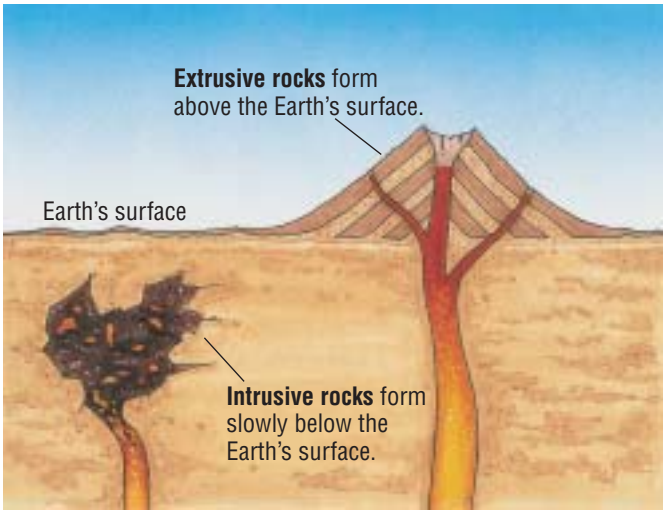


If the batholith is exposed to the environment, it will start to wear away along the cracks. Over time, the batholith may break down completely. The breakdown of rocks is called weathering.

The softer rocks and soil around the batholith may wear away.

Igneous rocks

Igneous rocks are those that form when magma or **lava** cools and turns into solid rock. **Extrusive** and intrusive rocks are both types of igneous rock. The word igneous comes from the Latin word for 'fire'.



A long walk to Tasmania?

Wilson's Promontory in Victoria is part of a huge granite batholith that measures about 300 km long and 50 km wide. The batholith links Tasmania to the rest of Australia and at one time created a land bridge between the two. About 15 000 years ago, rising sea levels covered the batholith and separated Tasmania from the rest of Australia.

Granite

Granite is a common intrusive rock. The crystals in granite form over long periods of time and grow large enough to make them easy to see with the naked eye.

Granite is very hard and can be used for building. Headstones and other monuments can be made from granite that has been polished to give it a glossy finish.

The crystals found in granite are a mixture of white, pink, grey, black and clear substances. The substances that make up rocks are called ***minerals****. Granite is made up of a few minerals, such as quartz (clear to grey), feldspars (white and pink) and mica (black). Some rocks are made up of only one mineral.*

Activities

REMEMBER

1. Describe one way in which intrusive rocks can become visible on the Earth's surface.
2. Complete this sentence: Intrusive and extrusive rocks are both types of _____ rock.
3. Complete these sentences: Intrusive rocks with large crystals form when _____ cools _____. Smaller crystals form in extrusive rocks when _____ cools _____.

THINK

4. True or false?
   (a) All intrusive rocks form batholiths.
   (b) Rocks are made up of substances called minerals.
   (c) All igneous rocks are extrusive rocks.
   (d) Intrusive rocks are more likely to have larger crystals than extrusive rocks.
   (e) Batholiths come from volcanoes.
5. Suggest reasons why granite has been chosen for this headstone.
6. Describe how granite and basalt are different from each other.

INVESTIGATE

7. Most towns and cities have buildings, or parts of buildings made from granite. Find out if there are any buildings or other structures such as statues or memorials in your area that are made from granite. You may find your local council helpful or you could try an Internet search. Describe the granite in one of the buildings or structures and explain why granite has been chosen. If a digital camera is available, you could present your findings in a multimedia report.

checklist

I can:

- ☐ describe how intrusive rocks form
- ☐ explain that all rocks are made up of minerals
- ☐ explain why the crystals in intrusive rocks are larger than those in extrusive rocks.

Wearing away

Volcanoes continue to erupt, leaving **igneous rocks** on the Earth's surface, yet the surface of the Earth's **crust** isn't a huge mass of solid rock. Wind, water, ice and other natural actions constantly break down rocks on the surface of the Earth.

The process of breaking down rocks is called **weathering**. Weathering is a slow process, but varies depending on the type of rock and the natural action involved. Hard rocks can take longer to break down than soft rocks. Winds that carry sand weather rocks more quickly than winds that don't carry sand. Some of the **minerals** found in rocks are easily dissolved by water, which speeds up the weathering process.

Wave Rock in Western Australia has been shaped by water and wind, which have also carried weathered rock away. The lines on the **granite** surface have been caused by water running down the face of the rock. The water has dissolved some of the minerals in the granite. This is an example of **chemical weathering**.

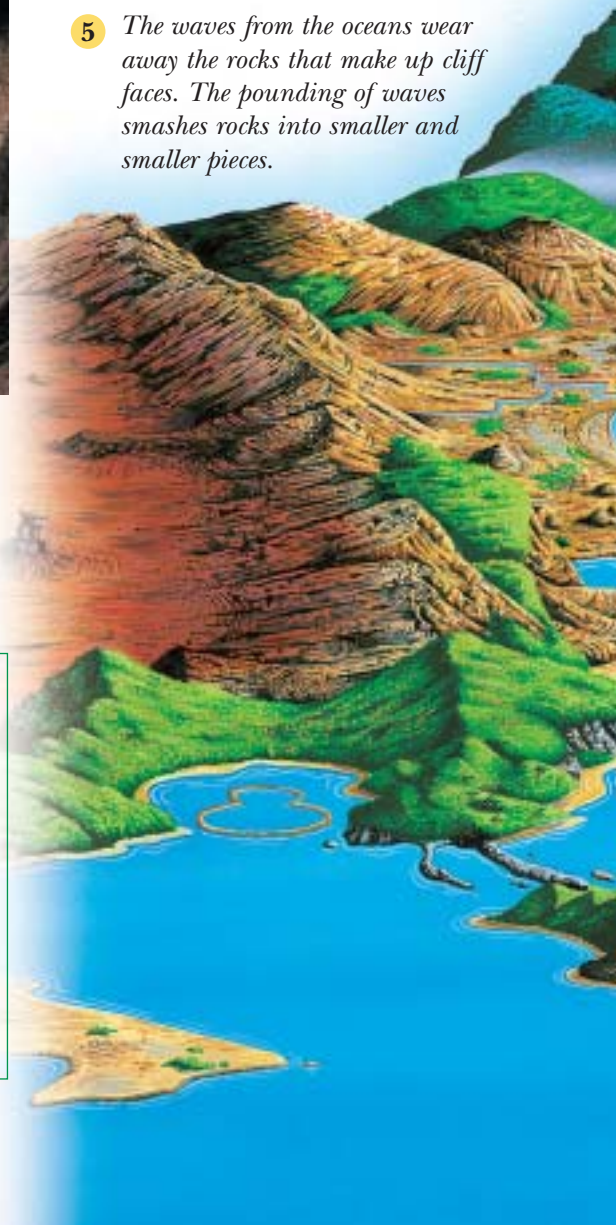
1 *Fast-flowing water can move sand, soil and even big rocks over long distances. All creeks and rivers flow to the sea or to inland lakes, but by the time they reach the seas or lakes, the water flows much slower.*

2 *As the water slows down, the bigger rocks are deposited.*

3 *By the end of the river's journey, all but very fine sediments have been deposited.*

4 *Coastlines can change quite quickly as a result of weathering, erosion and the depositing of sediments.*

5 *The waves from the oceans wear away the rocks that make up cliff faces. The pounding of waves smashes rocks into smaller and smaller pieces.*

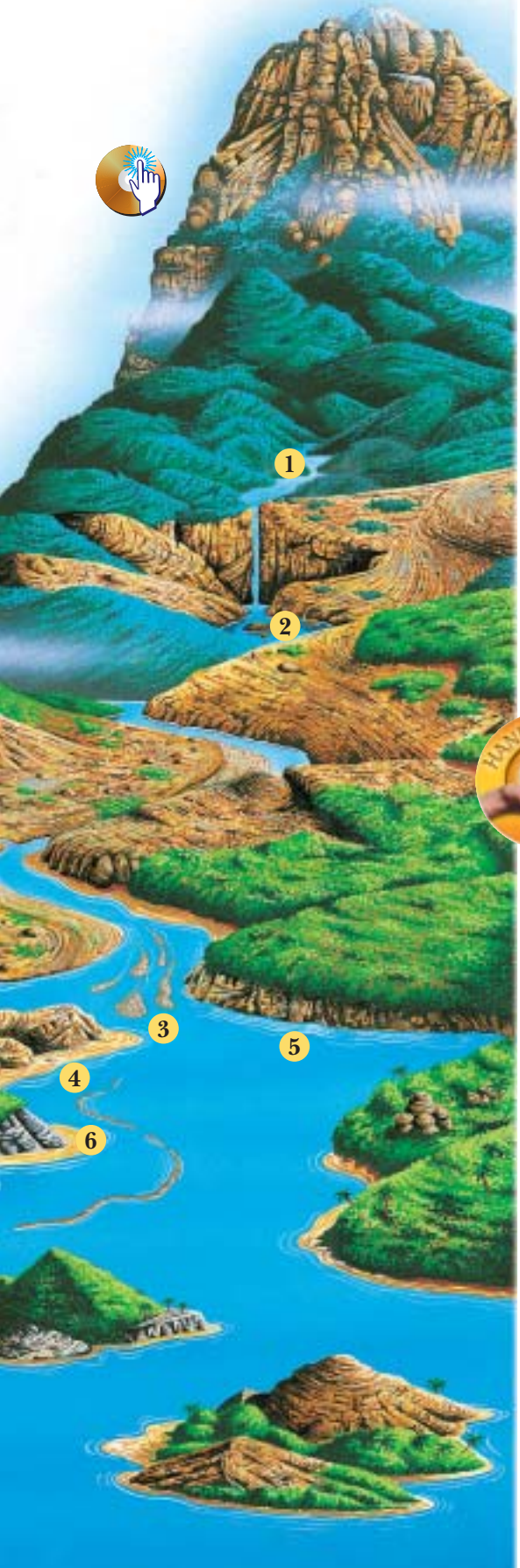


Cracking up

Some objects, like glass, crack if their temperature changes quickly. Rocks can do the same. The cracking occurs because the outer part of the rock cools more quickly than the inside of the rock after a hot day. Cracking can also occur when it rains on a hot day. The cracks gradually get larger, until large flakes begin to fall off. Granite often weathers this way.

Trees can begin to grow within cracks in rocks. As the trees grow, the cracks are forced apart, eventually splitting the rocks. This speeds up weathering.

6 *Sand is picked up by currents in the waves along one beach and deposited on other beaches. Strong winds have enough energy to pick up sand and carry it inland.*



Erosion

After weathering has broken down the rock, the wind, water or ice can then move it around. The process of moving weathered rocks and soil is called **erosion**. Eventually the wind, water or ice **deposits** the weathered material. The deposited materials are called **sediments**.

The wind is especially effective in wearing away rocks in dry, dusty areas. The wind can pick up tonnes of sand and soil and dump it many kilometres away. A dust storm is a very strong wind that carries huge amounts of weathered material. Dust storms are quite common in Australia.

Glaciers are large bodies of moving ice. They move very slowly compared to water and wind, but like these, collect weathered materials as they move. The difference between glacial erosion and other acts of erosion is that little of the sediment is dumped along the way. Glaciers deposit most of their sediment in one lot. That means that sediments of many different sizes are found in one place.

How ice speeds up weathering

You will need:
plastic bottle with a lid
water.

!CAUTION: Do not use a glass bottle.

- Fill the bottle with water right to the top.
- Freeze the bottle overnight.
- Observe what happens to the bottle.

1. What do your observations tell you about what happens to water as it freezes?
2. How do you think ice makes the breakdown of rocks happen faster?

Activities

REMEMBER

1. What is weathering?
2. What is erosion?
3. List three natural actions that break down rocks.

THINK

4. The Twelve Apostles, shown at right, were originally named because 12 'stacks' were close together along the western Victorian coast. Some of the stacks have since collapsed into the sea.
   (a) What is probably the main cause of the weathering of these stacks?
   (b) Which part of each stack is likely to weather most quickly? Give a reason for your answer.
   (c) Has this coastal feature been caused by weathering, erosion or both? Explain your answer.
5. Why are larger sediments deposited before finer ones in river systems?
6. How can you tell the difference between sediments deposited by a river and those deposited by a glacier after it melts?

INVESTIGATE

7. Find out if weathering and erosion occur on other planets or the Moon.

✓ checklist

I can:
- ☐ describe the weathering process
- ☐ understand how rocks can crack
- ☐ describe the process of erosion.

The human contribution

Without **weathering** and **erosion**, the rocks that rise to the Earth's surface would keep building up. Both weathering and erosion are natural processes. But what happens when humans disturb the natural process?

Imagine a world where acid falls from the sky, a place where deserts replace fertile land and where the beaches are vanishing. These are some of the effects that humans have already had on the Earth.

Why save the trees?

The roots of trees help to hold the soil together. Removing trees exposes good, fertile soil to wind and water. The soil is blown or washed away, leaving the land destroyed. Early Australian settlers originally cut down trees to create farmland. As the population grew larger, more trees were cleared to provide space for industrial areas and housing. Since then, industrial areas have grown larger and the forests smaller. Trees are still being cleared for wood and wood products like paper.

Over the past 200 years, over two thirds of Australian forests have been cleared.

Deforestation is a worldwide problem.

Acid rain

Every day many harmful chemicals are pumped into the **air**. Some are naturally forming chemicals, but many are from cars, factories or from human activity. The chemicals in the air can dissolve in water, much like salt in hot water. The dissolved chemicals return to the ground in rainwater, snow or fog and the combination is called **acid rain**.

Acid rain can poison trees, soil and water supplies. It even eats away at rocks, including those used in buildings and statues.

Coasts under threat

Coastal areas can be badly affected by erosion. Bare sand is easily washed away by water and blown inland by the wind. Vegetation that binds the sand together has been torn up by recreational vehicles. Vegetation near beaches in tourist areas such as the Gold Coast has been removed and replaced with huge buildings. Barriers such as sea walls, mesh fences and **groynes** are built to hold sand on the beaches.

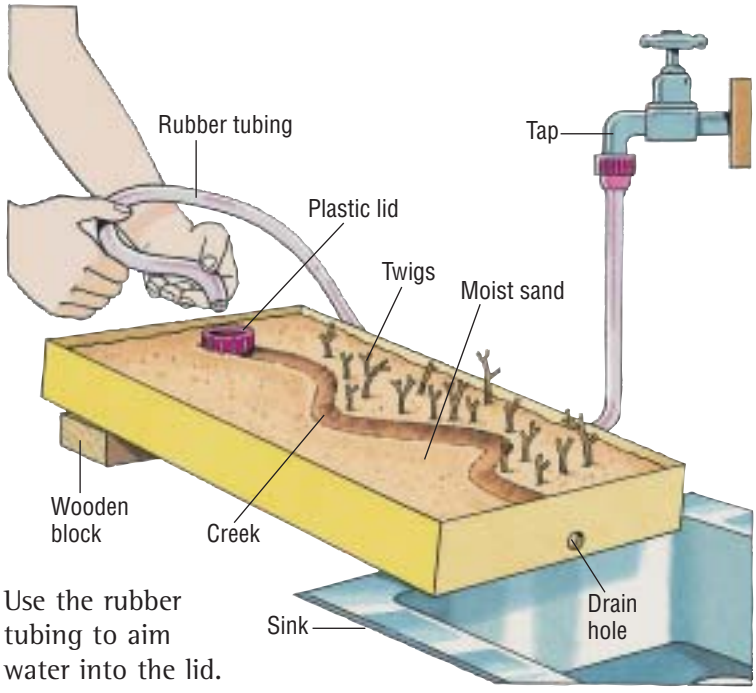
Go to worksheet 17

Why plant trees?

You will need:

stream tray or box
wooden block
plastic lid from a soft drink bottle
damp sand
rubber tubing
several small twigs.

- Pack the sand into the tray.
- Make a groove in the sand to represent a creek or river.
- Set up the equipment as shown in the diagram below. Make sure to 'plant trees' on one side of the 'creek' only.



- Use the rubber tubing to aim water into the lid.
- Allow water to flow slowly but steadily into the lid and then overflow into the 'creek'.

1. Where does most of the erosion occur along the 'creek'?
2. What effect do the 'trees' have on erosion?

On the mend

Scientists, conservation groups and government bodies play an important part in improving the environment. The aim is to reduce the impact of human activity and repair past damage. Some methods for reducing erosion and repairing the damage already caused by erosion include:

- farmers ploughing their fields around hills rather than up and down the slope. This prevents rain from washing soil down hills.
- sealing roads and gutters to direct water into proper drains
- controlling numbers of livestock
- replacing trees that have been removed
- fencing off large sections of beaches and banning recreational vehicles in many coastal areas
- reducing the impact of introduced animals, such as rabbits, on native vegetation.

Activities

REMEMBER

1. What is acid rain?
2. Give two reasons why forests have been cut down.
3. How does cutting down trees speed up erosion?
4. Describe two ways that farmers can reduce erosion.

THINK

5. Why has a barrier been placed at the back of the beach in the photo on page 84?
6. The photograph below shows an example of tunnel erosion. Suggest what has caused the erosion and how it may be stopped.

CREATE

7. Prepare a flow chart to show how deforestation can occur.

IMAGINE

8. Imagine you work for a local council in an area that has both national parks and coastlines. Your job is to educate people in the area about erosion and land care. Design a leaflet that gives the reasons for not using recreational vehicles in the area.

checklist

I can:

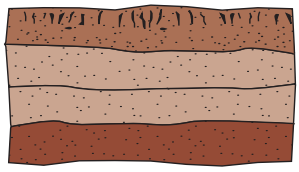
- ☐ describe how human activity can speed up erosion
- ☐ discuss ways to minimise erosion and improve the condition of the land.

Sedimentary rocks

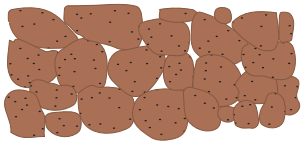
The Earth's **crust** is not made entirely of **igneous rocks** because these rocks break down. The process of **erosion deposits** broken-down rocks as **sediments**. But the Earth isn't full of sediment resulting from **weathering** either. It may take millions of years, but eventually sediments from weathering will form new rocks. Rocks formed from sediments are called **sedimentary rocks**.

Sedimentary rocks

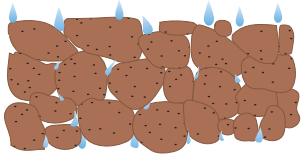
Many sedimentary rocks are made up of the grains of weathered rock. The grains cement together over time.



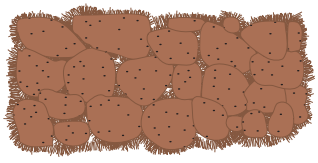
Sediments are deposited in layers called beds.



The grains of sediment in lower layers begin to squash together.



Chemicals that are dissolved in water can soak into the sediments.



The chemicals help cement the grains together once the water has evaporated.

Naming sedimentary rocks

Sandstone, mudstone and siltstone are types of sedimentary rock. Their names give us clues about the sediments they formed from. The types of sediment tell us what the environment was like when the sediments were deposited.

*Sandstone is made up of grains that are easy to see. Sand-sized grains are deposited in changing environments like deserts and beaches. The grains are carried by **air** in the deserts and by the surf at beaches.*

Siltstone has finer grains than sandstone. The grains are hard to see, but can be easily felt. Silt is deposited in quiet environments like slow flowing creeks and near the end of rivers. You are not likely to find silt on beaches, but it can be deposited out at sea.

Most of the grains in mudstone are too small to see or feel. Mud is deposited only in very still environments, like lakes and swamps.

Conglomerate rocks have grains in a wide range of sizes. Some grains are so small that they cannot be seen. Other grains are as big as pebbles. The sediments in conglomerate rocks are deposited in fast flowing rivers, during floods or by glaciers.

Go to worksheet 18

Biological rocks

Sedimentary rocks are not always formed from other rocks. Living things break down and are deposited as sediments too. Under special conditions, the remains of plants and animals form biological sedimentary rocks.

Limestone

The shells and hard parts of sea organisms break down in a similar way to rocks. Shells and shell fragments deposited in layers on the ocean floor can cement together to form limestone.

Coal

When plants and animals die, they break down and rot. **Coal** is formed from dead plants and animals that became buried before rotting completely. Even though the remains are squeezed together over millions of years, coal may still look like the trees it formed from.

Chemical rocks

Chemical sedimentary rocks are those that form when water evaporates, leaving behind a solid substance. When salt lakes or seabeds dry up, they leave behind solid salt. If the layer of salt is squeezed under other sediments, it can eventually form **rock salt**.

When ground water or rainwater passes over limestone, it can pick up some of the chemicals from the limestone. When the water evaporates, it leaves behind the limestone chemicals again. Limestone re-forms like this in caves. It looks a bit different from the limestone formed from the remains of sea animals, but is made up of the same chemicals.

Activities

REMEMBER

1. What helps to cement the grains of sediments together to form sedimentary rocks?
2. How are the environments that form sandstone different from those that form mudstone?
3. Why are limestone and coal called biological sedimentary rocks?
4. How are chemical sedimentary rocks formed?

THINK

5. Explain why not all plant and animal remains form coal.
6. What role do weathering and erosion play in the formation of sedimentary rocks?

INVESTIGATE

7. Find out what coal and diamonds have in common.

The chalk age?

You may have heard of the stone age, the bronze age and the iron age. But have you heard of the 'chalk age'? The period in history called the Cretaceous lasted from about 145 million years ago to about 65 million years ago. It was the time when flowering plants first existed on Earth. It was also the time when huge amounts of chalk were deposited on the ocean floors all around the world. Chalk, like other forms of limestone, is made from the remains of sea animals. The word cretaceous comes from the Latin word *creta* meaning chalk.

checklist

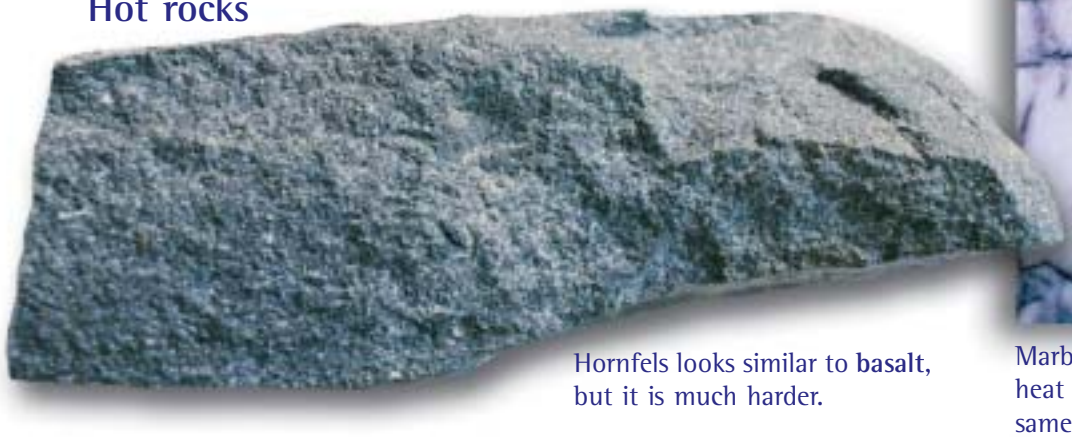
I can:

- ☐ explain the formation of sedimentary rocks
- ☐ describe the environments in which sedimentary rocks may form and how this affects the appearance of the rock.

Metamorphic rocks

Searing hot **magma** can push through the Earth's **crust** wherever there is a weakness. To get through to the surface, it needs to push through layers of other rocks. When rocks are heated to extreme temperatures by magma pushing through, their appearance and structure can change. Changes can also occur in rocks that are under **pressure** from layers of rocks above them. **Metamorphic rocks** are those rocks that have been changed by heat, pressure or both.

Hot rocks



Hornfels looks similar to **basalt**, but it is much harder.

When **igneous** and **sedimentary rocks** come into direct contact with rising magma, they are heated to very high temperatures. **Hornfels** is one rock that forms under these conditions. It is sometimes hard to tell what rock hornfels came from because it forms under such high temperatures.

Under pressure

Bands can sometimes be seen in metamorphic rocks formed under high pressure. The bands tell us that crystals were squeezed together to form new crystals in the new rock. Sometimes, the crystals are squeezed together so tightly that they partially melt and form fewer, but larger crystals.

Gneiss can be formed when **granite** is exposed to high pressures. Notice the bands in the new rock.

Old and new

The types of rock that are exposed to heat and pressure determine what the metamorphic rock will be. When sandstone containing the **mineral** quartz is heated, the metamorphic rock formed also contains quartz. This metamorphic rock is called quartzite.

Marble forms from **limestone** under heat and pressure. It contains the same minerals as limestone.

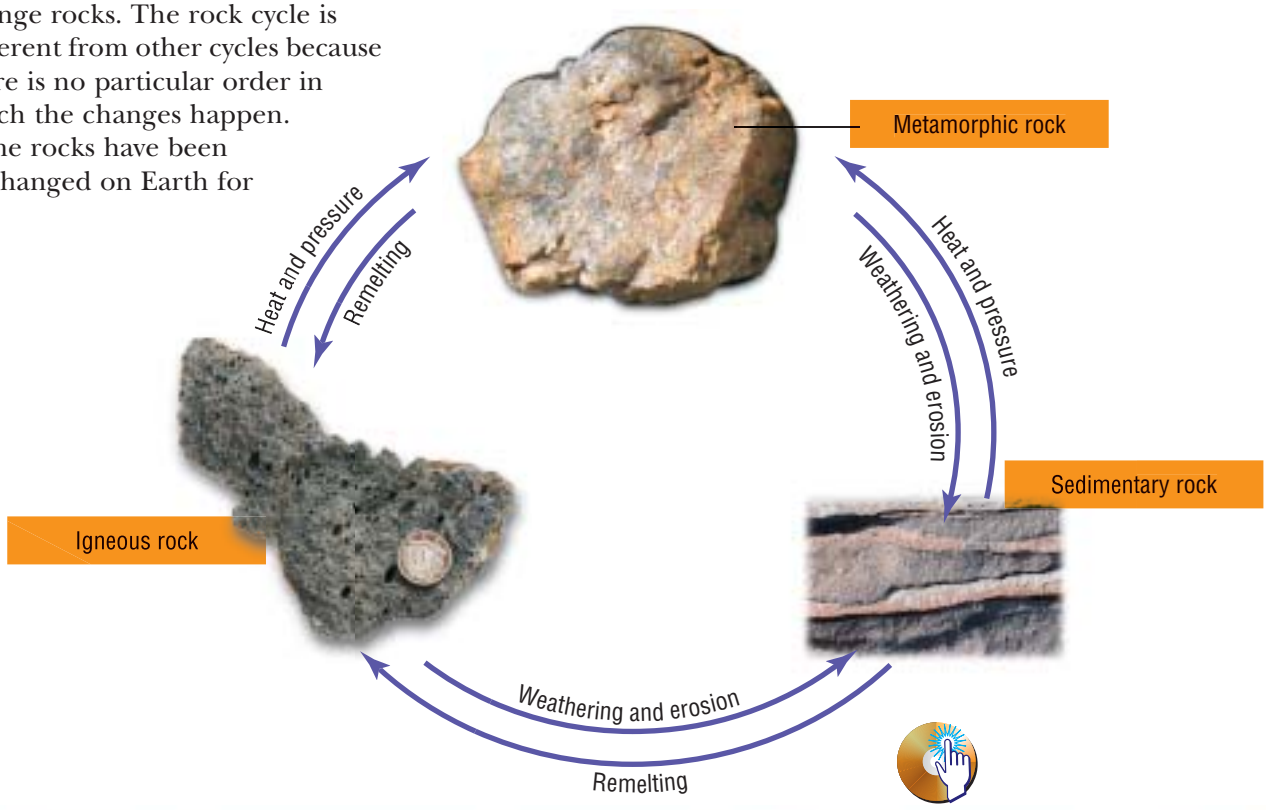
Shale is a common type of sedimentary rock. It has fine grains and crumbles easily along its layers. When shale is exposed to moderate heat and pressure, it forms slate.

Go to worksheet 19

MI VISUAL/SPATIAL

Recycling rocks

The **rock cycle** describes how rocks can change from one type to another. **Weathering**, **erosion**, heat, pressure and remelting are processes that help change rocks. The rock cycle is different from other cycles because there is no particular order in which the changes happen. Some rocks have been unchanged on Earth for millions of years and may not change for millions more. Some rocks change very quickly, especially near the edges of the plates that make up the Earth's crust.



Activities

REMEMBER

1. What do the bands in metamorphic rocks tell us about how the rocks were formed?
2. When sandstone is under heat and pressure, what metamorphic rock might it form?
3. From what rock has slate formed?

THINK

4. How could you tell the difference between basalt and hornfels?
5. Metamorphic rocks form only when the **parent rocks** do not melt completely from the heat and pressure. If the parent rocks do melt, the new rock is an igneous rock.
   (a) What do you think is meant by the term 'parent rock'?
   (b) Give an example of a metamorphic rock and its parent rock.
   (c) Why do metamorphic rocks form only when the parent rocks don't melt completely?
6. How are the bands in metamorphic rocks different from the layers in some sedimentary rocks?
7. Suggest why slate has been chosen for the surface (under the felt cover) of this billiard table.

✓ checklist

I can:

- ☐ describe how metamorphic rocks form
- ☐ explain how the rock cycle works
- ☐ list some common metamorphic rocks and their parent rocks.

How old is that rock?

Scientists believe that the Earth is about 4600 million years old. The clues held in rocks have helped them to calculate the Earth's age. Rocks also give scientists an idea about how the climate on Earth has changed over millions of years. But what types of clue do scientists need to look for in rocks?

Layers of clues

Sedimentary rocks form in layers. Each new layer forms on top of the older layers below. Sedimentary rocks are therefore useful for giving scientists clues about the order in which events have happened.

Sudden events, like exploding volcanoes, are recorded in the layers. Slow movements beneath the surface can also be seen in the layers.

7 *These layers were deposited last. They have started to weather and erode.*

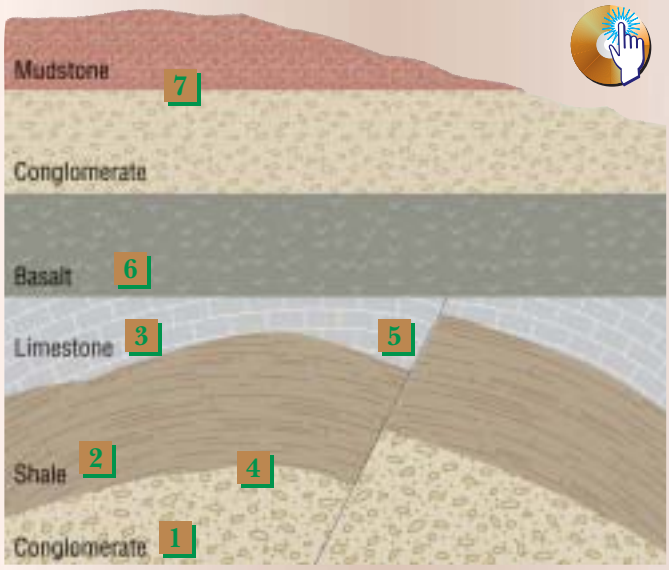
6 *A long period of **weathering** and **erosion** left the layer of **limestone** with a flat surface. When a volcano then erupted nearby, **lava** from the volcano cooled to form **basalt** on the flat surface.*

5 *A sudden event, like an earthquake, has occurred to break the layers of rocks like this. This event took place after the lower layers were folded. A break like this is called a **fault**.*

4 *A slow event has caused the lower levels to buckle. This is called **folding**. Folding can occur when rock layers are under **pressure** from both sides.*

3 *The third event to occur was the deposition of limestone. It tells us that there were probably sea **organisms** present in the area during this time.*

2 *This is the second layer deposited. Shale is a fine-grained rock that is deposited in a quiet environment such as a swamp, lake or the slow flowing part of a river.*



1 *Conglomerate was deposited first in this rock sample. This layer was deposited by a glacier or an active environment — like a very fast flowing river.*

Living in the past

The layers in rocks are useful for finding out about the order of events in a particular area. Finding out about the order of events, or comparing the ages of rocks without actually knowing their age in years is called **relative dating**.

Palaeontologists study **fossils**. A fossil is evidence of life in the past. Fossils can be used to compare the ages of rocks all around the world. Sedimentary rocks that contain the same fossils have usually formed at about the same time.

Fossil of a reptile

Go to worksheet 20

Fossils can form from the remains of plants and animals that have been preserved before they could decay. They can also be impressions, or **imprints**, left by rotting animals as a rock forms. In fact, even dinosaur footprints are fossils because they provide evidence of their existence. Fossils are not created very often because of the special conditions needed, and many fossils are destroyed if rocks are remelted in the **rock cycle**. Scientists can rely on fossils and rock layers only to *compare* the ages of rocks.

Finding out the actual age of a rock in years is called **absolute dating**. Scientists use a technique called **radiometric dating** to work out how old a rock is. Rocks, like everything else on Earth, have a small amount of radioactivity. Some substances in the rock release tiny radioactive particles. Radiometric dating relies on knowing how quickly those substances lose these particles. By measuring how much of these radioactive substances is left in a rock, scientists can work out its age.

A fatal fall . . . or was it murder?

In 1991, some German hikers found a body preserved in ice near the Italy–Austria border. Scientists used radiometric dating and found that the body was about 5300 years old! They thought that the person, known now as the Iceman, had died of hypothermia (extreme cold). Ten years later, another group of scientists using high-tech X-rays found the remains of an arrowhead lodged near his left lung. Specialists have not yet confirmed whether the Iceman fell back onto his arrow, or if he was murdered. And without any witnesses to question, the truth may never be known!

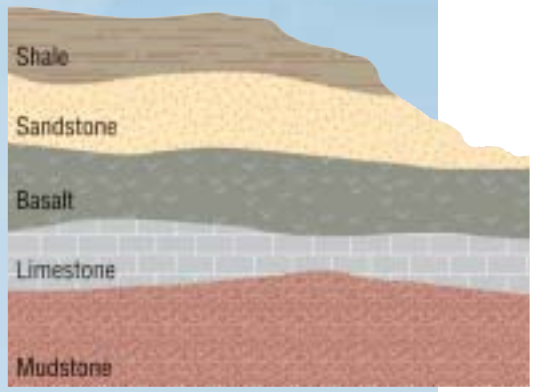
Activities

REMEMBER

1. Complete the following sentence:
   The buckling of rocks is called _____.
   It occurs when rocks are under _____.
2. What is a palaeontologist?
3. What is the difference between relative dating and absolute dating?

THINK

4. At right is a section of a rock face exposed in a quarry.



   (a) Make a list of the rocks in the cutting, from oldest to youngest.
   (b) Which rock is not a sedimentary rock? How did it form?
   (c) Describe the evidence for erosion in this rock face.
5. Why are fossils so rare?
6. How can fossil records help us to link rocks from different parts of the world?

CREATE

7. Use plasticine to create a sample of sedimentary rocks. Apply a gentle force to the sides of the layers. Observe how the layers fold under gentle pressure.

CONNECT

8. Even an animal's droppings can become a fossil. Go to www.jaconline.com.au/science/weblinks and click on the Fossilised Droppings link for this textbook to find out:
   (a) Which animal was responsible for a huge fossilised dropping found in Canada in 1998?
   (b) How long was the dropping?
   (c) What can palaeontologists find out from it?

✓ checklist

I can:

- ☐ understand that rocks provide clues to the past
- ☐ describe and compare ways to date rocks
- ☐ describe how folding occurs.

THE INVERLOCH INVESTIGATOR

9 March 1997 Weather: Perfect for the beach

Prehistoric mammal uncovered

Just yesterday, crowds of onlookers witnessed a most amazing find. A tiny 115 million-year-old jawbone, believed to be that of a placental mammal, was exposed. Until this find, scientists thought that placental mammals had reached Australia only 5 million years ago. The jawbone was found at the Dinosaur Dreaming site, a five-minute drive from the township of Inverloch on the Flat Rocks platform. A large group of volunteers is expected at the site tomorrow to search for more remains of the prehistoric mammal.

Every minute counts …

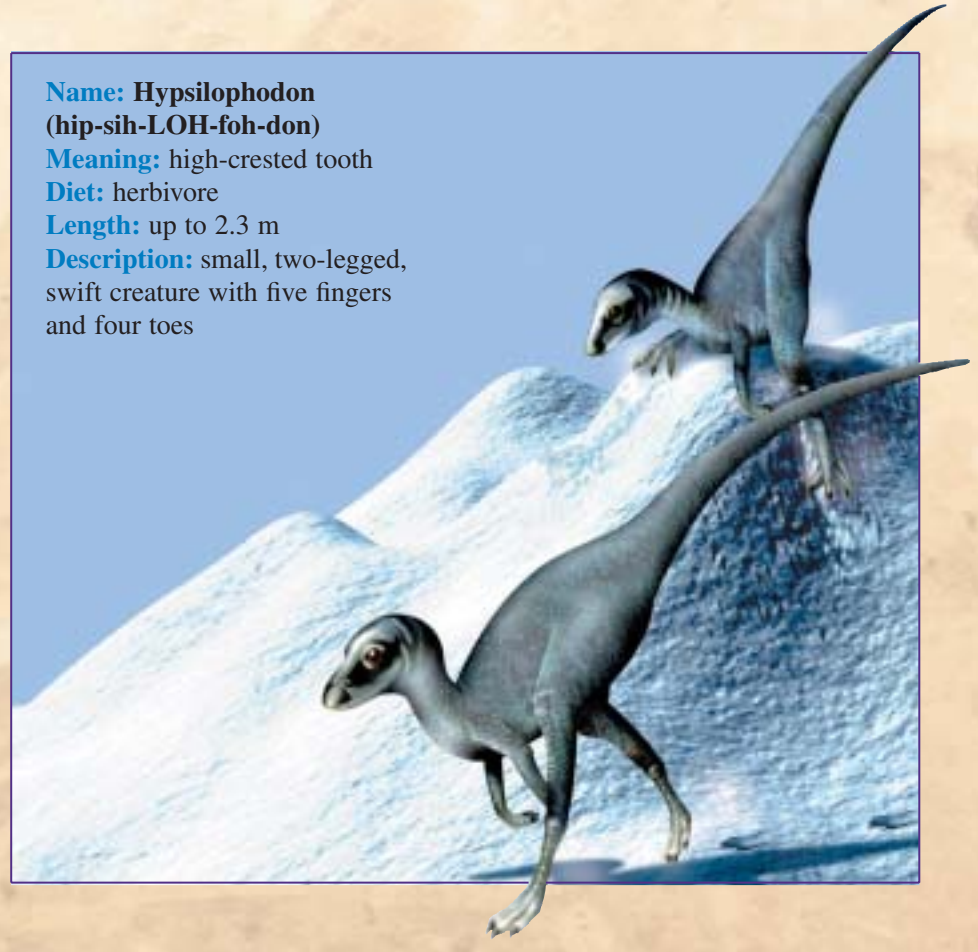
The site on the Flat Rocks platform is accessible for only three to four hours each day, when the tide is low. Removing the sand brought in at high tide can take a while, even though new mesh and tarpaulins have been installed to stop sand filling the diggings. With time so precious, a large team of volunteer diggers is expected at the site tomorrow.

They will work with a small group of **palaeontologists**, who will ensure that the fragile **fossil** is not destroyed by an overenthusiastic digger. The fossils at Flat Rocks are embedded in sandstone and mudstone. These rocks are quite hard, so tools like rock saws, hammers and chisels have to be used to locate fossils without damaging them.

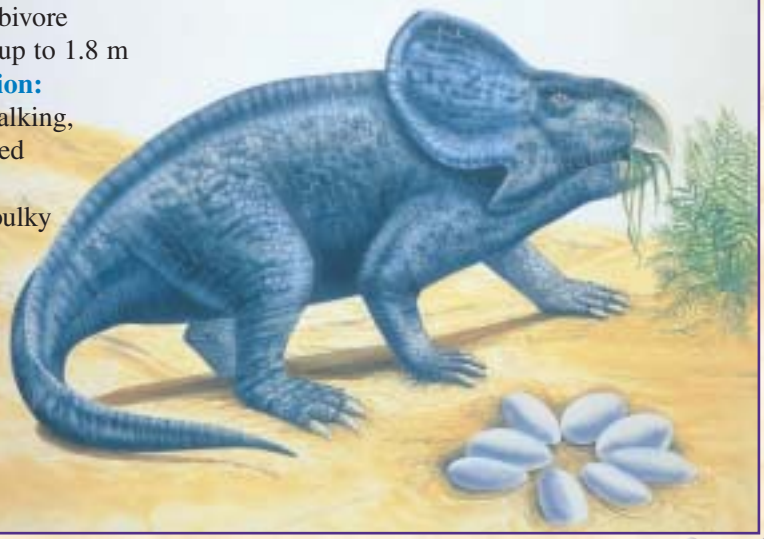
Looking back 115 million years

If palaeontologists are right about the jawbone uncovered yesterday, the mammal it belonged to lived in a world of dinosaurs. About 115 million years ago, a wide river valley was all that separated Australia from Antarctica. Hypsilophodons (shown below) ran between the trees, protoceratopsids grazed on the plains and labyrinthodonts lay in the water awaiting their prey. The distant roar of an allosaurus sent fear through to the core of every other dinosaur. Clouds of ash churned out of volcanoes that formed in the valley between Australia and Antarctica. The bones of dead dinosaurs lay scattered along the muddy banks of the rivers, a perfect place for preserving fossils.

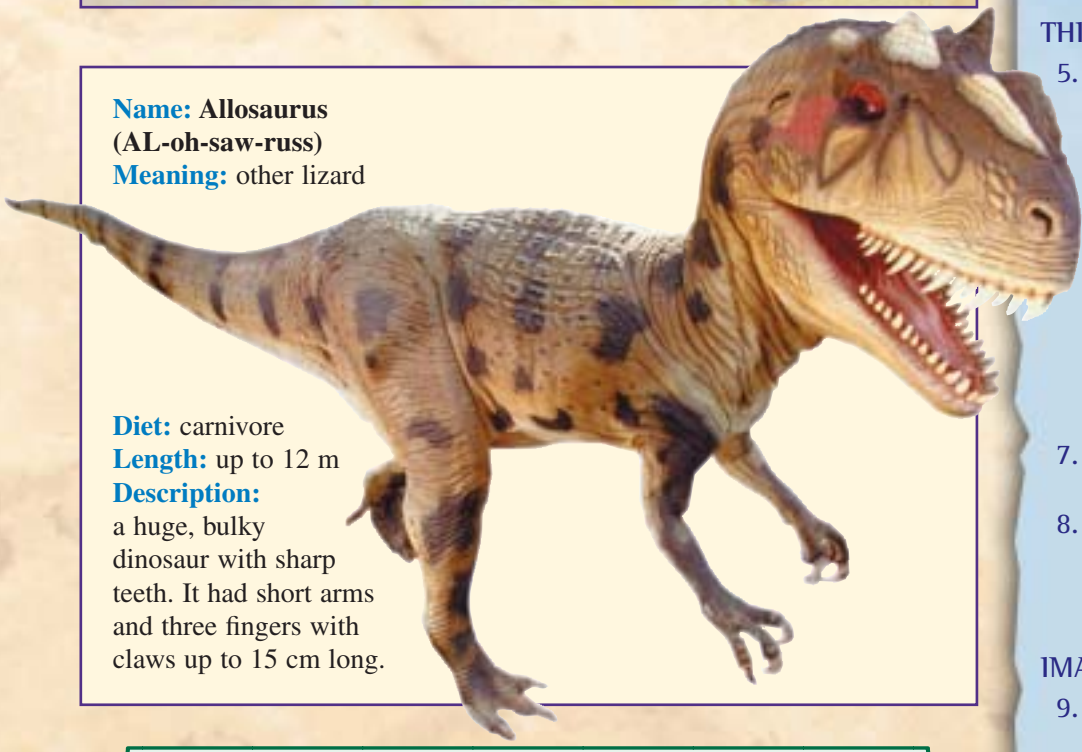
Name: **Hypsilophodon (hip-sih-LOH-foh-don)**
Meaning: high-crested tooth
Diet: herbivore
Length: up to 2.3 m
Description: small, two-legged, swift creature with five fingers and four toes



Name: **Protoceratopsid (pro-toe-ser-ah-top-sid)**
Meaning: first horned face
Diet: herbivore
Length: up to 1.8 m
Description:
a slow-walking, four-legged dinosaur. It had a bulky body and a small frill on its head.



Name: **Allosaurus (AL-oh-saw-russ)**
Meaning: other lizard



Diet: carnivore
Length: up to 12 m
Description:
a huge, bulky dinosaur with sharp teeth. It had short arms and three fingers with claws up to 15 cm long.

Accommodation

Be the first on site to dig for bones — stay at
The Caves Hotel!

- Get away from the city lights.
- Sleep in the sands that dinosaurs once roamed.
- Marvel at the spectacular view of the sea.
- Sleep under millions of stars — not just five.

The Caves Hotel is located right on the shore at Flat Rocks, a new dinosaur dig site.

BYO rubber dinghy or waterproof sleepwear.
Management will not be held responsible for customers who disappear in the rising tide.

Activities

REMEMBER

1. Why was the discovery of the jawbone so important?
2. Why is time precious when looking for fossils at Flat Rocks?
3. List two plant-eating animals whose fossils can be found in south-eastern Victoria.
4. Which dinosaur, found in Victoria, had a frill on its head?

THINK

5. The dinosaurs at Flat Rocks are found in mudstone and sandstone. What do these rocks tell us about the environment they formed in?
6. How might scientists have decided that the fossil layer of rock near Inverloch is about 115 million years old?
7. What would the claws of an allosaurus be used for?
8. Describe two ways in which fossils at Flat Rocks could become damaged or lost.

IMAGINE

9. Imagine you are a hypsilophodon. Write a short story that describes the environment you live in, what you eat and some of the dangers you face each day.

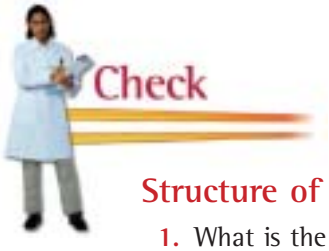
✓ checklist

I can:

- ☐ appreciate the importance of the jawbone discovery
- ☐ understand how a palaeontologist works
- ☐ imagine what prehistoric Australia was like.

Check and challenge

The changing Earth

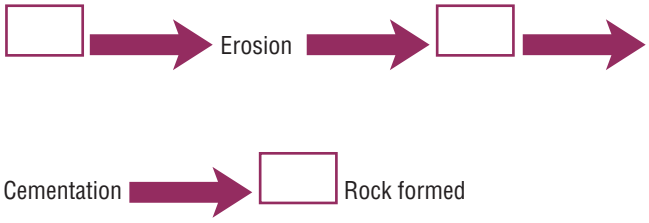


Structure of the Earth

1. What is the name of the layer in the Earth where rocks are formed?
2. From where in the Earth does magma come?
3. Which layer in the Earth is the thickest?
4. Scientists believe that the Earth's crust is made up of plates that are constantly moving. Describe how the movement of plates can lead to the formation of mountains.
5. On which of the Earth's plates is Australia situated?

Sedimentary, igneous and metamorphic rocks

6. How are igneous rocks formed?
7. Fill in the blanks on this flow chart.

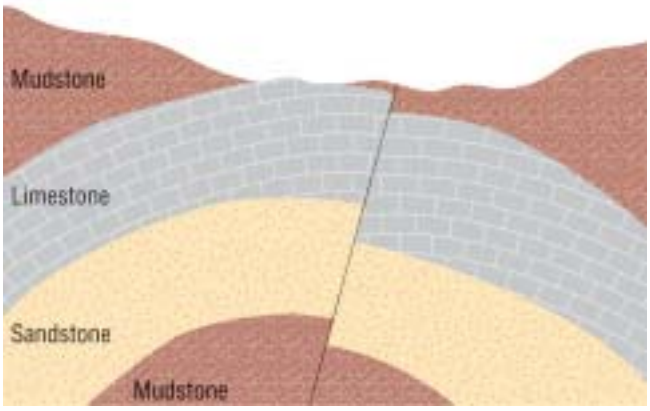


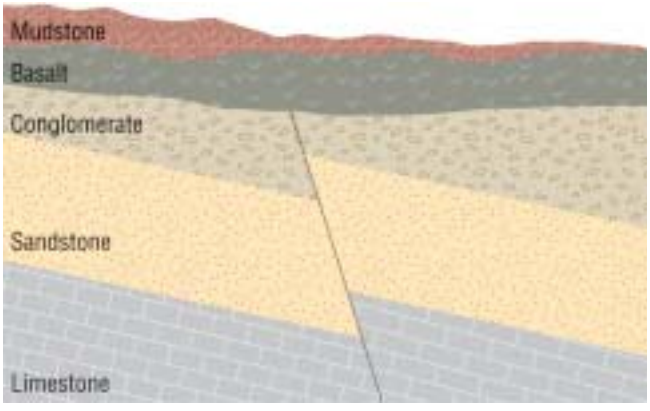
8. Why is limestone sometimes referred to as a biological sedimentary rock and at other times as a chemical sedimentary rock?
9. What two conditions can change an igneous or sedimentary rock into a metamorphic rock?
10. The parent rock of marble is:
    A sandstone?
    B mudstone?
    C limestone?
    D basalt?
11. Divide the following list into two columns. Label the columns 'extrusive igneous rocks' and 'intrusive igneous rocks'.
    Scoria
    Basalt
    Granite
    Obsidian
    Pumice
12. Decide if the word in *italic* makes the statement true or false. If the statement is false, replace the word in *italic* to make it true.
    (a) Extrusive rocks form *above* the Earth's surface.
    (b) The *faster* the cooling time, the larger the crystal size in igneous rocks.
    (c) *Weathering* is the process of moving broken-down rock or soil from one place to another.
13. Link the rock type with the environment it most likely formed in.

Mudstone (fine grains)	Very fast flowing river
Sandstone (sand-sized grains)	Swamp
Conglomerate (mixture of very large and small grains)	Desert

Dating rocks

14. Why can fossils be used to estimate the age of rocks around the world?
15. What information does radiometric dating give us that clues from fossils cannot?
16. Write a description of the order of events that occurred to form these rock samples.



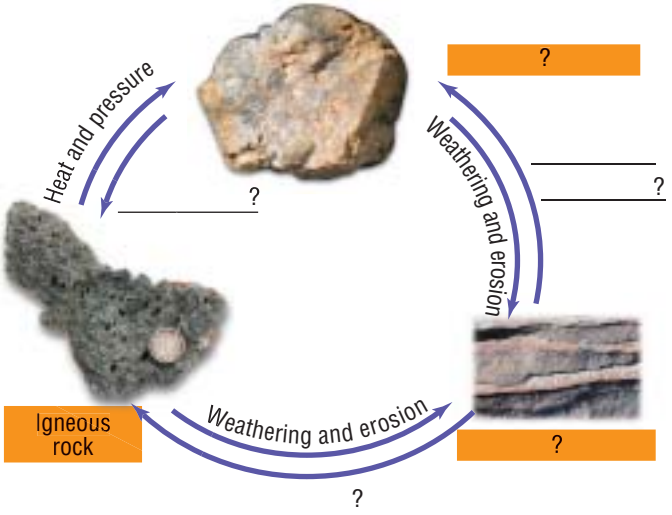


Volcanoes

17. Describe the difference between extinct, dormant and active volcanoes.
18. Describe how lava is different from magma.
19. When Mt Pinatubo erupted in 1991, it caused a drop in the Earth's temperature. Explain, in simple terms, how this could have happened.
20. Often, when lava cools, the rocks formed near the edge of the lava flow have different-sized crystals than the rocks formed in the middle of the flow.
    (a) Where in the flow would the rocks with the smallest crystals form?
    (b) Suggest a reason for this.

The rock cycle

21. Fill in the blanks on the following diagram.



Wearing away

22. Draw diagrams to explain the difference between weathering and erosion.
23. What most likely caused the rocks on this building to wear away?

24. Deforestation is a worldwide problem.
    (a) How does deforestation speed up the process of weathering and erosion?
    (b) List some measures that are being taken to reduce erosion and improve the environment.

Which rock?

1. Identify the rocks in the photos as igneous, sedimentary or metamorphic. If you are not certain, write down the types of rock they are most likely to be. In each case, state a reason for your answer.

(a)

(c)

(b)

(d)

2. When a weak acid is dropped onto limestone, the limestone fizzes. Would the same thing happen if the acid were dropped onto marble? Explain your answer.

Spanning time

3. Fossils can give us clues about the age of rocks. They can also tell us about the plants and animals that lived and the conditions they lived in. Suggest what type of information could be obtained from studying the following fossils:
    (a) fossilised teeth
    (b) footprints
    (c) impressions of leaves left in mud that formed a rock
    (d) fossilised droppings (called coprolites).
4. The Grand Canyon in Arizona has been forming over millions of years. It once formed the lower slopes of a mountain range that was twice as high as Mount Everest. Today, it is the largest gorge on Earth.
    (a) Find out how deep the gorge is.
    (b) Describe how the climate at the top differs from the climate at the bottom of the gorge.
    (c) By observing the picture below, make a list of the processes involved in creating this canyon. What types of weathering and erosion have occurred? What types of rock are found here?

Summary of key terms

abrasive: a property of a material or substance that easily scratches another

absolute dating: finding out the actual age of a rock

acid rain: rainwater, snow or fog that contains dissolved chemicals that make it acidic. Acid rain can cause rock to weather faster than pure rain.

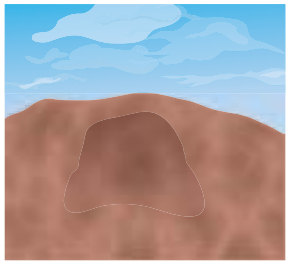
active: describes a volcano that has recently erupted and is likely to erupt again in the near future

air: the mixture of gases that surround the Earth

band: flat strip of differently coloured minerals that is sometimes found in metamorphic rocks formed under high pressure

basalt: a dark igneous rock with small crystals. It sometimes contains holes that once contained volcanic gases.

batholith: intrusive rock mass that measures more than 100 kilometres across



chemical weathering: the breakdown of rocks into small parts by the chemicals in water, air and other substances

coal: a sedimentary rock formed from dead plants and animals that became buried before rotting completely

coprolite: fossilised dropping

crust: the outermost layer of the Earth. The surface of the Earth is the very top part of the crust.

deposit: (*noun*) the sediment left as a result of erosion; (*verb*) to release and put down

dormant: not currently active, but not regarded as extinct. Dormant volcanoes show no signs of erupting in the near future.

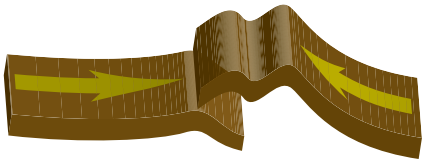
erosion: the process of moving weathered rock or soil from one place to another

extinct: no longer active. Extinct volcanoes have not erupted for thousands of years and show no signs of future eruption.

extrusive rock: igneous rock that forms when lava cools above the Earth's surface

fault: a break in a rock structure. The rock on either side of the break can move.

folding: the buckling of rocks. It is caused when rocks are under pressure from both sides



fossil: evidence of life in the past

granite: a hard igneous rock with crystals large enough to see. It forms below the Earth's surface.

groyne: a jetty built into the sea to prevent the erosion of the beach

hornfels: a dark metamorphic rock formed under extreme heat

igneous rock: rock that forms when magma or lava cools to a solid

imprint: the outline of an object, left as a record in a rock

intrusive rock: igneous rock that forms when magma cools below the Earth's surface

lava: a mixture of molten rock that loses gas as it erupts from a volcano

limestone: a sedimentary rock formed from the remains of sea organisms

magma: a mixture of molten rock and gases that comes from the top of the mantle

mantle: the thick layer in the Earth, below the crust. Most of the mantle is solid rock, although the upper part is molten rock called magma.

metamorphic rock: rock that forms from other rocks that are under great heat or pressure (or both)

minerals: the substances that make up rocks. Each mineral has its own chemical make-up.

molten: the liquid state of a substance that is usually a solid at room temperature

organism: living thing

palaeontologist: a scientist who studies fossils

parent rock: the original form of a rock before it was changed by heat, pressure, weathering, erosion or all of these

plates: the large pieces that make up the Earth's crust

pressure: a push or squeeze on an object

pumice: a pale rock that forms when frothy lava cools in the air. Pumice is very light and often floats on water.

radiometric dating: a technique in which radioactive substances are used to calculate the age of rocks or dead plants and animals

relative dating: comparing the age of rocks without actually knowing their age in years

rock cycle: a model that describes how a rock can change from one type to another

rock salt: a sedimentary deposit formed when a salt lake or seabed dries up

scoria: a dark igneous rock formed from gassy lava that cools quickly

sediment: substance deposited on land or in rivers, lakes or oceans

sedimentary rock: rock formed from sediments deposited by water, wind or ice. The sediments are cemented together in layers, under pressure.

volcanic bomb: large rock fragment that is blown out of erupting volcanoes

weathering: the process of breaking down rocks by elements of the weather

CHAPTER 5

Forensic science

Mysterious Mary

In 1872, the *Mary Celeste* left New York. Captain Briggs took along his wife, young daughter and a small crew to deliver a cargo of barrels to Italy.

A few weeks later, the crew of another ship found the *Mary Celeste* drifting. When the second crew boarded the *Mary Celeste*, they found nobody on board. The captain, his wife, daughter and crew were all missing without a trace! An inspection of the ship found no evidence of a struggle with pirates.

Any evidence on the ship was destroyed when it was deliberately sunk in the 1880s.

A modern forensic scientist looks for clues.

1 What could have happened to the passengers and crew of the *Mary Celeste*?
2 What kind of evidence would scientists or investigators look for if this mystery had happened more recently?
3 How can evidence be used to find out what happened in a mysterious situation such as the one shown in the photo above?
4 What kinds of scientist are involved in solving mysteries and crimes?

You will discover

How dentists and insect experts help to solve crimes

That it's not just fingers that leave behind prints

How criminals can be identified

If an identical twin can get away with a crime more easily

Forensic science

Forensic science assists judges and juries to make decisions about crimes. The word 'forensic' comes from the word '**forum**' — the buildings that housed the ancient Roman courts of law. Forensic scientists solve crimes and investigate accidents. Many branches of science are involved in forensic science. Forensic scientists can be chemists, biologists, medical doctors, dentists, physicists, geologists, or any other type of scientist.

A forensic scientist might work:
- at the scene of the crime or accident
- in a laboratory
- in an office
- or at a combination of these places.

They sometimes **testify** in court.

Insects time death

Entomologists are scientists who study insects. So what does a forensic entomologist do? The answer lies in flies — or more precisely, their newly hatched eggs known as **maggots**.

Flies can lay eggs in a dead body within an hour of death, when it is still warm. The eggs hatch very quickly — usually in less than a day. When fully grown, each maggot forms a case around itself to become a **pupa**. It transforms into an adult fly which breaks out of the case.

It usually takes between one and two weeks for the egg to develop into a pupa. Therefore, a forensic entomologist can use the age of the oldest maggots or pupae in a dead body to work out when the eggs were laid. A good estimate of the time of death can then be made.

At the scene of a crime

A forensic scientist's main task at the crime scene is to gather **evidence**. Any material left at the crime scene could be a clue.

When a crime is discovered, the area is sealed off. Whenever possible, nothing is moved. Close observation is very important. Any evidence that cannot be taken away must be photographed and every detail carefully recorded in writing. Diagrams and measurements of the scene are made. **Statements** are taken from **witnesses**. This detailed recording at the crime scene is called **documenting**.

During most criminal acts there is an exchange of material between the victim, the criminal and the site. After the site is documented, evidence is collected and clearly labelled with details of where it was found. Gloves must be worn so that the evidence is not **contaminated**. Items that might be collected from a crime scene include soil, blood, **fibres** from clothing, hair, plant and animal material, flesh, fingernails and anything else that looks out of place. All **physical evidence** is sent to a laboratory to be **analysed**.

Forensic scientists examine the physical evidence in the laboratory. The examination

Go to worksheet 21

involves scientific skills like sorting (classifying), observing, comparing, analysing and documenting.

When a crime or accident results in death, the body is taken to a **mortuary**. A forensic **pathologist** examines the body in an **autopsy**. The purpose of an autopsy, which is also called a post-mortem, is to find out the cause of death.

Using the evidence

After the physical evidence is examined, the results are given to detectives in charge of the investigation. These results are considered together with the other evidence, such as photographs, diagrams, measurements and witnesses' statements. The detectives, often working with the forensic scientists, make **inferences** based on all of the evidence. An inference is a proposed explanation and is not always correct. For example, if an empty wallet is found on a murder victim, at least three different inferences could be made:

- The victim was robbed.
- The victim had an empty wallet before being murdered.
- The wallet was emptied to make the crime look like a robbery gone wrong.

It is possible that none of these inferences are correct. To prove whether an inference is correct, it needs to be tested against other evidence. For example, if the victim had been seen withdrawing money at an automatic bank teller five minutes before the murder, it is more likely that the first inference is correct. But this extra evidence does not prove that the victim was robbed.

Activities

REMEMBER

1. What is forensic science?
2. How can a scientist who studies insects help a murder investigation?
3. What is physical evidence?

THINK

4. What is documenting and why is it important in the investigation of a crime?
5. Why must the scene of a crime be sealed off immediately after its discovery?
6. How is an inference different from a wild guess?
7. What is the difference between evidence and an inference?
8. What makes an inference different from proof?

LIST

9. Make a list of the items that a crime investigator would need at the crime scene to be able to gather and document evidence. Draw each of these and write about what they would be used for.

IMAGINE

10. Picture the scene of an accident where two cars have collided at a busy intersection. One car drives off immediately after the accident.
    (a) Make a list of the evidence that you think would be collected from the crime scene.
    (b) Which evidence in your list is physical evidence?
11. Forensic scientists have a wide range of scientific backgrounds. They usually have special knowledge and skills in one area of science. These areas include pathology, dentistry, psychiatry, entomology, toxicology, anthropology and serology. Select one of these fields of science that you don't know about and research it. Imagine you are a forensic scientist working in this field. Write about what you do and how you do it.

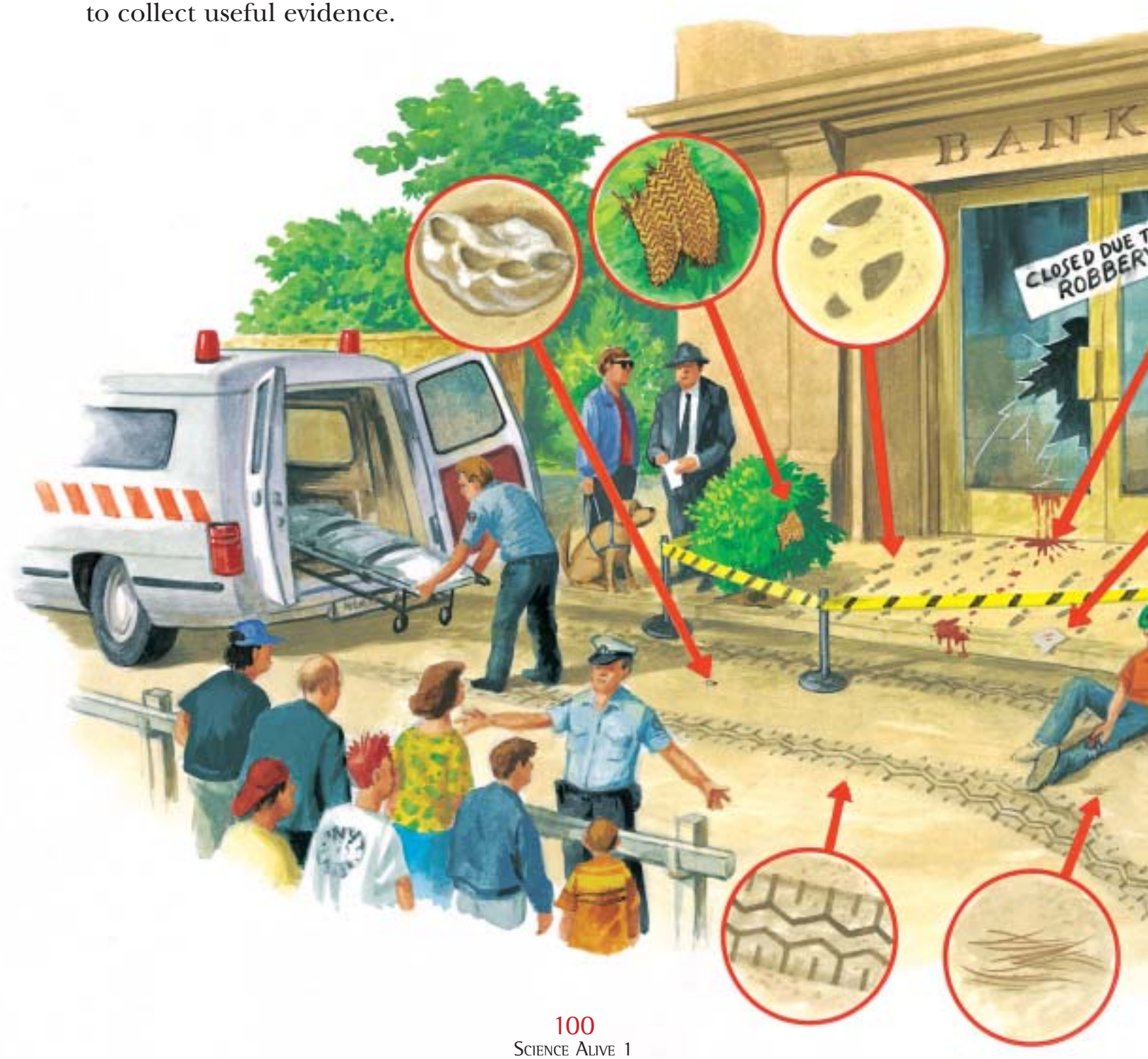
✓ checklist

I can:

- ☐ state what forensic science is
- ☐ describe how evidence is gathered at a crime scene
- ☐ define the terms evidence and inference.

Be the detective

Detectives must be thorough in their search for **evidence**. Any item found at the scene of a crime or accident could provide the police with a clue about who was there and what happened. The area is sealed off while evidence is collected so that the public cannot enter. A crime or accident scene can be **contaminated** by people removing or adding items to the scene. Once the scene is contaminated, it is more difficult to collect useful evidence.



Go to worksheet 22

MI VISUAL/SPATIAL

Comparison microscopy of the paint from the tree and paint from the victim's bike.

Activities

THINK

1. Why are the police paying so much attention to a bicycle accident that involves only minor injuries?
2. Why are the people behind the barrier being kept away from the scene?
3. After investigating the crime scene, police put out an alert to panel beaters in the area. What might the alert ask panel beaters to watch out for?

IMAGINE

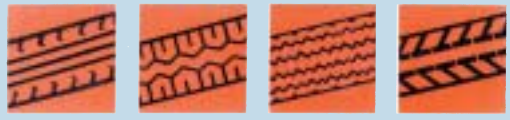
4. Imagine you are a detective at this crime scene. Write a step-by-step report detailing what you believe happened here.

BRAINSTORM

5. Write a list of information that the blind person might be able to provide.

COMPARE

6. Look at the different lip prints on page 108. What type of lip print was found on the tissue at the crime scene?
7. Which of these tyres may have left the tracks at the crime scene?



8. Which of these shoes may have left the footprints at the crime scene?

CREATE

9. Imagine you are a detective at this crime scene. Put together a multimedia presentation that includes:
   (a) a list of the people you will interview
   (b) a list of the questions you will ask each of the people you interview
   (c) a list of evidence that you can collect and physically take back to the laboratory
   (d) the items that you will need to photograph or take casts of.

✓ checklist

I can:

- ☐ understand why it is important to seal off a crime scene
- ☐ list the different types of evidence that can be gathered at a crime scene.

Eyewitness

Police and detectives use **statements** from eyewitnesses to piece together information related to a crime. **Witnesses** can be called on to tell courts and juries about what they have seen or heard. They can also help police and detectives to create computer reconstructions of criminals' faces.

Memory

Different people see things differently. Witnesses can confuse colours, vehicles, clothing and behaviour. This is because of the way we store information in our memories. Information is first stored in our short-term memory. This part of our memory holds information for only about 20–30 seconds. To be able to remember details for longer, we need to think about the information or use it. Writing down the information or telling it to someone else may help store it in our long-term memory. For this reason, an **eyewitness account** given in court needs to be backed up with other **evidence** — especially if the event occurred very quickly.

Being observant

You will need:

pen
paper
pencil case.

Part A The art room

- From memory, describe the art room in your school in as much detail as possible. Write for no more than five minutes.
- Compare your observations with those of the person next to you.
- Share your results with the class.

1. List any differences in your observations.
2. What sorts of things did everyone recall? Give reasons why these things were remembered.

Part B The pencil case

- Work in pairs. Tip out the contents of a pencil case.
- Look at the contents for one minute then cover them over.
- Write down all of the contents that you remember.
- Check how accurate you were.

Part C Another pencil case

- Repeat part B with the other person's pencil case. This time write a list of what is in the pencil case while still looking at the contents. Make sure you observe and list the items for only one minute.
- Cover the pencil case items and the written list.
- From memory, rewrite the list of objects in the pencil case.

1. Were you able to list more or fewer items using this second method?
2. The aim of part C was to see if you could remember more items after writing a list of, as well as observing, the items in the pencil case. Was this part of the experiment a fair test? Explain your answer.

Go to worksheet 23

Making faces

In the past, artists created hand-drawn impressions of a suspect. In more recent years, an identification kit has often been used to help witnesses remember the facial features of a criminal.

Photofit identification kits work by building an image of a **suspect**'s face in stages. The witness needs to separately select eyes, mouth, hair, nose and so on.

Today, computers create an image that looks like a photograph. A witness selects hairstyle, chin shape, eye colour and other facial features. Other items, including hats, facial hair, wrinkles, freckles and glasses, can be added. The computer program used for this is called the 'Facial Automated Composition Editing' or F.A.C.E. system.

Investigators present images of people suspected of being involved in the 2002 Bali bombings. The images were created using the F.A.C.E. system. F.A.C.E. produces coloured, realistic images.

Activities

REMEMBER

1. For how long are we likely to remember information stored in our short-term memory?
2. How can we remember things for longer than our short-term memory usually allows?

THINK

3. What was each member of your family wearing when you last saw them?
4. Why might eyewitness accounts from different people witnessing the same event disagree?
5. List some of the advantages of the F.A.C.E. program over Photofit and hand-drawn impressions.

BRAINSTORM

6. When police ask witnesses to recall what a person at a crime scene looked like, they use certain basic questions. Create a list of basic questions that you think a police officer would ask witnesses at the scene of a road accident between two cars at an intersection.

INVESTIGATE

7. What is a mug shot?

DEBATE

8. Provide arguments for and against requiring all Australian citizens to have their photographs on record when they turn 18, and updated every few years.

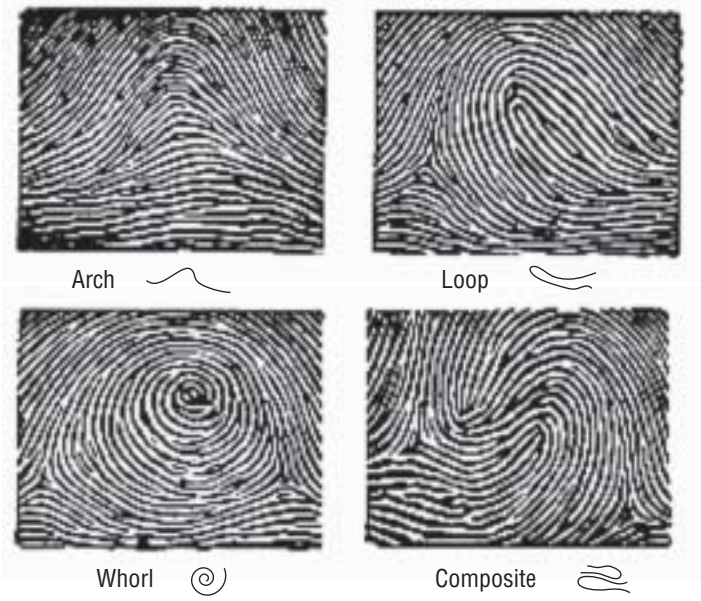
checklist

I can:

- ☐ recognise that different people remember things differently
- ☐ explain how police, with help from witnesses, put together pictures of suspects.

Fingerprints

Movies often show criminals wearing gloves to stop their fingerprints giving them away. Files of fingerprints were first set up in 1901. Since then, fingerprinting has become one of the techniques most widely used when investigating a crime.



Fingerprints are sorted into four main categories: arch, loop, whorl, and composite. Composite patterns are a combination of the other three patterns.

Patterned skin

The patterns of **friction ridges** that make up fingerprints are different for everybody — even identical twins. In fact, there are unique patterns on most areas of our skin. Fingerprints are generally used to identify people because we use our fingers for touching and holding things more than other parts of our body.

Fingerprints are formed before a baby leaves its mother's womb and they don't change as a person ages. After a person dies, their fingerprints remain until their body **decomposes**. Formed deep under the skin's surface, they will grow back even if scars **disfigure** the skin. Only damage to the deep layers, where the fingerprints form, will permanently change a person's fingerprints.

When are fingerprints useful as evidence?

Fingerprint **evidence** is very useful if there is a close match between prints found at a crime scene and those taken from a **suspect**. Accurate matching depends on a fingerprint expert identifying small details, called **minutiae.** In some countries, ten minutiae must be matched before a fingerprint can be used as evidence. In other countries, only eight are required. This isn't easy, as fingerprints found at crime scenes are often smudged.

Until recently, fingerprint experts could take weeks, or even months, to match the fingerprints left at a crime scene with those of a suspect. But computers have made the task much faster. The Australian National Automated Fingerprint Identification System (NAFIS) can compare 400 000 prints per second. The Australian system is one of the best in the world because it has a record of all the fingerprints taken across the country. In many other countries, each state or area has its own computer register; so print matching can take longer.

Leaving our mark

We leave behind fingerprints because our skin constantly releases sweat and body oils. Forensic scientists use fine, black powder to dust for prints because it sticks to the oil and sweat left behind by fingers. Prints on paper and cloth cannot be dusted, so chemicals are used to reveal these prints. Lasers and UV light can make some fingerprints easier to see.

Ancient fingerprints

Fingerprint samples, if left undisturbed, can provide clues even after a long period of time. Clear fingerprints have been taken inside 4000-year-old Egyptian pyramids and from the fingers of 2000-year-old mummies.

Go to worksheet 24

MI BODILY/KINAESTHETIC

Making fingerprints

You will need:
inkpad
table like the one below
soap and towel
magnifying glass.

Right hand

Thumb	Index	Middle	Ring	Little

Left hand

Little	Ring	Middle	Index	Thumb

- Place the inkpad and the piece of paper with your fingerprint table on the edge of a bench.
- Gently roll your left thumb over the inkpad. Without pressing too hard, roll your thumb over the space on your left-hand table marked thumb.
- Repeat this for all fingers on both hands.
- Wash your hands thoroughly with soap.
- Use the magnifying glass to examine your prints.
- Draw one of your prints and label the pattern types that you have in your prints.

1. What patterns are you able to identify on your fingerprints?
2. Are any of your prints the same?
3. Compare your prints to others in your class. What do you discover?

Activities

REMEMBER

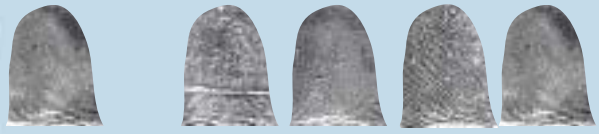
1. Why is fingerprinting a reliable way to identify a person?
2. Why do we leave fingerprints behind when we touch something?
3. How often do a person's fingerprints change in their lifetime?

THINK

4. A man in England was found guilty of burglary when his ear prints were discovered on the front doors of several homes that had been burgled. Why could the ear prints be used as proof of his identity?
5. In movies, criminals often wear gloves to avoid their fingerprints giving them away. How could gloves discarded from a crime scene be used to identify a criminal?
6. Imagine that a stolen car has been found. What parts of the car would you dust for prints?

COMPARE

7. Police fingerprinted four suspects. By comparing the fingerprints of the suspects with fingerprints found at the crime scene, decide which suspect was at the scene.



Fingerprint from scene | Suspect 1 | Suspect 2 | Suspect 3 | Suspect 4

INVESTIGATE

8. Find out why the ridges that form our fingerprints are called 'friction ridges'.
9. Find out how many minutiae need to be matched up before a fingerprint can be used as evidence in Australian courts.
10. Find out what DNA fingerprinting is. How effective is it in identifying people?

✓ checklist

I can:
- ☐ name and sketch the four main categories of fingerprints
- ☐ explain how fingerprints are left behind
- ☐ describe, in simple terms, methods used to reveal fingerprints.

Let's magnify

At a crime scene, forensic scientists might use magnifying glasses. Back in the laboratory, scientists also use microscopes to analyse evidence.

Monocular light microscope

A monocular light microscope passes light through the specimen to allow scientists to see details that are not visible through a magnifying glass. Only very thin objects, like microscopic organisms in water or blood, can be viewed. Monocular means one eyepiece. A monocular light microscope can magnify objects by up to 1500 times their real size.

Stereo light microscope

A stereo light microscope has two eyepieces and gives a three-dimensional view of the specimen. It uses visible light that is reflected off the specimen. This means that objects that are too thick to view through a monocular light microscope can be seen through a stereo light microscope.

Red blood cells transport oxygen throughout the body.

This view of human blood comes from an electron microscope. It is magnified to 5270 times its actual size.

Comparison microscope

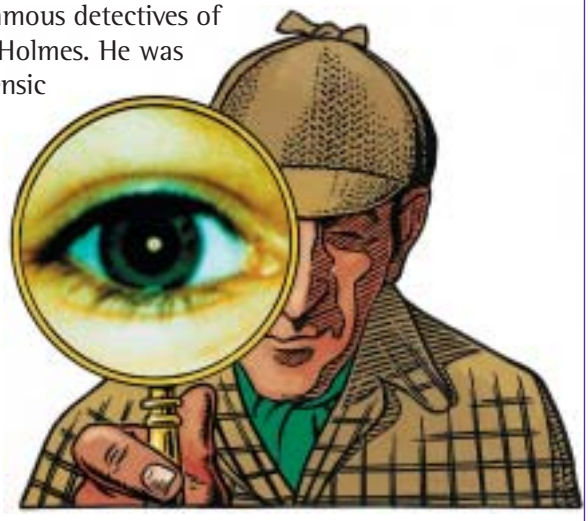
A **comparison microscope** is like two light microscopes in one. It allows two objects to be viewed at the same time. A comparison microscope is very useful for comparing specimens like hair, **fibres** from clothing, and paint fragments. It can also be used to compare the marks left on a bullet with the **cartridge** from which it was fired.

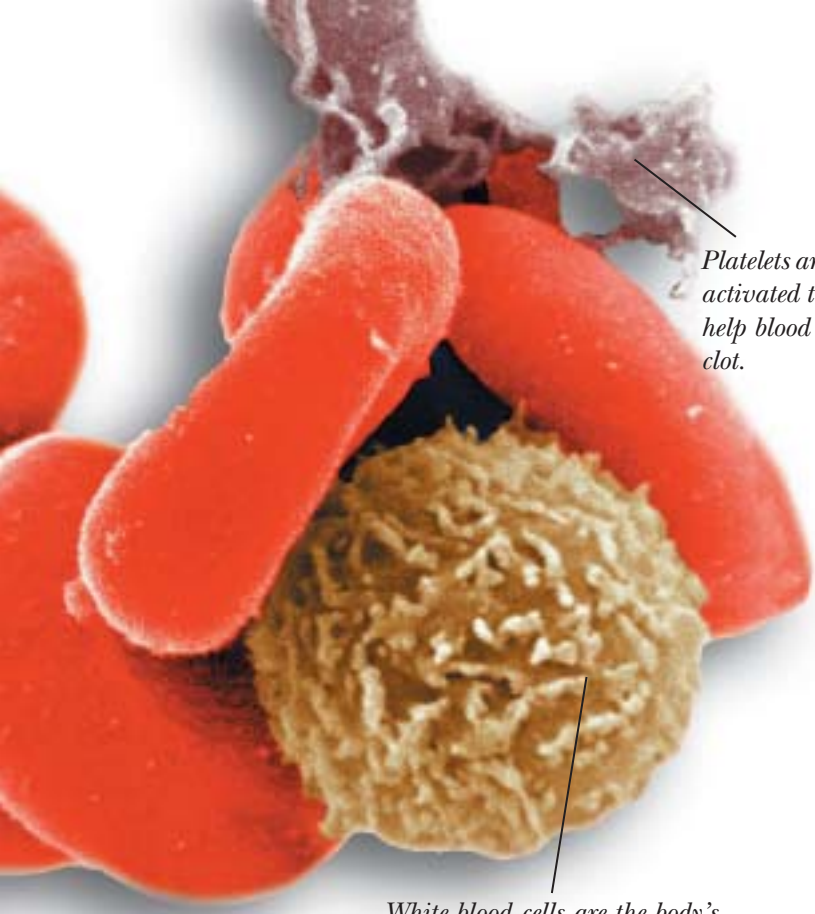
Electron microscope

An electron microscope uses beams of electrons instead of light and can magnify objects by up to one million times their real size. The residue left by a bullet might be examined under an electron microscope. Tiny traces of metals could help identify the type of bullet. Electron microscopes may also be used to examine lung tissue from a body found in water. If the victim drowned in a river, lake or the sea, **diatoms** will be found. If the victim was placed in the water after death, no diatoms will be found in the tissue.

Sherlock Holmes

One of the most famous detectives of all time was Sherlock Holmes. He was the first well-known forensic scientist. Together with his friend Dr Watson he used skills of observation and clever **inferences** to solve many crimes. One of Sherlock's 'tools of the trade' was a simple magnifying glass. Of course, Sherlock didn't really exist — or did he?





Platelets are activated to help blood clot.

White blood cells are the body's defence system against disease.

What's in your pocket?

You will need:
old toothbrush microscope slide
magnifying glass monocular microscope
stereo light microscope.

- Empty out your pockets.
- Use an old toothbrush to remove **fibres** from the seam of a pocket in your clothing.
- Carefully transfer what you have onto a microscope slide.
- View the specimen with a magnifying glass.

1. Draw what you see through the magnifying glass.

- View the slide with fibres under the monocular microscope. You can find information about using a monocular microscope in the Laboratory Toolbox on pages 241–8.

2. Draw what you see through the microscope. Remember to state the **magnification**.
3. Label anything that you can identify on your diagram.

- View the slide under a stereo light microscope.

4. How is your view through the stereo light microscope different from your view through the monocular microscope?

Activities

REMEMBER

1. List two differences between a stereo light microscope and a monocular light microscope.
2. List two differences between a light microscope and an electron microscope.
3. What makes a comparison microscope useful for comparing objects like clothing fibres?
4. List two uses for an electron microscope in forensic science.

THINK

5. What might forensic scientists look for in blood samples?
6. Which type of microscope is most likely to be used to observe:
   (a) a small, solid object (like a door key)?
   (b) the tip of a fly's wing?
   (c) a human hair (to see if it matches a hair from the crime victim)?

IMAGINE

7. Imagine you are a forensic scientist comparing a bullet found at a crime scene with a gun recovered from a suspect.
   (a) How can you get another fired bullet to compare?
   (b) What type of microscope will you use to compare the bullets?
   Look at the two bullet sections below.

   (c) Draw a sketch that highlights the marks on each bullet which suggest they were fired from the same gun.

✓ checklist

I can:
- ☐ distinguish between the different types of microscope
- ☐ relate the way a microscope works to its uses
- ☐ use a monocular light microscope and a stereo light microscope.

Prints of all kinds

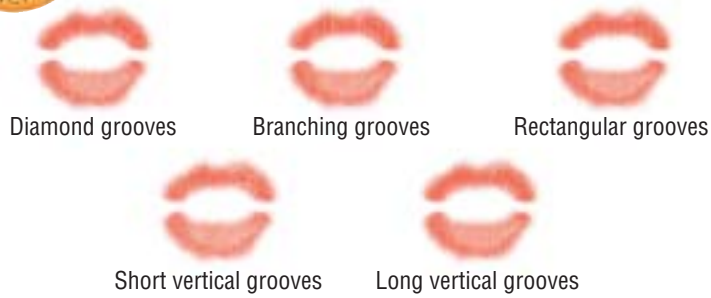
It's not just fingerprints that can be matched up and used as **evidence**. A criminal can be given away by lip prints, shoe prints or the tyre marks their car leaves behind. **Ballistics experts** and dentists are often called on to match up evidence found at the crime scene with a **suspect** or a victim.

Forensic dentists

Each person's teeth are slightly different. As we age, our teeth are made even more different by fillings and other dental work. Dentists keep detailed records of the work they perform on their patients' teeth. They may also have X-rays or **impressions** of a patient's teeth. Dental records can be used to match teeth or bite marks with those of the victim or suspect.

*A badly decayed or burnt body is difficult to identify. Teeth take longer to decay and are often the only part of the body left. Forensic dentists, also called **odontologists**, are able to match teeth recovered from crime scenes with dental records to identify a victim. If the age of the deceased person is unclear, a chemical can be added to the teeth to highlight growth lines. The number of lines provides a clue about the age of the victim.*

Lip prints



Lip prints are as unique as fingerprints and do not change during a person's life. This means that criminals can also be identified by their lip prints. **Cheiloscopy** is the study of lip prints.

You will need:
lipstick or coloured lip gloss
clean white tissues.

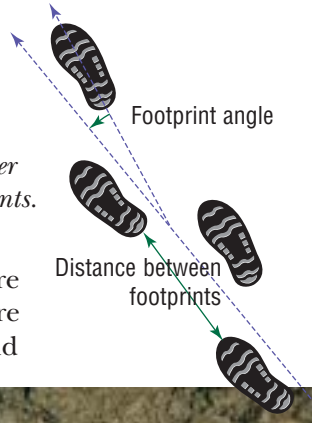
- Put some lipstick or lip gloss on your finger and wipe it onto your lips.
- Fold the tissue in half and press it firmly between your lips.

1. Look at the patterns and identify which category your lip prints fall into.
2. Would all members of your family have the same lip prints as you? Explain your answer.

Leaving the crime scene

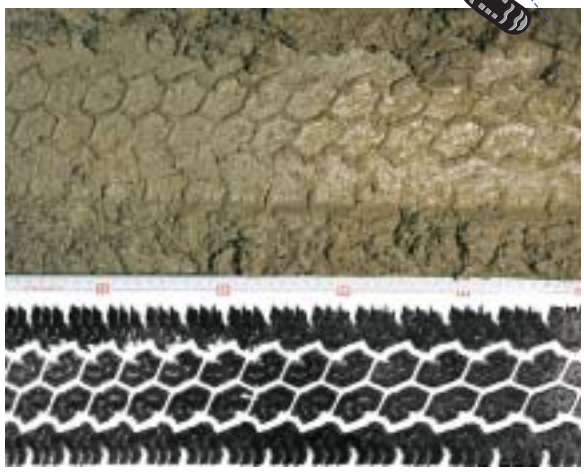
The depth of footprints in soil can tell us about a person's weight. It can even reveal whether a person was limping or carrying something.

This is a print taken from a suspect's shoe. The size, brand and wear pattern in this print were compared with footprints found at a crime scene. The results were presented in court.



The angle of people's footprints can be different. The distance between footprints shows how quickly the person was moving. Running steps are further apart and deeper than walking footprints.

The size of a getaway car and the make of a tyre can be determined by looking at the vehicle's tyre tracks. Evidence of tyre treads can also be found as bruises on hit-and-run victims.



As tyres and shoes wear out, they develop distinctive wearing patterns. Unique patterns from a tyre or shoe can be used to link evidence to a suspect. This tyre track was traced back to this tyre because the wear patterns matched up perfectly.

Making a plaster cast

Footprints and tyre tracks are important evidence. A **cast** of the **impression** can be made and taken back to the crime laboratory for analysis. Casts are more useful than photographs because they record information about the depth of the prints.

You will need:
shoebox or sand tray
shoes
soil or damp sand
prepared plaster of Paris
paintbrush
newspaper.

- Place the newspaper on the bench to catch spills.
- Place the shoebox or sand tray on the newspaper.
- Pack the shoebox or sand tray with damp soil or sand.
- Press the shoe firmly in the sand or soil to make an imprint of the sole of the shoe.
- Without disturbing the footprint, make a 2-cm-high ridge of soil or sand about 5 cm away from the print.
- Pour the plaster into the imprint, covering it completely.

!CAUTION: Dispose of the plaster in the bin, not down the sink.

- Leave the plaster to dry overnight.
- Remove the cast. Dust off soil or sand with the paintbrush.
- Observe the cast and record anything that you think would be useful in an investigation.

1. Why is a cast of a footprint more useful than a photograph?
2. What would happen if you tipped the plaster down the sink?

Activities

REMEMBER

1. What types of print from a crime scene can be collected and used as evidence?
2. Why can teeth be used to identify a burnt or decayed body?
3. What information can the depth of a footprint provide?

THINK

4. Where would you look for lip prints at a crime scene?
5. Would lip prints be as easily available as fingerprints? Explain your answer.
6. How could you make a copy of a lip print at a crime scene?

IMAGINE

7. Imagine you are the forensic scientist investigating tyre tracks found near a crime scene.
   (a) How will you make a copy of the tyre track?
   (b) What evidence will you look for on the tyre track?

INVESTIGATE

8. Design a way to check which category of lip print is more common in your class. Conduct your investigation. Record your results in a spreadsheet and display your data in a graph.

checklist

I can:
- ☐ list four types of print that could be left at a crime scene
- ☐ explain how prints can be used in investigations
- ☐ explain how and why casts of imprints are made.

Paper evidence

Even a handwritten note on a scrap of paper can be useful in solving a crime. It could be a ransom note, a signature, an address or a phone number. The handwriting can be **analysed** by looking at its size, spacing, slope, and even how hard the writer pressed on the paper. Other clues provided by a handwritten note include the type of paper and the type of pen or pencil used to write the note.

Paper chase

Forensic scientists can analyse paper to check for quality and the chemicals used to whiten or colour the paper.

Torn paper can be used as direct **evidence** to link a piece of paper to a **suspect**. No two pieces of paper tear in exactly the same way. A **comparison microscope** can be used to match torn paper found at a crime scene with a piece of paper or a note pad from which it was torn.

Handwriting analysis

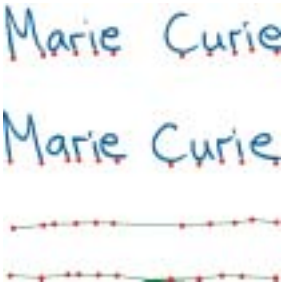
You will need:

pen or sharp pencil — sheet of paper
tracing paper — ruler.

- Sign your name twice on a sheet of paper.
- Place the tracing paper over the top of your signature.
- Make a small mark on the tracing paper at the highest point on each letter.
- Use the ruler to join each of the marks to make a zigzag pattern.

1. Compare the zigzags for each signature. How similar are they?

- Remove the tracing paper and place another sheet of tracing paper over the top of the signatures.
- Mark the lowest point on each letter.
- Use a ruler to join the marks.



2. Compare the patterns for each signature. How similar are they?

- Remove the tracing paper and place another sheet of tracing paper over the signatures.
- Make a small mark at the beginning and end of each letter in the signature.

- Use the ruler to join the mark at the end of a letter with the mark at the beginning of the next letter. The ruled lines show the length of the spaces between the letters.

3. Compare the spaces between letters in the two signatures. How similar are they?

- Remove the tracing paper and add another sheet.
- Use the ruler to make a slash through each letter of the signature on the tracing paper so that each slash has the same slant as each letter.

4. Compare the series of slanted lines for both signatures. How similar are they?
5. Is a person's signature exactly the same each time they sign it? Explain your answer.

Ink clues

Ink is made up of a number of different colours, called **pigments**. For example, black ink in a pen may be made up of blue, red and yellow. Black ink in a different brand of pen may contain a different combination of colours. To find out which pigments have been used to make the ink, scientists separate the different colours in the ink.

Chromatography can be used to separate small amounts of substances that have been mixed together, like the colours in ink. Read more about chromatography on pages 38–9.

Investigating paper and ink

You will need:

2 different types of pen of the same colour (e.g. felt and ball point)
several types of paper (copy paper, tissues, glossy paper, writing pad paper)
filter paper
eyedropper
magnifying glass
methylated spirits.

- Look closely at each piece of paper through the magnifying glass. Look for **fibres** in the paper. You may need to tear a small part off the paper to see the fibres.

1. In which types of paper are the fibres closest together?

- Write your name on each of the pieces of paper, with both pens.
- Using a magnifying glass, look closely at the writing. Notice how the ink spreads through the different paper fibres.

2. Which of the inks spread furthest through the paper fibres?
3. (a) Which type of paper allowed the ink to spread the most?
   (b) How does your answer to 3(a) relate to your answer to question 1?

- Draw a small spot in the middle of a piece of filter paper with one of the pens.
- Draw another spot on a separate piece of filter paper with the other pen.
- Add drops of methylated spirits, one at a time, to the spots. Wait for the methylated spirits to spread a little before adding the next drop.

4. What colour pigments are in each of the pens?
5. Suggest a reason why filter paper is used for this experiment. *Hint:* How easily does ink spread through filter paper?

Activities

REMEMBER

1. What are four ways that handwriting can be analysed?
2. What is chromatography?

THINK

3. Why can't an untorn piece of paper, found at a crime scene, be linked directly to the same paper found at the home of a suspect?
4. A forensic scientist compared handwriting in two signatures in four separate ways. Each way showed that the signatures were exactly alike. But the scientist came to the conclusion that the signatures were not from the same person. Give a possible reason for the scientist's conclusion.
5. List three tests that could be used to find out if these two pieces of paper (shown below) were once part of the same handwritten note.

This piece of torn paper was found at the scene of a crime.

This piece of torn paper was found in a suspect's house.

OBSERVE

6. Work in groups of four. Select one person to be the detective and one person to write a sample of handwriting. The person writing the sample writes the word RANSOM on a piece of paper. While the person selected as the detective turns away, the other three members of the group, including the sample writer, try to make copies of the sample word. The detective needs to work out which two words belong to the sample writer and which two are copies. Take turns in playing the role of the detective and the sample writer.

✓ checklist

I can:

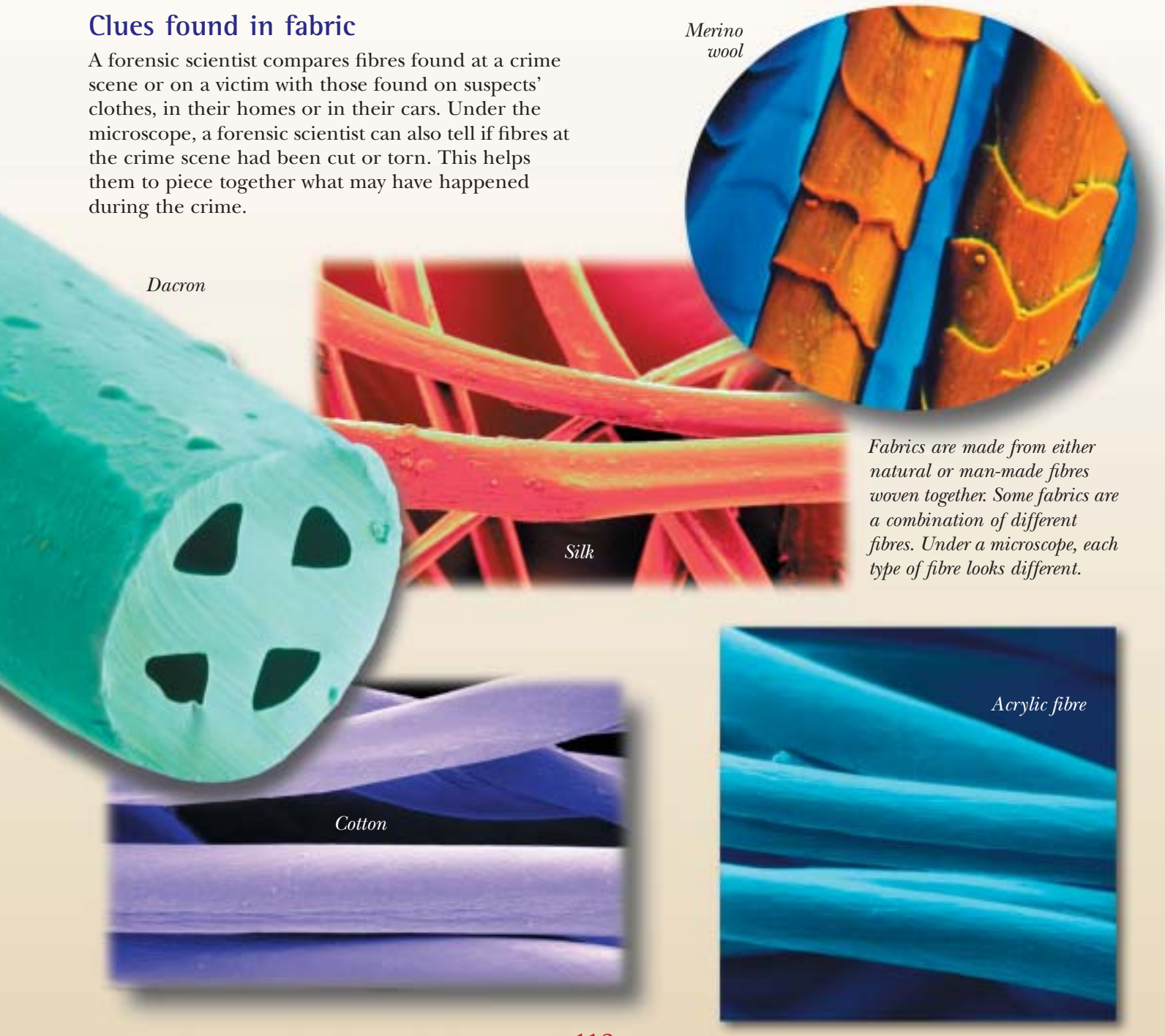
- ☐ describe four ways of analysing handwriting
- ☐ explain how analysing paper and ink can help solve a crime.

Hair and fibres

Wherever we go we leave some **evidence** behind and take some with us. Strands of hair or **fibres** from clothes, furniture and carpets can provide strong evidence that a **suspect** has been at a crime scene. Even the most careful criminals can't stop microscopic fibres sticking to their shoes. They may not realise that they have left a single strand of hair behind at a crime scene. Many hairs and fibres look the same, until they are examined under a microscope.

Clues found in fabric

A forensic scientist compares fibres found at a crime scene or on a victim with those found on suspects' clothes, in their homes or in their cars. Under the microscope, a forensic scientist can also tell if fibres at the crime scene had been cut or torn. This helps them to piece together what may have happened during the crime.



Fabrics are made from either natural or man-made fibres woven together. Some fabrics are a combination of different fibres. Under a microscope, each type of fibre looks different.

Hair

By looking at hair samples under a microscope, forensic scientists can tell whether the sample belongs to an animal or a human. The scaly, outer covering called a **cuticle** is different in each animal species.

With a microscope, the thickness, **coarseness,** colour and structure of hair can be checked. Scientists can even tell what type of shampoo has been used to wash the hair. A strand of hair found at a crime scene can be checked with a **comparison microscope** against hair from a suspect. A match between the hairs could be used as evidence to show that the suspect was at the scene.

Hair that has been pulled out can have skin or other substances stuck to it. **DNA** testing can link these hairs directly to a suspect.

Forensic scientists can tell if a hair sample has come from a person with curly, wavy or straight hair. Scientists can even sometimes tell the ethnic background of the person from their hair.

Comparing animal and human hair

You will need:
microscope, lamp and slides
tweezers
tape
animal hair
feather
human hair.

- Set up a microscope according to the instructions on page 247 of the Laboratory Toolbox.
- Tape a sample of animal hair, human hair and a feather to a microscope slide. You may need tweezers to help position the hairs on the slide.
- View the slides under the microscope.

1. Draw a diagram of each of the samples. Label your diagrams.
2. What are the main features of the three samples?

Activities

REMEMBER

1. Why do hairs or fibres found at a crime scene need to be compared under a microscope?
2. What is a cuticle?

THINK

3. Which hair sample (human or dog, shown at top left) is similar to one of the fabric samples (far left) shown? Suggest why this might be the case.
4. A detective found a small clump of human hairs at a crime scene. The hairs all showed evidence that they had been pulled out rather than falling out naturally.
   (a) What might this suggest about what happened at the scene?
   (b) What other tests could be applied to these hairs to help solve the crime?

A hair that has been pulled out

INVESTIGATE

5. Design an experiment to compare the properties of different fibres or fabrics. Properties that could be compared include:
   (a) strength
   (b) elasticity
   (c) absorption ability.

✓ checklist

I can:
- ☐ use a microscope to observe the differences between feathers, human hair, and hair or fur from other animals
- ☐ explain how hair and fibres can be used to help solve crimes.

Blood and saliva

Even a tiny amount of blood left at a crime scene can give scientists and detectives valuable information. Blood tells us who was at the crime scene and what might have happened. Saliva and dead skin cells can also be traced back to the person from whom they came.

Splat!

Movies often show criminals cleaning up after a crime. They wash their clothes and wipe up blood spills. But forensic scientists can still detect blood, even if it has been washed away. If crime-scene investigators believe that there may have been blood around, the area is treated with a chemical. The treated blood is easy to see under **ultraviolet light**.

Perpendicular impact

Blood falling straight down from a low height leaves an almost perfect circular drop of blood.

When blood drops from a greater height, there is a ring of small drops around the central drop of blood.

Angled impact

Blood that drops from an angle leaves a trail. The trail shows the direction in which the blood travelled.

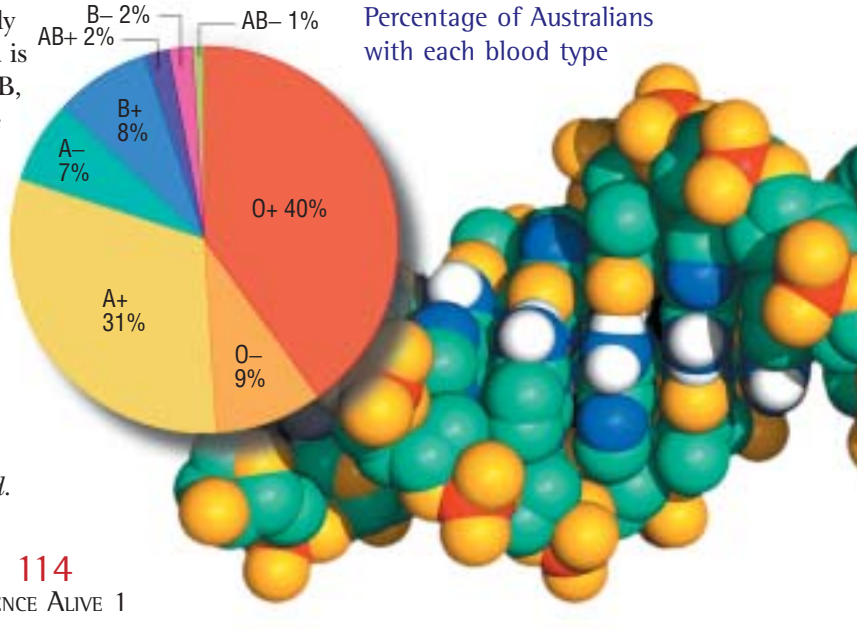
An arc of blood spatters forms when the victim pulls away from an impact. The number of arcs tells how many impacts there were to the victim. Forensic scientists can even tell if the **assailant** was left- or right-handed.

Types of blood

All human blood looks the same, but there are actually several different types of blood. In a laboratory, blood is tested to find out which of the four main groups — A, B, AB and O — it belongs to. Each of these groups can be further sorted into positive or negative. For example, a person could be A+ or A–, B+ or B– and so on.

*Finding a blood type at a crime scene that matches the blood type of a **suspect** does not mean that the suspect was there. Many people share the same blood types. But comparing blood types does narrow the search. Blood can also be **analysed** for diseases or other features that link the sample more closely to a suspect.*

If the blood types of samples from the crime scene don't match the suspect's blood type, then the suspect can be cleared.



Percentage of Australians with each blood type

Human codes

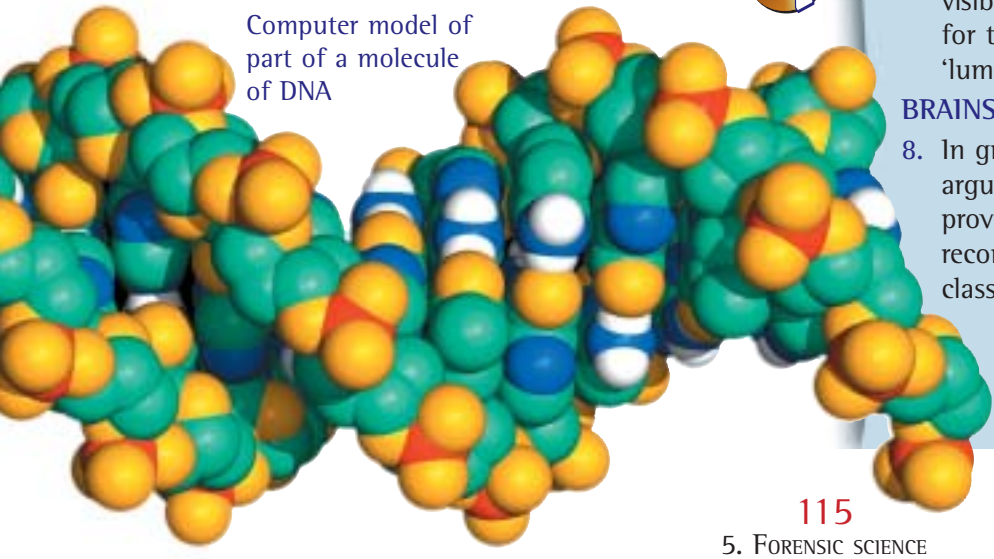
The cells that make up living things contain information. For the living thing it belongs to, this information is like a recipe. Information is stored in **DNA** (**d**eoxyribo**n**ucleic **a**cid). In humans, DNA contains information about the person's hair colour, eye colour and so on. DNA is different for different people, but the DNA in all of the cells of any one person is the same. Only identical twins have the same DNA as each other.

A single drop of blood found at a crime scene contains information about the person it came from. DNA profiling is a test that compares blood from a crime scene with that of a suspect. Actually, it's not just blood that can be used in this test. Body fluids, like saliva, hair roots and dead skin cells, can be tested as well. Everybody drops hair and skin cells all the time and criminals often leave DNA **evidence** behind. Scientists can even trace the path of paper from the DNA left by people who have touched it.

DNA Profiling

- Samples of blood, body fluids or skin that are collected from the crime scene are taken to a laboratory.
- DNA is removed from the sample. Chemicals cut the DNA into small pieces.
- The pieces of DNA from different samples are lined up across a jelly-like substance.
- When electricity is passed through the sample, the pieces of DNA spread across the jelly-like substance. As the pieces spread, they form a pattern.
- DNA samples collected from a suspect go through the same process as the sample from the crime scene.
- People look for matching pieces of DNA from the different samples.

The pattern that DNA from the crime scene forms is compared to the patterns of DNA from the suspects. Matching patterns can be used as evidence in court because it is very unlikely that two different people will have the same pattern.



Computer model of part of a molecule of DNA

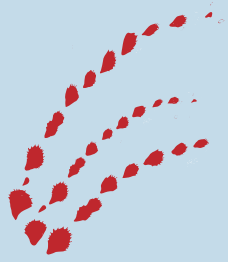
Activities

REMEMBER

1. A suspect has the same blood type as a sample found at a crime scene. Why is this not enough evidence to prove that the suspect was actually at the scene?
2. List the eight blood types that are possible with the two classification methods mentioned.
3. (a) What is the most common blood type in the Australian population?
   (b) What is the least common blood type in the Australian population?
4. Which people have the same DNA?

THINK

5. The diagram shows the blood spatter pattern at a crime scene.
   (a) Redraw the diagram. Use an arrow to show the direction in which the blood travelled.
   (b) How many blows did the victim receive? Explain your answer.



6. Explain how blood at the scene could be used to show that a suspect was not at a crime scene.

INVESTIGATE

7. The chemical used to make bloodstains visible is called luminol. Suggest a reason for this name. *Hint:* Look up the word 'luminous' in the dictionary.

BRAINSTORM

8. In groups of three or four, create a list of arguments for and against all people providing a DNA sample to keep on police records. Share your list with the rest of the class during a class discussion.

checklist

I can:

- ☐ list 8 blood groups
- ☐ explain how DNA profiling is used to solve crimes
- ☐ explain why blood at a crime scene is useful as evidence.

The Daily Goss

27 June 1996 Weather: 14°C

Celebrity caught out

Yesterday afternoon detectives questioned Julia Parker, lead actress from the movie *Fifteen Minutes of Fame*. Ms Parker was found to be in possession of a **counterfeit** $100 note.

Eva Goodeye, sales assistant in the Fab Gear fashion boutique, noticed the suspect note yesterday. She said that it was easy to tell the difference between the counterfeit and a real note. As soon as she realised the note was a fake, the sales assistant phoned police. They told her not to touch the note until they arrived at the scene.

This counterfeit claim has surfaced only a month after the release of the new $100 note. The new note is the last in the changeover from paper to **polymer** (plastic). Polymer notes will last longer, can go through the wash and are harder to counterfeit than paper money.

Detective Sargent Walker, who is in charge of the investigation, expressed surprise. 'Most counterfeit money is printed on paper. Before the introduction of polymer notes, lots of copied notes were floating through the system. People were just copying them on good quality colour photocopiers. It's a different story now', he said.

The media pack greeted Ms Parker as she entered the police station for questioning. She made a brief statement. 'I have never knowingly handled counterfeit money. It seems ridiculous that someone with my wealth and fame would need to make or use fake money. I'm just here to help the police with their investigation. That's all, I have nothing more to say.'

If Julia Parker is found guilty of knowingly using counterfeit money, she will be charged under the *Crimes (Currency) Act 1981*. *The Daily Goss* will keep you up to date on the police investigation.

Bank Victoria

Bank Victoria is looking for experienced bank tellers for three new branches. The tellers must be well presented and have excellent people skills. All tellers must be able to detect counterfeit notes and will be required to sit a test before they are accepted into a position. The test will check observation skills and the applicant's response to possible counterfeit and robbery situations.

Applications should be addressed to:
Human Resource Manager,
Bank Victoria, Box 123,
Melbourne, Vic., 3000.

Go to worksheet 25

How to tell a counterfeit note from a genuine note

- Check that the clear window of the note has the correct image:
  - $5 — eucalypt flower
  - $10 — windmill
  - $20 — nautical compass
  - $50 — Southern Cross
  - $100 — lyrebird.
- The clear window should feel smooth. It should be a part of the note and not just attached to the surface.
- There is a small diamond pattern on each side of a genuine note. Hold the note up to the light. Together, the diamond patterns from each side will form a perfect seven-pointed star in a circle.
- The printing on the main parts of a real note is raised, so you should be able to feel it.
- Genuine notes have an image that appears under the printing when the note is held up to the light.
- All notes contain very small writing around the edge ($5) or near the portrait (all other notes). The writing is visible with a magnifying glass. In a genuine note, the writing is sharp and clear.

Activities

REMEMBER

1. List two advantages polymer notes have over paper notes.

THINK

2. What is counterfeit money?
3. Coins are not usually counterfeited. Suggest a reason for this.
4. Why was the sales assistant asked not to touch the note?

USING DATA

Issue dates		Size (mm)	Average life of notes (months)	
			Polymer	Paper
$5	07/07/92 24/04/95 01/01/01	130 × 65	40	6
$10	01/11/93	137 × 65	40	8
$20	31/10/94	144 × 65	50	10
$50	04/10/95	151 × 65	About 100	24
$100	15/05/96	158 × 65	About 450	104

Reserve Bank of Australia

5. Why do the sizes of Australian currency notes differ from each other?
6. Roughly how much longer do the polymer notes last than the paper notes?
7. Suggest a reason why $5 and $10 notes do not last as long as other notes.

Australian note production (millions of notes)

1991–92	533	1996–97	195
1992–93	360	1997–98	180
1993–94	364	1998–99	189
1994–95	260	1999–00	190
1995–96	285	2000–01	73

Reproduced with permission of the Reserve Bank of Australia

8. How did the number of notes produced each year change from 1991 to 2001?
9. Suggest a reason for the change in the number of notes produced between 1991 and 2001.

✓ checklist

I can:

- ☐ state the meaning of the word counterfeit
- ☐ list methods used to prevent or reduce the counterfeiting of money.

Check and challenge

Forensic science

Gathering evidence

1. Make a list of the different branches of science involved in forensic science.
2. How do entomologists help to solve crimes?
3. Why is the public stopped from entering a crime scene?
4. Match the objects with the microscopes that are suitable for viewing them.

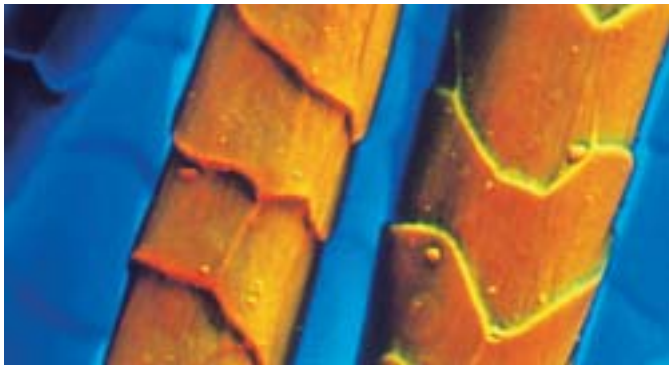
Two scratched bullets	Stereo light microscope
Thin layer of skin cells	Electron microscope
Tiny trace of dust	Comparison microscope
Iron nails	Monocular light microscope

5. Why do eyewitness accounts need to be backed up with other evidence?
6. What is the F.A.C.E. system?
7. Explain the difference between evidence and an inference.
8. Read the following sentences. Decide if the word in *italic* makes the sentence true or false. If the sentence is false, replace the word in *italic* with another word that makes the sentence true.
   (a) A *cast* of a footprint is more useful than a photograph.
   (b) A forensic *odontologist* studies insects.
   (c) The examination of a dead body is called a *mortuary*.
   (d) Ink is made up of colours called *chromatography*.
9. Look at the crime scene on pages 100–1. List:
   (a) the substances that could be DNA tested and where samples of the substances can be collected
   (b) the evidence that can be used as direct proof and the evidence that just adds to the body of evidence.

Identification

10. Identify the four patterns shown in these fingerprints.

11. Explain, in simple terms, how fingerprints are left behind.
12. How can a badly burnt body be identified?
13. What is cheiloscopy and how does it help to solve crimes?
14. The best clue to tell whether a hair belongs to an animal or a human is:
    A the roots?
    B the colour pigment?
    C the coarseness?
    D the cuticle?
    E substances stuck to the hair?
15. What type of evidence found at a crime scene can be analysed through DNA profiling?
16. Explain why matching the blood type of a suspect to that found at a crime scene does not prove that the suspect was at the scene.
17. (a) Which of these hairs has most likely come from an animal and which from a plant?



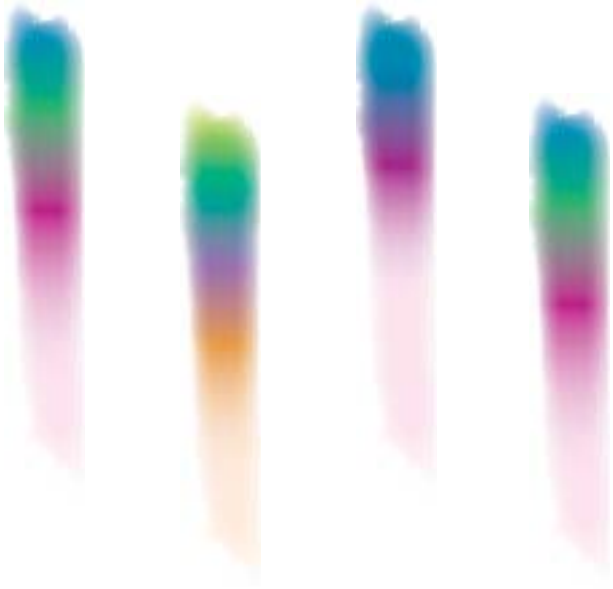
(b) Explain your answer to (a).

18. Name and describe a method police use to produce an image of a suspect from a description given by a witness.

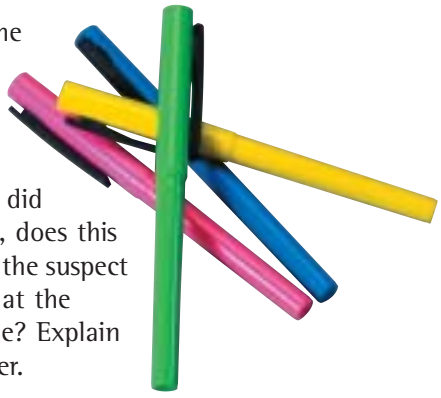
19. If identical twins were suspects in a crime, could DNA profiling be used to find out which was the criminal? Explain your answer.
20. Can fingerprinting be used to identify a criminal in a group of suspects that are all related? Explain your answer.
21. In the 1930s in America, the FBI arrested gangster John Dillinger by matching his fingerprints with those found at a crime scene. Dillinger had earlier attempted to surgically remove, and then burn off the prints from his fingers. Why was the FBI still able to fingerprint him?
22. The fingerprints of a person were found at the scene of a crime. Is that person guilty of committing the crime? Explain your answer.

Pen and paper

23. What technique can be used to compare the colours found in pen ink?
24. Which of the following inks may have come from the same pen?



25. Ink found on a ransom note exactly matched ink found in a pen at the home of a suspect.
    (a) Is this enough evidence to prove that the suspect was at the scene of the crime? Explain your answer.
    (b) If the inks did not match, does this mean that the suspect was never at the crime scene? Explain your answer.



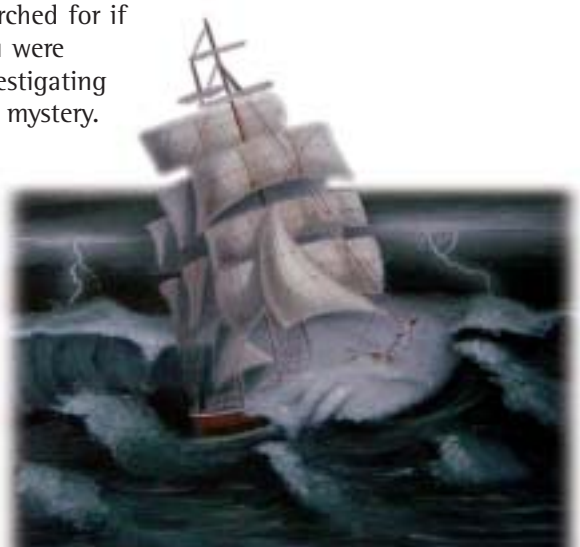
Getting away

1. The tracks left by a getaway car were identified as belonging to a particular make of tyre. All three suspects in the crime have the same make of tyre on their cars. How can the tracks be used to find out which car was at the scene of the crime?
2. What inferences can be made if black tyre marks are found at the scene of an accident where a car has hit a pedestrian?

Checking history

Forensic scientists were called in to analyse a photo of Ned Kelly that sold for over $19 000. For 40 years, the photo was believed to be of Ned Kelly. But some people were not convinced. Forensic scientists used computers to measure the ears, nose, forehead and eyebrow ridge of the person in the photo. The scientists compared the measurements with each other and with other photos of Ned Kelly. All of the forensic scientists independently decided that the photo was not Ned Kelly and the money was returned to the buyer.

3. Measurements taken from a photo are not the same as real-life measurements. A photo is much smaller than a real person. Explain why forensic scientists have to make a comparison between the measurements on one photo before comparing them with those in other photos. *Hint:* How do measurements change from one photo to the next, if one photo is close up and the other far away?
4. Forensic scientists often use techniques such as fingerprinting and DNA testing to solve crimes.
    (a) Suggest some uses, other than for solving crimes, for these techniques.
    (b) Suggest how other forensic techniques or skills could be used for purposes other than solving crimes.
5. Read the short description of the mystery of the *Mary Celeste* on page 97.
    (a) Use the Internet and other resources to find out about expert theories surrounding the mystery of the *Mary Celeste*.
    (b) Write a short story about what you believe may have happened to the *Mary Celeste* and her crew.
    (c) Make a list of the evidence you would have searched for if you were investigating the mystery.



Summary of key terms

analyse: examine closely to answer a question or solve a problem
assailant: a person who attacks another person
autopsy: examination of a dead body. An autopsy is performed to find out how the deceased person died.
ballistics experts: people who analyse bullets, firearms and their effects on victims and shooters. A ballistics expert can match bullets with the firearm from which they were shot.
cartridge: container in a firearm from which bullets are fired
cast: a form that is made by pouring liquid plaster into a mould. The cast is removed from the mould when it becomes solid.
cheiloscopy: the study of lip prints

chromatography: a separation technique that uses the difference between the solubilities of substances to separate them. This technique separates small traces of substances, such as the colours in ink.
coarseness: roughness due to having a surface made up of large particles
comparison microscope: a microscope that allows two objects to be viewed at the same time so that the objects can be compared
contaminated: spoiled by adding, removing or changing something at a crime scene
counterfeit: a copy
cuticle: the scaly outer covering of a hair
decomposes: rots or breaks down
diatom: microscopic organism found in fresh and salt water
disfigures: deforms or ruins
DNA: (deoxyribonucleic acid), a substance found in all living things that contains information about hair colour, eye colour etc. Only identical twins have the same DNA as each other.
documenting: recording of details at a crime scene
entomologist: a scientist who studies insects
evidence: clues that can be used to help solve crimes

eyewitness account: a description of an event, place or person supplied by an observer
facial: relating to the face
fibres: fine threads that make up fabric and paper
forum: the buildings which housed the ancient Roman courts of law
friction ridges: the pattern of ridges and valleys that form fingerprints. They help fingers to grip objects.

image: a picture or visual representation
impression: print or mark formed under pressure. Impressions include marks like fingerprints, footprints and lip prints.
inferences: reasonable conclusions made based on available evidence
maggots: newly hatched flies. Maggots have a worm-like appearance.
magnification: the number of times an object has been enlarged using a lens or lens system. For example, a magnification of two means the object has been enlarged to twice its actual size.
minutiae: small points of detail that are matched up in a fingerprint analysis
mortuary: the place where deceased bodies are kept before their burial
odontologist: a scientists who studies teeth
pathologist: a scientist who studies the cause of death and disease
physical evidence: clues that can be collected from a crime scene and analysed in a laboratory
pigment: a substance that gives an object or fabric a particular colour
polymer: a substance made up of smaller units. All plastics are polymers.
pupa: a maggot that is transforming into a fly
specimen: sample used for investigation or scientific examination
statements: verbal or written accounts
suspect: a person believed to be involved in committing a crime

testify: give evidence
ultraviolet light: light that cannot be seen with the eye
witness: a person who was present at or who knows something about an event

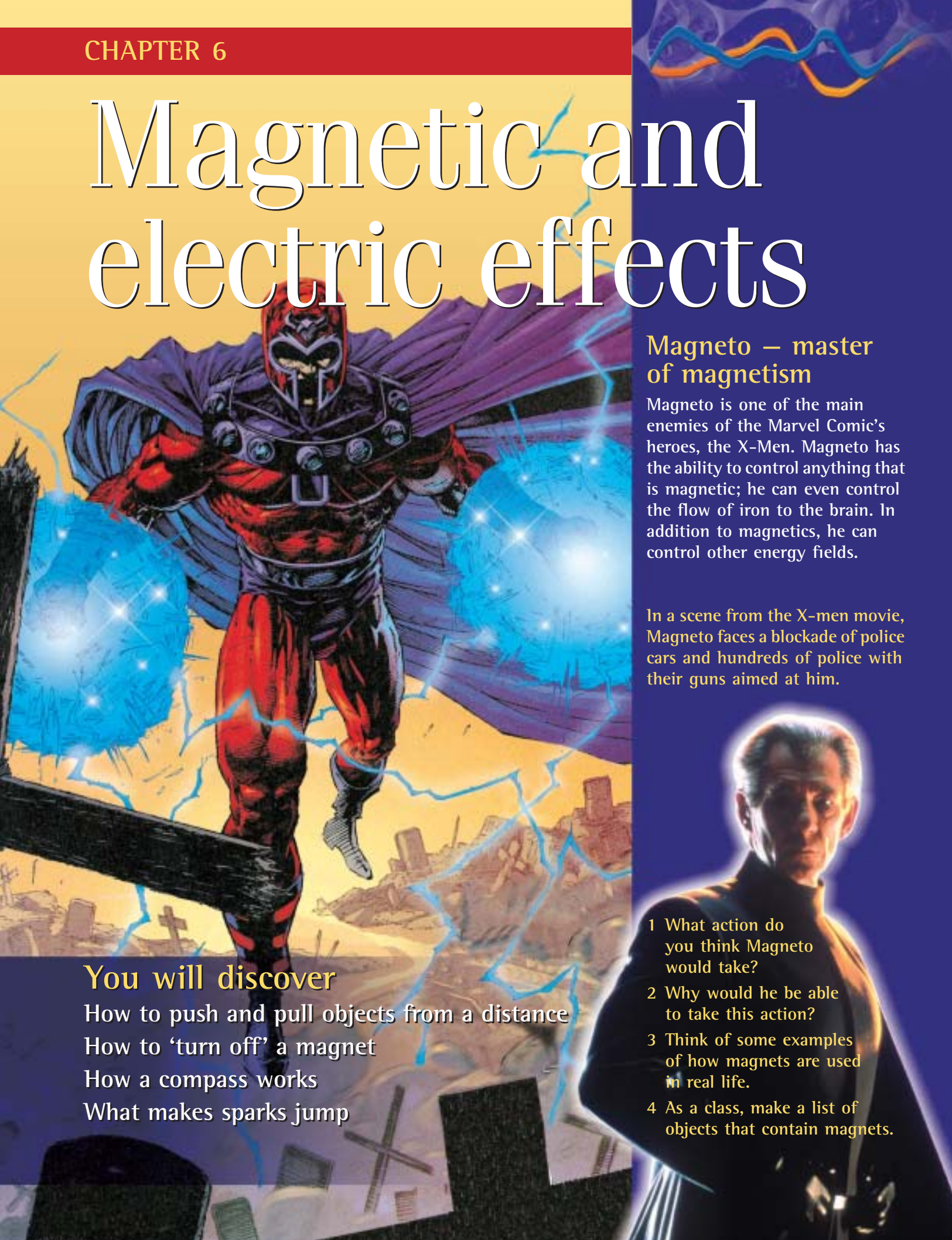
CHAPTER 6

Magnetic and electric effects

Magneto – master of magnetism

Magneto is one of the main enemies of the Marvel Comic's heroes, the X-Men. Magneto has the ability to control anything that is magnetic; he can even control the flow of iron to the brain. In addition to magnetics, he can control other energy fields.

In a scene from the X-men movie, Magneto faces a blockade of police cars and hundreds of police with their guns aimed at him.



You will discover

How to push and pull objects from a distance
How to 'turn off' a magnet
How a compass works
What makes sparks jump

1 What action do you think Magneto would take?
2 Why would he be able to take this action?
3 Think of some examples of how magnets are used in real life.
4 As a class, make a list of objects that contain magnets.

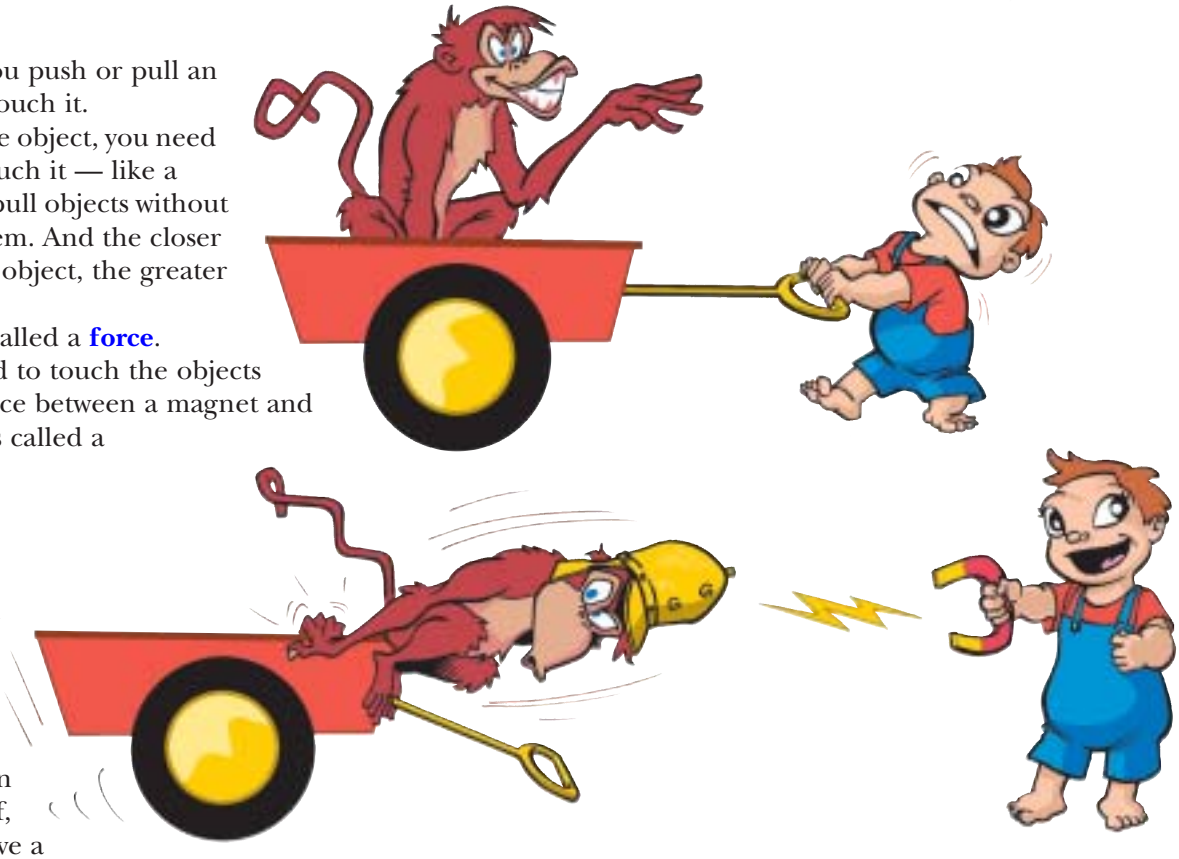
Attractive metals

Magnets come in many shapes and sizes, and **attract** some metal objects (like the fridge door). Magnets can even attract each other. The closer the magnet is to another magnet or metal object, the more strongly they pull together. The effect of a magnet can be a pull — but can it also be a push?

Out of touch

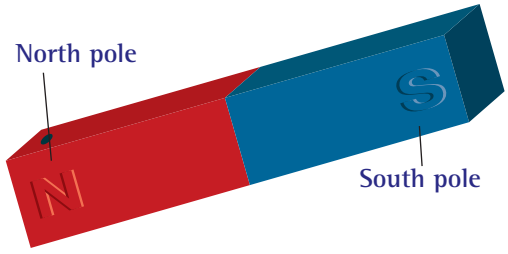
Almost every time you push or pull an object, you have to touch it.
If you don't touch the object, you need something else to touch it — like a string. Magnets can pull objects without actually touching them. And the closer the magnet is to the object, the greater the size of the pull.

A push or pull is called a **force**. Magnets do not need to touch the objects they pull. So, the force between a magnet and an attracted object is called a **non-contact force**.



Poles

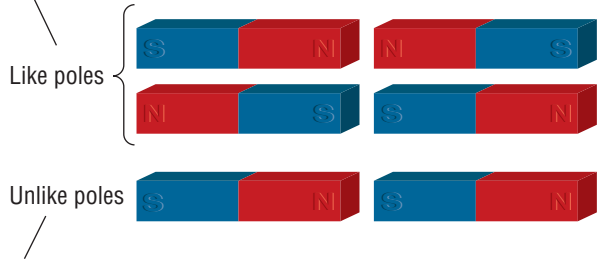
The pulling force of a magnet is strongest at its ends, or **poles**. All magnets have a north pole and a south pole. Even when a magnet is cut in half, each half will still have a north and a south pole. If you can keep cutting a magnet in half over and over again, each half will always have both north and south poles.



If you hang a magnet by its middle, it always lines up with the North and South poles of the Earth. The north pole of a magnet actually gets its name from the term 'north-seeking pole' because it lines up with or 'seeks' the North Pole of the Earth.

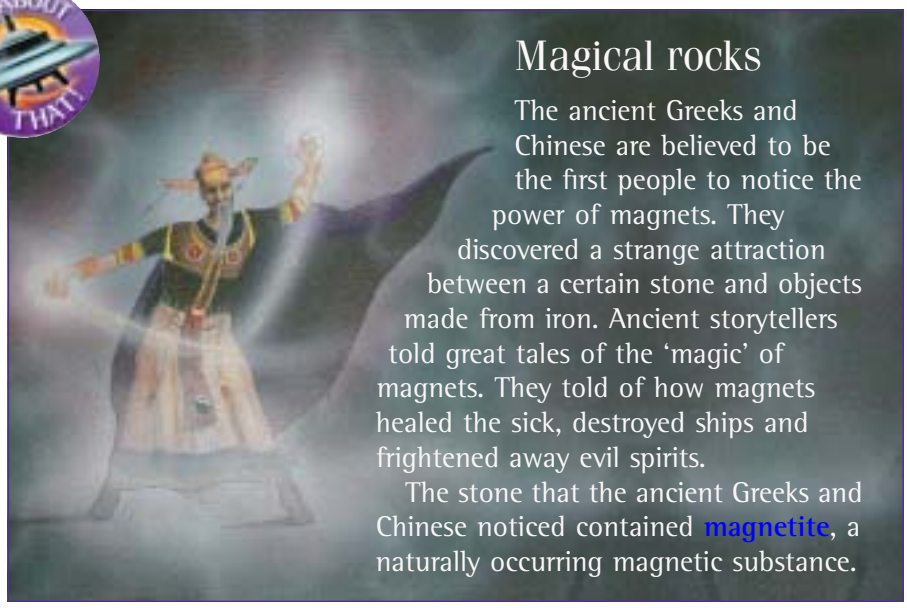
Opposites attract

*When the north poles of two magnets are brought close together, the magnets push away or repel each other. The same type of push (**repulsion**) is felt between two south poles.*



When the north pole of a magnet comes close to the south pole of another magnet, the opposite happens. They pull on each other, or attract.

Go to worksheet 26



Magical rocks

The ancient Greeks and Chinese are believed to be the first people to notice the power of magnets. They discovered a strange attraction between a certain stone and objects made from iron. Ancient storytellers told great tales of the 'magic' of magnets. They told of how magnets healed the sick, destroyed ships and frightened away evil spirits.

The stone that the ancient Greeks and Chinese noticed contained **magnetite**, a naturally occurring magnetic substance.

Can magnets push as well as pull?

You will need:
2 bar magnets
ruler.

- Hold two magnets about 10 cm apart, so that the north ends are facing each other. Slowly bring the magnets together.

1. What do you feel when the north ends are brought closer?

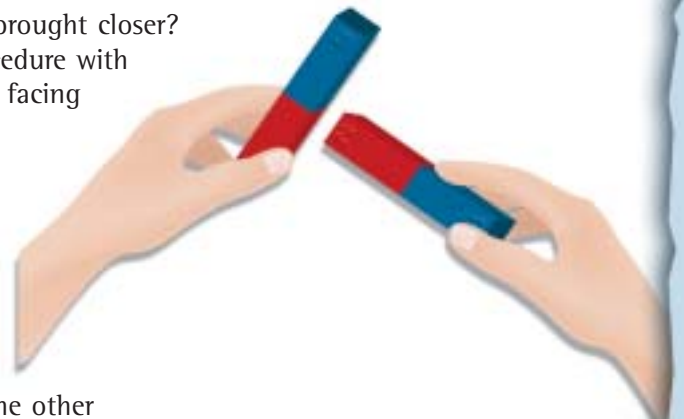
- Hold the magnets 10 cm apart, so that this time a north and south end are facing each other. Bring them together slowly.

2. What do you notice when a north and south pole are brought closer?

- Repeat the procedure with two south ends facing each other.

3. Record your observations.

- Hold the magnets so that the north pole of one magnet meets the middle of the other magnet at right angles. Slowly move the magnets apart, then together.



4. Is the push or pull as strong at the middle as at the poles?

- Try this again with the south pole at right angles to the other magnet.

5. Complete these sentences:
Like poles _____.
Unlike poles _____.

6. Where is the pulling force of a magnet the greatest?

Activities

REMEMBER

1. What are the ends of a magnet called?
2. How should two bar magnets be placed so that they will attract each other?
3. State whether the following are true or false:
   (a) The pulling force of a magnet is strongest in the middle.
   (b) Magnets can only attract other magnets.
   (c) Magnets can push each other apart.
   (d) The closer two magnets are, the weaker the force between them.

THINK

4. How is a magnetic force different from most other pushes and pulls?
5. Suggest a reason why the north pole of a magnet would be attracted to the North Pole of the Earth.
6. What happens to the poles of a magnet when a magnet is cut in half?

IMAGINE

7. Imagine you were the one who first noticed the effect of stones containing magnetite. Write a story about how you discovered the stone. In your story, describe the effect and suggest whom you would tell about your discovery.

INVESTIGATE

8. Design and perform an investigation to find out which objects magnets are attracted to.

✓ checklist

I can:
- ☐ label the poles of a magnet
- ☐ explain how magnets push and pull each other
- ☐ identify the parts of the magnet with the greatest pulling force.

Make it into a magnet

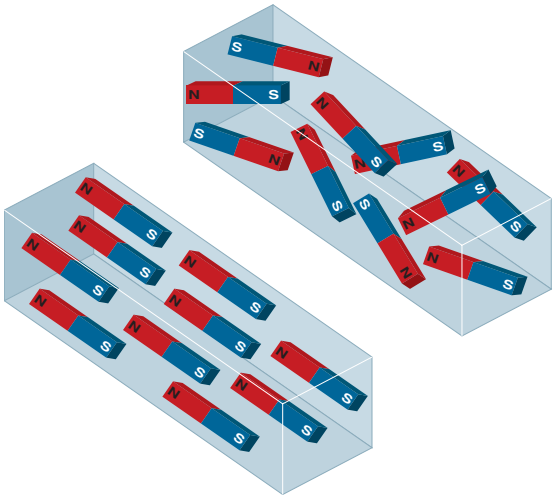
Once a **magnet**, always a magnet? Not necessarily. Some magnets keep their magnetic effect for years. Others lose their magnetic effect in just minutes. There are also magnets that can be turned on and off at the flick of a switch. No matter what type of magnet, the materials that form them have something in common.

Three special metals

Not all objects are attracted to magnets. Magnets affect only materials made with iron, nickel or cobalt. These three metals are made up of small parts that behave like mini-magnets. These small parts are called **domains**. Each of these domains or 'mini-magnets' has a north **pole** and a south pole.

As well as being affected by magnets, iron, nickel and cobalt can also be made into magnets. When the domains inside the metals face the same direction, the metal turns into a magnet.

If the domains inside magnetic materials are facing different directions, the pushes and pulls of the 'mini-magnets' have no effect. This material is not a magnet.



If the domains are lined up facing the same direction, the material has an overall north pole and an overall south pole. The material *does* have a magnetic effect. All magnets have domains that face the same direction.

Permanent and temporary magnets

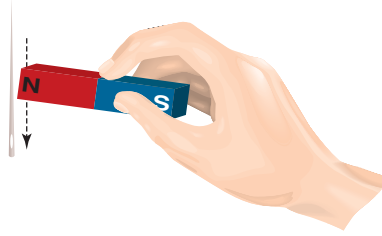
A needle can be **magnetised** by stroking it with a bar magnet in the same direction many times. The domains in the needle are lined up only temporarily and quickly go back to their original directions. Objects that lose their magnetism after a short period of time are called **temporary magnets**.

Permanent magnets keep their magnetism for long periods of time. Bar and horseshoe magnets are permanent magnets. They do not lose their magnetism easily, but can be damaged by being dropped or by being heated to very high temperatures.

How is a magnet made?

You will need:
steel needle
strong, permanent bar magnet
several pins.

- Use the north pole of a magnet to stroke the needle thirty to forty times. Stroke the needle in the same direction each time and lift the magnet well away from the needle between strokes.



1. Why do you need to stroke the needle in the same direction each time?

- Test your magnet by trying to pick up the pins.

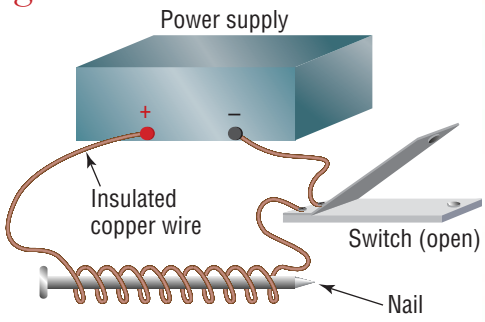
2. How many pins does your magnet attract? Compare your results with those of your classmates.
3. Have you made a permanent or temporary magnet? How do you know?
4. How could you tell which is the north pole of your new magnet?
5. Draw a diagram of the needle. Use an arrow to show the direction that the needle was stroked in. Label the north and south poles of your new magnet.

Switched on or off?

A magnet's pulling **force** can be very useful. But sometimes it gets in the way.
An **electromagnet** is a magnet that can be turned on and off with the flick of a switch. It is made up of a coil of wire wrapped around a piece of iron. The piece of iron turns into a magnet when electricity passes through the coil. The iron stops being magnetic as soon as the electricity is turned off.
Electromagnets are used in many machines and appliances.

Making electromagnets

You will need:
power supply
2 insulated wires, one short, the other 1.5 m long
large nail
switch
paperclips.



- Set up the circuit shown on the right.
- Wind the long wire neatly around the nail 15 times.
- Set the power supply to two volts and close the switch. Test the nail to see if it will pick up any paperclips.

1. Record your results in a table like the one below:

Voltage of power supply (V)	Number of turns of wire	Number of paperclips picked up
2	15	
2	20	
2	25	
2	30	
4	15	
4	20	
4	25	
4	30	

- Wind five more turns of wire onto the nail.

2. How many paperclips does the electromagnet pick up now?

- Keep winding the wire onto the nail. Record the number of paperclips picked up for 25 and 30 turns of wire.
- Raise the voltage to four volts. Repeat the steps above.

3. Record the number of paperclips picked up when the voltage is four volts.
4. What effect does increasing the number of turns of wire have?
5. What effect does raising the voltage have?

Activities

REMEMBER

1. What is the difference between a permanent and a temporary magnet?
2. How is an electromagnet different from a bar magnet?
3. What substance or substances does a needle contain if it can be magnetised?
4. What is the scientific name for the 'mini-magnets' that make up larger magnets?

THINK

5. Why doesn't iron always behave like a magnet with a north and south pole?
6. State two ways to increase the strength of an electromagnet.

CREATE

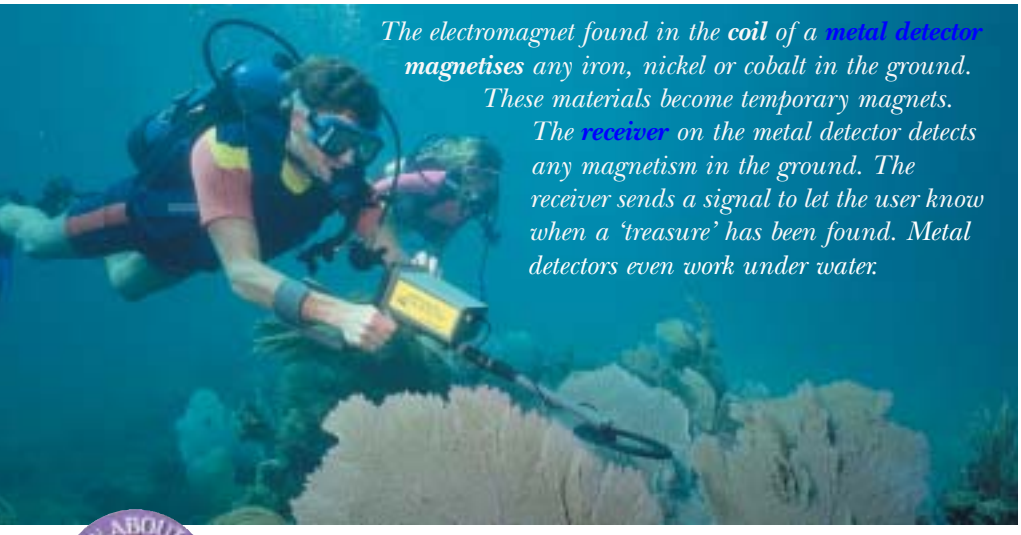
7. Design and build a device that uses an electromagnet to burst a balloon.

checklist

I can:
- ☐ explain magnetism in terms of domains in magnetic materials
- ☐ compare temporary and permanent magnets
- ☐ compare electromagnets with other magnets.

Using magnets

Permanent **magnets** and **electromagnets** are found in many places, including telephones, doorbells and liquid-crystal displays in calculators. They are even useful for finding treasure and creating pictures.



*The electromagnet found in the **coil** of a **metal detector** **magnetises** any iron, nickel or cobalt in the ground. These materials become temporary magnets. The **receiver** on the metal detector detects any magnetism in the ground. The receiver sends a signal to let the user know when a 'treasure' has been found. Metal detectors even work under water.*

Magnetic pictures

Magnets show up in the most amazing places! Four Japanese engineers invented the Magna Doodle in 1974. They were trying to make a blackboard that didn't create dust. The toy they came up with was the Magna Doodle. Since then, over 40 million of these toys have been sold around the world.

*The Magna Doodle is made up of a thick plastic screen, a magnetic eraser and a magnetic pen. The screen is made up of small sections that look like the honeycomb found in some chocolate bars. Each of these sections is filled with a thick white liquid and some fine, powdered, black **magnetite**.*

As the magnetic pen slides along the top of the plastic screen, the black magnetite is attracted towards it. The magnetite lies at the top of the thick liquid where it can be seen through the screen.

To remove the picture, the magnetic eraser slides over the back of the screen. The attraction to the eraser pulls all of the magnetite in the honeycomb sections to the back of the screen, leaving the front white.

Videotapes

Although magnetic videotape has the advantages of being cheap and easy to record and re-record on, it is easily damaged when stored near magnets. Magnets can change the pattern that has been stored on the tape.

The films that you see at the cinema are different from videotapes. Chemicals create the picture on the cinema film. The film used in cinemas, like that used in normal cameras, cannot be re-recorded on and is more expensive to make. Cinema films last much longer and produce higher quality pictures.

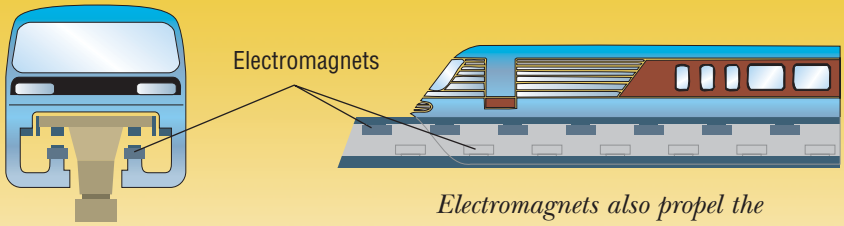
500 km/h magnets

The maglev train gets its name from MAGnetic LEVitation train. It reaches speeds of up to 500 km/h and doesn't even need a normal engine to run! Once the train is at full speed, passengers enjoy a fast, smooth and quiet ride.

The inside of a video recorder has two electromagnets, called heads. The recording head creates the magnetic pattern on the tape. The playback head reads the pattern and turns it into pictures and sound.

The plastic ribbon inside a videotape is coated with substances that contain 'mini-magnets' or **domains**.

When the ribbon passes through a magnetic field, the domains are forced into a pattern.



*This train seems to float above the train tracks. It uses pushing **forces** between electromagnets on the track and on the train to keep them apart. The train touches the track only while it is building up speed before lift-off.*

Electromagnets also propel the train forward. Magnets ahead of the train pull the train forward. Magnets behind the train push it forward.

Activities

REMEMBER

1. Why should magnets be kept away from videotapes?
2. How does a maglev train float above the track?
3. Which part of a metal detector is an electromagnet?
4. Why is magnetite used in a Magna Doodle?

THINK

5. Why is a maglev train able to travel so fast?
6. The coils in metal detectors need to be blocked off from the receiver. Why is this important?
7. Could you use a metal detector to find a pure-gold nugget? Explain your answer.

INVESTIGATE

8. Find out what combination of magnets are required for the maglev train to be pulled forward at the front and pushed forward from the back. You might find some clues by going to www.jaconline.com.au/science/weblinks and clicking on the Maglev Train link for this textbook.
9. Find out about the Magnetic Resonance Imaging (MRI) machine. What is it and how does it work?

✓ checklist

I can:

- ☐ list and describe some of the uses of magnets
- ☐ explain how electromagnets are used in videotapes, maglev trains and metal detectors.

Making magnetic maps

A **magnetic field** is like a map of the pushes and pulls of a **magnet**. You can't actually see the magnetic pushes and pulls because the magnetic field is invisible. But there is a way to make it visible using iron filings. When the magnetic field is made visible you can see how big the pushes and pulls are. When a **compass** is placed in different positions around the magnet it reveals the direction of the pushes and pulls.

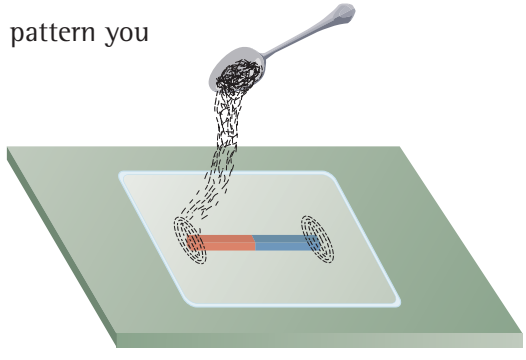
These iron filings show the magnetic field around a magnet. The iron filings are closer together where the pushes and pulls are strongest. Notice that the magnetic field is strongest near the ends (poles) of the magnet.

Making a magnetic map visible

You will need:
2 bar magnets
horseshoe magnet
small plastic bags
iron filings
large plastic lid (for example, ice-cream lid).

- Place one of the bar magnets in a plastic bag under the centre of the lid.
- Carefully sprinkle some of the iron filings over the lid. You may need to spread them a little by lightly tapping the lid.

1. Draw a diagram of the pattern you see in the iron filings.
2. What does this pattern represent?
3. Label the strongest parts of the magnet with an X.
4. Do the iron filings tell you which is the north pole of the magnet?



Which way is north?

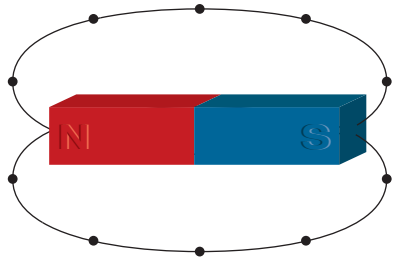
A compass is a simple tool for letting us know where north is. The compass needle always points to the magnetic North Pole of the Earth. As long as you have a compass you'll always know which way is north. This is especially handy if you are lost!

The compass needle moves freely around the centre point until it points north. It is pushed and pulled by ***magnetic forces*** *so that it lines up with the Earth's magnetic field, just like a hanging magnet. In fact, a compass needle is a magnet. The part of the needle that points or 'seeks' north must be the north pole of the magnet.*

Finding the direction of a magnetic field

You will need:
bar magnet
compass.

- Place the compass at 10 different points around the magnet. Use this diagram as a guide:
- At each of the ten points, mark the direction that the compass needle points in. Use arrows to show the direction.



1. Describe the pattern of arrows noting that the arrows show the direction of the pushes and pulls around a magnet.

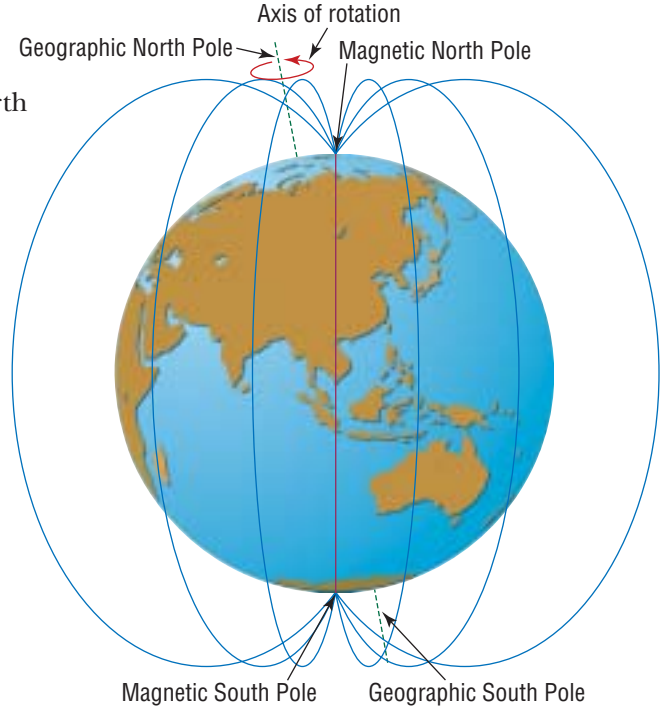
The Earth's magnetic field

The Earth is surrounded by a huge magnetic field. Scientists are still not sure what causes the Earth's magnetic field. Some believe that the large amounts of iron and nickel in the ground create the magnetic field. Others believe that, as the Earth spins, the liquid parts deep inside it move and create the Earth's magnetic field.

The magnetic forces around the Earth act on some objects in the same way that other magnets do. The force causes such objects to line up with the Earth's magnetic field. That is why a bar magnet hanging from its middle lines up with the North and South poles of the Earth.

Notice that there are two north poles and two south poles marked on the diagram at right.

The north poles of magnets do not point exactly to the North Pole, but actually line up with the magnetic North Pole of the Earth. The magnetic North Pole is nearly 2000 km from the geographic North Pole. The **magnetic South Pole** is found just over 2000 km from the geographic South Pole.



Activities

REMEMBER

1. How can you tell where the magnetic field is strongest around a magnet?
2. Why do hanging magnets line up with the North and South poles of the Earth?
3. A compass needle does not point exactly to the geographic North Pole. Where does it actually point?

THINK

4. Why is it important that the magnets used in compasses are small and light?
5. Why should magnets be kept away from compasses?
6. Draw a picture to represent the Earth with an imaginary magnet inside that helps us to think about the Earth's magnetic field. Label the north and south poles of the magnet and the North and South poles of the Earth.
7. The compass needle is missing in this diagram. Redraw the diagram showing the correct direction of the needle.

CREATE

8. Make a working model of a compass using a bar magnet as the compass needle.

checklist

I can:
- ☐ describe a magnetic field in terms of a map
- ☐ locate the strongest part of a magnetic field
- ☐ describe how a compass works.

The Sci-News Times

6 November 1997 — Weather: Earth — The usual. Sun — High solar-flare activity

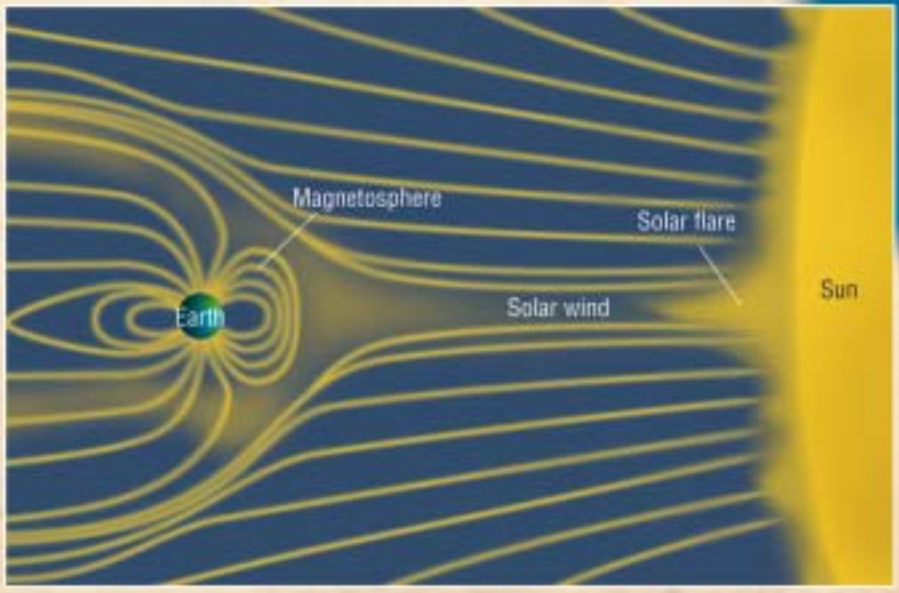
Homing pigeons grounded

All homing-pigeon races scheduled for the coming weekend have been cancelled due to **geomagnetic storms**. All owners of homing pigeons have been warned to avoid releasing them until the magnetic disturbance has passed. Two separate geomagnetic storms will strike the Earth within 48 hours on the weekend. They will distort the Earth's **magnetic field** and make it impossible for the homing pigeons to find their way home.

Why the pigeons get lost

Scientists have shown that homing pigeons use the magnetic field of the Earth to find their way over long distances. However, they still don't know exactly how the pigeons do this. An organ (like the eye or ear) that senses magnetism has not yet been discovered. It is possible that tiny crystals of **magnetite** found in the pigeons' brains act like mini-magnets. This could help pigeons to line up with the magnetic field — just like a **compass**. When the Earth's magnetic field is distorted by a geomagnetic storm, the pigeons don't know which way to fly.

Magnetically disturbed



Geomagnetic storms are caused by huge outbursts of energy on the Sun called **solar flares**. The flares release billions of tonnes of particles, which are carried away by the solar wind. The particles speed towards Earth at about three million kilometres per hour. When they arrive, they change the shape of the Earth's magnetic field.

It's not just pigeons!

Magnetite has been found in other animals, including some types of ant and bee. It is possible that they also lose their way in geomagnetic storms. Magnetite has also been found in the brains of dolphins. Even human brains contain small amounts of magnetite. Scientists are yet to find out how, or even if, we use it. There is still a lot to find out about animal magnetism.

Warning!

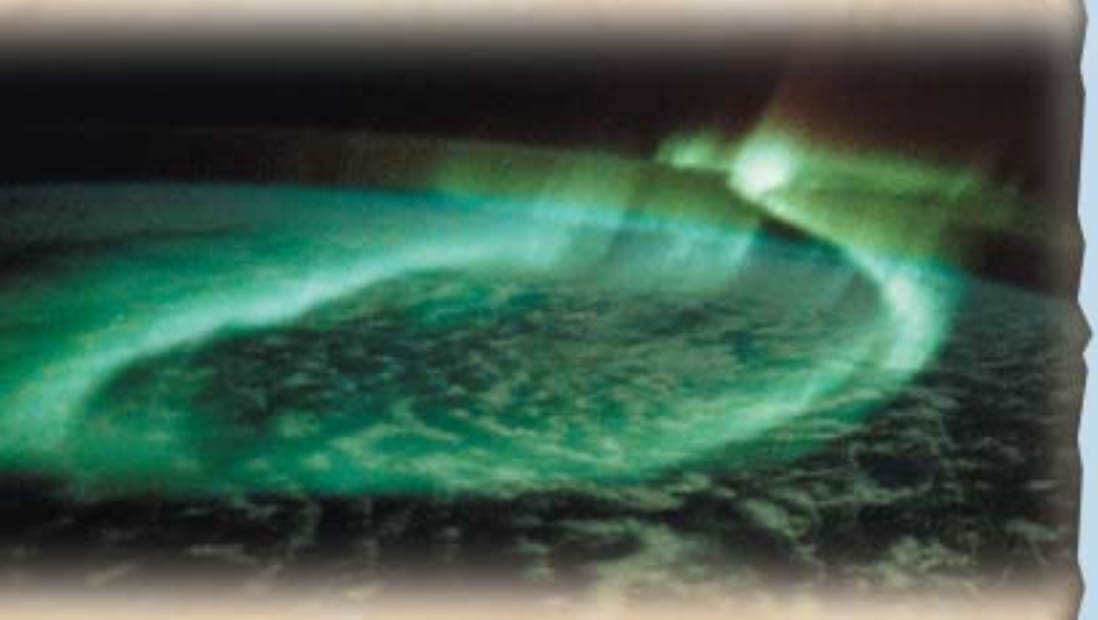
Some geomagnetic storms have caused millions of dollars of damage to business and industry. Emergency services are preparing for a busy weekend. The storms can affect:

- radio, television and telephone signals
- air and marine satellite-navigation instruments
- pipelines (corrosion).

The worst geomagnetic storms can even cut power supplies. In 1989, a geomagnetic storm knocked out the power supply to six million Canadians for nine hours. Experts predict that the two weekend storms will not be as serious as the 1989 storm. However, residents are advised not to panic if phones don't work or power is lost. Phone services and power will be restored within a few hours. Mobile-phone services are likely to be affected for the whole weekend.

The bright side of geomagnetic storms

Although geomagnetic storms can cause a lot of disruption, there is a bright side. The **aurora australis** (also called the southern lights) may be visible at night from parts of southern Australia. Usually this spectacular light show is visible only close to the South Pole. It is caused when the Earth's magnetic field captures particles caught in the solar wind. During a geomagnetic storm there are many more particles, so this natural light show is bigger and brighter. This photo (below) of the aurora australis was taken by a satellite camera.



Don't let a geomagnetic storm ruin your day . . . get

MagNacoat

MAGNACOAT — THE ULTIMATE IN PROTECTION

A single coating protects against the most severe geomagnetic storms.

With magnacoat you can say goodbye to power blackouts, power surges, corrosion, radiation poisoning and electrical interference — guaranteed.

Magnacoat can be painted on anything — gas and oil pipelines, satellites, electrical cables, skin, pigeon feathers, and even your mobile phone.

Buy now and receive a free magnetometer.*

*While stocks last

Activities

REMEMBER

1. What causes geomagnetic storms?
2. How do geomagnetic storms affect the Earth's magnetic field?
3. Why do scientists believe that homing pigeons get lost during geomagnetic storms?
4. What causes the aurora australis?

THINK

5. The particles released by solar flares travel three million kilometres every hour towards Earth. If the Earth is 150 million kilometres from the Sun, how long does it take the particles to reach the Earth?

BRAINSTORM

6. During a geomagnetic storm, compass needles might not point towards the North Pole. Make a list of human activities that this might affect.

INVESTIGATE

7. Find out what the magnetosphere is.

IMAGINE

8. Imagine that there was such a thing as magnacoat:
   (a) What do you think it would be made from?
   (b) How do you think it might work?

✓ checklist

I can:

- ☐ explain how geomagnetic storms on the Sun affect the Earth's magnetic field
- ☐ describe some of the hazards of geomagnetic storms.

Inside atoms

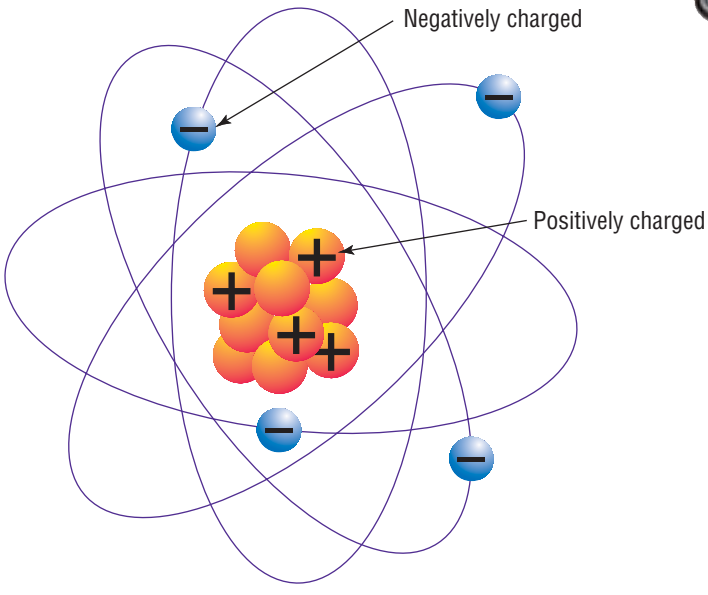
Imagine breaking up an object into smaller and smaller pieces. Eventually, the object cannot be broken up any further. It is a bit like breaking down a brick wall. In the end you just have a pile of bricks.

The small particles that are eventually formed by breaking up objects are called **atoms**. Atoms are the building blocks of all substances, just as bricks are the building blocks of walls.

Atoms are very small particles. They are so small that they cannot be seen with a magnifying glass or a normal microscope. Scientists have studied these particles and found that they are made up of even smaller parts. Some of these smaller parts have an electric **charge**. There are two different types of electric charge: positive and negative.

In and around atoms

Benjamin Franklin, an American scientist who lived in the 1700s, invented many scientific words that are still used when talking about electricity. He was the first person to use the word 'charge'. He also named the two charges found in atoms positive and negative. These charges are similar to the north and south **poles** of **magnets** because they are opposites of each other.



This is a simple model of an atom. The positive charge is in the centre of each atom and the negative charge is carried by very light particles that move around the outside of each atom.

HOW ABOUT THAT!

The central part of the atom is called the nucleus. The nucleus is very small compared with the overall size of the atom. To give you an idea of the size of the nucleus compared with the whole atom, imagine this: If the nucleus of an atom was the size of a marble, the overall size of the atom would be about as big as the MCG!

THEN AND NOW

Democritus

Democritus was a famous philosopher (thinker) from ancient Greece. He was one of the first people to try to describe what all things were made up of. Democritus named the parts that make up all things '*atomon*', which means 'indivisible'. Unfortunately, his ideas about the atom faded away for almost 2000 years. It was not until the 1600s that serious thought went into investigating the atom again. Scientists are still discovering new things about atoms – how they behave and what's inside them.

Go to worksheet 27

Benjamin Franklin — ahead of his time

Apart from 'charge', Benjamin Franklin 'invented' the words battery, **conductor**, **discharge**, uncharged, electrician, and even electric shock. He used the term 'electric shock' after performing many experiments with electricity. One of Franklin's most well-known experiments involved flying a kite in a thunderstorm. A metal key was attached to the kite's string. During the storm, Franklin was able to make a spark jump from the key to his finger. This proved that the storm was electrical.

A balancing act

In most atoms, the amount of positive charge is the same as the amount of negative charge. Atoms are usually **neutral**. We could also say that atoms usually have *no overall charge*. When there is more positive charge than negative charge in an atom, the atom has an overall positive charge. If an atom has more negative charge than positive, it has an overall negative charge.

Tipping the scales of balance in a neutral atom is easy to do. Remember that the negative charges are on the outer of the atom. This part of the atom is the easiest part to change. Removing some of the outside negative charge tips the balance. The atom would then have more positive charge than negative charge.

How about making an atom negative overall? It is difficult to remove the positive charge from the centre of the atom. It is far easier to add more negative charge to the outside of the atom.

Activities

Go to worksheet 28

REMEMBER

1. Why can't atoms be seen with the naked eye?
2. How are atoms similar to magnets?
3. What is balanced in a neutral atom?

THINK

4. Negative charges are found around the outside of atoms. What keeps the negative charges around the positive charges in the centre of the atom?

INVESTIGATE

5. There have been many scientists who have made significant discoveries about the atom. Find out who they are and what their ideas were.

CONNECT

6. Benjamin Franklin began as a newspaper publisher, but at the age of 42 started his scientific adventures. During this time he discovered the lightning rod, lightning bells, bifocal lenses, the Franklin stove, the catheter and many other devices. Choose at least one of Franklin's discoveries and write about how he discovered it, what it does and how it works. To find out more, go to www.jaconline.com.au/science/weblinks and click on the Benjamin Franklin's Inventions link for this textbook.

✓ checklist

I can:

- ☐ list the types of charged particle found in an atom and describe where in the atom they are found
- ☐ describe the difference between positive, negative and neutral atoms.

Lightning strikes

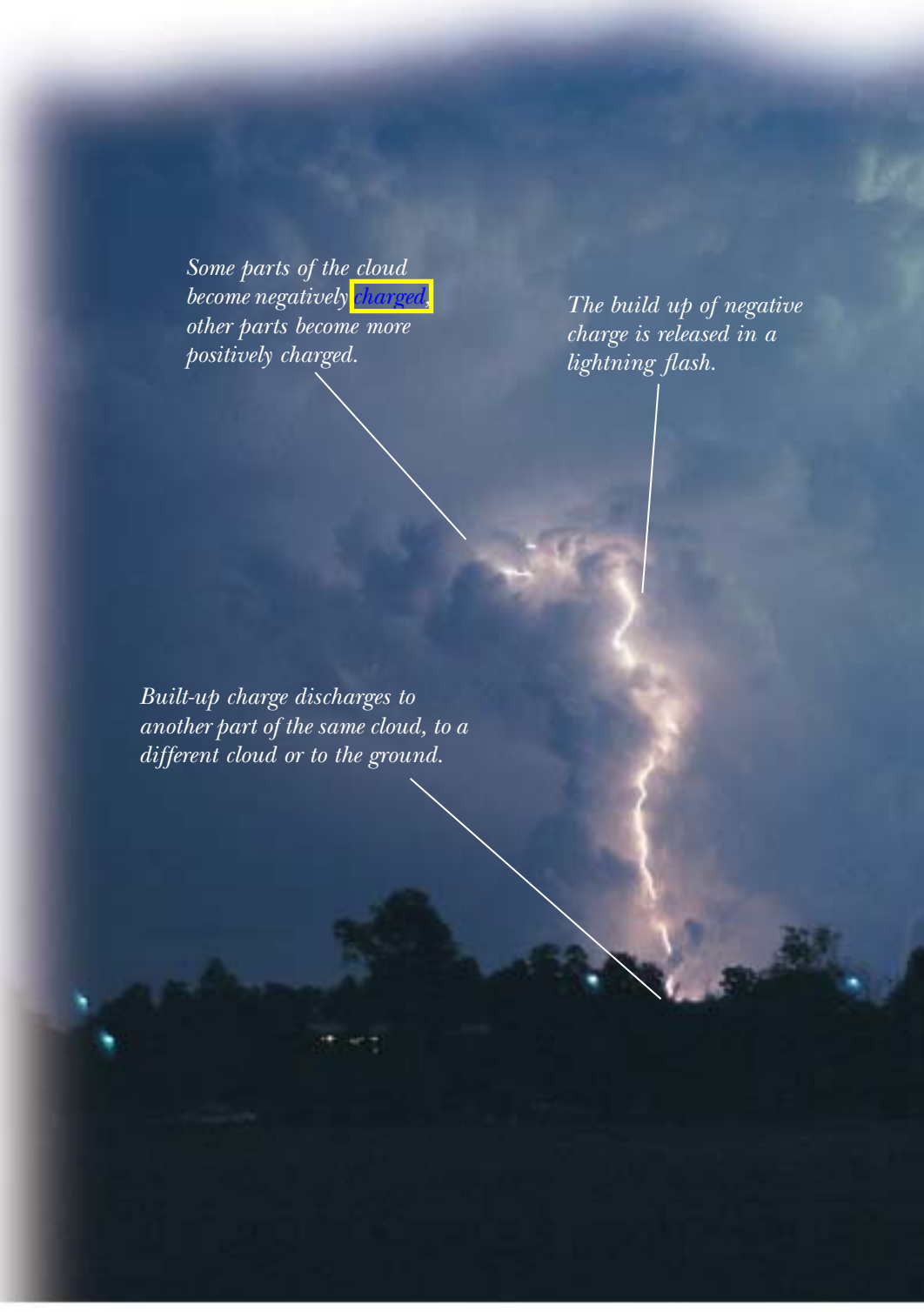
Have you ever taken off your jumper and heard a crackling noise? Ever been 'zapped' by a door handle after walking on carpet? Has your hair ever stood on end after being combed? If so, you have felt the effects of **static electricity**.

Build-up

Static electricity is a build-up of **charge** on an object. The build-up is created by tipping the balance between the amount of positive and negative charge in the **atoms** of the object. When two objects are rubbed together, it is possible for some of the **negative charge** to shift from one object to the other. That leaves one object with an overall negative charge and the other with an overall **positive charge**. This movement of charge is slow and not noticeable.

When sparks fly

*The 'zap' you feel when you reach for the door handle after rubbing the carpet comes from releasing built-up charge quickly. The build-up of negative charge jumps between your hand and the handle in the form of a spark. After the 'zap' you are **neutral** again. The release of built-up charge is called **discharge**.*



Go to worksheet 29

Static electricity on a massive scale!

The particles of water and ice inside clouds are constantly moving against each other. Their movement causes charge to build up in the cloud. Some parts of the cloud become more negative, while other parts become more positive.

The charges keep building up. Eventually, there is so much negative charge built up in part of the cloud that it quickly discharges to another cloud or to the ground below. The result is the spectacular spark we call **lightning**.

A bolt of lightning can measure 20 000 °C, which is much hotter than the surface of the Sun. If it strikes a building, it can cause a huge amount of damage. It is known that lightning takes the easiest path to the ground, so **lightning rods** are attached to the top of tall buildings. It is more likely that lightning will strike the rod, keeping the rest of the building safe.

Although lightning is spectacular to watch, it can also be very dangerous. Make sure you do not talk on the telephone during an electrical storm. Lightning can strike the phone line and travel to every phone on the line. Mobile or cordless phones are much safer. It is also unsafe to be outside during an electrical storm. Take shelter inside a building or in a car. Never take shelter under trees, as they are often struck by lightning.

The heat created during a lightning strike heats the nearby air to a very high temperature. The air suddenly expands and produces the crashing sound we know as thunder.

Activities

REMEMBER

1. What is static electricity?
2. What is the release of built-up charge called?
3. What causes the spark you sometimes feel when you reach for a door handle after rubbing your feet on the carpet?
4. Why should you avoid standing under trees in a thunderstorm?

THINK

5. Explain why a crackling noise is sometimes heard when you take off a jumper.
6. It is possible for cars to build up an overall charge, particularly after a long drive. Suggest a reason for this.
7. Why is talking on a mobile phone safer than talking on a normal cord phone during a thunderstorm?

CONNECT

8. Search the Internet to find out how many people are struck by lightning each year in Australia.
9. During ancient times, there were many myths about lightning. Search the Internet to find out about these and other myths that people believe nowadays.

✓ checklist

I can:

- ☐ explain what static electricity is
- ☐ describe discharge in terms of the release of built-up charge
- ☐ list some safety measures to avoid injuries due to lightning.

The electric force

The spark between objects is not the only sign that **static electricity** is at work. The crackling of sparks is caused by the fast movement of negative **charges**. Sometimes charge between objects can move more slowly towards each other and not make a spark at all.

The charge moves because of an **electric force**. The electric force can pull on charges without touching them. The pull is greatest when such objects are close together.

Two ways to charge

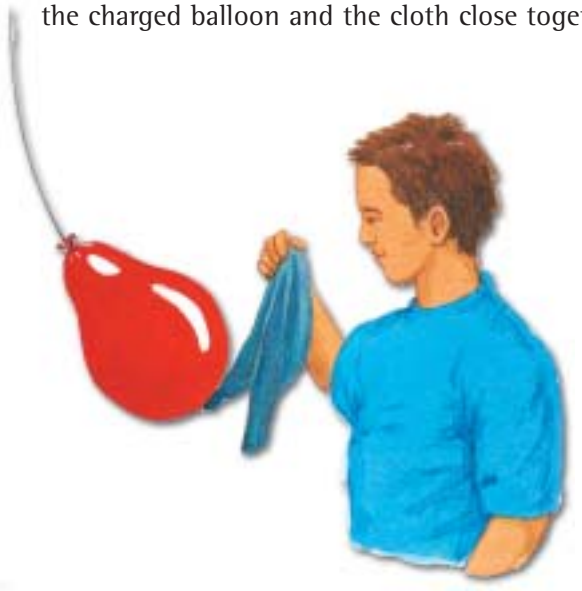
When two objects rub together, some of the negative charges from one object can move to the other. Both objects gain an overall charge.

Another way to transfer charges between objects is to allow them to touch. When a charged object touches a neutral one, negative charges can move between them. The neutral object now has an overall charge.

Can charges push as well as pull?

You will need:
2 inflated balloons
string
rod or stick (approximately half a metre in length)
piece of cloth (wool or silk).

- Tie the string to one of the balloons and hang it over the rod.
- Charge one of the balloons with the cloth. Hold the charged balloon and the cloth close together.



1. What happens to the balloon and the cloth when they are near each other?
2. Do the balloon and cloth have the same charge?

- Charge both balloons with the same cloth. Bring the two balloons close together.

3. Do the two balloons attract or repel this time?
4. (a) If the cloth is positively charged after rubbing, what is the charge on the balloons?
   (b) What if the cloth was negatively charged?
5. Why is it important to use the same cloth to rub both of the balloons?
6. Complete the sentences and label the charges as + or –.

_____ charges repel.

_____ charges attract.

Can a charged object attract a neutral one?

You will need:
perspex rod
silk or wool cloth
thin stream of running water.

- Bring the uncharged rod close to the stream of water.

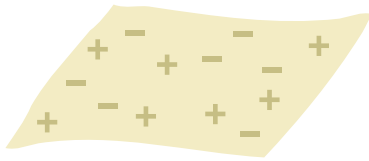
1. Describe what happens.

- Charge the rod with the cloth. Bring the charged rod near the water.

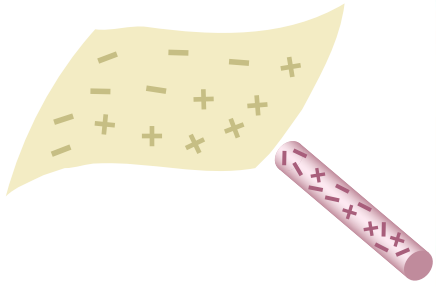
2. (a) Describe what happens to the water now.
   (b) Why does this happen?
3. Why is it important to try this experiment first with an uncharged rod?

Attracting neutral objects

Sometimes uncharged objects, like water and paper, are attracted to charged objects.



Neutral objects have equal amounts of negative and positive charge.



When a charged object comes close to the edge of a neutral object, the charges in the neutral object rearrange themselves. The charged object is attracted to the opposite charges in the neutral object. So, a ***negatively charged*** *object is attracted to the positive charges in a neutral object.*

Activities

REMEMBER

1. When a plastic rod is rubbed with a certain cloth, the rod becomes positively charged. What is the charge on the cloth?
2. Explain how charged objects can be attracted to uncharged objects.
3. State whether the following are true or false:
   (a) Objects with like charges attract.
   (b) Charged objects can attract uncharged objects.
   (c) Two neutral objects repel each other.
   (d) Neutral objects contain both negative and positive charges.
   (e) Objects with an overall negative charge still contain some positive charges.
   (f) If two objects repel, they must be **positively charged.**

THINK

4. How is an electric force like a magnetic force?
5. Describe two different ways to charge a plastic comb.

INVESTIGATE

6. Design an experiment to find out which materials are more likely to build up charge.

Sticky clothes

Have you ever worn clothes straight from the clothes dryer and found that they stick to you? As clothes rub together in a clothes dryer, they can build up charge. If this happens, the charged clothes will be attracted to you when you wear them. If this is not the look you are going for, fabric softeners can help. Fabric softeners coat the fibres of clothes with substances that do not allow static build-up. They make the charge flow away so that it cannot build up.

checklist

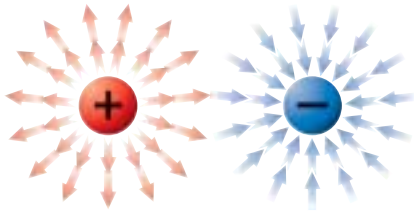
I can:
- ☐ describe how to charge up an object
- ☐ describe the forces between charged objects
- ☐ explain how neutral objects are attracted to charged objects.

Making electric maps

How much does a **charge** push or pull? And in what direction will charged objects move? Like **magnetic fields**, **electric fields** can provide information about the pushes and pulls near a charged object. Electric fields are invisible, but are revealed by other objects affected by charge.

Mapping charges

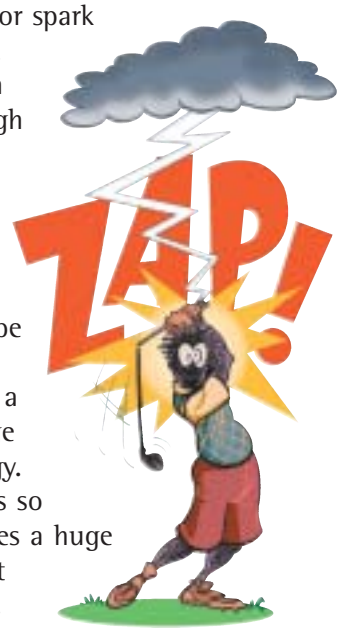
An electric field is like a map of the pushes and pulls caused by the electric charge of an object. Like a magnetic field, an electric field is invisible unless it is revealed by objects that are affected by the charge. When it is revealed, the electric field shows the direction of the pushes and pulls around a charge. It also shows how big the pushes and pulls are. The electric field lines are closest together near the charged object. This is where the **force** is strongest.



This diagram shows arrows pointing in the direction of the electric field around a positive and a negative charge. The electric field or 'map' shows the direction that a positive charge would be pushed or pulled by the **electric force.**

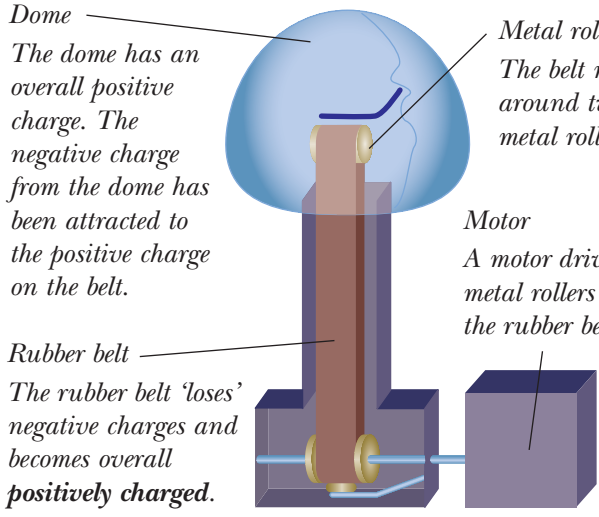
A deadly combination

The 240 V in your home can kill. However, a 100 000 V Van de Graaff generator spark is much safer. Even though each electric charge in a Van de Graaff generator spark carries a lot of energy, there is not very much charge jumping through the air. At home, each electric charge carries less energy, but there is a very large amount of charge moving through the wires. To be dangerous, the electricity has to have a large amount of charge carrying a lot of energy. That's why lightning is so dangerous. It discharges a huge amount of charge that carries a lot of energy.



The Van de Graaff generator

The Van de Graaff generator is a machine that makes **static electricity.** It has a large rubber belt held tightly between two metal rollers. When the motor is turned on, the belt begins to rotate. As it moves, the belt rubs against the metal rollers. Charges move between the belt and the metal rollers. The belt passes on the charge to the large metal dome. The dome becomes charged. Bringing a metal object near the dome will release the build-up of charge on the dome and produce a spark. So, the spark you see is the dome **discharging.**



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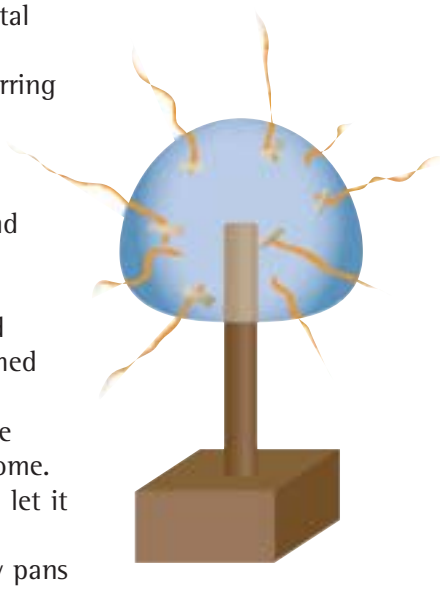
The electric field around a Van de Graaff generator

!CAUTION: Your teacher will carry out this activity. Do not touch the charged dome of a Van de Graaff generator. Always use an earthed testing rod to discharge. Carry out the demonstration while standing on a plastic tray.

You will need:
Van de Graaff generator
aluminium-foil patty pans
several strands of wool.

- Turn the Van de Graaff generator on and let it charge up. Bring the metal testing rod near it.

1. What do you observe occurring between the rod and the dome?
2. Explain your observation. Use words like charging and discharging in your explanation.



- Turn the generator off and discharge it using the earthed metal rod.
- Place the patty pans upside down on the top of the dome. Turn the generator on and let it charge up.

3. What happens to the patty pans once the generator charges up?
4. Discuss and then write down some possible reasons for your observations.

- Turn the generator off again and discharge it.
- Tape several strands of wool onto the dome. Make sure they are spread out over the surface of the dome. Turn the generator on and let it charge up once more.

5. What happens to the wool?
6. Explain why this happens in terms of the charges on the dome and the wool.
7. The wool forms a pattern around the dome. What do you think this pattern represents?

Conductors and insulators

The teacher needs to stand on a plastic tray in the experiment above because the tray is an **insulator**. Insulators do not allow charge to pass through them very easily. By standing on the tray, the charge from the Van de Graaff generator will pass through the metal rod, not the teacher.

The metal rod is a **conductor**. The built-up charge can flow easily through the metal rod, keeping the teacher safe from sparks.

Activities

REMEMBER

1. What two things does an electric field 'map' show?
2. How can you tell where an electric field is strongest?
3. How are insulators different from conductors?
4. Why is a spark from a Van de Graaff generator not as dangerous as lightning?
5. What charge – positive or negative – builds up on the dome of a Van de Graaff generator?
6. How do you discharge a Van de Graaff generator safely?

THINK

7. Water can be a good conductor of electricity. Why do you think it would be unsafe to operate a Van de Graaff generator while standing in a puddle of water?
8. Why do lightning rods need to be made out of materials that are conductors of electricity?

IMAGINE

9. Imagine you are a tiny particle with a small positive charge. Describe what will happen to you:
   (a) near a large positive charge
   (b) near a large negative charge
   (c) near a large neutral object.

checklist

I can:
- ☐ describe the electric field around a charge or charged object
- ☐ explain the behaviour of objects near a Van de Graaff generator
- ☐ explain what conductors and insulators are.

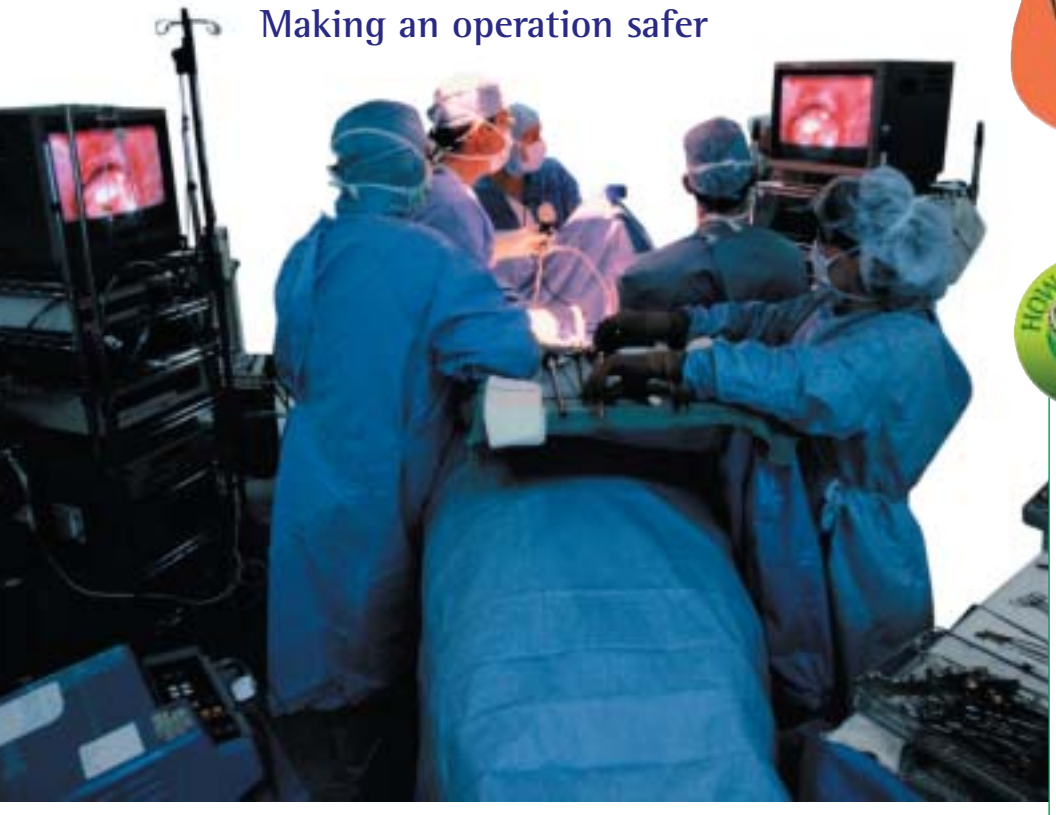
Static: good and bad

Static electricity can be a real nuisance sometimes, but it does have some benefits as well. Sparks caused by static electricity can trigger explosions, but without static-cling, printers and copiers would never work.

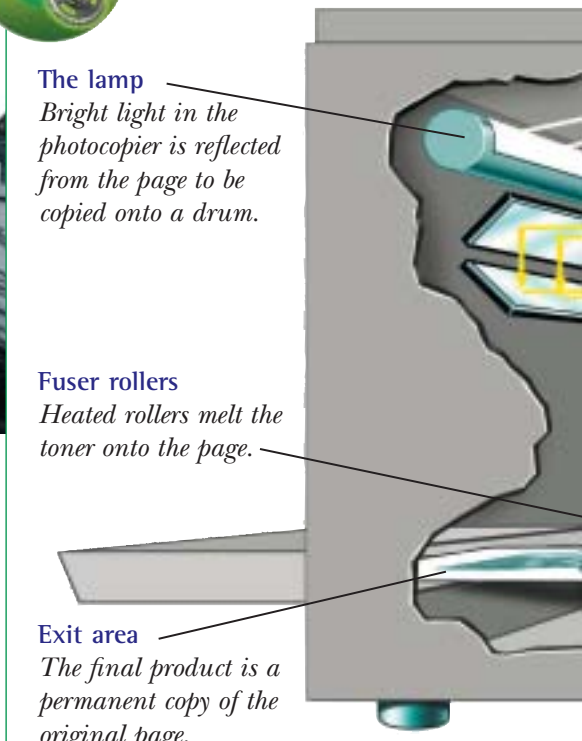
When getting out of a car is a hazard

A moving car is a great **charge** builder. As a car moves, its body 'rubs' against the air and its tyres 'rub' against the road. The rubbing can cause charge to build up on the car and its passengers. As you get out of the car and go to touch the metal body, a spark crosses the small gap between your hand and the metal just before you touch it. Zap!

Making an operation safer



Static electricity is a hazard in an operating theatre. Charge can build up on blankets and **discharge** quickly, causing a spark. Many of the instruments used in an operating theatre can also create sparks. This is very dangerous because gases that could easily explode are used in operating theatres. Doctors and nurses wear gowns made from natural fibres that do not build up electric charge easily. The patient and all of the equipment are **earthed**. An object is earthed when it is making contact with the ground. By earthing the patient and any equipment, charge flows to the ground before it can build up and cause a spark.



The lamp
Bright light in the photocopier is reflected from the page to be copied onto a drum.

Fuser rollers
Heated rollers melt the toner onto the page.

Exit area
The final product is a permanent copy of the original page.

When cleaning makes things dusty

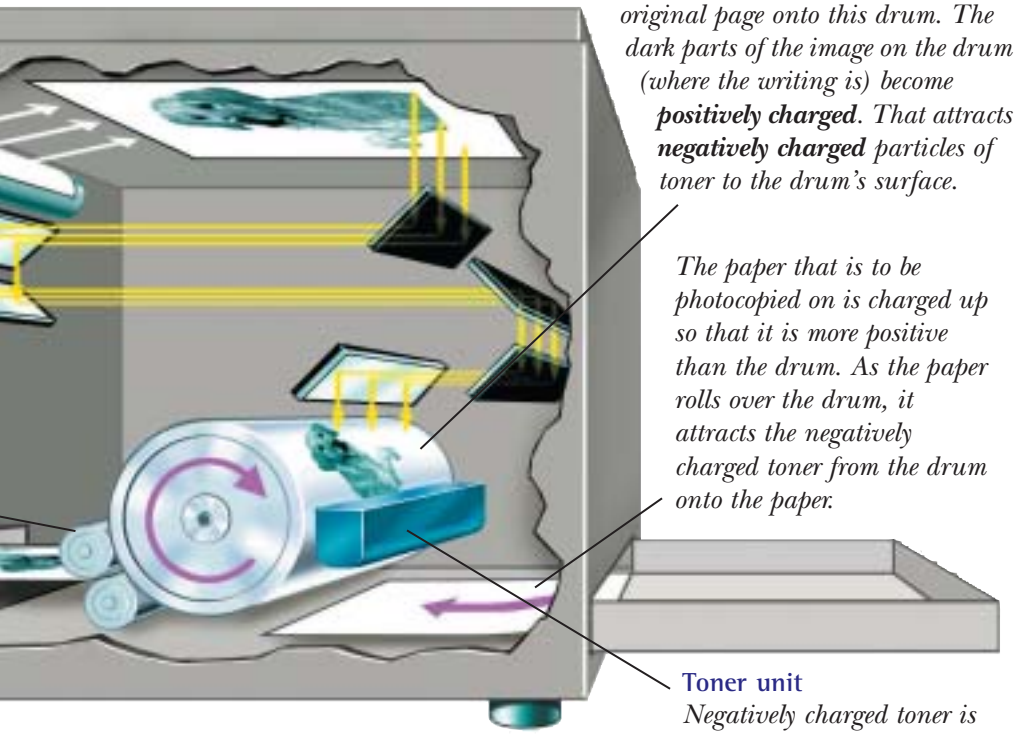
When you use a cloth to wipe over furniture it can sometimes make matters worse. Rubbing leaves the surface with an electric charge that can **attract** small dust particles in the air. The dust particles are **neutral** and will be attracted to either a positive or negative charge left on the furniture.

Using a furniture polish reduces the attraction between the furniture and dust particles by helping built-up charge leak into the air. Charge is less likely to build up and the dust won't come back so fast.

Have you ever noticed that your computer screen is dustier than the rest of the computer? Television and computer screens are excellent dust collectors. Charges build up on the screen, attracting dust particles.

Powder coating

Have you noticed the hard, painted finish on metal toys, bicycles, car-bumper bars and fridge doors? The paint is a long-lasting powder coating. Static charges are used to apply coloured powder to metal surfaces. As the powder shoots out of a nozzle, it is charged up and attracted to the metal's surface. Special ovens heat the powder and turn it into a smooth, tough surface. This method of painting gives an even coat of colour. One difficulty is that dust can also be attracted to the newly painted surface. Great care must be taken to keep dust away when the coating is applied.



Drum
Bright light is reflected from the original page onto this drum. The dark parts of the image on the drum (where the writing is) become ***positively charged****. That attracts* ***negatively charged*** *particles of toner to the drum's surface.*

The paper that is to be photocopied on is charged up so that it is more positive than the drum. As the paper rolls over the drum, it attracts the negatively charged toner from the drum onto the paper.

Toner unit
Negatively charged toner is stored here.

Activities

REMEMBER

1. Why are patients earthed during an operation?
2. How does charge build up in a car?
3. Why are heated rollers used in photocopiers?
4. Why can the screen of a television be dustier than the top of a television?

THINK

5. Why do some cars sometimes have strips attached to the bumper bar that touch the ground as the car moves?

6. Planes build up a lot of charge as they fly through the air. Explain why planes need to be earthed when they are re-fuelled.

INVESTIGATE

7. Plan and carry out an experiment that uses static electricity to separate salt and pepper. Find out why static electricity works for separating these mixtures and not others.
8. A static precipitator is used to remove soot particles from smoke. Find out how a static precipitator works. Draw a diagram to help with your explanation.

checklist

I can:

- ☐ list some of the dangers of static electricity
- ☐ list some practical uses of static electricity
- ☐ explain how and why objects are earthed.

Check and challenge

Magnetic and electric effects

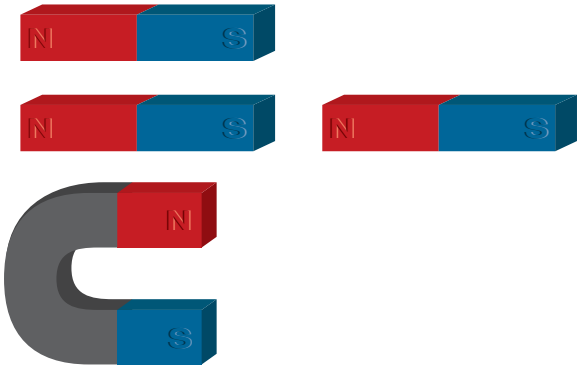
Magnetic pieces

1. Label the poles formed when this magnet was cut in half.

2. What is a domain?
   - A The space around a magnet or magnetic material
   - B The end of a magnet
   - C The edge of a magnet
   - D A 'mini magnet' found in magnetic materials
   - E A 'map' of the pushes and pulls around a magnet.

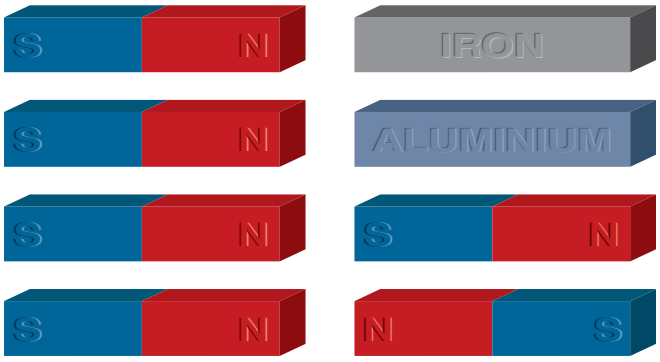
Magnetic field patterns

3. Copy these magnets and draw a map of the pushes and pulls around them.



Attraction and repulsion near magnets

4. Which of the following objects will:
   - (a) pull together?
   - (b) push apart?
   - (c) remain as they are?



5. Which part of a magnet has the greatest pulling effect?
   - A The edges
   - B The poles
   - C The domains
   - D The middle
6. By looking at a map of pushes and pulls around a magnet, how can you tell where the greatest pulling or pushing effect will occur?

The magnetic compass

7. Describe how a magnet could be used to find the magnetic North Pole of the Earth.
8. Re-draw this diagram. Show clearly the direction that a compass needle would point if it were placed in position A and position B.

9. Would a compass work if the needle were replaced with a piece of aluminium foil? Explain your answer.

Experimenting with magnets

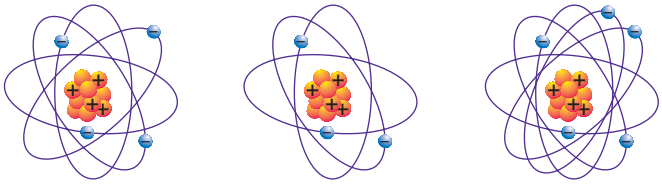
10. You can make your own compass by stroking a steel needle with a permanent magnet.
    Explain why you need to stroke the needle in one direction only. Use the word 'domains' in your explanation.
11. The following information refers to electromagnets made from wire coiled around an iron nail.
    List the electromagnets in order from strongest to weakest.

Electromagnet	Voltage (V)	Turns (coils) of wire
A	5	15
B	2	15
C	5	20

Atoms

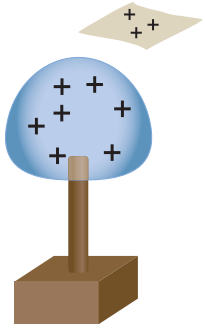
12. Why are the negative charges of an atom easier to remove than the positive charges?

13. Which of the following atoms is positively charged, which is negatively charged and which is neutral?



Attraction and repulsion in electric fields

14. This Van de Graaff generator has an overall positive charge.
    (a) In which direction would a positively charged piece of paper move if it was brought near the dome?
    (b) In what direction would a negatively charged piece of paper move?
    (c) Draw electric-field lines around the Van de Graaff generator.



15. Explain, with the aid of a diagram, how a neutral piece of paper is attracted to a charged plastic pen.

Static electricity

16. Imagine rubbing your shoes on the carpet, then reaching for the door. Why do you feel a spark if you reach for the metal door handle, but not if you reach for the wooden door?
17. Some objects produce a spark during static discharge and others don't. Which types of material tend to produce a spark?
18. Look up the word 'static' in the dictionary and then explain how static electricity gets its name.

Aeroplanes need to be earthed before they are refuelled because they build up large amounts of charge when they fly. If sparks were to jump between the plane and the fuel hose, a massive explosion could occur.

19. (a) Petrol tankers could build up charge travelling on the road. Describe how this could happen.
    (b) Describe how the tanker could safely release any built-up charge.

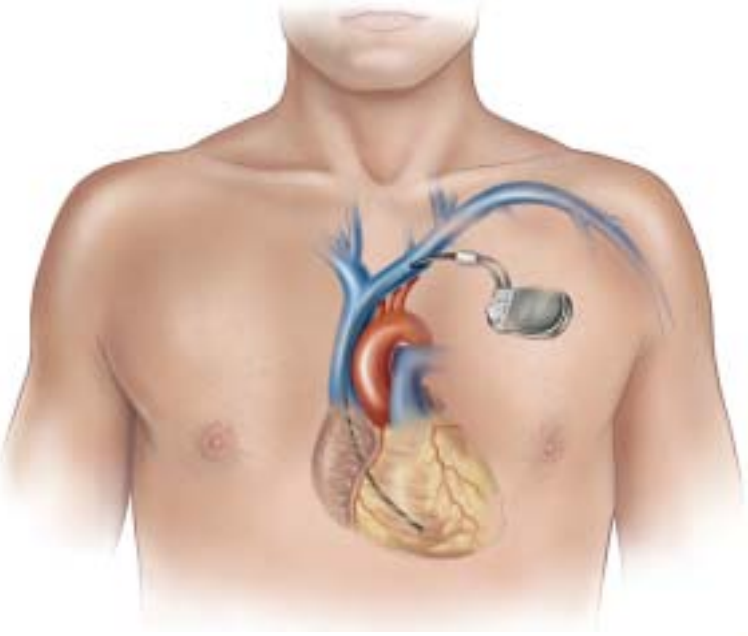
20. The spark between a cloud and the ground is caused by the release of built-up charge in the cloud.
    (a) What is the release of built-up charge called?
    (b) Where does the built-up charge in clouds come from?

Comparing magnetic and electric effects

1. Why do you think magnetic and electric forces are sometimes called 'non-contact forces'?
2. How are the poles of a magnet similar to the charges in an atom?
3. Make a list of the similarities and differences between magnetic and electric fields.

Medical magnets

4. A Magnetic Resonance Imaging (MRI) machine is a very strong magnet that is used to scan for diseases in the human body. Would the magnets used in this type of machine be permanent, temporary or electromagnetic? Give a reason for your answer.

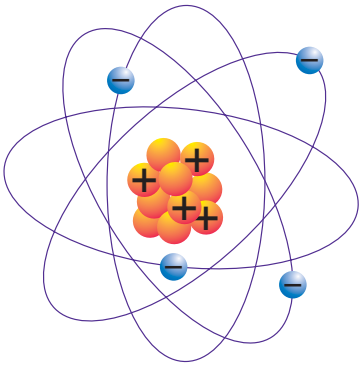


An artificial pacemaker is a device used by people with abnormal heartbeat patterns. The pacemaker works by giving the heart a small electric shock to 'remind' it to beat. Pacemakers are programmed like a computer. If the information programmed into the pacemaker needs to be changed, it can be wiped using a strong magnet.

5. Why do you think people with pacemakers should avoid touching a Van de Graaff generator?
6. Why must people with pacemakers avoid MRI machines?
7. Why is it important for doctors, patients and equipment to be earthed in an operating theatre?

Summary of key terms

atom: a very small particle that makes up all things. Atoms have the same properties as the objects they make up.



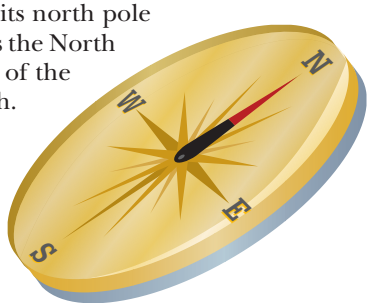
attract: a pull towards another object

aurora australis: a spectacular light show mainly visible from close to the South Pole. It is caused by solar particles interacting with the Earth's magnetic field. It is sometimes called the southern lights.

charge: (*noun*) a property of all objects. Charge can be positive or negative. There are some particles inside an atom that have no charge; (*verb*) to give an object an overall electric charge by adding or removing negative charges. Objects can be charged by rubbing.

charged: describes an object that has more of one type of charge than the other.

compass: a small magnet that is free to move in a circle. The magnet turns so that its north pole faces the North Pole of the Earth.



conductor: materials that allow charge to flow through them

discharge: to release built-up charge

domain: a 'mini-magnet' found in magnetic materials

earthed: to provide a path from an object to the ground for charge to flow through

electric field: lines that show the size and direction of an electric force. The size of the electric force is shown by how close the lines are together.

electric force: a push or pull towards or away from a charged object

electromagnet: a magnet formed by wrapping a coil of wire around an iron core. When electricity passes through the coil, the iron core becomes magnetic.

engineer: a person who uses scientific ideas to design and build new technology and make it work

force: a push, pull or twist

geomagnetic storm: an outburst of energy from the Sun that causes particles to speed towards Earth and interact with the Earth's magnetic field

insulator: material that does not allow charge to flow through it

lightning: the spark caused when built-up charges in a cloud discharge quickly to other clouds or the ground

lightning rod: conductive metal rod, often attached to the top of tall buildings. Lightning rods help to keep buildings safe from lightning strikes.

magnet: a piece of iron, cobalt or nickel (or combination of these) that pushes or pulls on other objects containing these same metals

magnetic field: lines that show the size and direction of a magnetic force. The size of the magnetic force is shown by how close the lines are together.

magnetic force: a push or pull between magnets and magnetic objects

magnetic North Pole: the place on Earth that the north pole of a magnet is attracted to

magnetic South Pole: the place on Earth that the south pole of a magnet is attracted to

magnetise: to make into a magnet

magnetite: a naturally-occurring magnetic rock

metal detector: a machine used to locate underground materials containing iron, nickel or cobalt

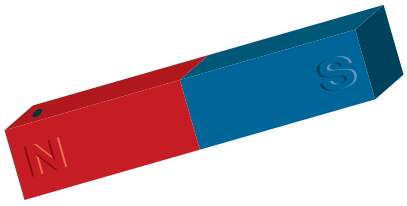
negatively charged: having more negative charges than positive charges

neutral: having the same amount of positive and negative charge

non-contact force: a push or pull felt between objects that are not touching each other

permanent magnet: a magnet that retains its magnetic effect for many years

pole: the ends of a magnet. One pole is the south pole, the other the north pole.



positively charged: having more positive charges than negative charges

receiver: a device that receives magnetic or other signals

repulsion: a push away from another object

solar flare: an outburst of energy from the Sun

static electricity: a build-up of charge

temporary magnet: a magnet that stays magnetic while it touches a permanent magnet, or one that is magnetic for a very short time

CHAPTER 7

Cells

When you play a computer game, you move a hero about, drive a car, fly a plane, or even speed through space. If you are good at the game, your hero survives, beats the clock or wins the race. The images you see on the screen are made up of tiny boxes, or pixels, of colour on the screen. These boxes have different colours. Groups of boxes of the same colour work together to make one object — like a rock. Or the tiny boxes of colour may just form part of an object — like a cloud.

Your body is made up of very tiny 'boxes'. These are called cells. You have about 10 trillion of them! To carry out the various tasks your body needs, you have more than 200 different types of cell, including muscle cells, nerve cells, liver cells and skin cells. Similar types of cell are grouped together in your body. They form the tissues and organs that carry out everyday tasks like breathing, walking, digesting food and even sleeping.

You will discover

What a cell is
Ways to see cells
How cells keep you alive
Why cells are so small
How and why plant and animal cells are different

1 How is your body like the image on a computer screen?
2 What do the cells in your body really look like?
3 Why do different types of cell have different shapes?
4 What happens inside your cells to keep you alive?

Health & Beauty Weekly

28 November 2008 Weather: Fine, 22°C

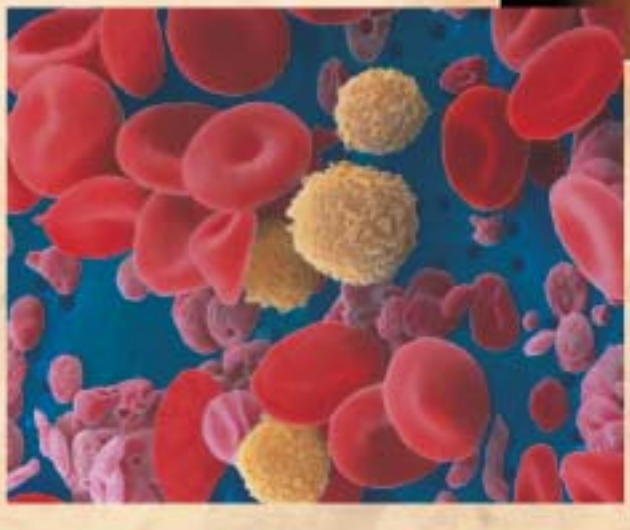
Inside Naomi

by Mike Roscopic

Supermodel and fashion designer Naomi Daniel's latest exhibition called *The Beauty Inside* promises to be a huge success. I headed off to the gala opening last week to find the answer to the question that everybody is asking, 'Is beauty just skin deep?'

Naomi's manager Art Buckle explained as he met me at the entrance. 'Naomi is just like every other human being … on the inside. We are all made up of tiny **cells**. Each of us has trillions of these cells in our body. And each human cell is a beautiful packet of life', he said.

The exhibition is packed with photographs that show what Naomi looks like inside. There are poster-size images of blood cells, skin cells, muscle cells, brain cells, lung cells and stomach cells. Art explains, 'Most people don't know what they look like on the inside. Each one of these cells helps keep Naomi alive'.



According to Art, the inside of Naomi's body, just like all other humans, is at least as beautiful as the outside. As we admired the huge **photomicrograph** of human blood (shown at left), he pointed out the white blood cells, red blood cells and platelets. He explained, 'The red blood cells deliver **oxygen** to every cell in your body, the white blood cells fight disease and the smaller platelets clump together to plug up any holes that are made in your blood vessels'.

He added, 'Each cell in your body is constantly taking in food and oxygen, and removing wastes'.

Go to worksheet 31

Living takes teamwork

As we moved onto a photograph of a soccer team I asked Art what this has to do with cells inside Naomi's body. His answer made sense. 'Your cells work together as a team to keep you alive', he said. He told me that cells in a body are a bit like that soccer team. 'There are lots of different kinds of cells in your team. Some are your running cells — your muscles. Others are your thinking cells — your brain cells. They all work together as a team, each playing their role in helping you move, eat, drink, sleep, play soccer, sing or play the guitar.'

It turns out that if you are looking at the inside of a human body, you just can't tell who's who. Art stood in front of two photographs of skin cells under a microscope. He asked, 'Which one is Naomi's?'. They both looked the same to me so I couldn't answer. Just as well — the skin cells belonged to a newborn baby! I didn't feel so silly when Art said, 'The amazing truth is that, on the inside, we all look just the same'. He added, 'The muscle cells of a newborn baby are about the same size as your muscle cells, and the same as those of supermodel Naomi. You and Naomi just have many more muscle cells than a baby'.

Is your skin dull and dry?

Overexposure to the Sun or air-conditioning can dry out your skin cells.

New, improved **CELL GELL** replaces water in your skin cells with its deep, penetrating 'hydrospheres'.

*If you order this month, you'll receive a **free** packet of **CELL SCRUB**, our special soap that helps remove dead skin cells to expose fresh, new skin cells.*

Big ones, small ones

The exhibition shows some of the largest and smallest cells that exist. Art says you can often work out what job a cell does by looking at its shape and size. In front of us was a display of a real egg from an ostrich. The label on the ostrich egg said:

> *Most cells can be seen only with a microscope. But in this egg is the world's largest single cell. The egg weighs 1.8 kg. The cell inside is more than 10 cm across.*

Art asked, 'Why do you think this one cell is so large?'. I just took a guess and crossed my fingers. I said, 'An egg cell has to store lots of food'. Luckily I was right. Art explained in more technical terms: 'You're right. A bird's egg cell has lots of **cytoplasm** which contains the stored food'. He went on to ask, 'Did you know that an ostrich egg is 1500 times bigger than a human egg?'. This makes sense too. A bird lays eggs. Their eggs need to have many days worth of food inside them to help the chick grow and hatch out of the egg. Women don't lay eggs because the growing baby stays inside the body. Human egg cells don't have to store as much food as bird eggs because the growing human baby gets its food directly from its mother and not from inside an egg.

Naomi's exhibition is well worth a visit. I had a great time at the museum and learnt heaps! I came away feeling confident that I knew the answer to that big question. *Beauty is certainly NOT just skin deep!*

Activities

LOGICAL/MATHEMATICAL

REMEMBER

1. What is Naomi and every other human (and every animal and plant) made up of?
2. Which cells are your thinking cells?
3. Which animal has the largest cell in the world?
4. Why does the egg cell of a bird need to be large?
5. Why is a human egg cell smaller than an ostrich egg cell?

THINK

6. How are your cells like a soccer team?
7. Why can't you tell the difference between Naomi's, a newborn baby's and your own skin cells?
8. If a newborn baby's cells are the same size as Naomi's cells, why is the baby smaller?

INVESTIGATE

9. Use the 'Summary of key terms' on page 168 to find out what a photomicrograph and cytoplasm are.
10. Find out what a human sperm cell looks like and suggest a reason for its shape.

CREATE

11. Form a small group and create your own exhibition that shows that there is beauty below the skin.

✓ checklist

I can:

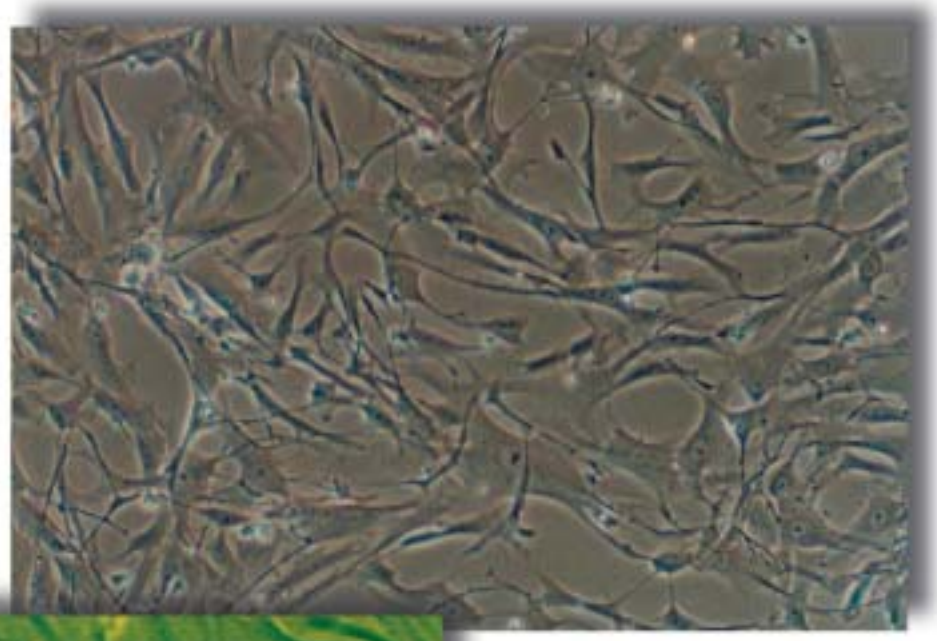
- ☐ explain that humans are made up of many tiny cells
- ☐ recognise that different types of cell in the human body work together as a team.

Living building blocks

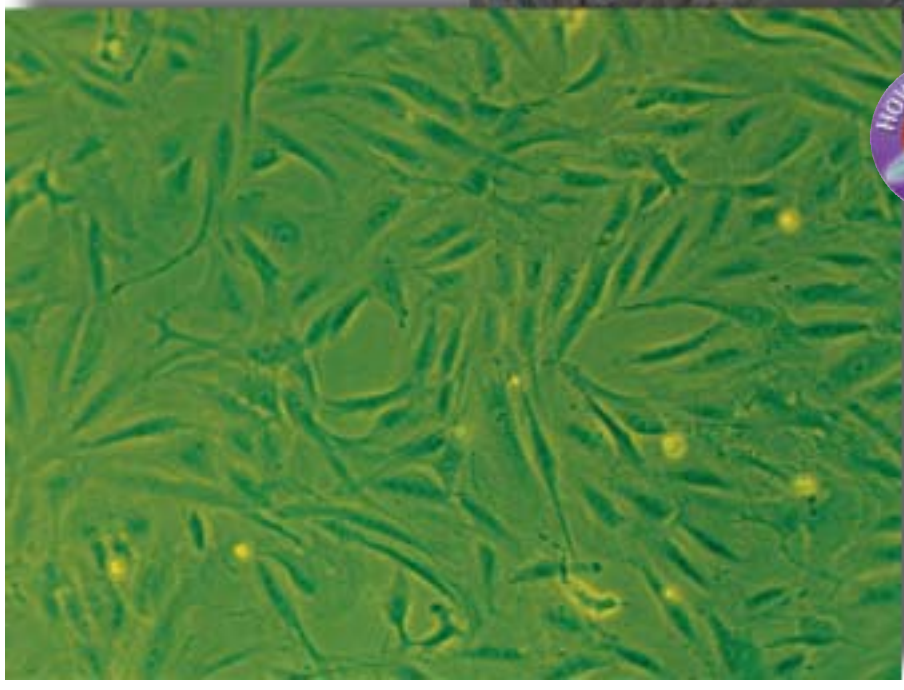
You, and every animal and every plant, are made of **cells**. Cells are the building blocks of life — the smallest units of living things. The cells of an ant, mouse and elephant are all quite similar to your cells. In fact, the cells of an elephant are about the same size as your cells, and those of a mouse or an ant. The elephant is bigger than you simply because it has many more cells, and a great many more than an ant.

The cells in all animals and plants work together to perform the many jobs needed to keep them alive.

There are many different types of cell in your body. Their size and shape depends on which job they do. All cells are very, very small. The cells in your blood would need to be magnified ten times before you could ever see that they were there.



Human muscle cells look very similar to the muscle cells in an elephant, a dog or a rat. That's because they do a similar job. These cells come from the muscles in an artery — human on the top, rat on the bottom.

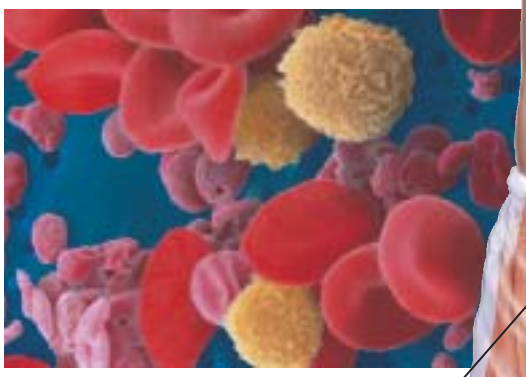


Cell count

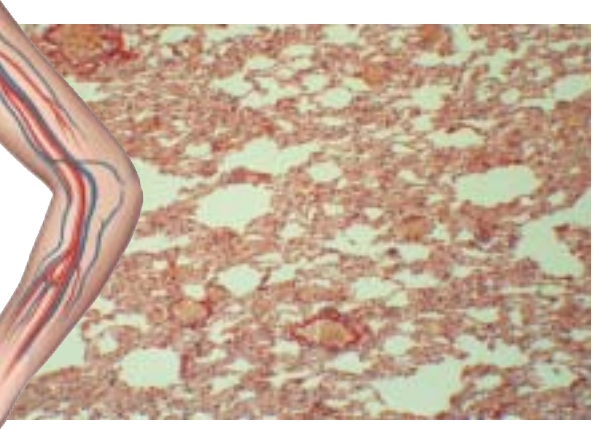
An adult human body is made up of more than one million million cells. That's amazing when you consider that a human life begins in the mother's **womb** with only two cells. New cells are being produced all the time, even when you've stopped growing. That's just as well, because while you are alive, about 50 million of your cells die every second and need to be replaced.

Oxygen enters your body through your respiratory system: the nose, windpipe and lungs. Your blood collects oxygen from the lungs before taking it to other cells. There are lots of air spaces between the groups of cells.

*The circulatory **system** moves blood around your body. It includes your heart, veins and arteries. Most of the cells in your blood are red blood cells. These cells carry **oxygen** to all of the other cells in your body.*

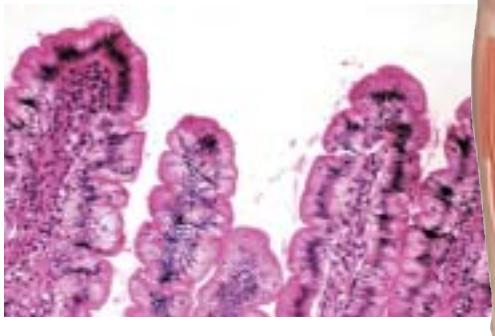


Human red blood cells



Human lung cells

The digestive system involves the mouth, liver, stomach, small intestine, large intestine, rectum and anus. In your small intestine, cells are lined up along the edges of finger-like structures. Digested food passes through these cells into your blood.



Human small intestine cells

Activities

REMEMBER

1. How does the size of your cells compare with those of an elephant?
2. Which has more cells – an adult elephant or an adult human?
3. Why aren't all cells in your body the same shape and size?
4. Describe a human red blood cell.

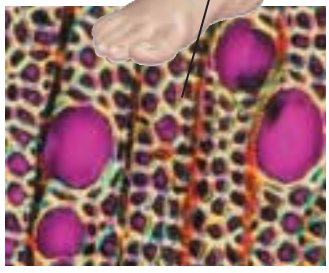
THINK

5. Would you expect the red blood cells in a mouse to be the same shape as human red blood cells? Give a reason for your answer.
6. Explain why cells are sometimes called the building blocks of life.

BRAINSTORM

7. List at least 10 types of cell found in your body.

Bones and muscles help support and move the body. The bone cells work together to provide strength. The muscle cells work together to stretch or relax.



Human bone cells

Human muscle cells

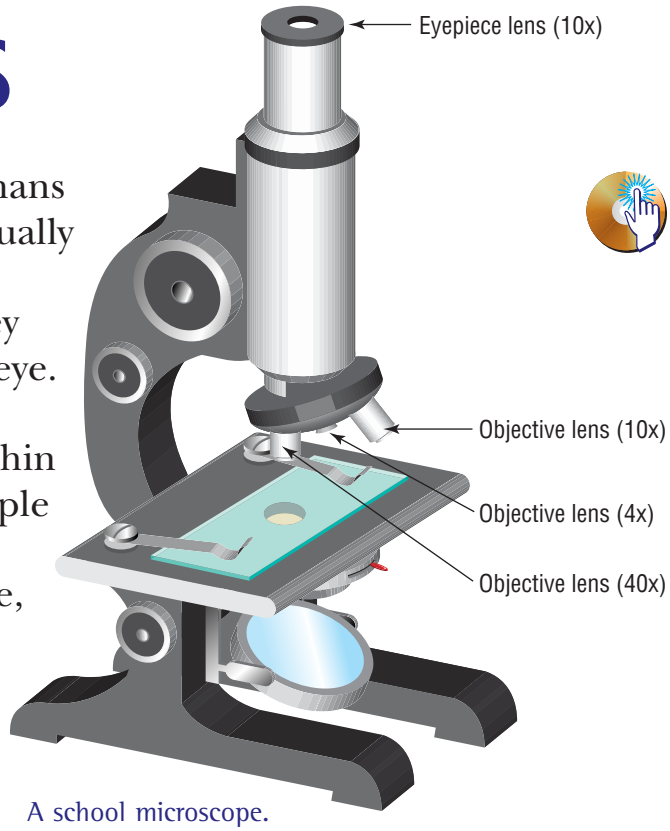
✓ checklist

I can:

- ☐ explain that cells are the smallest living units of plants and animals
- ☐ recognise that the shape and size of cells depends on the job they do.

Seeing cells

Cells existed long before the first humans walked the Earth … but no-one actually saw a cell until the seventeenth century. This is because cells are so small that they cannot be seen with the unaided human eye. In 1665, Robert Hooke became the first person to see cells when he looked at a thin slice of cork through one of the first simple **light microscopes**. Seeing the cork cells magnified to about 30 times their real size, he described them as 'many little boxes'. The microscopes used in schools today can magnify things to about 400–500 times their real size.

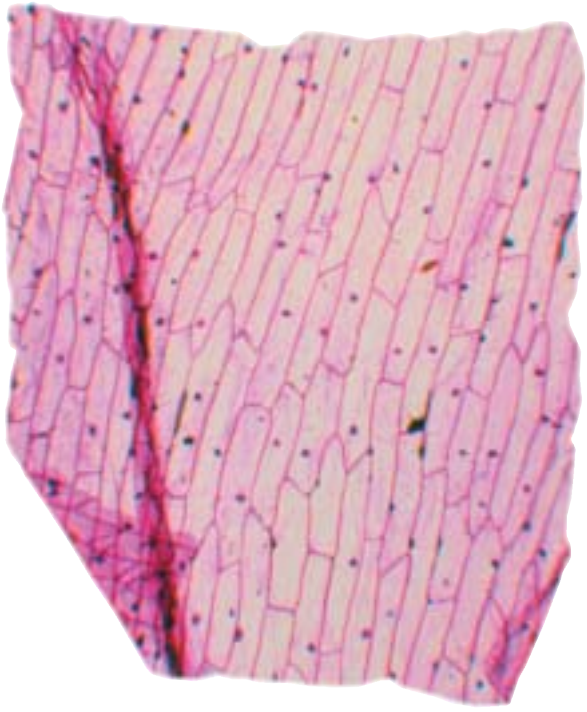


A school microscope.
Not all school microscopes look exactly like this. Your school's microscopes might not have all of the features of this one. They might even have additional features.

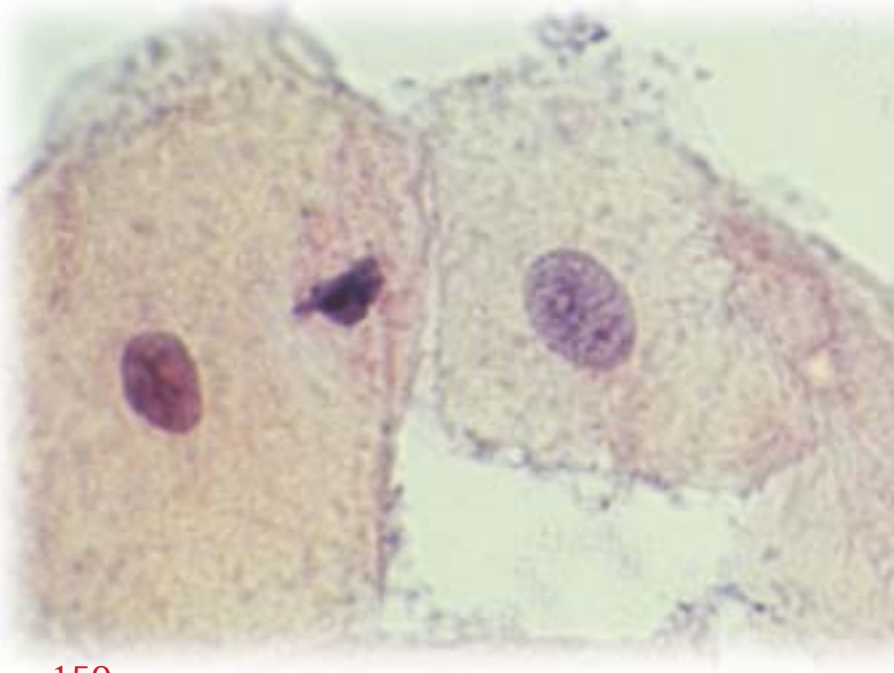
The microscopic world of cells

To calculate the total magnification of a microscope, multiply the magnification written on the eyepiece by the magnification written on the objective lens. For example, a 10× eyepiece and a 40× objective lens would provide 10 multiplied by 40 = 400× magnification. This combination magnifies things to 400 times their actual size. You can use different combinations of microscope eyepieces and objective lenses to get different magnifications.

These onion cells form the skin of the onion. They have been magnified to nine times their real size.



Human cheek cells are about 0.05 mm wide. The smallest object visible to the unaided eye is one-tenth of a millimetre or 0.1 mm across. Because cells are smaller than 0.1 mm across, we need a microscope to see them. This photograph (below) shows the cheek cells magnified to 1000 times their real size.

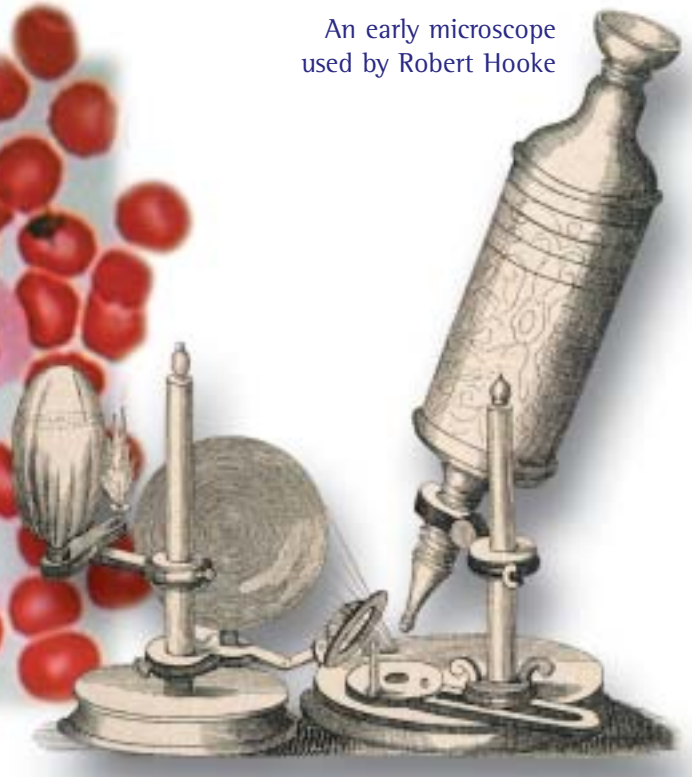


Go to worksheet 32

The first light microscope magnified about 30 times. Today's light microscopes magnify up to 1500 times. **Electron microscopes** are even more powerful and magnify up to 1 million times.

Human blood cells as seen through a light microscope

An early microscope used by Robert Hooke



An electron microscope

Human red blood cells as seen through an electron microscope

Making a wet mount: Onion cells

Read all the instructions before you begin. You can read more about using a microscope in the Laboratory Toolbox (pages 247–8).

You will need:

microscope
clean microscope slide
coverslip
dropping bottle of water
forceps (tweezers)
scalpel
toothpick
small section of a peeled onion
blotting paper.

- Use the dropper to place a drop of water on a microscope slide.

- Use forceps to peel a small piece of the very thin, almost transparent onion skin from the inside surface of the onion.

- Put the piece of the onion skin into the drop of water on the microscope slide.

1. Set the microscope to low magnification (see page 247). What is the total magnification?
2. Draw and label a group of cells. Is your drawing large enough to see clearly?
3. Set the microscope to a higher magnification. What is the total magnification?
4. Find the same group of cells that you were looking at on low magnification. Draw them again at this higher magnification. Add any extra detail that you can now see.

- Place a coverslip over the top of the water containing the onion skin. Use a toothpick to lower the coverslip gently to avoid air bubbles. Use blotting paper to soak up any excess water outside the coverslip.

Drawing cells – size matters!

A key skill in using microscopes is accurately drawing what you see.

- Your drawings of cells should be large enough to show detail.
- Look down the microscope with one eye but keep both eyes open. Don't squint! Squinting is hard work for your eye muscles.
- Choose two or three typical cells to draw.
- Use a sharp HB pencil.
- Give your drawing a title and include the date (e.g. Onion cells, 2 May 03).
- Include the magnification (e.g. 400×).

Staining onion cells

Read all the instructions before you begin. You can read more about using a microscope in the Laboratory Toolbox (pages 247–8).

!CAUTION: Take care when using stains. Stains can stain your clothing and books. Always keep stains where they won't be knocked over. Replace droppers on bottles promptly after use. Use newspaper or other covering to protect tables.

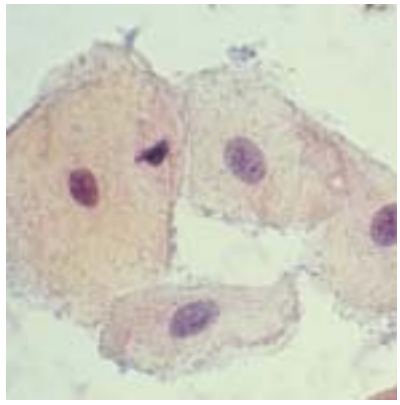
You will need:
microscope — microscope slide
dropping bottle of methylene blue or other similar stain (check with your teacher)
coverslip — forceps
small section of a peeled onion
dropping bottle of water — toothpick.

- Use the dropper to place a drop of methylene blue (or other stain) in the centre of a microscope slide.
- Use forceps to place the section of onion in the drop of stain. Allow the stain to soak in for about three minutes. Then use blotting paper to soak up any excess stain.
- Place one drop of water on the section of onion.
- Use a toothpick to lower the coverslip gently onto the slide.
- View the sample with the microscope on low power, then on high power.

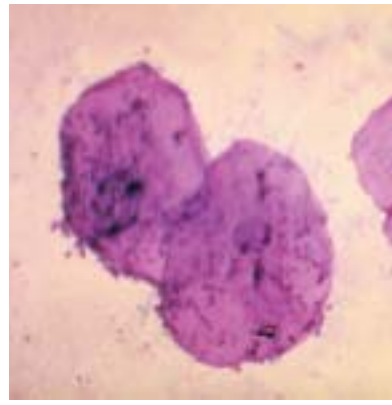
1. Make a drawing of a group of four or five cells. Include as much detail in your drawing as possible.
2. Give your drawing a title and date. Don't forget to include the magnification.
3. What difference does the stain make to your observation of onion cells?

Adding colour cells

It is much easier to see the details of cells if they are coloured with stains. Unstained cells are harder to see because cells do not normally have much colour in them.



Unstained human cheek cells



Stained human cheek cells

Activities

REMEMBER

1. What is the smallest size we can see with the unaided eye?
2. In which century was the light microscope invented?
3. Which type of microscope can magnify objects up to:
   (a) 1 million times?
   (b) 1500 times?
4. Why are stains used to make microscope slides?

THINK

5. What total magnification do a 10× eyepiece and a 100× objective provide?

INVESTIGATE

6. Find out how a light microscope works.
7. Find out how an electron microscope works.
8. Make a table to compare the light and electron microscope using these headings: 'Magnification', 'Advantages of use', 'Disadvantages of use'.
9. Observe the following under the light microscope: cat hair, dog hair and human hair. Suggest why hair can be used to identify some animals.

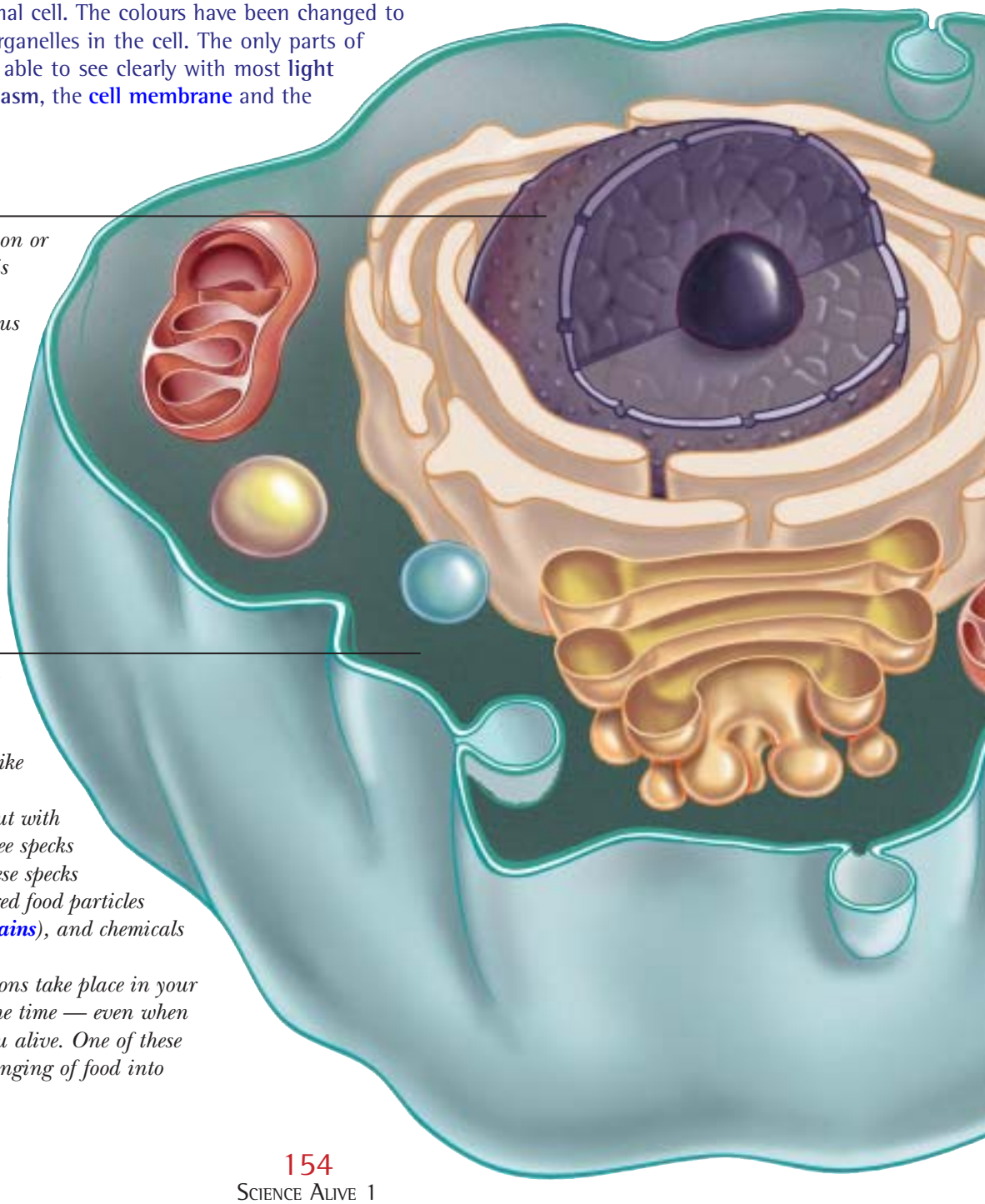
✓ checklist

I can:
- ☐ explain why microscopes are used to look at cells
- ☐ use a microscope to see cells
- ☐ make a wet mount
- ☐ stain cells
- ☐ make accurate drawings of cells.

Inside cells

Most of the **cells** in your body take in food and **oxygen**, remove wastes, and produce new living cells. Your cells control the amount of water they hold, produce important chemicals, react to changes in the environment and reproduce to give more cells. So can the cells of every animal and plant. Every cell contains 'little organs' called **organelles**. There are different types of organelle and each type plays a role in helping the cell survive.

A look inside a typical animal cell. The colours have been changed to make it easier to see the organelles in the cell. The only parts of the cell that you would be able to see clearly with most **light microscopes** are the **cytoplasm**, the **cell membrane** and the **nucleus**.



Nucleus

*The nucleus is the information or control centre of the cell. This roundish organelle controls what the cell does. The nucleus contains **chromosomes** that carry information that is passed on to newly created cells. The nucleus looks like a clear globule in a living cell. It appears much darker than the cytoplasm when the cell is stained.*

Cytoplasm

*The cytoplasm is the jelly-like material inside a cell. In a living cell, the cytoplasm is usually moving about. It is like mucus in texture — slimy! Cytoplasm is mostly water, but with a light microscope you can see specks and blobs floating in it. These specks and blobs are organelles, stored food particles (like fat globules or **starch grains**), and chemicals such as salts and proteins.*

Important chemical reactions take place in your cytoplasm. These occur all the time — even when you are asleep. They keep you alive. One of these chemical reactions is the changing of food into energy for your cells to use.

Go to worksheet 33

MI VISUAL/SPATIAL

Why are cells so small?

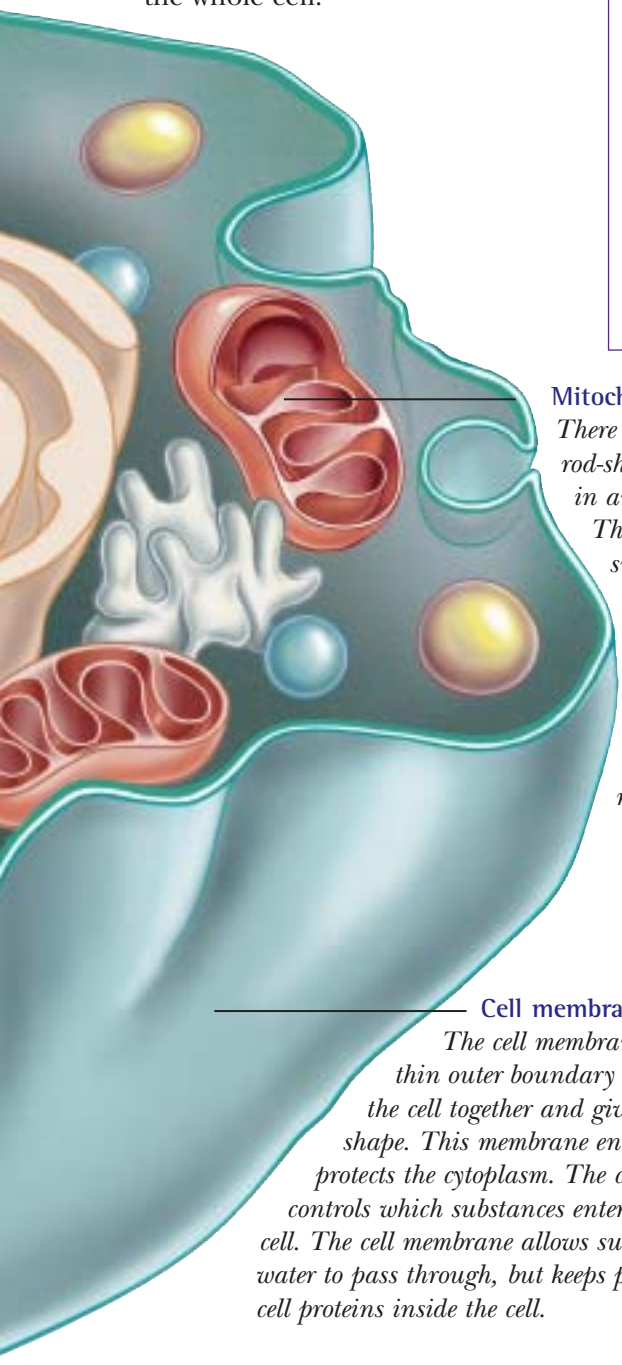
Cells have to be very small because they must be able to take up the substances they need and remove wastes quickly. The bigger a cell is, the further the centre is from the edge and the longer it takes to move material in and out. If a cell was too big, it would not be able to take up or remove materials fast enough to support itself. In a cell that was too big, the nucleus would not be able to pass on information and control the whole cell.

HOW ABOUT THAT!

When cells die

Most of the cells in your body don't live for as long as you do. Usually, when they die they are replaced. The cells that make up your skin live for only between 20 and 35 days. Skin cells are able to replace themselves before they die. Luckily they don't all die at the same time! The dead cells are rubbed off, or just fall from your body. They land on the floor, on furniture and in your bed. In fact, most of the 'dust' that you sweep up or vacuum is actually dead skin cells. Snakes, on the other hand, usually shed their dead skin cells all at once.

Young snakes shed their skin every six to eight weeks. Adult snakes shed their skin only once every year or two.



Mitochondrion
There are many of these rod-shaped organelles in an animal cell. The ***mitochondria*** *supply energy wherever it is needed in the cell. They are too small to be seen clearly with a light microscope.*

Cell membrane
The cell membrane is the very thin outer boundary of a cell. It holds the cell together and gives the cell its shape. This membrane encloses and protects the cytoplasm. The cell membrane controls which substances enter and leave the cell. The cell membrane allows substances such as water to pass through, but keeps particles such as cell proteins inside the cell.

Activities

REMEMBER

1. What are organelles? List two examples of organelles.
2. What substance fills a cell?
3. Why is the nucleus important to a cell?
4. What is the role of the cell membrane?
5. Why do cells need to be so small?

THINK

6. Why do mitochondria move around in the cytoplasm?
7. If most of our cells don't live as long as we do, how do we stay alive?

SKILLBUILDER

8. Look through a microscope at prepared stained slides of animal cells under high power. Make accurate drawings of cells you can see. Label any features you can identify.

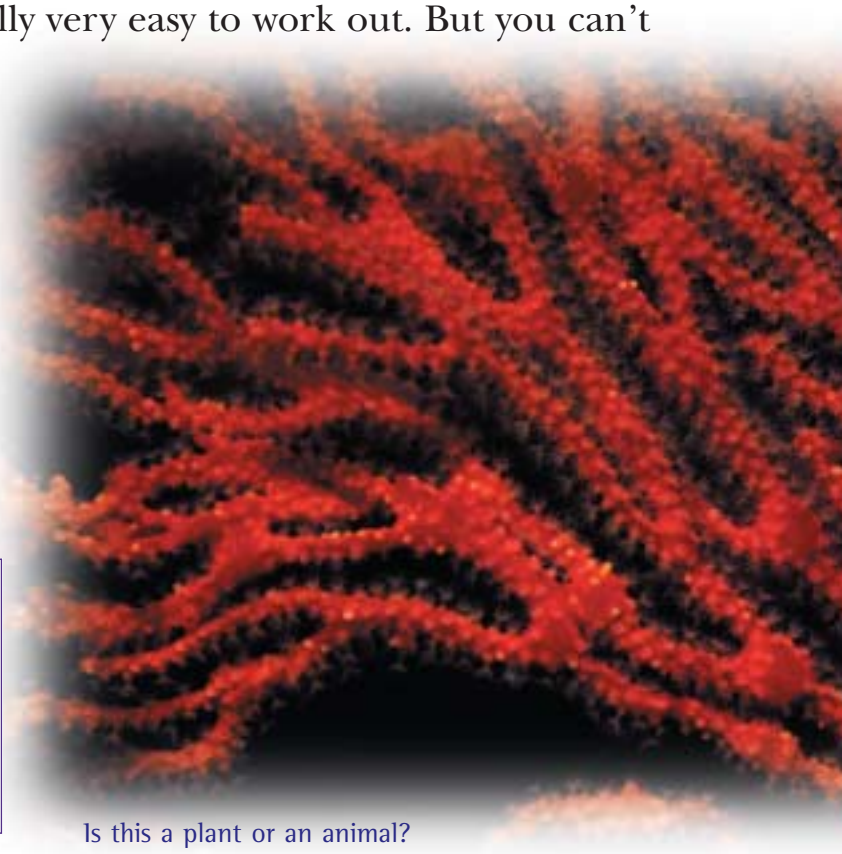
✓ checklist

I can:
- ☐ name the main parts of cells
- ☐ state the role of each of the main parts of cells
- ☐ identify structures in cells
- ☐ make accurate drawings of cells.

Plant or animal?

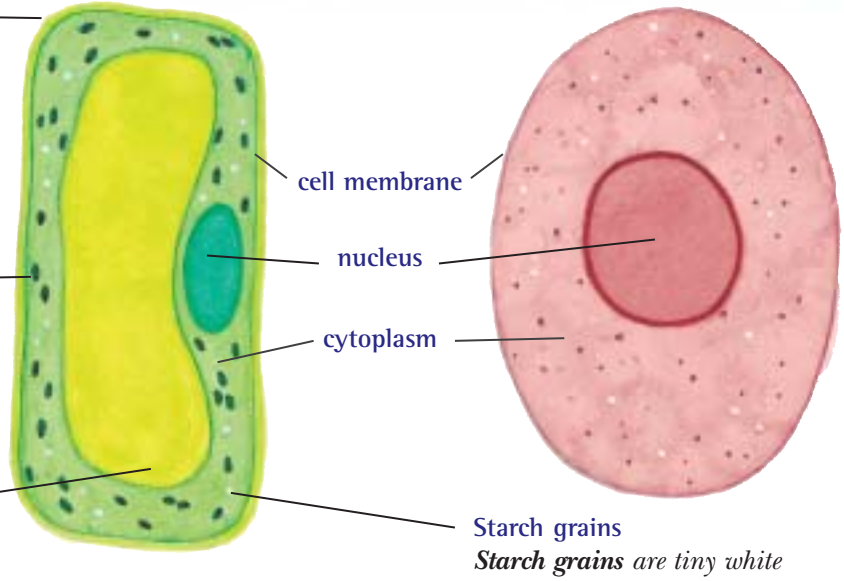
The organism in the photograph below is a coral on the Great Barrier Reef. Is it a plant or an animal? It's usually very easy to work out. But you can't always tell with a quick look. You can decide whether an organism is a plant or animal by looking at its basic building blocks — its **cells**.

All plant and animal cells have a **cell membrane**, **cytoplasm** and a **nucleus** at some stage in their life. That's because all plant and animal cells need food for energy, water and a 'control centre'. Plant cells also have **cell walls**, **chloroplasts** and **vacuoles**. Plants need those extra features in their cells to make and store food, and to keep their shape. Some animal cells have vacuoles, but they are very small.



Is this a plant or an animal?

The cellulose fibres in cell walls in trees are used to make paper. The cork used in wine bottles is also made of cell walls. The cork cells have died and only the wall is left.



Cell wall
*The tough covering around plant cells is the cell wall. It gives plant cells strength and holds them in shape. Cell walls are made of a substance called **cellulose**. Water and dissolved substances can pass through the cell wall. Animal cells do not have a cell wall.*

Chloroplasts
*Chloroplasts are the oval-shaped **organelles** found only in plant cells. Chloroplasts contain a green substance called **chlorophyll**. Chloroplasts use energy from the Sun to make food. Not all plant cells contain chloroplasts. They are found only in leaf and stem cells.*

Vacuole
*The vacuole is an organelle used to store water and dissolved substances. Vacuoles also store food and waste material. Vacuoles can look empty, like an air bubble. Plant cells usually have one large vacuole. The mixture inside a plant's vacuoles is called **cell sap**. The red, blue and violet colours that you often see in plant leaves and flowers are due to the substances stored in vacuoles. Most animal cells don't have vacuoles.*

Starch grains
***Starch grains** are tiny white granules in the cytoplasm. These granules are not organelles. They are a form of food. The food made by the plants is stored as starch for later use.*

Comparing some features of plant and animal cells

Feature	Plant cells	Animal cells
Cell membrane	Yes	Yes
Cytoplasm	Yes	Yes
Nucleus	Yes	Yes
Cell wall	Yes	No
Vacuoles	Yes, large	Sometimes, but are small
Chloroplasts	All plants have some cells with these.	No

The coral shown in the photograph on the left is actually a colony of tiny animals called **coral polyps**. Most coral polyps are only a couple of centimetres in diameter. Each cell of a coral polyp has a cell membrane, cytoplasm and a nucleus. None of the coral polyps have a cell wall or chloroplasts.

Comparing plant and animal cells

You will need:
microscope
prepared slide of leaf epidermal cells
prepared slide of skin cells.

- Look at a prepared slide of the cells on the surface of a leaf (epidermal cells).

1. What shape are the cells? Draw three or four cells clearly. What features can you see that tell you these are plant cells? Label the features of the cell.

- Look at a prepared slide of the skin cells.

2. What shape are the cells? Draw three or four cells clearly. What features can you see that tell you these are animal cells? Label the features of the cell.
3. Why do these plant and animal cells have a similar shape?

Activities

REMEMBER

1. List three features that are found in both plant and animal cells.
2. List two features of plant cells that are not found in animal cells.
3. How do plant cells obtain food?
4. What material are cell walls made of?
5. Where in a cell do you find cell sap?
6. What happens inside chloroplasts?

THINK

7. Why do animal cells need to be more flexible than plant cells?
8. Which cell structure could be described as a 'rigid jacket'? Would you find this in a plant or an animal cell?
9. How do we know that a coral polyp is an animal?

SKILLBUILDER

10. Look through a microscope at prepared stained slides of animal cells under high power. Make accurate drawings of cells you can see. Label any features you can identify.

INVESTIGATE

11. Research different products made from plant cell walls. In what ways do people use them?

CREATE

12. Make a model of a plant and animal cell using edible materials like jelly, lollies and sponge cake. Think carefully about what you should use for each part. Devise a suitable way of labelling all the parts of your model cells.

!CAUTION: Do not eat any of the parts if your model has been made in the science laboratory!

✓ checklist

I can:

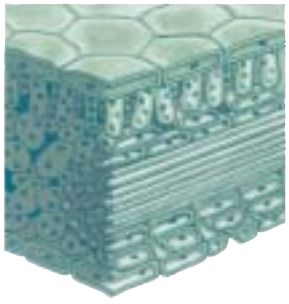
- ☐ describe the differences between plant and animal cells
- ☐ describe features that plant and animal cells have in common
- ☐ explain why there are differences between plant and animal cells.

Plants up close

The job of keeping a plant alive is shared between different **cells**. Some cells mainly make food while others move water and minerals around the plant. There are even special cells that shut the pores in a leaf to stop a plant drying up in hot weather. Because each of these types of cell does a different job, they look different. A cell's shape, size and features all depend on the job it has to do.

The outer lining

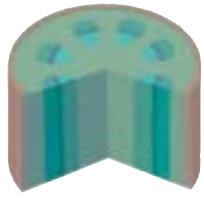
*The upper and lower **epidermal cells** protect the top and bottom surfaces of the leaf. These cells have a flattened shape so that they can easily cover the whole leaf. Leaf epidermal cells do not usually have **chloroplasts** because their role is to protect, not to make food. Stems and roots also have an outer lining of cells for protection.*



Plant highways

The stem contains long hollow tubes. These tubes run all the way from the end of each leaf, through the stem, down to the deepest root tips.

*One set of tubes transports water and dissolved minerals up from the roots to the stem and leaves. This set of tubes is made up of long narrow cells called **xylem cells**. Xylem cells have to die before they can link together end to end to form tubes. When each xylem cell dies, its top and bottom walls dissolve, forming a long tube made up of dead cells. Xylem cells have very thick walls with lots of **cellulose**. This makes them very strong. Tiny holes in their dead cell walls allow substances to move in and out of the tube. Xylem cells form the inner parts of stems, trunks and branches. Xylem cells form the bulk of the wood that makes up a tree trunk.*



*A different set of tubes, made up of **phloem cells**, transports food made in the leaves to other parts of the plant. Like xylem cells, the phloem cells join end to end to form long hollow vessels.*

On guard

*Among the lining cells are pairs of special cells called **guard cells**. These control the movement of gases into and out of the leaf though pores called **stomata**. When water enters the guard cells, they expand and pull open the pore. One leaf pore is called a 'stoma'; the plural is 'stomata'. When these stomata are open, water vapour and gases (carbon dioxide and **oxygen**) can move into and out of the leaf. During hot, dry weather the guard cells close the pores so that the plant does not lose too much water.*

As well as the guard cells, most leaves also have a waxy coating that helps reduce the loss of water and gas.

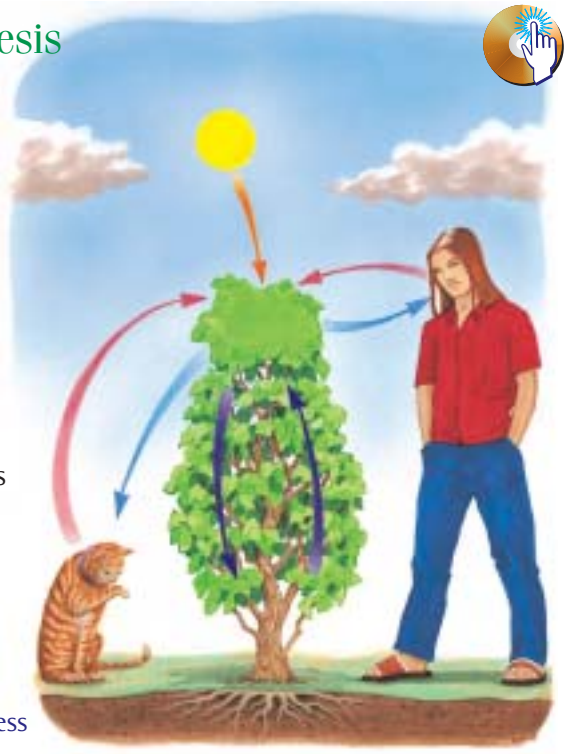
Roots

Go to worksheet 34

Photosynthesis

Plants make their own food using a process called **photosynthesis**. This happens because **chlorophyll** in the chloroplasts is able to absorb energy from the Sun.

Plants use the Sun's energy, carbon dioxide in the air (which animals breathe out) and water and minerals in the soil to create sugars (their food) and oxygen (which we breathe in).



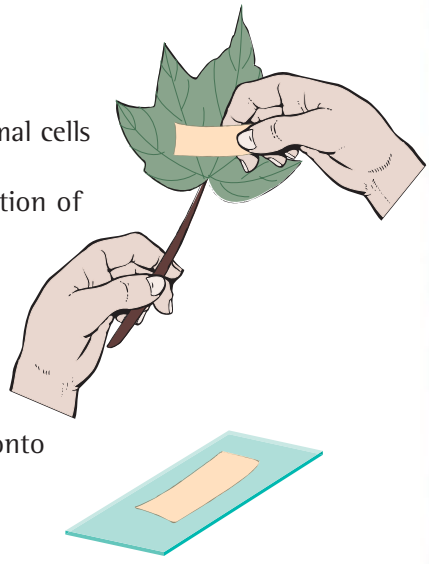
The photosynthesis process

Observing leaf epidermal cells

You will need:
leaf
clear sticky tape
microscope slide
microscope.

You can make a 'slide' of leaf epidermal cells with sticky tape.

- Place some sticky tape over a section of the underside of a leaf.
- Press the sticky tape firmly onto the leaf.
- Tear the tape off. Some of the lining cells should come off with the sticky tape.
- Press the tape, sticky side down onto a microscope slide.
- View the sticky tape under the microscope.
- Try to find a pair of guard cells and one of the stomata.



1. Is the stoma (the opening) open or closed?
2. Make a drawing of a group of cells, including the guard cells. Include as much detail in your drawing as possible.
3. Label the guard cells and stomata.
4. Title and date your drawing. Write down the magnification used.

Activities

REMEMBER

1. Why do the epidermal cells in leaves have a flattened shape?
2. How does the waxy coating found on most leaves protect them?
3. How do xylem cells change into long tubes?
4. Which cells make up the tubes that transport food in the leaves down through the stem?

THINK

5. How did guard cells get their name?
6. Guard cells and stomata mainly occur only on the lower part of the leaf, away from direct sunlight. Suggest why.
7. Would you expect to find chloroplasts in roots? Give a reason for your answer.
8. Why are all plant cells not the same?

CREATE

9. Draw a flow chart to show the processes that occur during photosynthesis.

INVESTIGATE

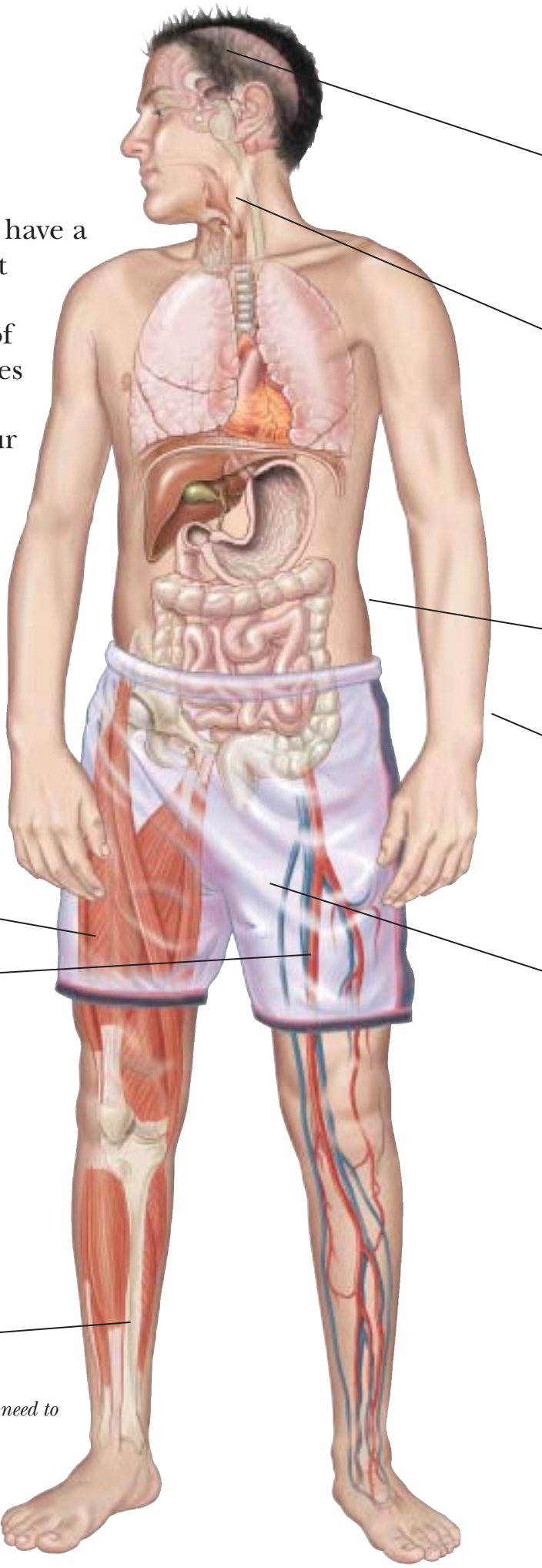
10. Look at prepared microscope slides of plant leaves, roots or stems. For each slide, accurately draw a group of cells and label the cell types.

checklist

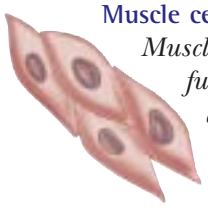
I can:
- ☐ describe the function (job) of different types of plant cell
- ☐ explain how the shape and features of plant cells help them do their job.

Animal cells

Like all other animals, humans have **cells** that have a **cell membrane**, **cytoplasm** and a **nucleus** at some stage in their life. You have about 200 different types of cell in your body. Each type of cell does a specific job. The cells in your muscles look a bit different from the cells in your skin because they do different tasks. The cells in your blood do not look the same as the cells in your bones. Every cell must be the right size and shape for its job!



Every cell has its purpose



Muscle cells

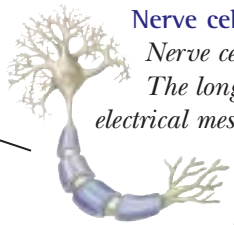
*Muscle cells are long and elastic. Long thin cells can slide further over each other to allow you to move. There are even different types of muscle cell. The walls of your blood vessels and parts of your digestive **system** have 'smooth muscle' cells. The muscles that are joined to your bones are called 'skeletal muscles'. Skeletal muscles work in pairs — one muscle contracts (shortens) and pulls the bone in one direction while the other muscle relaxes.*

Red blood cells

*Red blood cells carry **oxygen** around the body. Their round shape helps them move easily through blood vessels. The nucleus in a red blood cell dies soon after the cell is made. Without a nucleus, red blood cells live for only a few weeks. The body keeps making new blood cells to replace those that have died. Red blood cells are made in **bone marrow** at the rate of 17 million cells per minute! This is why most people can donate some of their blood to the Red Cross without harm. White blood cells, which are larger than red blood cells, are also made in the bone marrow. Their job is to rid the body of disease-causing organisms and foreign material.*

Bone cells

Minerals such as calcium surround your bone cells. The minerals help make bone cells hard and strong. Bone cells need to be hard so that they can keep you upright.



Nerve cells

Nerve cells are very long and have a star shape at one end. The long shape of nerve cells helps them detect and send electrical messages through the body at the speed of a Formula 1 racing car. There are nerve cells all over your body. They allow you to detect touch, smell, taste, sound, light and, unfortunately, pain.

Living filters

*The cells that line your nose, throat and windpipe are a type of lining cell. They have hair-like tips called **cilia**. These cells help protect you by stopping dust and fluid from getting down your windpipe. The cilia can also move these substances away from your lungs. You remove some of these unwanted substances whenever you sneeze, cough or blow your nose.*

Energy storage

Some cells store fat. Fat stores a lot of energy for cells to use later. Round shapes are good for holding a lot of material in a small space. Fat cells are mostly found underneath your skin, especially in the chest, waist and buttocks.

Body liners

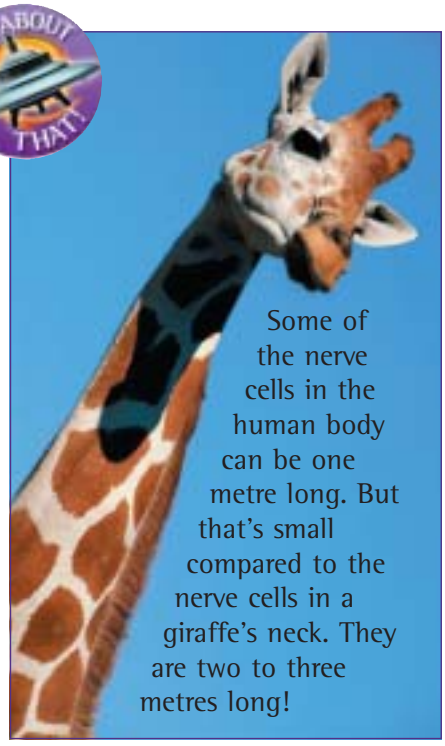
Special cells line the inside and outside surfaces of your body. These are the cells that form your skin and the lining of your mouth, nose, throat, stomach and intestine. These cells have a flattened shape so they can better cover and protect your body.

Made for swimming

Sperm cells have a long tail which helps them swim towards egg cells. Only males have sperm cells.

Egg cells

*Egg cells are some of the largest cells in a human body. Their large round shape helps them store plenty of food. Only females have egg cells. When a sperm cell moves into an egg cell, the egg cell is **fertilised**.*



Some of the nerve cells in the human body can be one metre long. But that's small compared to the nerve cells in a giraffe's neck. They are two to three metres long!

Activities

REMEMBER

1. What three features do all animal cells have at some stage in their life?
2. Which type of animal cell spends most of its life without a nucleus?
3. Which type of cell is found in the walls of blood vessels?
4. What special job is done by:
   (a) blood cells?
   (b) skin cells?
   (c) nerve cells?
5. How do the cilia in your nose, throat and windpipe protect your lungs?
6. Why are egg cells so large?

THINK

7. Redraw the table below with the correct shape next to each type of cell.

Cell	Shape
(a) Muscle cells	Disc shaped
(b) Egg cells	Star shaped
(c) Red blood cells	Flat
(d) Nerve cells	Long, thin
(e) Skin cells	Spherical

8. Explain how the shape of each of the cells in the table above helps the cell do its job.

SKILLBUILDER

9. Compare the cells from different animals. Use a microscope to look at prepared slides of blood cells from a human and a frog. Draw and describe the cells.

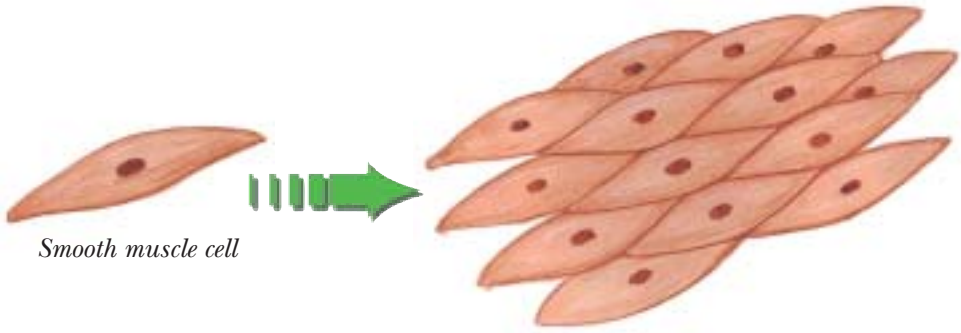
✓ checklist

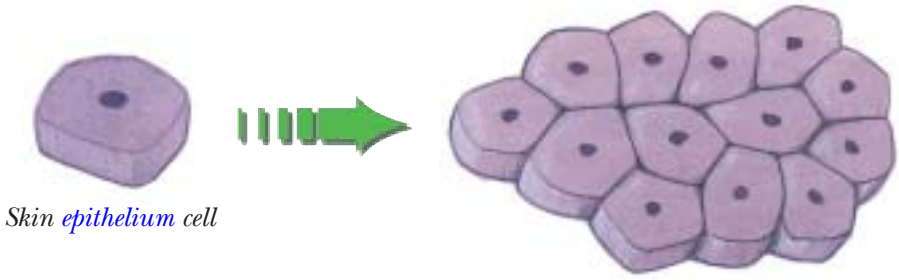
I can:

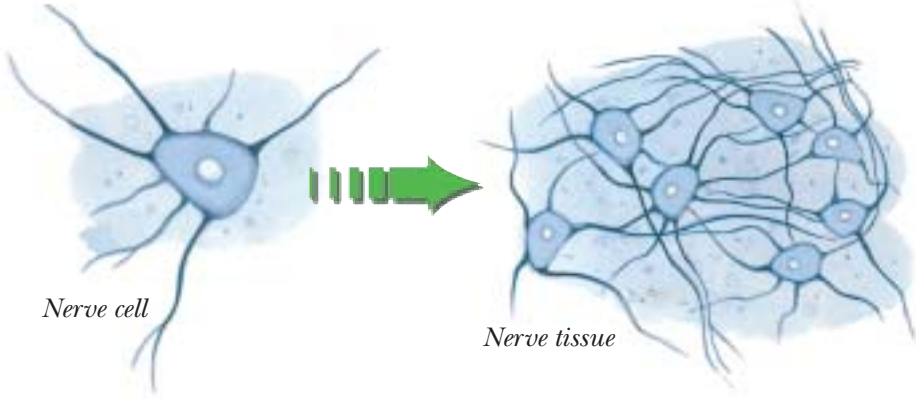
- ☐ describe features of animal cells
- ☐ describe the function (job) of different types of animal cell
- ☐ explain how the shape and features of animal cells help them do their job.

Tissues and organs

In animals and plants, **cells** work together in teams. Each team has a particular job to do. As long as all of the teams do their job properly, the animal or plant stays alive and healthy. If one or more of the teams doesn't do its job, the animal or plant becomes sick. It could even die.







Teams working together

Your **organs** are made of different types of tissue. Your brain, heart, liver and stomach are just some of the organs in your body. Each organ has a very important job to do. The tissues in the organ work together so that the job is done properly. For example, each of your lungs is an organ in which **oxygen** enters your blood and carbon dioxide is removed from your blood. Each lung is made of muscle tissue joined to lining tissue. Within this lining tissue are blood vessels which carry the connective tissue that we call blood. Nerve tissue in the lung lining sends and receives messages from your brain to help you breathe.

The different types of cell in the human body are grouped into four main types of tissue:

Type of tissue	What it does	Example
Epithelial tissue (or lining tissue)	Forms a lining around other body parts to protect them	Skin surface (epidermis), stomach lining, lung lining
Muscle tissue	Tightens and loosens itself to move other body parts	Biceps in your arm, cardiac muscle (heart muscle)
Nerve tissue	Carries messages around your body	Optic nerve (from your eye), spinal cord
Connective tissue	Holds other tissues together	Bone, cartilage, blood

All animals and plants are **multicellular**. That means that they have many cells. Plants and animals can have billions of cells. Cells of the same type group together to form teams of cells called **tissue**. For example, your muscle cells form muscle tissue. Smooth muscle cells group together to form the smooth muscular tissue in your blood vessels and your digestive **system**. Other types of muscle cell group together to form the cardiac muscle that keeps your heart beating. Your skin cells form skin tissue and your nerve cells form nerve tissue.

Skinny stuff

Your skin is an organ. It protects your body from germs and water, helps control your body temperature and releases some of your waste products. It senses warmth (or lack of it), pressure and pain. It even uses sunlight to make a vitamin. Your skin contains lining tissue, nerve tissue and connective tissue.

The skin of an adult human weighs about 5 kg. The thinnest part of your skin is on your eyelids (about 0.5 mm thick). The thickest part of your skin is on the soles of your feet (about 4 mm thick).

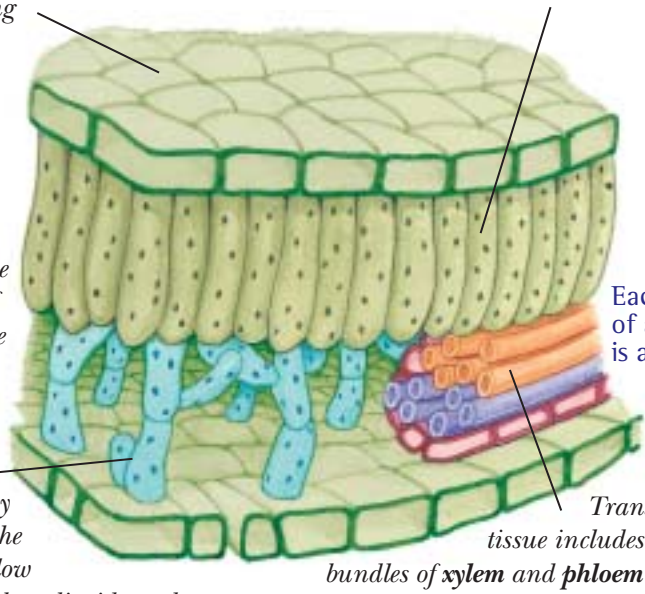
The elephant and rhinoceros are the most thick-skinned animals – the skin on their back can be 2.5 cm thick.

Plants have organs too!

It's not just animals that have organs. Each leaf, flower, stem and root is an organ. Each organ is made up of different types of tissue. Each type of tissue has its own job that helps the organ work properly.

*Lining tissue (a layer of **epidermal cells**) forms a lining around the leaf to protect it.*

*Food-making tissue is usually found on the top side of the leaf. It contains most of the **chloroplasts**.*



Each leaf of a plant is an organ.

Support tissue gives the leaf its shape. The spongy cells that make up this support tissue are surrounded by air spaces. The air spaces allow gases like carbon dioxide and oxygen to flow in and out of these cells.

*Transport tissue includes the bundles of **xylem** and **phloem cells** that carry water and minerals from the roots to the rest of the plant, and food from the leaves to the rest of the plant.*

Activities

REMEMBER

1. What is meant by the term multicellular?
2. Which types of cell work together to form muscle tissue?
3. Name four types of tissue that can be found in your lungs.
4. In animals, what is the main job of:
   (a) epithelial tissue?
   (b) muscle tissue?
   (c) nerve tissue?
   (d) connective tissue?
5. What important jobs are done by your skin?
6. List four organs that you could find in a plant.
7. What is the main job of transport tissue in plants?

THINK

8. How do you know that skin is an organ rather than a tissue?
9. Why is the food-making tissue in a leaf usually found on its top side?

IMAGINE

10. Imagine that you are a tree. You need to get as much water and as much sunlight as you can – but you can't move to another location. What features do your organs need to have?

INVESTIGATE

11. What jobs are done by the human brain, liver and stomach?

checklist

I can:

- ☐ explain the links between organs, tissues and cells
- ☐ describe the jobs done by different tissues in animals and plants.

Putting it all together

Some eucalypt trees grow over 100 m tall. The leaves at the top of the tree need to get water from the soil so they can make food. That's a long way to move water straight upwards! It requires teamwork — and some of the tree's **organs** work together to make it possible.

Reaching great heights

*The leaves, stem and roots are some of the tree's organs. **Cells** work together to form **tissue**. Tissues work together in organs. Organs work together as **systems**. The leaves work together as a system to make enough food to keep the tree alive. The roots work together as a system to collect water and minerals from the soil. The roots, stem and leaves work together as a transport system. This system moves water up from the ground and food all the way down to the root tips.*

Animal systems

The systems of animals are also made up of organs working together. For example, your own respiratory system includes your windpipe and lungs. The respiratory system takes **oxygen** from the air to your blood and gets carbon dioxide out of your blood back into the air. Another system, your circulatory system, moves food and gases, like oxygen and carbon dioxide, around your body. The respiratory system and circulatory system work together with other systems to keep you alive and healthy.

Some systems in the human body

System	Main organs	Job
Digestive system	Stomach, liver, intestines	To digest and move food into your circulatory system
Respiratory system	Trachea (windpipe), lungs	To take oxygen from the air and return carbon dioxide to the air
Circulatory system	Heart, blood vessels	To move food and gases around the body
Nervous system	Brain, spinal cord	To send messages around the body

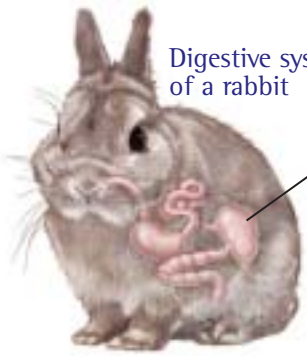
Variety is the spice of life

The digestive system in all animals does the same job. It digests and moves food into the blood so that it can be carried to every cell. But the organs that make up the system don't always look the same in different animals. The animals shown on the right each have:

- a mouth to take in food
- an oesophagus to move food away from the mouth
- intestines to absorb food and water
- an anus to remove waste material.

Apart from the earthworm, they all have a stomach where food is mixed with digestive juices to break it down. The rabbit uses its stomach to store food.

Digestive system of a human



Digestive system of a rabbit

Caecum

The caecum is an organ found in animals that eat plants. Rabbits eat leaves. The ***cellulose*** *in the cell walls is digested in the caecum. Humans also have a caecum but it doesn't play a big role in digestion.*

Digestive system of a pigeon

Crop

The crop is an organ found in birds, insects and worms. Food is stored in the crop before passing through to the gizzard.

Digestive system of an earthworm

Gizzard

The gizzard is a grinding organ found in birds, reptiles, insects and worms. In grain-eating birds like pigeons, the gizzard contains small stones that grind hard seeds. In earthworm gizzards, sand and dirt are used to grind food.

Go to worksheet 35

MI VISUAL/SPATIAL

Activities

REMEMBER

1. Which organs make up the food-making system of a tree?
2. Which organs work together in a plant to move water up from the ground and food down to the roots?
3. What is the main job of your:
   (a) respiratory system?
   (b) circulatory system?
4. List two of the organs that make up your:
   (a) digestive system
   (b) nervous system.
5. What is the job of the digestive system in animals?
6. Complete the following sentences about animals and plants.
   (a) _____work together to form tissues.
   (b) Tissues work together to form _____.
   (c) _____ work together to form systems.
7. Which two organs can be found in birds and earthworms, but not humans?
8. Which organ in a rabbit is used to digest the cellulose in plant cell walls?

THINK

9. The sensory system in humans is the system that allows us to sense the world around us. List five organs that are part of the sensory system.
10. Why do all animals need a digestive system?
11. Why is a caecum less important to the digestive system of humans than it is to the digestive system of rabbits?
12. Name a well-known Australian native animal that is likely to have a caecum and explain your choice.

INVESTIGATE

13. Research and report on the digestive system of one of the animals listed below. Include a diagram of the digestive system in your report, along with a short description of what each organ does. Choose your animal from: frog, grasshopper, ant, shark, cow, horse, pelican, chicken, crocodile, lizard, whale, elephant, tiger.

✓ checklist

I can:

- ☐ explain the links between systems, organs, tissues and cells
- ☐ describe the jobs done by different systems in animals and plants.

Check and challenge

Cells

Check

Sizing up cells

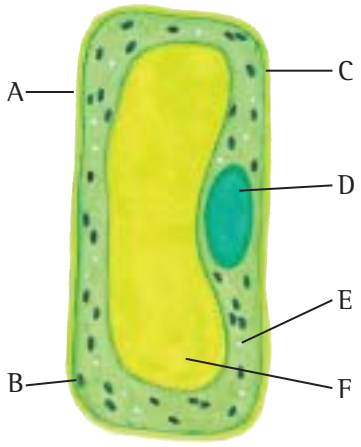
1. Cells are described as the building blocks of life. Why is this a good description?
2. Draw a labelled set of diagrams to show how to make a wet mount and how to stain cells.
3. What magnification does a 10× eyepiece used with a 10× objective lens in a microscope give?
4. Why is a microscope needed to see cells and the parts inside them?
5. Copy and complete: The lens of a microscope that is closest to the object that you are looking at is called the _____ lens.

The structure of cells

6. Copy this table and complete the first column by referring to the diagram in question 7 and inserting the correct label (A, B, C, D, E or F) next to each feature in the table. Use the third column to briefly describe the main job done by each feature.

Correct label	Cell feature	Main job done by feature
	Cell wall	
	Cytoplasm	
	Vacuole	
	Chloroplast	
	Cell membrane	
	Nucleus	

7. Is the cell shown at right a plant cell or an animal cell? State three reasons for your answer.



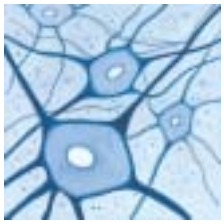
8. Draw an animal cell and a plant cell, showing and labelling the parts that can be seen with a normal school microscope.

The function of cells

9. Copy and complete the table below. Identify each of the human cell types shown below. Describe the job done by each cell type and explain the reason for its shape.

A

B

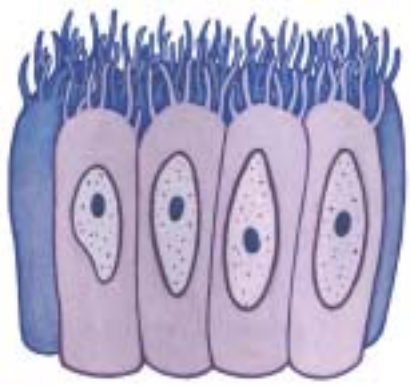


C

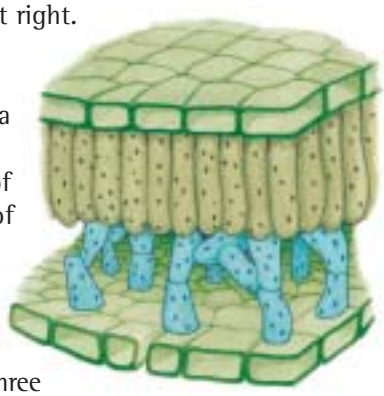
D

Cell type	Name of cell type	Job and reason for shape
Type A		
Type B		
Type C		
Type D		

10. Cells like the ones shown below can be found in your windpipe.
    (a) What are the hair-like structures on the cells and what do they do?
    (b) Name one other place in your body where you might find cells with these structures.



11. Copy the diagram at right.
    (a) Label a chloroplast, a guard cell and a stoma.
    (b) In which part of a plant are all of these features found?
    (c) State the major job of each of these three features.



Teamwork – cells, tissues and organs

12. Groups of similar cells that carry out the same job are called _____.
13. The main jobs in the table below have been placed incorrectly. Redraw the table so that the jobs correctly match the tissue.

Tissue	Main job
Lining tissue	To move
Bone tissue	To send messages
Muscle tissue	To support
Nerve tissue	To protect

14. True or false?
    (a) Tissues are made of different types of organ.
    (b) Lining tissue can be found in animal and plant cells.
    (c) A system is made of organs working together.
    (d) The skin is made up of one type of tissue.
    (e) Blood is an example of connective tissue.
15. Name four different organs that can be found in plants.
16. Copy and complete the table below. The table describes some of the systems in animals and plants.

System	Organs	Main job of system
Animal digestive system		
Animal circulatory system		
Root system		
Plant transport system		

17. Complete the flow chart below to show how systems, cells, organs and tissues are related to each other.

Animal and plant cells

1. Which organelles will definitely not be found in any of the cells shown in the figures on the top right of page 166?
2. How can the root cells of a plant survive without having chloroplasts?
3. Plant cells have a vacuole whereas most animal cells have no vacuoles.
    (a) Why do plant cells need to have a large vacuole?
    (b) Suggest why some animals might need to have vacuoles in their cells.

Stem cells in animals?

The stem cells that have been in the news in recent years have nothing to do with plants. Animals with bony skeletons have stem cells in their bone marrow. Stem cells are important because blood cells are produced from them. One type of blood cell – the white blood cell – helps fight diseases like cancer. Scientists are studying these stem cells to find ways to make them better at fighting diseases. The stem cells for this research come from people who donate their cells.

Another type of stem cell is the embryonic stem cell. These stem cells come from unborn babies. These special stem cells can be used to research white blood cells, but have another use too. The stem cells from unborn babies can be used to produce any type of body part. Embryonic stem cells could eventually be used to replace any damaged or injured body parts.

4. Where are stem cells found in animals?
5. Why would some people be unhappy about using embryonic stem cells to make human body parts?
6. Imagine a time when every part of the human body could be replaced by growing 'spare parts' from embryonic stem cells. Explain how it would change human life. Describe any problems that would be caused by using 'spare parts'.
7. Scientists are trying to grow body parts from adult animal stem cells. Their research is being done on small animals like mice. List some arguments for and against this type of research.

Summary of key terms

bone marrow: a substance inside bones in which blood cells are made

cell: the smallest unit of life. Cells are the building blocks of life.

cell membrane: the thin outer boundary of a cell. This membrane encloses and protects the cytoplasm.

cell sap: the mixture of water, dissolved substances, food and waste material that can be found in the vacuoles of plant cells.

cell wall: a wall around the cell membrane in plant cells, providing a tough extra covering which gives strength and support to the plant cell

cellulose: the cell walls of plants are made of this

chlorophyll: the green pigment that occurs in chloroplasts

chloroplast: oval-shaped organelles found only in plant cells. Chloroplasts contain the pigment chlorophyll. They are the 'factories' in which carbon dioxide and water are changed by sunlight and water into food through the process of photosynthesis.

chromosome: tiny thread-like structure inside the nucleus of a cell. Chromosomes contain genes which carry information that is passed on when an organism reproduces itself.

cilia: hair-like tips on cells. The cilia that line your windpipe and lungs help stop germs, dust and fluid getting to your lungs.

coral polyp: tiny animal that mostly lives in colonies and forms corals, some of which look like plants

cytoplasm: the jelly-like material inside a cell. It contains organelles such as the nucleus, vacuoles and many others.

electron microscope: tool for viewing very small objects. An electron microscope is much more powerful than a light microscope and can magnify things up to a million times.

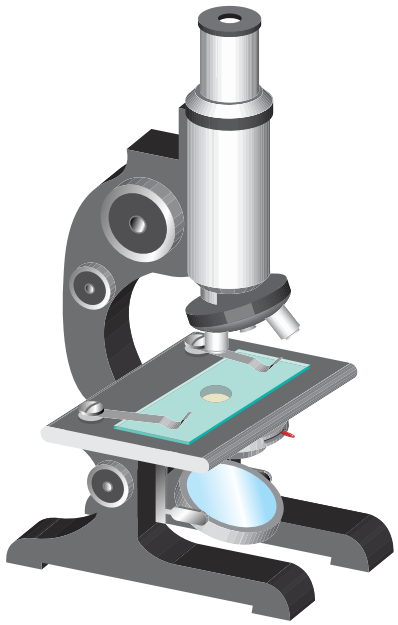
epidermal cell: flattened type of cell that protects the top and bottom surfaces of leaves. Stems and roots also have an outer ring of these cells for protection.

epithelial tissue: the lining cells that form the outside or inside surfaces of a plant or animal

fertilised: an egg cell is fertilised when a live sperm cell enters it. A fertilised egg eventually grows into a new organism.

guard cell: special cell that occurs among the lining cells in a leaf. They control the movement of gases into and out of the leaf through pores called stomata (singular = 'stoma'). When these pores are open, water vapour, carbon dioxide and oxygen gases can move into and out of the leaf. The guard cells close the pores during hot weather so that the plant does not wilt.

light microscope: tool for viewing very small objects. A light microscope can magnify things up to 1500 times.



mitochondria: small rod-shaped organelles that supply energy to other parts of the cell. Mitochondria are usually too small to be seen with light microscopes. Singular = mitochondrion.

multicellular: having many cells. All animals and plants are multicellular.

nucleus: the roundish structure inside cells which acts as the control centre for the cell

organ: group of tissues working together to carry out a particular job

organelles: the 'little organs' found in the cytoplasm of a cell. Each organelle performs a particular job for the cell.

oxygen: a gas in the air (and water) that animals need to breathe in. Plants produce oxygen as part of photosynthesis.

phloem cells: long, narrow cells that are joined together to form long tubes in a plant. The tubes made from phloem cells move the food made in the leaves to other parts of the plant such as its roots and storage areas.

photomicrograph: photograph taken through a microscope

photosynthesis: the food-making process in plants that takes place in chloroplasts within cells. The process uses carbon dioxide, water and energy from the sun.

starch grains: tiny white granules in the cytoplasm. Starch is stored energy for plant cells.

stem cells: the cells in your bone marrow that make your blood cells

stomata: pores located on leaf surfaces. These pores are opened and closed by guard cells. Singular = stoma.

system: groups of organs work together in a system. Each system performs a particular job. Examples of systems include the circulatory system and the digestive system.

tissue: a group of similar cells that work together to do a job

vacuole: a sac within a cell used to store food and wastes. Plant cells usually have one large vacuole. Animal cells may have several small vacuoles or none at all.

womb: the organ in humans and most other mammals in which a fertilised egg cell develops into a baby

xylem cells: long narrow cells that are joined together to form long tubes in a plant. The tubes, made from xylem cells, move water and dissolved minerals up from the roots to the stem and leaves. The wood in a tree trunk consists mostly of dead xylem cells.

CHAPTER 8

Classification

You will discover

How living things are classified by scientists

That most animals with skeletons have them outside their bodies

That not all living things are plants or animals

Why mushrooms and toadstools are not plants

You want to borrow the latest action movie from your local video store. Where do you look? Certainly not under 'Romance' in the 'Weekly' section. You know where to start your search because the store has classified the videos. To help us organise our world we classify many things.

Scientists classify, or group, living things using their similar features. Fortunately, most scientists throughout the world use the same system for classifying living things.

Knowing how a living thing is classified helps us understand how it survives in the environment. If it is dangerous or poisonous, your life could depend on your ability to classify it.

1 Explain in your own words why videos and DVDs in a video store are classified.

2 Make a list of the features of the animal in the photograph below.

3 Explain how you know that the animal is not an insect.

4 What feature enables you to classify this particular spider as dangerous?

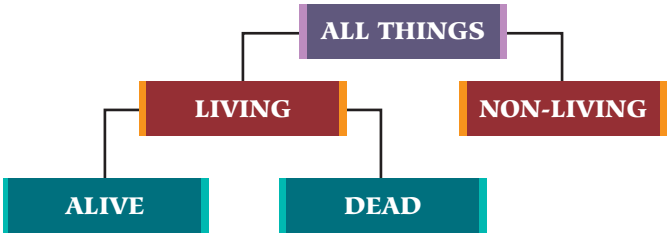
Imagine that you were bitten by this redback spider. Your life might depend on knowing how it is classified.

It's alive!

Classification means sorting things into groups. Before sorting living things into groups we need to make sure we know what 'living' means. We can sort all things into two groups — **living** or **non-living**. The features, or characteristics, that something has tell us to which group it belongs. The 100 million or so different kinds of living things on Earth, called **organisms**, have some features in common.

Living or non-living?

A 'living' thing is alive right now, or was alive at some time. You are a living thing. Phar Lap is a famous horse that once lived, but is now dead. Things that were never living are described as 'non-living'. Non-living things are objects such as drink cans, CD-ROMs, stones and nose studs.



Some living things are easy to classify as living or non-living. For example, it's clear that your classmates are living things. So is a family pet. So are the plants, weeds and fungi growing in your garden. The mould on stale bread and the bacteria that give you food poisoning are also living things. It's harder to see the features that show they are alive. This makes them more difficult to classify as living or non-living.

Living things

Need energy

All living things that are alive need energy. Animals get their energy by eating food made up of other animals and plants. Plants use the Sun's energy to make their own food and, hence, their own energy supply. This process is called ***photosynthesis****. Fungi such as mushrooms and mould can't make their own food but have to get their food by growing on other organisms (living or dead).*

Need air

All living things that are alive absorb air, use some of the oxygen and other gases in it, and release it, along with some waste gases. Many land animals breathe the air in and out through their nose and mouth. Some animals, including insects and worms, absorb and release air through their skin. Gases dissolved in water move in and out of sharks and other fish through gills. Gases in the air move in and out of plants through tiny holes in their leaves.

Produce waste products

All living things that are alive produce waste. Chemical reactions take place inside organisms when they make or digest food and whenever they use energy. The waste products are the substances left over from chemical reactions inside the organism. Others are substances that are taken in but not used. Not all waste can be seen. For example, we cannot see the carbon dioxide that plants produce as waste. Other wastes are more obvious!

Can move

All living things that are alive can move by themselves. They do not need to be pushed or pulled. The movement of plants is hard to see. The whole plant doesn't move, but parts of it do. The growth of stems towards the Sun, the growth of roots into the ground and the opening and closing of flowers are examples of plant movement.

Respond to changes in their environment

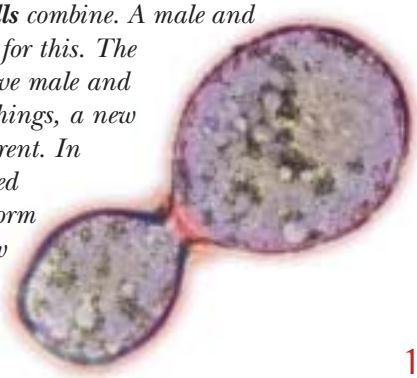
All living things that are alive respond (or react) to things that happen to them or around them. Humans shiver if it suddenly gets colder. A dog might bark when it sees a stranger. The pores in the leaves of plants close when it gets hot to reduce water loss.

Grow as they age

All living things that are alive grow. Usually this means they get bigger and sometimes change in shape as they get older. Some living things grow very quickly (e.g. a teenager having a growth spurt). Others grow much more slowly.

Can reproduce themselves

All living things that are alive produce others like them. In some living things, including most animals and plants, new individuals form when two ***cells*** *combine. A male and female cell are usually needed for this. The flowers of flowering plants have male and female parts. In some living things, a new individual forms from one parent. In bacteria and other singled-celled organisms, new individuals form when a cell splits into two new cells that are identical.*



Go to worksheet 36

MI VERBAL/LINGUISTIC

Activities

REMEMBER

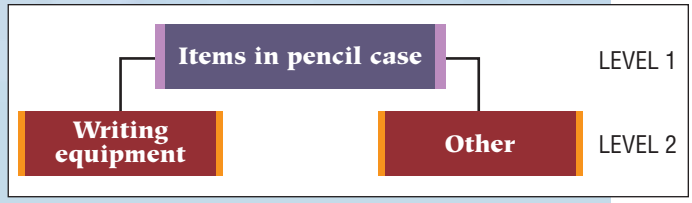
1. What is the difference between a living thing and a non-living thing?
2. List the features that all living things have.

THINK

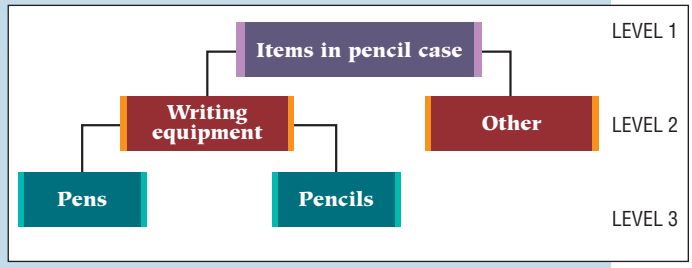
3. Non-living things often have some, but not all, of the features of living things. What features do the following non-living things share?
   (a) A hot-air balloon
   (b) Your television set

SKILLBUILDER

4. Classify all the things in your pencil case into two groups. Group them as either 'writing equipment' or 'other'.



5. Now classify the writing equipment in your pencil case to a third level. Group them as either pens, pencils or other.



6. Find a way of going further and classifying pens, pencils or both to a fourth level.
7. Walk around your garden. List everything you see as either 'living' or 'non-living'. If some of your living items are dead, mark them with a cross.

CREATE

8. Create a colourful collage that illustrates as many examples as you can find of one of the typical features of living things.

✓ checklist

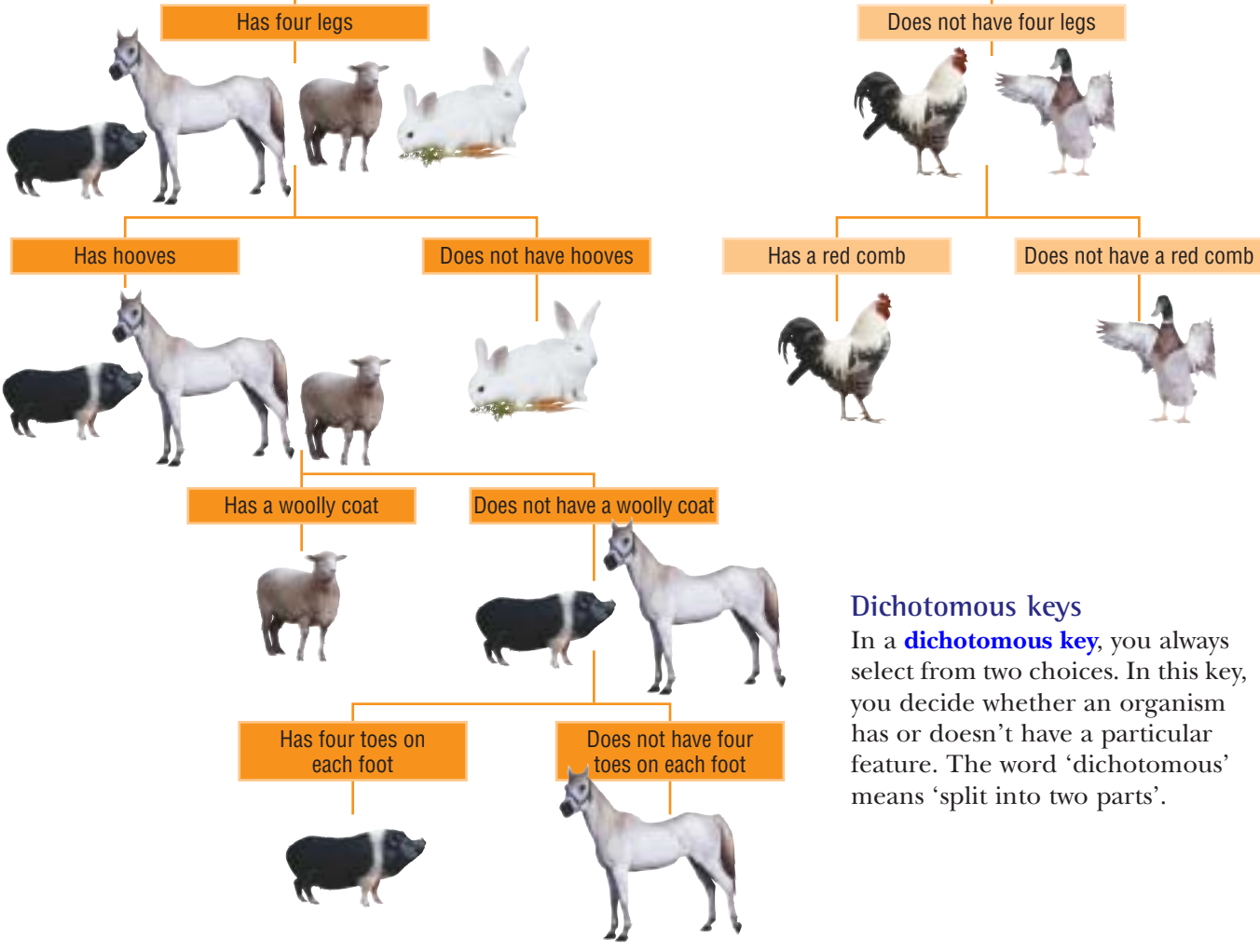
I can:

- ☐ explain why classification is helpful
- ☐ describe the difference between living and non-living things
- ☐ develop different ways of classifying the same items.

This or that?

Think about what you do when you find a strange object. You are curious. You want to know what it is. You note its features — its size, shape, colour, feel and smell. You compare it in your mind with objects that have the same or similar features. Scientists follow a process like this one when they try to identify and classify a newly discovered **organism**. But they often use **keys** like the one below to help them.

Keys provide a simple way to classify organisms. There are many different types of key. When you want to classify an organism, you need to observe its characteristics or features. Be careful if using colour or size as characteristics. This is because these may change over time. For example, the Arctic fox changes colour from season to season; a chameleon (a type of lizard) changes colour according to its mood; and animals and plants change in size and shape during their lifetime.



Dichotomous keys

In a **dichotomous key**, you always select from two choices. In this key, you decide whether an organism has or doesn't have a particular feature. The word 'dichotomous' means 'split into two parts'.

Go to worksheet 37

Tabular keys

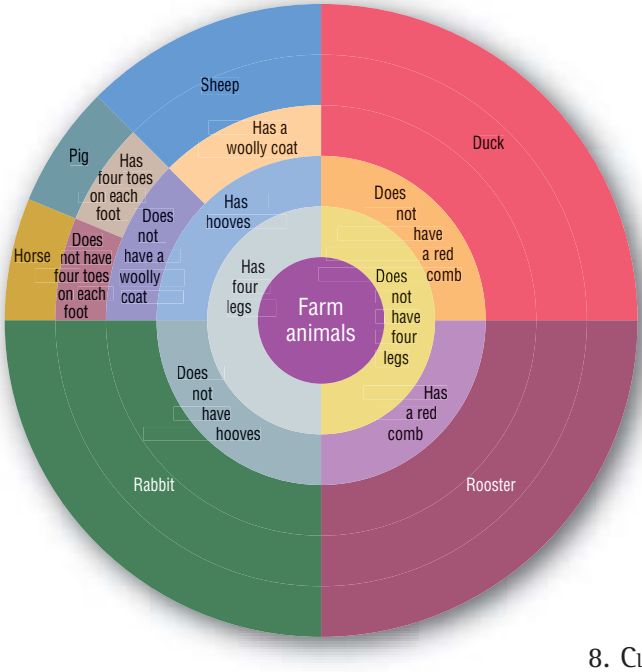
The same information about farm animals can be set out using a **tabular key**. In a tabular key, you read through the numbered items in order. Each item presents two options. As you proceed, more and more information is given until one of the animals shown below can be identified. When this happens, the animal's name is stated. If there is not enough information to identify an animal, you are told to go to another numbered item and make another choice.

Key

1.	Has four legs	*Go to 3*
	Does not have four legs	*Go to 2*
2.	Has a red comb	Rooster
	Does not have a red comb	Duck
3.	Has hooves	*Go to 4*
	Does not have hooves	Rabbit
4.	Has a woolly coat	Sheep
	Does not have a woolly coat	*Go to 5*
5.	Has four toes on each foot	Pig
	Does not have four toes on each foot	Horse

Circular keys

Circular keys are also very useful. To read a circular key, start in the middle and work outwards. As you go, choose one of the options given at each layer. When you get to the outer layer of the circle, you will have identified the organism.



Activities

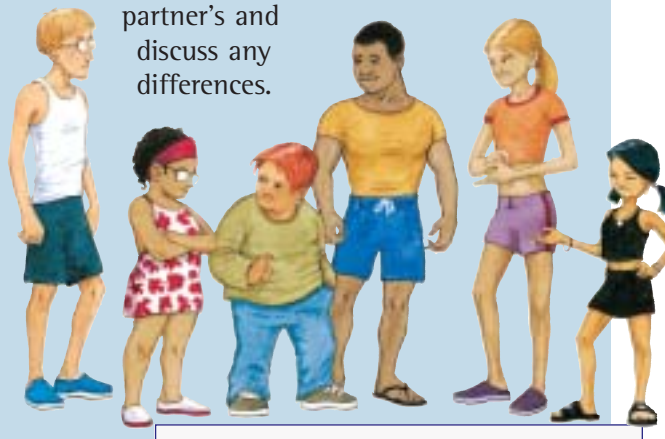
THINK

1. Suggest why the ability to classify organisms is important in the following situations:
   (a) You have been bitten by a spider.
   (b) You are very hungry and find a bush with berries in the forest.
2. Imagine that you have landed on another planet. Weird creatures live there. You noted the characteristics of some that you saw and prepared the circular key shown below. Use it to help you classify the creature you have just found, shown here.

Alien creature

CREATE

3. Construct a dichotomous key to classify the following people. Compare your key with a partner's and discuss any differences.



checklist

I can:

- ☐ explain what keys are and describe their uses
- ☐ draw a dichotomous key and a tabular key
- ☐ use a key to identify an unknown creature.

Living kingdoms

Imagine how difficult it would be to identify one of the millions of living things in the world if people couldn't agree on one system for classifying (or grouping) them. Fortunately, there is a worldwide classification system that most scientists do agree on. It groups living things according to the structural features they have in common. Structural features include skull shape, teeth, number of legs, as well as the structure of the **cells** that make them up.

Sorting things out

Swedish naturalist Carl von Linné (1707–1778) — also known as Linnaeus — was the first person to develop a proper system for classifying living things. It is known as the Linnaean system.

Using the Linnaean system, only two names are needed to identify an **organism** — a genus name and a **species** name (see the diagram below). This two-part name, called a scientific name, is always written in Latin. The genus name starts with a capital letter. For example, *Felis catus* is the scientific name of the organism with the common name (or everyday name) of house cat. *Musa acuminata* is the scientific name for a banana tree.

Taxonomists are scientists who decide how organisms are classified. They divide **living** things up into smaller and smaller groups based on their common features or characteristics. The smaller the group, the more features its members have in common. For example, members of the same family will have more features in common than members of the same class. Members of a species have more in common than members of a genus.

* 'Phylum' for Kingdom Animalia
** 'Division' for Kingdom Plantae

How living things are classified, using the Linnaean system

When members of the same species mate, they are able to produce **fertile** young. This is possible even though the individuals are not identical. For example, not all domestic cats (*Felis catus*) have the same colour, type of fur or ear shape. But all cats can mate with each other to produce fertile young. In the same way, not all banana trees look the same size and shape but they are still one species. They can reproduce to give more banana plants, which can then reproduce and give more banana plants and so on.

If members of different species mate, their offspring will not be fertile.

The five kingdoms into which organisms are classified

Go to worksheet 38

The five kingdoms

Most scientists today accept that there are five different kingdoms: Animalia, Plantae, Monera, Protoctista and Fungi.

Animalia includes worms, snails, fish, frogs, insects, reptiles, birds and mammals (such as dogs, camels and humans). They are complex organisms made up of many, often millions, of cells. They obtain food by eating or absorbing other living (alive or dead) things.

Plantae includes mosses, grasses, flowering plants, shrubs and trees. They are made up of many cells that contain **chlorophyll**. Chlorophyll allows plants to use the energy of sunlight to make their own food from carbon dioxide and water. Oxygen is released as a waste product. This food-making process is called **photosynthesis**.

Fungi includes mushrooms, toadstools, moulds, mildew and **yeasts**. They are usually made up of many cells, but may be only one. Unlike plants, they have no true leaves, flowers, stems — or chlorophyll. They obtain their food by growing on other organisms (alive or dead).

Protoctista includes green **algae** (from single-celled **diatoms** to multi-celled organisms such as **kelp**) and the single-celled **amoeba**. Although they may have some of the features of plants or animals, they cannot be classified as either of these. They mostly live in water.

Monera includes bacteria and **cyanobacteria**. They are the simplest organisms on Earth, being made up of one cell. Most get their food from other organisms such as dead animals and plants. Some bacteria cause disease; other bacteria are used to make foods such as yoghurt.

Activities

REMEMBER

1. List each of the five kingdoms and give two examples of each.
2. Although they are not identical to each other, what ability do members of the same species have?
3. Explain the difference between a common name and a scientific name.

THINK

4. Describe one way in which members of Kingdom Fungi are different from members of Kingdom Plantae.
5. To which kingdom do each of the following organisms belong: lizard, *E. coli* bacteria, brown seaweed, toadstool, moss.
6. The red kangaroo (*Macropus rufus*) and the grey kangaroo (*Macropus giganteus*) are members of the same genus. Can they breed with each other to produce fertile young? Give a reason for your answer.

CONNECT

7. Search the Internet to find more information (including pictures) about each of the five kingdoms. Use this information to create your own 'kingdom wheel' similar to that shown on the left. Use an A3 sheet of paper.

✓ checklist

I can:

- ☐ explain why Linneaus is famous
- ☐ show in a diagram how an organism is classified
- ☐ explain how organisms are classified into five kingdoms.

The animal kingdom

How many different **species** of animal can you think of? Not even scientists know how many there are on Earth. Over one million animal species have been classified so far, but hundreds of new species are being discovered every year. The task of classifying newly discovered animals begins with carefully observing their body characteristics.

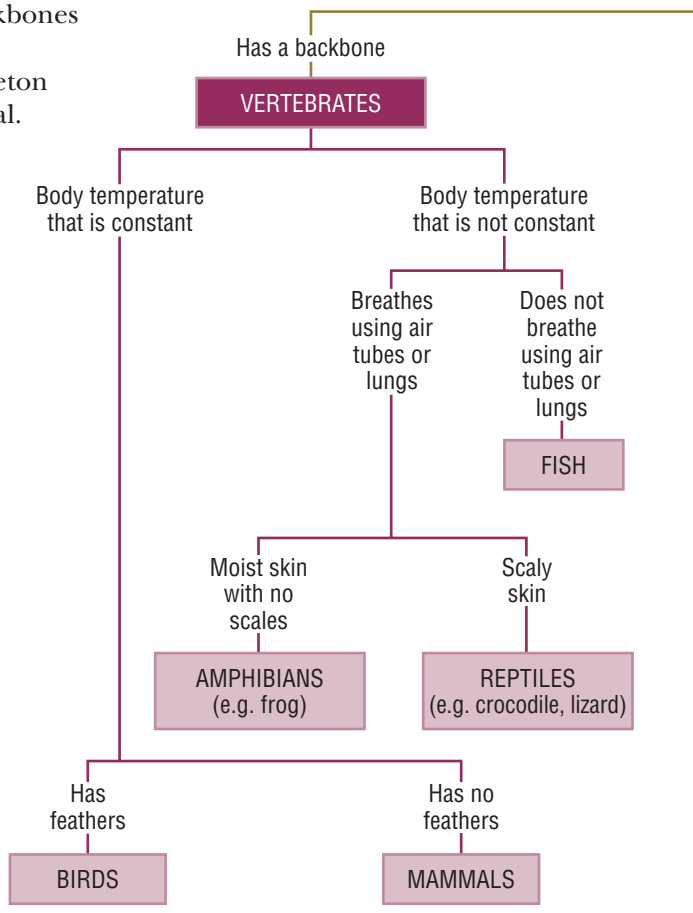
There is a huge variety of animals on Earth. They range from the massive blue whale (about 24 m long and 100 000 kg in weight) down to creatures that that are so small that we need a microscope to see them. Animals hop, swim, fly, walk and slide through their environments. Some live all their lives in water; others live in the soil, on icy slopes or in tree tops. Each type of animal has characteristics that best suit the environment in which it lives.

Vertebrates

There are many ways in which the massive animal kingdom could be broken down into smaller groups of animals that are more alike. Most scientists today start grouping them by whether or not they have a backbone. Animals with backbones are called **vertebrates**. The backbone is part of the skeleton inside the body of the animal. The scientific name for the vertebrates is Phylum Chordata.

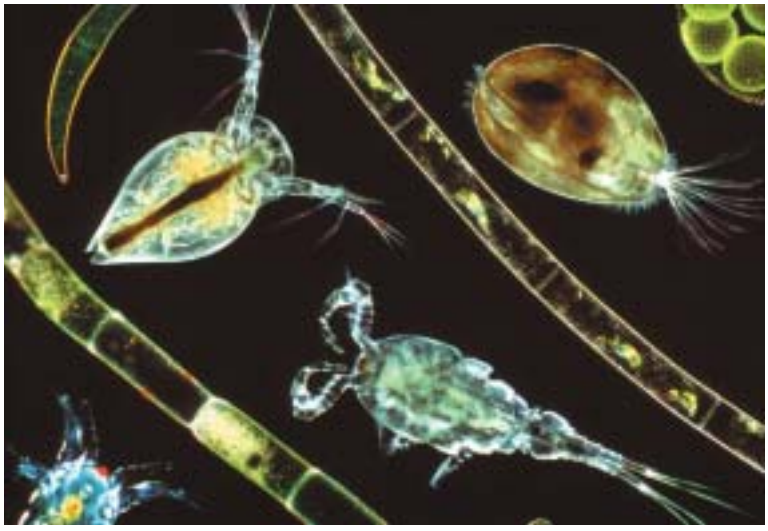
Invertebrates

Animals without a backbone are called **invertebrates**. These animals may have an external skeleton (called an **exoskeleton**), or no skeleton at all. An exoskeleton may be a shell or a hard, jointed covering that encloses the body. A growing animal with an exoskeleton sheds it from time to time. It is then replaced by a larger exoskeleton. This shedding is necessary because an exoskeleton does not grow like the bones in our skeleton do.



Dichotomous key showing how the main groups of animals are classified

The largest animal on Earth, the blue whale, feeds on some of the smallest animals on Earth. In the summer feeding season an average-sized blue whale feeds on up to 4 tonnes of tiny animals like those shown below every day.



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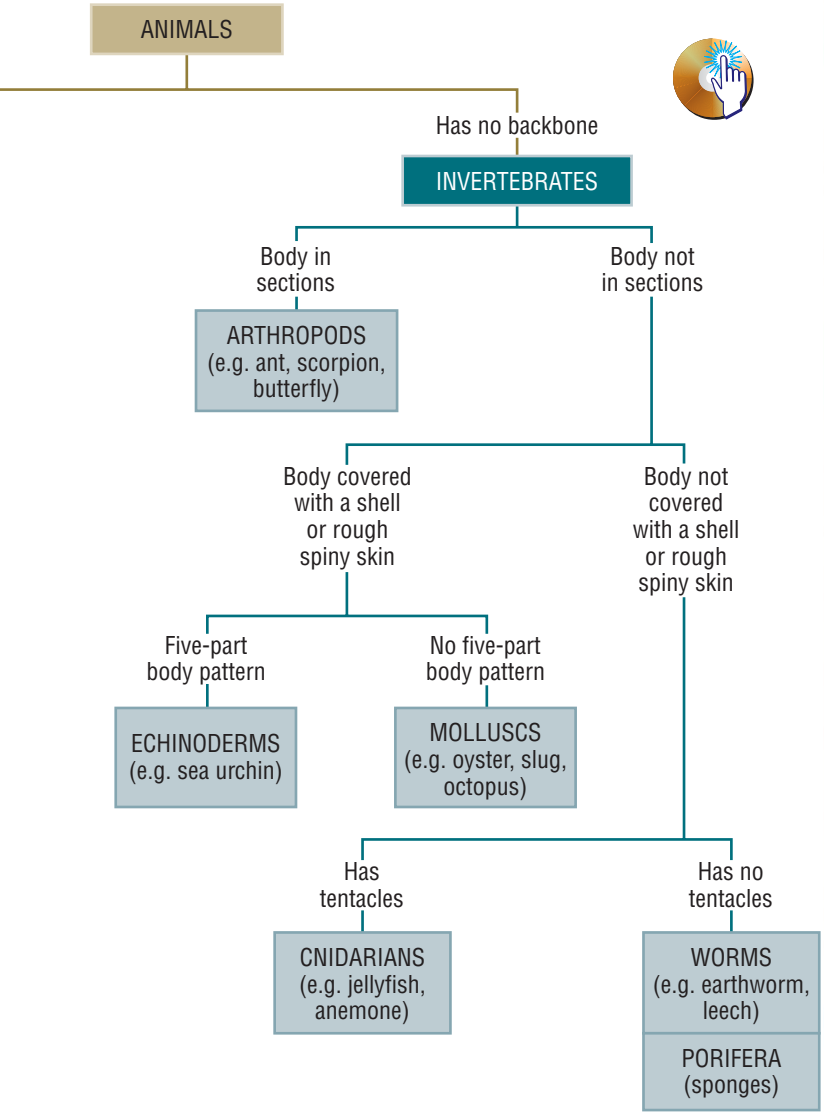
Three-quarters of all animal species have an exoskeleton. Most of these are **arthropods** (e.g. insects, spiders). Animals with exoskeletons are usually not as flexible as those with skeletons inside the body. Exoskeletons prevent an animal from bending and twisting easily because the muscles are attached to the inside of the hard skeleton. The muscles of animals with an internal skeleton are attached outside the skeleton and can pull on parts of the skeleton to move it in many directions.

Some animals with no skeleton at all have strong muscles (e.g. worms, jellyfish). Their muscles contract (get shorter) and stretch to allow their bodies to move through their environment.



Activities

REMEMBER

1. Describe the difference between vertebrates and invertebrates.
2. What is an exoskeleton?
3. What is the largest group of animals with an exoskeleton?

THINK

4. Which group of animals is the most common – vertebrates or invertebrates?
5. Is a snail a vertebrate or an invertebrate? Give a reason for your answer.
6. Worms have muscles around and along their bodies. These allow them to become long and thin one moment and short and fat the next. How might this help them move through the earth?
7. The system most scientists use for classifying animals that can be seen without a microscope starts by dividing them into those with and without a backbone. Suggest at least one other way to divide them into two groups.
8. Select a characteristic for the following group of animals to divide it into:
   (a) two groups
   (b) three groups.
   dolphin, slug, beetle, horse, jellyfish, dog, spider, ant
   Explain your choice of characteristic in each case.

SKILLBUILDER

9. Study the dichotomous key opposite.
   (a) To which group does each of these animals (below) belong?
   - One with a backbone, a changing body temperature and that does not use air tubes or lungs
   - One with no backbone, legs or covering shell and that has tentacles with stinging cells.

   (b) Work through the key backwards to list as many characteristics as you can for:
   • birds • molluscs • reptiles.

✓ checklist

I can:
- ☐ understand how animals are commonly classified
- ☐ explain the difference between vertebrates and invertebrates
- ☐ use data about animals in a dichotomous key to come to conclusions.

Inner backbones

Most animals have a skeleton on the outside of their body. This external type of skeleton is found in insects, for example. Other animals have no skeleton at all, such as worms. But when we think of animals, we usually think of those that have a skeleton with a backbone inside the body. These animals with a backbone, much like yours, are called **vertebrates**.

Vertebrates have the following characteristics:

- When they first start developing, they have a **cartilage** rod in their back. Cartilage is softer than bone. Your nose and ears are made mostly of cartilage.
- As they develop, the cartilage rod is replaced by a bony backbone made up of bones called **vertebrae**.
- They have gill slits near the throat when they start developing. In air-breathing vertebrates, these slits change into other structures or disappear completely. In humans, the gill slits disappear before birth.

Vertebrates can be 'broken down' further into the five groups described in the key opposite.

VERTEBRATES

Body temperature that is constant

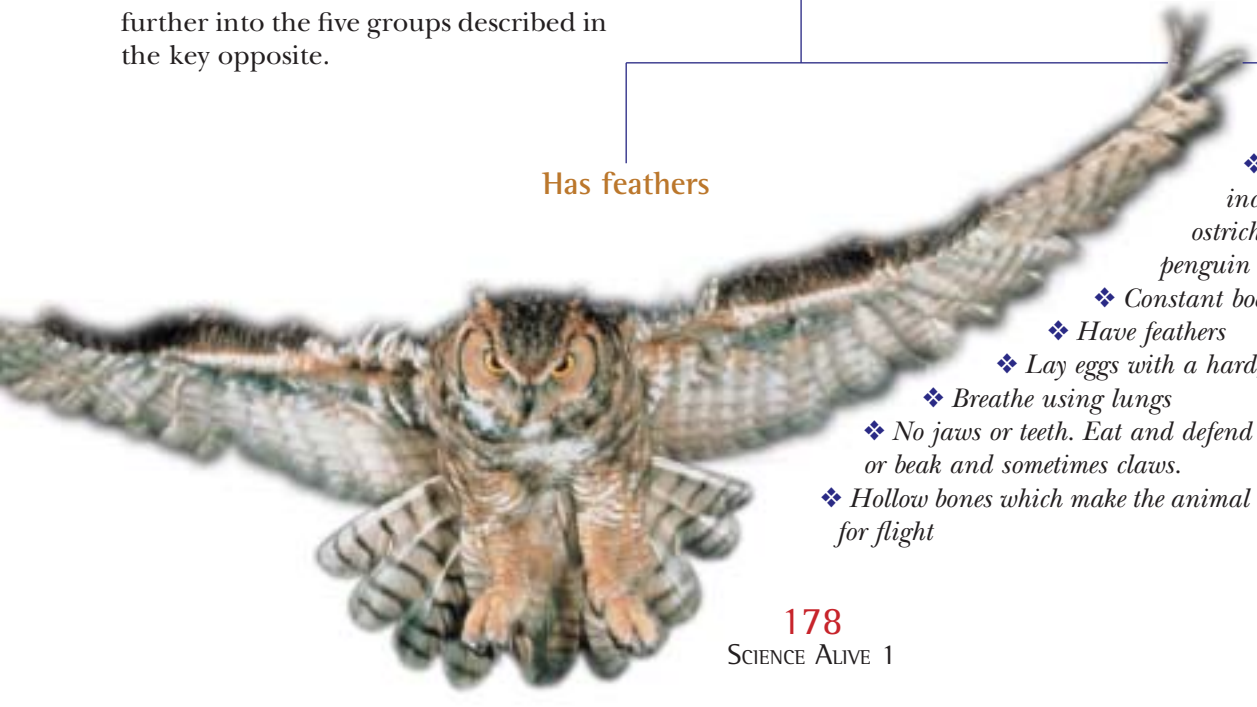
Breathes air using tubes or lungs

Moist skin with no scales

Amphibians

- *About 3200 **species** including toad, frog and salamander*
- *Changing body temperature*
- *Thin, smooth skin with no hair or scales*
- *Reproduce and lay eggs in water (mostly) or wet earth. Eggs have no shell.*
- *Usually start their life in water (with gills) and live as adults both on land and in water (with lungs)*

Has feathers



Aves (birds)

- *About 9300 species including owl, finch, ostrich, duck, parrot and penguin*
- *Constant body temperature*
- *Have feathers*
- *Lay eggs with a hard, brittle shell*
- *Breathe using lungs*
- *No jaws or teeth. Eat and defend themselves with bill or beak and sometimes claws.*
- *Hollow bones which make the animal light and well suited for flight*

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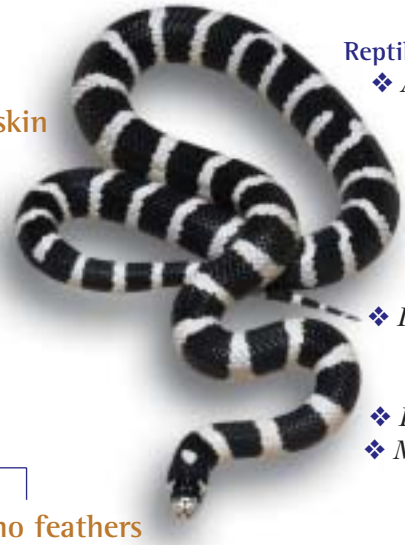
Body temperature that is not constant

Does not breathe air using tubes or lungs

Fish

- *Group has more species than all other kinds of vertebrates put together. Some have bony skeletons, like salmon and goldfish. Others, like sharks and stingrays, have skeletons of cartilage.*
- *Live only in water*
- *Most have changing body temperature*
- *Most have scales and a slimy coating that make the body waterproof*
- *Most have fins to help them turn when swimming*
- *Breathe through gills*



Scaly skin

Reptiles

- *About 6000 species including turtle, crocodile, snake and lizard*
- *Changing body temperature*
- *Skin has dry scales that overlap. These help to retain the body's moisture. Many reptiles shed their skin a number of times each year.*
- *Lay eggs mostly on land. Shells are leathery, making the eggs watertight and less likely to break open.*
- *Breathe using lungs*
- *Most have very good eyesight*

Has no feathers

Mammals

- *About 4000 species including dolphin, rabbit, giraffe, platypus, rat and you*
- *Constant body temperature*
- *Skin has fur or hair, even if only a few bristles*
- *Breathe using lungs*
- *Newborns feed on their mother's milk, produced by* ***mammary glands***
- *Three types — placental mammals that give birth to fully developed young, marsupials whose young develop in a pouch and monotremes that lay eggs.*

Activities

REMEMBER

1. Apart from an internal skeleton with a backbone, what other features do all vertebrates have?
2. Name the five groups into which vertebrates are commonly divided. To which group do you belong?

THINK

3. Which groups of vertebrates have a changing body temperature? Explain what this means and how this characteristic affects the animal's behaviour.
4. Which group of vertebrates has the following characteristics?
   (a) Changing body temperature; lays eggs in water; thin, smooth skin
   (b) Dry overlapping scales, breathes using lungs, lays eggs mostly on land

SKILLBUILDER

5. To which group of vertebrates do the following animals belong?
   (a) Turtle
   (b) Emu
   (c) Human
   (d) Dolphin

INVESTIGATE

6. Fish can be divided into three separate groups. Find out what the groups are, the characteristics of each group and list two examples from each group.

✓ checklist

I can:

- ☐ describe the features of a vertebrate
- ☐ list the sub-groups into which vertebrates are divided.

London Science

London Town: 23 November 1799 Weather: Overcast, drizzle, 30 °F

Platypus Hoax Exposed!

Examining the so-called Platypus (an Artist's Impression)

The so-called Platypus is no more than a clever Hoax. This is the firm Opinion of Dr Shaw, a highly respected Scientist with the British Museum.

Dr Shaw made his Announcement last Week after inspecting a stuffed Creature sent to England from the Colony of New South Wales. Although he was not able to cut the Bill off with Scissors, Dr Shaw said it did not take long to find the Hidden Stitches.

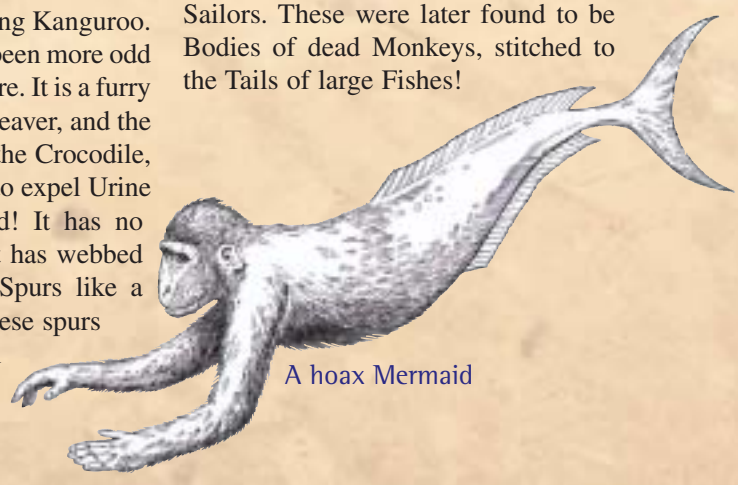
'It is the practical Joke of some **Taxidermist**', he said. 'This Skin is just a Combination of Skins of other Animals. They have all been sewn together to look like they belong to a single, absurd Beast. It insults our Intelligence!'

The stuffed Specimen and a Sketch of the Creature were sent to Dr Shaw earlier this Year. A Number of bizarre and ugly Animals have already been found in the remote Colony since it was settled eleven Years ago. There was the fierce Koala Bear. And, of course, the strange hopping Kanguroo.

None, however, have been more odd than this so-called Creature. It is a furry Beast with a Tail like a Beaver, and the Bill of a Duck. Yet, like the Crocodile, it has only one Opening to expel Urine and Faeces. How absurd! It has no Feathers or Scales. Yet it has webbed Feet like a Goose and Spurs like a Rooster. What's more, these spurs contain poisonous Venom like that of a Snake.

Dr Shaw's Verdict was quite clear. 'Of course, a Creature that has Hair and gives Milk, but with a Bill and webbed Feet, could not possibly exist. It could not be classified as a Mammal, Bird, Reptile or Amphibian — nor for that matter any other Class of Animal. These People in the Colony are suggesting it belongs in a Sub-class of its own. The very Idea is ridiculous! You would think someone who wanted to fool us would have made a more believable Animal', he said.

Dr Shaw has good Reason to suspect a Trick. Recently, there have been a number of Cases where smart Taxidermists, eager to improve their business Prospects, have been caught selling 'Mermaids' to our British Sailors. These were later found to be Bodies of dead Monkeys, stitched to the Tails of large Fishes!



A hoax Mermaid

SENIOR TAXIDERMIST BRITISH MUSEUM

The British Museum wishes to employ an experienced Taxidermist to prepare animal specimens for its Collection from the Colonies.

Experience required in:

- skinning Animals
- preparing plaster casts of animal Bodies
- sewing Skins together
- mounting Animals in a lifelike Pose.

Abilities needed:
a very good knowledge of Anatomy, an Eye for Detail, artistic Ability, and a strong Stomach.

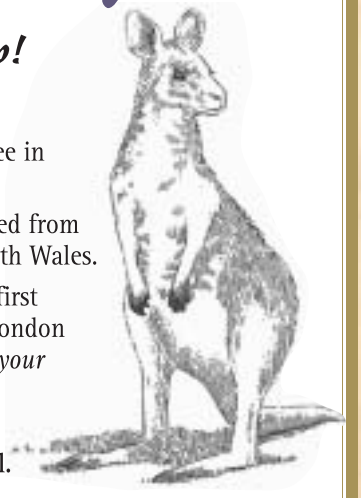
In Search of 'the Watermole'

Ever since this strange 'Platypus' was sent to London, it is reported that Colonists everywhere are looking for more specimens. They call it a 'Watermole' in the Colony.

Captain Hunter, the current Governor, says he first saw one in 1797. He says he watched an Aborigine sit for over an Hour beside a Lagoon near the Hawkesbury River. He had his Spear poised, waiting for the Creature to surface. You never know what sort of Stories you are going to get out of these faraway Colonies!

See a Kanguroo!

Roll up . . . roll up!



See one of the strangest Creatures you will ever see in your Life.

It's alive – and just arrived from the Colony of New South Wales.

If you missed seeing the first Kanguroo brought to London town in 1796 – *now's your Chance.*

It does not bite or spit – but watch out for its Tail.

Only three pence!

Long Queues expected, so hurry!

Activities

REMEMBER

1. Why did Dr Shaw think the platypus was so peculiar?
2. What had previously happened to make Dr Shaw suspicious about the strange skin he was sent?
3. What other 'strange' animals had already been found in New South Wales by 1799?

THINK

4. What features of the platypus are:
   (a) like those of other mammals?
   (b) unlike those of other mammals?

INVESTIGATE

5. Today, the platypus and the echidna are classified as a special type of mammal known as a monotreme. Find out which feature separates these two animals from all other mammals?

CREATE

6. Make a model of the platypus using materials such as fur, plastic and paint. We now know that only the male platypus has spurs. They are on the inner side of each hind 'leg'.

COMMUNICATE

7. Conduct a class role-play of what it must have been like for people of London town to see a kangaroo for the first time. Imagine what would be said. Some can be members of the public (adults and children), some can be museum staff and some can be scientists like Dr Shaw.

CONNECT

8. Go to www.jaconline.com.au/science/weblinks and click on the Platypus link for this textbook. Look for facts about where the platypus is found, what it eats, and what sort of home it makes. Complete a poster that includes diagrams, sketches, a map and, if possible, pictures.

✓ checklist

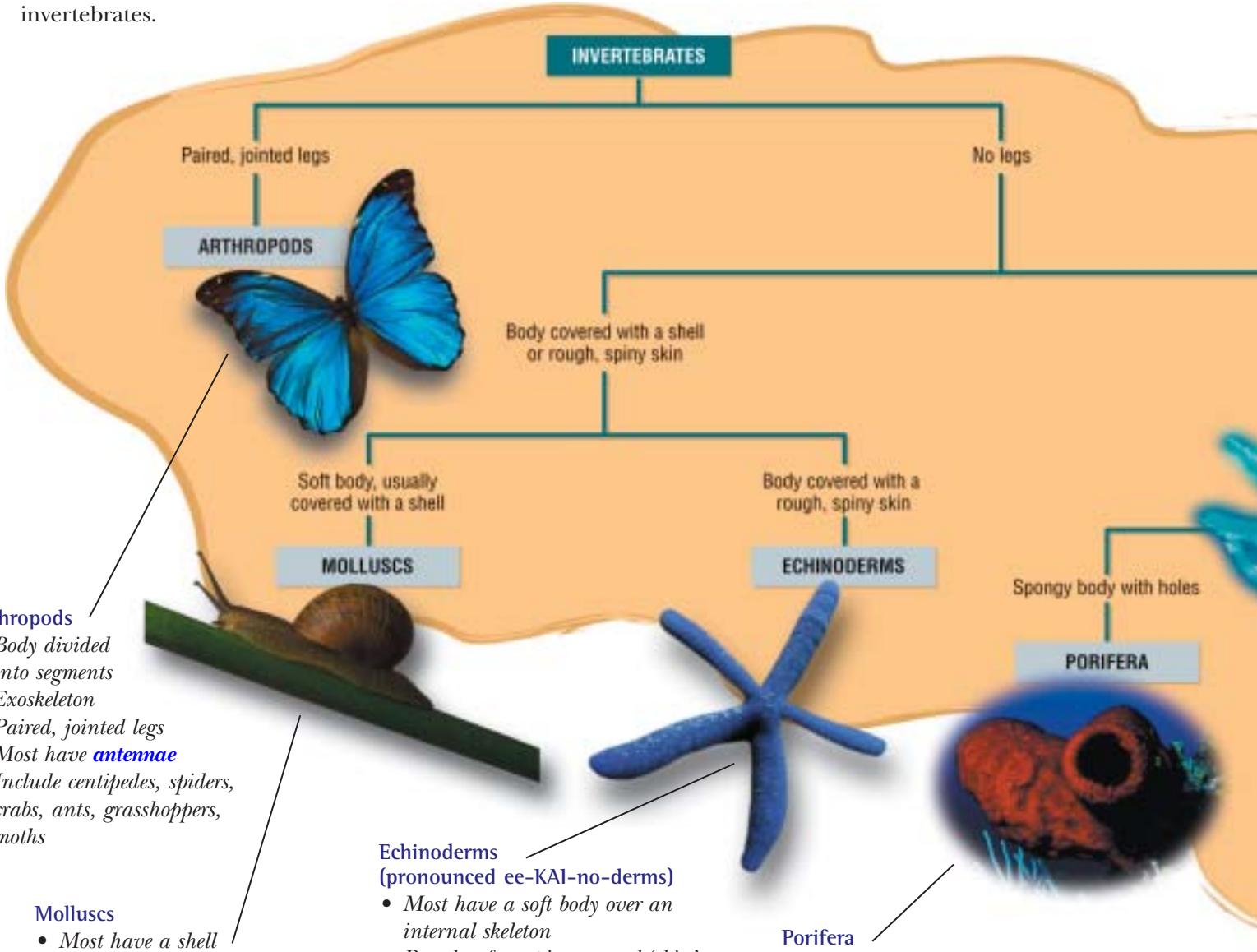
I can:

- ☐ understand why it was hard to classify the platypus
- ☐ explain why a platypus is like and unlike other mammals
- ☐ appreciate the need for adequate evidence before accepting or rejecting ideas.

Spineless wonders

The main characteristic of **invertebrates** is that they don't have a backbone. Most of them have an **exoskeleton** — a skeleton on the outside of their body. Some of them have no skeleton at all. Some of them, like sea stars, have a skeleton (but no backbone) inside their bodies.

No-one knows how many **species** of animal there are on Earth. What is known for sure is that most of them are invertebrates. The **dichotomous key** below describes some of the characteristics of invertebrates.



Arthropods
- *Body divided into segments*
- *Exoskeleton*
- *Paired, jointed legs*
- *Most have **antennae***
- *Include centipedes, spiders, crabs, ants, grasshoppers, moths*

Molluscs
- *Most have a shell*
- *Soft body, not divided into segments*
- *No legs, but may have tentacles*
- *Have a strong 'foot' muscle to help them move*
- *Include oysters, octopus, scallops, slugs, snails*

Echinoderms
(pronounced ee-KAI-no-derms)
- *Most have a soft body over an internal skeleton*
- *Rough, often spine-covered 'skin'*
- *Body has a five-part pattern*
- *Move through water by taking water in and pushing it out of tubes in their bodies*
- *Include sea stars, sea urchins, sea cucumbers*

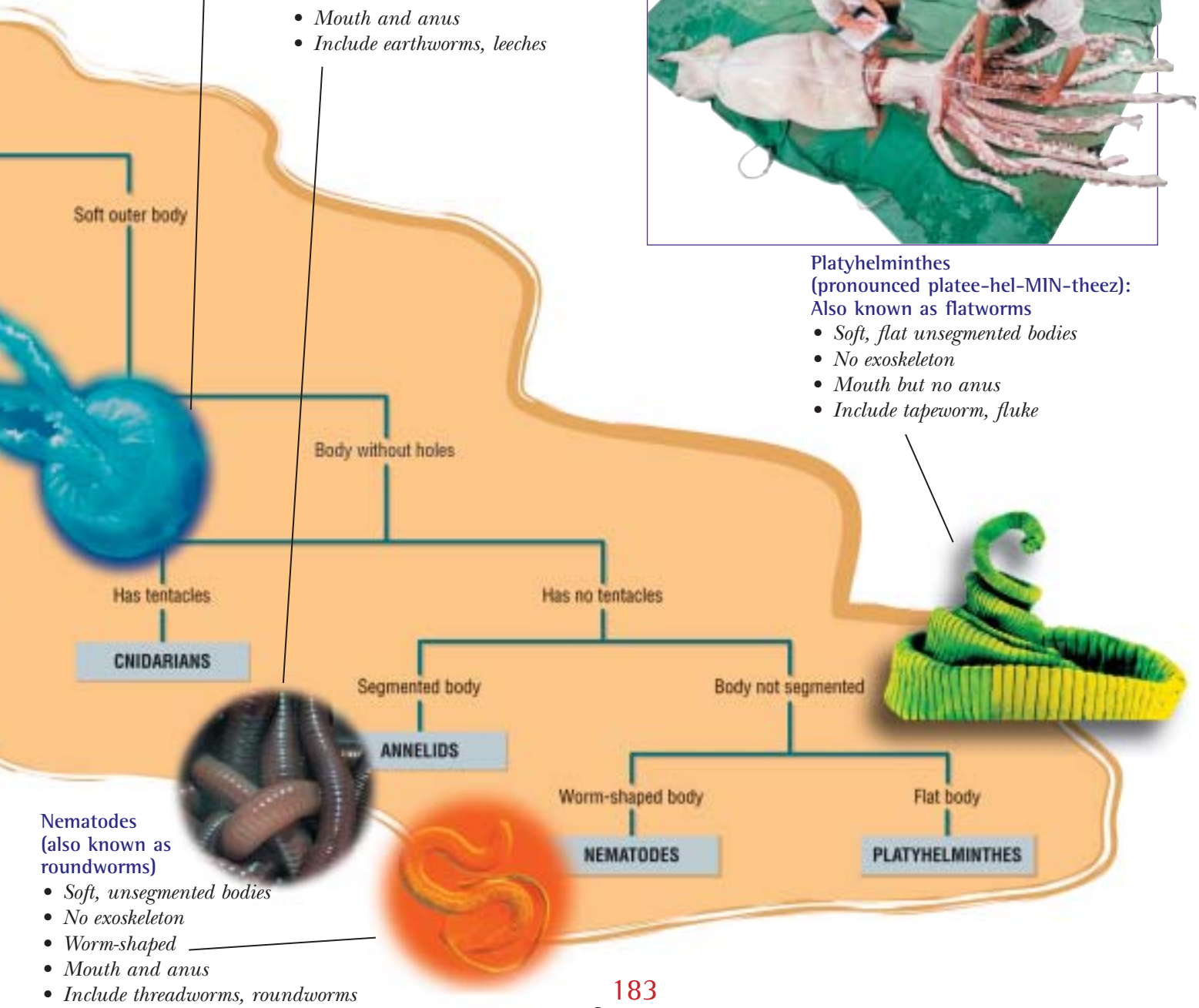
Porifera
- *Spongy body with no body organs or tissue*
- *Exoskeleton made of fibres or pointed 'needles'*
- *Water and food enter through tiny pores (holes) in body*
- *Wastes pass out through one big opening*
- *Include barrel sponges, glass sponges, tube sponges*

Cnidarians
(pronounced ny-DAIR-ee-ins – the 'C' is silent)

- *Hollow, soft body*
- *No body organs*
- *Take in food and pass out waste through one opening*
- *Have tentacles containing stinging* ***cells****, which fire shots of toxin*
- *Include box jellyfish, sea anemone, Portuguese man-of-war, coral*

Annelids
(also known as segmented worms)

- *Soft, segmented bodies with an obvious head*
- *No exoskeleton*
- *Mouth and anus*
- *Include earthworms, leeches*



Nematodes
(also known as roundworms)

- *Soft, unsegmented bodies*
- *No exoskeleton*
- *Worm-shaped*
- *Mouth and anus*
- *Include threadworms, roundworms*

Platyhelminthes
(pronounced platee-hel-MIN-theez): Also known as flatworms

- *Soft, flat unsegmented bodies*
- *No exoskeleton*
- *Mouth but no anus*
- *Include tapeworm, fluke*

The largest invertebrate is a mollusc called the giant squid. Its scientific name is *Architeuthis dux*. It lives in the very deep parts of the ocean. Its eyes are as large as soccer balls. Its beak-like mouth can cut through steel cables. The giant squid can be as long as 18 m. Two of its ten tentacles are much longer than the others. These are used to catch food.

In 1966, two lighthouse keepers in South Africa watched a giant squid wrap its tentacles around a baby whale to drown it. The whale's mother could do nothing to save her calf.

Arthropods

Mosquito

About 80 per cent of invertebrates are **arthropods**. The insect is the most common arthropod. There are about six million insect species known to exist. Many insects **pollinate** flowers. Some provide us with food (e.g. bees provide honey). Insects are a food source for many animals such as other insects, birds and fish. Some insects feed by chewing; others, like the mosquito, suck up their food (sometimes human blood!) through a long thin tube called a **proboscis**. The proboscis of some insects rolls up at the end when not in use (a bit like a party whistle). All insects have three pairs of legs. An insect's legs are connected to the middle section of its body, called a **thorax**.

Spider

Crab

Centipede

Like the mosquito, all other insects have:

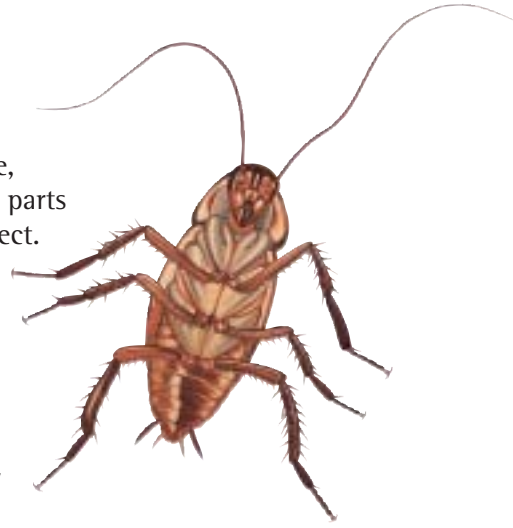
- an exoskeleton
- a body made up of three segments — head, thorax and **abdomen**
- one pair of antennae. Most insects smell using their antennae. (Some insects use their feet to taste things.)
- internal tubes that end in openings in their sides, through which they breathe.

Millipede

The other arthropods shown in this column are not insects. The spider, crab, millipede and centipede all have more legs than insects. There are also other significant differences.

What body features can I see on an insect?

You will need:
preserved or freshly killed cockroach or grasshopper
hand lens
disposable gloves
probe
pen
paper.



- Place the insect upside down on a viewing slide. With the unaided eye, see if you can identify all the body parts labelled in the diagram of your insect.
- Try to identify the three main segments of the insect – the head, the thorax and the abdomen. Remember that the insect's legs are attached to its thorax.
- Some insects have **cerci**. The cerci of a cockroach are very large. They detect the tiniest motion and help warn the cockroach of approaching danger. Some insects have cerci that are more like stumps than hairs. The cerci of some insects are too small to see or not present at all.

1. Does your insect have cerci that you can see with a hand lens? If it does, sketch them.
2. Based on what you see without the hand lens, sketch the detail of one of the insect's back legs.

- Now use a hand lens to look at the back legs.

3. What feature can you observe more clearly on the insect's back leg? What purpose do you think these serve?

- Use the hand lens to look at the head of the insect.

4. Does it have a proboscis or chewing mouthparts? Draw a labelled diagram of the insect's feeding parts.

- Insects breathe through tiny holes called **spiracles**. Look at the side of your insect with a hand lens to see if you can find a 'line' of spiracles. You are more likely to see these on a grasshopper.
- Look at one of the eyes of the insect. There are many lenses in each eye. (You have only one in each eye.)

5. How do you think these extra lenses might help the insect's vision?

- Hold the insect with two hands. Without snapping it in two, see if you can gently bend the tail end of the body sideways (not upwards) towards the head.

6. What feature of the insect makes it difficult to bend its body?

The cockroach is an amazing insect. It has been around for about 350 million years. If you cut off its head, it will stay alive for about a week. It dies only because it has no mouth to drink through. It can also run faster than any other insect – almost four kilometres per hour. It can also change direction very quickly. If a cockroach loses a leg, a replacement will appear next time it sheds its exoskeleton.

Activities

REMEMBER

1. What is an invertebrate?
2. Give examples of two invertebrates that have an exoskeleton, and two that have no skeleton at all.
3. What group of invertebrates do each of the following animals belong in: spider, leech, sea star, moth.
4. List the main characteristics of all insects.

THINK

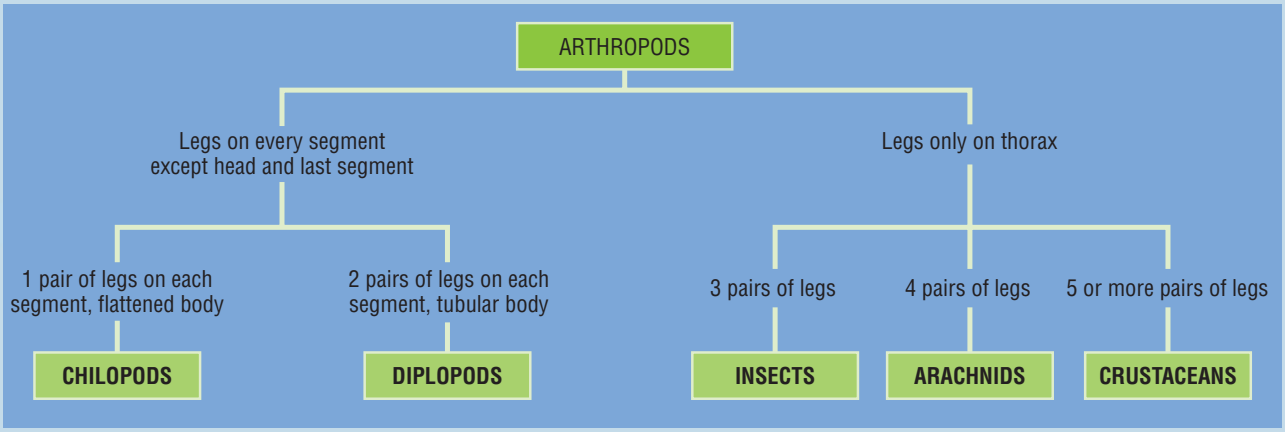
5. Write a paragraph listing as much evidence as you can to support the statement: 'The cockroach is an arthropod'.
6. A snail is a mollusc. So is the giant squid. In what ways are they alike and in what ways are they different?

SKILLBUILDER

7. Describe the characteristics of the following animals using the labels used to classify them in the dichotomous key on pages 182–3: coral, earthworm, fluke, centipede.
8. If you found an animal with a soft, segmented body, no legs or tentacles or hard external covering, how would you classify it, based on the data given in the dichotomous key?
9. (a) Use the key below to classify the five arthropods shown opposite.
   (b) Why is the key below not a dichotomous key?

CONNECT

10. Go to www.jaconline.com.au/science/weblinks and click on the Giant Squid link for this textbook to discover more about this amazing creature.
11. Go to www.jaconline.com.au/science/weblinks and click on the Cockroach link for this textbook to discover more about this amazing creature.



checklist

I can:

- ☐ describe the features of invertebrates
- ☐ use a dichotomous key to classify invertebrates
- ☐ describe the features of arthropods.

The plant kingdom

We must have plants to live. They provide the oxygen we breathe, the food we eat and the food eaten by animals we eat. They also provide medicines, and raw materials such as timber and cloth fibres. Scientists think there may be about 350 000 **species** of plants, but no-one is sure. Plants are classified on the basis of such features as the presence of roots, stems, leaves, seeds and flowers.

Like most members of the animal kingdom, plants are complex **multicellular organisms**. Unlike animals, they:

- have structures such as roots, stems and leaves
- have thick **cell** walls
- make their own food through a process called **photosynthesis**
- are not free to move from place to place.

Classifying plants

One of the main ways plants can be grouped is by whether or not they have transport tissue. Known as **vascular tissue**, this transport tissue consists of two sets of tubes. One set, made of **phloem cells**, transports food from the leaves to the rest of the plant. The other set, made of **xylem cells**, transports water and minerals from the soil to the rest of the plant.

Plants with vascular tissue are called **tracheophytes**. They have roots, stems and leaves. They include flowering plants, conifers and ferns. Plants that do not have vascular tissue are called **bryophytes**. They are usually very small plants. Bryophytes include mosses and liverworts.

Plant group	Location
Flowering plants (e.g. roses, fruit trees, grasses, eucalyptus trees)	Mostly on land
Conifers (e.g. firs, pines and spruces)	On land
Ferns (e.g. maidenhair fern, fishtail fern)	Damp, shady, cool regions
Mosses (e.g. sphagnum moss)	Damp, shady, cool regions
Liverworts	Damp, shady, cool regions

Stem and roots	Leaves	Flowers	Seeds
Yes	Yes	Yes	Yes. Reproduce from seeds. Seeds form inside flower, which develops into a fruit.
Yes	Yes. Mostly fine and needle-shaped.	No	Yes. Form on scales of cones.
Yes	Leaves are fronds which uncurl as they get bigger.	No	No. Reproduce from **spores** on leaves. These are released from brown spore cases that form on the underside of leaves.
No stem. 'Roots' are more like fine hairs.	Yes, but simple structure. Tiny and dainty.	No	No. Reproduce from spores.
No stem. 'Roots' are more like fine hairs.	Yes, but simple structure. Flat, thick and leathery.	No	No. Reproduce from spores.

Activities

REMEMBER

1. Explain the ways that plants differ from animals.
2. What is the main difference between bryophytes and tracheophytes?
3. Which groups of plants are called tracheophytes?
4. Which groups of plants are called bryophytes?
5. State two differences between conifers and flowering plants.

THINK

6. Suggest why plants with vascular tissue are usually larger than ones without it.
7. A description of three plant specimens follows. Name the plant group to which each may belong.
   (a) A plant found beside a rainforest creek with no seeds or clear root system and with thick leaves
   (b) A plant with an obvious root system, a stem, leaves and with fruit containing seeds
   (c) A plant found in a cool, shady rainforest, with a horizontal stem and curled up new leaves with rows of brown spots underneath them

SKILLBUILDER

8. Use the information in this spread to create a key that shows one way to classify plants. Start by deciding which feature best divides plants into two groups. For each group, decide what features they have in common, and how they differ. This will help you decide how to break them down further into sub-groups. If you need some help with drawing keys, check some shown earlier in this chapter.

CONNECT

9. Go to www.jaconline.com.au/science/weblinks and click on the Corpse Flower link for this textbook. Find out what purpose the flower's dreadful smell serves.

✓ checklist

I can:

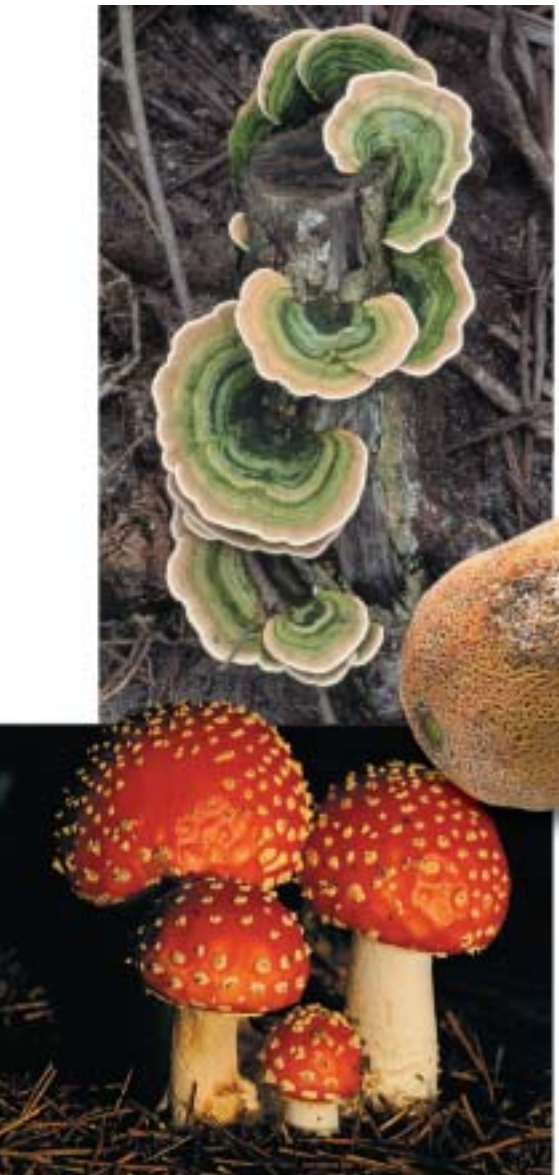
- ☐ explain some of the main features of plants
- ☐ use data to create a classification key.

The other kingdoms

Most of the **living** things that we recognise are plants or animals. But some of the most spectacular and unusual living things belong to the other three kingdoms of living things — Kingdom Fungi, Kingdom Monera and Kingdom Protoctista. Within these lesser-known kingdoms are **organisms** that can keep us alive, make us sick or even kill us.

Kingdom Fungi

Fungi come in an amazing variety of shapes and colours. Perhaps the most familiar are the mushrooms we eat. But fungi also include toadstools, truffles, mould, mildew and **yeast**.



Fungi used to be classified as plants. However, unlike plants, they have no true roots, leaves, stems or flowers. Also, they do not contain **chlorophyll**. This means they cannot make their own food. Instead, they produce chemicals to break down food from outside sources. The broken-down food is then absorbed into the fungi. Different sorts of fungi feed on different sorts of food. Some grow on or in dead animal or plant matter (e.g. vegetable scraps, cow dung, decaying fruit) and slowly break it down (or decompose it). Some grow on or in living **organisms**. Such fungi are called **parasites**.

Fungi grow from tiny **spores** released by a 'parent' fungus. These are blown through the air, or carried by animals. Some fungi have interesting ways of releasing their spores. The *Pilobolus*, which lives in cow dung, releases its spores by exploding. Spores can be shot up to two metres high by the force of the explosion, which is set off by sunlight.

Some fungi cause plant diseases such as stem rot, and painful infections such as **tinea** and ringworm. The antibiotic penicillin is made from a fungus. So is yeast, which is used in the making of bread and beer.

Under the microscope

Members of two of the kingdoms — Monera and Protoctista — are generally so small that they can be seen properly only under a microscope. To see some monerans you would even need to use the more powerful **electron microscope**.

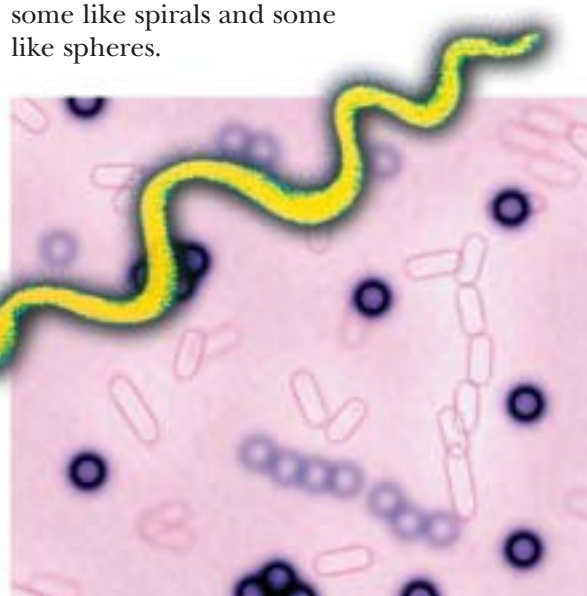
Kingdom Monera

Monerans are thought to be the first form of life to exist on Earth. They are very simple organisms consisting of one **cell** without a **nucleus**. They are everywhere — in water, in soil, in the air and in your body. You might know them as bacteria.

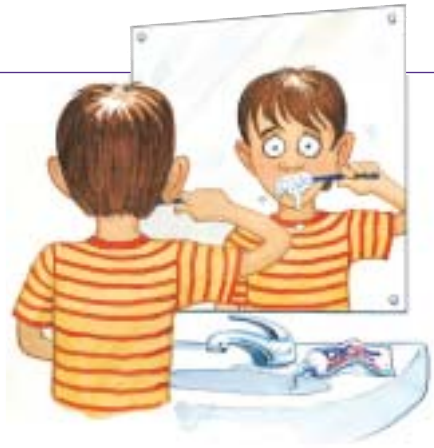
Monerans can be both helpful and harmful. Some cause illnesses such as cholera and pneumonia. Some cause tooth decay. Some, such as *Salmonella*, can give you food poisoning. On the positive side, other monerans that live in your intestine help you to digest food and make vitamins. Others are used to make foods such as yoghurt and vinegar.

Cyanobacteria, sometimes called blue-green **algae**, are another member of Kingdom Monera. Like some plants, this group makes its own food using **photosynthesis**.

Bacteria are classified by their shape. Some are shaped like rods, some like spirals and some like spheres.



Did you know that toothpaste contains the remains of lots of crushed diatom shells? When diatoms die, their microscopic shells pack down in layers to form diatomaceous earth (which is what is used in toothpaste). This is why toothpaste is a bit gritty.



Kingdom Protoctista

This kingdom could be described as the 'leftovers'. It includes organisms that don't belong in any of the other four kingdoms. Most protoctists are made up of only one cell. Unlike bacteria cells, the cells of protoctists have a nucleus.

Some protoctists (e.g. **diatoms** and algae) have features that are a bit like plants. Algae, for example, make their own food using photosynthesis, like a plant. Yet algae have no true roots, leaves or stems and no **vascular tissue**. Other protoctists, such as the blob-like ***amoeba*** and *plasmodium* which causes malaria, are more animal-like. Others are a bit like fungi, such as the one that destroyed the potato crop in Ireland in 1845–46, causing the death of about 30 per cent of the population.

An amoeba — animal-like, but is only one cell

How do I classify lichen?

You will need:
piece of lichen (You will find it growing on rocks and tree trunks in colder, wetter regions, especially towards the tops of ranges and hills.)
stereomicroscope
probe.

Look carefully at the lichen under a microscope.

- Identify any true roots, stems, leaves or flowers.
- Identify any thread-like parts.

1. Describe the appearance of the lichen. Can you see two different sorts of organism?
2. Lichen is actually made up of a fungus and an alga, growing together. Which part do you think is the fungus, and which the alga? Why?
3. What benefits do you think the algal cells in lichen provide for the fungus part?
4. What benefits do you think the fungus provides for the algal cells?
5. How would you classify lichen? Why?
6. What does this suggest about the difficulties that scientists sometimes face in trying to classify organisms?

Activities

REMEMBER

1. List at least two characteristics of:
   (a) fungi
   (b) monerans
   (c) protoctists.

THINK

2. Some scientists argue that algae, lichen and fungi are plants. Give some reasons why they might believe this.
3. Ringworm is a little worm that gets under your skin and burrows around in the shape of a ring. Is this statement true or false? Explain.
4. Algae is a member of Kingdom Protoctista. Why do you think it is misleading, and a bit confusing, to refer to cyanobacteria as blue-green algae?

INVESTIGATE

5. Place each of the following items in its own sealed plastic bag: a thick slice of bread, a tomato, and a damp piece of bark from a moist part of the garden. Leave each bag in a warm place for about a week, or until you notice mould or mildew starting to form on each item. Look carefully at the different moulds that form. If possible, also observe them with a stereomicroscope. **Do not open the plastic bag.** List features of the moulds that would help you to classify them. Consider colour, texture, shape, etc.

✓ checklist

I can:
- ☐ describe the characteristics of fungi, monerans and protoctists
- ☐ explain why it is hard to classify some organisms.

Check and challenge

CLASSIFICATION

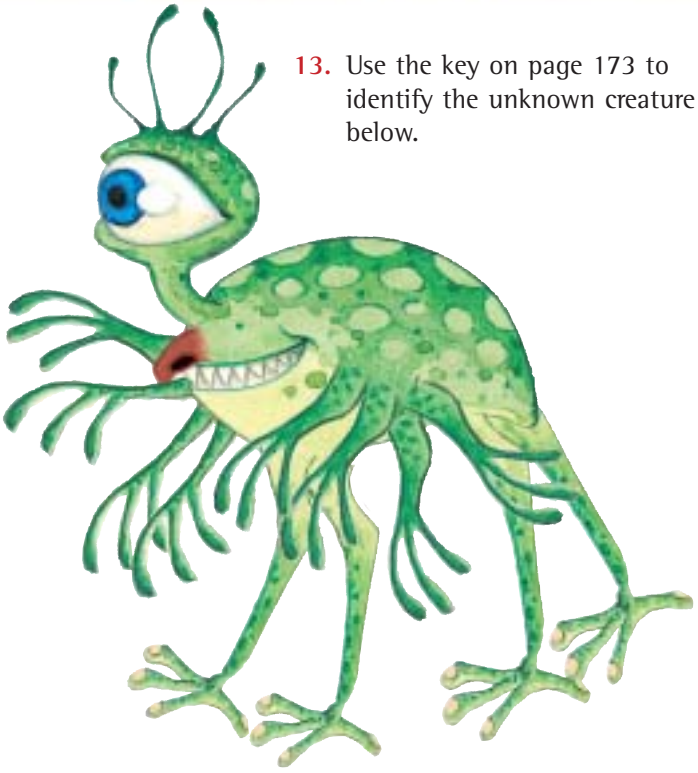
Living things – similarities and differences

1. List the characteristics of all living things.
2. Why is it helpful to classify living things?
3. Briefly describe the different ways in which:
   (a) you and a plant take in and release air
   (b) you and a plant take in water
   (c) a wattle tree and a dog get their food
   (d) a mushroom and a single-celled protoctist reproduce.
4. Copy and complete the following table.

Groups of organisms	Ways they are alike	Ways they are different
Amphibians and reptiles		
Molluscs and echinoderms		
Rose bushes and ferns		

Sorting organisms using dichotomous keys

5. (a) Explain the difference between vertebrates and invertebrates.
   (b) Why are these two groups commonly the first groups identified when drawing a dichotomous key for animals?
   (c) List the five groups of the vertebrates.
6. Do all invertebrates have an exoskeleton? Explain.
7. Which feature of arthropods makes it possible to divide them into five different groups?
8. Why was it so difficult to classify the platypus when the animal was first found? What makes it different from the other two groups of mammals?
9. Mushrooms were once classified as plants.
   (a) Suggest why they could have been classified as plants.
   (b) What features of mushrooms are different from those of plants?
10. Draw up a dichotomous key to classify the animals pictured below. First, list the featues of each animal, based on what you can see here and what you know about the animal. Think about their shape, the way they move, eat and reproduce. How many sub-groups of animals are you able to identify from your key?
11. Draw up a tabular key to classify the pictured animals.
12. To which group of invertebrates does the sea star shown below belong?



13. Use the key on page 173 to identify the unknown creature below.

Five kingdoms

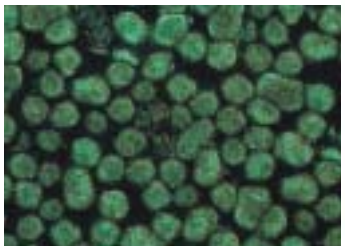
14. Name the five kingdoms into which living things are usually grouped. List examples of two organisms from each of the kingdoms. What characteristics does each pair of organisms have in common?
15. To which kingdom do the following organisms belong? Explain why in each case.
algae, ant, *Salmonella* bacteria, mildew, moss, tapeworms
16. Imagine that you are Linnaeus speaking for the first time to a group of naturalists about the new system of classification you have developed. Be prepared to role-play this presentation for the class if asked.

Viruses — living things?

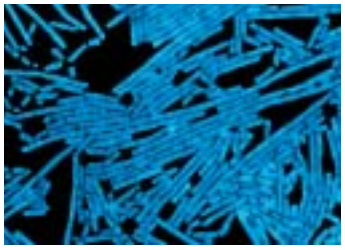
Viruses are very small — much smaller than bacteria. The influenza virus, for example, is only 100 nanometres in diameter. (One nanometre is one-millionth of a millimetre.) Viruses can be seen only with an electron microscope.

Viruses are not cells and therefore have no **nucleus**. They are made up of a strand of **genetic** material within an outer coat of **protein**. This genetic material is the same as that in the nucleus of living cells. It contains the instructions for reproducing the virus.

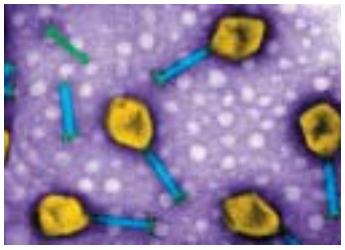
Viruses cannot function on their own. They do not eat or pass waste. In fact, a virus may stay **dormant** for many years. Like some chemicals, viruses can form crystals. But like living things, they can also reproduce. The big difference is that they can only reproduce inside a living cell. In this respect, they are like **parasites**. Once inside **a host cell**, they force it to reproduce millions of new virus particles. Eventually, the host cell may die and burst open.



Influenza A virus



Tobacco mosaic virus (TMV, RNA virus)



T4 bacteriophage (DNA virus)

1. Do all viruses look the same? Consider their different shapes and sizes.
2. In what ways are viruses like living things, and not like living things. List as many points as you can. In particular, think how viruses are like members of the Kingdom Monera.
3. Many seeds of desert plants lie dormant in the ground — sometimes for years. They burst into life only following rain. Discuss the following questions as a class:
   (a) Are the seeds of desert plants alive when they are lying dormant?
   (b) Are viruses alive when they lie dormant, waiting for a host cell to invade?
   (c) What are the similarities and differences between such a plant seed and a virus?

Summary of key terms

abdomen: the end part of an insect's body behind the thorax

algae: commonly known as seaweed. An organism belonging to Kingdom Protoctista which lives in water. It contains chlorophyll, and may range from a single-celled organism to a multi-celled long structure such as seaweed.

amoeba: microscopic organism consisting of one cell which has a thin membrane. It belongs to Kingdom Protoctista but has animal-like features.

antennae: the pair of fine probing feelers on the head of many invertebrates (e.g. flies, crabs)

arthropod: animal that has an exoskeleton, a segmented body and jointed legs (e.g. insects, crabs)

bryophyte: plant that does not contain vascular tissue (e.g. mosses and liverworts)

cartilage: a waxy, whitish, flexible substance that lines or connects bone joints or, in some animals such as sharks, replaces bone in supporting body tissue. Your ears and the tip of your nose are shaped by cartilage.

cell: the microscopic 'building block' of which all living things are made

cerci: a pair of movement detectors at the rear of many insects. They may look like antennae.

chlorophyll: a green substance found in plants which enables them to use the Sun's energy to make food from carbon dioxide and water

cyanobacteria: a single-celled organism belonging to Kingdom Monera. Although it is known as blue-green algae, cyanobacteria are related to bacteria rather than algae, which belong to Kingdom Protoctista.

diatom: microscopic organism, consisting of one cell, which lives in water. Belongs to Kingdom Protoctista, but has plant-like features (e.g. it contains chlorophyll).

dichotomous key: a diagram used to classify things by grouping them into smaller and smaller groups that are more and more alike, based on choosing one of two features

dormant: inactive — but capable of being active again at some time in the future

electron microscope: a microscope for viewing very small objects. An electron microscope is much more powerful than a light microscope and can magnify things up to a million times.

exoskeleton: skeleton or shell that lies outside the body

fertile: able to successfully breed with members of its own species

genetic: describes material in living things which contains information about features that can be passed on to the next generation

host cell: a cell which is being used by a virus to reproduce itself

invertebrate: an animal without a backbone

kelp: a large, brown seaweed

key: a way of showing how a group of things can be broken up into smaller and smaller groups that are more and more alike. A key can be drawn as a diagram or table.

living: being alive or once alive

mammary glands: milk-producing glands in a female mammal

multicellular: made up of more than one cell

non-living: not being alive and not ever having been alive

nucleus: the control centre of a cell. It also holds the 'codes' responsible for producing new cells.

organism: any living thing

parasite: an organism that survives by being inside or attached to another living organism from which it obtains its food

phloem cell: long, narrow cell that forms long tubes in plants with true leaves, stems and roots. The tubes made from phloem cells move the food made in the leaves to other parts of the plant.

photosynthesis: process whereby an organism uses the Sun's energy to make food (sugars) from carbon dioxide and water

pollinate: to transfer pollen from the stamen of a flower (the male part) to the pistil of a flower (the female part). Often insects help to pollinate flowers by transporting the sperm (male cells) on their body parts.

proboscis: a long feeding tube attached to the head of some insects. It sometimes rolls up when not in use.

protein: a chemical that is needed for the growth and repair of the cells that make up living things

species: a group of animals with many features in common. Members of the same species can mate with each other to produce fertile young under natural conditions.

spiracles: small openings in the body through which some animals breathe. All insects have spiracles.

spore: a reproductive cell of some organisms (e.g. ferns and fungi) that is protected within the organism until the time is ready for it to be released

tabular key: a way of classifying things by selecting one of the characteristics given for each consecutive number in a list of numbers. A choice either identifies the item or refers you to a later number in the list so you can make another choice.

taxidermist: a person who stuffs the skins of dead animals so that they are lifelike

taxonomist: a scientist who classifies living things by sorting them into groups that have more and more similar characteristics

thorax: the section of an insect's body between its head and its abdomen

tinea: a painful skin disease caused by a fungus that often occurs between the toes

tracheophyte: plant that contains vascular tissue

vascular tissue: tissue that allows food, water and nutrients to be carried through a plant. The tissue consists of tubes consisting of phloem cells or xylem cells.

vertebrae: the bones that make up the backbone

vertebrate: an animal that has a backbone, or internal skeleton

xylem cell: long narrow cell that forms long tubes in plants with true leaves, stems and roots. The tubes made from xylem cells move water and dissolved minerals up from the roots to the rest of the plant.

yeast: a fungus which causes certain foods (e.g. dough) to rise

CHAPTER 9

Neighbours in space

The Sojourner rover was only 63 cm long. It sent back images from Mars for three months.

You will discover

What causes the tides on Earth

Why the Moon seems to change its shape

Why we have night and day, and different seasons

Why the stars seem to move in circles in the sky

Exploring Mars

The planet Mars is one of Earth's closest neighbours. No reliable evidence of past or present life on Mars has yet been found. But the curiosity about Mars remains.

The first close-up television pictures of the surface of Mars were taken in 1965 by the space probe ***Mariner 5***. These pictures showed a very dry surface with lots of thick red dust. There was no sign of life or water. There were, however, icecaps at the poles.

The first soft landing on Mars was made in 1971 by the Russian space probe ***Mars 3***. It sent back TV pictures from the surface. Much more information about Mars was gathered in 1997 as part of NASA's ***Mars Pathfinder*** mission. For three months, a small Earth-controlled rover called ***Sojourner*** explored the surface and sent back images.

Other space probes continue to take images of Mars. And perhaps it won't be too long before a human lands on Mars!

1 Why do you think Mars is sometimes called the 'red planet'?

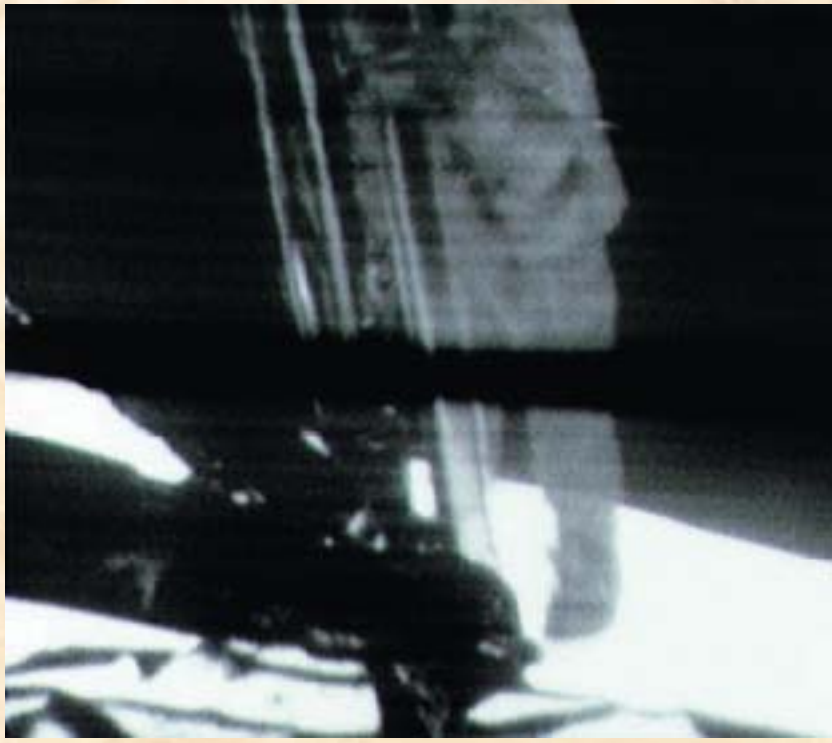
2 Can you locate the icy poles of Mars?

3 In what way are our space missions like the journeys of the world's past explorers?

THE DAILY SATELLITE

MONDAY 21 JULY 1969 WEATHER: WARM 63°F

MAN ON MOON



'That's one small step for man'.

This picture is from the live television broadcast of Neil Armstrong's first walk on the Moon. The remote-controlled camera was attached to one of the legs of the landing module. The black bar in the photo was caused by a communications fault back on Earth.

At 10.56 pm last night, millions of people throughout the world watched on television as Neil Armstrong slowly climbed down the ladder of the lunar landing module to become the first human to set foot on the Moon. As his first foot touched the surface, he said, 'That's one small step for man'. Less than a second later, when his second foot touched the surface, he added, 'One giant leap for mankind'.

Armstrong described the surface as fine and powdery as he kicked it, sending a cloud of dust upwards. It fell slowly, but straight down — not a sign of wind. He also described the surface as being covered with lots of 'little impact craters'. About 20 minutes later, Buzz Aldrin climbed down the ladder to join his fellow astronaut on the surface. They spent over two hours collecting rocks and setting up scientific experiments before returning to the landing module for a rest.

Both astronauts took quite a while to get used to the smaller pull of **gravity** on the Moon. It is about one-sixth as much as it is here on Earth. In fact, they had to lean forward to balance and had to walk slowly so that they could stop quickly if they needed to.

While Armstrong and Aldrin were 'moon-walking', the third astronaut of the *Apollo 11* team, Michael Collins, was in orbit **about** 100 km above the surface. Even though he was closer to the Moon than the millions of people on Earth watching television, he did not see his companions on the surface. He had no video monitor and could only listen by radio.

After some well-earned rest and a chat to President Nixon, Armstrong and Aldrin will climb down the ladder again to set up some more experiments. They'll also begin collecting samples of soil and rocks to bring back to Earth with them for analysis.

ASTRONAUTS WANTED

- Must be fit and have a good sense of balance
- Must be able to work closely with others in a confined space
- Scientific and engineering qualifications highly desirable
- Successful applicants will be given training in handling both high 'g' forces and working in low-gravity conditions
- Ability to work in very high and very low temperatures is essential
- Ability to stay alive with small amounts of water will be an advantage
- Professional runners need not apply

Apply in person at:
Lunar Exploration Headquarters, Crater Street, Moon River at 3.30 am.

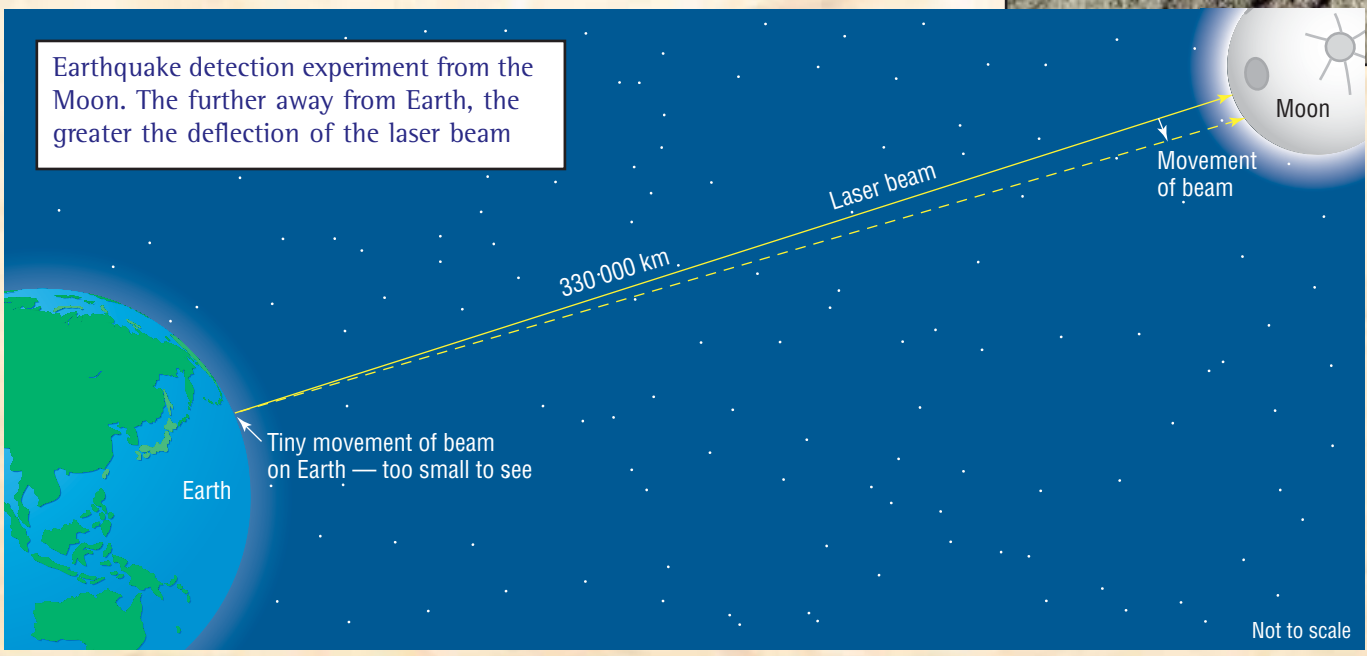
Go to worksheet 41

Checking for earthquakes

One of the experiments the astronauts are setting up will detect movements on both the Earth and the Moon. The experiment uses an earth-tremor recorder to detect small earthquakes on the Earth. The recorder receives a narrow laser beam coming from Earth. Because of the huge distance between the Earth and the Moon, even a tiny movement on Earth causes a large movement in the laser beam by the time it reaches the Moon.

Footstep record of history

According to space scientists, the first footprint ever made on the Moon will last for millions of years. Buzz Aldrin will photograph many footprints left by himself and Neil Armstrong before the lunar module returns to the command module and then heads back to Earth. The photographs are part of an experiment to find out how the dust on the Moon behaves when it is put under pressure.



Earthquake detection experiment from the Moon. The further away from Earth, the greater the deflection of the laser beam

Activities

REMEMBER

1. Describe the surface of the moon.
2. Why is it so easy to detect a tiny movement on Earth with a laser beam pointed at the Moon?

THINK

3. What do you think Neil Armstrong meant when he said 'That's one small step for man, one giant leap for mankind'?
4. The first footprint on the Moon will remain there for millions of years. How can it last that long?
5. Photographs of footprints are part of an experiment to find out how the dust on the Moon behaves when it is put under pressure. How could the results of this experiment be useful?
6. Why do you think that so much money was spent sending humans to the Moon?

IMAGINE

7. Imagine that a small colony is set up on the Moon. Up to 50 people at a time are allowed to live and work on the Moon for one year:
   (a) Sketch one way of making it easier for people to adjust to the gravity on the Moon.
   (b) Describe some of the difficulties that would have to be solved to allow the colony to be set up.

checklist

I can:

- ☐ describe the surface of the Moon
- ☐ discuss how the reduced gravity on the Moon affects movement
- ☐ discuss the purpose of a manned mission to the Moon.

The neighbourhood

In the film *Titanic*, Leonardo DiCaprio said he felt like he was 'king of the world'. Think what a huge kingdom that would be — the whole world! Now, think again — the world, Earth, is just a tiny, tiny part of a very large neighbourhood.

The neighbours

The Earth is the fifth biggest of nine planets in our **solar system**. All of the planets **orbit** the Sun, our nearest star. The Sun weighs more than 300 000 times as much as Earth. It is made up mostly of hydrogen and helium. The hot gas swirling near its surface has a temperature of around 6000 °C. The temperature in the centre of the Sun is believed to be about 15 000 000 °C.

The orbit of all planets is not quite circular. All planets except Pluto, usually the most distant, orbit the Sun in roughly the same flat plane, or level. Pluto's orbit is at an angle to this plane. Its orbit is also more oval-shaped than the other planets. Sometimes Pluto can be as far away from the sun as 7304 million kilometres. At other times it can be inside the orbit of the planet Neptune, making Neptune the furthest planet.

Some of our neighbours are much smaller than the planets. **Asteroids** are small, irregular rocks that, like the planets, orbit the Sun. Most asteroids are found between the orbits of Mars and Jupiter.

Comets are balls of dust, ice and frozen gases. They orbit the Sun, but their paths are not circular. As they approach the Sun, they develop a tail as the frozen gases heat up.

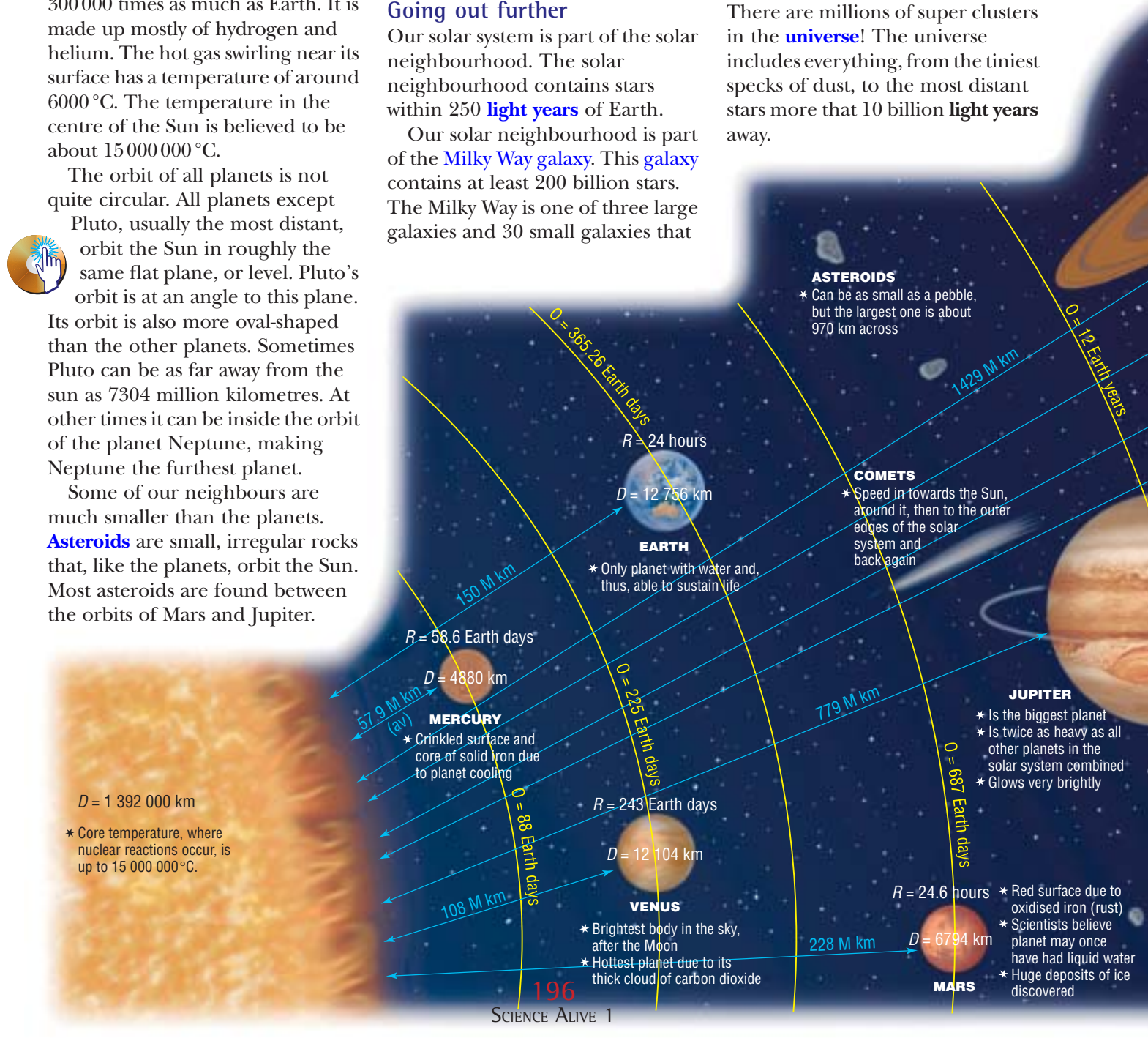
Going out further

Our solar system is part of the solar neighbourhood. The solar neighbourhood contains stars within 250 **light years** of Earth.

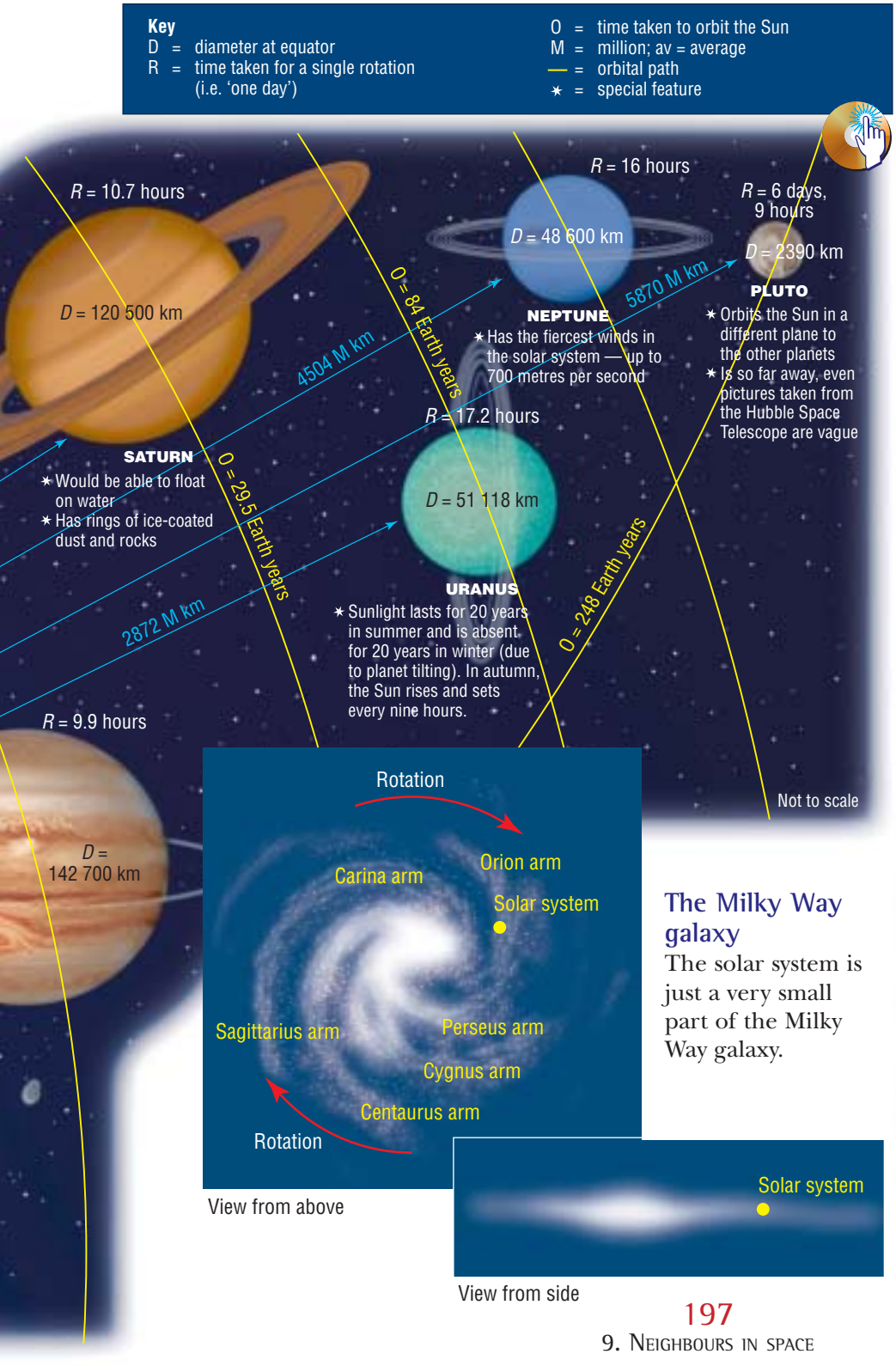
Our solar neighbourhood is part of the Milky Way galaxy. This galaxy contains at least 200 billion stars. The Milky Way is one of three large galaxies and 30 small galaxies that make up the Local Group. Some groups have over 2000 galaxies!

And further!

The Local Group is, in turn, part of a super cluster. And, guess what? There are millions of super clusters in the **universe**! The universe includes everything, from the tiniest specks of dust, to the most distant stars more that 10 billion **light years** away.



A star which is 250 light years away doesn't sound too far away. Let's see, though, what this means. A light year is the distance light travels in one year. We know that light travels at close to 300 000 km/s and that there are 31 536 000 seconds in a year (365 days × 24 hours × 60 minutes × 60 seconds). Multiplying these two figures gives us the number of kilometres in one light year — about 9 500 000 000 000 kilometres. So a star 250 light years away is about 2 400 000 000 000 000 kilometres away.



View from above

View from side

The Milky Way galaxy

The solar system is just a very small part of the Milky Way galaxy.

Activities

REMEMBER

1. What is a light year?
2. Describe the differences between asteroids and comets.

INVESTIGATE

3. Where is the solar system in the Milky Way? Describe how big it is compared to the Milky Way.

USING DATA

4. Look at the illustration of the solar system.
   (a) How far is Earth from:
      (i) Mars?
      (ii) Venus?
      (iii) Jupiter?
   (b) How much closer to the Sun is Mars than Neptune?
   (c) How much longer does Pluto take to orbit the Sun than Uranus?
   (d) Which planet takes
      (i) the longest and
      (ii) the shortest time to complete one rotation?
   (e) What is so unusual about the seasons on Uranus?
5. A star is six light years away from Earth. How far is this in kilometres?

CREATE

6. Work together as a class for this activity. Use balls of varying sizes, wire and string to create a model of the solar system. Try to give some idea of the different sizes of planets and their different distances from the sun. Remember, though, it won't be to scale.

✓ checklist

I can:

- ☐ compare the size of Earth and the solar system with other parts of the universe
- ☐ appreciate the relative sizes of galaxies, super clusters and the universe.

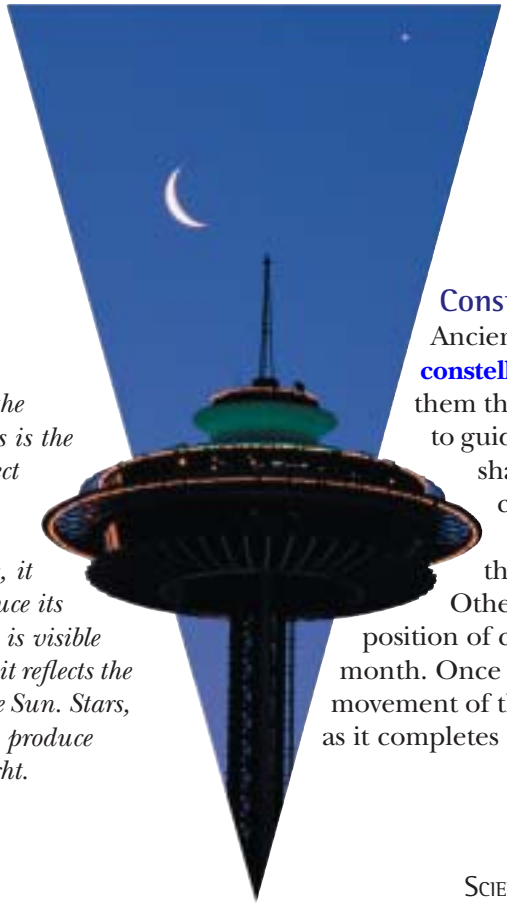
Twinkle . . . twinkle

Twinkling stars have been the stuff of many fairytales, nursery rhymes and folk songs. There may be more than 200 billion billion stars. They are grouped in **galaxies** throughout the **universe**. The stars that we see in the night sky seem to be on the move. In fact it is the Earth that moves — not the stars.

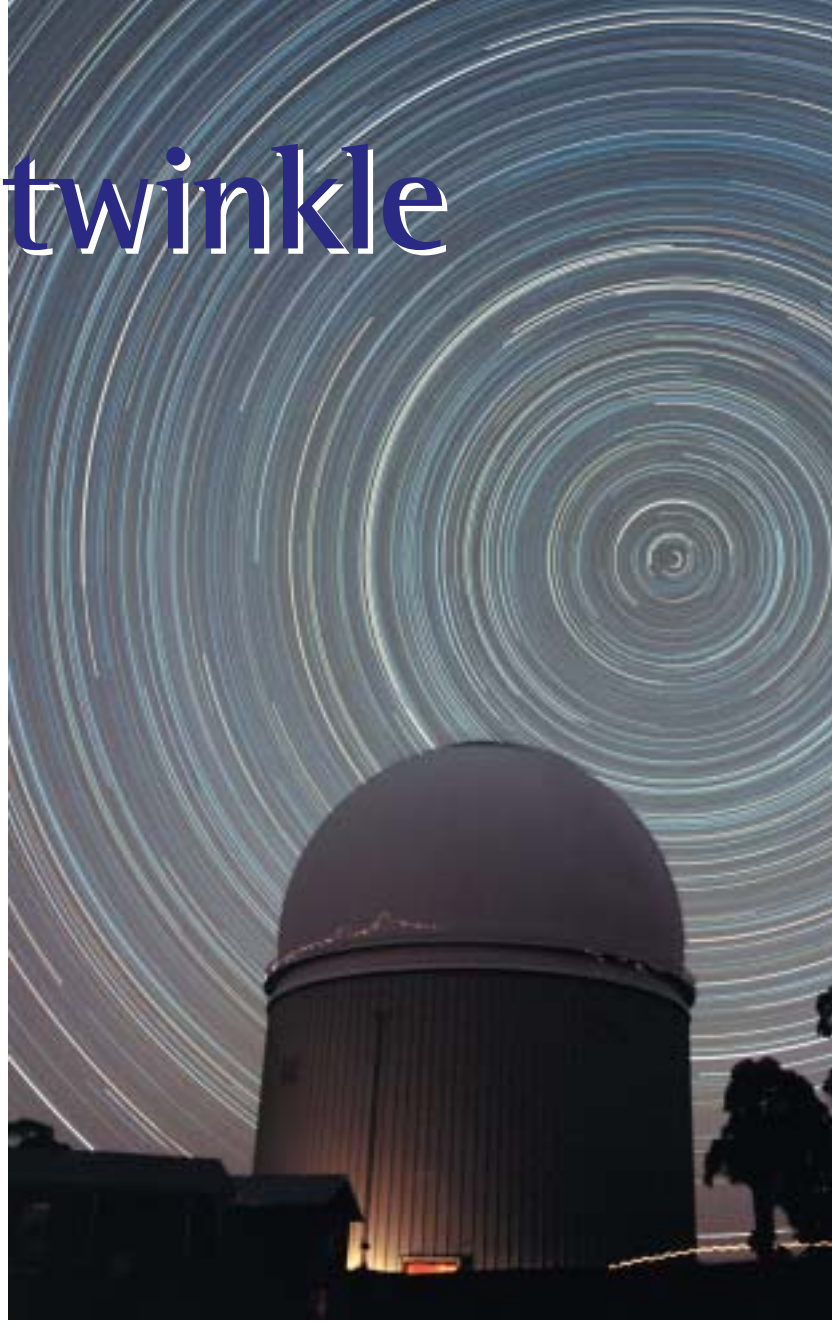
Star spotting

Stars shine all the time. However, we can see them only at night when the sun has set. During the day, the light from the Sun is so bright that no other star is visible.

If you had the patience to watch the sky all night, you would notice the stars seem to move. From Australia, they seem to move in a circular path about a point in the southern sky. The movement is too slow to see unless you watch for a couple of hours. In the northern hemisphere, they seem to move about a point in the northern sky. The stars are not moving. The apparent circular movement of stars is due to the **rotation** of the Earth.



Not a star
Apart from the Moon, Venus is the brightest object in the night sky. But like other planets, it doesn't produce its own light. It is visible only because it reflects the light from the Sun. Stars, like the Sun, produce their own light.



The apparent circular movement of the stars. The camera's shutter was left open for some hours to 'catch' the trail of light.

Constellations

Ancient people knew that different groups of stars, called **constellations**, appeared in the sky at different times. This reminded them that the seasons were changing. Some used the constellations to guide them when travelling. They were named according to the shapes their stars seemed to form in the sky. There are 88 constellations with names agreed on by scientists.

We can't see all of the constellations. Some of them, like the Southern Cross, can be seen from Australia all year round. Others are visible in the night sky for only part of the year. The position of constellations in the sky at nightfall changes from month to month. Once again, this apparent change in position is due to the movement of the Earth. This time, it is due to the movement of the Earth as it completes its year-long **orbit** of the Sun.

How can I track the apparent movement of constellations?

You will need:

torch covered in red cellophane (to help your night vision)

star chart for the southern hemisphere night sky (Go to www.jaconline.com.au/science/weblinks and click on the Sky Map link for this textbook to access the star charts for the most recent month, and the following month.)

Make your observations on clear nights. Pick a flat area, well away from bright lights.

- Check the star chart for your starting month to find out where the constellations Crux (Southern Cross) and Triangulum Australe should be at 10.00 pm around the middle of the month. Also note the position of the bright star Alpha Centauri.
- Try to locate these two constellations and star in the sky. Check the sky at 10.00 pm around the middle of the month. Let your eyes get used to the dark for about 15 min before searching.
- Repeat this task for the following month.

1. In which direction did the constellations and star you located seem to move between one month and the next?
2. Explain why the movement occurred.
3. Why was it important to check the sky at the same time as in the sky chart?

Activities

REMEMBER

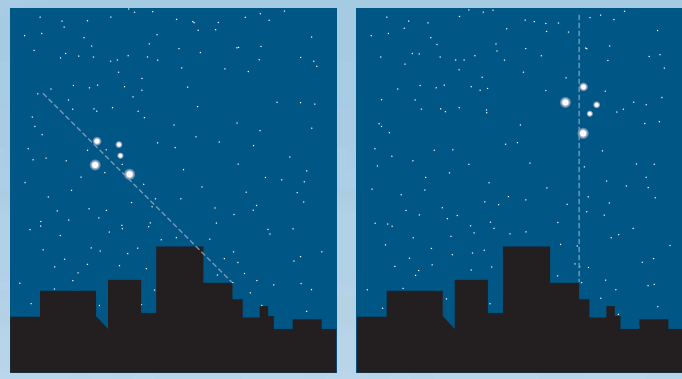
1. Why do the stars appear to move slowly in a circle at night?
2. How is a star different from a constellation?
3. Why does the position of constellations change from month to month.

THINK

4. Explain why you can see stars only at night.
5. Why can certain constellations only be seen:
   (a) in one hemisphere or the other?
   (b) at certain times of the year, looking either south or north?
6. Ancient people used the stars the way we use calendars or clocks. Suggest what activities might have depended on their observation of the stars.

INVESTIGATE

7. Investigate the movement of the constellation Crux (Southern Cross) over a three-hour period. Use landmarks on the horizon to help you estimate the approximate angle the constellation has moved through in this time. Sketch your observations. (See the example below.)



7.00 pm 10.00 pm

These diagrams show how the Southern Cross appears to move early in May.

CREATE

8. Set up a model to explain, in simple terms, why the stars appear to move in circular arcs each night.

checklist

I can:

- ☐ measure and explain the apparent nightly movements of stars
- ☐ explain why constellations appear to move from month to month
- ☐ use a star chart to locate constellations in the night sky.

Spacetrek

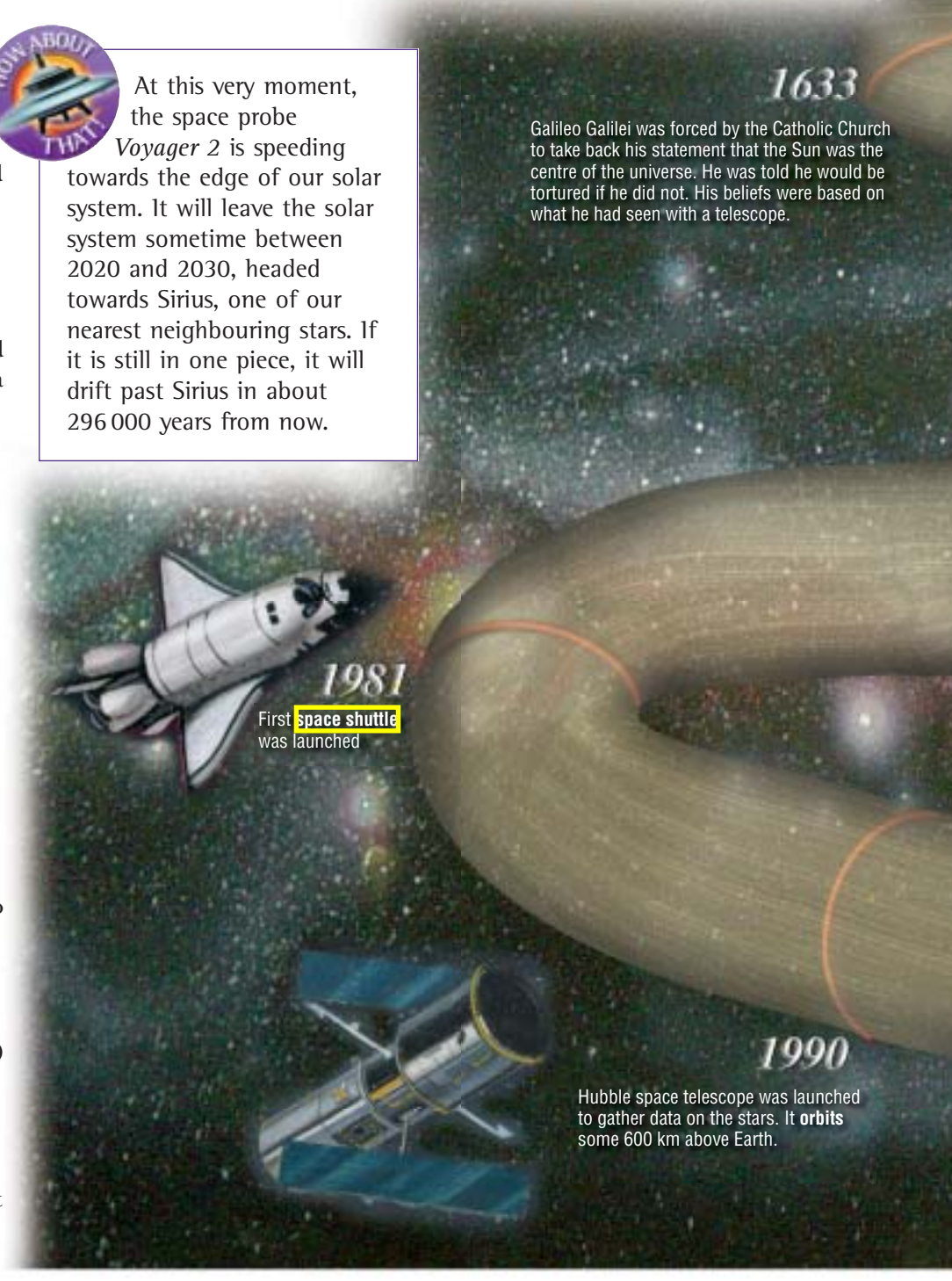
Ancient people tried to explain what they saw in the skies. To them, the stars, the Moon, the Sun and the change of seasons had to be the gods at work! Today, we know a lot more. Over the last few hundred years, many important things have been found out about space. We've used telescopes and **space probes** to explore the **solar system** and beyond. Yet, there is still so much more to learn.

The Sun was a great mystery to ancient people. To them, it looked like a moving ball of fire. It provided warmth and light and made their crops grow. It rose and fell in the sky each day. In some parts of the world it even went away for months at a time. Many treated the Sun as a god.

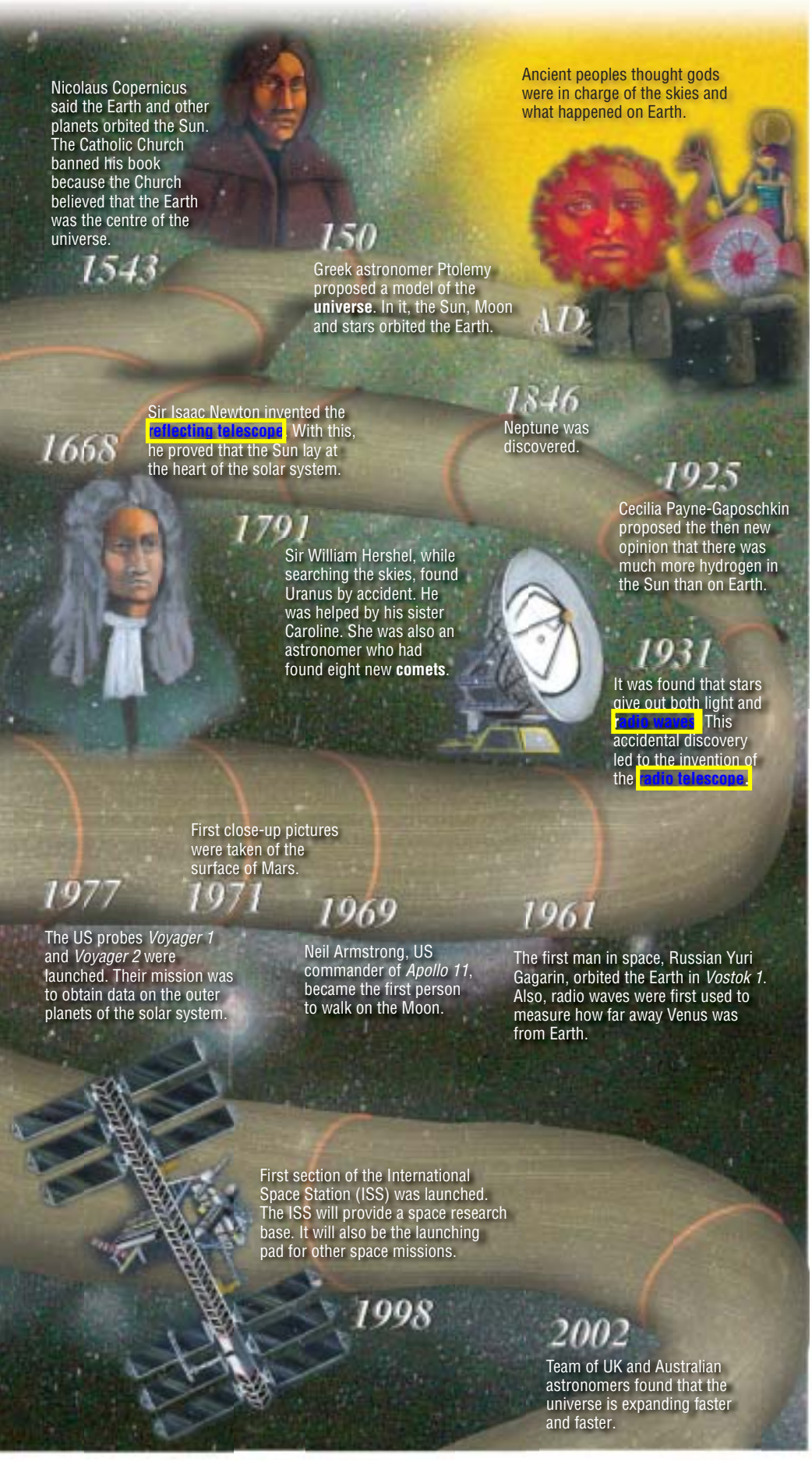
- To the ancient Egyptians, the Sun was their god Ra, who sailed across the heavens each day in a fiery boat.
- The Indians in North and Central America, and the Vikings of Scandinavia, all held special feast days for the Sun. At these, they danced and sometimes killed animals.
- The Sun god was one of the most important of the Hindu gods.
- The Japanese once believed their emperor came from their Sun god.
- Many Aboriginal stories tell of people who became the Sun.

It was not until the seventeenth century, after the invention of the telescope, that people were able to see detail in the night sky. Knowledge about space grew quickly after that time. But most things we know about space today have been found out in the last 50 years. Space probes and satellites have given us much of this information. Some finds, though, have been made by amateurs. Other things have been found out by accident!

At this very moment, the space probe *Voyager 2* is speeding towards the edge of our solar system. It will leave the solar system sometime between 2020 and 2030, headed towards Sirius, one of our nearest neighbouring stars. If it is still in one piece, it will drift past Sirius in about 296 000 years from now.



Go to worksheet 42



Activities

REMEMBER

1. Why did the knowledge about space grow more quickly after the seventeenth century?
2. Name one space discovery found by accident.
3. Name one way in which a woman has helped us to find out more about space.

THINK

4. Why do you think so much effort is being made today to find out what lies beyond our Earth? Do you think it is worth it? Why?
5. Suggest why the Sun might have been such a common part of the beliefs and practices of ancient peoples.

IMAGINE

6. Imagine you were visited by an alien. How do you think people might react if you said you had seen an alien?
7. After stating that the Earth orbited the Sun, Galileo was forced by the Church to deny this in public. He spent the rest of his life under house arrest in his home. Write a letter that he might have written to a fellow astronomer during this time.

CONNECT

8. Use the Internet to find out more about one of the people or events shown on the timeline opposite. Write a short report of about 200 words on your findings.

✓ checklist

I can:

- ☐ understand why the Sun was an important part of ancient religions
- ☐ justify an opinion about the value of space research.

Mysteries in the sky

Some people today confuse the terms ‘astronomy’ and ‘astrology’. Perhaps this is not so strange. Until a few hundred years ago there was hardly any difference between them. But now they are quite different. **Astronomy** is the scientific study of our **solar system**, and its place in the **universe**. **Astrology** is a study of how the positions of the Sun, Moon and planets are said to affect our lives.

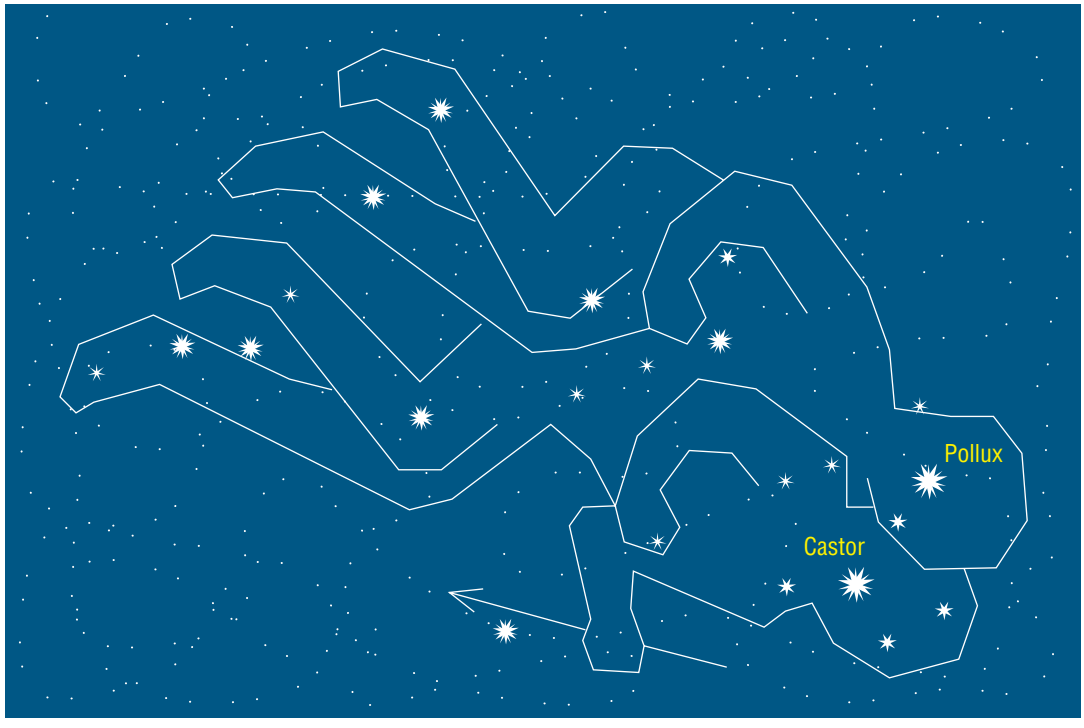
Before people began to understand the science of space, they thought events in the skies were the work of gods. **Eclipses**, **comets**, a full Moon, or the return of the Sun after a long winter were signs from the gods for people to take action, or to make decisions.

A horoscope prediction for a ‘Gemini’ — someone born between 22 May and 21 June

Zone of the zodiac

Looking at the sky from Earth, the Sun, Moon and planets follow a set path in the sky. This path is called the **ecliptic**. It crosses the heavens in front of 12 well-known **constellations** — Aries, Taurus, Gemini, Cancer, Leo, Virgo, Scorpio, Libra, Sagittarius, Capricorn, Aquarius and Pisces. It also passes in front of the lesser known constellation Ophiuchus. The part in space that includes the ecliptic and these constellations is called the **zodiac**.

The ancient Greeks believed that events on Earth (such as a person’s birth) were influenced by whatever constellation the Sun was in front of at the time. These beliefs gave rise to what we call **horoscopes**. In a horoscope, each of the zodiac’s 12 constellations becomes a ‘star sign’. Each has its own symbol and special features.



The zodiac symbol for each ‘star sign’ comes from the way the stars lie in their constellation. The Gemini symbol — the twins — is formed from the Gemini constellation’s two brightest stars, Castor and Pollux. Greek legend has it that the twins Castor and Pollux were born to a woman named Leda after she was visited by the god Zeus. He came to her as a swan.

The signs of the zodiac

Comparing astrology and astronomy

Astronomers are scientists. They make and use observations to learn more about the universe, including our solar system. Astrologers are not scientists. They make predictions about people's lives. Their predictions are based on star signs and observations of the Sun, Moon and planets. The dates listed in horoscopes are based on when the Sun seemed to pass in front of each constellation over 2000 years ago. Those dates have changed, but the dates in horoscopes remain the same.

In 1979, a French psychologist offered free horoscopes to 150 people who agreed to complete a survey about their accuracy. He sent the horoscope for the birth date of Dr Petiot, a mass murderer, to all 150 people. When they were asked if the horoscope made accurate predictions about themselves, 94 per cent said yes.

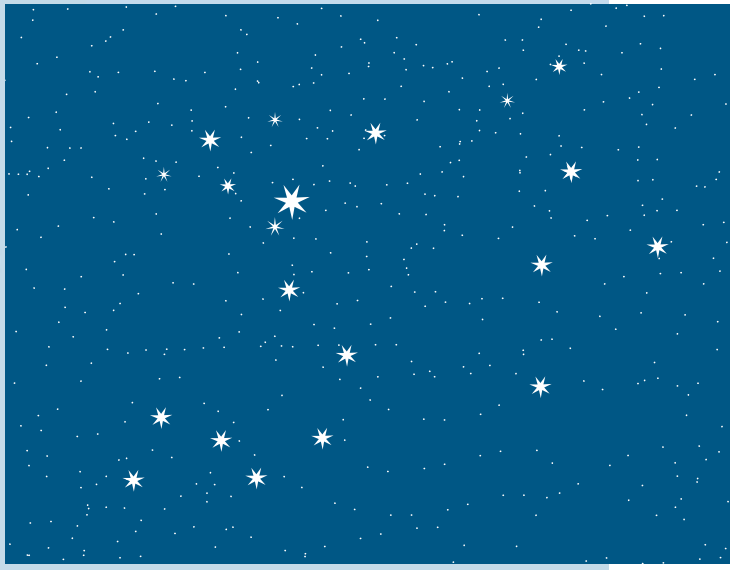
Activities

REMEMBER

1. What is the main difference between astronomy and astrology?
2. What is a horoscope?
3. Name the 12 constellations in the zodiac.
4. There are 88 named constellations. What is special about the 12 constellations of the zodiac?

THINK

5. Suggest some ways that astrologers helped the growth of knowledge about the stars and planets.
6. The following diagram shows how the stars in the Leo constellation look from Australia. First, check the horoscope opposite to find the symbol for Leo. Then suggest how the stars could be joined with straight lines to form a shape that looks like the Leo symbol.



Note that the diagram that you draw should be upside down. Because we live in the southern hemisphere, this is how we see the constellation Leo. The ancient people who created the star signs lived in the northern hemisphere.

CREATE

7. Draw the star sign symbols for a horoscope to be included in a teenage magazine. Be creative!

✓ checklist

I can:

- ☐ explain the difference between astrology and astronomy
- ☐ show how a zodiac symbol is related to the layout of a constellation's stars.

Days and seasons

Have you ever wondered why it gets dark? Or why the Sun rises later in Perth than in Melbourne? Or why Australia has a cold winter when Greece is having a hot summer? These things have all got something to do with the way the Earth moves and its position in space at a given time.

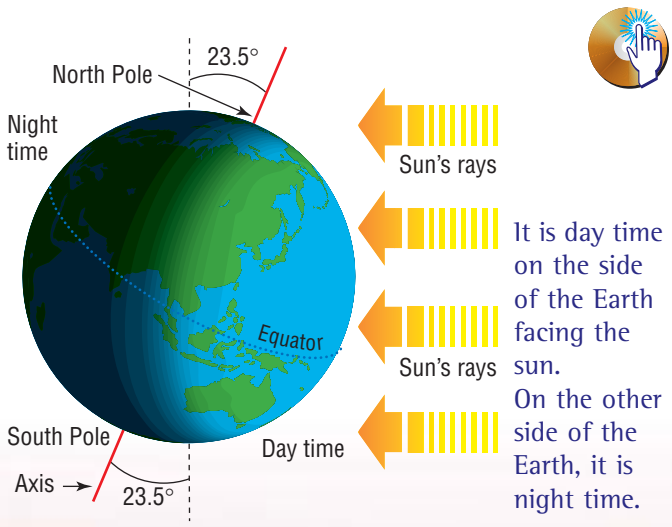
Three important things you need to remember about the movement of the Earth:

1. The Earth spins about an imaginary **axis**. It takes 24 hours to complete one **rotation**.
2. The Earth's axis is tilted at an angle of 23.5° to the vertical.
3. The Earth **revolves** once around the Sun every 365 and a quarter days.

Day and night

To us on Earth it seems that the Sun rises each day in the east and sets in the west. In fact, the Sun doesn't move across the sky at all. It is the Earth that moves — and quite fast, too! People living on the equator are moving at close to 1670 kilometres per hour! We don't sense we are moving as everything around us moves at the same speed.

Like a spinning top, the Earth rotates — spinning from west to east about its axis. One rotation takes 24 hours. We call the time for a complete rotation one day. As the Earth spins around, first one side and then the other faces the Sun and experiences daytime. The side facing away from the Sun gets no sunlight, so experiences night.

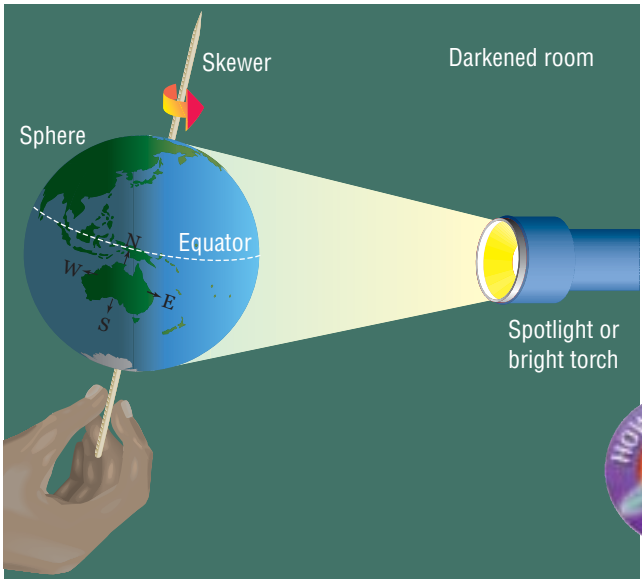


It is day time on the side of the Earth facing the sun. On the other side of the Earth, it is night time.

The Sun sets in the west behind a Melbourne skyline.

Why is it day time in Australia when it's night time in Africa?

You will need:
polystyrene (or similar) sphere (about the size of a small rockmelon)
metal or wooden skewer
pen
spotlight or bright torch.

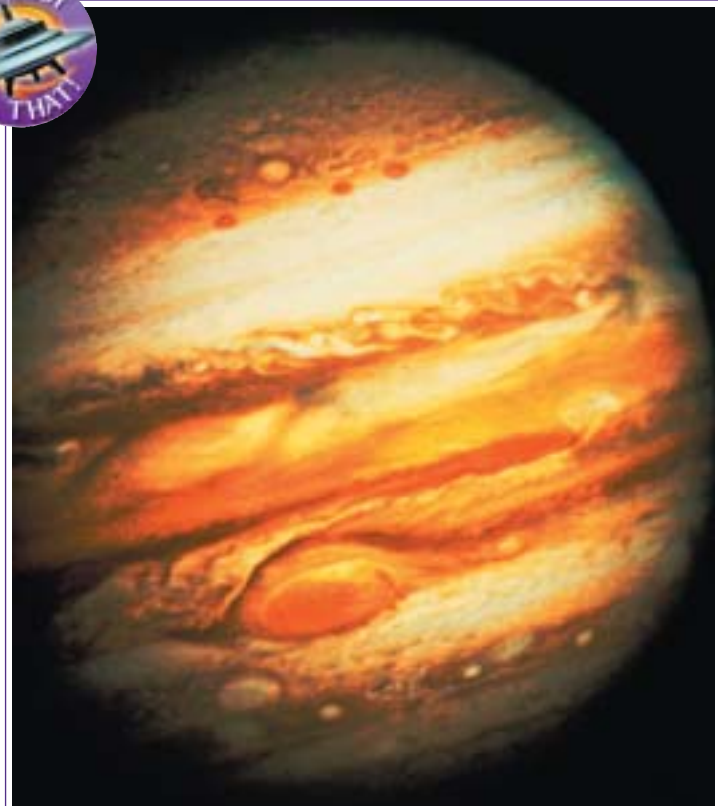


- Your sphere represents the Earth. Draw a line around the centre to represent the equator. Label the northern and southern hemispheres and mark in the North and South poles.
- Draw an outline of Australia and Africa on your sphere. Use an atlas to check the positions and approximate shape of each continent. Also note the position of north.
- Mark the four compass directions — north, south, east and west — around the outlines of each continent.
- Gently push a skewer through the centre of your sphere from bottom to top through the 'polar regions'. This skewer represents the Earth's imaginary axis.

Carry out this experiment in a darkened room. This will help you see more clearly the contrast between light and dark.

- Turn on the spotlight in a dark room. Its light represents the Sun's light. Hold the skewer so it leans a little away from the vertical. This represents the Earth's tilt.
- Turn your sphere very slowly in the light, making sure you keep the skewer slightly tilted all the time. Turn it in an anticlockwise direction (as seen from above). Watch what happens from side on.

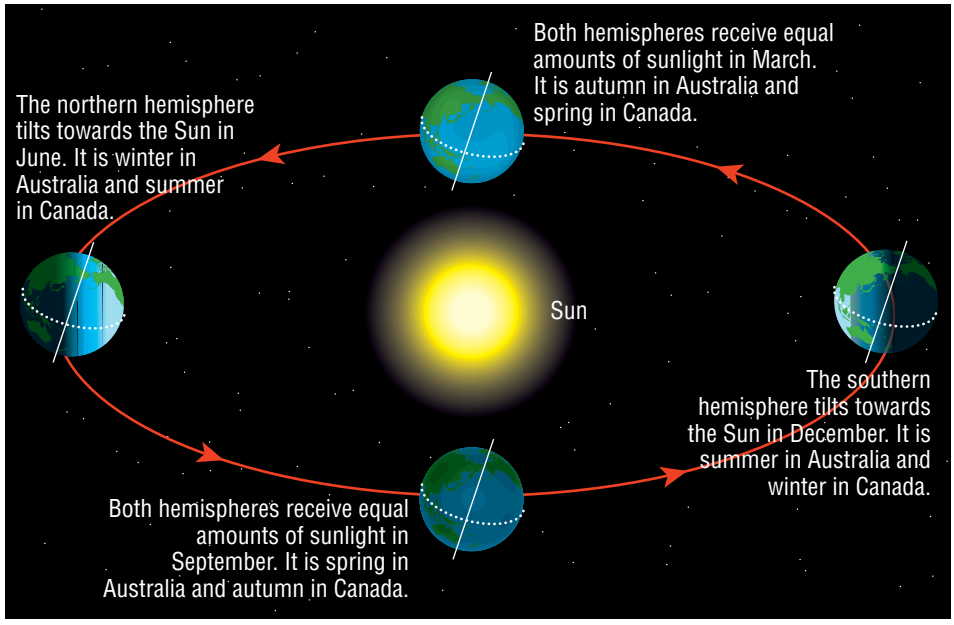
1. In which direction is the 'Earth' rotating — from east to west or west to east? Check the compass directions you marked on your sphere.
2. In which direction does the 'Sun's' light seem to move around the 'Earth'? How does this explain the apparent movement of the Sun across the sky?
3. Where is Africa when Australia is lit up? Where is Australia when Africa is lit up? Explain why this occurs.
4. How does this experiment help to explain why night falls in Perth about two hours later than in Melbourne?



A day on Jupiter is less than 10 hours. This means it takes under 10 hours to complete one rotation. But this giant planet, made mostly of gas, is about 13 000 times bigger than Earth. So when it rotates, its outermost clouds move at close to 45 000 kilometres every hour!

The seasons

The Earth takes 365 days (or 365 and a quarter days to be more exact) to complete one **revolution** of the Sun. As the Earth completes its **orbit**, the tilt of its axis does not change. (It leans either to the right or the left, depending from which way you are 'looking' at the orbit.) This means that at certain times during the Earth's orbit, one hemisphere or the other is leaning more towards the Sun.



Because of the tilt of the Earth, seasons change as the Earth completes its orbit of the Sun.

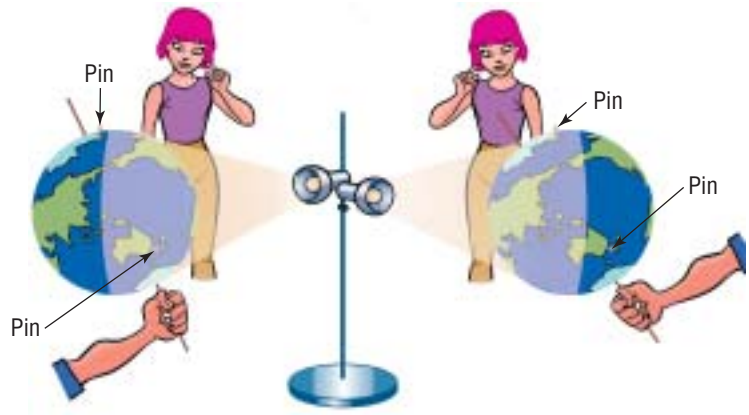
When a hemisphere tilts towards the Sun, it is 'hit' more directly by the Sun's rays. So it heats up faster and the days are warmer. This hemisphere experiences summer. When a hemisphere tilts away from the sun, the Sun's rays 'hit' it at more of an angle. So the heat from the Sun is spread out more. This means it takes longer for that hemisphere to heat up. Days are colder. This hemisphere experiences winter. When neither hemisphere tilts towards the Sun, which happens in autumn and spring, each receives the same amount of the Sun's rays. So there is not much difference between, say, a northern hemisphere spring and a southern hemisphere autumn.

(a) In January, when the northern hemisphere tilts away from the Sun, it is winter in Canada.

(b) In January, when the southern hemisphere tilts towards the Sun, it is summer in Australia.

Why do days not always have the same amount of daylight and darkness?

You will need:
the equipment you used for the earlier experiment on page 205
2 pins with coloured heads.



As you did before, carry out this experiment in a darkened room. This will help you see more clearly the contrast between light and dark.

- Hold the skewer vertically. Push two pins into your sphere — one about where Sydney is and the other directly above it at the top of the sphere, near the skewer.
- Set the spotlight up in a central place (e.g. on a table you can move around).
- Stand to the left of the spotlight. Turn on the spotlight. Hold the skewer so it leans away to the left from the vertical. The southern half of your sphere should be leaning more towards the light.
- Slowly turn your sphere in the light, making sure you keep the skewer slightly tilted. Turn it in an anticlockwise direction. Watch what happens from side on. Watch the side of the sphere you can see as you turn it. A partner should watch the other side.

1. Which pin comes into the light first? Ask your partner which pin moves out of the light first?
2. What does this tell you about the number of daylight hours in each hemisphere when the southern hemisphere tilts towards the Sun?

- Now stand to the right of the spotlight. Keep your skewer tilted to the left as before. This time the northern half of your sphere should be leaning more towards the light. Repeat what you did for the last dot point above.

3. Which pin comes into the light first this time? Ask your partner which pin moves out of the light first.
4. What does this tell you about the number of daylight hours in each hemisphere when the northern hemisphere tilts towards the Sun?

- Repeat the whole procedure above two more times. The first time, look at what happens at each of the poles. The second time, look at what happens at the equator.

5. What is the length of day and night at the equator in each season?
6. Suggest why the Sun never sets at certain times of year at the North and South Poles. What season is the southern hemisphere experiencing when the South Pole has six months of darkness?

Activities

REMEMBER

1. Explain how you would use two balls — one representing the Earth and the other the Sun — to explain the difference between the Earth's rotation and revolution.
2. Why do we have day and night?
3. During which season in Australia does the southern hemisphere tilt towards the Sun?
4. Explain why it is warmer on a summer's day that it is on a winter's day.

THINK

5. If the axis of the Earth were vertical and not tilted:
   (a) How would this affect weather patterns at the equator and the poles?
   (b) How would this affect where you live?
6. What season are we having in Australia when:
   (a) it is autumn in England?
   (b) it is summer in Canada?
   (c) the Sun does not set at the South Pole?
7. What season is Australia shown as having in the figure on the bottom of the opposite page?

INVESTIGATE

8. Check the Internet or library resources to find out when and why we have leap years. Write a paragraph in your notebook to explain this.

checklist

I can:
- ☐ explain the ways in which the Earth moves
- ☐ understand why we have days and seasons
- ☐ contribute to a class discussion on life in the polar regions.

Moon shapes

The Moon is our nearest 'space neighbour'. It is about 384 000 kilometres from Earth. Its changing shape in the night sky was a great mystery to ancient people. Today, we know that the **phases of the Moon** are caused by light from the Sun falling on the Moon as it **orbits** the Earth.

Of all the objects we can see from Earth in the night sky, the Moon is the biggest and brightest. In fact, the Moon produces no light of its own. When it shines, it is simply reflecting the light of the Sun.

Like the Earth, the Moon spins (or rotates) around its own **axis** while revolving around a larger object.

A full Moon as viewed from Earth

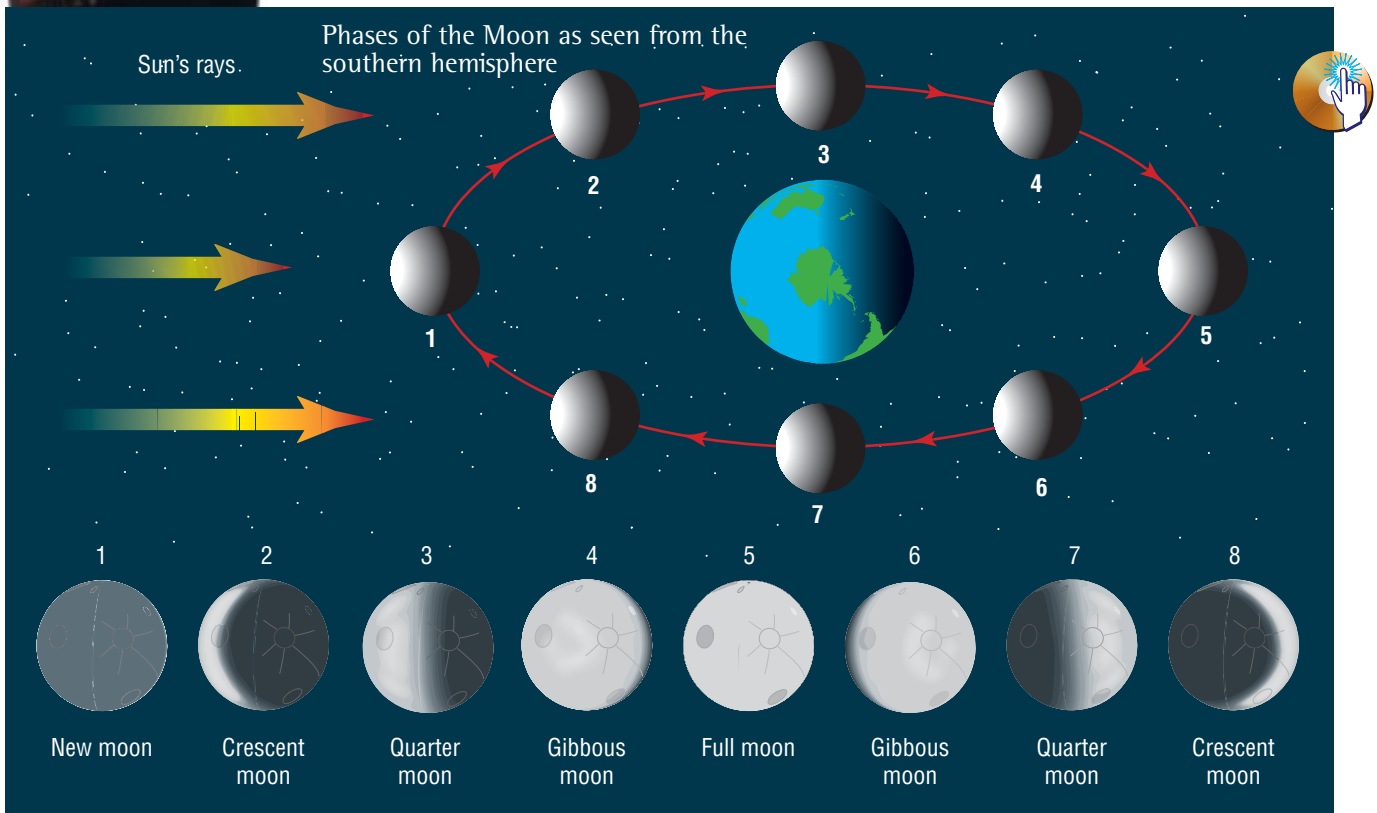
Far side of the Moon

When you look at the Moon from Earth you only ever see the same side. It is called the **near side**. It was not until 1959 that scientists first saw **space-probe** images of the other side, called the **far side**. The far side is the most mountainous of the two, and has more craters than the near side. The reason you can't see the far side from Earth is because the moon takes about the same time to complete one **rotation** as it does to complete one **revolution** of the Earth.

Phases of the Moon

The Moon doesn't change its shape. What does keep changing, however, is how much of it we see lit up by the Sun. This, in turn, depends on where the Moon is on its orbit around the Earth. The different views we see of the Moon are called phases of the Moon. There are eight of them.

A **new Moon** occurs when the Moon lies between the Sun and the Earth. The side of the Moon we see, the near side, is dark. The far side is the one lit by the Sun. As the Moon moves further around the Earth, the near side is lit up more and more by the Sun. Finally, it is completely bright. We call this a **full Moon**. This occurs when the Earth lies between the Sun and the Moon. In the second half of the Moon's orbit, the lit-up part of the near side gets less and less wide. Finally, the near side is completely dark again and there is another new Moon. It takes 29.5 days from one new Moon to the next.

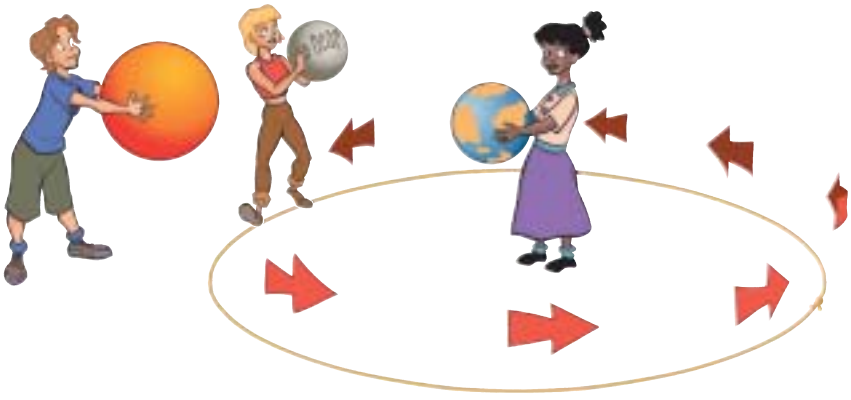


Go to worksheet 43

Why can't I see the far side of the Moon from Earth?

You will need:
3 polystyrene balls of different sizes:
- the largest one painted yellow (to represent the Sun)
- the medium one painted blue and green (to represent the Earth)
- the smallest one marked by the words 'near' on one side and 'far' on the other.

3 people, one to hold each of the balls
6 m or more of string.



- Lay out the string in a circle. It represents the Moon's orbit of the Earth.
- The 'Sun' stands at a point well outside the circle. The 'Earth' stands in the centre of the circle. The 'Moon' stands on the string between the Sun and the Earth, with the side labelled 'near' facing the Earth. This is the start position.

1. What phase of the Moon does the start position represent? What would people on Earth see, and why? Which side of the 'Moon' is lit by the Sun?

- The 'Moon' slowly follows the string around, back to the starting point. Hold the 'Moon' so that the 'near' label always faces Earth.

2. At what position around the circle would people on Earth see a full Moon? Why? Which side of the 'Moon' would now be lit up by the Sun?
3. As the 'Moon' completes one orbit of the 'Earth', count the number of rotations the 'Moon' makes. What do you notice?

- Repeat this experiment a second time. This time as the 'Moon' moves around the string, always make sure that the 'Far' side is facing the Sun. This means that although the 'Moon' is orbiting the 'Earth', it is not rotating.

4. If the Moon did not rotate, what side of the moon would face Earth during (a) a new Moon and (b) a full Moon?
5. If the Moon rotated very quickly — say 40 times during every orbit of the Earth — what do you think people on Earth (with telescopes) might see during a full Moon?

Activities

REMEMBER

1. Name and draw the eight phases of the Moon as we see them from Australia.
2. What phase of the Moon do we see when:
   (a) the Earth is between the Sun and the Moon?
   (b) the Moon is between the Sun and the Earth?
   Explain why each of these phases occurs.
3. How many days are there between one new Moon and the next?

THINK

4. Why are there more craters on the Moon than the Earth, even though the Earth is a bigger target?
5. What is the far side of the Moon, and why can't we see it?
6. How much of the far side of the Moon is lit by the Sun when we can see a quarter Moon?

INVESTIGATE

7. Start your investigation a day or two after a new Moon. Check the sky soon after sunset every second night for a month. Record in words and sketches:
   - where the Sun set (it might help to pick out landmarks on the horizon)
   - where the Moon rose
   - the shape of the lit part of the Moon.

   (a) Did you see all phases of the Moon?
   (b) What links were there between the Sun's position and phases of the Moon?

✓ checklist

I can:
- ☐ describe the phases of the Moon and explain why they occur
- ☐ explain the difference between the near and far sides of the Moon.

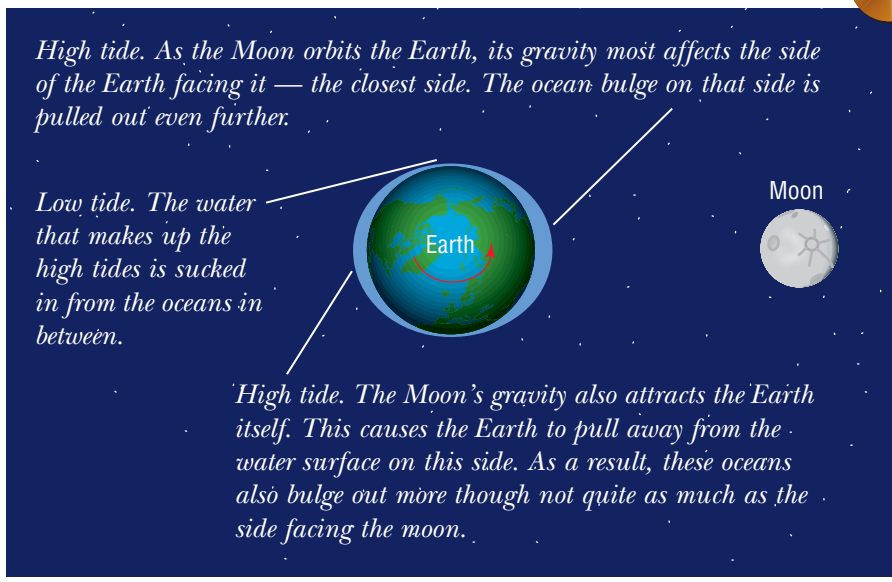
Ebb and flow

Day after day, the waters of the Earth's oceans rise and fall against the coastlines of islands and continents. These changes in sea level are called tides. They are caused by the **gravitational attraction** of the Moon and Sun on the Earth.

When the Earth rotates about its **axis**, its waters spin outwards. It is a bit like the way clothes fling out during the spin cycle in a washing machine. This action creates a bulge of water around the Earth. Why, you might ask, does the bulge not fly out into space? It is held back by the Earth's **gravity**.

Bulge in ... bulge out

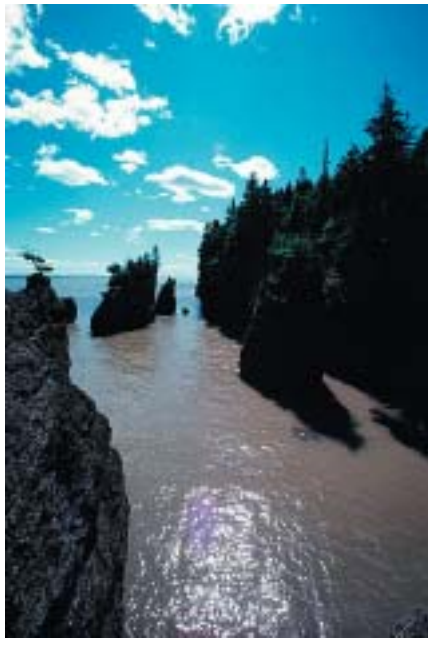
The size of the bulge is not the same everywhere. The bulge in the oceans gets larger and smaller because of the pull of gravity of the Sun and the Moon. It is the change in size of the bulge that we call tides. High tide occurs where the bulge is large. Low tide occurs where the bulge is small.



Looking down on the Earth from above the North Pole. As the Earth rotates different places experience high tide.

Gravity is a force of attraction between any two bodies in the **universe** that have **mass**. How big this force is depends on two things: the mass of the bodies and how close they are. The Sun and the Moon each pull on the Earth. Even though the mass of the Moon is 27 million times less than the Sun, it pulls more on the Earth than the Sun because it is very much closer to Earth.

In theory, every place on Earth has two high tides and two low tides on most days. Sometimes, though, other factors cause strange events to happen. For example, the extremes of tides in the Bay of Fundy in Nova Scotia are caused by its geography.



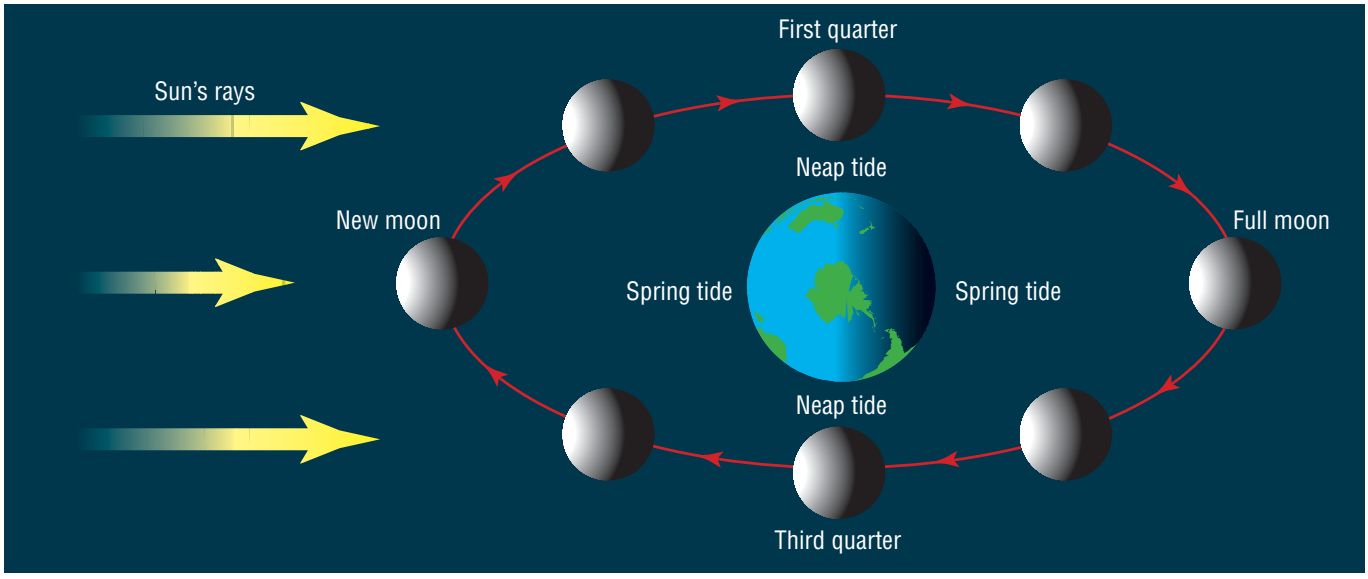
High and low tide in the Bay of Fundy. Its **tidal range** — over 16 m — is the biggest in the world. The bay has a very wide mouth that allows a lot of water to rush in as the tide rises. But the bay gets much narrower as it cuts into the land. The huge volume of water has nowhere to go but up!

Go to worksheet 44

Spring and neap tides

Twice each month the Moon, Sun and Earth line up. This occurs when there is a full Moon and a new Moon. At these times, the pull on the Earth and its waters is much stronger as the gravity of the Sun and Moon combine. Hence, the ocean bulges contain even more water. This means the **tidal range** is much greater. These tides are called **spring tides**.

About seven days after a spring tide, the Moon and Sun are at right angles to each other with respect to the Earth. In this position, their forces of gravity work against one another, rather than together. So, the tidal range is narrow. These tides are called **neap tides**.



During a **tidal month**, there will be two spring tides and two neap tides.

Activities

REMEMBER

1. What is gravity?
2. Why does the Moon's gravity have a stronger pull on Earth than the Sun's?
3. Explain with the aid of a diagram the difference between a spring and a neap tide.

THINK

4. What sort of tide will occur when there is a full Moon? Why?
5. If the height of the highest tide on a particular day was 6.5 m and the tidal range was 4.2 m, what was the height of the lowest tide?
6. Suggest why one high tide on any given day is always higher than the other one.
7. Imagine that you tied a small fishing boat to a pier in the Bay of Fundy using a two-metre rope. The pier is two metres above the water level at high tide. Draw a labelled diagram to show the position of the boat at high and low tides.

CREATE

8. Design a role-play involving at least four people that shows how the movement of the Earth around the Sun, and the Moon around the Earth, cause tides. A narrator could be used to give a commentary of the role-play.

checklist

I can:

- ☐ explain how the Moon and the Sun cause tides on the Earth
- ☐ explain why there are two high tides and two low tides on most days
- ☐ describe the difference between spring and neap tides.

Block out!

When you stand in the Sun, you cast a shadow. That's because your body is blocking some of the Sun's light. This is similar to what happens when an **eclipse** occurs. Something is blocking the light of the Sun. Sometimes the Moon blocks the light, casting a shadow on the Earth. At other times, the Earth blocks the light, casting a shadow on the Moon.

A lunar eclipse may last for about four hours.

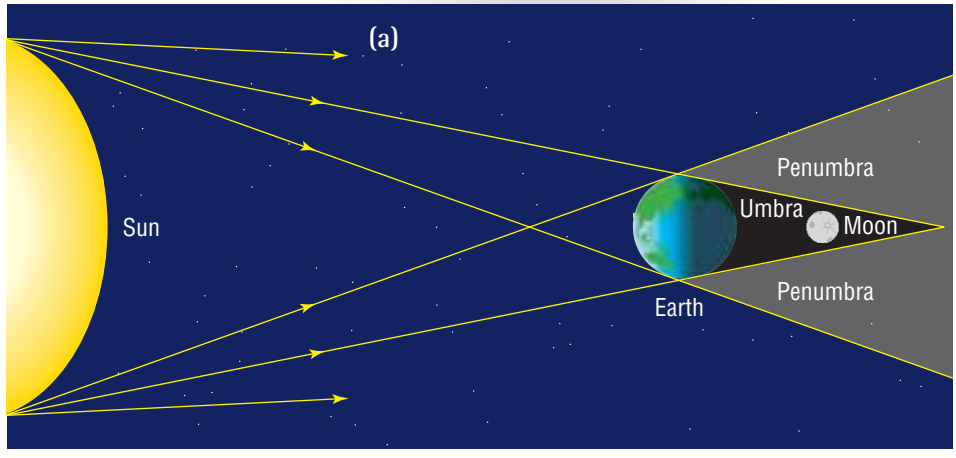
The darkest part of a shadow is in the centre, where most light is blocked out. Towards the edges, shadows are less dark. The dark middle section is called the **umbra**. The lighter part of the shadow is called the **penumbra**.

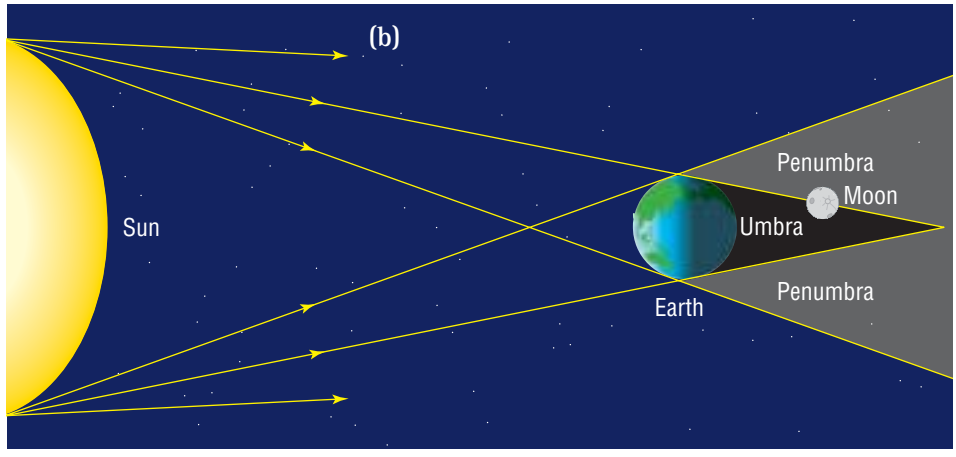
When we see a lunar eclipse, we are seeing the Earth's shadow on the Moon. When we experience a solar eclipse, we on Earth are in the shadow of the Moon. Eclipses happen once or twice a year — and some years not at all. They occur only when the Earth and Moon are in certain positions in their **orbits** with respect to the Sun.

Lunar eclipses

Lunar eclipses occur when the Earth's shadow falls on the Moon's surface. This happens when the Earth lies between the Sun and the Moon. The Earth fully or partly blocks the sunlight that makes the Moon's surface shine with reflected light.

A total lunar eclipse occurs when the Moon is totally covered by the darkest part of the Earth's shadow (the umbra). This happens only when the Sun, Moon and Earth lie in a straight line. A total eclipse can occur only during a **full Moon**. A partial eclipse occurs when the Moon is partly covered by the umbra of the Earth's shadow and partly by the penumbra.





(a) Total lunar eclipse (b) Partial lunar eclipse

Go to worksheet 45

Solar eclipses

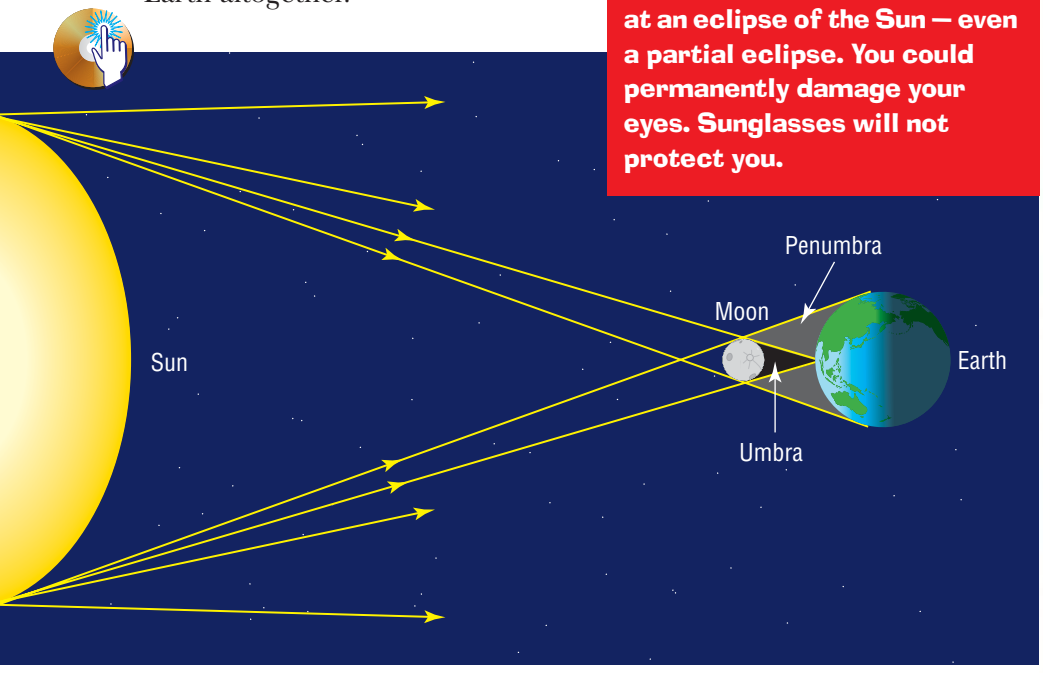
Solar eclipses occur when the Moon lies between the Sun and the Earth. This means that the Moon's shadow falls on the Earth. People on Earth within the umbra of the Moon's shadow see a total eclipse of the Sun. Those within the penumbra see a partial eclipse.

Total eclipses occur only when there is a **new Moon**. They are not seen often as the Moon casts only a narrow shadow on Earth. The umbra may be only about 100 km wide. It may fall in the middle of an ocean. It may even miss the Earth altogether.

King Louis of Bavaria became so terrified at seeing an eclipse of the Sun in AD 840 that he died of fright. His three sons fought to gain control of his throne. To keep the peace, the kingdom was divided into three regions, which are known today as Germany, France and Italy.

WARNING

You must NEVER look directly at an eclipse of the Sun — even a partial eclipse. You could permanently damage your eyes. Sunglasses will not protect you.



Total and partial solar eclipse

During a total eclipse, the area within the umbra on Earth becomes quite dark for a few minutes. You might even see some stars. The Sun's **corona**, or atmosphere, can still be seen. The corona is not normally seen because the Sun is so bright.

A total solar eclipse — the Sun's light is blocked as the Moon passes in front of it.

Activities

REMEMBER

1. What is the difference between a solar eclipse and a lunar eclipse?
2. What can be seen from Earth during a total eclipse of the (a) Moon and (b) Sun?
3. Why must you never look directly at a solar eclipse?

THINK

4. Why do you think the Moon does not block out the Sun's corona during a total eclipse of the Sun?
5. Explain, using a labelled diagram, the difference between a shadow's umbra and penumbra.
6. Suggest why a total lunar eclipse occurs only when there is a full Moon.
7. Suggest why a solar eclipse only occurs when there is a new Moon.

IMAGINE

8. Imagine that you are an astronaut on the Moon during a total lunar eclipse. Prepare a one-minute speech to Earth control to describe what it's like on the Moon, and what you see when you look towards Earth.

INVESTIGATE

9. Work in small groups to model what happens during a total lunar eclipse. Your 'Sun' can be a spotlight or bright torch. Your Earth could be a basketball and your Moon a tennis ball. Use the diagram of the total lunar eclipse opposite to help you. Discuss what you see happening.

checklist

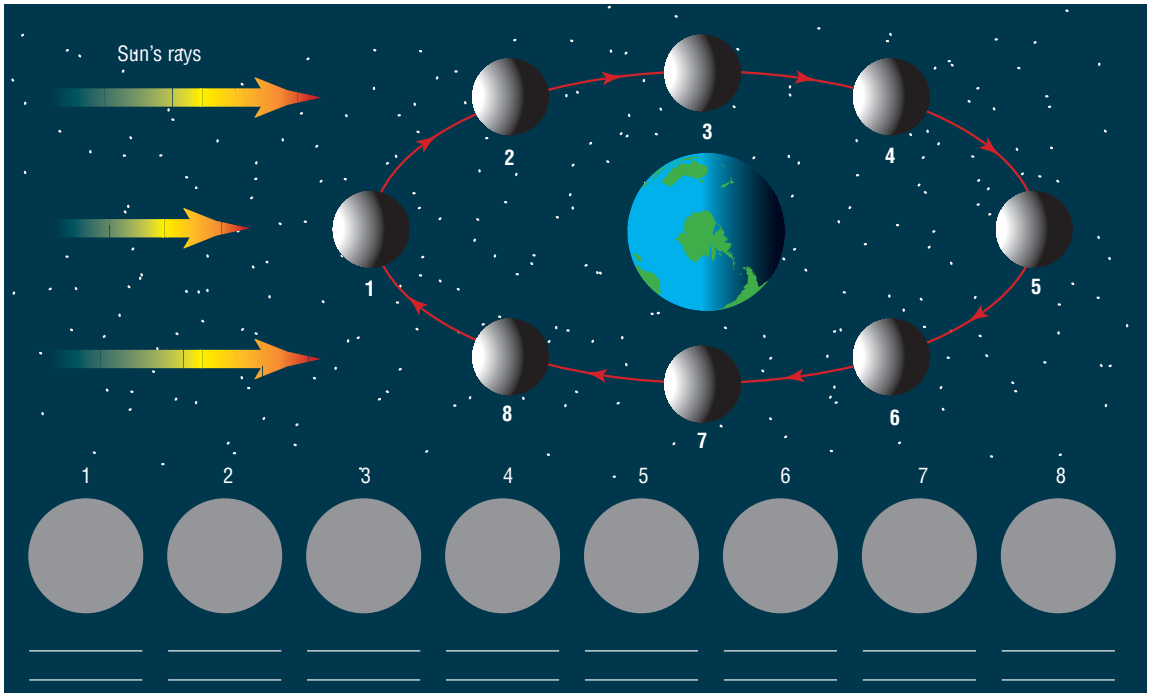
I can:

- ☐ describe and explain what happens during lunar and solar eclipses
- ☐ illustrate the difference between a shadow's umbra and penumbra.

Check and challenge

Neighbours in space

Tides and the phases of the Moon



1. Copy the diagram on the right in your workbook.
   (a) Fill in and label each of the phases of the Moon.
   (b) Put a red circle around the phases of the Moon that produce spring tides.
   (c) Put a blue circle around the phases of the Moon that produce neap tides.
   (d) Add labels to explain spring and neap tides and why they occur.
   (e) Label the near side of one of the Moons in orbit.
2. Explain why we have high and low tides on Earth.

Star moves

3. The stars appear to change their positions during each night and during each year:
   (a) Why do the stars appear to move in circular arcs during the night?
   (b) Why do constellations change their positions in the night sky during the year?

Space stories

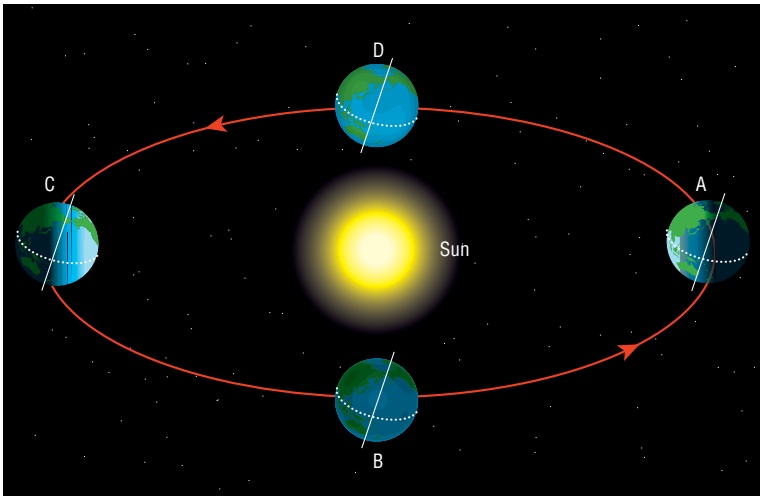
4. Why was the Sun such a key part of the myths and legends of ancient cultures?
5. How do you think people on Earth would react if news broke that another planet just like Earth had been found outside our solar system? Write a short story to describe the discovery and what happened next.

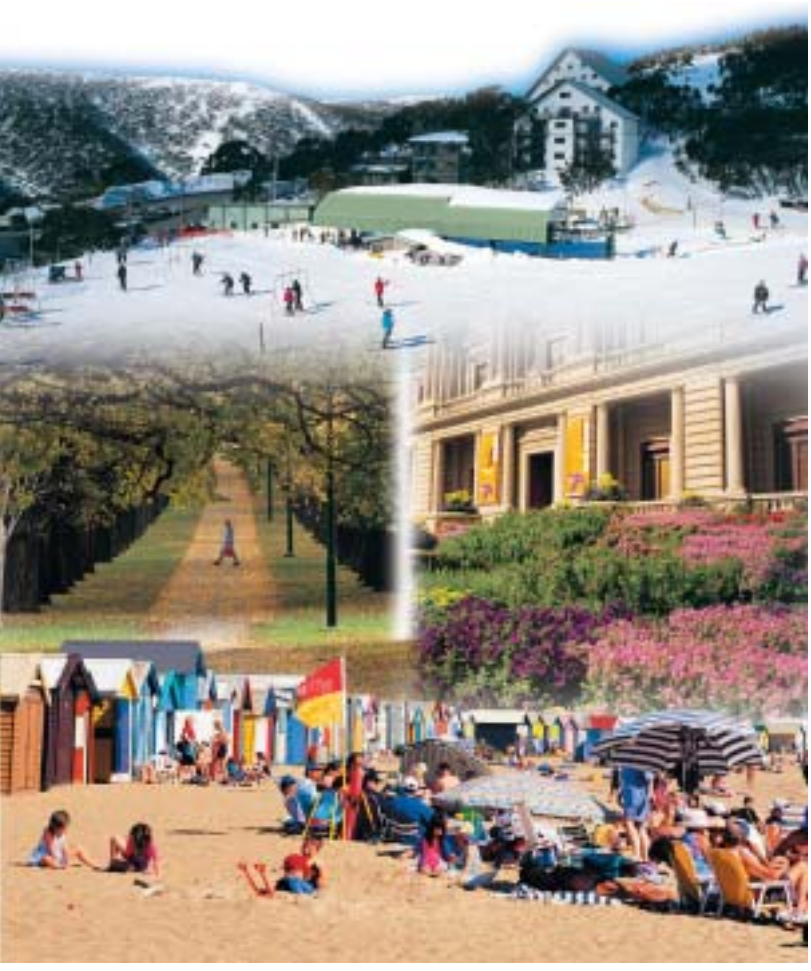
Our solar system and beyond

6. Name the nine planets in our solar system.
   Go to www.jaconline.com.au/science and follow the links to Neighbours in Space. Click on the web site given for the 'Solar system model'. Use the scaled-down distances given to demonstrate on the school oval or another safe place (using class members) how far away from the Sun each 'planet' is.
7. What do you think is the most interesting fact that has been discovered so far about space? Why?
8. Decide whether the following statements are true or false:
   (a) Buzz Aldrin was the first human to walk on the surface of the Moon.
   (b) Our solar system is in the Orion arm of the Milky Way galaxy.
   (c) Neptune was discovered 135 years before the first space shuttle was launched.
   (d) The first close-up pictures of the surface of Mars were taken in 1961.
   (e) The two smallest planets in our solar system are Pluto and Mercury.

The seasons

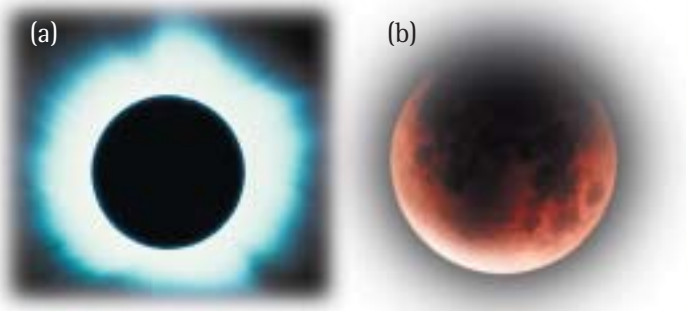
9. Discuss as a class what it would be like to live in the world's polar regions. There are places where for six months of the year the Sun never sets, and for the other six months there is constant darkness. Think how this would affect what you and your family and friends could do, and how you would feel about this.
10. Label each of the four pictures (showing the four seasons in Victoria) A, B, C or D to match the diagram below.





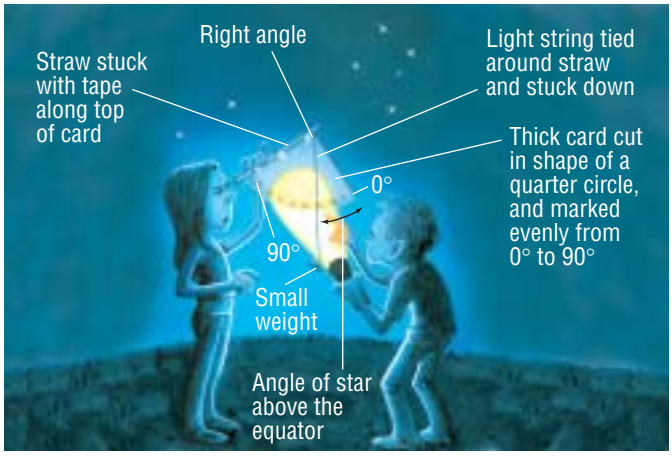
Earth on the move

1. Decide which of these photographs shows an eclipse of (a) the Sun and (b) the Moon. Use labelled diagrams to explain how each eclipse happens.



2. In ancient times, sailors used an instrument called an astrolabe to help them find their way at sea using the stars. They measured the angle of a star above the horizon (usually the North Star as it did not seem to move. It was directly above the North Pole). Once sailors knew the angle, they could work out how far away they were from the equator.

   Make your own astrolabe. Use it to demonstrate how the angle of the Southern Cross (or Crux constellation) changes over a three-hour period. (Check every half hour.) Measure the angle using the 'top' star of the constellation.



The universe

3. Cover a black sheet of A3 card with tiny white dots of different sizes. Find one interesting group of 'stars':
   (a) Join them with connecting lines to form the shape of a 'constellation'.
   (b) Give your 'constellation' a name and create a myth about that name.
   (c) Write a horoscope for your 'constellation' or 'star sign'.
4. Explain why it is impossible for us to draw the universe, even if scaled-down.

Summary of key terms

asteroid: one of the millions of rocky bodies that orbit the Sun. Some asteroids are like tiny pebbles; others may be hundreds of kilometres in diameter.

astrology: the study of how the positions of the Sun, Moon and planets are said to affect our lives. Astrologers claim that these predictions depend on where these bodies were in space on the day we were born.

astronomy: the scientific study of space, including the solar system and the stars. In fact, astronomy can involve anything outside the Earth's atmosphere.

axis: an imaginary line running through the centre of the Earth from north to south, about which the Earth rotates

comet: a body composed of rock, dust, ice. When close to the Sun, a comet has a tail which points away from the Sun.

constellation: a group of stars which has been given a particular name because of the shape the stars seem to form in the sky when viewed from Earth

corona: the Sun's bright, hazy atmosphere. It is only obvious when there is a total solar eclipse. You must never look at the Sun even for a brief moment. You could permanently damage your eyes.

eclipse: an eclipse occurs when a body in space blocks the light of the sun for a short while. A lunar eclipse occurs when the Earth blocks the Sun's light and makes a shadow on the Moon. A solar eclipse occurs when the Moon lies between the Earth and the Sun's light, casting a shadow on the Earth.

ecliptic: the invisible path across the sky that is followed by the Sun, Moon and planets, as seen from Earth

far side: the far side of the Moon. It is never seen from Earth because the Moon takes about the same time to spin around once as it does to orbit the Earth.

full Moon: the view of the Moon seen from Earth when the whole of its near side is in sunlight

galaxy: a very large group of stars held together as a system by gravity

gravitational attraction: an attraction that exists between any two bodies in the universe that have mass

gravity: the force of attraction that exists between any two bodies in the universe that have mass. The gravity at the Earth's surface is the pull of the Earth on objects at its surface.

horoscope: a set of predictions about what will happen to someone based on that person's star sign. There is no scientific basis for horoscopes.

light year: the distance that light travels in a year (9 500 000 000 000 km)

mass: the quantity of matter in an object (often measured in kilograms)

Milky Way: the galaxy of which our solar system is a very, very small part

neap tide: the tide that occurs when the Sun and Moon are at right angles with respect to the Earth (during the Moon's first and third quarters). The **gravitational attraction** of the Moon and Sun work against each other.

near side: the near side of the Moon, which is the side we always see from Earth (*see* **far side**)

new Moon: the phase of the Moon that occurs when the part facing the Earth is facing away from the sun. During a new Moon, the Moon's face is not visible from Earth, even when it is overhead.

orbit: the path that a body follows when it moves all the way around another body

penumbra: the lighter part of a shadow

phases of the Moon: eight different shapes the Moon seems to have (when viewed from Earth) while it completes one orbit of the Earth

radio telescope: a telescope that includes a huge dish, or a collection of smaller dishes, to pick up radio waves (not light waves) from distant stars and galaxies

radio waves: a type of energy that is created when electric and magnetic fields move together

reflecting telescope: a telescope that uses a curved mirror instead of a lens to collect light

revolves: what a body (e.g. the Moon) does when it moves right around, or orbits another body (e.g. the Earth or the Sun). One completed orbit is called a **revolution**.

rotation: one complete turn by a body around its imaginary axis. Such a body (e.g. the Earth), is said to rotate.

solar system: our Sun and everything that orbits it, including the nine planets (including Earth) and their moons, asteroids and comets

space probe: unmanned, remote controlled spacecraft which has a special mission (e.g. to find out more about Mars). Space probes carry many scientific instruments used to investigate objects in space.

space shuttle: a spacecraft with a plane-like orbiter that can be re-used. It is made up of the orbiter (the part which returns to Earth), liquid fuel tank and solid rocket booster.

spring tide: the very high tide that occurs when there is a new or full Moon. At such times, the Earth experiences the combined gravitational attraction of the Sun and the Moon because all three bodies are in the same straight line.

tidal month: the time it takes from one full Moon to the next. A tidal month is not the same as a calendar month (e.g. August).

tidal range: the difference between the highest of the high tides in a given day and the lowest of the low tides

umbra: the darker central part of a shadow

universe: every part of space

zodiac: the region that the Sun, Moon and planets seem, from Earth, to occupy as they pass in front of 12 constellations that make up what we call 'star signs'

CHAPTER 10

Light and sound

You will discover

How we see and hear
The differences between sound and light
How mirrors work
Different ways to measure sound

Lights are flashing and the music is so loud that you can feel the vibrations. The band has been playing for only 15 minutes and your ears are beginning to ring. Maybe you should have brought along some earplugs. But the atmosphere is electric. The lights move with the music — and the music is just what you came to hear.

When the concert is over, your ears are still ringing. You're also having trouble hearing. Even after you go to bed and the house is silent, the ringing is still there. The ringing in your ears is called tinnitus (pronounced tin-eye-tus). Some of the cells in your inner ear — the ones that detect vibrations — have been damaged. Fortunately, they are likely to recover. The ringing will stop and your hearing will return to normal — hopefully in a few hours — but maybe in a day or two. If you listen to loud music for too long or too often, the cells don't recover. Your hearing can be permanently damaged.

1 What caused the vibrations I could feel?
2 How loud does the sound have to be to cause ringing — or even pain?
3 Why did the singer's lips start moving before I heard the sound?
4 How come I could see the light beams?

Light energy

Like all stars, the Sun changes the **energy** stored inside it into heat and light energy. A burning candle converts the chemical energy in wax into heat and light energy. Some living things are also able to change the chemical energy in their bodies into light energy.

Luminous objects

Light can be produced only from other forms of energy. All objects that convert energy to produce their own light are said to be **luminous**. An example of a luminous living thing is the firefly. When a male firefly wants to attract a mate, it flashes its light and performs a dance. Females watch from near the ground and respond by flashing their lights.

Most of the living things that produce their own light live in the ocean. Some fish living in the dark depths of the ocean produce their own light to attract prey.

The firefly's light comes from a chemical reaction that takes place in special light cells in its abdomen.

Non-luminous objects

The Moon (on the right) and the statue (on the left) are not luminous. In fact, most things that you see are not luminous. We see **non-luminous** objects because light from luminous objects bounces from them. The bouncing of light from an object is called **reflection**. You see the Moon because it reflects light from the Sun — and some of that reflected light enters your eyes. You see the statue because it reflects light from the Sun or (if it were indoors) the lights in the room.

We are able to see things when light coming from them enters our eyes. The light energy is then changed into electrical energy by special **cells** called **receptors** at the back of each eye. That energy is then sent to the brain so that we know what we are looking at.

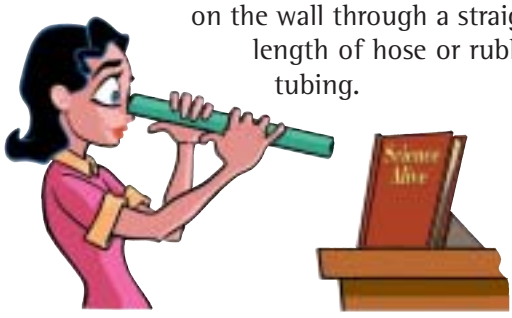
Glowing in the dark

Glow-in-the-dark stickers and toys are made with a chemical called phosphor, which **absorbs** light energy. It then slowly releases the light energy as one colour — usually green. Because the light energy is released more slowly than it is absorbed, when you turn off the lights in your room, the sticker or toy continues to release light for quite some time. This process of absorption and slow release is called **phosphorescence**.

Does light travel in straight lines?

You will need:
small length of hose or rubber tubing.

- Look at a few letters on this page or a poster on the wall through a straight length of hose or rubber tubing.



- Bend the hose or rubber tubing slightly until you can no longer see the letters.

1. What do you see when you bend the tubing?
2. Explain in words why you can no longer see the letters.
3. Draw a diagram to show why you can't see the letters.
4. Does the light reflected from the letters travel in straight lines?

Activities

REMEMBER

1. How is light energy produced?
2. What type of energy is used to produce light in a firefly?
3. Why do fireflies need light energy?

THINK

4. Which of the following objects are luminous?
   (a) the Sun
   (b) the Moon
   (c) a human eye
   (d) an unlit candle
   (e) a burning candle
5. You can see luminous objects in the dark. Why can't you see most non-luminous objects in the dark?
6. Name at least two non-luminous objects that you can see in the darkness of night and explain why you can see them.
7. What form of energy allows light to be produced in a glow-in-the-dark sticker?
8. Light energy travels through empty space and air at a speed of 300 000 km/s. How long does light take to travel the 150 million km from the Sun to the Earth?

INVESTIGATE

9. Find out what the word 'incandescent' means. Make a list of luminous objects that are incandescent.

CONNECT

10. Light produced by living things is called **bioluminescence**. Go to www.jaconline.com.au/science/weblinks and click on the Living Lights link for this textbook to find out more about the many fascinating creatures that use chemicals in their body to produce light.

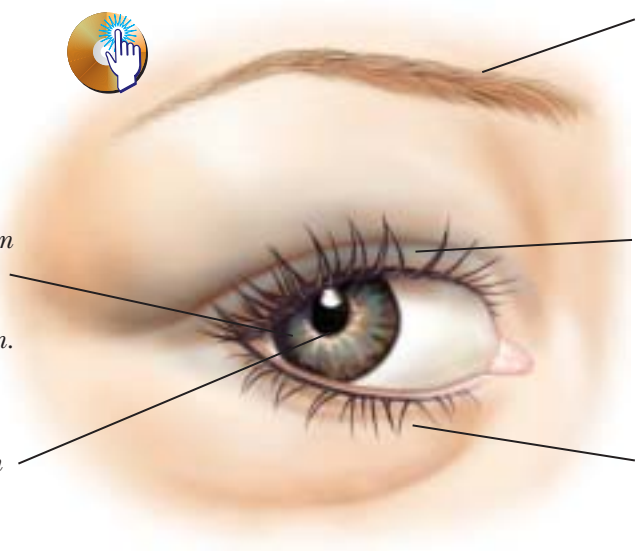
checklist

I can:
- ☐ describe light as a form of energy
- ☐ explain how light is produced
- ☐ explain how we see objects that don't produce their own light.

Seeing things

Your eyes are light detectors. They collect light from **luminous** and **non-luminous** objects. Your eyes form **images** of these objects which they then send to your brain. Your brain makes sense of the images so that you can see.

Look into my eye



Iris
*The coloured part of the eye that opens and closes to let the right amount of light into the eye. In dark conditions, the **iris** opens to allow more light in so that you can see as much as possible. In bright conditions, the iris closes to make sure that not too much light gets in.*

Pupil
The hole through which light gets into the eye. Its size changes when the iris opens and closes.

Eyebrow
Helps stop dust and other small particles from getting into your eye. The eyes are set back below your eyebrow. This helps stop water and sweat from dripping into your eye.

Eyelid
Your eyelids close to protect the eye from wind, dust and very bright light. They also keep your eye moist. Every time you blink, your eyelids spread liquid over the surface of your eyes.

Eyelashes
Short hairs that trap dust and other small particles that could damage your eye

What gives an eyeball its shape?

The part behind the lens is filled with a completely clear, jelly-like substance. This substance is what gives the eyeball its shape. It is as clear as glass, which is how it got its name – vitreous humour. Vitreous comes from the Latin word for glassy. Humour comes from the Latin word for moisture. The substance at the front of the eyeball is also clear, but more watery. It's called aqueous humour.

Light from a distant object

Light comes directly from luminous objects or is **reflected** from non-luminous objects.

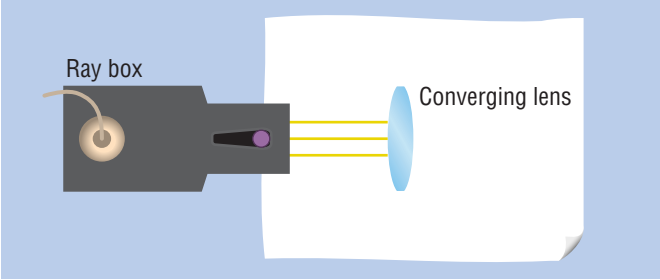
Cornea
*The clear outer surface of the front of the eye. It is curved so that it bends light through the **pupil**.*

Lens
*Bends light towards the **retina**. The **lens** is clear and flexible. It is connected to muscles that change its shape to make it thinner or thicker. The lens changes shape so that you can get sharp **images** whether you are looking at distant or nearby objects.*

What does a lens do to light?

You will need:
ray-box kit
sheet of white paper
pencil.
12 V power supply
ruler

- Place the ray box on the edge of a sheet of white paper and connect it to the power supply.



- Put a thin convex lens about 5 cm in front of the ray box.

A convex lens

- Use a pencil to trace the shape of the lens and mark the paths of the **light rays** on both sides of the lens. Remove the lens and complete your ray diagram by showing the path of the rays through the lens. Use a ruler to draw the rays.

1. How does the thin convex lens change the path of the light rays?

- Replace the thin convex lens with a thicker convex lens. Trace the shape of the lens and the light rays on a different part of the white paper.

2. What difference does it make to the path of the light rays when they pass through a thicker convex lens?

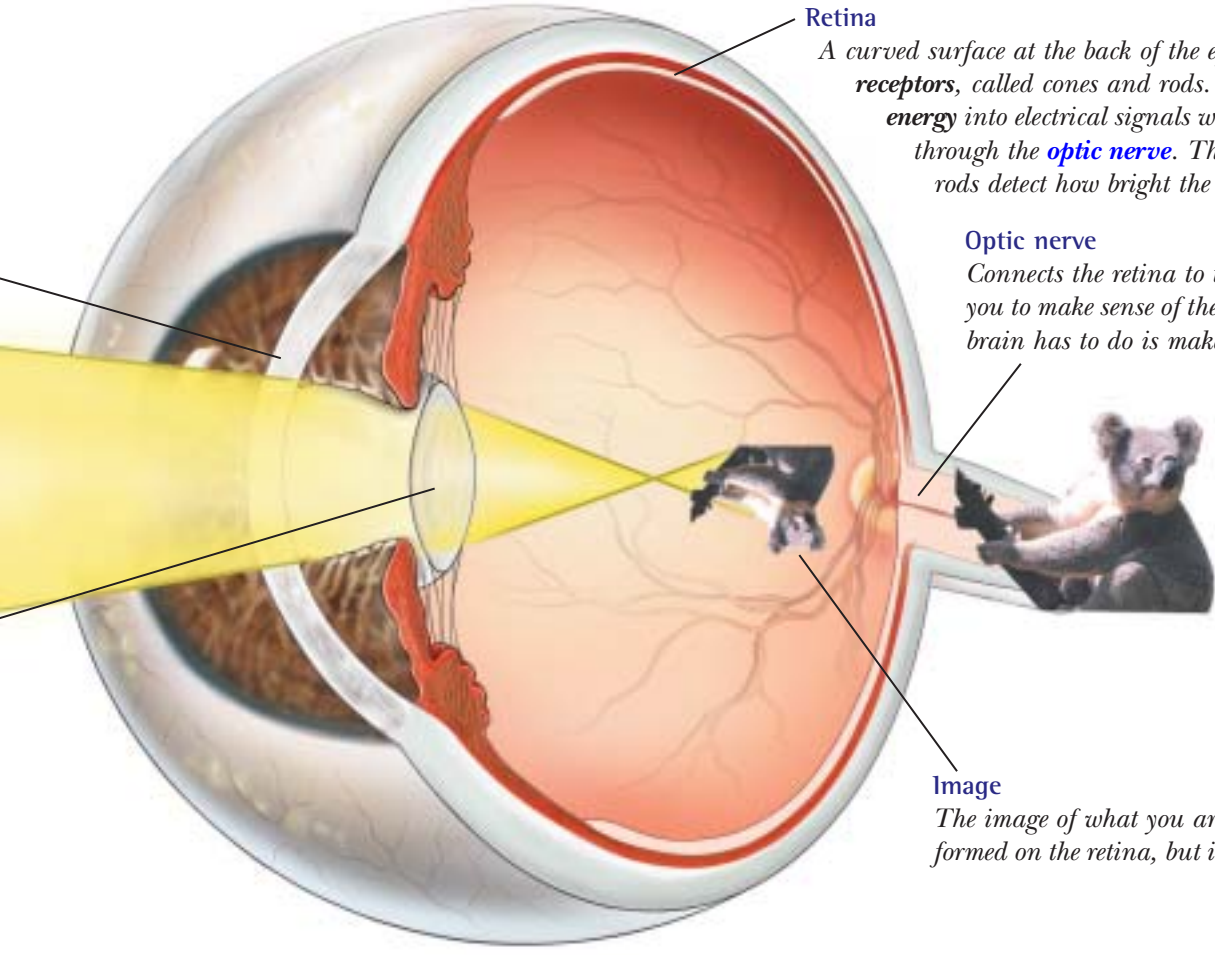
- Repeat the steps above with a thin concave lens.

3. How does a thin concave lens change the path of the light rays?

A concave lens

- Replace the thin concave lens with a thicker concave lens.

4. What difference does the extra thickness make to the path of the light rays?



Retina
A curved surface at the back of the eye. It is lined with the sight ***receptors****, called cones and rods. The receptors change light* ***energy*** *into electrical signals which are then sent to the brain through the* ***optic nerve****. The cones detect colour and the rods detect how bright the light is.*

Optic nerve
Connects the retina to the brain. The brain allows you to make sense of the image. One of the things the brain has to do is make the image 'right-side up'.

Image
The image of what you are looking at is formed on the retina, but it is upside down.

What's in a bullock's eye?

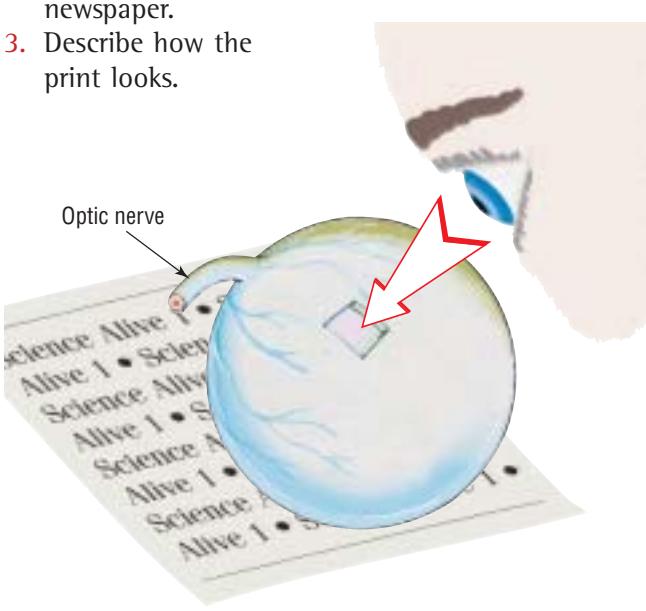
You will need:

safety glasses
fresh bullock's eye
newspaper
sharp-nose scissors
disposable gloves and lab coat or apron
dissection board
paper towel
forceps.

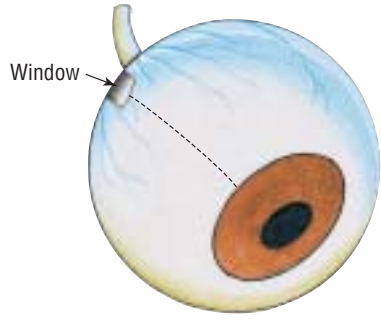
!CAUTION: Before beginning this experiment, make sure you read all of the initial instructions below.

1. Before beginning the experiment, make a list of safety rules that you should follow.
- Put on a lab coat or apron, gloves and safety glasses.
- Cover a dissection board with newspaper. Place a sheet of paper towel on the newspaper.
- Place the bullock's eye in the centre of the paper towel.
- Find the optic nerve, the cornea, the iris and the pupil.

2. Draw a diagram of the eye, labelling the parts that you found.
- Carefully cut a square hole in the white coating at the back of the eye near the optic nerve.
- Hold the eye just above some newspaper with the pupil facing some words on the paper. Look through the square hole and try to read the newspaper.
3. Describe how the print looks.



- Carefully cut from the edge of the square hole to the iris.
- Peel back the white coating and use the forceps to remove the lens and wash it with water.
4. Describe the appearance of the lens.



- Place the lens on the newspaper and look through it at some of the printed words.
- Push down gently on the lens with your finger and look for changes in the way the print looks.
5. Describe how the look of the words changes as you push down on the lens.
6. How does the lens feel when you push down on it?

Meeting new substances

When light energy travelling through the air meets a new substance, three things can happen to it:

1. it can be **transmitted**; that is, it can travel through the substance
2. it can be reflected from the surface of the substance, or reflected (scattered) from small particles inside the substance
3. it can be **absorbed**; that is, the light energy is **transferred** to the particles inside the substance.

All substances can be described by what happens to light energy when it meets them. Substances can be **transparent**, **translucent** or **opaque**.

This sheet of glass is transparent. You can see Mick through the glass because the light reflected from his face travels through the glass. That is, the glass transmits light. You can also see an image of something in the glass. That's because some of the light striking the glass is reflected — just as it would be from a mirror.

This sheet of glass is translucent. Tiny particles inside the glass scatter the light reflected from Mick's face in many directions. However, because some of the light gets through the glass without being scattered, you can still see the general shape of Mick's face.

This piece of wood is opaque. None of the light reflected from Mick's face travels through the wood. Opaque objects absorb or reflect light. Dark-coloured objects absorb more light than light-coloured objects made from the same material.

Can you see a torch beam?

You can't usually see light as it travels through the air. You don't usually see beams of light rays coming from the Sun or the lights in a room. But you can sometimes see a torch beam on a dark night. When there are lots of tiny droplets of water in the air (called fog) or small particles of smoke or dust in the air, the light from the torch hits the droplets or particles and is reflected in all directions. This type of reflection is called **scattering**. Some of the scattered light reaches your eyes. You are not actually seeing the light. You are seeing the small droplets or particles. Those droplets or particles show you the path taken by the light.

Go to worksheet 46

Activities

Go to worksheet 47

REMEMBER

1. What important job does your eye do?
2. Which three features near your eye protect it from dust and other small particles?
3. Which two parts of the eye bend light?
4. What does the iris do when you walk from bright light into a dark room?
5. What is the main job of (a) the pupil and (b) the optic nerve?
6. What special name is given to the reflection of light from many tiny particles?
7. Describe what can happen to light travelling through the air when it meets a new substance.
8. What type of substance allows the least amount of light through to the other side — transparent, translucent or opaque?

THINK

9. Why can't you usually see a beam of light from a torch at night?
10. The image on your retina is upside down. Why do you actually see things 'right-side up'?
11. How do you know that the piece of wood held by 'Mick' in the figure above left reflects some of the light that strikes it?

✓ checklist

I can:

- ☐ describe the parts of the eye
- ☐ describe how the eye forms an image of an object
- ☐ explain how light behaves when it meets a new substance.

Mirror, mirror

When you look in a mirror, you see an **image** of yourself. That's because light from the Sun or other **luminous** object bounces off your body. If some of that light is **reflected** from a smooth surface, like a mirror, you see an image of yourself. But are you seeing yourself exactly as others see you?

Flat mirrors like this are also known as **plane mirrors**. When you look at yourself in a plane mirror, you see an image.

The image in the photo above is formed because light coming from the baby is reflected from a smooth, shiny surface behind a thin sheet of glass.

The image is the same size and shape as the baby. But the way the baby sees himself in the mirror is not exactly the same as the way others see him. For example, when you look at the baby directly, his right arm is touching the mirror. When you look at the image in the mirror, it looks like his left arm is touching the mirror. His image is **laterally inverted**. In other words, it is reversed sideways.

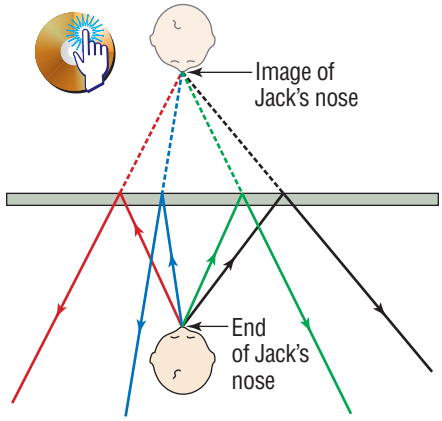
The Law of Reflection

Jenny needs to hit the centre ball with the white ball without disturbing the black ball. Luckily, she has remembered the Law of Reflection — anything that bounces from a surface comes away at the same angle from which it went in. Jenny has to work out which part of the cushion to aim for so that the 'in' angle and 'out' angle are the same.

The same law applies to light **energy** — and anything else that bounces, or is reflected from a surface.

Two kinds of reflection

You see a painted wall when you look at it because light is reflected from it. So, why can't you see an image of yourself in the wall like you can in a mirror? To answer this question you need to look very, very closely at the wall.



Light energy from the lights in the room bounces from baby Jack's nose. It is reflected from the shiny, smooth, back surface of the mirror.

The light reflected from the mirror all appears to be coming from one place. And that is exactly where the image of Jack's nose is.

*This type of reflection is called **regular reflection**.*

This type of reflection is called regular reflection.

Go to worksheet 48

One-way mirrors

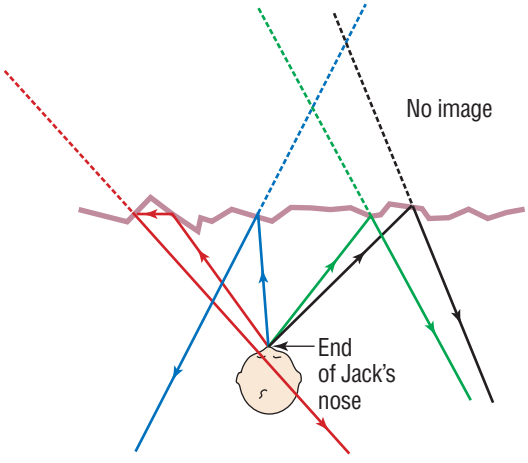
You may have seen TV shows in which a criminal is being questioned while detectives watch through a 'one-way' mirror.

A 'one-way' mirror is made from a darker type of glass than a normal mirror. The metal coating on the glass is much thinner than the coating on a normal mirror. It allows half of the light to go through. The other half is reflected. If the rooms on both sides of the mirror are well lit, you can actually see through it from both sides. The mirror becomes a one-way mirror only when the lighting is right.

The interview room is brightly lit. There is plenty of light bouncing off the suspect, the two detectives and everything else in the room. Enough reflected light goes through the mirror to allow the detectives in the viewing room to see what's happening.

The rest of the light is reflected from the mirror. This allows the people in the interview room to see a clear image of themselves.

The viewing room on the other side of the mirror is only dimly lit. There is very little light bouncing off the detectives and the room. So these detectives can't see themselves in the mirror. More importantly, there is not enough light going through the mirror to allow the suspect in the interview room to see who's watching.



This type of reflection is called diffuse reflection.

Light energy from the lights in the room bounces from Jack's nose. It is reflected from the surface of the painted wall. Notice that the Law of Reflection is obeyed wherever light meets the surface.

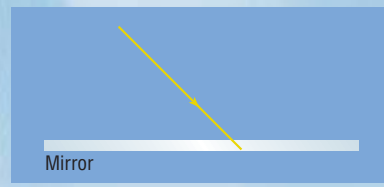
Because the surface is not smooth, the light is reflected in all directions. It doesn't all appear to be coming from a single point behind the wall. There is no image of Jack's nose.

This type of reflection is called ***diffuse reflection****.*

Activities

REMEMBER

1. Copy the diagram and complete it by carefully drawing the path of the light after it is reflected from the mirror.



2. Why can't you see your image when you look at a painted wall even though it reflects light?

THINK

3. The image of the baby is formed because some of the light coming from him is reflected from the mirror. Where does this light originally come from?
4. Which of the baby's ears is visible in the photo — left or right?
5. Write down three capital letters of the alphabet that (if you wrote them on a piece of card and held it up to a mirror) would not look any different in the mirror.
6. Why is the word 'fire' on the fire truck shown below printed in two different ways?

✓ checklist

I can:

- ☐ describe images in plane mirrors
- ☐ draw a diagram to illustrate the Law of Reflection
- ☐ explain why mirrors produce images while most other surfaces do not.

Mirrors with a curve

Curved mirrors can produce some very strange **images**. If you stand in front of a curved mirror, your image can be large, small or even upside down. Some curved mirrors even allow you to see around corners. Curved mirrors obey the Law of Reflection — just as flat mirrors do. But their shape causes the image they produce to be quite different from a flat-mirror image.

Mirrors that are curved outwards are called **convex mirrors**. These mirrors collect light from a wide angle and **reflect** it towards your eyes. The images in convex mirrors are smaller than those in a plane mirror. But there is one other difference. What is it?

Mirrors that are curved inwards are called **concave mirrors**. The images in concave mirrors can be large, small or upside down. Dentists use a concave mirror on a handle to get a magnified view of your teeth. A dentist's mirror is not curved as much as this soup spoon.

Describing mirror images

You will need:
convex and concave mirror or a polished spoon
2 plane mirrors
glue stick or similar small object.

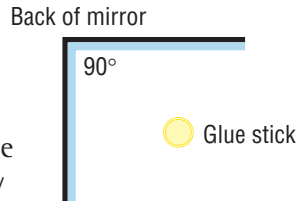
- Look at an image of yourself in a convex mirror. Move the mirror towards yourself and then further away to see how the image changes.

1. How is the image in the convex mirror different from your image in a plane mirror?

- Repeat the same observations with a concave mirror.

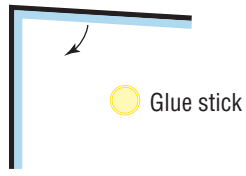
2. How is the image in the concave mirror different from your image in a plane mirror?
3. How does the image in a concave mirror change as your face gets closer to the mirror?

- Place two plane mirrors at right angles to each other and stand a glue stick between them. Look at the images of the glue stick.



4. How many images are there?
5. Are all of the images exactly the same? If not, explain how they are different.
6. Why do you think there are more than two images? (You'll need to think hard about this one!)

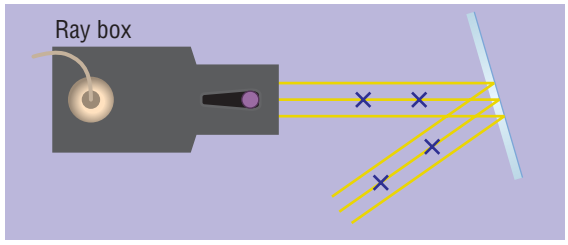
- Move the mirrors to gradually make the angle between them smaller.



7. What happens to the number of images as the angle between the mirrors gets smaller?

How does light behave when it meets different mirrors?

You will need:
ray-box kit
12 V power supply
sheet of white paper
ruler
pencil
protractor.



- Place the ray box on the edge of a sheet of white paper and connect it to the power supply.
- Put a plane mirror about 5 cm in front of the ray box.
- Draw a pencil line on the paper to show where the back of the glass part of the mirror is. You will need to move the mirror while you do this. Make sure that you put it back in exactly the same place.
- Use the pencil to mark two points on one **light ray** going into the mirror. Do the same for the ray as it goes away from the mirror.
- Remove the mirror and complete your ray diagram by drawing the path of the rays towards and away from the mirror. Use a ruler to draw the light rays.
- Use a protractor to measure the angle between the incoming ray and the mirror. Then use it to measure the angle between the outgoing ray and the mirror.

1. Are the angles the same? Is the Law of Reflection obeyed? If the angles are not the same, explain why you think they are different.

- Turn the sheet of paper over. Replace the plane mirror with a convex mirror and observe how the three rays are reflected.
- Look closely at where each ray meets the surface of the mirror.

2. Does it look like the Law of Reflection is obeyed at the surface of the convex mirror?

- Replace the convex mirror with a concave mirror.
- Again, look closely at where each ray meets the mirror surface.

3. Does it look like the Law of Reflection is obeyed at the surface of the concave mirror?
4. Trace the shape of the front of the concave mirror and the paths of the rays towards and away from the mirror.

- The point where the reflected rays meet is called the **focus**, or focal point. Label the focus on your diagram. The **focal length** of the mirror is the distance between the mirror and the focus. Measure the focal length of the concave mirror and write it down on your diagram.
- Replace the concave mirror with another one that is either more curved or less curved.

5. What happens to the focus if you use a concave mirror that is more curved?

- Turn off the light box and let it cool down while you answer the next two questions.

6. Which shaped mirror reflects light energy so that it spreads out?
7. If a mirror were used to reflect sunlight to cook a sausage, what shape would it need to be — plane, convex or concave?

Activities

REMEMBER

1. In which shaped mirror/s:
   (a) can you see an upside-down image of yourself?
   (b) can you always see an image that is smaller than the image in a plane mirror?
   (c) is the image always right-side up?
   (d) is the Law of Reflection obeyed?
2. Which shaped mirror:
   (a) spreads light out?
   (b) brings light rays together?
3. What is special about the focus, or focal point, of a concave mirror?

THINK

4. Describe the shape of the mirror in the photo at the top of the previous page.
5. Which shaped mirror could be used:
   (a) to make a solar reflector to heat food?
   (b) to allow you see around corners?
6. Why is a convex mirror used at the worksite shown at far left instead of a plane mirror?
7. A searchlight is designed to produce a beam that does not spread out. The lamp of a searchlight is placed in front of a large curved mirror. Would you expect the mirror of a searchlight to be concave or convex?

✓ checklist

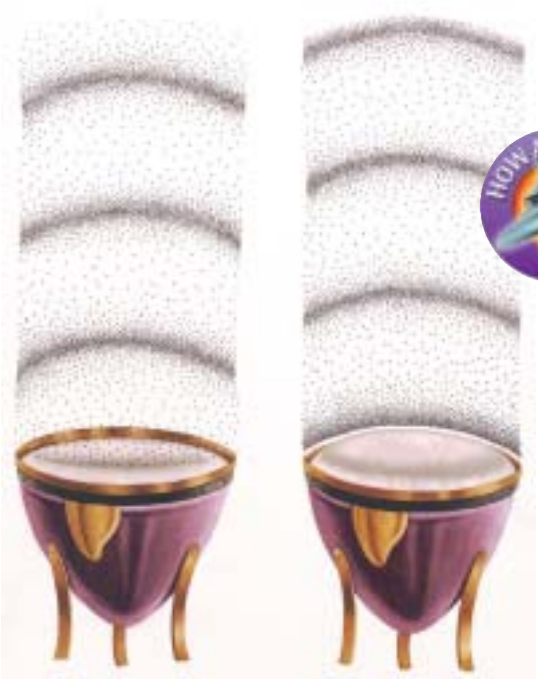
I can:
- ☐ describe the images in convex and concave mirrors
- ☐ list some uses of convex and concave mirrors, and relate their use to the way they reflect light.

Sound energy

In 1883, the Indonesian island of Krakatoa was blown apart by a volcanic explosion. The sound of the explosion was probably the loudest that human ears have ever detected. It was heard as far away as South Australia. That's over 3000 kilometres away! How did the sound travel such a huge distance?

Thunder and lightning

Vibrations everywhere

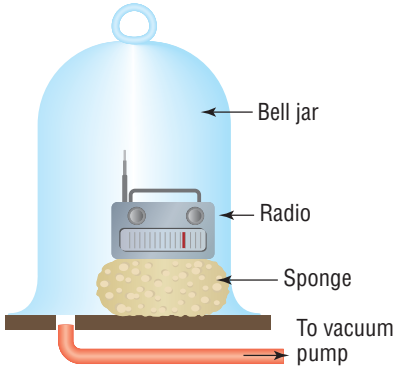


Sound is caused by fast, back-and-forth movements called **vibrations**. When you strike a drum, the drumskin vibrates. The movements of the drumskin cause air particles to be pulled back and spread out. A fraction of a second later, the air particles are pushed together. This pulling and pushing of air particles continues until the drumskin stops vibrating. The energy of the vibrating drumskin is **transferred** to the nearby air particles. So the air particles vibrate just as quickly as the drumskin vibrates.

The vibrating air particles bump into other air particles, making them vibrate as well. If there is enough **energy** in the vibrating air, the vibrations are passed on all the way to your ear. If the vibrations reach your ear, you hear sound.

Sound silenced

When a loud radio is played in a bell jar, the sound can be heard clearly. When the air in the bell jar is sucked out by a vacuum pump, the sound fades. If all of the air is removed, no sound can be heard at all. Sound cannot travel through empty space. The energy of vibrating objects can travel only by making particles vibrate. In empty space, there are no particles to vibrate. Unlike sound, light can travel through empty space. It doesn't need particles. So you can still see the radio, even if you can't hear it.



The vibrations caused by the volcanic explosion on Krakatoa in 1883 had enough energy to travel through the air for at least 3000 km.

This photo shows Krakatoa erupting more recently in 1995.

When lightning strikes during a thunderstorm, a giant electric spark heats the air around it. The hot air quickly expands, crashing into the cold air around it. The sound of that crash is thunder. So why do you always hear thunder AFTER you see the lightning? The answer lies in one of the differences between sound energy and light energy. Sound travels through air at a speed of about 340 m/s. Light travels through air at a speed of 300 000 km/s.

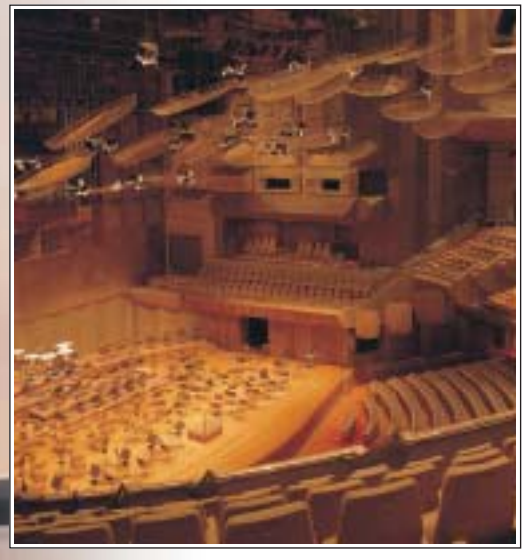
The delay between seeing lightning and hearing thunder is about three seconds for each kilometre that you are away from the lightning.

Sounding great

Like light, sound can be **transmitted**, **reflected** and **absorbed**:

- All materials transmit some sound, some better than others. That's why you can sometimes hear conversations from another room through the walls.
- Sound is reflected by hard surfaces like the tiles in bathrooms and showers. Each note that you sing in the shower lasts longer because its sound is reflected. This effect is called **reverberation**.
- Soft materials, like curtains and carpet, absorb much more sound than walls of tiles or plaster.

The Melbourne Concert Hall is designed to control the transmission, reflection, and absorption of sound. The inner walls are designed to make sure that unwanted noise is not transmitted from outside. The 30 perspex plates above the stage control how much sound is reflected to different parts of the hall. To control the amount of sound absorbed there are 22 woollen banners around the hall. The banners are lowered to allow more sound to be absorbed. Heavy curtains behind the audience can also be closed to increase the amount of sound absorbed.



Activities

REMEMBER

1. What causes sound?
2. Why can't sound travel through empty space?
3. Which form of energy travels faster in air — light or sound?
4. What types of surface cause reverberation?

THINK

5. When sound energy reaches a wall, there are three things that can happen to it.
   (a) What are those three things?
   (b) Which one or more of these things can also happen to light energy?
6. If you see lightning, then hear thunder 9 seconds later, how far are you from the lightning strike?
7. How could two astronauts repairing the outside of a space shuttle talk to and hear each other without using radios?
8. Imagine if light could not travel through empty space. What would you observe if air was removed from a bell jar containing a radio playing loudly?

INVESTIGATE

9. You can feel your vocal cords vibrate if you place your hand gently over your throat while you talk. Say a long 'hummmm' in a deep voice and feel the vibrations. How do the vibrations change when you say 'hummmm' in:
   (a) a louder voice?
   (b) a higher voice?

✓ checklist

I can:

- ☐ explain how sound is produced and how it travels through the air
- ☐ describe two differences between the way light and sound travel
- ☐ give examples of the transmission, reflection and absorption of sound.

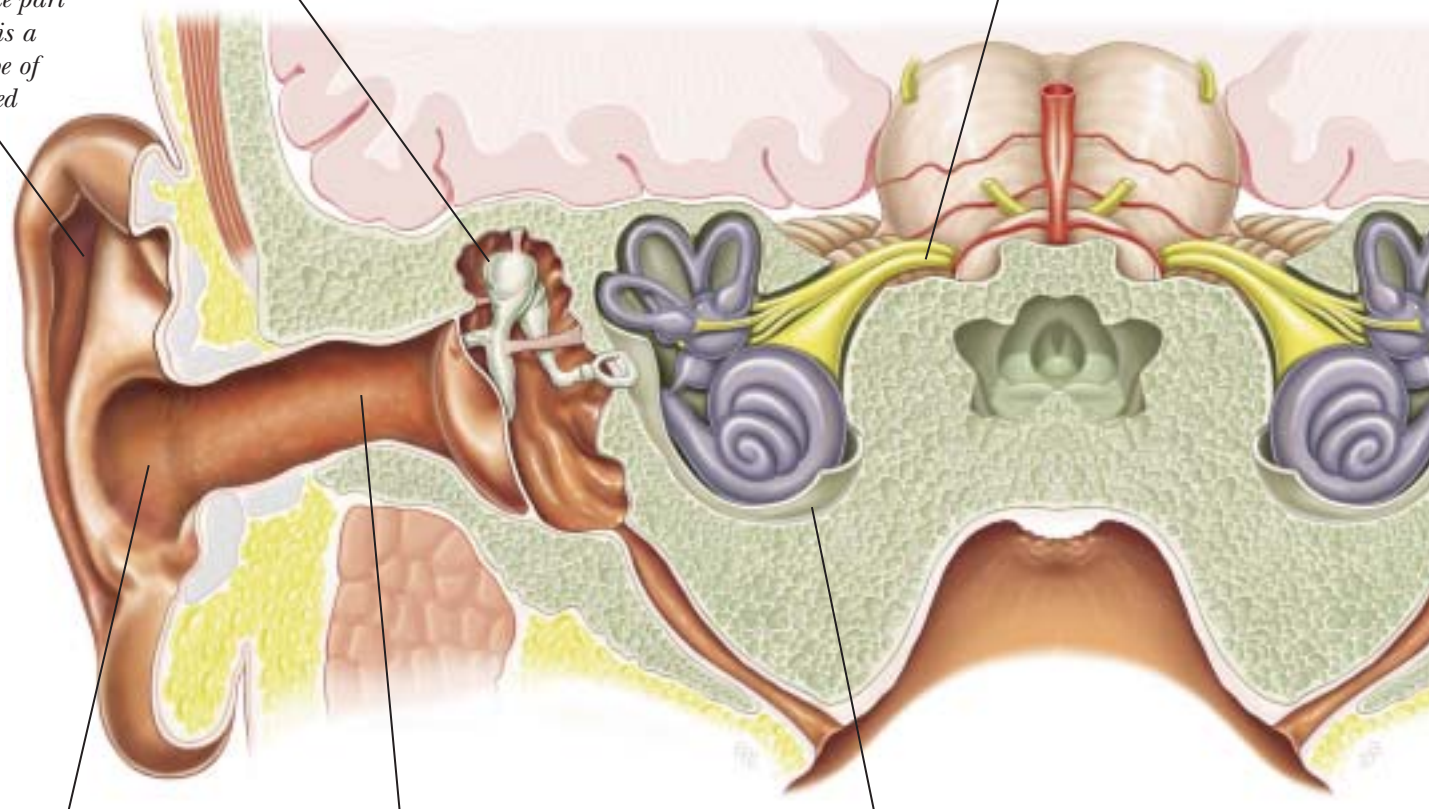
Your ears and hearing

The main job of your ears is to detect sound. They collect the **energy** of vibrating air and change it into electrical signals that are sent to your brain. Each ear has three main parts — the outer ear, the middle ear and the inner ear. Each part has its own special job to do.

Middle ear
*The middle ear contains the three smallest bones in your body. Together, they are known as the **ossicles**. These tiny bones send **vibrations** from the **eardrum** to the inner ear. They also make the vibrations larger. One of the ossicles (the stirrup) presses against a thin layer of skin called the **oval window** at the entrance to the inner ear.*

Auditory nerve
*Nerves from the **receptors** in the **cochlea** merge together to form this large nerve that sends signals to the brain.*

Auricle
The outside part of the ear is a spongy type of tissue called cartilage.



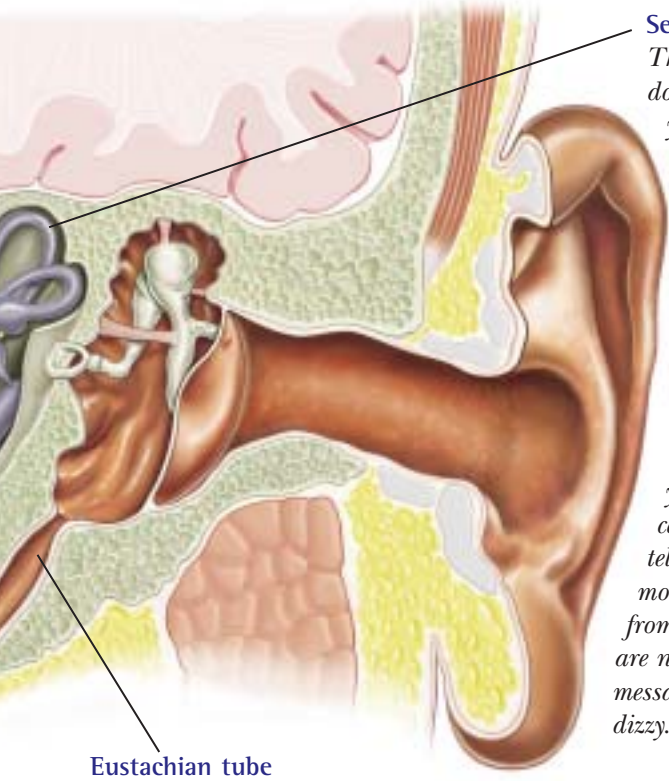
Outer ear
*The outer ear collects the energy of the vibrating air and funnels it along the **ear canal**. The air along the ear canal vibrates. That makes the eardrum vibrate. High-pitched sounds make the eardrum vibrate quickly. Low-pitched sounds make the eardrum vibrate slowly.*

Ear canal
Contains wax and tiny hairs to trap dust so that it doesn't get to your eardrum. Don't try to remove the wax. It's there to protect your eardrum. If it builds up enough to block your ear canal, a doctor can remove it for you.

Inner ear
*The inner ear is filled with fluid. The vibrations are passed along the fluid into a snail-shaped tube called the cochlea. The inside of the cochlea is lined with millions of tiny hairs. Each hair is attached to a receptor. When the fluid vibrates, the hairs move. The receptors change the energy of the moving hairs into electrical energy and send signals through the **auditory nerve** to the brain. You interpret those signals as sound.*

Aye-aye

The aye-aye is a rare animal that lives on the island of Madagascar. It feeds at night and has goggle eyes and huge ears. The aye-aye searches for food by tapping one of its stick-like fingers on tree trunks. It listens to the sound as vibrations go through the wood. The sound tells it where gaps, cracks and hollows are under the bark and where tasty grubs are hiding. Then it chews through the wood and hooks out the grub with its long middle finger.



Semicircular canals

These three tubes have nothing to do with hearing. They control your sense of balance. When you move, fluid in the tubes flows past cells that sense the movement. These cells send signals to the brain. The signals tell you when you are moving and whether you are up, down or on your side. When you move around in circles quickly, the fluid moves quickly — even for a while after you stop. The messages from the cells in the ***semicircular canals*** *tell your brain that you are still moving. However, the messages from your eye tell the brain that you are not moving. These mixed messages to the brain make you feel dizzy.*

Eustachian tube

This tube joins the middle ear to the nose and throat. It is usually closed. When the air ***pressure*** *on the eardrum is not the same on both sides, the tube opens. Air then moves either into or out of the middle ear until the pressure is balanced again.*

When the air pressure on one side of the eardrum changes quickly, you can feel a 'pop' as the ***Eustachian tube*** *opens and air rushes through it. This happens when you are in a plane that is climbing steeply. The air pressure in the plane becomes less than the air pressure in your middle ear. The Eustachian tube then opens and some air moves from the middle ear to the nose and throat so that the air pressure on your eardrum is balanced. You can also feel the 'pop' just before you land, because the air pressure in the plane increases. The tube opens again and air rushes into your middle ear from your nose and throat to balance the pressure outside your eardrum.*

Activities

REMEMBER

1. Why does the ear canal contain wax and hairs?
2. What is the job of the outer ear?
3. How do the ossicles change vibrations?
4. In which part of the inner ear are the receptors for hearing?
5. How do receptors in the inner ear detect the vibrations that arrive from the outer ear?
6. Part of the inner ear has nothing to do with hearing:
   (a) Which part is it?
   (b) What important job does this part do?

THINK

7. Why shouldn't you try to remove wax from your ear canal yourself?
8. Why do you think there are three semicircular canals rather than just one?

SKILLBUILDER

9. Design an experiment to find out the direction from which a sound is coming. Try your experiment with high-pitched sounds and low-pitched sounds to see if the pitch makes any difference.

INVESTIGATE

10. Apart from the aye-aye, what animals have big ears? What are they used for apart from hearing?

checklist

I can:

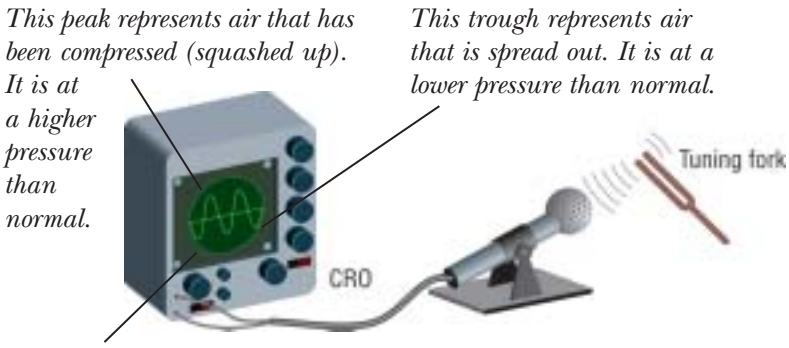
- ☐ describe the main parts of the human ear and what they do
- ☐ explain how the human ear changes the energy of vibrating air into signals that we hear as sound.

Measuring sound

You can't see sound, but that doesn't mean you can't measure it. Fortunately, sound **energy** can be converted into electrical energy. That means that we can measure how quickly it makes the air **vibrate** and how much the air is squashed up as sound passes. These measurements tell us how high or low a sound is and how loud it might sound.

Pictures of sound

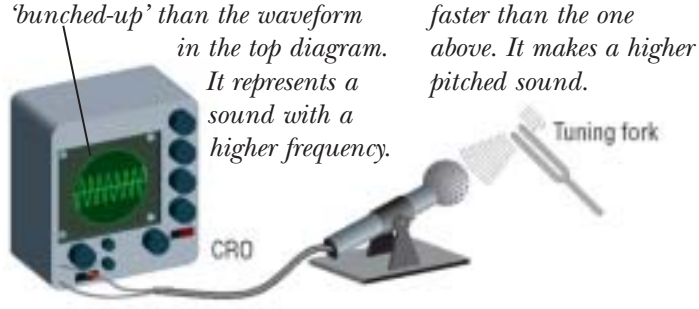
A **cathode ray oscilloscope** (CRO) changes sound energy into electrical energy. The shape on each screen is called a **waveform**. It shows how the air **pressure** changes when the sound passes the microphone.



Taller waveforms represent louder sounds. That's because louder sounds change the air pressure more than soft sounds do.

The **pitch** of a sound depends on how quickly it makes the air vibrate. High-pitched sounds make the air vibrate quickly, so the air pressure changes quickly. As a result, they produce 'bunched-up' waveforms. Low-pitched sounds make the air vibrate less quickly, so the air pressure changes more slowly. The waveforms are more spread out.

The number of times that a sound makes air vibrate every second is called its **frequency**. Frequency is measured in a unit called **Hertz** (Hz). Higher frequencies sound more highly pitched than low-frequency sounds.



How does the size of vibrating objects affect the sound they make?

You will need:
ruler
2 clean, different-sized test tubes.

- Hold a ruler firmly over the edge of a table so that about half of it overhangs the table. Flick the ruler so that it vibrates.
- Change the position of the ruler so that less of it overhangs the table and flick it again.

1. What happens to the pitch of the sound when the length of the vibrating part of the ruler becomes shorter?

- Blow across the mouth of a large, empty test tube and listen to the sound.

2. What happens to the air inside the test tube when you blow across its mouth?

- Blow across the mouth of a smaller test tube.

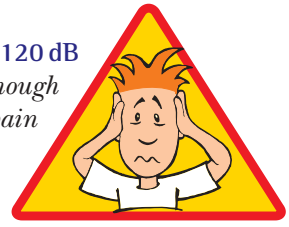
3. How is the sound of the small test tube different from that of the larger test tube?
4. How does the size of vibrating objects appear to affect pitch?

Turn that music down!

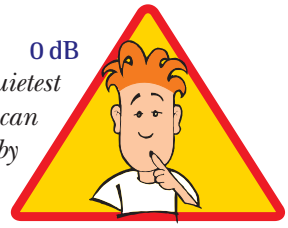
The loudness of a sound is sometimes a matter of opinion. But there is a way to measure how much the air is disturbed by a sound. This measurement, called **sound level**, is made in **decibels** (dB). The number of decibels gives a good indication of loudness.

Go to worksheet 49

150 dB
At this level, your ***eardrum*** *could rupture*



120 dB
Just loud enough to cause pain



0 dB
The quietest sound that can be heard by most people

Very loud noises (greater than 120 dB) can cause pain and damage your ear. Even long periods of listening to noise with a sound level more than about 80 dB can damage your inner ear. You should always wear earmuffs when using noisy machines like chainsaws.

Activities

REMEMBER

1. Why do louder sounds produce taller waveforms on the screen of a cathode ray oscilloscope?
2. What is the unit of frequency?
3. What happens to the pitch of a sound when its frequency is increased?
4. What is the name of the quantity that is measured in decibels?
5. What is the sound level of the softest sound that most people can hear?

THINK

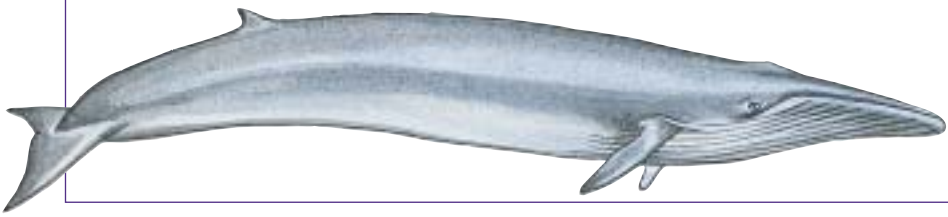
6. Which of the two waveforms (top or bottom) displayed on the cathode ray oscilloscope screens in the figures on the far left is caused by:
   (a) the loudest sound?
   (b) the highest pitched sound?
7. If a dog barks to produce a sound with a frequency of 400 Hz, how many times does the air near the dog vibrate every second?
8. The sound level of chainsaws is below the threshold of pain. Why is it necessary to wear ear protection while using a chainsaw?

SKILLBUILDER

9. Use a sound-level meter to measure the sound level in decibels at a range of places and times around your school. Your results could be displayed in a graph like the one on this page.

HOW ABOUT THAT!

The calls of the blue whale can be even louder than the launch of a space shuttle, with sound levels of more than 180 dB. That's not surprising when you consider that its tongue can weigh more than an elephant and its heart can be the size of a small car.



checklist

I can:

- ☐ use a variety of equipment to measure sound
- ☐ explain what is meant by the frequency of a sound
- ☐ recognise situations in which ear protection is necessary.

Sound technology

We all know that sound **energy** allows us to hear, but did you know that we can use sound to look inside things — including the human body? **Ultrasound** is used to treat muscle injuries and other medical problems. We also use sound to map the depths of the ocean floor.

Ultrasound

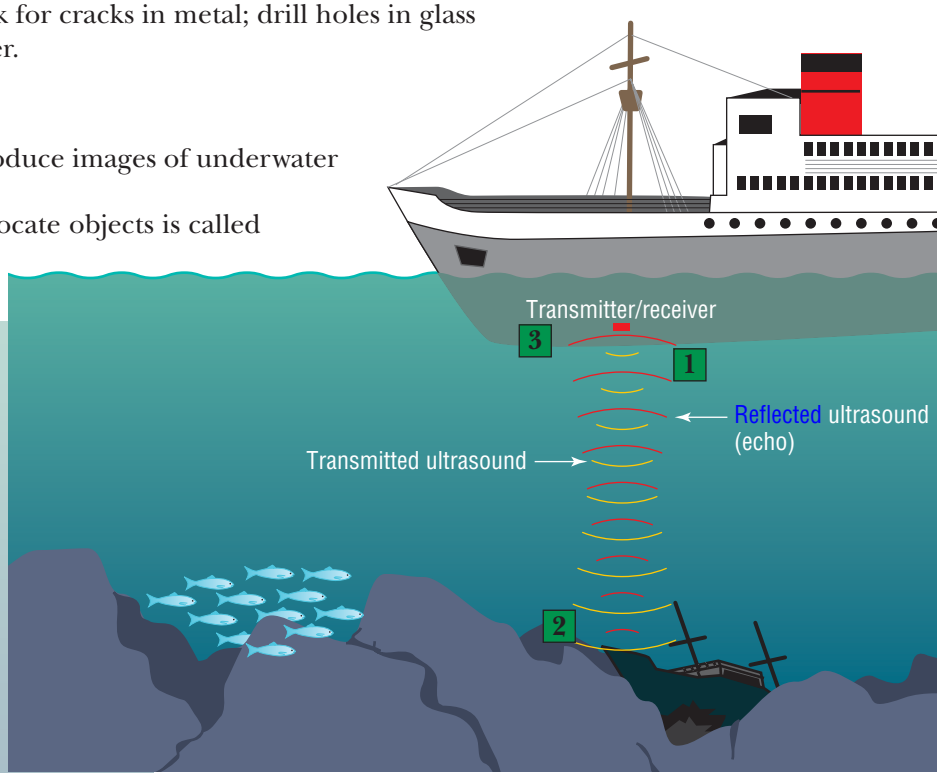
Sound with **frequencies** higher than those that humans can hear is called ultrasound. This **image** of an unborn baby was produced with ultrasound. To produce images like this, ultrasound is sent through the mother's body. Some of it is **reflected** from the baby. A computer is used to change the reflected ultrasound into an image. The images are used to check for problems during pregnancy.

Ultrasound is also used to check for cracks in metal; drill holes in glass and steel; and join metals together.

Seeing without seeing

Ultrasound is used in **sonar** to produce images of underwater objects or the ocean floor.

The use of reflected sound to locate objects is called **echolocation**.



1 *Ultrasound is sent down into the water.*

2 *Objects under the water (and the ocean floor) reflect some of the ultrasound.*

3 *A receiver detects the reflected ultrasound.*

A computer uses the time taken for the reflected ultrasound to return to the ship to calculate the depth of objects in the water. It can also map the ocean floor.

HOW ABOUT THAT!

Catching prey in the dark

Bats use ultrasound to find their prey in the dark. The ultrasound they emit from their nostrils is reflected from insects. Bats can tell from the reflected ultrasound, or **echo**, exactly where their prey is. The further away the insects are, the longer it takes the echo to return to the bat. The echo from more distant insects is also fainter. This use of echolocation by bats is called **biosonar**.

Go to worksheet 50

The bionic ear

The **cochlear implant**, also known as the bionic ear, has allowed some people with inner-ear problems to hear sound for the first time. When deafness results from serious inner-ear damage, no sounds are heard at all. Normal hearing aids that make sound louder will not help because the **cochlea** cannot detect the **vibrations**. However, in many cases, the cochlear implant can help. The cochlear implant changes sound energy from outside the ear into electrical signals that can be sent to the brain.

An enlarged X-ray of the cochlea showing the experimental electrode array inside.

3. *The electrical code is sent through a cable to the transmitting coil. Radio waves are then used to send the code through the skin.*

4. *The receiver-stimulator is implanted in a bone behind the ear. It decodes the signal and sends electric pulses through wires towards the cochlea.*

1. *A microphone is worn behind the ear.*

2. *The speech processor changes the sound into an electrical code. It can be worn on a belt or a smaller version can be built into the microphone and worn behind the ear.*

6. *The hearing receptors send messages through the* ***auditory nerve*** *to the brain. The sound heard by the user is not completely natural because there are only 22 electrodes replacing the tens of thousands of hair cells in the cochlea of a normal ear.*

5. *Electrodes placed inside the cochlea receive the decoded signals. The 22 electrodes allow a range of different pitches to be detected. The electrodes stimulate the* ***hearing receptors****.*

Activities

REMEMBER

1. What is the difference between sound that we hear and ultrasound?
2. List two medical uses of ultrasound.
3. How does ultrasound enable bats to locate insects?
4. What do the electrodes in the cochlear implant replace?

THINK

5. Why is the use of sound to locate objects called echolocation?
6. In the case of bats, which sense does the use of biosonar replace?
7. Why can't hearing aids help people with a seriously damaged cochlea?

CONNECT

8. Go to www.jaconline.com.au/science/weblinks and click on the Bionic Ear link for this textbook to find out more about the cochlear implant.

checklist

I can:

- ☐ describe ultrasound and list some of its uses
- ☐ explain what is meant by echolocation
- ☐ explain how the bionic ear works.

Solar System Gazette

Monday 9 September 1999 Weather: Jupiter surface — cold, –140 °C

Extra-terrestrial life?

Washington: Scientists have released evidence of what they think could be life on Europa, the fourth largest moon of the planet Jupiter. Until 1979, when the *Voyager* space probes photographed Europa, little was known about its surface. But recent photographs taken by NASA's *Galileo* spacecraft have provided enough detail to make scientists think that there could be an ocean, and maybe even life, under the sheet of ice.

NASA is planning to launch a new space probe to try to solve the mystery of what is below the icy surface of Europa. The probe, called *Europa Orbiter*, is due to leave Earth in 2008. After a three-year journey from Earth, *Europa Orbiter* will go into orbit around Europa. It is not designed to land on the surface. It will begin by finding out whether the ocean exists, how deep it is and how thick the sheet of ice is.

If water is found beneath the ice on Europa, NASA is likely to send space probes that can land. They would carry special robots. A cryobot, an ice-melting robot, would drill down into the ice. Then a hydrobot with a camera and chemical sensors would explore the mysterious ocean.

If there is water on Europa, it is possible that there might be some form of life. Every year more and more unusual micro-organisms are discovered living in unexpected places on Earth. Some found near the vents of hot, underwater volcanoes live without sunlight or oxygen, at temperatures of 105 °C. Not that long ago, people thought this was impossible. Perhaps similar life exists on the strange and distant moon of Europa?

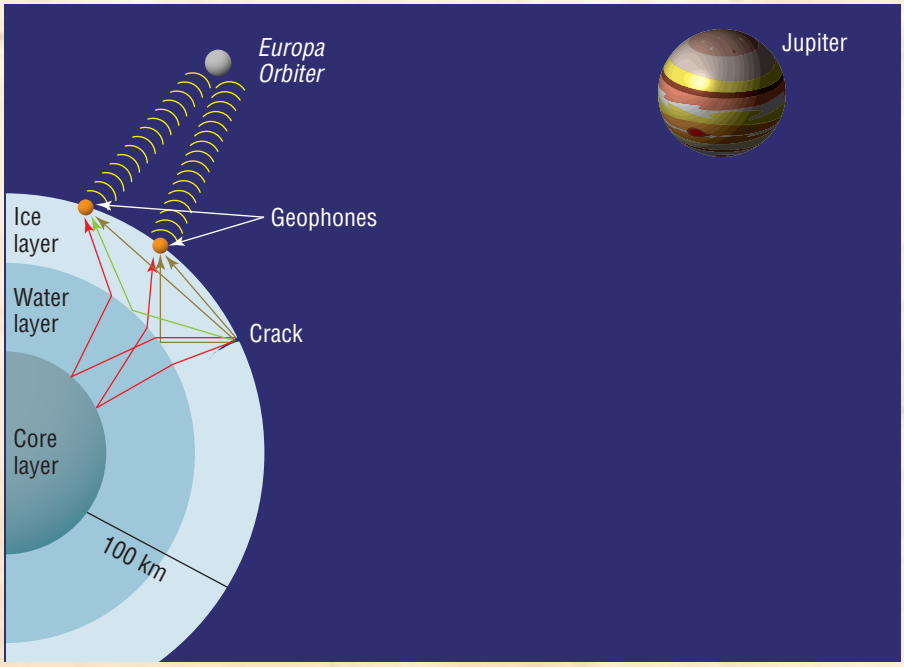
Eavesdropping from 680 million km

The challenge facing NASA scientists is how to find out if there is water under a sheet of ice that could be 10, 50 or even 100 km thick. Even on Earth that would be difficult. But Jupiter and its moons are 680 million kilometres away!

Europa Orbiter's first task will be to use radar to find out if there is water under the ice. If it does find water, the next part of the challenge begins. It must explore a thick sheet of ice and the ocean below it without landing.

Exploring with sound

Giant cracks appear in the ice on the surface of Europa about twice every minute. Each crack creates a very loud noise on the surface. The sound then travels down through the ice. Some of it bounces back when it reaches the ocean, causing an **echo** at the surface. Some of the sound bounces back when it reaches the moon's hard core, causing a second echo. The amount of time that passes between the original cracking sound and the two different echoes can be used to work out the thickness of the ice and the depth of the ocean.



There will be two echoes — one from the bottom of the ice and another from the bottom of the ocean.

To listen to these echoes, scientists will put special microphones (called geophones) in the ice on Europa's surface. Because the *Europa Orbiter* will not land on Europa, the geophones will probably be carried to the surface by rockets.

Astronomically delicious!

Icy poles delivered to your home — all the way from Europa, Jupiter

All natural, all pure ice

Available in three taste sensations
* Surface bliss
* Shallow mallow
* Deeply delicious

The deeper its icy origins, the deeper the flavour

Non-stick astropoles — gentle on your tongue

Activities

REMEMBER

1. Where on Europa is the ocean believed to be?
2. How long will it take the space probe *Europa Orbiter* to travel from Earth to Europa?
3. Explain how sound is used to measure the thickness of the ice on Europa and the depth of the ocean beneath it.
4. What is a geophone?

THINK

5. How many loud noises will a geophone detect each time the ice surface cracks?
6. What discoveries on Earth make scientists more hopeful of finding life on Europa?

INVESTIGATE

7. Radar will be used to check if there is an ocean under Europa's ice before geophones are sent down to detect sound. Radar uses radio waves rather than sound.
   (a) Find out what radar is and how it works.
   (b) What is the difference between radar and sonar?

CONNECT

8. Go to www.jaconline.com.au/science/weblinks and click on the Jupiter link for this textbook to find out more about the discoveries about the moons and rings of the planet Jupiter made by the *Galileo* space probe.

checklist

I can:

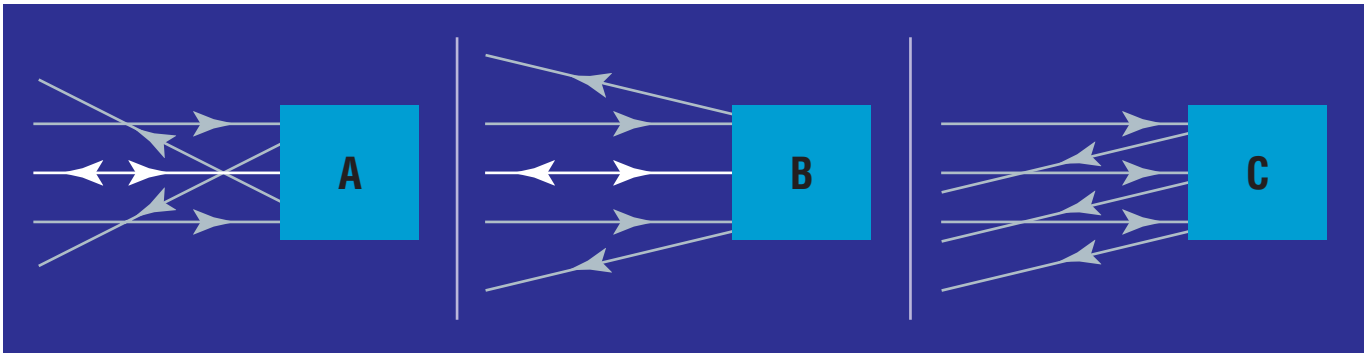
- ☐ explain how sound waves can show what lies beneath thick ice
- ☐ describe the surface of Europa
- ☐ explain why scientists think that Europa could support life.

Check and challenge

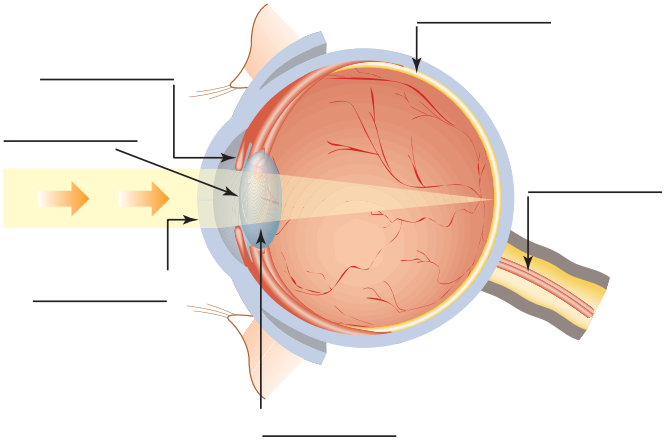
Light and sound

Changing the path of light

1. Three narrow beams of light are directed at some mystery objects. The objects are covered with a card so light can get in and out, but you can't see them. There is only one object under each square. Name and describe the object under each square.



2. (a) Why is a concave mirror useful as a solar reflector?
   (b) Use a diagram to show how a convex mirror can be used to see around corners.
3. Identify the parts of the eye that enable you to see images of the objects you look at.

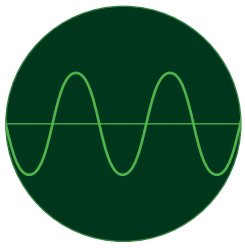


Looking at images

4. Write your own name in capital letters so that it would read correctly when you are looking at it in a plane mirror. Test it to see if you wrote it correctly.
5. Explain why you see images in a mirror while you can't see images in most other flat surfaces.
6. Which shaped mirror (plane, concave or convex) can produce images that are:
   (a) upside down compared to the object in front of the mirror?
   (b) always smaller than the object in front on the mirror?
   (c) the same shape and size as the object in front of the mirror?
   (d) larger than the object in front of the mirror?

Sound

7. The waveform shown has been produced by plucking a string on an electric guitar.
   (a) Make a copy of the waveform.
   (b) In another circle of the same size, draw a waveform that shows a louder sound.
   (c) In another circle of the same size, draw a waveform that has a higher pitch than the original waveform.



8. (a) What quantity is a measure of the number of times an object vibrates every second?
   (b) How does a sound change when this quantity increases?
9. What is ultrasound?
10. List at least five uses of ultrasound.

Comparing light and sound

11. Copy and complete the table below and use ticks to show which statements refer to light and which refer to sound. Some of the statements apply to both light and sound.

	Light	Sound
Travels through empty space at 300 000 km/s		
Can be reflected		
Always caused by vibrating objects or substances		
Can travel through transparent substances		
Cannot travel through opaque objects		
Can be measured in decibels		
Can only be produced from another form of energy		
Is detected by receptors in the human body		
Travels faster than a speeding bicycle		

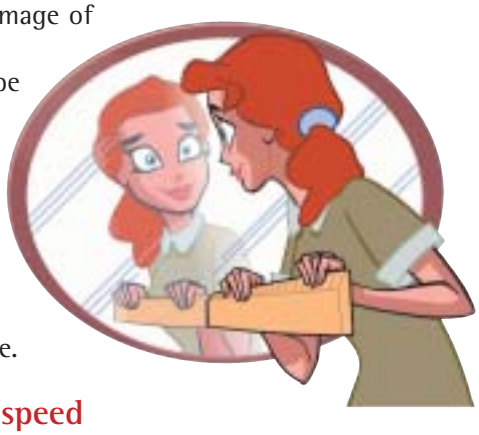
Seeing and hearing

12. How do we see objects that don't produce their own light?
13. When you blow across the top of a plastic drinking straw, you hear a sound. What could you do to the straw to make a more highly pitched note?
14. Copy and complete the table below to describe the main parts of the eye and ear and what they do.

Part	Eye or ear?	What it looks like	What it does
Auditory nerve			
Auricle			
Cochlea			
Cornea			
Eardrum			
Iris			
Lens			
Optic nerve			
Ossicles			
Pupil			
Retina			

Where is it?

1. Where is your image in a plane mirror? Stand in front of a mirror and hold a length of measuring tape or ruler to the mirror. Look at the image of you AND the measuring tape or ruler. How far behind the mirror is your image? Compare this to how far in front of the mirror you are.



The issue of speed

2. (a) How far does light travel through empty air in one hour?
   (b) How far does sound travel through empty air in one hour?
3. How could you persuade someone that light travelled much faster than sound? Include at least two examples of evidence in your answer.

Two kinds of waves

Read the paragraph below before answering questions 4–7.

Light and sound energy travel as waves. That means that energy travels from one place to another, but no material moves from one place to another. Sound waves are made up of vibrating particles. But the particles only move backwards and forwards 'on the spot'. They don't actually go anywhere. Light travels as electromagnetic waves. That is, light is made up of electric and magnetic pushes and pulls. Light waves can travel through empty space because those electric and magnetic pushes and pulls can happen in empty space.

4. When you want to attract someone's attention you could yell out 'hey you' or walk over to them and tap the person on the shoulder.
   (a) Which method is an example of energy travelling in waves?
   (b) Why is the other method NOT an example of energy travelling in waves?
5. Why are sound waves sometimes called compression waves?
6. Why can't sound waves travel through completely empty space?
7. Why are light waves able to travel through empty space?

Summary of key terms

absorb: take in something such as light, sound or heat energy. The energy taken in can be stored or released as a different form of energy.

aqueous humour: the clear substance at the front of the eyeball between the lens and the cornea

auditory nerve: a large nerve that sends signals to the brain from the hearing receptors in the cochlea

auricle: the outside part of the ear

bioluminescence: the release of light energy from a living thing

biosonar: the use of echolocation by living things

cathode ray oscilloscope: an instrument with a small screen that can be used to display the waveform of a sound. It does this by converting sound energy into electrical signals.

cells: the tiny units that make up all living things. There are many differently sized and shaped cells in the human body.

cochlea: the snail-shaped part of the inner ear. It is lined with tiny hairs and the hearing receptors.

cochlear implant: a device inserted in the ear to allow vibrations caused by sound to reach receptors in the inner ear. Also known as the bionic ear.

concave mirror: a mirror with its reflecting surface curved inwards

convex mirror: a mirror with its reflecting surface curved outwards

cornea: the clear, curved outer surface of the front of the eye

decibels: the unit of sound level. Abbreviation is dB. The sound level is measured with a sound-level meter.

diffuse reflection: the type of reflection that happens at a surface that is not smooth. Diffuse reflection does not produce an image.

ear canal: the tube that leads from the outside of the ear to the eardrum

eardrum: a thin piece of stretched skin and muscle that vibrates when sound energy reaches it

echo: a sound that is heard a second time when it is reflected

echolocation: the use of sound to locate objects by detecting echoes

energy: there are many forms of energy, including light, sound and heat. All forms of energy can cause change or make things happen.

Eustachian tube: a tube that joins the middle ear to the throat. It opens and closes to make sure that the air pressure is the same on both sides of the eardrum.

focal length: the distance between the centre of a curved mirror or lens and the focus

focus: the point at which light rays coming from one direction meet after being reflected or bent

frequency: the number of vibrations each second

hearing receptor: tiny cell in the cochlea that detects vibrations caused by sound and sends signals to the brain

Hertz: the unit of frequency. Abbreviation is Hz. One Hertz is equal to one vibration each second. So, for example, 500 Hz is equal to 500 vibrations each second.

image: a view of an object

iris: the coloured part of the eye that opens and closes to control the amount of light that gets into the eye

laterally inverted: reversed sideways

lens: a transparent curved object that bends light towards, or away from a point called a focus

light ray: very thin beam of light

luminous: releasing its own light

non-luminous: describes objects that don't give off their own light. Non-luminous objects can only be seen because they reflect light.

opaque: describes substances that don't allow any light to pass through

optic nerve: a large nerve that sends signals to the brain from the sight receptors in the retina

ossicles: a set of three tiny bones that send the vibrations from the eardrum to the inner ear. They also make the vibrations larger.

oval window: an egg-shaped hole covered with a thin tissue. It is the entrance from the middle ear to the inner ear.

phosphorescence: the slow release of light of one colour from a chemical after it is exposed to light

pitch: the highness or lowness of a sound. The pitch that you hear depends on the frequency of the vibrating air.

plane mirror: a flat mirror

pressure: the amount of push, or force, on a certain area

pupil: the hole through which light gets into the eye

receptors: special cells that detect energy and convert it to electrical energy that is sent to the brain

reflect: bounce off

regular reflection: the type of reflection that happens at a smooth, shiny surface, like glass or a flat mirror. Regular reflection produces an image.

retina: the curved surface at the back of the eye. It is lined with the sight receptors.

reverberation: the longer lasting sound caused by repeated reflection from hard surfaces

scattering: reflection of light in many directions from small particles or droplets

semicircular canals: three large, curved tubes in your inner ear that control your sense of balance

sonar: the use of reflected sound waves to locate objects under water

sound level: a measurement of how much the air is disturbed by a sound. Sound level is measured in decibels.

transferred: moved from one place to another

translucent: describes a substance that scatters some light and allows some light to pass through. It is possible to make out large shapes through translucent substances, but it isn't possible to see clearly through them.

transmitted: the passing through of something, for example, light or sound

transparent: describes a substance that allows most light to pass through. You can see clearly through transparent substances.

ultrasound: sound with frequencies too high for humans to hear

vibrations: repeated back-and-forth movements

waveform: a graph that shows how the amount of energy a sound has changes with time

The laboratory toolbox

A laboratory contains many specialised pieces of equipment. It is important to know how to use the equipment safely and accurately before attempting any practical experiments. You must also be familiar with techniques for heating, mixing, measuring and pouring before beginning. After completing an experiment, you need to record results and explain your observations.

Use this chapter as a guide to safety and laboratory techniques.

Working safely

Safety rules

Safety rules

Always…

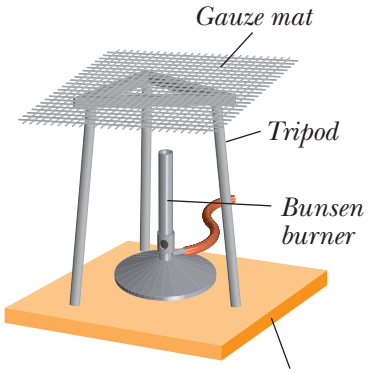
- Follow the teacher's instructions
- Push in chairs and keep walkways clear during experiments
- Tie back long hair
- Wear safety glasses and laboratory coats, unless instructed not to by the teacher
- Keep books and papers away from heating equipment and chemicals
- Inform your teacher immediately of spills or broken equipment
- Inform your teacher immediately of any cuts or burns
- Point the mouths of test tubes away from yourself and other students
- Place heating equipment and chemicals on heatproof mats
- Wait until heating equipment has cooled down before putting it away
- Clean up and wash your hands thoroughly after doing an experiment.

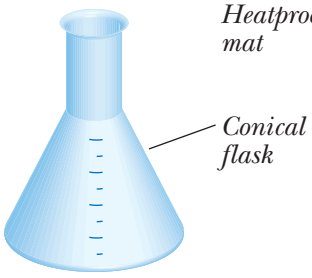
Never…

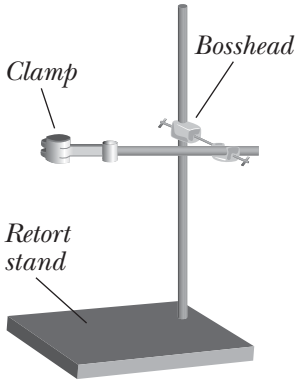
- Enter the laboratory without a teacher present
- Enter the preparation rooms
- Eat or drink in the laboratory
- Sit on the benches or tables
- Run or push in the laboratory
- Conduct experiments or touch equipment unless your teacher tells you to
- Put solid objects, like matches, in the sink
- Pour hazardous chemicals down the sink (check with your teacher)
- Smell or taste chemicals unless your teacher says it's OK
- Leave an experiment unattended
- Operate electrical switches with wet hands.

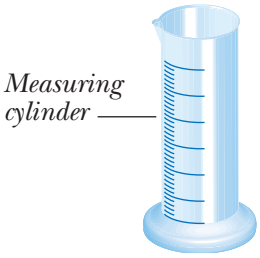
Laboratory equipment

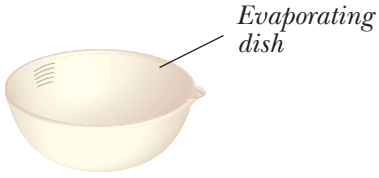
EQUIPMENT	USE
Beaker	Container for mixing or heating substances
Bosshead	Holds the clamp to a retort stand
Bunsen burner	Heats substances
Clamp	Holds objects at a required height along a retort stand
Conical flask	Container for mixing substances or collecting filtered substances
Evaporating dish	Container for evaporating small amounts of substances over a Bunsen burner
Filter funnel	Used with filter paper to filter substances
Gauze mat	Supports containers over a Bunsen burner while they are heated. Spreads heat evenly under the container
Heatproof mat	Protects benches from damage
Measuring cylinder	Used to measure volume accurately
Retort stand	Used with clamps and bossheads to hold substances at a required height
Safety glasses	Protects eyes
Spatula	Used to pick up small amounts of solid substances
Stirring rod	Used to stir mixtures
Test tube	Container for holding, heating or mixing small amounts of substances
Test-tube holder	Holds test tubes while they are being heated
Test-tube rack	Holds test tubes upright
Tongs	Used to pick up and hold small solids while they are heated
Tripod	Supports gauze mats over Bunsen burners

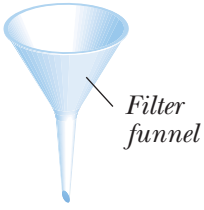


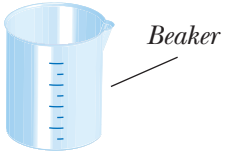


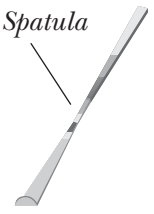


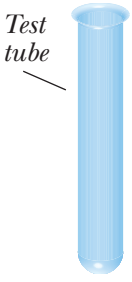


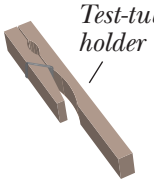


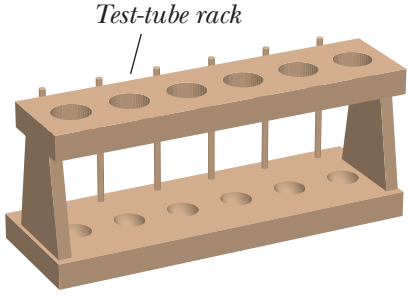






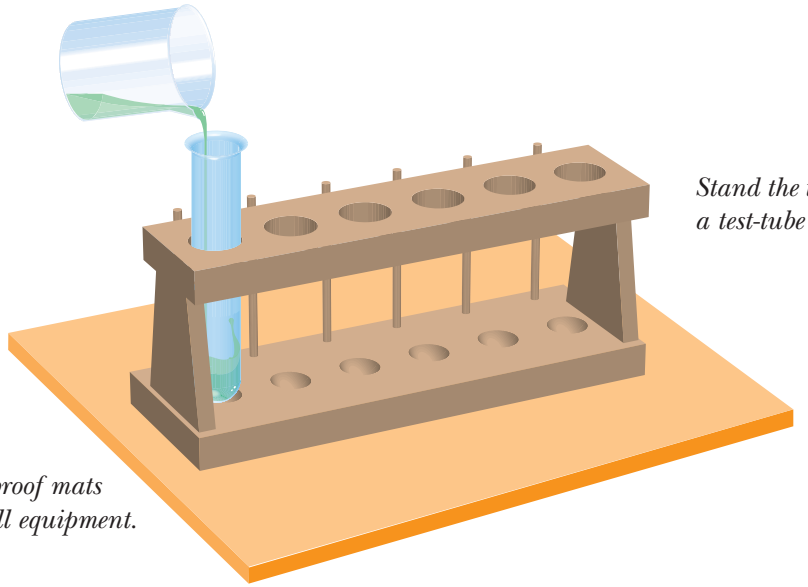






Pouring a liquid into a test tube

Pour liquids carefully into the test tube from a beaker or measuring cylinder. Use a filter funnel when pouring from bottles or containers without a lip.

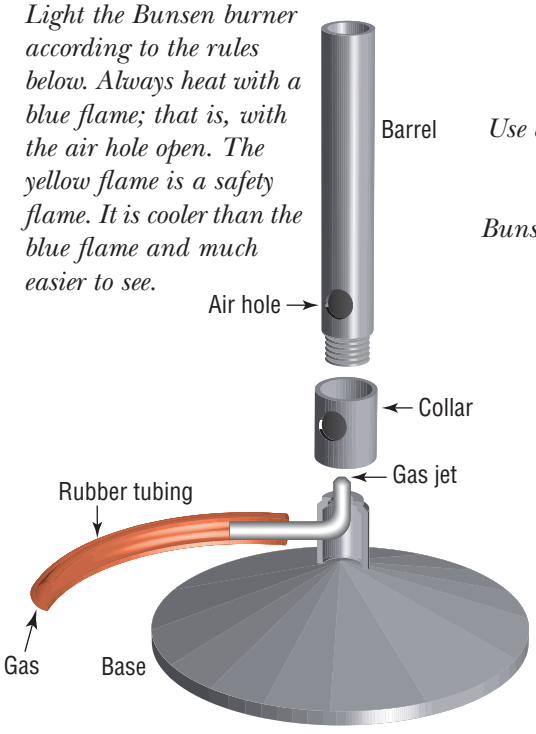


Stand the test tube in a test-tube rack.

Use heatproof mats under all equipment.

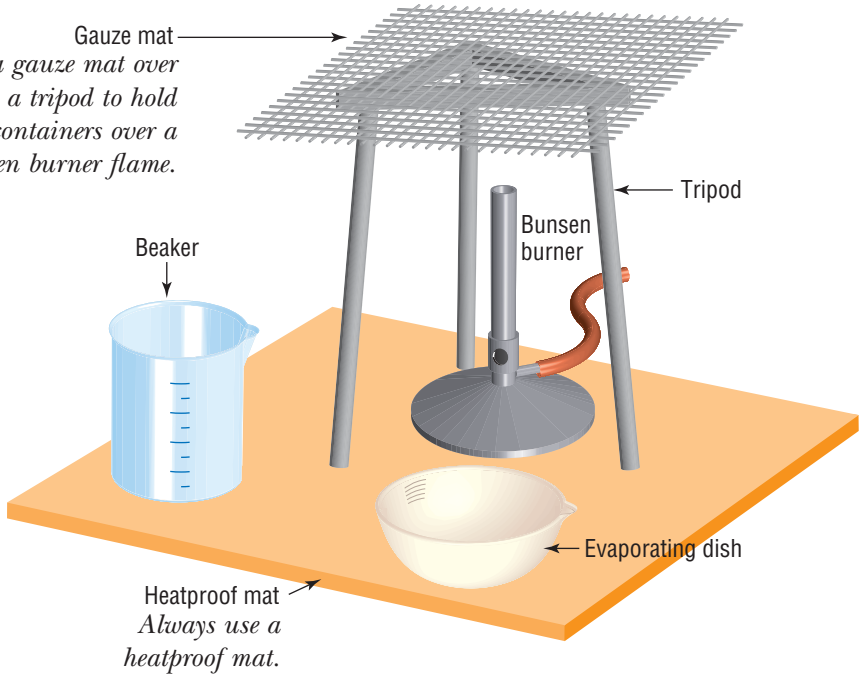
Using a Bunsen burner

Light the Bunsen burner according to the rules below. Always heat with a blue flame; that is, with the air hole open. The yellow flame is a safety flame. It is cooler than the blue flame and much easier to see.



Beakers and evaporating dishes can be placed straight onto a gauze mat for heating. Never look directly into a container while it is being heated. Wait until the equipment has cooled properly before handling it.

Gauze mat
Use a gauze mat over a tripod to hold containers over a Bunsen burner flame.



1. Place the Bunsen burner on a heatproof mat.
2. Check that the gas tap is in the *off* position.
3. Connect the rubber hose to the gas tap.
4. Close the air hole of the Bunsen burner.
5. Light the match and hold it over the barrel of the Bunsen burner.
6. Turn on the gas tap.
7. Adjust the flame.
8. Remember to keep long hair tied back. Wear safety glasses and a laboratory coat at all times.

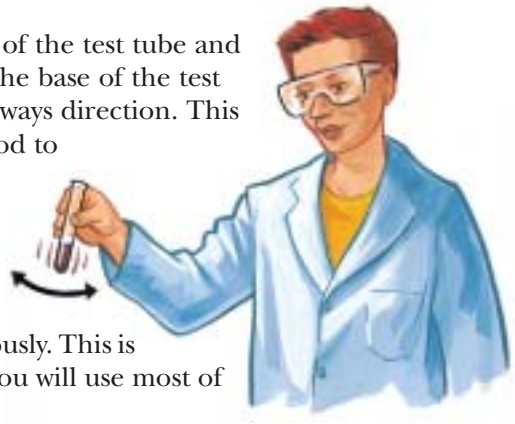
Heating a test tube

Tripods and gauze mats are not used when heating test tubes. Hold the test tube with the test-tube holder. Keep the base of the test tube above the flame. Make sure that the test tube points away from you and other students.

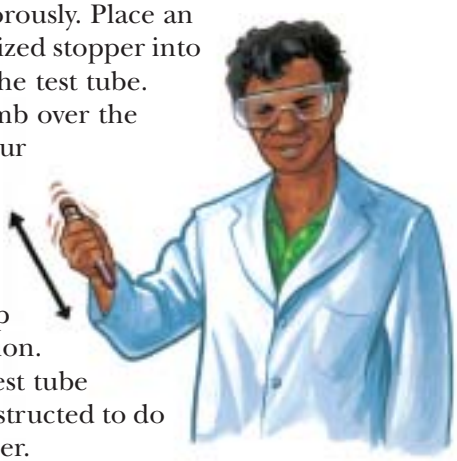
Shaking a test tube

There are two ways to shake substances in a test tube.

Method 1:
Hold the top of the test tube and gently move the base of the test tube in a sideways direction. This method is good to use with non-hazardous substances that do not need to be shaken vigorously. This is the method you will use most of the time.



Method 2:
Use a stopper when a substance needs to be mixed by shaking vigorously. Place an appropriately sized stopper into the mouth of the test tube. With your thumb over the stopper and your hand securely around the test tube, shake the test tube with an up and down motion. Only shake a test tube in this way if instructed to do so by the teacher.

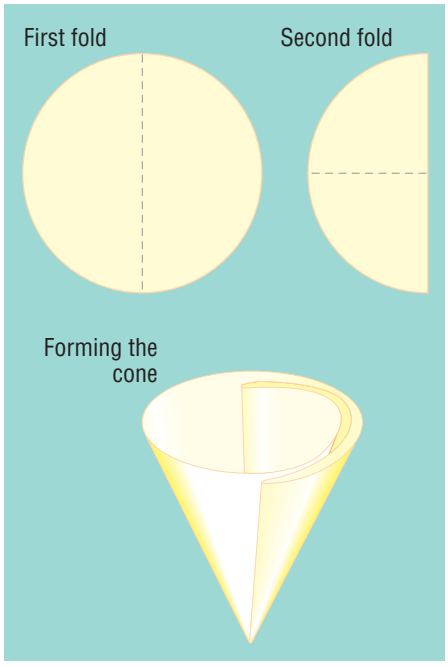


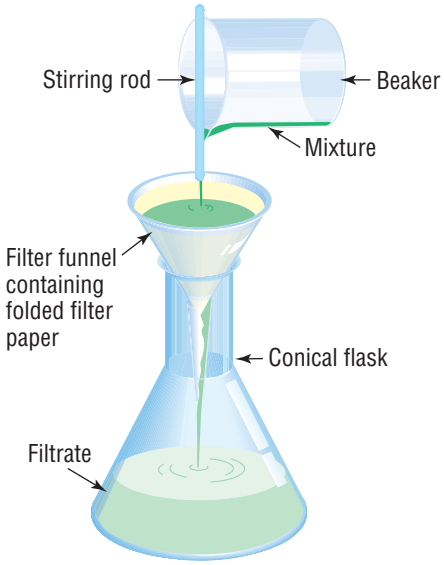
Filtering mixtures

Place the folded filter paper in the filter funnel.

Moisten the filter paper or filter funnel with clean water, to hold it in place.

Gradually pour the mixture into the filter funnel. Be careful not to overflow the filter funnel or allow the mixture to be poured between the filter paper and the funnel.





Using electricity safely

Never:

- place heavy appliances on the edge of a bench
- allow water near electrical cords or plugs
- place objects, other than electrical plugs, into an electrical switch
- use appliances with damaged cords or exposed wires.

Disposing of chemicals safely

Always:

- follow the teacher's directions for the safe disposal of different chemicals.

Never:

- put solid materials down the sink
- put hot objects in the bin. Wait for them to cool down first.
- put broken glass in a classroom bin. The teacher will dispose of it in a separate bin using a dustpan or gloves.

Measuring things

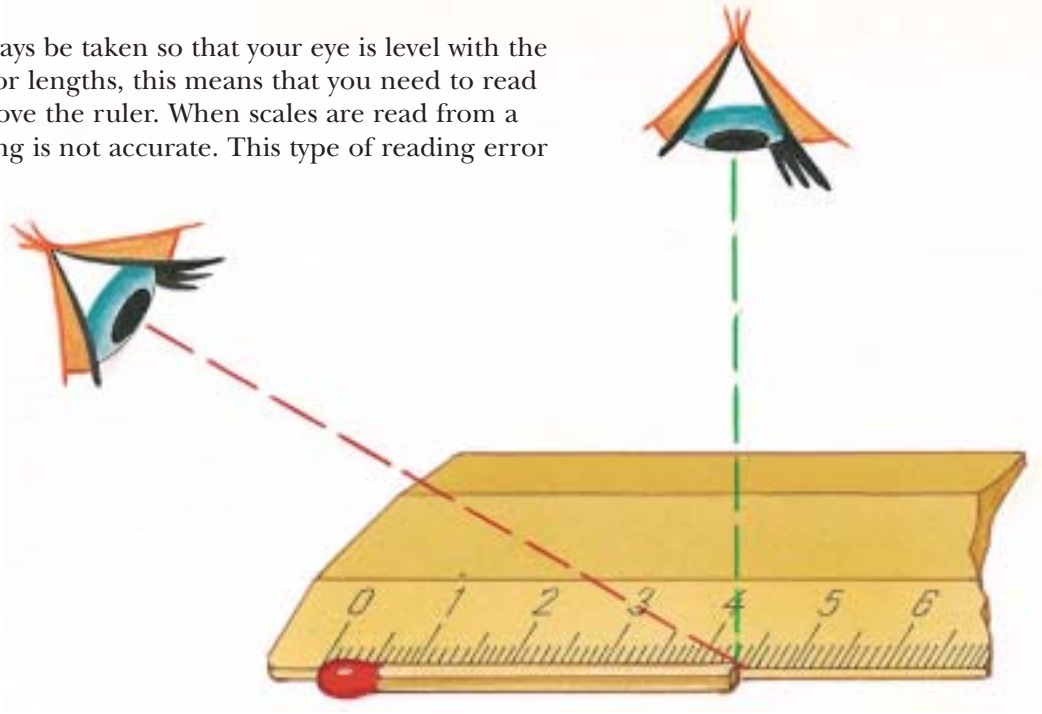
Measuring length

The standard unit for measuring length is the metre (m). But length can also be measured in millimetres (mm), centimetres (cm) or kilometres (km).

Make sure you read a ruler from directly above it.
The length of this match is 4.4 cm.

Parallax error

Measurements should always be taken so that your eye is level with the reading you are taking. For lengths, this means that you need to read the scale from directly above the ruler. When scales are read from a different angle, the reading is not accurate. This type of reading error is called **parallax error**.



Measuring temperature

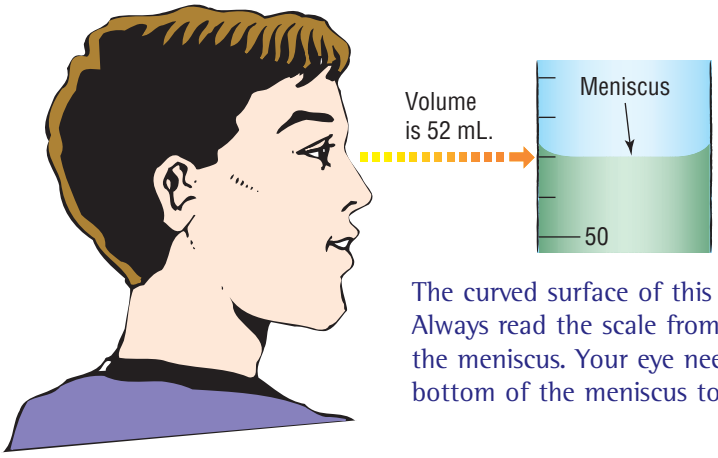
A thermometer is used to measure temperature. Temperature is often measured in degrees Celsius (°C).

The thermometers used in schools are filled with alcohol, dyed red so that it is easier to see. When using thermometers remember these points:

- Ensure that the bulb of the thermometer is fully covered by the substance you are taking the temperature of.
- Never rest the bulb on the bottom of a container; the bottom of a beaker being heated may be hotter than its contents.
- Never use a thermometer as a stirring rod.
- Read a thermometer as soon as the level of the alcohol stops rising or falling.
- Read the thermometer with your eye level at the top of the alcohol column. A reading taken from any other angle will result in parallax error.
- Do not rest a thermometer on a bench where it is likely to roll off.
- Immediately report any breakage of thermometers to the teacher. Broken thermometers are very sharp.

Measuring volume

Volume measures how much space a substance takes up. The standard unit for volume is litres (L), but in a laboratory, it is usually measured in millilitres (mL) with a measuring cylinder.

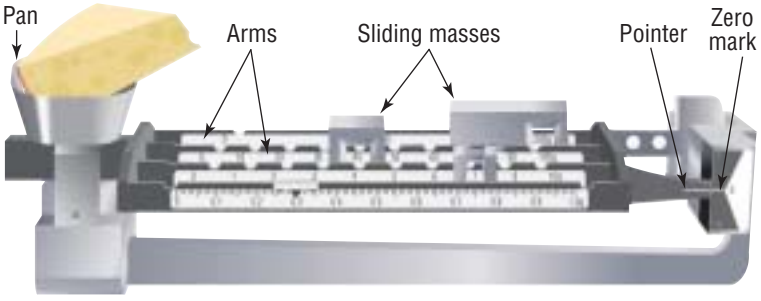


The curved surface of this liquid is called a **meniscus.** Always read the scale from the bottom, flat section of the meniscus. Your eye needs to be level with the bottom of the meniscus to avoid parallax error.

Measuring mass

Mass measures the amount of matter in a substance. The standard unit for mass is kilograms (kg), but in the laboratory, you are most likely to measure mass in grams (g). Mass is measured with a balance. It may be either an electronic balance or a beam balance. When using balances:

- Make sure that the balance is 'zeroed' before using it. An electronic balance is zeroed by pressing the zero or tare button. Zero a beam balance by sliding all of the masses to the zero notches.
- Place the object on the electronic or beam balance.
- Read the mass. If you are using an electronic balance, you can read the mass straight from the screen. If you are using a beam balance, the pointer will have moved. By 'zeroing' the pointer, the mass can be determined. Do this by moving the heaviest sliding mass towards the pointer. Slide it as far as possible without the arm overbalancing the zero, or balance, mark. Repeat this for the other arms, except the one with the smallest sliding mass. The smallest mass should balance the pointer, so that it lines up with the zero or balance mark. Add the masses on each of the arms to determine the total mass.



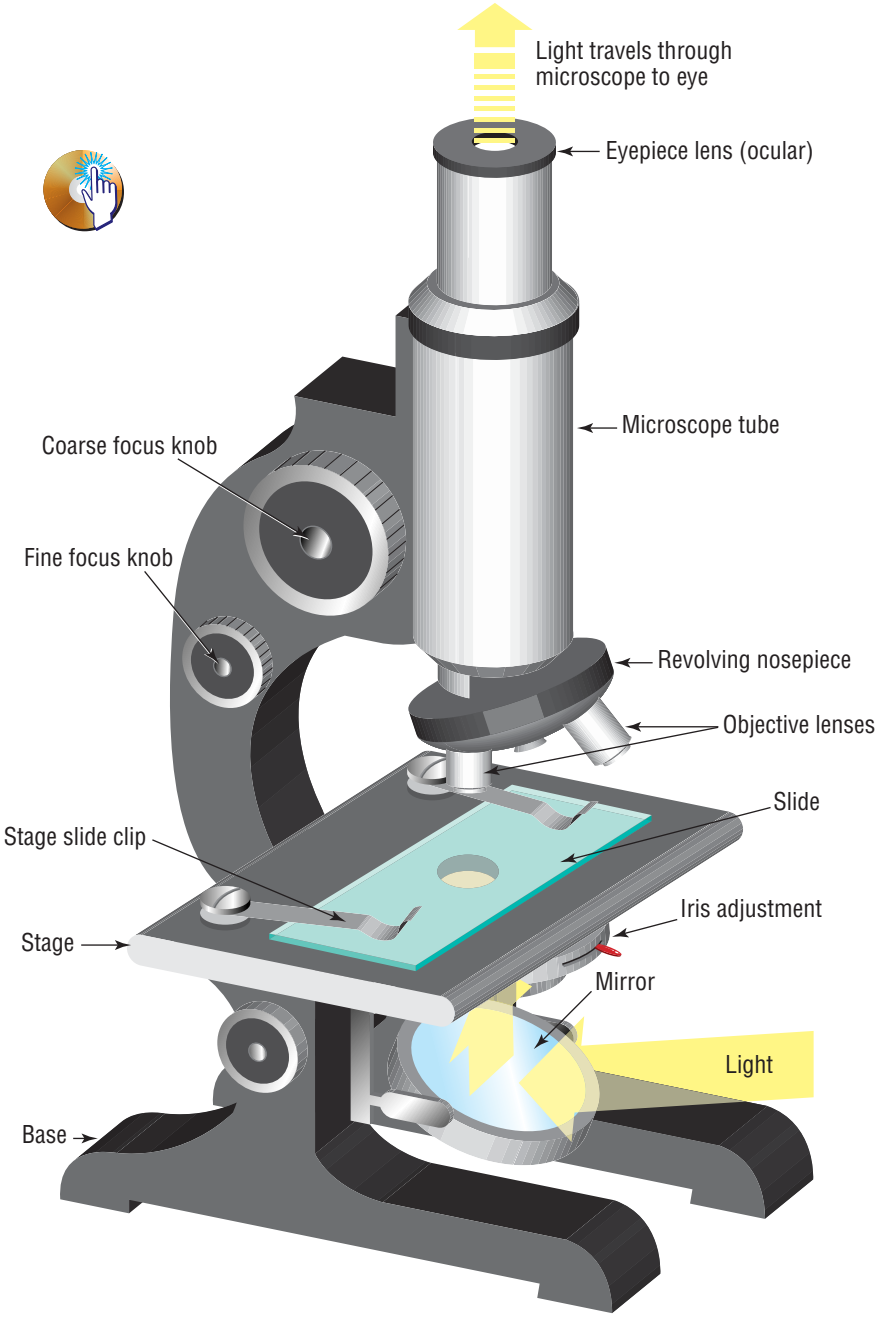
200 g + 40 g + 7 g + 0.3 g = 247.3 g

Sensitive equipment

Using a microscope

Microscopes make small objects easier to see. The microscopes commonly used in schools can magnify objects up to 400 times. The total magnification of a microscope can be calculated by multiplying the magnification of the eyepiece lens by the magnification of the objective lens. For example, a 10× eyepiece and a 20× objective lens would provide: 10 multiplied by 20 = 200 times magnification.

Microscopes are very expensive and are easily damaged if they are not handled carefully.



The microscopes at your school may look slightly different from this one. Some microscopes have a built in light. Microscopes with built in lights do not have a mirror and do not require a separate microscope lamp.

Rules for handling a microscope:

- Always use two hands when carrying a microscope: one on the arm of the microscope and the other under the base.
- Place the microscope securely on a flat surface, away from the edge.
- Never shine sunlight directly up the microscope tube. You could damage your eyes.
- Use only lens tissues to clean microscope lenses: never use your fingers.

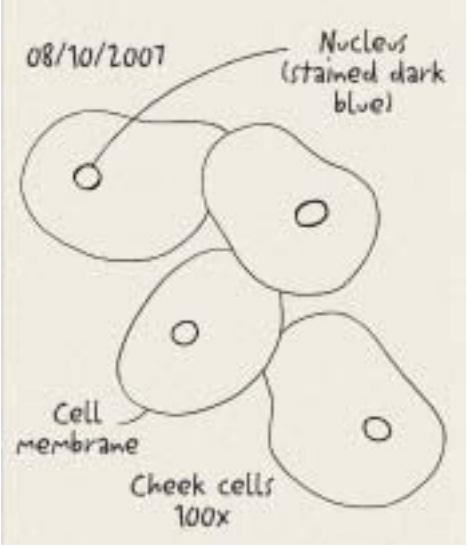
More hints for using a microscope:

- Begin focusing a microscope on the lowest magnification.
- Focus a microscope by beginning with the coarse focus. Look from the side and adjust the objective lens so that it is just above the microscope slide.
- Turn the coarse focus knob to move the tube up until the object comes into view.
- Turn the fine focus to make the image of the object as clear as possible.

Drawing what you see

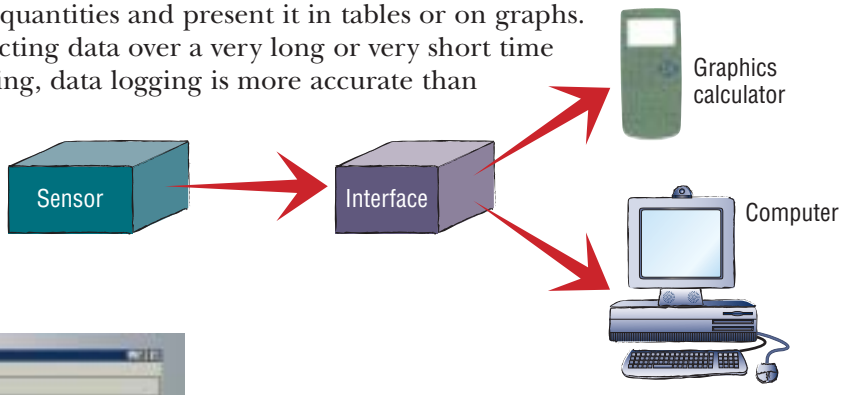
To effectively draw what you see through a microscope, follow these instructions:

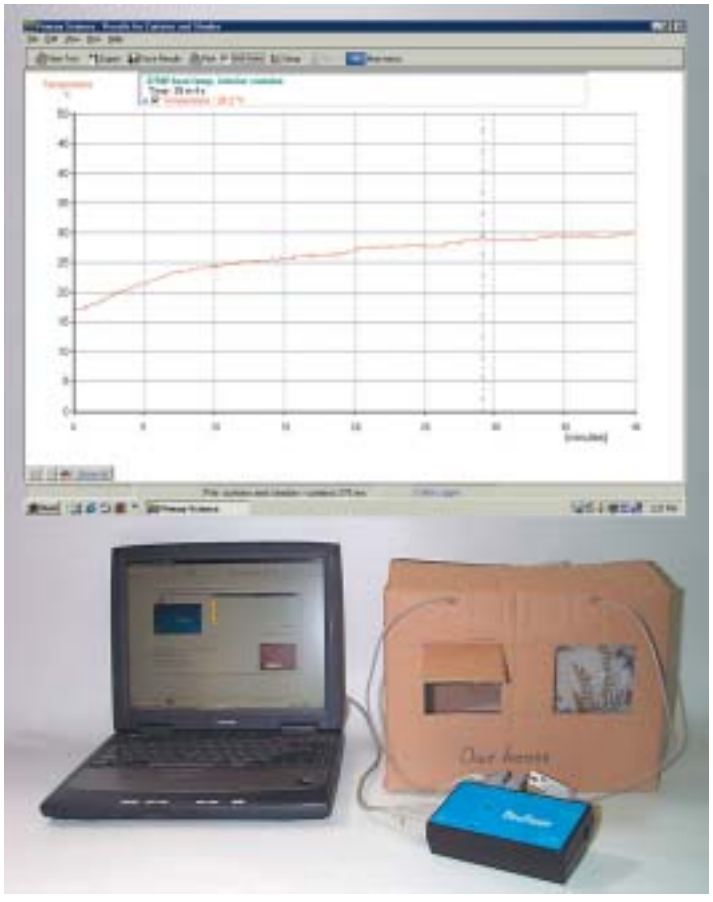
- Always use pencil.
- Keep the diagrams simple. Don't include shading or colours. Draw only the lines and shapes you see.
- Do not attempt to draw everything you see, especially if there are many cells. Draw three or four representative, or typical, cells.
- The diagram should be quite large — about a third of a page.
- Label the structures you can identify.
- Record the date, the name of the specimen and the magnification setting on the microscope.



Using data-logging equipment

Data-logging equipment is used to measure quantities and present it in tables or on graphs. Data loggers are particularly useful for collecting data over a very long or very short time span. In many cases, for example, when timing, data logging is more accurate than measuring time with a clock, watch or stopwatch. Data loggers consist of two main parts, a sensor (or probe) and an interface. The interface can be plugged into either a graphics calculator or a computer.





Data loggers can be used to measure temperature, time, distance, magnetism, force, sound level and much more. Once a data logger has collected data, it is important to analyse the data. What does the data mean? How can it be used?

Keeping a record

Constructing a table

Always include the units used.

The heading for each column is a clear label of what has been measured.

Distance (cm)	Time for ant to travel between markers (s)
0	0
2	3
4	7
6	8
8	12

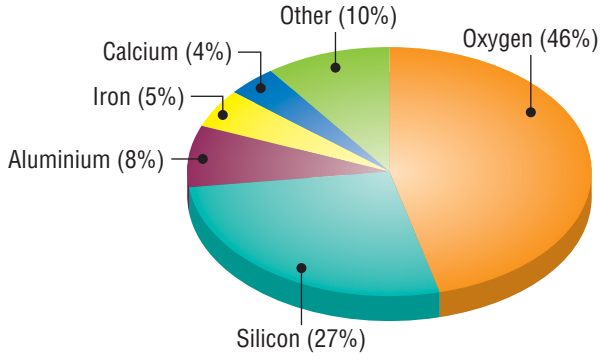
Use a ruler to draw lines for rows, columns and borders.

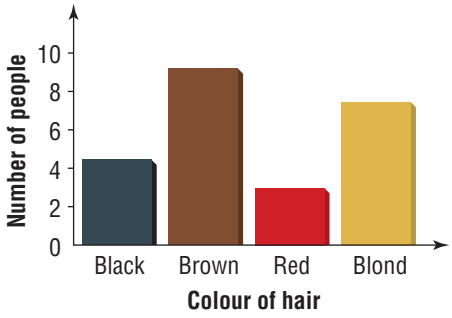
Enter the data in the body of the table. Do not use units in this part of the table.

Graphing

Graphs are used to make data easier to interpret. The type of graph used depends on the type of data to be displayed.

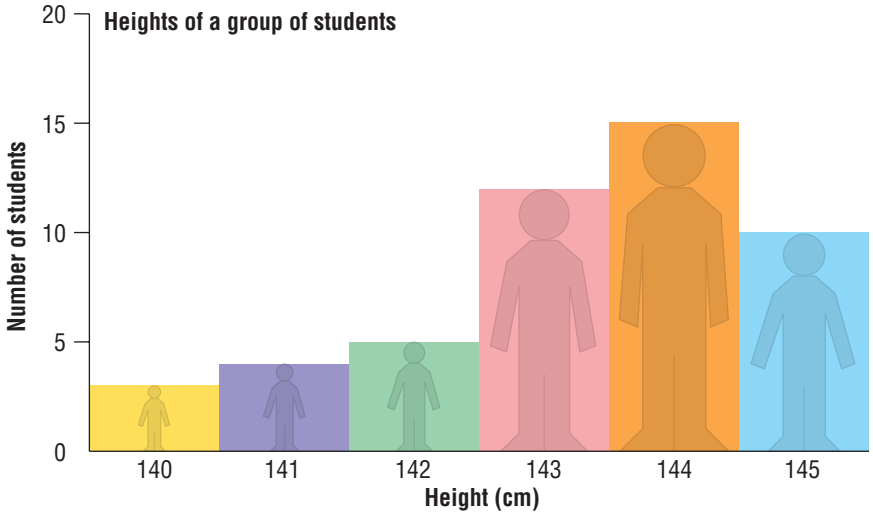
Pie charts are useful for showing the parts that make up a whole. For example, a pie chart can be used to show the percentages of different substances in the Earth's crust.



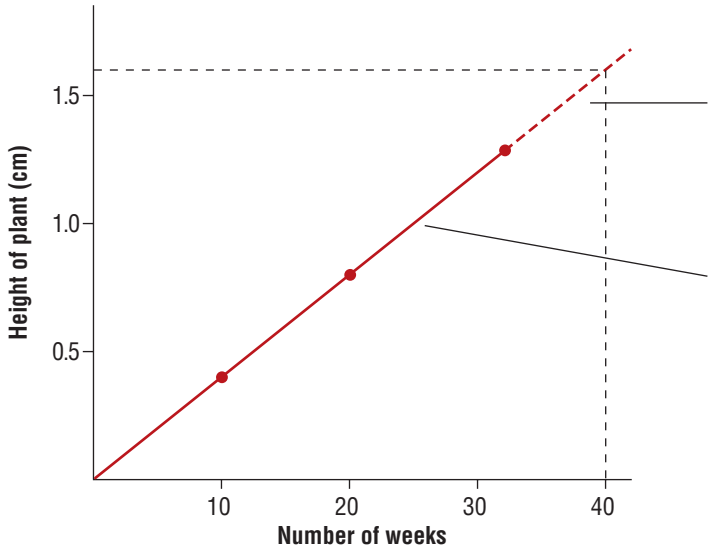


Bar and column graphs are used to display data that is not **continuous**; one piece of data does not relate exactly to the next. For example, a bar graph can be used to show the number of students in a class with each hair colour.

Histograms are bar graphs with the bars touching. They are often used to display the results of a survey. The heights of a group of students could be represented this way. Notice that the heights are recorded along the axis and in the middle of the bar.



Lines graphs have a horizontal *x*-axis and a vertical *y*-axis. They are often used to represent data gathered in laboratory experiments that are continuous. A line graph is used to show how something *changes*. For example, line graphs could be used to show how quickly a plant grows.



A line graph can be used to predict what might happen in the future.

Line graphs are useful for working out values between those that you actually observed.

Reporting your findings

Writing a laboratory report

All practical reports should include your name, your partners' names, the date and the title of the experiment. Each of the following must also be included in the report.

Aim

One or two short sentences that describe the reason for doing the experiment. The aim can also include an **hypothesis —** a guess about the results of an experiment. The aim may begin something like:

To find out . . .

To investigate . . .

To compare . . .

To make . . .

Materials

A list of the equipment and chemicals used

Method

The set of steps you followed to do the experiment. Scientific diagrams are used to describe how equipment was set up. The method needs to be clear enough so that anyone reading the report would be able to repeat your experiment.

Results

Measurements and observations made during the experiment. Measurements can be recorded in a table or on a graph.

Discussion

What the results show and a possible explanation of your results. Any difficulties you had or ideas for improving the experiment are also written in this section.

Conclusion

A short statement describing what you found out. The conclusion needs to relate back to the aim.

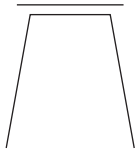
Drawing laboratory equipment

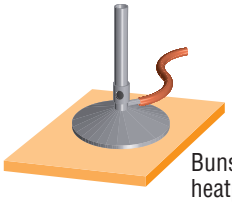
Scientific drawings can be used in laboratory reports to show how equipment was set up. It is important for the drawings to be clear and easy to understand. When drawing scientific diagrams, you should:

- Always draw in pencil.
- Use a ruler to draw straight lines.
- Label the equipment drawn.
- Draw only a cross-section of the equipment.
- Do not put lines across the top of glassware.

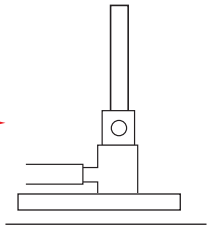
Here are some ideas for simplifying scientific diagrams.

Tripod and gauze mat

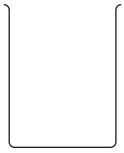




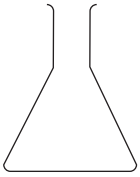
Bunsen burner and heatproof mat



Beaker

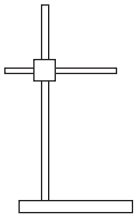


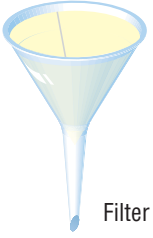
Conical flask



Test tube

Retort stand, bosshead and clamp





Filter funnel and filter paper